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# Contents

Intellectual Property Rights ..... 2
Legal Notice ..... 2
Modal verbs terminology ..... 2
Foreword ..... 14
1 Scope ..... 15
2 References ..... 15
3 Definitions, symbols and abbreviations ..... 17
3.1 Definitions ..... 17
3.2 Symbols ..... 18
3.3 Abbreviations ..... 18
4 General ..... 19
4.1 Procedure specification principles ..... 19
4.2 Forwards and backwards compatibility ..... 19
4.3 Specification notations ..... 19
5 X2AP services ..... 20
5.1 X2AP procedure modules ..... 20
5.2 Parallel transactions ..... 20
6 Services expected from signalling transport ..... 20
7 Functions of X2AP ..... 20
8 X2AP procedures ..... 23
8.1 Elementary procedures ..... 23
8.2 Basic mobility procedures ..... 27
8.2.1 Handover Preparation ..... 27
8.2.1.1 General ..... 27
8.2.1.2 Successful Operation ..... 27
8.2.1.3 Unsuccessful Operation ..... 31
8.2.1.4 Abnormal Conditions ..... 31
8.2.2 SN Status Transfer ..... 32
8.2.2.1 General ..... 32
8.2.2.2 Successful Operation ..... 33
8.2.2.3 Abnormal Conditions ..... 34
8.2.3 UE Context Release ..... 34
8.2.3.1 General ..... 34
8.2.3.2 Successful Operation ..... 34
8.2.3.3 Unsuccessful Operation ..... 36
8.2.3.4 Abnormal Conditions ..... 36
8.2.4 Handover Cancel ..... 36
8.2.4.1 General ..... 36
8.2.4.2 Successful Operation ..... 36
8.2.4.3 Unsuccessful Operation ..... 37
8.2.4.4 Abnormal Conditions ..... 37
8.2.5 Handover Success ..... 37
8.2.5.1 General ..... 37
8.2.5.2 Successful Operation ..... 37
8.2.5.3 Unsuccessful Operation ..... 37
8.2.5.4 Abnormal Conditions ..... 38
8.2.6 Conditional Handover Cancel ..... 38
8.2.6.1 General ..... 38
8.2.6.2 Successful Operation ..... 38
8.2.6.3 Unsuccessful Operation ..... 38
8.2.6.4 Abnormal Conditions ..... 38

8.2.7 Early Status Transfer ..... 39
8.2.7.1 General ..... 39
8.2.7.2 Successful Operation ..... 39
8.2.7.3 Abnormal Conditions ..... 40
8.3 Global Procedures ..... 40
8.3.1 Load Indication ..... 40
8.3.1.1 General ..... 40
8.3.1.2 Successful Operation ..... 41
8.3.1.3 Unsuccessful Operation ..... 42
8.3.1.4 Abnormal Conditions ..... 42
8.3.2 Error Indication ..... 42
8.3.2.1 General ..... 42
8.3.2.2 Successful Operation ..... 43
8.3.2.3 Unsuccessful Operation ..... 44
8.3.2.4 Abnormal Conditions ..... 44
8.3.3 X2 Setup ..... 44
8.3.3.1 General ..... 44
8.3.3.2 Successful Operation ..... 44
8.3.3.3 Unsuccessful Operation ..... 46
8.3.3.4 Abnormal Conditions ..... 46
8.3.4 Reset ..... 46
8.3.4.1 General ..... 46
8.3.4.2 Successful Operation ..... 47
8.3.4.3 Unsuccessful Operation ..... 47
8.3.4.4 Abnormal Conditions ..... 47
8.3.5 eNB Configuration Update ..... 48
8.3.5.1 General ..... 48
8.3.5.2 Successful Operation ..... 48
8.3.5.3 Unsuccessful Operation ..... 50
8.3.5.4 Abnormal Conditions ..... 50
8.3.6 Resource Status Reporting Initiation ..... 50
8.3.6.1 General ..... 50
8.3.6.2 Successful Operation ..... 51
8.3.6.3 Unsuccessful Operation ..... 52
8.3.6.4 Abnormal Conditions ..... 52
8.3.7 Resource Status Reporting ..... 53
8.3.7.1 General ..... 53
8.3.7.2 Successful Operation ..... 53
8.3.7.3 Unsuccessful Operation ..... 54
8.3.7.4 Abnormal Conditions ..... 54
8.3.8 Mobility Settings Change ..... 54
8.3.8.1 General ..... 54
8.3.8.2 Successful Operation ..... 54
8.3.8.3 Unsuccessful Operation ..... 55
8.3.8.4 Abnormal Conditions ..... 55
8.3.9 Radio Link Failure Indication ..... 55
8.3.9.1 General ..... 55
8.3.9.2 Successful Operation ..... 55
8.3.9.3 Unsuccessful Operation ..... 56
8.3.9.4 Abnormal Conditions ..... 56
8.3.10 Handover Report ..... 56
8.3.10.1 General ..... 56
8.3.10.2 Successful Operation ..... 56
8.3.10.3 Unsuccessful Operation ..... 57
8.3.10.4 Abnormal Conditions ..... 57
8.3.11 Cell Activation ..... 57
8.3.11.1 General ..... 57
8.3.11.2 Successful Operation ..... 57
8.3.11.3 Unsuccessful Operation ..... 58
8.3.11.4 Abnormal Conditions ..... 58
8.3.12 X2 Removal ..... 58
8.3.12.1 General ..... 58

8.3.12.2 Successful Operation....58
8.3.12.3 Unsuccessful Operation....59
8.3.12.4 Abnormal Conditions....59
8.3.13 Retrieve UE Context....59
8.3.13.1 General....59
8.3.13.2 Successful Operation....59
8.3.13.3 Unsuccessful Operation....61
8.3.13.4 Abnormal Conditions....61
8.3.14 EN-DC X2 Removal....61
8.3.14.1 General....61
8.3.14.2 Successful Operation....61
8.3.14.3 Unsuccessful Operation....62
8.3.14.4 Abnormal Conditions....63
8.3.15 Data Forwarding Address Indication....63
8.3.15.1 General....63
8.3.15.2 Successful Operation....63
8.3.15.3 Unsuccessful Operation....64
8.3.15.4 Abnormal Conditions....64
8.3.16 Access and Mobility Indication....65
8.3.16.1 General....65
8.3.16.2 Successful Operation....65
8.3.16.3 Abnormal Conditions....65
8.4 X2 Release....65
8.4.1 General....65
8.4.2 Successful Operation....65
8.4.3 Unsuccessful Operation....66
8.4.4 Abnormal Condition....66
8.5 X2AP Message Transfer....66
8.5.1 General....66
8.5.2 Successful Operation....66
8.5.3 Unsuccessful Operation....66
8.5.4 Abnormal Condition....66
8.6 Procedures for Dual Connectivity....67
8.6.1 SeNB Addition Preparation....67
8.6.1.1 General....67
8.6.1.2 Successful Operation....67
8.6.1.3 Unsuccessful Operation....69
8.6.1.4 Abnormal Conditions....69
8.6.2 SeNB Reconfiguration Completion....70
8.6.2.1 General....70
8.6.2.2 Successful Operation....70
8.6.2.3 Abnormal Conditions....70
8.6.3 MeNB initiated SeNB Modification Preparation....70
8.6.3.1 General....70
8.6.3.2 Successful Operation....71
8.6.3.3 Unsuccessful Operation....73
8.6.3.4 Abnormal Conditions....73
8.6.4 SeNB initiated SeNB Modification....74
8.6.4.1 General....74
8.6.4.2 Successful Operation....74
8.6.4.3 Unsuccessful Operation....75
8.6.4.4 Abnormal Conditions....75
8.6.5 MeNB initiated SeNB Release....76
8.6.5.1 General....76
8.6.5.2 Successful Operation....76
8.6.5.3 Unsuccessful Operation....76
8.6.5.4 Abnormal Conditions....77
8.6.6 SeNB initiated SeNB Release....77
8.6.6.1 General....77
8.6.6.2 Successful Operation....77
8.6.6.3 Unsuccessful Operation....77
8.6.6.4 Abnormal Conditions....77

8.6.7 SeNB Counter Check ..... 78
8.6.7.1 General ..... 78
8.6.7.2 Successful Operation ..... 78
8.6.7.3 Unsuccessful Operation ..... 78
8.6.7.4 Abnormal Conditions ..... 78
8.7 Procedures for E-UTRAN-NR Dual Connectivity ..... 78
8.7.1 EN-DC X2 Setup ..... 78
8.7.1.1 General ..... 78
8.7.1.2 Successful Operation ..... 79
8.7.1.3 Unsuccessful Operation ..... 81
8.7.1.4 Abnormal Conditions ..... 82
8.7.2 EN-DC Configuration Update ..... 82
8.7.2.1 General ..... 82
8.7.2.2 Successful Operation ..... 82
8.7.2.3 Unsuccessful Operation ..... 85
8.7.2.4 Abnormal Conditions ..... 86
8.7.3 EN-DC Cell Activation ..... 86
8.7.3.1 General ..... 86
8.7.3.2 Successful Operation ..... 86
8.7.3.3 Unsuccessful Operation ..... 87
8.7.3.4 Abnormal Conditions ..... 87
8.7.4 SgNB Addition Preparation ..... 87
8.7.4.1 General ..... 87
8.7.4.2 Successful Operation ..... 87
8.7.4.3 Unsuccessful Operation ..... 92
8.7.4.4 Abnormal Conditions ..... 92
8.7.5 SgNB Reconfiguration Completion ..... 93
8.7.5.1 General ..... 93
8.7.5.2 Successful Operation ..... 93
8.7.5.3 Abnormal Conditions ..... 93
8.7.6 MeNB initiated SgNB Modification Preparation ..... 93
8.7.6.1 General ..... 93
8.7.6.2 Successful Operation ..... 94
8.7.6.3 Unsuccessful Operation ..... 99
8.7.6.4 Abnormal Conditions ..... 99
8.7.7 SgNB initiated SgNB Modification ..... 100
8.7.7.1 General ..... 100
8.7.7.2 Successful Operation ..... 101
8.7.7.3 Unsuccessful Operation ..... 103
8.7.7.4 Abnormal Conditions ..... 103
8.7.8 SgNB Change ..... 104
8.7.8.1 General ..... 104
8.7.8.2 Successful Operation ..... 104
8.7.8.3 Unsuccessful Operation ..... 105
8.7.8.4 Abnormal Conditions ..... 105
8.7.9 MeNB initiated SgNB Release ..... 105
8.7.9.1 General ..... 105
8.7.9.2 Successful Operation ..... 106
8.7.9.3 Unsuccessful Operation ..... 107
8.7.9.4 Abnormal Conditions ..... 107
8.7.10 SgNB initiated SgNB Release ..... 107
8.7.10.1 General ..... 107
8.7.10.2 Successful Operation ..... 107
8.7.10.3 Unsuccessful Operation ..... 108
8.7.10.4 Abnormal Conditions ..... 108
8.7.11 SgNB Counter Check ..... 108
8.7.11.1 General ..... 108
8.7.11.2 Successful Operation ..... 108
8.7.11.3 Unsuccessful Operation ..... 109
8.7.11.4 Abnormal Conditions ..... 109
8.7.12 RRC Transfer ..... 109
8.7.12.1 General ..... 109

8.7.12.2 Successful Operation ..... 109
8.7.12.3 Abnormal Conditions ..... 110
8.7.13 Secondary RAT Data Usage Report ..... 110
8.7.13.1 General ..... 110
8.7.13.2 Successful Operation ..... 110
8.7.13.3 Unsuccessful Operation ..... 110
8.7.13.4 Abnormal Conditions ..... 110
8.7.14 Partial reset of EN-DC ..... 110
8.7.14.1 General ..... 110
8.7.14.2 Successful Operation ..... 111
8.7.14.3 Unsuccessful Operation ..... 112
8.7.14.4 Abnormal Conditions ..... 112
8.7.15 E-UTRA – NR Cell Resource Coordination ..... 112
8.7.15.1 General ..... 112
8.7.15.2 Successful Operation ..... 112
8.7.16 SgNB Activity Notification ..... 113
8.7.16.1 General ..... 113
8.7.16.2 Successful Operation ..... 113
8.7.16.3 Abnormal Conditions ..... 114
8.7.17 gNB Status Indication ..... 114
8.7.17.1 General ..... 114
8.7.17.2 Successful Operation ..... 114
8.7.17.3 Abnormal Conditions ..... 114
8.7.18 EN-DC Configuration Transfer ..... 114
8.7.18.1 General ..... 114
8.7.18.2 Successful Operation ..... 114
8.7.18.3 Abnormal Conditions ..... 115
8.7.19 Trace Start ..... 116
8.7.19.1 General ..... 116
8.7.19.2 Successful Operation ..... 116
8.7.19.3 Abnormal Conditions ..... 116
8.7.20 Deactivate Trace ..... 116
8.7.20.1 General ..... 116
8.7.20.2 Successful Operation ..... 116
8.7.20.3 Abnormal Conditions ..... 117
8.7.21 EN-DC Resource Status Reporting Initiation ..... 117
8.7.21.1 General ..... 117
8.7.21.2 Successful Operation ..... 117
8.7.21.2.1 Successful Operation - eNB-initiated ..... 117
8.7.21.2.2 Successful Operation - en-gNB-initiated ..... 118
8.7.21.3 Unsuccessful Operation ..... 119
8.7.21.4 Abnormal Conditions ..... 119
8.7.22 EN-DC Resource Status Reporting ..... 119
8.7.22.1 General ..... 119
8.7.22.2 Successful Operation ..... 120
8.7.22.3 Unsuccessful Operation ..... 120
8.7.22.4 Abnormal Conditions ..... 120
8.7.23 Cell Traffic Trace ..... 120
8.7.23.1 General ..... 120
8.7.23.2 Successful Operation ..... 120
8.7.24 UE Radio Capability ID Mapping ..... 121
8.7.24.1 General ..... 121
8.7.24.2 Successful Operation ..... 121
8.7.24.3 Unsuccessful Operation ..... 121
8.7.25 Conditional PSCell Change Cancel ..... 121
8.7.25.1 General ..... 121
8.7.25.2 Successful Operation ..... 122
8.7.25.3 Unsuccessful Operation ..... 122
8.7.25.4 Abnormal Conditions ..... 122
8.8 IAB Procedures ..... 122
8.8.1 F1-C Traffic Transfer ..... 122
8.8.1.1 General ..... 122

8.8.1.2 Successful Operation ..... 122
8.8.1.3 Unsuccessful Operation ..... 123
8.8.1.4 Abnormal Conditions ..... 123

9 Elements for X2AP Communication ..... 123
9.0 General ..... 123
9.1 Message Functional Definition and Content ..... 123
9.1.1 Messages for Basic Mobility Procedures ..... 123
9.1.1.1 HANDOVER REQUEST ..... 123
9.1.1.2 HANDOVER REQUEST ACKNOWLEDGE ..... 126
9.1.1.3 HANDOVER PREPARATION FAILURE ..... 128
9.1.1.4 SN STATUS TRANSFER ..... 128
9.1.1.5 UE CONTEXT RELEASE ..... 131
9.1.1.6 HANDOVER CANCEL ..... 131
9.1.1.7 HANDOVER SUCCESS ..... 132
9.1.1.8 CONDITIONAL HANDOVER CANCEL ..... 132
9.1.1.9 EARLY STATUS TRANSFER ..... 133
9.1.2 Messages for global procedures ..... 135
9.1.2.1 LOAD INFORMATION ..... 135
9.1.2.2 ERROR INDICATION ..... 136
9.1.2.3 X2 SETUP REQUEST ..... 137
9.1.2.4 X2 SETUP RESPONSE ..... 138
9.1.2.5 X2 SETUP FAILURE ..... 139
9.1.2.6 RESET REQUEST ..... 140
9.1.2.7 RESET RESPONSE ..... 140
9.1.2.8 ENB CONFIGURATION UPDATE ..... 140
9.1.2.9 ENB CONFIGURATION UPDATE ACKNOWLEDGE ..... 143
9.1.2.10 ENB CONFIGURATION UPDATE FAILURE ..... 143
9.1.2.11 RESOURCE STATUS REQUEST ..... 143
9.1.2.12 RESOURCE STATUS RESPONSE ..... 145
9.1.2.13 RESOURCE STATUS FAILURE ..... 147
9.1.2.14 RESOURCE STATUS UPDATE ..... 148
9.1.2.15 MOBILITY CHANGE REQUEST ..... 148
9.1.2.16 MOBILITY CHANGE ACKNOWLEDGE ..... 149
9.1.2.17 MOBILITY CHANGE FAILURE ..... 149
9.1.2.18 RLF INDICATION ..... 149
9.1.2.19 HANDOVER REPORT ..... 151
9.1.2.20 CELL ACTIVATION REQUEST ..... 153
9.1.2.21 CELL ACTIVATION RESPONSE ..... 153
9.1.2.22 CELL ACTIVATION FAILURE ..... 153
9.1.2.23 X2 RELEASE ..... 153
9.1.2.24 X2AP MESSAGE TRANSFER ..... 154
9.1.2.25 X2 REMOVAL REQUEST ..... 154
9.1.2.26 X2 REMOVAL RESPONSE ..... 154
9.1.2.27 X2 REMOVAL FAILURE ..... 154
9.1.2.28 RETRIEVE UE CONTEXT REQUEST ..... 155
9.1.2.29 RETRIEVE UE CONTEXT RESPONSE ..... 157
9.1.2.30 RETRIEVE UE CONTEXT FAILURE ..... 159
9.1.2.31 EN-DC X2 SETUP REQUEST ..... 160
9.1.2.32 EN-DC X2 SETUP RESPONSE ..... 162
9.1.2.33 EN-DC X2 SETUP FAILURE ..... 162
9.1.2.34 EN-DC CONFIGURATION UPDATE ..... 163
9.1.2.35 EN-DC CONFIGURATION UPDATE ACKNOWLEDGE ..... 165
9.1.2.36 EN-DC CONFIGURATION UPDATE FAILURE ..... 166
9.1.2.37 EN-DC CELL ACTIVATION REQUEST ..... 167
9.1.2.38 EN-DC CELL ACTIVATION RESPONSE ..... 167
9.1.2.39 EN-DC CELL ACTIVATION FAILURE ..... 167
9.1.2.40 EN-DC X2 REMOVAL REQUEST ..... 168
9.1.2.41 EN-DC X2 REMOVAL RESPONSE ..... 168
9.1.2.42 EN-DC X2 REMOVAL FAILURE ..... 168
9.1.2.43 DATA FORWARDING ADDRESS INDICATION ..... 169
9.1.2.44 EN-DC CONFIGURATION TRANSFER ..... 171

9.1.2.45 EN-DC RESOURCE STATUS REQUEST ..... 171
9.1.2.46 EN-DC RESOURCE STATUS RESPONSE ..... 173
9.1.2.47 EN-DC RESOURCE STATUS FAILURE ..... 173
9.1.2.48 EN-DC RESOURCE STATUS UPDATE ..... 174
9.1.2.49 CELL TRAFFIC TRACE ..... 175
9.1.2.50 ACCESS AND MOBILITY INDICATION ..... 176
9.1.3 Messages for Dual Connectivity Procedures ..... 176
9.1.3.1 SENB ADDITION REQUEST ..... 176
9.1.3.2 SENB ADDITION REQUEST ACKNOWLEDGE ..... 178
9.1.3.3 SENB ADDITION REQUEST REJECT ..... 180
9.1.3.4 SENB RECONFIGURATION COMPLETE ..... 180
9.1.3.5 SENB MODIFICATION REQUEST ..... 181
9.1.3.6 SENB MODIFICATION REQUEST ACKNOWLEDGE ..... 184
9.1.3.7 SENB MODIFICATION REQUEST REJECT ..... 186
9.1.3.8 SENB MODIFICATION REQUIRED ..... 187
9.1.3.9 SENB MODIFICATION CONFIRM ..... 187
9.1.3.10 SENB MODIFICATION REFUSE ..... 188
9.1.3.11 SENB RELEASE REQUEST ..... 188
9.1.3.12 SENB RELEASE REQUIRED ..... 189
9.1.3.13 SENB RELEASE CONFIRM ..... 190
9.1.3.14 SENB COUNTER CHECK REQUEST ..... 191
9.1.4 Messages for E-UTRAN-NR Dual Connectivity Procedures ..... 192
9.1.4.1 SGNB ADDITION REQUEST ..... 192
9.1.4.2 SGNB ADDITION REQUEST ACKNOWLEDGE ..... 196
9.1.4.3 SGNB ADDITION REQUEST REJECT ..... 199
9.1.4.4 SGNB RECONFIGURATION COMPLETE ..... 199
9.1.4.5 SGNB MODIFICATION REQUEST ..... 200
9.1.4.6 SGNB MODIFICATION REQUEST ACKNOWLEDGE ..... 206
9.1.4.7 SGNB MODIFICATION REQUEST REJECT ..... 210
9.1.4.8 SGNB MODIFICATION REQUIRED ..... 210
9.1.4.9 SGNB MODIFICATION CONFIRM ..... 213
9.1.4.10 SGNB MODIFICATION REFUSE ..... 215
9.1.4.11 SGNB RELEASE REQUEST ..... 215
9.1.4.12 SGNB RELEASE REQUEST ACKNOWLEDGE ..... 217
9.1.4.13 SGNB RELEASE REQUEST REJECT ..... 217
9.1.4.14 SGNB RELEASE REQUIRED ..... 218
9.1.4.15 SGNB RELEASE CONFIRM ..... 218
9.1.4.16 SGNB COUNTER CHECK REQUEST ..... 220
9.1.4.17 SGNB CHANGE REQUIRED ..... 220
9.1.4.18 SGNB CHANGE CONFIRM ..... 222
9.1.4.19 SGNB CHANGE REFUSE ..... 224
9.1.4.20 SECONDARY RAT DATA USAGE REPORT ..... 225
9.1.4.21 RRC TRANSFER ..... 225
9.1.4.22 PARTIAL RESET REQUIRED ..... 227
9.1.4.23 PARTIAL RESET CONFIRM ..... 227
9.1.4.24 E-UTRA – NR CELL RESOURCE COORDINATION REQUEST ..... 228
9.1.4.25 E-UTRA – NR CELL RESOURCE COORDINATION RESPONSE ..... 229
9.1.4.26 SGNB ACTIVITY NOTIFICATION ..... 230
9.1.4.27 GNB STATUS INDICATION ..... 231
9.1.4.28 TRACE START ..... 231
9.1.4.29 DEACTIVATE TRACE ..... 232
9.1.4.30 UE Radio Capability ID Mapping Request ..... 232
9.1.4.31 UE Radio Capability ID Mapping Response ..... 232
9.1.4.32 CONDITIONAL PSCELL CHANGE CANCEL ..... 232
9.1.5 Messages for IAB Procedures ..... 233
9.1.5.1 F1-C TRAFFIC TRANSFER ..... 233
9.2 Information Element definitions ..... 233
9.2.0 General ..... 233
9.2.1 GTP Tunnel Endpoint ..... 234
9.2.2 Trace Activation ..... 234
9.2.3 Handover Restriction List ..... 236
9.2.4 PLMN Identity ..... 239

9.2.5 DL Forwarding ....239
9.2.6 Cause ....239
9.2.7 Criticality Diagnostics ....244
9.2.8 Served Cell Information....245
9.2.9 E-RAB Level QoS Parameters....250
9.2.10 GBR QoS Information ....250
9.2.11 Bit Rate ....252
9.2.12 UE Aggregate Maximum Bit Rate....252
9.2.13 Message Type ....253
9.2.14 ECGI....253
9.2.15 COUNT Value ....253
9.2.16 GUMMEI....254
9.2.17 UL Interference Overload Indication....254
9.2.18 UL High Interference Indication....254
9.2.19 Relative Narrowband Tx Power (RNTP)....255
9.2.20 GU Group Id ....258
9.2.21 Location Reporting Information ....258
9.2.22 Global eNB ID....258
9.2.23 E-RAB ID ....258
9.2.24 eNB UE X2AP ID ....259
9.2.25 Subscriber Profile ID for RAT/Frequency priority....259
9.2.25a Additional RRM Policy Index ....259
9.2.26 EARFCN ....259
9.2.27 Transmission Bandwidth ....260
9.2.28 E-RAB List ....260
9.2.29 UE Security Capabilities....260
9.2.30 AS Security Information....261
9.2.31 Allocation and Retention Priority ....261
9.2.32 Time To Wait....262
9.2.33 SRVCC Operation Possible ....262
9.2.34 Hardware Load Indicator ....262
9.2.35 S1 TNL Load Indicator....263
9.2.36 Load Indicator....263
9.2.37 Radio Resource Status ....263
9.2.38 UE History Information ....263
9.2.39 Last Visited Cell Information ....264
9.2.40 Last Visited E-UTRAN Cell Information....264
9.2.41 Last Visited GERAN Cell Information....265
9.2.42 Cell Type ....265
9.2.43 Number of Antenna Ports ....265
9.2.44 Composite Available Capacity Group ....265
9.2.45 Composite Available Capacity ....265
9.2.46 Cell Capacity Class Value ....266
9.2.47 Capacity Value....266
9.2.48 Mobility Parameters Information....266
9.2.49 Mobility Parameters Modification Range....266
9.2.50 PRACH Configuration....267
9.2.51 Subframe Allocation ....267
9.2.52 CSG Membership Status....267
9.2.53 CSG ID ....267
9.2.54 ABS Information ....268
9.2.55 Invoke Indication ....270
9.2.56 MDT Configuration ....270
9.2.57 Void ....273
9.2.58 ABS Status....273
9.2.59 Management Based MDT Allowed ....274
9.2.60 MultibandInfoList....275
9.2.61 M3 Configuration ....275
9.2.62 M4 Configuration ....275
9.2.63 M5 Configuration ....275
9.2.64 MDT PLMN List ....276
9.2.65 EARFCN Extension....276

9.2.66 COUNT Value Extended ..... 276
9.2.67 Extended UL Interference Overload Info ..... 276
9.2.68 RNL Header ..... 277
9.2.69 Masked IMEISV ..... 277
9.2.70 Expected UE Behaviour ..... 278
9.2.71 Expected UE Activity Behaviour ..... 278
9.2.72 SeNB Security Key ..... 278
9.2.73 SCG Change Indication ..... 279
9.2.74 CoMP Information ..... 279
9.2.75 CoMP Hypothesis Set ..... 279
9.2.76 RSRP Measurement Report List ..... 280
9.2.77 Dynamic DL transmission information ..... 281
9.2.78 ProSe Authorized ..... 281
9.2.79 CSI Report ..... 281
9.2.80 Wideband CQI ..... 282
9.2.81 Subband CQI ..... 282
9.2.82 COUNT Value for PDCP SN Length 18 ..... 283
9.2.83 LHN ID ..... 283
9.2.84 Correlation ID ..... 283
9.2.85 UE Context Kept Indicator ..... 283
9.2.86 eNB UE X2AP ID Extension ..... 284
9.2.87 M6 Configuration ..... 284
9.2.88 M7 Configuration ..... 284
9.2.89 Tunnel Information ..... 284
9.2.90 X2 Benefit Value ..... 285
9.2.91 Resume ID ..... 285
9.2.92 Bearer Type ..... 285
9.2.93 V2X Services Authorized ..... 286
9.2.94 Offset of NB-IoT Channel Number to EARFCN ..... 286
9.2.95 WT ID ..... 286
9.2.96 WT UE XwAP ID ..... 286
9.2.97 UE Sidelink Aggregate Maximum Bit Rate ..... 286
9.2.98 NR Neighbour Information ..... 287
9.2.99 Extended Bit Rate ..... 290
9.2.100 en-gNB UE X2AP ID ..... 290
9.2.101 SgNB Security Key ..... 291
9.2.102 Target SgNB ID Information ..... 291
9.2.103 SCG Configuration Query ..... 291
9.2.104 Delivery Status ..... 291
9.2.105 Void ..... 291
9.2.106 NR Frequency Info ..... 291
9.2.107 NR UE Security Capabilities ..... 292
9.2.108 EN-DC Resource Configuration ..... 293
9.2.109 PDCP Change Indication ..... 293
9.2.110 Served NR Cell Information ..... 294
9.2.111 NR CGI ..... 297
9.2.112 Global en-gNB ID ..... 298
9.2.113 Void ..... 298
9.2.114 NR Transmission Bandwidth ..... 298
9.2.115 Cell Assistance Information ..... 298
9.2.116 MeNB Resource Coordination Information ..... 299
9.2.117 SgNB Resource Coordination Information ..... 301
9.2.118 UL Configuration ..... 303
9.2.119 RLC Mode ..... 303
9.2.120 Secondary RAT Usage Report List ..... 304
9.2.121 UE Application layer measurement configuration ..... 305
9.2.122 DRB ID ..... 305
9.2.123 SUL Information ..... 306
9.2.124 Packet Loss Rate ..... 306
9.2.125 Protected E-UTRA Resource Indication ..... 306
9.2.126 Data Traffic Resource Indication ..... 310
9.2.127 Data Traffic Resources ..... 311

9.2.128 Reserved Subframe Pattern ..... 312
9.2.129 Aerial UE subscription information ..... 313
9.2.130 User plane traffic activity report ..... 313
9.2.131 RLC Status ..... 313
9.2.132 RRC config indication ..... 314
9.2.133 PDCP SN Length ..... 314
9.2.134 Bluetooth Measurement Configuration ..... 314
9.2.135 WLAN Measurement Configuration ..... 314
9.2.136 Subscription Based UE Differentiation Information ..... 315
9.2.137 Duplication activation ..... 316
9.2.138 LCID ..... 317
9.2.139 MeNB Coordination Assistance Information ..... 317
9.2.140 SgNB Coordination Assistance Information ..... 317
9.2.141 Desired Activity Notification Level ..... 317
9.2.142 Location Information at SgNB ..... 317
9.2.143 Interface Instance Indication ..... 318
9.2.144 NB-IoT UL DL Alignment Offset ..... 318
9.2.145 Lower Layer presence status change ..... 318
9.2.146 Cell and Capacity Assistance Information ..... 318
9.2.147 Maximum Cell List Size ..... 319
9.2.148 Message Oversize Notification ..... 319
9.2.149 TNL Transport Layer Address Info ..... 319
9.2.150 CP Transport Layer Information ..... 320
9.2.151 TNL Association Usage ..... 320
9.2.152 RAN UE NGAP ID ..... 320
9.2.153 EPC Handover Restriction List Container ..... 320
9.2.154 DAPS Request Information ..... 320
9.2.155 DAPS Response Information ..... 321
9.2.156 Maximum Number of CHO Preparations ..... 321
9.2.157 Ethernet Type ..... 321
9.2.158 NR V2X Services Authorized ..... 321
9.2.159 NR UE Sidelink Aggregate Maximum Bit Rate ..... 321
9.2.160 PC5 QoS Parameters ..... 322
9.2.161 TNL Capacity Indicator ..... 322
9.2.162 NR Radio Resource Status ..... 323
9.2.163 NR Composite Available Capacity Group ..... 324
9.2.164 NR Composite Available Capacity ..... 324
9.2.165 NR Cell Capacity Class Value ..... 324
9.2.166 NR Capacity Value ..... 324
9.2.167 SSB Index ..... 325
9.2.168 NR Carrier List ..... 325
9.2.169 SSB Positions In Burst ..... 326
9.2.170 NPRACH Configuration ..... 327
9.2.171 UE Radio Capability ID ..... 328
9.2.172 QoS Mapping Information ..... 328
9.2.173 UE Radio Capability ..... 328
9.2.174 URI ..... 328
9.2.175 SFN Offset ..... 328
9.2.176 Global RAN Node ID ..... 329
9.2.177 SCG UE History Information ..... 329
9.2.178 SCG Activation Status ..... 329
9.2.179 SCG Activation Request ..... 329
9.2.180 Served Cell Specific Info Request ..... 329
9.2.181 Security Indication ..... 330
9.2.182 Security Result ..... 330
9.2.183 Sensor Measurement Configuration ..... 330
9.3 Message and Information Element Abstract Syntax (with ASN.1) ..... 332
9.3.1 General ..... 332
9.3.2 Usage of Private Message Mechanism for Non-standard Use ..... 332
9.3.3 Elementary Procedure Definitions ..... 332
9.3.4 PDU Definitions ..... 347
9.3.5 Information Element definitions ..... 441

9.3.6 Common definitions ....513
9.3.7 Constant definitions ....514
9.3.8 Container definitions ....526
9.4 Message transfer syntax ....530
9.5 Timers ....530
10 Handling of unknown, unforeseen and erroneous protocol data ....530
**Annex A (informative): Change history ....531**
History ....540

# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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where:

- x the first digit:
  - 1 presented to TSG for information;
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- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

# 1 Scope

The present document specifies the radio network layer signalling procedures of the control plane between eNBs in E-UTRAN. X2AP supports the functions of X2 interface by signalling procedures defined in this document. X2AP is developed in accordance to the general principles stated in TS 36.401 [2] and TS 36.420 [3].

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 36.401: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Architecture Description".

[3] 3GPP TS 36.420: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 General Aspects and Principles".

[4] 3GPP TS 36.413: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)".

[5] ITU-T Recommendation X.691 (2002-07): "Information technology - ASN.1 encoding rules - Specification of Packed Encoding Rules (PER) ".

[6] 3GPP TS 32.422: "Telecommunication Management; Subscriber and Equipment Trace; Trace Control and Configuration Management".

[7] 3GPP TS 32.421: "Telecommunication Management; Subscriber and Equipment Trace; Trace concepts and requirements".

[8] 3GPP TS 36.424: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 data transport".

[9] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRAN); Radio Resource Control (RRC) Protocol Specification".

[10] 3GPP TS 36.211: "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical Channels and Modulation".

[11] 3GPP TS 36.213: "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical layer procedures ".

[12] 3GPP TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".

[13] 3GPP TS 23.203: "Policy and charging control architecture".

[14] 3GPP TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System; Stage 3".

[15] 3GPP TS 36.300: "Evolved Universal Terrestrial Radio Access (E-UTRA), Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; stage 2".

[16] 3GPP TS 36.104: "Base Station (BS) radio transmission and reception ".

[17] Void.

[18] 3GPP TS 33.401: "Security architecture".

[19] 3GPP TS 36.414: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 data transport".

[20] 3GPP TS 23.216: "Single Radio Voice Call Continuity (SRVCC)".

[21] 3GPP TS 36.422: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 signaling transport".

[22] 3GPP TS 36.314: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Layer 2 - Measurements".

[23] Void.

[24] 3GPP TS 25.413: "UTRAN Iu interface RANAP signalling"

[25] 3GPP TS 37.320: "Universal Terrestrial Radio Access (UTRA) and Evolved Universal Terrestrial Radio Access (E-UTRA); Radio measurement collection for Minimization of Drive Tests (MDT);Overall description; Stage 2".

[26] 3GPP TS 29.281: "General Packet Radio Service (GPRS); Tunnelling Protocol User Plane (GTPv1-U)".

[27] ITU-T Recommendation X.680 (2002-07): "Information technology – Abstract Syntax Notation One (ASN.1): Specification of basic notation".

[28] ITU-T Recommendation X.681 (2002-07): "Information technology – Abstract Syntax Notation One (ASN.1): Information object specification".

[29] 3GPP TS 23.003: "Technical Specification Group Core Network and Terminals; Numbering, addressing and identification".

[30] 3GPP TR 25.921 (version.7.0.0): "Guidelines and principles for protocol description and error handling".

[31] 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol Specification".

[32] 3GPP TS 37.340: "Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity; Stage 2".

[33] 3GPP TS 38.323: "NR; Packet Data Convergence Protocol (PDCP) specification".

[34] 3GPP TS 38.401: "NG-RAN; Architecture description".

[35] IETF RFC 5905: "Network Time Protocol Version 4: Protocol and Algorithms Specification".

[36] 3GPP TS 26.247: "Transparent end-to-end Packet-switched Streaming Service (PSS); Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH)".

[37] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception".

[38] 3GPP TS 23.501: "System Architecture for the 5G System"

[39] 3GPP TS 38.413: "NG Radio Access Network (NG-RAN); NG Application Protocol (NGAP)".

[40] 3GPP TS 36.322: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Link Control (RLC) protocol specification".

[41] 3GPP TS 23.285: " Technical Specification Group Services and System Aspects; Architecture enhancements for V2X services".

[42] 3GPP TS 38.211: "NR; Physical channels and modulation".

[43] 3GPP TS 38.213: "NR; Physical layer procedures for control".

[44] 3GPP TS 38.473: "NG-RAN; F1 application protocol (F1AP)".

[45] 3GPP TS 38.314: "NR; Layer 2 Measurements".

[46] Void

[47] 3GPP TS 38.300: "NR; Overall description; Stage-2".

[48] 3GPP TS 38.472: "NG-RAN; F1 signalling transport"

[49] 3GPP TS 38.423: "NG-RAN; Xn Application Protocol (XnAP)".

[50] 3GPP TS 33.501: "Security architecture and procedures for 5G System".

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

**ACL functionality:** A functionality controlling the access to network nodes. In case of Access Control Lists (ACL) functionality is applied in a network node the network node may only accept connections from other peer network nodes once the source addresses of the sending network node is already known in the target node.

**Elementary Procedure:** X2AP protocol consists of Elementary Procedures (EPs). An X2AP Elementary Procedure is a unit of interaction between two eNBs. An EP consists of an initiating message and possibly a response message. Two kinds of EPs are used:

- Class 1: Elementary Procedures with response (success or failure),
- Class 2: Elementary Procedures without response.

**E-RAB:** Defined in TS 36.401 [2].

**CSG Cell:** as defined in TS 36.300 [15].

**Dual Connectivity:** as defined in TS 36.300 [15].

**E-UTRA-NR Dual Connectivity**: as defined in TS 37.340 [32].

**E-UTRAN node**: either an eNB or an en-gNB.

**Hybrid cell:** as defined in TS 36.300 [15].

**Master eNB:** as defined in TS 36.300 [15].

**Secondary Cell Group:** as defined in TS 36.300 [15].

**Secondary eNB:** as defined in TS 36.300 [15].

**en-gNB**: as defined in TS 37.340 [32].

**Conditional Handover:** As defined in TS 36.300 [15].

**DAPS HO:** As defined in TS 36.300 [15].

**Conditional PSCell Change:** As defined in TS 37.340 [32].

**Immediate Handover:** Used in the context of Conditional Handover, to refer to a handover that is executed immediately after the UE receives the Handover Command.

**IAB-node:** as defined in TS 38.300 [47].

## 3.2 Symbols

For the purposes of the present document, the following symbols apply:

<symbol> <Explanation>

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

| | |
|---|---|
| ABS | Almost Blank Subframe |
| ARPI | Additional RRM Policy Index |
| ACL | Access Control List |
| BBF | Broadband Forum |
| BL | Bandwidth reduced Low complexity |
| CCO | Cell Change Order |
| CE | Coverage Enhancement |
| CHO | Conditional Handover |
| CoMP | Coordinated Multi Point |
| CPA | Conditional PSCell Addition |
| CPAC | Conditional PSCell Addition or Change. |
| CPC | Conditional PSCell Change |
| DAPS | Dual Active Protocol Stacks |
| DC | Dual Connectivity |
| DL | Downlink |
| EARFCN | E-UTRA Absolute Radio Frequency Channel Number |
| E-CID | Enhanced Cell-ID (positioning method) |
| eNB | E-UTRAN NodeB |
| EN-DC | E-UTRA-NR Dual Connectivity |
| EP | Elementary Procedure |
| EPC | Evolved Packet Core |
| E-RAB | E-UTRAN Radio Access Bearer |
| E-UTRAN | Evolved UTRAN |
| GNSS | Global Navigation Satellite System |
| GUMMEI | Globally Unique MME Identifier |
| HFN | Hyper Frame Number |
| IAB | Integrated Access and Backhaul |
| IE | Information Element |
| L-GW | Local GateWay |
| LWA | LTE-WLAN Aggregation |
| MCG | Master Cell Group |
| MDT | Minimization of Drive Tests |
| MeNB | Master eNB |
| MME | Mobility Management Entity |
| MTSI | Multimedia Telephony Service for IMS |
| NAICS | Network-Assisted Interference Cancellation and Suppression |
| NR | New Radio |
| PDCP | Packet Data Convergence Protocol |
| PLMN | Public Land Mobile Network |
| ProSe | Proximity Service |
| QMC | QoE Measurement Collection |
| QoE | Quality of Experience |
| SCG | Secondary Cell Group |
| S-GW | Serving Gateway |
| SeNB | Secondary eNB |
| SgNB | Secondary gNB |
| SIPTO | Selected IP Traffic Offload |
| SIPTO@LN | Selected IP Traffic Offload at the Local Network |
| SN | Sequence Number |

| | |
|---|---|
| SSID | Service Set Identifier |
| TAC | Tracking Area Code |
| UE | User Equipment |
| UL | Uplink |
| V2X | Vehicle-to-Everything |
| WLAN | Wireless Local Area Network |
| WT | WLAN Termination |

# 4 General

## 4.1 Procedure specification principles

The principle for specifying the procedure logic is to specify the functional behaviour of the terminating eNB exactly and completely. Any rule that specifies the behaviour of the originating eNB shall be possible to be verified with information that is visible within the system.

The following specification principles have been applied for the procedure text in clause 8:

- The procedure text discriminates between:

  1) Functionality which "shall" be executed

     The procedure text indicates that the receiving node "shall" perform a certain function Y under a certain condition. If the receiving node supports procedure X but cannot perform functionality Y requested in the initiating message of a Class 1 EP, the receiving node shall respond with the message used to report unsuccessful outcome for this procedure, containing an appropriate cause value.

  2) Functionality which "shall, if supported" be executed

     The procedure text indicates that the receiving node "shall, if supported," perform a certain function Y under a certain condition. If the receiving node supports procedure X, but does not support functionality Y, the receiving node shall proceed with the execution of the EP, possibly informing the requesting node about the not supported functionality.

- Any required inclusion of an optional IE in a response message is explicitly indicated in the procedure text. If the procedure text does not explicitly indicate that an optional IE shall be included in a response message, the optional IE shall not be included. For requirements on including *Criticality Diagnostics* IE, see section 10.

## 4.2 Forwards and backwards compatibility

The forwards and backwards compatibility of the protocol is assured by a mechanism where all current and future messages, and IEs or groups of related IEs, include ID and criticality fields that are coded in a standard format that will not be changed in the future. These parts can always be decoded regardless of the standard version.

## 4.3 Specification notations

For the purposes of the present document, the following notations apply:

| | |
|---|---|
| Procedure | When referring to an elementary procedure in the specification the Procedure Name is written with the first letters in each word in upper case characters followed by the word "procedure", e.g. Handover Preparation procedure. |
| Message | When referring to a message in the specification the MESSAGE NAME is written with all letters in upper case characters followed by the word "message", e.g. HANDOVER REQUEST message. |
| IE | When referring to an information element (IE) in the specification the *Information Element Name* is written with the first letters in each word in upper case characters and all letters in Italic font followed by the abbreviation "IE", e.g. *E-RAB ID* IE. |

Value of an IE When referring to the value of an information element (IE) in the specification the "Value" is written as it is specified in sub clause 9.2 enclosed by quotation marks, e.g. "Value".

# 5 X2AP services

The present clause describes the services an eNB offers to its neighbours.

## 5.1 X2AP procedure modules

The X2 interface X2AP procedures are divided into two modules as follows:

1. X2AP Basic Mobility Procedures;
2. X2AP Global Procedures;

The X2AP Basic Mobility Procedures module contains procedures used to handle the UE mobility within E-UTRAN.

The Global Procedures module contains procedures that are not related to a specific UE. The procedures in this module are in contrast to the above module involving two peer eNBs.

## 5.2 Parallel transactions

Unless explicitly indicated in the procedure specification, at any instance in time one protocol peer shall have a maximum of one ongoing X2AP procedure related to a certain UE.

# 6 Services expected from signalling transport

The signalling connection shall provide in sequence delivery of X2AP messages. X2AP shall be notified if the signalling connection breaks.

X2 signalling transport is described in TS 36.422 [21].

# 7 Functions of X2AP

The X2AP protocol provides the following functions:

- Mobility Management. This function allows the eNB to move the responsibility of a certain UE to another eNB. Forwarding of user plane data, Status Transfer and UE Context Release function are parts of the mobility management.
- Dual Connectivity. This function allows the eNB to request another eNB to provide radio resources for a certain UE while keeping responsibility for that UE.
- E-UTRA-NR Dual Connectivity. This function allows the eNB to request another en-gNB to provide radio resources for a certain UE while keeping responsibility for that UE.
- Load Management. This function is used by eNBs to indicate resource status, overload and traffic load to each other.
- Reporting of General Error Situations. This function allows reporting of general error situations, for which function specific error messages have not been defined.
- Resetting the X2. This function is used to reset the X2 interface.
- Setting up the X2. This function is used to exchange necessary data for the eNB or en-gNB for setup the X2 interface and implicitly perform an X2 Reset.

- eNB Configuration Update. This function allows updating of application level data needed for two eNBs to interoperate correctly over the X2 interface.
- Mobility Parameters Management. This function allows the eNB to coordinate adaptation of mobility parameter settings with a peer eNB.
- Mobility Robustness Optimisation. This function allows reporting of information related to mobility failure events.
- Energy Saving. This function allows decreasing energy consumption by enabling indication of cell activation/deactivation over the X2 interface.
- X2 Release. This function allows an eNB to be aware that the signalling connection to a peer eNB is unavailable.
- Message Transfer. This function allows indirect transport of X2AP messages to a peer eNB.
- Registration. This function allows registration of eNB in case indirect transport of X2AP messages is supported.
- Removing the X2. This function allows removing the signalling connection between two eNBs or between eNB and en-gNB in a controlled manner.
- Inter-eNB UE Context Retrieval. This function allows retrieval of a UE context in case of resumption or re-establishment of an RRC connection.
- Secondary RAT Data Usage Report. This function allows eNB to get the uplink and downlink data volumes for the Secondary RAT on a per E-RAB basis.
- E-UTRA - NR Spectrum Sharing. This function allows uplink and downlink spectrum sharing between a number of E - UTRA and a number of NR cells with overlapping coverage.
- EN-DC Configuration Transfer. This function supports en-gNB X2 TNL address discovery.
- EN-DC Load Management. This function is used by MeNB/en-gNB to indicate resource status, overload and traffic load to each other.
- UE Radio Capability ID Mapping.

The mapping between the above functions and X2 EPs is shown in the table below.

**Table 7-1: Mapping between X2AP functions and X2AP EPs**

| Function | Elementary Procedure(s) |
|---|---|
| Mobility Management | a) Handover Preparation<br>b) SN Status Transfer<br>c) UE Context Release<br>d) Handover Cancel<br>e) Handover Success<br>f) Conditional Handover Cancel |
| Dual Connectivity | a) SeNB Addition Preparation<br>b) SeNB Reconfiguration Completion<br>c) MeNB initiated SeNB Modification Preparation<br>d) SeNB initiated SeNB Modification<br>e) MeNB initiated SeNB Release<br>f) SeNB initiated SeNB Release<br>g) SeNB Counter Check |
| E-UTRA-NR Dual Connectivity | a) SgNB Addition Preparation<br>b) SgNB Reconfiguration Completion<br>c) MeNB initiated SgNB Modification Preparation<br>d) SgNB initiated SgNB Modification<br>e) SgNB change<br>f) MeNB initiated SgNB Release<br>g) SgNB initiated SgNB Release<br>h) SgNB Counter Check<br>i) RRC transfer<br>j) EN-DC X2 Setup<br>k) EN-DC Configuration Update<br>l) EN-DC Cell Activation<br>m) SgNB Activity Notification<br>n) EN-DC X2 Removal<br>o) gNB Status Indication<br>p) EN-DC Resource Status Reporting Initiation<br>q) EN-DC Resource Status Reporting<br>r) F1-C Traffic Transfer |
| Load Management | a) Load Indication<br>b) Resource Status Reporting Initiation<br>c) Resource Status Reporting |
| Reporting of General Error Situations | Error Indication |
| Resetting the X2 | Reset |
| Setting up the X2 | X2 Setup |
| eNB Configuration Update | a) eNB Configuration Update<br>b) Cell Activation |
| Mobility Parameters Management | Mobility Settings Change |
| Mobility Robustness Optimisation | a) Radio Link Failure Indication<br>b) Handover Report |
| Energy Saving | a) eNB Configuration Update<br>b) Cell Activation |
| X2 Release | X2 Release |
| Message Transfer Registration | X2AP Message Transfer |
| Removing the X2 | X2 Removal |
| Inter-eNB UE Context Retrieval | a) Retrieve UE Context<br>b) Data Forwarding Address Indication |
| Secondary RAT Data Usage Report | Secondary RAT Data Usage Report |
| E-UTRA – NR Spectrum Sharing | E-UTRA - NR Cell Resource Coordination |
| EN-DC Configuration Transfer | EN-DC Configuration Transfer |
| UE Radio Capability ID Mapping | UE Radio Capability ID Mapping |

# 8 X2AP procedures

## 8.1 Elementary procedures

In the following tables, all EPs are divided into Class 1 and Class 2 EPs.

**Table 8.1-1: Class 1 Elementary Procedures**

| Elementary Procedure | Initiating Message | Successful Outcome | Unsuccessful Outcome |
|---|---|---|---|
| | | Response message | Response message |
| Handover Preparation | HANDOVER REQUEST | HANDOVER REQUEST ACKNOWLEDGE | HANDOVER PREPARATION FAILURE |
| Reset | RESET REQUEST | RESET RESPONSE | |
| X2 Setup | X2 SETUP REQUEST | X2 SETUP RESPONSE | X2 SETUP FAILURE |
| eNB Configuration Update | ENB CONFIGURATION UPDATE | ENB CONFIGURATION UPDATE ACKNOWLEDGE | ENB CONFIGURATION UPDATE FAILURE |
| Resource Status Reporting Initiation | RESOURCE STATUS REQUEST | RESOURCE STATUS RESPONSE | RESOURCE STATUS FAILURE |
| Mobility Settings Change | MOBILITY CHANGE REQUEST | MOBILITY CHANGE ACKNOWLEDGE | MOBILITY CHANGE FAILURE |
| Cell Activation | CELL ACTIVATION REQUEST | CELL ACTIVATION RESPONSE | CELL ACTIVATION FAILURE |
| SeNB Addition Preparation | SENB ADDITION REQUEST | SENB ADDITION REQUEST ACKNOWLEDGE | SENB ADDITION REQUEST REJECT |
| MeNB initiated SeNB Modification Preparation | SENB MODIFICATION REQUEST | SENB MODIFICATION REQUEST ACKNOWLEDGE | SENB MODIFICATION REQUEST REJECT |
| SeNB initiated SeNB Modification | SENB MODIFICATION REQUIRED | SENB MODIFICATION CONFIRM | SENB MODIFICATION REFUSE |
| SeNB initiated SeNB Release | SENB RELEASE REQUIRED | SENB RELEASE CONFIRM | |
| X2 Removal | X2 REMOVAL REQUEST | X2 REMOVAL RESPONSE | X2 REMOVAL FAILURE |
| Retrieve UE Context | RETRIEVE UE CONTEXT REQUEST | RETRIEVE UE CONTEXT RESPONSE | RETRIEVE UE CONTEXT FAILURE |
| SgNB Addition Preparation | SGNB ADDITION REQUEST | SGNB ADDITION REQUEST ACKNOWLEDGE | SGNB ADDITION REQUEST REJECT |
| MeNB initiated SgNB Modification Preparation | SGNB MODIFICATION REQUEST | SGNB MODIFICATION REQUEST ACKNOWLEDGE | SGNB MODIFICATION REQUEST REJECT |
| SgNB initiated SgNB Modification | SGNB MODIFICATION REQUIRED | SGNB MODIFICATION CONFIRM | SGNB MODIFICATION REFUSE |
| SgNB change | SGNB CHANGE REQUIRED | SGNB CHANGE CONFIRM | SGNB CHANGE REFUSE |
| MeNB initiated SgNB Release | SGNB RELEASE REQUEST | SGNB RELEASE REQUEST ACKNOWLEDGE | SGNB RELEASE REQUEST REJECT |
| SgNB initiated SgNB Release | SGNB RELEASE REQUIRED | SGNB RELEASE CONFIRM | |
| EN-DC X2 Setup | EN-DC X2 SETUP REQUEST | EN-DC X2 SETUP RESPONSE | EN-DC X2 SETUP FAILURE |
| EN-DC Configuration Update | EN-DC CONFIGURATION UPDATE | EN-DC CONFIGURATION UPDATE ACKNOWLEDGE | EN-DC CONFIGURATION UPDATE FAILURE |
| EN-DC Cell Activation | EN-DC CELL ACTIVATION REQUEST | EN-DC CELL ACTIVATION RESPONSE | EN-DC CELL ACTIVATION FAILURE |
| E-UTRA - NR Cell Resource Coordination | E-UTRA - NR CELL RESOURCE COORDINATION REQUEST | E-UTRA - NR CELL RESOURCE COORDINATION RESPONSE | |

| Elementary Procedure | Initiating Message | Successful Outcome | Unsuccessful Outcome |
|---|---|---|---|
| | | Response message | Response message |
| EN-DC X2 Removal | EN-DC X2 REMOVAL REQUEST | EN-DC X2 REMOVAL RESPONSE | EN-DC X2 REMOVAL FAILURE |
| EN-DC Resource Status Reporting Initiation | EN-DC RESOURCE STATUS REQUEST | EN-DC RESOURCE STATUS RESPONSE | EN-DC RESOURCE STATUS FAILURE |
| UE Radio Capability ID Mapping | UE RADIO CAPABILITY ID MAPPING REQUEST | UE RADIO CAPABILITY ID MAPPING RESPONSE | |

**Table 8.1-2: Class 2 Elementary Procedures**

| Elementary Procedure | Initiating Message |
|---|---|
| Load Indication | LOAD INFORMATION |
| Handover Cancel | HANDOVER CANCEL |
| SN Status Transfer | SN STATUS TRANSFER |
| UE Context Release | UE CONTEXT RELEASE |
| Resource Status Reporting | RESOURCE STATUS UPDATE |
| Error Indication | ERROR INDICATION |
| Radio Link Failure Indication | RLF INDICATION |
| Handover Report | HANDOVER REPORT |
| X2 Release | X2 RELEASE |
| X2AP Message Transfer | X2AP MESSAGE TRANSFER |
| SeNB Reconfiguration Completion | SENB RECONFIGURATION COMPLETE |
| MeNB initiated SeNB Release | SENB RELEASE REQUEST |
| SeNB Counter Check | SENB COUNTER CHECK REQUEST |
| SgNB Reconfiguration Completion | SGNB RECONFIGURATION COMPLETE |
| SgNB Counter Check | SGNB COUNTER CHECK REQUEST |
| RRC Transfer | RRC TRANSFER |
| Secondary RAT Data Usage Report | SECONDARY RAT DATA USAGE REPORT |
| SgNB Activity Notification | SGNB ACTIVITY NOTIFICATION |
| Data Forwarding Address Indication | DATA FORWARDING ADDRESS INDICATION |
| gNB Status Indication | GNB STATUS INDICATION |
| EN-DC Configuration Transfer | EN-DC CONFIGURATION TRANSFER |
| Trace Start | TRACE START |
| Deactivate Trace | DEACTIVATE TRACE |
| Handover Success | HANDOVER SUCCESS |
| Conditional Handover Cancel | CONDITIONAL HANDOVER CANCEL |
| Early Status Transfer | EARLY STATUS TRANSFER |
| EN-DC Resource Status Reporting | EN-DC RESOURCE STATUS UPDATE |
| Cell Traffic Trace | CELL TRAFFIC TRACE |
| F1-C Traffic Transfer | F1-C TRAFFIC TRANSFER |
| Access and Mobility Indication | ACCESS AND MOBILITY INDICATION |
| Conditional PSCell Change Cancel | CONDITIONAL PSCELL CHANGE CANCEL |

# 8.2 Basic mobility procedures

## 8.2.1 Handover Preparation

### 8.2.1.1 General

This procedure is used to establish necessary resources in an eNB for an incoming handover. If the procedure concerns a conditional handover, parallel transactions are allowed. Possible parallel requests are identified by the target cell ID when the source UE AP IDs are the same.

The procedure uses UE-associated signalling.

### 8.2.1.2 Successful Operation

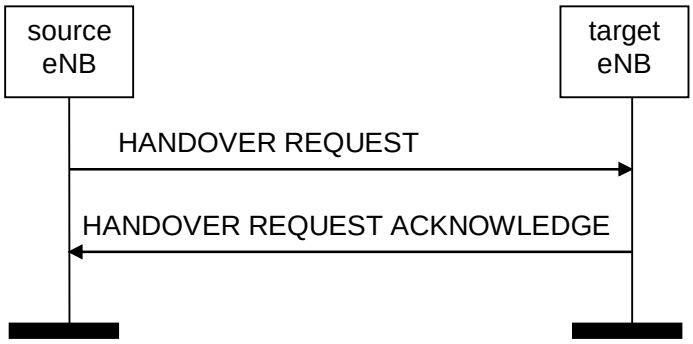



**Figure 8.2.1.2-1: Handover Preparation, successful operation**

The source eNB initiates the procedure by sending the HANDOVER REQUEST message to the target eNB. When the source eNB sends the HANDOVER REQUEST message, it shall start the timer $T_{RELOCprep}$.

If the *Conditional Handover Information Request* IE is contained in the HANDOVER REQUEST message, the target eNB shall consider that the request concerns a conditional handover and shall include the *Conditional Handover Information Acknowledge* IE in the HANDOVER REQUEST ACKNOWLEDGE message.

If the *New eNB UE X2AP ID* IE is contained in the *Conditional Handover Information Request* IE included in the HANDOVER REQUEST message, then the target eNB shall remove the existing prepared conditional HO identified by the *New eNB UE X2AP ID* IE and the *Target Cell ID* IE. It is up to the implementation of the target eNB when to remove the HO information.

The allocation of resources according to the values of the *Allocation and Retention Priority* IE included in the *E-RAB Level QoS Parameters* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [4].

The source eNB may include in the *GUMMEI* IE any GUMMEI corresponding to the source MME node.

If at least one of the requested non-GBR E-RABs is admitted to the cell indicated by the *Target Cell ID* IE, the target eNB shall reserve necessary resources, and send the HANDOVER REQUEST ACKNOWLEDGE message back to the source eNB. The target eNB shall include the E-RABs for which resources have been prepared at the target cell in the *E-RABs Admitted List* IE. The target eNB shall include the E-RABs that have not been admitted in the *E-RABs Not Admitted List* IE with an appropriate cause value.

At reception of the HANDOVER REQUEST message the target eNB shall:

- prepare the configuration of the AS security relation between the UE and the target eNB by using the information in the *UE Security Capabilities* IE and the *AS Security Information* IE in the *UE Context Information* IE.

For each E-RAB for which the source eNB proposes to do forwarding of downlink data, the source eNB shall include the *DL Forwarding* IE within the *E-RABs To be Setup Item* IE of the HANDOVER REQUEST message. The source eNB shall include the DL Forwarding IE if it requests a DAPS handover for that E-RAB. For each E-RAB that it has decided to admit, the target eNB may include the *DL GTP Tunnel Endpoint* IE within the *E-RABs Admitted Item* IE of the HANDOVER REQUEST ACKNOWLEDGE message to indicate that it accepts the proposed forwarding of

downlink data for this bearer. This GTP tunnel endpoint may be different from the corresponding GTP tunnel endpoint, i.e. the information contained in the *Transport Layer address* IE and *GTP TEID* IE in the *E-RAB To Be Switched in Downlink List* IE of the PATH SWITCH REQUEST message (see TS 36.413 [4]) depending on implementation choice.

For each bearer in the *E-RABs Admitted List* IE, the target eNB may include the *UL GTP Tunnel Endpoint* IE to indicate that it requests data forwarding of uplink packets to be performed for that bearer.

Upon reception of the HANDOVER REQUEST ACKNOWLEDGE message the source eNB shall stop the timer $T_{RELOCprep}$ and terminate the Handover Preparation procedure. If the procedure was initiated for an immediate handover, the source eNB shall start the timer TX2RELOCoverall. The source eNB is then defined to have a Prepared Handover for that X2 UE-associated signalling.

If the *Trace Activation* IE is included in the HANDOVER REQUEST message then the target eNB shall, if supported, initiate the requested trace function as described in TS 32.422 [6]. In particular, the target eNB shall, if supported:

- if the *Trace Activation* IE does not include the *MDT Configuration* IE, initiate the requested trace session as described in TS 32.422 [6];

- if the *Trace Activation* IE includes the *MDT Activation* IE, within the *MDT Configuration* IE, set to "Immediate MDT and Trace" initiate the requested trace session and MDT session as described in TS 32.422 [6];

- if the *Trace Activation* IE includes the *MDT Activation* IE, within the *MDT Configuration* IE, set to "Immediate MDT Only" initiate the requested MDT session as described in TS 32.422 [6] and the target eNB shall ignore *Interfaces To Trace* IE, and *Trace Depth* IE;

- if the *Trace Activation* IE includes the *MDT Location Information* IE, within the *MDT Configuration* IE, store this information and take it into account in the requested MDT session;

- if the *Trace Activation* IE includes the *Signalling based MDT PLMN List* IE, within the *MDT Configuration* IE, the eNB may use it to propagate the MDT Configuration as described in TS 37.320 [31];

- if the *Trace Activation* IE includes the *UE Application layer measurement configuration* IE, initiate the requested trace session and QoE Measurement Collection function as described in TS 36.300 [15].

- if the *Trace Activation* IE includes the *Bluetooth Measurement Configuration* IE, within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [31].

- if the *Trace Activation* IE includes the *WLAN Measurement Configuration* IE, within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [31].

- if the *Trace Activation* IE includes the *MDT Configuration NR* IE, store and forward the *MDT Configuration NR* IE to the SgNB, if the target eNB has configured EN-DC for the UE.

- if the *Trace Activation* IE includes the *Sensor Measurement Configuration* IE within the *MDT Configuration* IE, take it into account for MDT Configuration as described in TS 37.320 [31].

If the *Management Based MDT Allowed* IE only or the *Management Based MDT Allowed* IE and the *Management Based MDT PLMN List* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, store the received information in the UE context, and use this information to allow subsequent selection of the UE for management based MDT defined in TS 32.422 [6].

If the *Masked IMEISV* IE is contained in the HANDOVER REQUEST message the target eNB shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

The source eNB shall, if supported and available in the UE context, include the *Management Based MDT Allowed* IE and the *Management Based MDT PLMN List* IE in the HANDOVER REQUEST message, except if the source eNB selects a serving PLMN in the target eNB which is not included in the Management Based MDT PLMN List. If the *Management Based MDT PLMN List* IE is not present, the source eNB shall, if supported, include the *Management Based MDT Allowed* IE, if this information is available in the UE context, in the HANDOVER REQUEST message, except if the source eNB selects a serving PLMN in the target eNB different from the serving PLMN in the source eNB.

If the *Handover Restriction List* IE is

- contained in the HANDOVER REQUEST message, the target eNB shall

  - store the information received in the *Handover Restriction List* IE in the UE context;

- use this information to determine a target for the UE during subsequent mobility action for which the eNB provides information about the target of the mobility action towards the UE, except when one of the E-RABs has a particular ARP value (TS 23.401 [12]) in which case the information shall not apply;

- use this information to select a proper SCG during dual connectivity operation.

- not contained in the HANDOVER REQUEST message, the target eNB shall consider that no roaming and no access restriction apply to the UE.

If the *Location Reporting Information* IE is included in the HANDOVER REQUEST message then the target eNB should initiate the requested location reporting functionality as defined in TS 36.413 [4].

If the *SRVCC Operation Possible* IE is included in the HANDOVER REQUEST message, the target eNB shall store the content of such IE in the UE context and use it as defined in TS 23.216 [20].

If the *UE Security Capabilities* IE included in the HANDOVER REQUEST message only contains the EIA0 algorithm as defined in TS 33.401 [18] and if this EIA0 algorithm is defined in the configured list of allowed integrity protection algorithms in the eNB (TS 33.401 [18]), the eNB shall take it into use and ignore the keys received in the *AS Security Information* IE.

The HANDOVER REQUEST message shall contain the *Subscriber Profile ID for RAT/Frequency priority* IE, if available.

If the *Subscriber Profile ID for RAT/Frequency priority* IE is contained in the HANDOVER REQUEST message, the target eNB shall store this information and the target eNB should use the information as defined in TS 36.300 [15].

If the *Additional RRM Policy Index* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information and the target eNB should use the information as defined in TS 36.300 [15].

Upon reception of *UE History Information* IE in the HANDOVER REQUEST message, the target eNB shall collect the information defined as mandatory in the *UE History Information* IE and shall, if supported, collect the information defined as optional in the *UE History Information* IE, for as long as the UE stays in one of its cells, and store the collected information to be used for future handover preparations.

Upon reception of the *UE History Information from the UE* IE in the HANDOVER REQUEST message, the target eNB shall, if supported, store the collected information to be used for future handover preparations.

If the *Mobility Information* IE is provided in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information and use it as defined in TS 36.300 [15]. The target eNB shall, if supported, store the C-RNTI of the source cell received in the HANDOVER REQUEST message.

If the *Expected UE Behaviour* IE is provided in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information and may use it to determine the RRC connection time.

If the *ProSe Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the eNB shall, if supported, consider that the UE is authorized for the relevant ProSe service(s).

If the *V2X Services Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the eNB shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *UE Context Reference at the SeNB* IE is contained in the HANDOVER REQUEST message the target eNB may use it as specified in TS 36.300 [15]. In this case, the source eNB may expect the target eNB to include the *UE Context Kept Indicator* IE set to "True" in the HANDOVER REQUEST ACKNOWLEDGE message, which shall use this information as specified in TS 36.300 [15]. If the *UE Context Reference at the WT* IE is contained in the HANDOVER REQUEST message, the target eNB may use it as specified in TS 36.300 [15]. In this case, the source eNB may expect the target eNB to include the *WT UE Context Kept Indicator* IE set to "True" in the HANDOVER REQUEST ACKNOWLEDGE message; the source eNB shall use this information as specified in TS 36.300 [15].

If the *UE Context Reference at the SgNB* IE is contained in the HANDOVER REQUEST message the target eNB may use it as specified in TS 37.340 [32]. In this case, the source eNB may expect the target eNB to include the *UE Context Kept Indicator* IE set to "True" in the HANDOVER REQUEST ACKNOWLEDGE message, which shall use this information as specified in TS 37.340 [32].

If the *Bearer Type* IE is included in the HANDOVER REQUEST message and is set to "non IP", then the target eNB shall not perform IP header compression for the concerned E-RAB.

If the *Ethernet Type* IE is included in the HANDOVER REQUEST message and is set to "True", then the target eNB shall, if supported, take this into account to perform header compression appropriately for the concerned E-RAB.

If the *UE Sidelink Aggregate Maximum Bit Rate* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for V2X services.

If the *NR UE Security Capabilities* IE is included in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information in the UE context and send it to the respective peer node during subsequent handover preparations and/or EN-DC operations for the UE as defined in TS 33.401 [18].

If the *Aerial UE subscription information* IE is included in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information in the UE context and use it as defined in TS 36.300 [15].

If the *Subscription Based UE Differentiation Information* IE is included in the HANDOVER REQUEST message, the eNB shall, if supported, store this information in the UE context for further use according to TS 23.401 [12].

If the *DAPS Request Information* IE is included for an E-RAB to be setup in the HANDOVER REQUEST message, the target eNB shall consider that the request concerns a DAPS handover for that E-RAB, as described in TS 36.300 [15]. Accordingly, the target eNB shall include the *DAPS Response Information* IE in the HANDOVER REQUEST ACKNOWLEDGE message.

If the *Maximum Number of CHO Preparations* IE is included in *Conditional Handover Information Acknowledge* IE contained in the the HANDOVER REQUEST ACKNOWLEDGE message, then the source eNB should not prepare more candidate target cells for a CHO for the same UE towards the target eNB than the number indicated in the *Maximum Number of CHO Preparations* IE.

If the *Estimated Arrival Probability* IE is contained in the *Conditional Handover Information Request* IE included in the HANDOVER REQUEST message, then the target eNB may use the information to allocate necessary resources for the incoming CHO.

If the *EPC Handover Restriction List Container* IE is included in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information in the UE context and shall use it as specified in TS 36.300 [15].

If the *NR V2X Services Authorized* IE is contained in the HANDOVER REQUEST message and it contains one or more IEs set to "authorized", the eNB shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *NR UE Sidelink Aggregate Maximum Bit Rate* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for NR V2X services.

If the *PC5 QoS Parameters* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, use it for the concerned UE’s NR sidelink communication as specified in TS 23.285 [41].

If the *UE Radio Capability ID* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information in the UE context and use it as specified in TS 23.401 [12].

If the *IAB Node Indication* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, consider that the request is for an IAB node.

If the *IMS Voice EPS Fallback from 5G* IE is contained in the HANDOVER REQUEST message, the target eNB shall, if supported, store this information in the UE context and consider that the UE was previously handed over from NG-RAN to E-UTRAN due to an IMS voice fallback.

If the target eNB receives a HANDOVER REQUEST message containing the *Source DL Forwarding IP Address* IE as part of the *E-RABs To Be Setup Item* IE, the target eNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

For each E-RAB for which the *Security Indication* IE is included in the *E-RABs To Be Setup Item IEs* IE of the HANDOVER REQUEST message, and the EIA7 bit in the *Integrity Protection Algorithms* IE contained in *UE Security Capabilities* IE is set to "1":

- if the *Integrity Protection Indication* IE is set to "required", the target eNB shall, if supported, perform user plane integrity protection for the concerned E-RAB as specified in TS 33.401 [18], otherwise it shall reject the establishment of the concerned E-RAB with an appropriate cause value.

- if the *Integrity Protection Indication* IE is set to "preferred", the target eNB should perform user plane integrity protection for the concerned E-RAB as specified in TS 33.401 [18].

- if the *Integrity Protection Indication* IE is set to "not needed", the target eNB shall not perform user plane integrity protection for the concerned E-RAB.

**Interaction with SN Status Transfer procedure:**

If the *UE Context Kept Indicator* IE set to "True" and the *E-RABs transferred to MeNB* IE are included in the HANDOVER REQUEST ACKNOWLEDGE message, then the source eNB shall, if supported, include the uplink/downlink PDCP SN and HFN status received from the SgNB in the SN Status Transfer procedure towards the target eNB, as specified in TS 37.340 [32].

### 8.2.1.3 Unsuccessful Operation

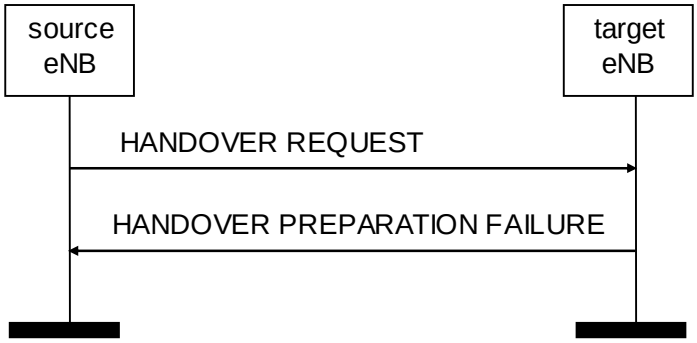



**Figure 8.2.1.3-1: Handover Preparation, unsuccessful operation**

If the target eNB does not admit at least one non-GBR E-RAB, or a failure occurs during the Handover Preparation, the target eNB shall send the HANDOVER PREPARATION FAILURE message to the source eNB. The message shall contain the *Cause* IE with an appropriate value.

If the target eNB receives a HANDOVER REQUEST message containing *RRC Context* IE that does not include required information as specified in TS 36.331 [9], the target eNB shall send the HANDOVER PREPARATION FAILURE message to the source eNB.

If the *Conditional Handover Information Request* IE is contained in the HANDOVER REQUEST message and the target eNB rejects the handover or a failure occurs during the Handover Preparation, the target eNB shall include the *Requested Target Cell ID* IE in the HANDOVER PREPARATION FAILURE message.

**Interactions with Handover Cancel procedure:**

If there is no response from the target eNB to the HANDOVER REQUEST message before timer $T_{RELOCprep}$ expires in the source eNB, the source eNB should cancel the Handover Preparation procedure towards the target eNB by initiating the Handover Cancel procedure with the appropriate value for the *Cause* IE. The source eNB shall ignore any HANDOVER REQUEST ACKNOWLEDGE or HANDOVER PREPARATION FAILURE message received after the initiation of the Handover Cancel procedure and remove any reference and release any resources related to the concerned X2 UE-associated signalling.

### 8.2.1.4 Abnormal Conditions

If the target eNB receives a HANDOVER REQUEST message containing multiple *E-RAB ID* IEs (in the *E-RABs To Be Setup List* IE) set to the same value, the target eNB shall not admit the corresponding E-RABs.

If the target eNB receives a HANDOVER REQUEST message containing a *E-RAB Level QoS Parameters* IE which contains a *QCI* IE indicating a GBR bearer (as defined in TS 23.203 [13]), and which does not contain the *GBR QoS Information* IE, the target eNB shall not admit the corresponding E-RAB.

If the supported algorithms for encryption defined in the *Encryption Algorithms* IE in the *UE Security Capabilities* IE in the *UE Context Information* IE, plus the mandated support of EEA0 in all UEs (TS 33.401 [18]), do not match any algorithms defined in the configured list of allowed encryption algorithms in the target eNB (TS 33.401 [18]), the target eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

If the supported algorithms for integrity defined in the *Integrity Protection Algorithms* IE in the *UE Security Capabilities* IE in the *UE Context Information* IE, plus the mandated support of the EIA0 algorithm in all UEs (TS 33.401 [18]), do not match any algorithms defined in the configured list of allowed integrity protection algorithms in the eNB (TS 33.401 [18]), the eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

If the target eNB receives a HANDOVER REQUEST message which does not contain the *Handover Restriction List* IE, and the PLMN to be used cannot be determined otherwise, the target eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

If the target eNB receives a HANDOVER REQUEST message containing the *Handover Restriction List* IE, and the serving PLMN is not supported by the target cell, the target eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

If the target eNB receives a HANDOVER REQUEST message which does not contain the *CSG Membership Status* IE, and the target cell is a hybrid cell, the target eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

If the target cell is a CSG cell and the target eNB has not received any CSG ID of the source cell, the target eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

If the target cell is a CSG cell with a different CSG from the source cell, the target eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

If the *CHO trigger* IE is set to "CHO-replace" in the HANDOVER REQUEST message, but there is no CHO prepared for the included *New eNB UE X2AP ID* IE, or the candidate cell in the *Target Cell ID* IE was not prepared using the same UE-associated signaling connection, the target eNB shall reject the procedure using the HANDOVER PREPARATION FAILURE message.

## 8.2.2 SN Status Transfer

### 8.2.2.1 General

The purpose of the SN Status Transfer procedure is to transfer the uplink PDCP SN and HFN receiver status and the downlink PDCP SN and HFN transmitter status either, from the source to the target eNB during an X2 handover, between the eNBs involved in dual connectivity and/or LWA, or between MeNB and en-gNB involved in EN-DC, for each respective E-RAB for which PDCP SN and HFN status preservation applies.

In case that the X2 handover is a DAPS handover, the SN Status Transfer procedure may also be used to transfer the uplink PDCP SN and HFN receiver status, and the downlink PDCP SN and HFN transmitter status for an E-RAB associated with RLC-UM and configured with DAPS as described in TS 36.300 [15].

If the SN Status Transfer procedure is applied in the course of dual connectivity, LWA, RRC connection re-establishment or EN-DC, in the subsequent specification text

- the behaviour of the eNB from which the E-RAB context is transferred, i.e., the eNB involved in dual connectivity, LWA, RRC connection re-establishment from which data forwarding, is specified by the behaviour of the "source eNB",

- the behaviour of the eNB to which the E-RAB context is transferred, i.e., the eNB involved in dual connectivity, LWA, RRC connection re-establishment to which data is forwarded, is specified by the behaviour of the "target eNB".

- in case of EN-DC, the behaviour of the node from which the E-RAB context is transferred, i.e., either the en-gNB or the MeNB from which data is forwarded, is specified by the behaviour of the "source eNB",

- in case of EN-DC, the behaviour of the node to which the E-RAB context is transferred, i.e., either the en-gNB or the MeNB to which data is forwarded, is specified by the behaviour of the "target eNB".

The procedure uses UE-associated signalling.

### 8.2.2.2 Successful Operation

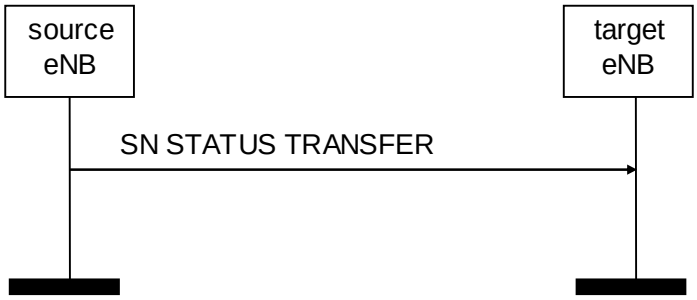



**Figure 8.2.2.2-1: SN Status Transfer, successful operation**

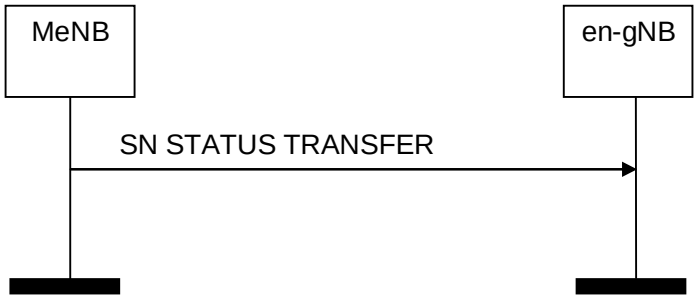



**Figure 8.2.2.2-2: MeNB initiated SN Status Transfer for EN-DC, successful operation**

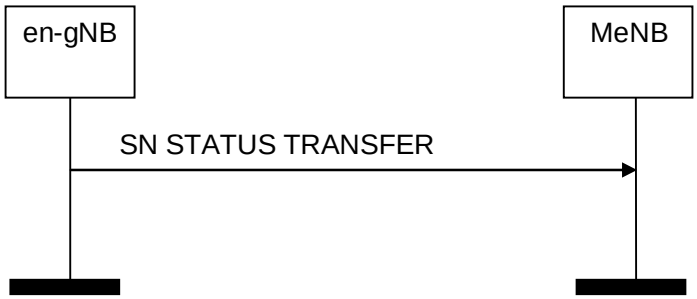



**Figure 8.2.2.2-3: en-gNB initiated SN Status Transfer for EN-DC, successful operation**

The source eNB initiates the procedure by stop assigning PDCP SNs to downlink SDUs and stop delivering UL SDUs towards the EPC and sending the SN STATUS TRANSFER message to the target eNB at the time point when it considers the transmitter/receiver status to be frozen. The target eNB using Full Configuration for this handover as per TS 36.300 [15] or for the EN-DC operations as per TS 37.340 [32] shall ignore the information received in this message. In case of EN-DC, if the target eNB performs PDCP version change or PDCP SN length change or RLC mode change for an E-RAB as specified in TS 37.340 [32], it shall ignore the information received for that E-RAB in this message.

In case that the X2 handover is a DAPS handover, the source eNB may continue assigning PDCP SNs to downlink SDUs and delivering uplink SDUs toward the EPC when initiating this procedure for E-RABs not configured with DAPS as in TS 36.300 [15].

The *E-RABs Subject To Status Transfer List* IE included in the SN STATUS TRANSFER message contains the E-RAB ID(s) corresponding to the E-RAB(s) for which PDCP SN and HFN status preservation shall be applied. In case that the X2 handover is a DAPS handover, this IE may contain the E-RAB ID(s) corresponding to the E-RAB(s) associated with RLC-UM.

If the source eNB includes in the SN STATUS TRANSFER message, the information on the missing and received uplink SDUs in the *Receive Status Of UL PDCP SDUs* IE or *Receive Status Of UL PDCP SDUs Extended* IE or *Receive Status Of UL PDCP SDUs for PDCP SN Length 18* IE for each E-RAB for which the source eNB has accepted the request from the target eNB for uplink forwarding, then the target eNB may use it in a Status Report message sent to the UE over the radio.

For each E-RAB for which the *DL COUNT Value* IE is received in the SN STATUS TRANSFER message, the target eNB shall use it to mark with the value contained in the *PDCP-SN* IE of this IE the first downlink packet for which there is no PDCP SN yet assigned. If the *DL COUNT Value Extended* IE or *DL COUNT Value for PDCP SN Length 18* IE is included in the *E-RABs Subject To Status Transfer Item* IE, the target eNB shall, if supported, use the value contained in the *PDCP-SN Extended* IE of the *DL COUNT Value Extended* IE or *PDCP-SN Length 18 IE of the DL COUNT Value for PDCP SN Length 18* IE instead of the value contained in the *PDCP-SN* IE of the *DL COUNT Value* IE.

For each E-RAB for which the *UL COUNT Value* IE is received in the SN STATUS TRANSFER message, the target eNB shall not deliver any uplink packet which has a PDCP SN lower than the value contained in the *PDCP-SN* IE of this IE. If the *UL COUNT Value Extended* IE or *UL COUNT Value for PDCP SN Length 18* IE is included in the *E-RABs Subject To Status Transfer Item* IE, the target eNB shall, if supported, use the value contained in the *PDCP-SN Extended* IE of the *UL COUNT Value Extended* IE or *PDCP-SN Length 18 IE of the UL COUNT Value for PDCP SN Length 18* IE instead of the value contained in the *PDCP-SN* IE of the *UL COUNT Value* IE.

**EN-DC**

If the en-gNB sends the message to the MeNB, then the *SgNB UE X2AP ID* IE shall be included in the SN STATUS TRANSFER message, while the *Old eNB UE X2AP ID* IE is ignored. The *SgNB UE X2AP ID* IE is used as the old UE ID.

If the MeNB sends the message to the en-gNB, then the *SgNB UE X2AP ID* IE shall be included in the SN STATUS TRANSFER message, while the *New eNB UE X2AP ID* IE is ignored. The *SgNB UE X2AP ID* IE is used as the new UE ID.

#### 8.2.2.3 Abnormal Conditions

If the target eNB receives this message for a UE for which no prepared handover exists at the target eNB, the target eNB shall ignore the message.

### 8.2.3 UE Context Release

#### 8.2.3.1 General

For handover, the UE Context Release procedure is initiated by the target eNB to indicate to the source eNB that radio and control plane resources for the associated UE context are allowed to be released.

For dual connectivity, UE Context Release procedure is initiated by the MeNB to finally release the UE context at the SeNB. For dual connectivity specific mobility scenarios specified in TS 36.300 [15] only resources related to the UE-associated signalling connection between the MeNB and the SeNB are released. For EN-DC, the UE Context Release procedure is initiated by the MeNB to finally release the UE context at the en-gNB. For EN-DC specific mobility scenarios specified in TS 37.340 [32] where SCG radio resources in the en-gNB are kept, only resources related to the UE-associated signalling connection between the MeNB and the en-gNB are released.

The procedure uses UE-associated signalling.

#### 8.2.3.2 Successful Operation




**Figure 8.2.3.2-1: UE Context Release, successful operation for handover**




**Figure 8.2.3.2-2: UE Context Release, successful operation for dual connectivity**




**Figure 8.2.3.2-3: UE Context Release, successful operation for EN-DC**

**Handover**

The UE Context Release procedure is initiated by the target eNB. By sending the UE CONTEXT RELEASE message the target eNB informs the source eNB of Handover success and triggers the release of resources.

Upon reception of the UE CONTEXT RELEASE message, the source eNB may release radio and control plane related resources associated to the UE context. For E-RABs for which data forwarding has been performed, the source eNB should continue forwarding of U-plane data as long as packets are received at the source eNB from the EPC or the source eNB buffer has not been emptied (an implementation dependent mechanism decides that data forwarding can be stopped). When the eNB supporting L-GW function for SIPTO@LN operation releases radio and control plane related resources associated to the UE context, it shall also request using intra-node signalling the collocated L-GW to release the SIPTO@LN PDN connection as defined in TS 23.401 [12].

**Dual Connectivity**

The UE Context Release procedure is initiated by the MeNB. By sending the UE CONTEXT RELEASE message the MeNB informs the SeNB that the UE Context can be removed.

Upon reception of the UE CONTEXT RELEASE message, the SeNB may release radio and control plane related resources associated to the UE context. For E-RABs for which data forwarding has been performed, the SeNB should continue forwarding of U-plane data as long as packets are received at the SeNB from the EPC or the SeNB buffer has not been emptied (an implementation dependent mechanism decides that data forwarding can be stopped). The SeNB supporting L-GW function for LIPA operation shall also request using intra-node signalling the collocated L-GW to release the LIPA PDN connection as defined in TS 23.401 [12]. If the *SIPTO Bearer Deactivation Indication* IE is received in the UE CONTEXT RELEASE message, the SeNB supporting L-GW function for SIPTO@LN operation shall also request using intra-node signalling the collocated L-GW to release the SIPTO@LN PDN connection as defined in TS 23.401 [12].

**EN-DC**

The UE Context Release procedure is initiated by the MeNB. By sending the UE CONTEXT RELEASE message the MeNB informs the en-gNB that the UE Context can be removed.

Upon reception of the UE CONTEXT RELEASE message, the en-gNB may release radio and control plane related resources associated to the UE context. For E-RABs for which data forwarding has been performed, the en-gNB should continue forwarding of U-plane data as long as packets are received at the en-gNB from the EPC or the en-gNB buffer has not been emptied (an implementation dependent mechanism decides that data forwarding can be stopped).

In the course of signalling for EN-DC, the *SgNB UE X2AP ID* IE shall be included in the UE CONTEXT RELEASE message, while the *Old eNB UE X2AP ID* IE is ignored. The *SgNB UE X2AP ID* IE is used as the new UE ID.

**Interaction with the MeNB initiated SeNB Release procedure:**

The SeNB may receive the SENB RELEASE REQUEST message including the *UE Context Kept Indicator* IE set to "True", upon which the SeNB shall, if supported, only release the resources related to the UE-associated signalling connection between the MeNB and the SeNB, as specified in TS 36.300 [15].

**Interaction with the MeNB initiated SgNB Release procedure:**

The en-gNB may receive the SGNB RELEASE REQUEST message including the *UE Context Kept Indicator* IE set to "True", upon which the en-gNB shall, if supported, only release the resources related to the UE-associated signalling connection between the MeNB and the en-gNB, as specified in TS 37.340 [32].

#### 8.2.3.3 Unsuccessful Operation

Not applicable.

#### 8.2.3.4 Abnormal Conditions

If the UE Context Release procedure is not initiated towards the source eNB from any prepared eNB before the expiry of the timer $TX2_{RELOCoverall}$, the source eNB shall request the MME to release the UE context.

If the UE returns to source eNB before the reception of the UE CONTEXT RELEASE message or the expiry of the timer $TX2_{RELOCoverall}$, the source eNB shall stop the $TX2_{RELOCoverall}$ and continue to serve the UE.

### 8.2.4 Handover Cancel

#### 8.2.4.1 General

The Handover Cancel procedure is used to enable a source eNB to cancel an ongoing handover preparation or an already prepared handover.

The procedure uses UE-associated signalling.

#### 8.2.4.2 Successful Operation




**Figure 8.2.4.2-1: Handover Cancel, successful operation**

The source eNB initiates the procedure by sending the HANDOVER CANCEL message to the target eNB. The source eNB shall indicate the reason for cancelling the handover by means of an appropriate cause value.

At the reception of the HANDOVER CANCEL message, the target eNB shall remove any reference to, and release any resources previously reserved to the concerned UE context.

The *New eNB UE X2AP ID* IE and, if available, the *New eNB UE X2AP ID Extension* IE shall be included if it has been obtained from the target eNB.

If the *Candidate Cells To Be Cancelled List* IE is included in the HANDOVER CANCEL message, the target eNB shall consider that the source eNB is cancelling only the handover associated to the candidate cells identified by the included ECGI and associated to the UE-associated signaling connection identified by the *Old eNB UE X2AP ID* IE (or the *Old*

*eNB UE X2AP ID Extension* IE if included) and, if included , also by the *New eNB UE X2AP ID* IE (or the *New eNB UE X2AP ID Extension* IE if included).

#### 8.2.4.3 Unsuccessful Operation

Not applicable.

#### 8.2.4.4 Abnormal Conditions

Should the HANDOVER CANCEL message refer to a context that does not exist, the target eNB shall ignore the message.

If the *Candidate Cells To Be Cancelled List* IE is included in the HANDOVER CANCEL message and the handover is not associated to a conditional handover, the target eNB shall ignore the *Candidate Cells To Be Cancelled List* IE.

If one or more candidate cells in the *Candidate Cells To Be Cancelled List* IE included in the HANDOVER CANCEL message were not prepared using the same UE-associated signaling connection, the target eNB shall ignore those non-associated candidate cells.

### 8.2.5 Handover Success

#### 8.2.5.1 General

The Handover Success procedure is used during a conditional handover or a DAPS handover to enable a target eNB to inform the source eNB that the UE has successfully accessed the target eNB.

The procedure uses UE-associated signalling.

#### 8.2.5.2 Successful Operation




**Figure 8.2.5.2-1: Handover Success, successful operation**

The target eNB initiates the procedure by sending the HANDOVER SUCCESS message to the source eNB.

If late data forwarding was configured for this UE, the source eNB shall start data forwarding using the tunnel information related to the global target cell ID provided in the HANDOVER SUCCESS message.

When the source eNB receives the HANDOVER SUCCESS message, it shall consider all other CHO preparations accepted for this UE under the same UE-associated signalling connection in the target eNB as cancelled.

**Interactions with other procedures**

If a CONDITIONAL HANDOVER CANCEL message was received for this UE prior the reception of the HANDOVER SUCCESS message, the source eNB node shall consider that the UE successfully executed the handover. The source eNB may initiate Handover Cancel procedure towards the other signaling connections or other candidate target eNBs for this UE, if any.

#### 8.2.5.3 Unsuccessful Operation

Not applicable.

### 8.2.5.4 Abnormal Conditions

If the HANDOVER SUCCESS message refers to a context that does not exist, the source eNB shall ignore the message.

## 8.2.6 Conditional Handover Cancel

### 8.2.6.1 General

The Conditional Handover Cancel procedure is used to enable a target eNB to cancel an already prepared conditional handover.

The procedure uses UE-associated signalling.

### 8.2.6.2 Successful Operation




**Figure 8.2.6.2-1: Conditional Handover Cancel, successful operation**

The target eNB initiates the procedure by sending the CONDITIONAL HANDOVER CANCEL message to the source eNB. The target eNB shall indicate the reason for cancelling the conditional handover by means of an appropriate cause value.

The *New eNB UE X2AP ID* IE and, if available, the *New eNB UE X2AP ID Extension* IE shall be included.

At the reception of the CONDITIONAL HANDOVER CANCEL message, the source eNB shall consider that the target eNB is about to remove any reference to, and release any resources previously reserved for candidate cells associated to the UE-associated signalling identified by the *Old eNB UE X2AP ID* IE (or the *Old eNB UE X2AP ID Extension* IE if included) and the *New eNB UE X2AP ID* IE (or the *New eNB UE X2AP ID Extension* IE if included).

If the *Candidate Cells To Be Cancelled List* IE is also included, the source eNB shall consider that only the resources reserved for the cells identified by the included ECGI are about to be released.

### 8.2.6.3 Unsuccessful Operation

Not applicable.

### 8.2.6.4 Abnormal Conditions

Should the CONDITIONAL HANDOVER CANCEL message refer to a context that does not exist, the source eNB shall ignore the message.

If one or more candidate cells in the *Candidate Cells To Be Cancelled List* IE included in the CONDITIONAL HANDOVER CANCEL message were not prepared using the same UE-associated signaling connection, the source eNB shall ignore those non-associated candidate cells.

## 8.2.7 Early Status Transfer

### 8.2.7.1 General

The purpose of the Early Status Transfer procedure is to transfer the COUNT of the first downlink SDU that the source eNB forwards to the target eNB or the COUNT for discarding already forwarded downlink SDUs for respective E-RAB during DAPS Handover or Conditional Handover.

For Dual Connectivity or EN-DC, the Early Status Transfer procedure is also used, during a Conditional Handover, from the SeNB to the MeNB as specified in TS 36.300 [15], or from the en-gNB to the MeNB as specified in TS 37.340 [32].

For Conditional PSCell Addition in EN-DC, the Early Status Transfer procedure is also used, from the MeNB to the en-gNB as specified in TS 37.340 [32].

For Conditional PSCell Change in EN-DC, the Early Status Transfer procedure is also used, from the source en-gNB to the MeNB, and from the MeNB to the target en-gNB, as specified in TS 37.340 [32].

The procedure uses UE-associated signalling.

### 8.2.7.2 Successful Operation

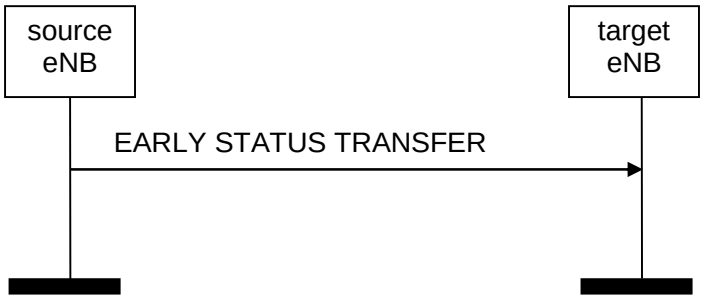



**Figure 8.2.7.2-1: Early Status Transfer during DAPS Handover or Conditional Handover, successful operation**

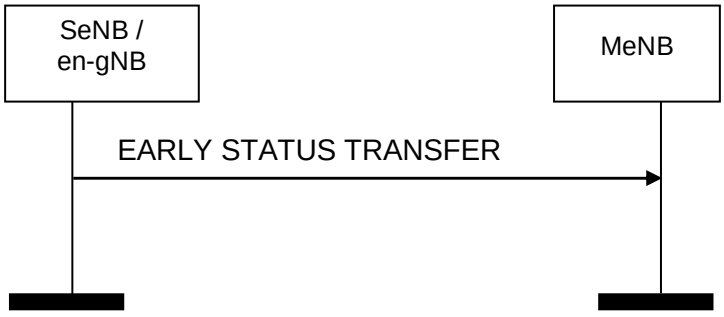



**Figure 8.2.7.2-2: Early Status Transfer during Conditional Handover in dual connectivity or EN-DC operation, successful operation**

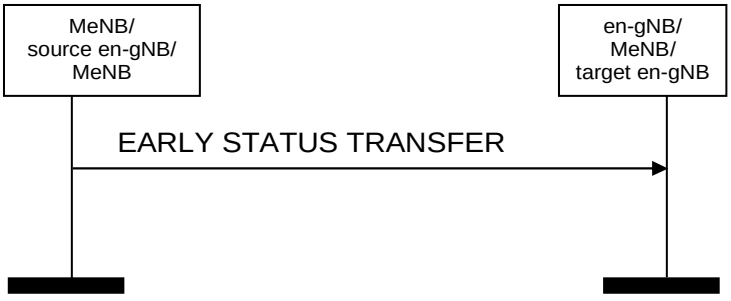



**Figure 8.2.7.2-3: Early Status Transfer during CPAC in EN-DC operation, successful operation**

**From source eNB to target eNB**

The *E-RABs Subject To Early Status Transfer List* IE included in the EARLY STATUS TRANSFER message contains the E-RAB ID(s) corresponding to the E-RAB(s) subject to be simultaneously served by the source and the target eNBs during DAPS Handover or the E-RAB(s) transferred during Conditional Handover.

For each E-RAB for which the *FIRST DL COUNT Value* IE is received in the EARLY STATUS TRANSFER message, the target eNB shall use it as the COUNT of the first downlink SDU that the source eNB forwards to the target eNB. If the *FIRST DL COUNT Value Extended* IE or *FIRST DL COUNT Value for PDCP SN Length 18* IE is included in the *E-RABs Subject To Early Status Transfer Item* IE, the target eNB shall, if supported, use this value instead of the value contained in the *FIRST DL COUNT Value* IE.

For each E-RAB for which the *DISCARD DL COUNT Value* IE is received in the EARLY STATUS TRANSFER message, the target eNB does not transmit forwarded downlink SDUs to the UE whose COUNT is less than the provided and discards them if transmission has not been attempted. If the *DISCARD DL COUNT Value Extended* IE or *DISCARD DL COUNT Value for PDCP SN Length 18* IE is included in the *E-RABs Subject To Early Status Transfer Item* IE, the target eNB shall, if supported, use this value instead of the value contained in the *DISCARD DL COUNT Value* IE.

**From SeNB (respectively, en-gNB) to MeNB, the source eNB for Conditional Handover**

**From MeNB to en-gNB for CPA**

**From source en-gNB to MeNB, and from MeNB to target en-gNB, for CPC**

The *E-RABs Subject To Early Status Transfer List* IE included in the EARLY STATUS TRANSFER message contains the E-RAB ID(s) corresponding to the E-RAB(s) transferred during Conditional Handover or during CPAC.

For each E-RAB in the *E-RABs Subject To Early Status Transfer List* IE, the source eNB shall forward to the target during Conditional Handover, or the sending node shall forward to the receiving node during CPAC, the value of the received *FIRST DL COUNT Value* IE or *DISCARD DL COUNT Value* IE. If the *FIRST DL COUNT Value Extended* IE or *FIRST DL COUNT Value for PDCP SN Length 18* IE is included, if supported, this value is forwarded instead of the value contained in the *FIRST DL COUNT Value* IE. If the *DISCARD DL COUNT Value Extended* IE or *DISCARD DL COUNT Value for PDCP SN Length 18* IE is included, if supported, this value is forwarded instead of the value contained in the *DISCARD DL COUNT Value* IE.

If the en-gNB sends the message to the MeNB, then the *SgNB UE X2AP ID* IE shall be included in the EARLY STATUS TRANSFER message, while the *Old eNB UE X2AP ID* IE is ignored. The *SgNB UE X2AP ID* IE is used as the old UE ID.

#### 8.2.7.3 Abnormal Conditions

If the target eNB receives this message for a UE for which no prepared DAPS Handover or Conditional Handover exists at the target eNB, the target eNB shall ignore the message.

## 8.3 Global Procedures

### 8.3.1 Load Indication

#### 8.3.1.1 General

The purpose of the Load Indication procedure is to transfer load and interference co-ordination information between eNBs controlling intra-frequency neighboring cells, and additionally between eNBs controlling inter-frequency neighboring cells for TDD.

The procedure uses non UE-associated signalling.

### 8.3.1.2 Successful Operation

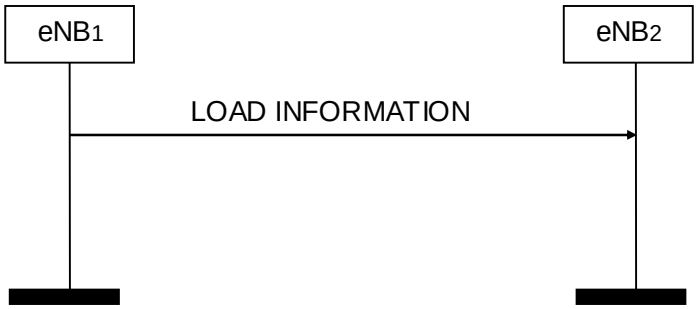



**Figure 8.3.1.2-1: Load Indication, successful operation**

An $eNB_1$ initiates the procedure by sending LOAD INFORMATION message to a peer $eNB_2$.

If the *UL Interference Overload Indication* IE is received in the LOAD INFORMATION message, it indicates the interference level experienced by the indicated cell on all resource blocks, per PRB. If the *Extended UL Interference Overload Info* IE is received in the LOAD INFORMATION message, the *UL Interference Overload Indication* IE indicates the interference level experienced by the indicated cell ignoring the UL subframe(s) represented as value "1" in the *Associated Subframes* IE. The receiving eNB may take such information into account when setting its scheduling policy and shall consider the received *UL Interference Overload Indication* IE value valid until reception of a new LOAD INFORMATION message carrying an update of the same IE.

If the *UL High Interference Indication* IE is received in the LOAD INFORMATION message, it indicates, per PRB, the occurrence of high interference sensitivity, as seen from the sending eNB. The receiving eNB should try to avoid scheduling cell edge UEs in its cells for the concerned PRBs. The *Target Cell ID* IE received within the *UL High Interference Information* IE group in the LOAD INFORMATION message indicates the cell for which the corresponding UL High Interference Indication is meant. The receiving eNB shall consider the value of the *UL High Interference Information* IE group valid until reception of a new LOAD INFORMATION message carrying an update.

If the *Relative Narrowband Tx Power (RNTP)* IE is received in the LOAD INFORMATION message, it indicates, per PRB or per subframe per PRB (Enhanced RNTP), whether downlink transmission power is lower than the value indicated by the *RNTP Threshold* IE. If the *Enhanced RNTP* IE is included in the *Relative Narrowband Tx Power (RNTP)* IE, it additionally indicates whether the downlink transmission power is lower than the value specified by the *RNTP High Power Threshold* IE. The receiving eNB may take such information into account when setting its scheduling policy and shall consider the received *Relative Narrowband Tx Power (RNTP)* IE value valid until reception of a new LOAD INFORMATION message carrying an update. If the *Enhanced RNTP* IE included in the *Relative Narrowband Tx Power (RNTP)* IE is present, the receiving eNB shall consider the received *Enhanced RNTP* IE value valid starting from the subframe indicated by the *Start SFN* IE and *Start Subframe Number* IE, if present.

If the *ABS Information* IE is included in the LOAD INFORMATION message, the *ABS Pattern Info* IE indicates the subframes designated as almost blank subframes by the sending eNB for the purpose of interference coordination. The receiving eNB may take such information into consideration when scheduling UEs.

The receiving eNB may use the *Measurement Subset* IE received in the LOAD INFORMATION message, for the configuration of specific measurements towards the UE.

The receiving eNB shall consider the received information as immediately applicable. The receiving eNB shall consider the value of the *ABS Information* IE valid until reception of a new LOAD INFORMATION message carrying an update.

If an ABS indicated in the *ABS pattern info* IE coincides with a MBSFN subframe, the receiving eNB shall consider that the subframe is designated as almost blank subframe by the sending eNB.

If the *Invoke Indication* IE is included in the LOAD INFORMATION message, it indicates which type of information the sending eNB would like the receiving eNB to send back. The receiving eNB may take such request into account.

If the *Invoke Indication* IE is set to "ABS Information", it indicates the sending eNB would like the receiving eNB to initiate the Load Indication procedure, with the LOAD INFORMATION message containing the *ABS Information* IE indicating non-zero ABS patterns in the relevant cells. If the *Invoke Indication* IE is set to "Start NAICS Information", it indicates the sending eNB would like the receiving eNB to initiate the Load Indication procedure with the LOAD

INFORMATION message containing the *Dynamic DL transmission information* IE. The first time the *Dynamic DL transmission information* IE is signalled after receiving the *Invoke Indication* IE set to "Start NAICS Information", all the NAICS parameters in the *NAICS Information* IE shall be included. If the *Invoke Indication* IE is set to "Stop NAICS Information", it indicates the sending eNB does not need NAICS information and therefore the receiving eNB should stop signalling NAICS parameters for the concerned cell.

If the *NAICS Information* IE is set to "NAICS Active", the receiving eNB may use it for the configuration of DL interference mitigation assistance information towards the UE. Information included in the *NAICS Information* IE shall replace corresponding NAICS information existing at the receiver. If the *NAICS Information* IE is set to "NAICS Inactive", the receiving eNB shall consider the existing NAICS information as invalid.

If the *Intended UL-DL Configuration* IE is included in the LOAD INFORMATION message, it indicates the UL-DL configuration intended to be used by the indicated cell. The receiving eNB may take such information into account when setting its scheduling policy and shall consider the received *Intended UL-DL Configuration* IE value valid until reception of a new LOAD INFORMATION message carrying an update of the same IE.

If the *Extended UL Interference Overload Info* IE is received in the LOAD INFORMATION message, the *Extended UL Interference Overload Indication* IE indicates the interference level experienced by the indicated cell on all resource blocks, per PRB, in the UL subframe(s) which is represented as value "1" in the *Associated Subframes* IE. The receiving eNB may take such information into account when setting its scheduling policy and shall consider the received *Extended UL Interference Overload Info* IE value valid until reception of a new LOAD INFORMATION message carrying an update of the same IE.

If the *CoMP Information* IE is received in the LOAD INFORMATION message, the receiving eNB may take the IE into account for RRM. The receiving eNB shall consider the *CoMP Information* IE valid starting in the subframe indicated by the *Start SFN* IE and *Start Subframe Number* IE, if present. If the *Start SFN* IE and *Start Subframe Number* IE are not present, then the receiving eNB shall consider the *CoMP Information* IE as immediately valid. The receiving eNB shall consider the *CoMP Information* IE valid until an update of the same IE, received in a new LOAD INFORMATION message, is considered valid.

#### 8.3.1.3 Unsuccessful Operation

Not applicable.

#### 8.3.1.4 Abnormal Conditions

Void.

### 8.3.2 Error Indication

#### 8.3.2.1 General

The Error Indication procedure is initiated by an eNB to report detected errors in one incoming message, provided they cannot be reported by an appropriate failure message.

If the error situation arises due to reception of a message utilising UE associated signalling, then the Error Indication procedure uses UE-associated signalling. Otherwise the procedure uses non UE-associated signalling.

### 8.3.2.2 Successful Operation

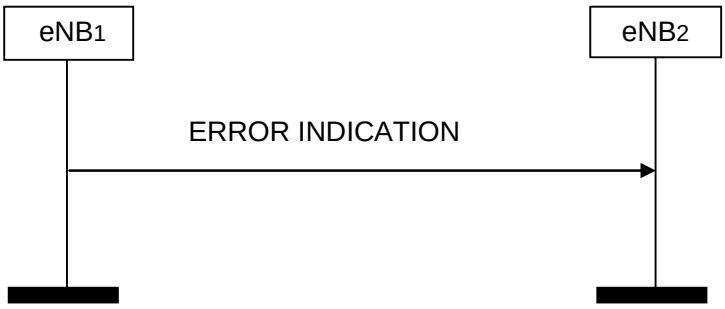



**Figure 8.3.2.2-1: Error Indication, successful operation.**

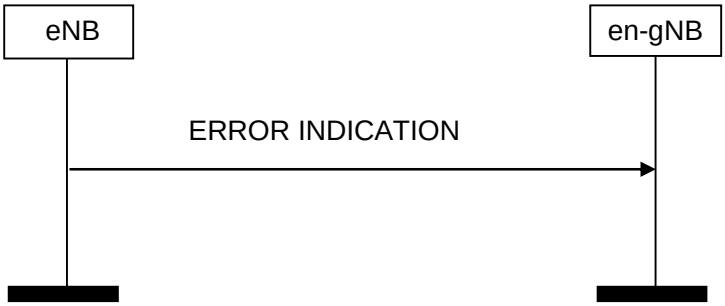



**Figure 8.3.2.2-2: eNB initiated Error Indication for EN-DC, successful operation.**

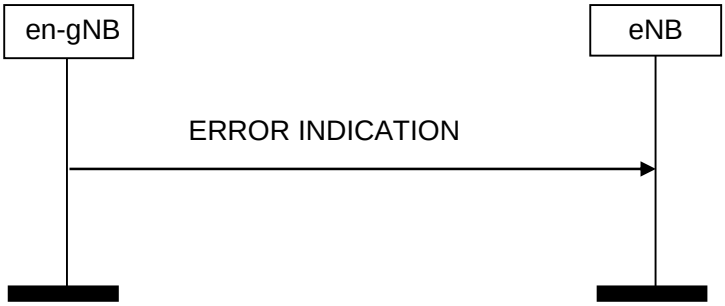



**Figure 8.3.2.2-3: en-gNB initiated Error Indication for EN-DC, successful operation.**

When the conditions defined in clause 10 are fulfilled, the Error Indication procedure is initiated by an ERROR INDICATION message sent from the node detecting the error situation.

The ERROR INDICATION message shall contain at least either the *Cause* IE or the *Criticality Diagnostics* IE.

In case the Error Indication procedure is triggered by UE associated signalling, in the course of handover signalling and signalling for dual connectivity, the *Old eNB UE X2AP ID* IE and the *New eNB UE X2AP ID* IE shall be included in the ERROR INDICATION message. In case the Error Indication procedure is triggered by UE associated signalling, in the course of signalling for EN-DC, the *Old en-gNB UE X2AP ID* IE and the *New eNB UE X2AP ID* IE shall be included in the ERROR INDICATION message. If any of *Old eNB UE X2AP ID* IE, *Old en-gNB UE X2AP ID* IE and *New eNB UE X2AP ID* IE is not correct, the cause shall be set to appropriate value e.g. "unknown Old eNB UE X2AP ID", "unknown Old en-gNB UE X2AP ID", "unknown New eNB UE X2AP ID" or "unknown pair of UE X2AP ID".

If the UE-associated signalling connection is identified by extended eNB UE X2AP IDs the specification text above is applicable for the UE X2AP ID Extension accordingly.

In case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], if the Error Indication procedure is triggered by non UE-associated signalling, the ERROR INDICATION message shall include the *Interface Instance Indication* IE to identify the corresponding interface instance.

#### 8.3.2.3 Unsuccessful Operation

Not applicable.

#### 8.3.2.4 Abnormal Conditions

Not applicable.

### 8.3.3 X2 Setup

#### 8.3.3.1 General

The purpose of the X2 Setup procedure is to exchange application level configuration data needed for two eNBs to interoperate correctly over the X2 interface. This procedure erases any existing application level configuration data in the two nodes and replaces it by the one received. This procedure also resets the X2 interface like a Reset procedure would do.

NOTE: Exchange of application level configuration data also applies between two eNBs in case the SN (i.e. the en-gNB) does not broadcast system information other than for radio frame timing and SFN, as specified in the TS 37.340 [32]. How to use this information when this option is used is not explicitly specified.

The procedure uses non UE-associated signalling.

#### 8.3.3.2 Successful Operation

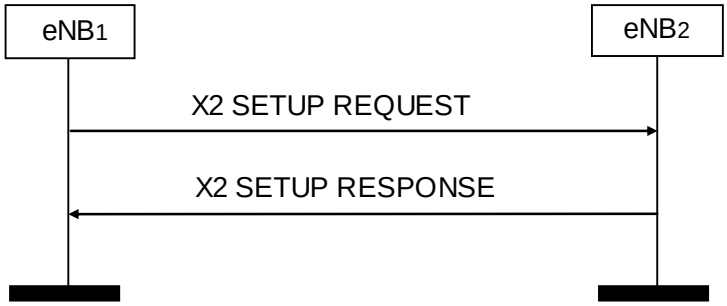



**Figure 8.3.3.2-1: X2 Setup, successful operation**

An $eNB_1$ initiates the procedure by sending the X2 SETUP REQUEST message to a candidate $eNB_2$. The candidate $eNB_2$ replies with the X2 SETUP RESPONSE message. The initiating $eNB_1$ shall transfer the complete list of its served cells and, if available, a list of supported GU Group Ids to the candidate $eNB_2$. The candidate $eNB_2$ shall reply with the complete list of its served cells and shall include, if available, a list of supported GU Group Ids in the reply.

If a cell is switched off for energy savings reasons, it should be activated before initiating or responding to the X2 Setup procedure and shall still be included in the list of served cells.

The initiating $eNB_1$ may include the *Neighbour Information* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *Neighbour Information* IE in the X2 SETUP RESPONSE message. The *Neighbour Information* IE shall only include E-UTRAN cells that are direct neighbours of cells in the reporting eNB. A direct neighbour of one cell of a given eNB may be any cell belonging to an eNB that is a neighbour of that given eNB cell e.g. even if the cell has not been reported by a UE. The initiating $eNB_1$ may include the *TAC* IE with the *Neighbour Information* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *TAC* IE with the *Neighbour Information* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use it according to TS 36.300 [15].

The initiating $eNB_1$ may include the *NR Neighbour Information* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *NR Neighbour Information* IE in the X2 SETUP RESPONSE message. The *NR Neighbour Information* IE shall only include NR cells capable of performing EN-DC with the corresponding served E-UTRA cell. The eNB receiving the *NR Neighbour Information* IE may use it according to TS 36.300 [15].

The initiating $eNB_1$ may include the *Number of Antenna Ports* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *Number of Antenna Ports* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use it according to TS 36.331 [9].

The initiating $eNB_1$ may include the *PRACH Configuration* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *PRACH Configuration* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use this information for RACH optimisation.

The initiating $eNB_1$ may include the *MBSFN Subframe Info* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *MBSFN Subframe Info* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use it according to TS 36.331 [9].

For each CSG cell or hybrid cell served by the initiating $eNB_1$ the X2 SETUP REQUEST message shall contain the *CSG ID* IE. For each CSG cell or hybrid cell served by the candidate $eNB_2$ the X2 SETUP RESPONSE message shall contain the *CSG ID* IE. The eNB receiving the IE shall take this information into account when further deciding whether X2 handover between the source cell and the target cell may be performed.

The initiating $eNB_1$ may include the *MBMS Service Area Identity List* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *MBMS Service Area Identity List* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use it according to TS 36.300 [15].

For each cell served by the initiating $eNB_1$ the X2 SETUP REQUEST message may contain the *MultibandInfoList* IE and may also contain the *FreqBandIndicatorPriority* IE. For each cell served by the candidate $eNB_2$ the X2 SETUP RESPONSE message may contain the *MultibandInfoList* IE and may also contain the *FreqBandIndicatorPriority* IE. The eNB receiving the *MultibandInfoList* IE shall, if supported, take this information into account when further deciding whether subsequent mobility actions between the source cell and the target cell may be performed, and use this IE and the *FreqBandIndicatorPriority* IE, if received, as specified in TS 36.331 [9].

The initiating $eNB_1$ may include the *LHN ID* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include *LHN ID* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use it according to TS 36.300 [15].

The initiating $eNB_1$ may include the *BandwidthReducedSI* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include *BandwidthReducedSI* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use it to determine a suitable target in case of subsequent outgoing mobility involving BL UEs or UEs requiring CE.

The initiating $eNB_1$ may include the *NPRACH Configuration* IE in the X2 SETUP REQUEST message. The candidate $eNB_2$ may also include the *NPRACH Configuration* IE in the X2 SETUP RESPONSE message. The eNB receiving the IE may use this information for RACH optimization.

If the *Served Cell Specific Info Request* IE is included in the X2 SETUP REQUEST message, the $eNB_2$ shall, if supported, include the *Additional Measurement Timing Configuration List* IE for the requested NR neighbour cells, capable of performing EN-DC with the served E-UTRA cell in the X2 SETUP RESPONSE message.

**Interaction with the EN-DC Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *NR Neighbour Information* IE in the X2 SETUP REQUEST message or in the X2 SETUP RESPONSE message to neighbouring en-gNBs by triggering the EN-DC Configuration Update procedure.

**Interaction with the eNB Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *NR Neighbour Information* IE in the X2 SETUP REQUEST message or in the X2 SETUP RESPONSE message to neighbouring eNBs by triggering the eNB Configuration Update procedure.

#### 8.3.3.3 Unsuccessful Operation

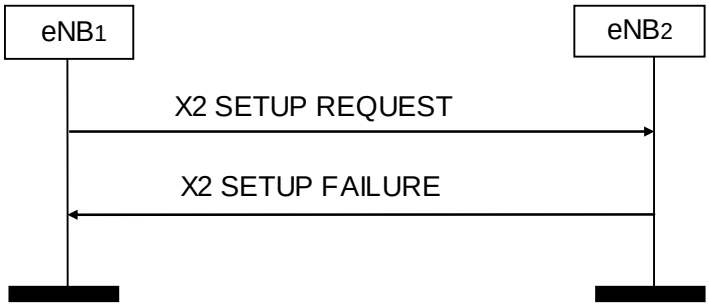



**Figure 8.3.3.3-1: X2 Setup, unsuccessful operation**

If the candidate $eNB_2$ cannot accept the setup it shall respond with an X2 SETUP FAILURE message with appropriate cause value.

If the X2 SETUP FAILURE message includes the *Time To Wait* IE the initiating $eNB_1$ shall wait at least for the indicated time before reinitiating the X2 Setup procedure towards the same $eNB_2$.

#### 8.3.3.4 Abnormal Conditions

If the first message received for a specific TNL association is not an X2 SETUP REQUEST, X2 SETUP RESPONSE, or X2 SETUP FAILURE message then this shall be treated as a logical error.

If the initiating $eNB_1$ does not receive either X2 SETUP RESPONSE message or X2 SETUP FAILURE message, the $eNB_1$ may reinitiate the X2 Setup procedure towards the same eNB, provided that the content of the new X2 SETUP REQUEST message is identical to the content of the previously unacknowledged X2 SETUP REQUEST message.

If the initiating $eNB_1$ receives an X2 SETUP REQUEST message from the peer entity on the same X2 interface:

- In case the $eNB_1$ answers with an X2 SETUP RESPONSE message and receives a subsequent X2 SETUP FAILURE message, the $eNB_1$ shall consider the X2 interface as non operational and the procedure as unsuccessfully terminated according to sub clause 8.3.3.3.
- In case the $eNB_1$ answers with an X2 SETUP FAILURE message and receives a subsequent X2 SETUP RESPONSE message, the $eNB_1$ shall ignore the X2 SETUP RESPONSE message and consider the X2 interface as non operational.

### 8.3.4 Reset

#### 8.3.4.1 General

The purpose of the Reset procedure is to align the resources in $eNB_1$ and $eNB_2$, or the resources in eNB and en-gNB involved in the EN-DC in the event of an abnormal failure. The procedure resets the X2 interface. This procedure doesn't affect the application level configuration data exchanged during, e.g., the X2 Setup procedure, EN-DC X2 Setup procedure.

The procedure uses non UE-associated signalling.

### 8.3.4.2 Successful Operation

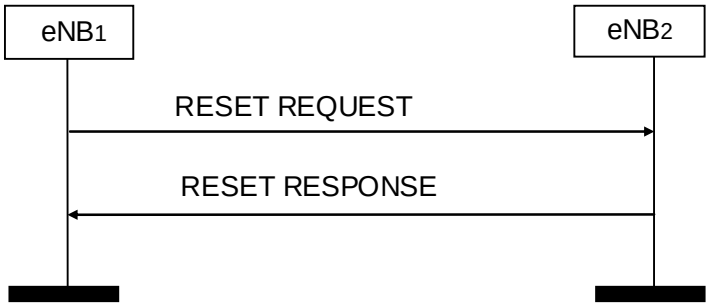



**Figure 8.3.4.2-1: Reset, successful operation**

The procedure is initiated with a RESET REQUEST message sent from the $eNB_1$ to the $eNB_2$. Upon receipt of this message, $eNB_2$ shall abort any other ongoing procedures over X2 between $eNB_1$ and $eNB_2$. The $eNB_2$ shall delete all the context information related to the $eNB_1$, except the application level configuration data exchanged during the X2 Setup or eNB Configuration Update procedures, and release the corresponding resources. After completion of release of the resources, the $eNB_2$ shall respond with a RESET RESPONSE message.

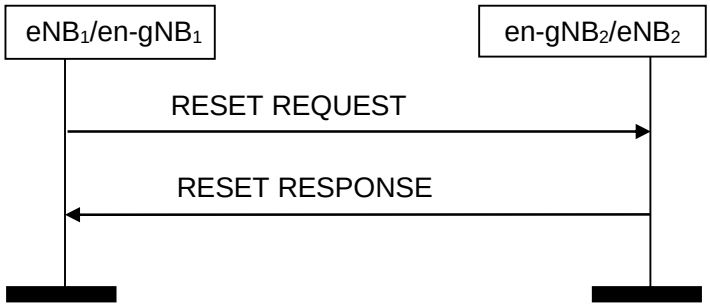



**Figure 8.3.4.2-2: Reset, successful operation for EN-DC.**

The procedure is initiated with a RESET REQUEST message sent from the $eNB_1$/en-$gNB_1$ to en-$gNB_2$/$eNB_2$. Upon receipt of this message, $eNB_2$/en-$gNB_2$ shall abort any other ongoing procedures over X2 between both nodes. $eNB_2$/en-$gNB_2$ shall delete all the context information related to $eNB_1$/en-$gNB_1$, except the application level configuration data exchanged during the EN-DC X2 Setup or EN-DC Configuration Update procedures, and release the corresponding resources. After completion of release of the resources, $eNB_2$/en-$gNB_2$ shall respond with a RESET RESPONSE message.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the RESET REQUEST and the RESET RESPONSE messages shall include the *Interface Instance Indication* IE to identify the corresponding interface instance.

### 8.3.4.3 Unsuccessful Operation

Void.

### 8.3.4.4 Abnormal Conditions

If the RESET REQUEST message is received, any other ongoing procedure (except another Reset procedure) on the same X2 interface shall be aborted.

If Reset procedure is ongoing and the responding node receives the RESET REQUEST message from the peer entity on the same X2 interface, it shall respond with the RESET RESPONSE message as described in 8.3.4.2.

If the initiating node does not receive RESET RESPONSE message, the initiating node may reinitiate the Reset procedure towards the same eNB/en-gNB, provided that the content of the new RESET REQUEST message is identical to the content of the previously unacknowledged RESET REQUEST message.

## 8.3.5 eNB Configuration Update

### 8.3.5.1 General

The purpose of the eNB Configuration Update procedure is to update application level configuration data needed for two eNBs to interoperate correctly over the X2 interface.

NOTE: Update of application level configuration data also applies between two eNBs in case the SN (i.e. the en-gNB) does not broadcast system information other than for radio frame timing and SFN, as specified in the TS 37.340 [32]. How to use this information when this option is used is not explicitly specified.

The procedure uses non UE-associated signalling.

### 8.3.5.2 Successful Operation

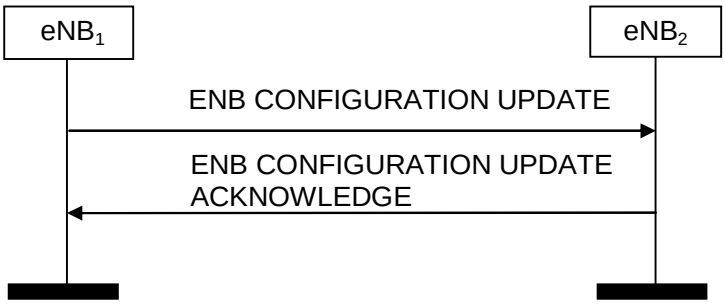



**Figure 8.3.5.2-1: eNB Configuration Update, successful operation**

An $eNB_1$ initiates the procedure by sending an ENB CONFIGURATION UPDATE message to a peer $eNB_2$. Such message shall include an appropriate set of up-to-date configuration data, including, but not limited to, the complete lists of added, modified and deleted served cells, that $eNB_1$ has just taken into operational use.

Upon reception of an ENB CONFIGURATION UPDATE message, $eNB_2$ shall update the information for $eNB_1$ as follows:

**Update of Served Cell Information:**

- If *Served Cells To Add* IE is contained in the ENB CONFIGURATION UPDATE message, $eNB_2$ shall add cell information according to the information in the *Served Cell Information* IE.

- If *Number of Antenna Ports* IE is contained in the *Served Cell Information* IE in the ENB CONFIGURATION UPDATE message, $eNB_2$ may use this information according to TS 36.331 [9].

- If the *PRACH Configuration* IE is contained in the *Served Cell Information* IE in the ENB CONFIGURATION UPDATE message, the eNB receiving the IE may use this information for RACH optimisation.

- If *Served Cells To Modify* IE is contained in the ENB CONFIGURATION UPDATE message, $eNB_2$ shall modify information of cell indicated by *Old ECGI* IE according to the information in the *Served Cell Information* IE.

- If *MBSFN Subframe Info* IE is contained in the *Served Cell Information* IE in the ENB CONFIGURATION UPDATE message, $eNB_2$ may use this information according to TS 36.331 [9]. If a MBSFN subframe indicated in the *MBSFN Subframe Info* IE coincides with an ABS, the $eNB_2$ shall consider that the subframe is designated as ABS by the sending eNB.

- If *BandwidthReducedSI* IE is contained in the *Served Cell Information* IE in the ENB CONFIGURATION UPDATE message, $eNB_2$ may use this information to determine a suitable target in case of subsequent outgoing mobility involving BL UEs or UEs requiring CE.

  When either served cell information or neighbour information of an existing served cell in $eNB_1$ need to be updated, the whole list of neighbouring cells, if any, shall be contained in the *Neighbour Information* IE.

  If the *Deactivation Indication* IE is contained in *Served Cells To Modify* IE, it indicates that the concerned cell was switched off to lower energy consumption.

The $eNB_2$ shall overwrite the served cell information and the whole list of neighbour cell information for the affected served cell.

- If *Served Cells To Delete* IE is contained in the ENB CONFIGURATION UPDATE message, $eNB_2$ shall delete information of cell indicated by *Old ECGI* IE.

- If *MBMS Service Area Identity List* IE is contained in the *Served Cell Information* IE in the ENB CONFIGURATION UPDATE message, the eNB receiving the IE may use it according to TS 36.300 [15].

  When the MBMS Service Area Identities of a cell in $eNB_1$ need to be updated, the whole list of MBMS Service Area Identities of the affected cell shall be contained in the *Served Cell Information* IE.

- If the *NPRACH Configuration* IE is contained in the *Served Cell Information* IE in the ENB CONFIGURATION UPDATE message, the eNB receiving the IE may use this information for RACH optimization.

**Update of GU Group Id List:**

- If *GU Group Id To Add List* IE is contained in the ENB CONFIGURATION UPDATE message, $eNB_2$ shall add the GU Group Id to its GU Group Id List.

- If *GU Group Id To Delete List* IE is contained in the ENB CONFIGURATION UPDATE message, $eNB_2$ shall remove the GU Group Id from its GU Group Id List.

If *Neighbour Information* IE is contained in the ENB CONFIGURATION UPDATE message, $eNB_2$ may use this information to update its neighbour cell relations, or use it for other functions, like PCI selection. The *Neighbour Information* IE shall only include E-UTRAN cells that are direct neighbours of cells in the reporting eNB. A direct neighbour of one cell of a given eNB may be any cell belonging to an eNB that is a neighbour of that given eNB cell e.g. even if that cell has not been reported by a UE. The *Neighbour Information* IE may contain the *TAC* IE of the included cells. The receiving eNB may use *TAC* IE, as described in TS 36.300 [15].

If the *NR Neighbour Information* IE is contained in the ENB CONFIGURATION UPDATE message, $eNB_2$ may use this information to update its neighbour cell relations or use it for other functions. The *NR Neighbour Information* IE shall only include NR cells capable of performing EN-DC with the corresponding served E-UTRA cell. The eNB receiving the *NR Neighbour Information* IE may use it according to TS 36.300 [15].

After successful update of requested information, $eNB_2$ shall reply with the ENB CONFIGURATION UPDATE ACKNOWLEDGE message to inform the initiating $eNB_1$ that the requested update of application data was performed successfully. In case the peer $eNB_2$ receives an ENB CONFIGURATION UPDATE without any IE except for *Message Type* IE it shall reply with ENB CONFIGURATION UPDATE ACKNOWLEDGE message without performing any updates to the existing configuration.

The $eNB_1$ may initiate a further eNB Configuration Update procedure only after a previous eNB Configuration Update procedure has been completed.

For each cell served by the initiating $eNB_1$ the ENB CONFIGURATION UPDATE message may contain the *MultibandInfoList* IE and may also contain the *FreqBandIndicatorPriority* IE. The eNB receiving the *MultibandInfoList* IE shall, if supported, take this information into account when further deciding whether subsequent mobility actions between the source cell and the target cell may be performed, and use this IE and the *FreqBandIndicatorPriority* IE, if received, as specified in TS 36.331 [9].

If the *Coverage Modification List* IE is present, $eNB_2$ may use the information in the *Cell Coverage State* IE to identify the cell deployment configuration enabled by $eNB_1$ and for configuring the mobility towards the cell(s) indicated by the *ECGI* IE, as described in TS 36.300 [15]. If the *Cell Deployment Status Indicator* IE is present in the *Coverage Modification List* IE, the $eNB_2$ shall consider the cell deployment configuration of the cell to be modified as the next planned configuration and shall remove any planned configuration stored for this cell. If the *Cell Deployment Status Indicator* IE is present and the *Cell Replacing Info* IE contains non-empty cell list, the $eNB_2$ may use this list to avoid connection or re-establishment failures during the reconfiguration, e.g. consider the cells in the list as possible alternative handover targets. If the *Cell Deployment Status Indicator* IE is not present, the $eNB_2$ shall consider the cell deployment configuration of cell to be modified as activated and replace any previous configuration for the cells indicated in the *Coverage Modification List* IE.

**Interaction with the eNB Configuration Update procedure:**

If an $eNB_2$ which has not stored a *FreqBandIndicatorPriority* IE received from $eNB_1$, but has signaled a *FreqBandIndicatorPriority* IE to $eNB_1$ after the TNL association has become available, receives an ENB

CONFIGURATION UPDATE message from $eNB_1$ containing the *FreqBandIndicatorPriority* IE, the $eNB_2$ shall initiate the eNB Configuration Update procedure towards $eNB_1$ including the *FreqBandIndicatorPriority* IE.

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *NR Neighbour Information* IE in the ENB CONFIGURATION UPDATE message to neighbouring eNBs by triggering the eNB Configuration Update procedure.

**Interaction with the EN-DC Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *NR Neighbour Information* IE in the ENB CONFIGURATION UPDATE message to neighbouring en-gNBs by triggering the EN-DC Configuration Update procedure.

### 8.3.5.3 Unsuccessful Operation

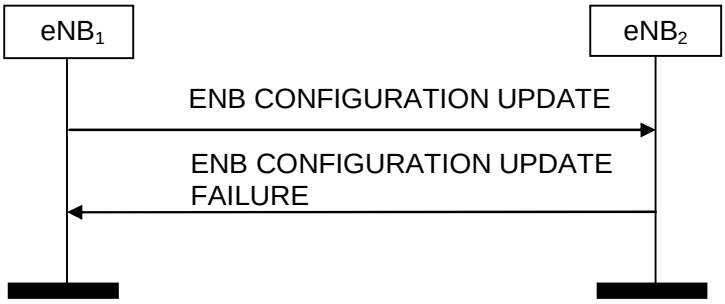



**Figure 8.3.5.3-1: eNB Configuration Update, unsuccessful operation**

If the $eNB_2$ can not accept the update it shall respond with an ENB CONFIGURATION UPDATE FAILURE message and appropriate cause value.

If the ENB CONFIGURATION UPDATE FAILURE message includes the *Time To Wait* IE the $eNB_1$ shall wait at least for the indicated time before reinitiating the eNB Configuration Update procedure towards the same $eNB_2$. Both nodes shall continue to operate the X2 with their existing configuration data.

### 8.3.5.4 Abnormal Conditions

If the $eNB_1$ after initiating eNB Configuration Update procedure receives neither ENB CONFIGURATION UPDATE ACKNOWLEDGE message nor ENB CONFIGURATION UPDATE FAILURE message, the $eNB_1$ may reinitiate the eNB Configuration Update procedure towards the same $eNB_2$, provided that the content of the new ENB CONFIGURATION UPDATE message is identical to the content of the previously unacknowledged ENB CONFIGURATION UPDATE message.

## 8.3.6 Resource Status Reporting Initiation

### 8.3.6.1 General

This procedure is used by an eNB to request the reporting of load measurements to another eNB.

The procedure uses non UE-associated signalling.

### 8.3.6.2 Successful Operation

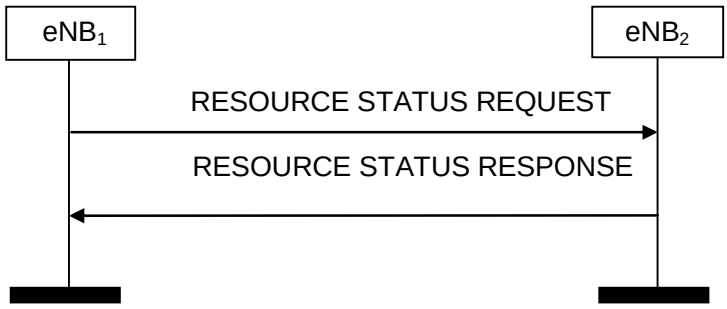



**Figure 8.3.6.2-1: Resource Status Reporting Initiation, successful operation**

The procedure is initiated with a RESOURCE STATUS REQUEST message sent from $eNB_1$ to $eNB_2$. Upon receipt, $eNB_2$:

- shall initiate the requested measurement according to the parameters given in the request in case the *Registration Request* IE set to "start"; or
- shall stop all cells measurements and terminate the reporting in case the *Registration Request* IE is set to "stop"; or
- if supported, stop cell measurements and terminate the reporting for cells indicated in the *Cell To Report* IE list, in case the *Registration Request* IE is set to "partial stop"; or
- if supported, add cells indicated in the *Cell To Report* IE list to the measurements initiated before for the given measurement IDs, in case the *Registration Request* IE is set to "add".

If the $eNB_2$ received a RESOURCE STATUS REQUEST message, which includes the *Registration Request* IE set to "stop", the *Cell To Report* IE list shall be ignored.

If the *Registration Request* IE is set to "start" then the *Report Characteristics* IE shall be included in RESOURCE STATUS REQUEST message. The $eNB_2$ shall ignore the *Report Characteristics* IE, if the *Registration Request* IE is not set to "start".

The *Report Characteristics* IE indicates the type of objects $eNB_2$ shall perform measurements on. For each cell, the $eNB_2$ shall include in the RESOURCE STATUS UPDATE message:

- the *Radio Resource Status* IE, if the first bit, "PRB Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1;
- the *S1 TNL Load Indicator* IE, if the second bit, "TNL Load Ind Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1;
- the *Hardware Load Indicator* IE, if the third bit, "HW Load Ind Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1;
- the *Composite Available Capacity Group* IE, if the fourth bit, "Composite Available Capacity Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1. If *Cell Capacity Class Value* IE is included within the *Composite Available Capacity Group* IE, this IE is used to assign weights to the available capacity indicated in the *Capacity Value* IE;
- the *ABS Status* IE, if the fifth bit, "ABS Status Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1 and $eNB_1$ had indicated the ABS pattern to $eNB_2$;
- the *RSRP Measurement Report List* IE, if the sixth bit, "RSRP Measurement Report Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1;
- the *CSI Report* IE, if the seventh bit, "CSI Report Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1.

- the *Radio Resource Status* IE within the *NR Neithbour Cell Measurement Result* IE, if the eighth bit, "Neighbour Cell CAC Periodic" of the *Report Characteristics* IE included in the RESOURCE STATUS REQUEST message is set to 1.

NOTE: In order to avoid duplication, the $eNB_2$ may include only one copy of *Measurement Result for NR Cells Possibly Aggregated Item* per RESOURCE STATUS UPDATE messages per NR cell, even if this NR cell possibly aggregated to multiple E-UTRA cells served by $eNB_2$. The $eNB_2$ should only include *Measurement Result for NR Cells Possibly Aggregated Item* for NR cells that may be aggregated with at least one cell served by $eNB_1$.

If the *Reporting Periodicity* IE is included in the RESOURCE STATUS REQUEST message, $eNB_2$ shall use its value as the time interval between two subsequent RESOURCE STATUS UPDATE messages that include the *Radio Resource Status* IE, *S1 TNL Load Indicator* IE, *Hardware Load Indicator* IE, *Composite Available Capacity Group* IE, or *ABS Status* IE.

If the *Reporting Periodicity of RSRP Measurement Report* IE is included in the RESOURCE STATUS REQUEST message, $eNB_2$ shall use its value as the minimum time interval between two subsequent RESOURCE STATUS UPDATE messages that include the *RSRP Measurement Report List* IE.

If the *Reporting Periodicity of CSI Report* IE is included in the RESOURCE STATUS REQUEST message, $eNB_2$ shall use its value as the minimum time interval between two subsequent RESOURCE STATUS UPDATE messages that include the *CSI Report* IE.

If $eNB_2$ is capable to provide all requested resource status information, it shall initiate the measurement as requested by $eNB_1$, and respond with the RESOURCE STATUS RESPONSE message.

If $eNB_2$ is capable to provide some but not all of the requested resource status information and the *Partial Success Indicator* IE is present in the RESOURCE STATUS REQUEST message, it shall initiate the measurement for the admitted measurement objects and include the *Measurement Initiation Result* IE in the RESOURCE STATUS RESPONSE message.

### 8.3.6.3 Unsuccessful Operation

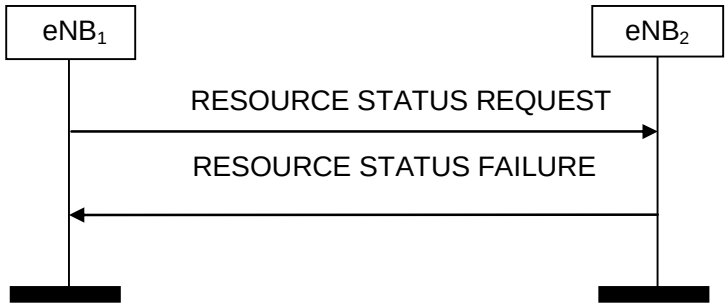



**Figure 8.3.6.3-1: Resource Status Reporting Initiation, unsuccessful operation**

If none of the requested measurements can be initiated, $eNB_2$ shall send a RESOURCE STATUS FAILURE message. The *Cause* IE shall be set to an appropriate value e.g. "Measurement Temporarily not Available" or "Measurement not Supported For The Object" for each requested measurement object. The eNB may use the *Complete Failure Cause Information* IE to enhance the failure cause information per measurement in the RESOURCE STATUS FAILURE message.

### 8.3.6.4 Abnormal Conditions

If the initiating $eNB_1$ does not receive either RESOURCE STATUS RESPONSE message or RESOURCE STATUS FAILURE message, the $eNB_1$ may reinitiate the Resource Status Reporting Initiation procedure towards the same eNB, provided that the content of the new RESOURCE STATUS REQUEST message is identical to the content of the previously unacknowledged RESOURCE STATUS REQUEST message.

If the initiating $eNB_1$ receives the RESOURCE STATUS RESPONSE message including the *Measurement Initiation Result* IE containing no admitted measurements, the $eNB_1$ shall consider the procedure as failed.

If the *Report Characteristics* IE bitmap is set to "0" (all bits are set to "0") in the RESOURCE STATUS REQUEST message then $eNB_2$ shall initiate a RESOURCE STATUS FAILURE message, the cause shall be set to appropriate value e.g. "ReportCharacteristicsEmpty".

If the *Reporting Periodicity* IE value is not specified when at least one of the bits of the *Report Characteristics* IE, for which semantics is specified, other than the sixth or seventh bit, is set to 1 then $eNB_2$ shall initiate a RESOURCE STATUS FAILURE message, the cause shall be set to appropriate value e.g. "NoReportPeriodicity".

If the *Reporting Periodicity of RSRP Measurement Report* IE value is not specified when the sixth bit of the *Report Characteristics* IE is set to 1, then $eNB_2$ shall initiate the RESOURCE STATUS FAILURE message and the cause shall be set to appropriate value e.g. "NoReportPeriodicity".

If the *Reporting Periodicity of CSI Report* IE value is not specified when the seventh bit of the *Report Characteristics* IE is set to 1, then $eNB_2$ shall initiate the RESOURCE STATUS FAILURE message and the cause shall be set to appropriate value e.g. "NoReportPeriodicity".

If the $eNB_2$ received a RESOURCE STATUS REQUEST message which includes the *Registration Request* IE set to "start" and the *eNB1Measurement ID* IE corresponding to an existing on-going load measurement reporting, then $eNB_2$ shall initiate a RESOURCE STATUS FAILURE message, the cause shall be set to appropriate value e.g. "ExistingMeasurementID".

If the *Registration Request* IE is set to "stop", "partial stop" or "add" and the RESOURCE STATUS REQUEST message does not contain *eNB2 Measurement ID* IE, $eNB_2$ shall consider the procedure as failed and respond with the RESOURCE STATUS FAILURE message, the cause shall be set to appropriate value e.g. "Unknown eNB Measurement ID".

If the *Registration Request* IE is set to "partial stop" and the *Cell To Report* IE contains cells that have not been initiated for the reporting before, $eNB_2$ shall consider the procedure as failed and respond with the RESOURCE STATUS FAILURE message, the cause shall be set to appropriate value e.g. "Cell not Available". If the *Registration Request* IE is set to "add" and the *Cell To Report* IE contains cells that have been initiated for the reporting before, $eNB_2$ shall consider the procedure as failed and respond with the RESOURCE STATUS FAILURE message, the cause shall be set to appropriate value e.g. "Cell not Available".

## 8.3.7 Resource Status Reporting

### 8.3.7.1 General

This procedure is initiated by $eNB_2$ to report the result of measurements admitted by $eNB_2$ following a successful Resource Status Reporting Initiation procedure.

The procedure uses non UE-associated signalling.

### 8.3.7.2 Successful Operation

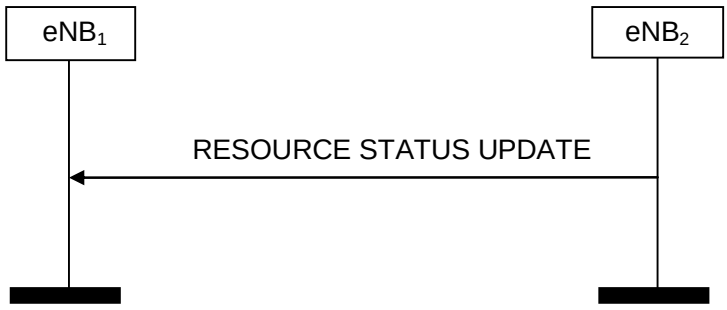



**Figure 8.3.7.2-1: Resource Status Reporting, successful operation**

The $eNB_2$ shall report the results of the admitted measurements in RESOURCE STATUS UPDATE message. The admitted measurements are the measurements that were successfully initiated during the preceding Resource Status Reporting Initiation procedure, and thus not reported in the *Measurement Failed Report Characteristics* IE for the concerned cell in the RESOURCE STATUS RESPONSE message.

If the $eNB_1$ receives the RESOURCE STATUS UPDATE message which includes the *UE ID* IE in the *RSRP Measurement Report List* IE, the $eNB_1$ may use the *UE ID* IE to link the associated RSRP measurement report with other measurement results (e.g. CSI reports, RSRP measurement reports) of the same UE.

If the *CSI Report* IE including the *CSI Process Configuration Index* IE is received, $eNB_1$ shall interpret this IE as an index identifying one of the CSI process configurations that can be configured for all UEs within the cell where the CSI measurements were collected. For all UEs within the cell, the maximum number of CSI process configurations is given by the maximum value of the *CSI Process Configuration Index* IE.

If the $eNB_1$ receives the RESOURCE STATUS UPDATE message, which includes the *Cell Reporting Indicator* IE set to "stop request" in one or more items of the *Cell Measurement Result* IE, the $eNB_1$ should initialise the Resource Status Reporting Initiation procedure to remove all or some of the corresponding cells from the measurement.

#### 8.3.7.3 Unsuccessful Operation

Not applicable.

#### 8.3.7.4 Abnormal Conditions

If the $eNB_1$ receives a RESOURCE STATUS UPDATE message which includes the *ABS Status* IE, and all bits in the *Usable ABS Pattern Info* IE are set to '0', the eNB1 shall ignore the *DL ABS Status* IE.

### 8.3.8 Mobility Settings Change

#### 8.3.8.1 General

This procedure enables an eNB to negotiate the handover trigger settings with a peer eNB controlling neighbouring cells.

The procedure uses non UE-associated signalling.

#### 8.3.8.2 Successful Operation

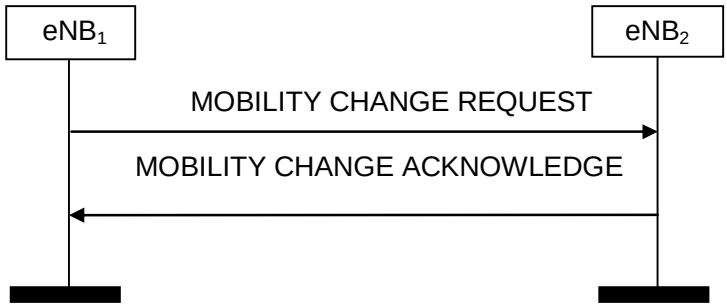



**Figure 8.3.8.2-1: Mobility Settings Change, successful operation**

The procedure is initiated with a MOBILITY CHANGE REQUEST message sent from $eNB_1$ to $eNB_2$.

Upon receipt, $eNB_2$ shall evaluate if the proposed $eNB_2$ handover trigger modification may be accepted. If $eNB_2$ is able to successfully complete the request it shall reply with MOBILITY CHANGE ACKNOWLEDGE.

#### 8.3.8.3 Unsuccessful Operation

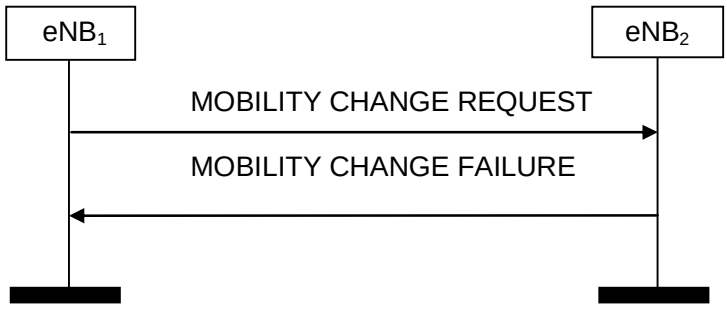



**Figure 8.3.8.3-1: Mobility Settings Change, unsuccessful operation**

If the requested parameter modification is refused by the $eNB_2$, or if the $eNB_2$ is not able to complete the procedure, the $eNB_2$ shall send a MOBILITY CHANGE FAILURE message with the *Cause* IE set to an appropriate value. The $eNB_2$ may include *eNB2 Mobility Parameters Modification Range* IE in MOBILITY CHANGE FAILURE message, for example in cases when the proposed change is out of permitted range.

#### 8.3.8.4 Abnormal Conditions

Void.

### 8.3.9 Radio Link Failure Indication

#### 8.3.9.1 General

The purpose of the Radio Link Failure Indication procedure is to transfer information regarding RRC re-establishment attempts, or received RLF Reports, between eNBs. The signalling takes place from the eNB at which a re-establishment attempt is made, or an RLF Report is received, to an eNB to which the UE concerned may have previously been attached prior to the connection failure. This may aid the detection of radio link failure and handover failure cases (TS 36.300 [15]).

The procedure uses non UE-associated signalling.

#### 8.3.9.2 Successful Operation

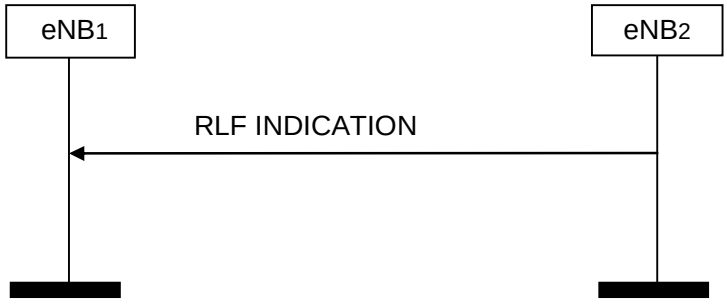



**Figure 8.3.9.2-1: Radio Link Failure Indication, successful operation**

$eNB_2$ initiates the procedure by sending the RLF INDICATION message to $eNB_1$ following a re-establishment attempt or an RLF Report reception from a UE at $eNB_2$, when $eNB_2$ considers that the UE may have previously suffered a connection failure at a cell controlled by $eNB_1$.

$eNB_2$ may include the *ShortMAC-I* IE in the RLF INDICATION message, e.g., in order to aid the $eNB_1$ to resolve a potential PCI confusion situation or to aid the $eNB_1$ to identify the UE.

$eNB_2$ may include the *UE RLF Report Container* IE and optionally also the *UE RLF Report Container for extended bands* IE in the RLF INDICATION message, which may be used by the $eNB_1$ to determine the nature of the failure. If the *UE RLF Report Container* IE is included in the RLF INDICATION message sent after successful re-establishment,

the $eNB_2$ shall use the *Re-establishment Cell ECGI* IE in the RLF INDICATION message to indicate the ECGI of the cell where the re-establishment was successful.

$eNB_2$ may include the *RRC Conn Setup Indicator* IE in the RLF INDICATION message, which indicates that the RLF Report is retrieved after an RRC connection setup or an incoming successful handover.

If the *RRC Conn Setup Indicator* IE is present in the RLF INDICATION message, the $eNB_1$ shall ignore the values in the *Failure cell PCI* IE, *Re-establishment cell ECGI* IE, *C-RNTI* IE and *ShortMAC-I* IE.

$eNB_2$ may include the *RRC Conn Reestab Indicator* IE in the RLF INDICATION message, which may be used by the $eNB_1$ to determine where the failure occurred.

$eNB_2$ may include the *NB-IoT RLF Report Container* IE in the RLF INDICATION message, which may be used by the $eNB_1$ to determine the nature of the failure. If the *NB-IoT RLF Report Container* IE is included in the RLF INDICATION message sent after successful re-establishment, the $eNB_2$ shall use the *Re-establishment Cell ECGI* IE in the RLF INDICATION message to indicate the ECGI of the cell where the re-establishment was successful.

#### 8.3.9.3 Unsuccessful Operation

Not applicable.

#### 8.3.9.4 Abnormal Conditions

Void.

### 8.3.10 Handover Report

#### 8.3.10.1 General

The purpose of the Handover Report procedure is to transfer mobility related information between eNBs.

The procedure uses non UE-associated signalling.

#### 8.3.10.2 Successful Operation




**Figure 8.3.10.2-1: Handover Report, successful operation**

An eNB initiates the procedure by sending an HANDOVER REPORT message to another eNB. By sending the message $eNB_1$ indicates to $eNB_2$ that a mobility-related problem was detected.

If the *Handover Report Type* IE is set to "HO too early" or "HO to wrong cell", then the $eNB_1$ indicates to $eNB_2$ that, following a successful handover from a cell of $eNB_2$ to a cell of $eNB_1$, a radio link failure occurred and the UE attempted RRC Re-establishment either at the original cell of $eNB_2$ (Handover Too Early), or at another cell (Handover to Wrong Cell). The detection of Handover Too Early and Handover to Wrong Cell events is made according to TS 36.300 [15].

If the UE-related information is available in $eNB_1$, the $eNB_1$ should include in HANDOVER REPORT message:

- the *Mobility Information* IE, if the *Mobility Information* IE was sent for this handover from $eNB_2$;

- the *Source cell C-RNTI* IE.

If received, the $eNB_2$ uses the above information according to TS 36.300 [15].

If the UE RLF Report received from the eNB sending the RLF INDICATION message, as described in TS 36.300 [15], is available, the $eNB_1$ may also include it in the HANDOVER REPORT as *UE RLF Report Container* IE and optionally also *UE RLF Report Container for extended bands* IE.

If the *Handover Report Type* IE is set to "InterRAT ping-pong", then the $eNB_1$ indicates to $eNB_2$ that a completed handover from a cell of $eNB_2$ to a cell in other RAT might have resulted in an inter-RAT ping-pong and the UE was successfully handed over to a cell of $eNB_1$ (indicated with the *Failure cell ECGI* IE).

If the *Handover Report Type* IE is set to "Inter-system ping-pong", then the $eNB_1$ indicates to $eNB_2$ that a completed handover from a cell of $eNB_2$ to a cell in NG-RAN might have resulted in an inter-system ping-pong and the UE was successfully handed over to a cell of $eNB_1$ (indicated with the *Failure cell ECGI* IE).

The report contains the source and target cells, and cause of the handover. If the *Handover Report Type* IE is set to "HO to wrong cell", then the *Re-establishment cell ECGI* IE shall be included in the HANDOVER REPORT message. If the *Handover Report Type* IE is set to "InterRAT ping-pong", then the *Target cell in UTRAN* IE shall be included in the HANDOVER REPORT message. If the *Handover Report Type* IE is set to "Inter-system ping-pong", then the *Target cell in NG-RAN* IE shall be included in the HANDOVER REPORT message.

#### 8.3.10.3 Unsuccessful Operation

Not applicable.

#### 8.3.10.4 Abnormal Conditions

Void.

### 8.3.11 Cell Activation

#### 8.3.11.1 General

The purpose of the Cell Activation procedure is to request to a neighbouring eNB to switch on one or more cells, previously reported as inactive due to energy saving reasons.

The procedure uses non UE-associated signalling.

#### 8.3.11.2 Successful Operation

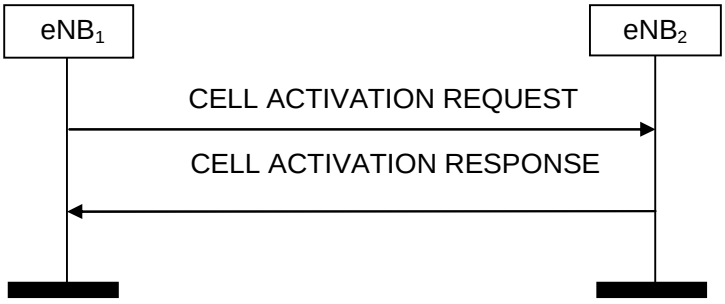



**Figure 8.3.11.2-1: Cell Activation, successful operation**

An $eNB_1$ initiates the procedure by sending a CELL ACTIVATION REQUEST message to a peer $eNB_2$.

Upon receipt of this message, $eNB_2$ should activate the cell(s) indicated in the CELL ACTIVATION REQUEST message and shall indicate in the CELL ACTIVATION RESPONSE message for which cells the request was fulfilled.

**Interactions with eNB Configuration Update procedure:**

$eNB_2$ shall not send an ENB CONFIGURATION UPDATE message to $eNB_1$ just for the reason of the cell(s) indicated in the CELL ACTIVATION REQUEST message changing state, as the receipt of the CELL ACTIVATION RESPONSE message by $eNB_1$ is used to update the information about cell activation state of $eNB_2$ cells in $eNB_1$.

### 8.3.11.3 Unsuccessful Operation

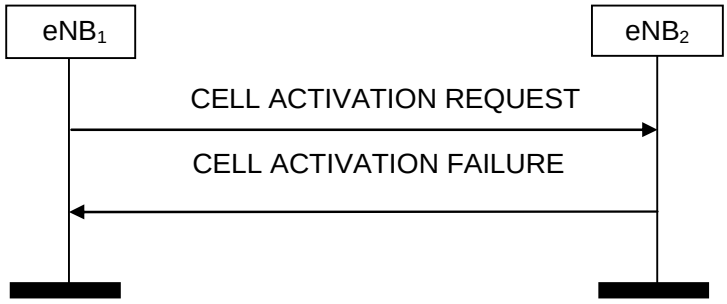



**Figure 8.3.11.3-1: Cell Activation, unsuccessful operation**

If the $eNB_2$ cannot activate any of the cells indicated in the CELL ACTIVATION REQUEST message, it shall respond with a CELL ACTIVATION FAILURE message with an appropriate cause value.

### 8.3.11.4 Abnormal Conditions

Not applicable.

## 8.3.12 X2 Removal

### 8.3.12.1 General

The purpose of the X2 Removal procedure is to remove the signaling connection between two eNBs in a controlled manner. If successful, this procedure erases any existing application level configuration data in the two nodes.

The procedure uses non UE-associated signaling.

### 8.3.12.2 Successful Operation

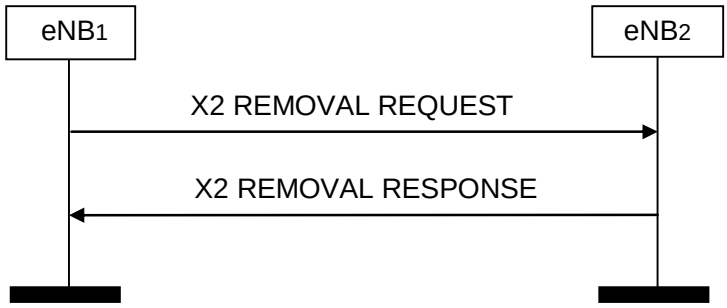



**Figure 8.3.12.2-1: X2 Removal, successful operation**

An $eNB_1$ initiates the procedure by sending the X2 REMOVAL REQUEST message to a candidate $eNB_2$. Upon reception of the X2 REMOVAL REQUEST message the candidate $eNB_2$ shall reply with the X2 REMOVAL RESPONSE message. After receiving the X2 REMOVAL RESPONSE message, the initiating $eNB_1$ shall initiate removal of the TNL association towards $eNB_2$ and may remove all resources associated with that signaling connection. The candidate $eNB_2$ may then remove all resources associated with that signaling connection.

If the *X2 Removal Threshold* IE is included in the X2 REMOVAL REQUEST message, the candidate $eNB_2$ shall, if supported, accept to remove the signalling connection with $eNB_1$ if the X2 Benefit Value of the signalling connection determined at the candidate $eNB_2$ is lower than the value of the *X2 Removal Threshold* IE.

### 8.3.12.3 Unsuccessful Operation

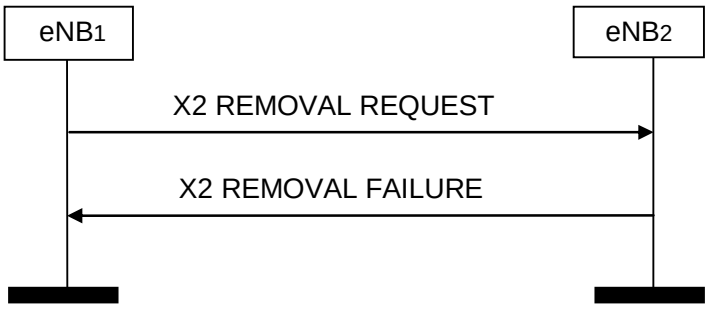



**Figure 8.3.12.3-1: X2 Removal, unsuccessful operation**

If the candidate $eNB_2$ cannot accept to remove the signaling connection with $eNB_1$ it shall respond with an X2 REMOVAL FAILURE message with an appropriate cause value.

### 8.3.12.4 Abnormal Conditions

Void.

## 8.3.13 Retrieve UE Context

### 8.3.13.1 General

The purpose of the Retrieve UE Context procedure is to retrieve the UE context from the eNB where the RRC connection has been suspended (old eNB) and transfer it to the eNB where the RRC Connection has been requested to be resumed (new eNB) or to retrieve the UE context for a UE which attempts to re-establish its RRC connection in an eNB (the new eNB) different from the eNB (the old eNB) where the RRC connection failed, e.g. due to RLF.

The procedure uses UE-associated signalling.

### 8.3.13.2 Successful Operation

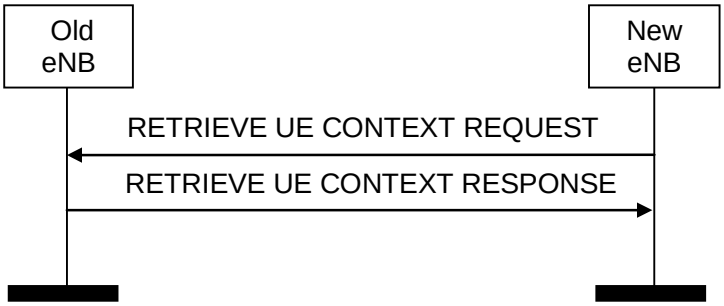



**Figure 8.3.13.2-1: Retrieve UE Context, successful operation**

The new eNB initiates the procedure by sending the RETRIEVE UE CONTEXT REQUEST message to the old eNB.

If the old eNB is able to identify the UE context and to successfully verify the UE by means of the Resume ID, the ShortMAC-I, optionally the C-RNTI, the failure cell PCI and the E-UTRAN Cell Identifier of the new cell contained in the RETRIEVE UE CONTEXT REQUEST message, it shall respond with the RETRIEVE UE CONTEXT RESPONSE message. The allocation of resources according to the values of the *Allocation and Retention Priority* IE included in the *E-RAB Level QoS Parameters* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [4].

If the *C-RNTI* IE is present in the RETRIEVE UE CONTEXT REQUEST, the old eNB shall ignore the *Resume ID* IE.

The old eNB may include in the *GUMMEI* IE any GUMMEI corresponding to the source MME node.

If the PLMN of the new cell is not the Serving PLMN stored in the UE Context the old eNB shall replace the Serving PLMN with the PLMN of the new cell and move the Serving PLMN to the equivalent PLMN list, before propagating the roaming and access restriction information to the new eNB.The new eNB shall act upon reception of the

- *UE Security Capabilities* IE,
- *AS Security Information* IE,
- *Subscriber Profile ID for RAT/Frequency priority* IE,
- *Additional RRM Policy Index* IE,
- *Handover Restriction List* IE,
- *Location Reporting Information* IE,
- *Management Based MDT Allowed* IE
- *Management Based MDT PLMN List* IE
- *Trace Activation* IE,
- *SRVCC Operation Possible* IE,
- *Masked IMEISV* IE
- *Expected UE Behaviour* IE,
- *ProSe Authorized* IE,
- *V2X Services Authorized* IE,
- *Aerial UE subscription information* IE,
- *Subscription Based UE Differentiation Information* IE,
- *EPC Handover Restriction List Container* IE,
- *Security Indication* IE,

within the RETRIEVE UE CONTEXT RESPONSE message as specified for the new eNB upon reception of the HANDOVER REQUEST message for the Handover Preparation procedure.

If the *UE Sidelink Aggregate Maximum Bit Rate* IE is contained in the RETRIEVE UE CONTEXT RESPONSE message, the new eNB shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for V2X services.

If the *Aerial UE subscription information* IE is included in the RETRIEVE UE CONTEXT RESPONSE message, the new eNB shall, if supported, store this information in the UE context and use it as defined in TS 36.300 [15].

For each E-RAB for which the old eNB proposes to do forwarding of downlink data, the old eNB shall include the *DL Forwarding* IE within the *E-RABs To Be Setup Item* IE of the RETRIEVE UE CONTEXT RESPONSE message.

If the *Bearer Type* IE is included in the RETRIEVE UE CONTEXT RESPONSE message and is set to "non IP", then the new eNB shall not perform IP header compression for the concerned E-RAB.

If the *Ethernet Type* IE is included in the RETRIEVE UE CONTEXT RESPONSE message and is set to "True", then the new eNB shall, if supported, take this into account to perform header compression appropriately for the concerned E-RAB.

If the *NR UE Sidelink Aggregate Maximum Bit Rate* IE is contained in the RETRIEVE UE CONTEXT RESPONSE message, the new eNB shall, if supported, use it for the concerned UE's sidelink communication in network scheduled mode for NR V2X services.

If the *NR V2X Services Authorized* IE is contained in the RETRIEVE UE CONTEXT RESPONSE message and it contains one or more IEs set to "authorized", the eNB shall, if supported, consider that the UE is authorized for the relevant service(s).

If the *PC5 QoS Parameters* IE is contained in the RETRIEVE UE CONTEXT RESPONSE message, the new eNB shall, if supported, use it for the concerned UE's NR sidelink communication as specified in TS 23.285 [41].

If the *UE Radio Capability ID* IE is contained in the RETRIEVE UE CONTEXT RESPONSE message, the new eNB shall, if supported, store this information in the UE context and use it as specified in TS 23.401 [12].

If the *IMS voice EPS fallback from 5G* IE is contained in the RETRIEVE UE CONTEXT RESPONSE message, the new eNB shall, if supported, store this information in the UE context and consider that the UE was previously handed over from NG-RAN to E-UTRAN due to an IMS voice fallback.

### 8.3.13.3 Unsuccessful Operation

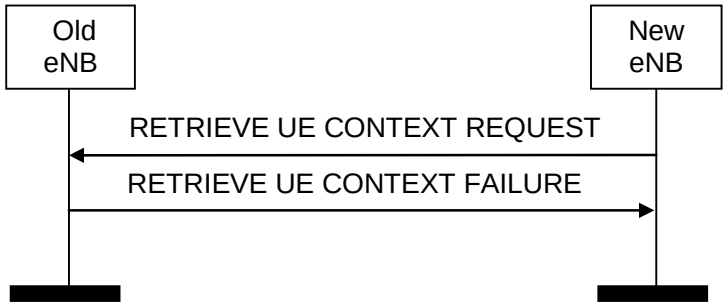



**Figure 8.3.13.3-1: Retrieve UE Context, unsuccessful operation**

If the old eNB is not able to identify the UE context by means of the Resume ID, or with the ShortMAC-I, C-RNTI, failed cell PCI and new E-UTRAN Cell Identifier contained in the RETRIEVE UE CONTEXT REQUEST message, it shall respond to the new eNB with the RETRIEVE UE CONTEXT FAILURE message.

### 8.3.13.4 Abnormal Conditions

Void.

## 8.3.14 EN-DC X2 Removal

### 8.3.14.1 General

The purpose of the EN-DC X2 Removal procedure is to remove the interface instance between eNB and en-gNB in a controlled manner. If successful, this procedure erases any existing application level configuration data in the two nodes.

NOTE: In case the signalling transport is shared among several X2-C interface instances, and the TNL association is still used by one or more X2-C interface instances, the initiating node should not initiate the removal of the TNL association.

The procedure uses non UE-associated signaling.

### 8.3.14.2 Successful Operation

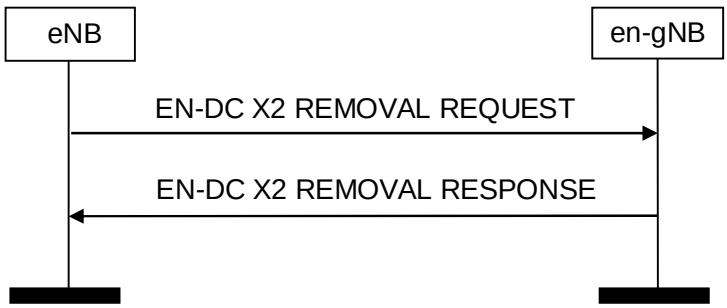



**Figure 8.3.14.2-1: eNB Initiated EN-DC X2 Removal, successful operation**

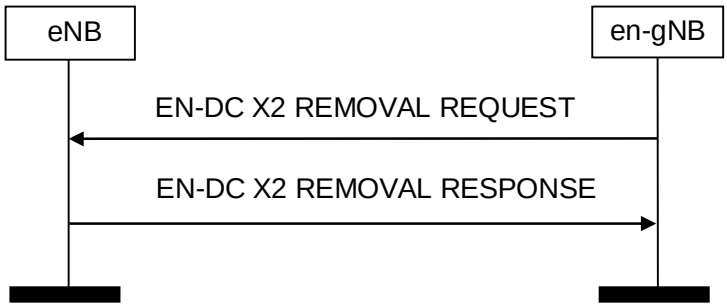



**Figure 8.3.14.2-2: en-gNB Initiated EN-DC X2 Removal, successful operation**

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC X2 REMOVAL REQUEST message and the EN-DC X2 REMOVAL RESPONSE message shall include the *Interface Instance Indication* IE to identify the corresponding interface instance.

**eNB initiated EN-DC X2 Removal:**

An eNB initiates the procedure by sending the EN-DC X2 REMOVAL REQUEST message to a candidate en-gNB. Upon reception of the EN-DC X2 REMOVAL REQUEST message the candidate en-gNB shall reply with the EN-DC X2 REMOVAL RESPONSE message. After receiving the EN-DC X2 REMOVAL RESPONSE message, the initiating eNB shall initiate removal of the TNL association towards en-gNB and may remove all resources associated with that interface instance. The candidate eNB may then remove all resources associated with that interface instance.

If the *X2 Removal Threshold* IE is included in the EN-DC X2 REMOVAL REQUEST message, the candidate en-gNB shall, if supported, accept to remove the interface instance with eNB if the X2 Benefit Value of the interface instance determined at the candidate en-gNB is lower than the value of the *X2 Removal Threshold* IE.

**en-gNB initiated EN-DC X2 Removal:**

An en-gNB initiates the procedure by sending the EN-DC X2 REMOVAL REQUEST message to a candidate eNB. Upon reception of the EN-DC X2 REMOVAL REQUEST message the candidate eNB shall reply with the EN-DC X2 REMOVAL RESPONSE message. After receiving the EN-DC X2 REMOVAL RESPONSE message, the initiating en-gNB shall initiate removal of the TNL association towards eNB and may remove all resources associated with that interface instance. The candidate eNB may then remove all resources associated with that interface instance.

If the *X2 Removal Threshold* IE is included in the EN-DC X2 REMOVAL REQUEST message, the candidate eNB shall, if supported, accept to remove the interface instance with en-gNB if the X2 Benefit Value of the interface instance determined at the candidate eNB is lower than the value of the *X2 Removal Threshold* IE.

### 8.3.14.3 Unsuccessful Operation

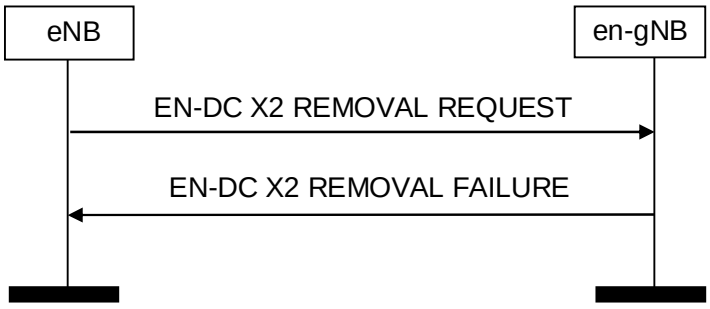



**Figure 8.3.14.3-1: eNB Initiated EN-DC X2 Removal, unsuccessful operation**

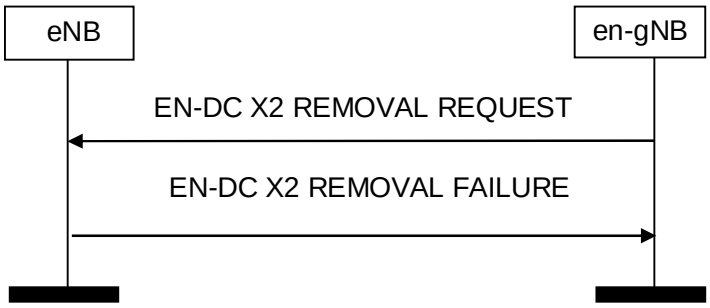



**Figure 8.3.14.3-2: en-gNB Initiated EN-DC X2 Removal, unsuccessful operation**

If the candidate receiving node cannot accept to remove the interface instance with initiating node it shall respond with an EN-DC X2 REMOVAL FAILURE message with an appropriate cause value.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC X2 REMOVAL REQUEST message and the EN-DC X2 REMOVAL FAILURE message shall include the *Interface Instance Indication* IE to identify the corresponding interface instance.

### 8.3.14.4 Abnormal Conditions

Void.

## 8.3.15 Data Forwarding Address Indication

### 8.3.15.1 General

The purpose of the Data Forwarding Address Indication procedure is to allow the new eNB to provide data forwarding addresses to the old eNB in case the RRC connection has been re-established, as specified in TS 36.300 [15].

For Dual Connectivity or EN-DC, the Data Forwarding Address Indication procedure is used during a Conditional Handover to provide data forwarding related information from the MeNB to the SeNB as specified in TS 36.300 [15], or from the MeNB to the en-gNB as specified in TS 37.340 [32].

The procedure uses UE-associated signalling.

### 8.3.15.2 Successful Operation




**Figure 8.3.15.2-1: Data Forwarding Address Indication, successful operation**




**Figure 8.3.15.2-2: Data Forwarding Address Indication for Conditional Handover, successful operation**

The new eNB initiates the procedure by sending a DATA FORWARDING ADDRESS INDICATION message to the old eNB.

For each E-RAB included in *E-RABs Data Forwarding Address List* IE, the new eNB indicates that it requests data forwarding of downlink packets to the GTP TEID indicated in the *DL GTP Tunnel Endpoint* IE.

If the DATA FORWARDING ADDRESS INDICATION message includes the *CHO DC Indicator* IE, the SeNB (respectively, the en-gNB for EN-DC) shall, if supported, consider that the DATA FORWARDING ADDRESS INDICATION message concerns a Conditional Handover, and act as specified in TS 36.300 [15] for dual connectivity (respectively, act as specified in TS 37.340 [32] for EN-DC).

If the DATA FORWARDING ADDRESS INDICATION message includes the *CHO DC Early Data Forwarding Indicator* IE set to "stop", the SeNB (respectively, the en-gNB for EN-DC) shall, if supported and if already initiated, stop early data forwarding for the provided E-RABs Data Forwarding Address information.

If the DATA FORWARDING ADDRESS INDICATION message includes the *CPC Data Forwarding Indicator* IE set to “triggered”, the en-gNB for EN-DC shall, if supported, consider that the DATA FORWARDING ADDRESS INDICATION message concerns a Conditional PSCell Change, and act as specified in TS 37.340 [32]. If the *CPC Data Forwarding Indicator* IE is set to "early data transmission stop", the en-gNB shall, if supported and if already initiated, stop early data forwarding for the provided Data Forwarding Address information.

**EN-DC**

If the MeNB sends the message to the en-gNB, then the *SgNB UE X2AP ID* IE shall be included in the DATA FORWARDING ADDRESS INDICATION message, while the *New eNB UE X2AP ID* IE is ignored. The *SgNB UE X2AP ID* IE is used as the new UE ID.

**Interaction with SeNB initiated SeNB Modification or SgNB initiated SgNB Modification procedure:**

If the *CHO MR-DC Indicator* IE or the *CPC Data Forwarding indicator* IE is set to "coordination-only" in the DATA FORWARDING ADDRESS INDICATION message and if any SCG reconfiguration is executed, then the SeNB (respectively, the en-gNB for EN-DC) shall, if supported, trigger the SeNB initiated SeNB Modification, or respectively SgNB initiated SgNB Modification procedure to inform the MeNB as specified in TS 37.340 [32].

### 8.3.15.3 Unsuccessful Operation

Not applicable.

### 8.3.15.4 Abnormal Conditions

Void.

### 8.3.16 Access and Mobility Indication

#### 8.3.16.1 General

The purpose of the Access and Mobility Indication procedure is to transfer Access and Mobility related information between E-UTRAN nodes.

#### 8.3.16.2 Successful Operation




**Figure 8.3.16.2-1: Access and Mobility Indication. Successful operation – eNB-initiated**

The eNB initiates the procedure by sending the ACCESS AND MOBILITY INDICATION message sent to the en-gNB.

#### 8.3.16.3 Abnormal Conditions

Not applicable.

## 8.4 X2 Release

### 8.4.1 General

The purpose of the X2 Release procedure is to inform an eNB that the signalling (i.e. SCTP) connection to a peer eNB is unavailable.

### 8.4.2 Successful Operation




**Figure 8.4.2-1: X2AP Release, successful operation**

$eNB_1$ initiates the procedure by sending the X2 RELEASE message to $eNB_2$. Upon the reception of X2 RELEASE message, $eNB_2$ shall consider that the signalling connection to an eNB indicated by the *eNB ID* IE is unavailable. $eNB_2$ may delete all the context information related to the indicated eNB.

### 8.4.3 Unsuccessful Operation

Not Applicable

### 8.4.4 Abnormal Condition

Not Applicable.

## 8.5 X2AP Message Transfer

### 8.5.1 General

The purpose of the X2AP Message Transfer procedure is to allow indirect transport of an X2AP message (except the X2AP MESSAGE TRANSFER message) between two eNBs and to allow an eNB to perform registration.

### 8.5.2 Successful Operation




**Figure 8.5.2-1: X2AP Message Transfer, successful operation**

$eNB_1$ initiates the procedure by sending the X2AP MESSAGE TRANSFER message to $eNB_2$.

Upon the reception of X2 MESSAGE TRANSFER message the target eNB may:

- Retrieve the X2AP message included in the *X2AP Message* IE;
- Consider the target eNB ID contained in the *Target eNB ID* IE, included in the *RNL Header* IE, as the destination for the X2AP message signaled in the *X2AP Message* IE;
- Consider the source eNB ID contained in the *Source eNB ID* IE, included in the *RNL Header* IE, as the source of the X2AP message signaled in the *X2AP Message* IE.

In case the included *RNL Header* IE does not contain the *Target eNB ID* IE, the receiving eNB shall consider the eNB ID included in the *Source eNB ID* IE as the eNB ID corresponding to the TNL address(es) of the sender and update its internal information.

### 8.5.3 Unsuccessful Operation

Not Applicable.

### 8.5.4 Abnormal Condition

Not Applicable.

# 8.6 Procedures for Dual Connectivity

## 8.6.1 SeNB Addition Preparation

### 8.6.1.1 General

The purpose of the SeNB Addition Preparation procedure is to request the SeNB to allocate resources for dual connectivity operation for a specific UE.

The procedure uses UE-associated signalling.

### 8.6.1.2 Successful Operation

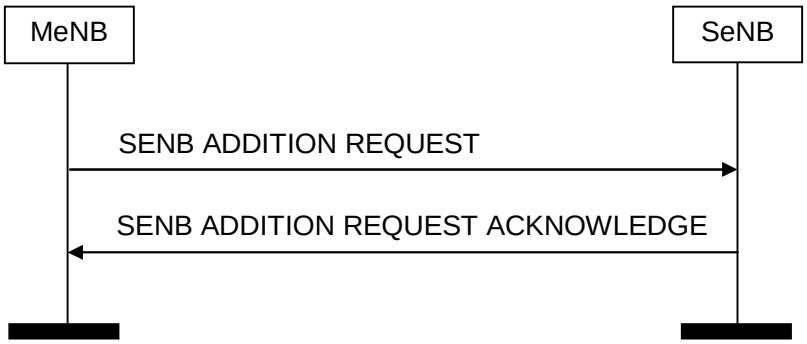



**Figure 8.6.1.2-1: SeNB Addition Preparation, successful operation**

The MeNB initiates the procedure by sending the SENB ADDITION REQUEST message to the SeNB. When the MeNB sends the SENB ADDITION REQUEST message, it shall start the timer $T_{DCprep}$.

The allocation of resources according to the values of the *Allocation and Retention Priority* IE included in the *E-RAB Level QoS Parameters* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [4].

If the SENB ADDITION REQUEST message contains the *Serving PLMN* IE, the SeNB may use it for RRM purposes.

If the SENB ADDITION REQUEST message contains the *Expected UE Behaviour* IE, the SeNB shall, if supported, store this information and may use it to optimize resource allocation.

The SeNB shall report to the MeNB, in the SENB ADDITION REQUEST ACKNOWLEDGE message, the result for all the requested E-RABs in the following way:

- A list of E-RABs which are successfully established shall be included in the *E-RABs Admitted To Be Added List* IE.
- A list of E-RABs which failed to be established shall be included in the *E-RABs Not Admitted List* IE.

NOTE: The MeNB may trigger the SeNB Addition Preparation procedure in the course of the Inter-MeNB handover without SeNB change procedure as described in 36.300 [15]. The deleted E-RABs are not included in the *E-RABs To Be Added List* IE in the SENB ADDITION REQUEST message, from MeNB point of view. If the SeNB reports a certain E-RAB to be successfully established, respective SCG resources, from an SeNB point of view, may be actually successfully established or modified or kept; if a certain E-RAB is reported to be failed to be established, respective SCG resources, from an SeNB point of view, may be actually failed to be established or modified or kept.

For each E-RAB configured with the SCG bearer option

- the SeNB shall choose the ciphering algorithm based on the information in the *UE Security Capabilities* IE and locally configured priority list of AS encryption algorithms and apply the key indicated in the *SeNB Security Key* IE as specified in the TS 33.401 [18].
- the MeNB may propose to apply forwarding of downlink data by including the *DL Forwarding* IE within the *E-RABs To be Added Item* IE of the SENB ADDITION REQUEST message. For each E-RAB that it has decided

to admit, the SeNB may include the *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs Admitted To Be Added Item* IE of the SENB ADDITION REQUEST ACKNOWLEDGE message to indicate that it accepts the proposed forwarding of downlink data for this bearer. This GTP tunnel endpoint may be different from the corresponding GTP tunnel endpoint, i.e the information contained in the *Transport Layer Address* IE and the *DL GTP TEID* IE in the *E-RAB To Be Modified List* IE of the E-RAB MODIFICATION INDICATION message (see TS 36.413 [4]) depending on implementation choice.

- the SeNB may include for each bearer in the *E-RABs Admitted To Be Added List* IE the *UL Forwarding GTP Tunnel Endpoint* IE to indicate that it requests data forwarding of uplink packets to be performed for that bearer.
- If the *Correlation ID* IE for the concerned E-RAB is received by the SeNB, the SeNB shall use this information for LIPA operation for the concerned E-RAB.
- If the *SIPTO Correlation ID* IE for the concerned E-RAB is received by the SeNB, the SeNB shall use this information for SIPTO@LN operation for the concerned E-RAB.
- If the *Bearer Type* IE for the concerned E-RAB is received by the SeNB and is set to "non IP", the SeNB shall, if supported, not perform IP header compression for the concerned E-RAB.
- If the *Ethernet Type* IE for the concerned E-RAB is received by the SeNB and is set to "True", the SeNB shall, if supported, take this into account to perform header compression appropriately for the concerned E-RAB.

If the *CSG Membership Status* IE is included in the SENB ADDITION REQUEST message, the SeNB shall act as specified in TS 36.300 [15].

Upon reception of the SENB ADDITION REQUEST ACKNOWLEDGE message the MeNB shall stop the timer $T_{DCprep}$.

If the *GW Transport Layer Address* IE is received in the SENB ADDITION REQUEST ACKNOWLEDGE message, the MeNB stores this information and use it according to TS 36.300 [15].

If the *SIPTO L-GW Transport Layer Address* IE is received in the SENB ADDITION REQUEST ACKNOWLEDGE message, the MeNB stores this information and use it according to TS 36.300 [15].

If the *SeNB UE X2AP ID* IE and/or *SeNB UE X2AP ID Extension* IE are contained in the SENB ADDITION REQUEST message, the SeNB shall, if supported, store this information and use it as defined in TS 36.300 [15].

If the *Tunnel Information for BBF* IE is received in the SENB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, transfer the tunnel information for BBF to the core network.

If the *Source DL Forwarding IP Address* IE is included in the SENB ADDITION REQUEST message, the SeNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If the *Source DL Forwarding IP Address* IE is included in the SENB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

**Interactions with the SeNB Reconfiguration Completion procedure:**

If the SeNB admits at least one E-RAB, the SeNB shall start the timer $T_{DCoverall}$ when sending the SENB ADDITION REQUEST ACKNOWLEDGE message to the MeNB. The reception of the SENB RECONFIGURATION COMPLETE message shall stop the timer $T_{DCoverall}$.

### 8.6.1.3 Unsuccessful Operation

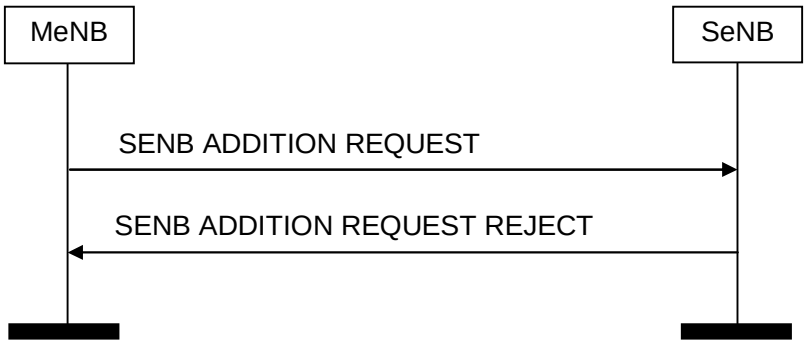



**Figure 8.6.1.3-1: SeNB Addition Preparation, unsuccessful operation**

If the SeNB is not able to accept any of the bearers or a failure occurs during the SeNB Addition Preparation, the SeNB sends the SENB ADDITION REQUEST REJECT message with an appropriate cause value to the MeNB.

### 8.6.1.4 Abnormal Conditions

If the SeNB receives a SENB ADDITION REQUEST message containing multiple *E-RAB ID* IEs (in the *E-RABs To Be Added List* IE) set to the same value, the SeNB shall consider the establishment of the corresponding E-RAB as failed.

If the SeNB receives a SENB ADDITION REQUEST message containing a *E-RAB Level QoS Parameters* IE which contains a *QCI* IE indicating a GBR bearer (as defined in TS 23.203 [13]), and which does not contain the *GBR QoS Information* IE, the SeNB shall consider the establishment of the corresponding E-RAB as failed.

If the supported algorithms for encryption defined in the *Encryption Algorithms* IE in the *UE Security Capabilities* IE, plus the mandated support of EEA0 in all UEs (TS 33.401 [18]), do not match any algorithms defined in the configured list of allowed encryption algorithms in the SeNB (TS 33.401 [18]), the SeNB shall reject the procedure using the SENB ADDITION REQUEST REJECT message.

If the SeNB receives a SENB ADDITION REQUEST message which does not contain the *CSG Membership Status* IE, and the SCell served by the SeNB is a hybrid cell, the SeNB shall reject the procedure using the SENB ADDITION REQUEST REJECT message.

If the SeNB receives a SENB ADDITION REQUEST message containing a *SeNB UE X2AP ID* IE that does not match any existing UE Context that has such ID, the SeNB shall reject the procedure using the SENB ADDITION REQUEST REJECT message.

If the SeNB receives a SENB ADDITION REQUEST message containing both the *Correlation ID* and the *SIPTO Correlation ID* IEs for the same E-RAB, the SeNB shall consider the establishment of the corresponding E-RAB as failed.

**Interactions with the SeNB Reconfiguration Completion and SeNB initiated SeNB Release procedure:**

If the timer $T_{DCoverall}$ expires before the SeNB has received the SENB RECONFIGURATION COMPLETE or the SENB RELEASE REQUEST message, the SeNB shall regard the requested RRC connection reconfiguration as being not applied by the UE and shall trigger the SeNB initiated SeNB Release procedure.

**Interactions with the MeNB initiated SeNB Release procedure:**

If the timer $T_{DCprep}$ expires before the MeNB has received the SENB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall regard the SeNB Addition Preparation procedure as being failed and shall trigger the MeNB initiated SeNB Release procedure.

## 8.6.2 SeNB Reconfiguration Completion

### 8.6.2.1 General

The purpose of the SeNB Reconfiguration Completion procedure is to provide information to the SeNB whether the requested configuration was successfully applied by the UE.

The procedure uses UE-associated signalling.

### 8.6.2.2 Successful Operation




**Figure 8.6.2.2-1: SeNB Reconfiguration Complete procedure, successful operation.**

The MeNB initiates the procedure by sending the SENB RECONFIGURATION COMPLETE message to the SeNB.

The SENB RECONFIGURATION COMPLETE message may contain information that

- either the UE has successfully applied the configuration requested by the SeNB. The MeNB may also provide configuration information in the *MeNB to SeNB Container* IE.
- or the MeNB has not triggered configuration requested by the SeNB. The MeNB shall provide information with sufficient precision in the included *Cause* IE to enable the SeNB to know the reason for an unsuccessful reconfiguration. The MeNB may also provide configuration information in the *MeNB to SeNB Container* IE.

Upon reception of the SENB RECONFIGURATION COMPLETE message the SeNB shall stop the timer $T_{DCoverall}$.

### 8.6.2.3 Abnormal Conditions

Void.

## 8.6.3 MeNB initiated SeNB Modification Preparation

### 8.6.3.1 General

This procedure is used to enable an MeNB to request an SeNB to modify the UE context at the SeNB.

The procedure uses UE-associated signalling.

### 8.6.3.2 Successful Operation

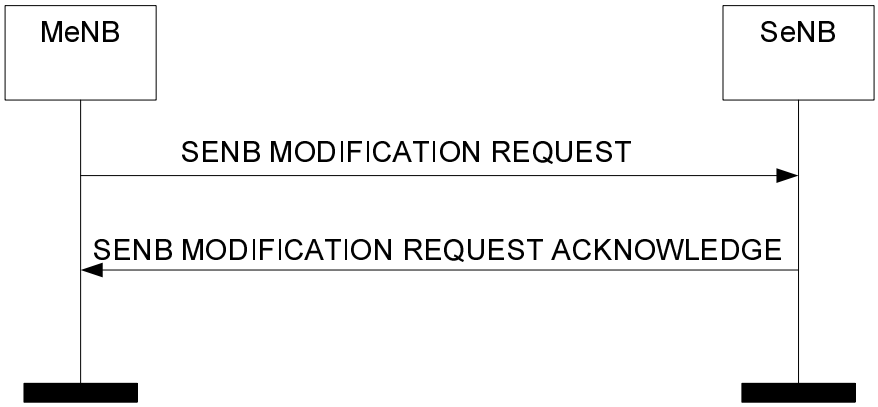



**Figure 8.6.3.2-1: MeNB initiated SeNB Modification Preparation, successful operation**

The MeNB initiates the procedure by sending the SENB MODIFICATION REQUEST message to the SeNB. When the MeNB sends the SENB MODIFICATION REQUEST message, it shall start the timer $T_{DCprep}$.

The SENB MODIFICATION REQUEST message may contain

- within the *UE Context Information* IE;
  - E-RABs to be added within the *E-RABs To Be Added Item* IE;
  - E-RABs to be modified within the *E-RABs To Be Modified Item* IE;
  - E-RABs to be released within the *E-RABs To Be Released Item* IE;
  - the *SeNB UE Aggregate Maximum Bit Rate* IE;
- the *MeNB to SeNB Container* IE;
- the *SCG Change Indication* IE;
- the *CSG Membership Status* IE.

If the SENB MODIFICATION REQUEST message contains the *Serving PLMN* IE, the SeNB may use it for RRM purposes.

If the *SeNB UE Aggregate Maximum Bit Rate* IE is included in the SENB MODIFICATION REQUEST message, the SeNB shall:

- replace the previously provided SeNB UE Aggregate Maximum Bit Rate by the received SeNB UE Aggregate Maximum Bit Rate in the UE context;
- use the received SeNB UE Aggregate Maximum Bit Rate for non-GBR Bearers for the concerned UE as defined in TS 36.300 [15].

The allocation of resources according to the values of the *Allocation and Retention Priority* IE included in the *E-RAB Level QoS Parameters* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [4].

If at least one of the requested modifications is admitted by the SeNB, the SeNB shall modify the related part of the UE context accordingly and send the SENB MODIFICATION REQUEST ACKNOWLEDGE message back to the MeNB.

The SeNB shall include the E-RABs for which resources have been either added or modified or released at the SeNB either in the *E-RABs Admitted To Be Added List* IE or the *E-RABs Admitted To Be Modified List* IE or the *E-RABs Admitted To Be Released List* IE. The SeNB shall include the E-RABs that have not been admitted in the *E-RABs Not Admitted List* IE with an appropriate cause value.

For each E-RAB configured with the SCG bearer option

- the SeNB shall, if included, choose the ciphering algorithm based on the information in the *UE Security Capabilities* IE and locally configured priority list of AS encryption algorithms and apply the key indicated in the *SeNB Security Key* IE as specified in the TS 33.401 [18].

- if applicable, the MeNB may propose to apply forwarding of downlink data by including the *DL Forwarding* IE within the *E-RABs To Be Added Item* IE of the SENB MODIFICATION REQUEST message. For each E-RAB that it has decided to admit, the SeNB may include the *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs Admitted To Be Added Item* IE of the SENB MODIFICATION REQUEST ACKNOWLEDGE message to indicate that it accepts the proposed forwarding of downlink data for this bearer. The MeNB may also provide for an applicable E-RAB to be released the *DL Forwarding GTP Tunnel Endpoint* IE and the *UL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE of the SENB MODIFICATION REQUEST message.

- if applicable, the SeNB may include for each bearer in the *E-RABs Admitted To Be Added List* IE in the SENB MODIFICATION REQUEST ACKNOWLEDGE message the *UL Forwarding GTP Tunnel Endpoint* IE to indicate that it requests data forwarding of uplink packets to be performed for that bearer.

- If the *Correlation ID* IE for the concerned E-RAB is received by the SeNB, the SeNB shall use this information for LIPA operation for the concerned E-RAB.

- If the *SIPTO Correlation ID* IE for the concerned E-RAB is received by the SeNB, the SeNB shall use this information for SIPTO@LN operation for the concerned E-RAB.

- If the *Bearer Type* IE for the concerned E-RAB is received by the SeNB and is set to "non IP", the SeNB shall, if supported, not perform IP header compression for the concerned E-RAB.

- If the *Ethernet Type* IE for the concerned E-RAB is received by the SeNB and is set to "True", the SeNB shall, if supported, take this into account to perform header compression appropriately for the concerned E-RAB.

For each E-RAB configured with the split bearer option to be modified, if the SENB MODIFICATION REQUEST message includes the *SCG Change Indication* IE and the *MeNB GTP Tunnel Endpoint* IE in the *E-RABs To Be Modified Item* IE, the SeNB shall act as specified in TS 36.300 [15].

For each E-RAB configured with the split bearer option to be modified (released)

- if applicable, the MeNB may provide for an applicable E-RAB to be released the *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE of the SENB MODIFICATION REQUEST message.

If the *E-RAB level QoS parameter* IE is included in the SENB MODIFICATION REQUEST message for an E-RAB to be modified the SeNB shall allocate respective resources and provide corresponding radio configuration information within the *SeNB to MeNB Container* IE as described in TS 36.300 [15].

If the SENB MODIFICATION REQUEST message contains for an E-RAB to be modified which is configured with the SCG bearer option the *S1 UL GTP Tunnel Endpoint* IE the SeNB shall use it as the new UL S1-U address.

If the SENB MODIFICATION REQUEST message contains for an E-RAB to be modified which is configured with the split bearer option the *MeNB GTP Tunnel Endpoint* IE the SeNB shall use it as the new UL X2-U address.

For an E-RAB to be modified which is configured with the SCG bearer option the SeNB may include in the SENB MODIFICATION REQUEST ACKNOWLEDGE message the *S1 DL GTP Tunnel Endpoint* IE.

For an E-RAB to be modified which is configured with the split bearer option the SeNB may include in the SENB MODIFICATION REQUEST ACKNOWLEDGE message the *SeNB GTP Tunnel Endpoint* IE.

If the *SCG Change Indication* IE is included in the SENB MODIFICATION REQUEST message, the SeNB shall act as specified in TS 36.300 [15].

If the *CSG Membership Status* IE is included in the SENB MODIFICAITON REQUEST message, the SeNB shall act as specified in TS 36.300 [15].

Upon reception of the SENB MODIFICATION REQUEST ACKNOWLEDGE message the MeNB shall stop the timer $T_{DCprep}$. If the SENB MODIFICATION REQUEST ACKNOWLEDGE message has included the *SeNB to MeNB Container* IE the MeNB is then defined to have a Prepared SeNB Modification for that X2 UE-associated signalling.

When the SeNB supporting L-GW function for LIPA operation releases radio and control plane related resources associated to the LIPA bearer, it shall also request using intra-node signalling the collocated L-GW to release the LIPA PDN connection as defined in TS 23.401 [12].

If the *Source DL Forwarding IP Address* IE is included in the SENB MODIFICATION REQUEST message, the SeNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If the *Source DL Forwarding IP Address* IE is included in the SENB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions to identify source TNL address for data forwarding in case of subsequent handover preparation, if such ACL functionality is deployed.

**Interactions with the SeNB Reconfiguration Completion procedure:**

If the SeNB admits a modification of the UE context requiring the MeNB to report about the success of the RRC connection reconfiguration procedure, the SeNB shall start the timer $T_{DCoverall}$ when sending the SENB MODIFICATION REQUEST ACKNOWLEDGE message to the MeNB. The reception of the SeNB RECONFIGURATION COMPLETE message shall stop the timer $T_{DCoverall}$.

### 8.6.3.3 Unsuccessful Operation

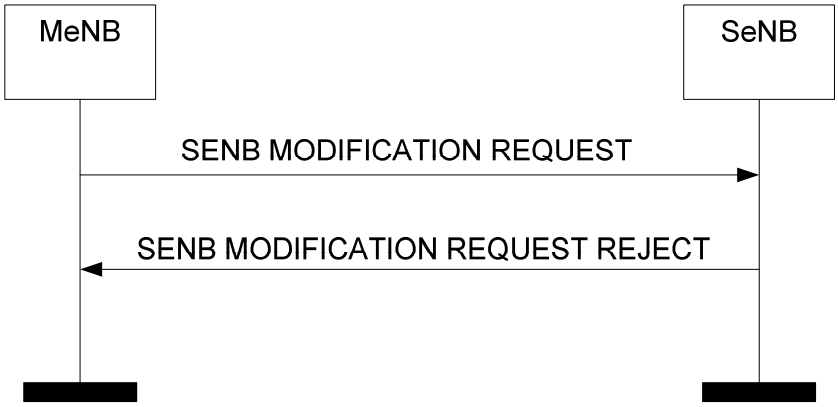



**Figure 8.6.3.3-1: MeNB initiated SeNB Modification Preparation, unsuccessful operation**

If the SeNB does not admit any modification requested by the MeNB, or a failure occurs during the MeNB initiated SeNB Modfication Preparation, the SeNB shall send the SENB MODIFICATION REQUEST REJECT message to the MeNB. The message shall contain the *Cause* IE with an appropriate value.

If the SeNB receives a SENB MODIFICATION REQUEST message containing the *MeNB to SeNB Container* IE that does not include required information as specified in TS 36.331 [9], the SeNB shall send the SENB MODIFICATION REQUEST REJECT message to the MeNB.

### 8.6.3.4 Abnormal Conditions

If the SeNB receives a SENB MODIFICATION REQUEST message containing multiple *E-RAB ID* IEs (in the *E-RABs To Be Added List* IE and/or the *E-RABs To Be Modified List* IE) set to the same value, the SeNB shall not admit the action requested for the corresponding E-RABs.

If the SeNB receives an SENB MODIFICATION REQUEST message containing multiple *E-RAB ID* IEs (in the *E-RAB To Be Released List* IE) set to the same value, the SeNB shall initiate the release of one corresponding E-RAB and ignore the duplication of the instances of the selected corresponding E-RABs.

If the SeNB receives a SENB MODIFICATION REQUEST message containing a *E-RAB Level QoS Parameters* IE which contains a *QCI* IE indicating a GBR bearer (as defined in TS 23.203 [13]), and which does not contain the *GBR QoS Information* IE, the SeNB shall not admit the corresponding E-RAB.

If the supported algorithms for encryption defined in the *Encryption Algorithms* IE in the *UE Security Capabilities* IE in the *UE Context Information* IE, plus the mandated support of EEA0 in all UEs (TS 33.401 [18]), do not match any

algorithms defined in the configured list of allowed encryption algorithms in the SeNB (TS 33.401 [18]), the SeNB shall reject the procedure using the SENB MODIFICATION REQUEST REJECT message.

If the timer $T_{DCprep}$ expires before the MeNB has received the SENB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall regard the MeNB initiated SeNB Modification Preparation procedure as being failed and shall release the UE Context at the SeNB.

If the SeNB receives a SENB MODIFICATION REQUEST message containing both the *Correlation ID* and the *SIPTO Correlation ID* IEs for the same E-RAB, the SeNB shall consider the establishment of the corresponding E-RAB as failed.

**Interactions with the SeNB Reconfiguration Completion and SeNB initiated SeNB Release procedure:**

If the timer $T_{DCoverall}$ expires before the SeNB has received the SENB RECONFIGURATION COMPLETE or the SENB RELEASE REQUEST message, the SeNB shall regard the requested modification RRC connection reconfiguration as being not applied by the UE and shall trigger the SeNB initiated SeNB Release procedure.

**Interaction with the SeNB initiated SeNB Modification Preparation procedure:**

If the MeNB, after having initiated the MeNB initiated SeNB Modification procedure, receives the SENB MODIFICATION REQUIRED message, the MeNB shall refuse the SeNB initiated SeNB Modification procedure with an appropriate cause value in the *Cause* IE.

If the MeNB has a Prepared SeNB Modification and receives the SENB MODIFICATION REQUIRED message, the MeNB shall respond with the SENB MODIFICATION REFUSE message to the SeNB with an appropriate cause value in the *Cause* IE.

## 8.6.4 SeNB initiated SeNB Modification

### 8.6.4.1 General

This procedure is used by the SeNB to modify the UE context in the SeNB.

The procedure uses UE-associated signalling.

### 8.6.4.2 Successful Operation

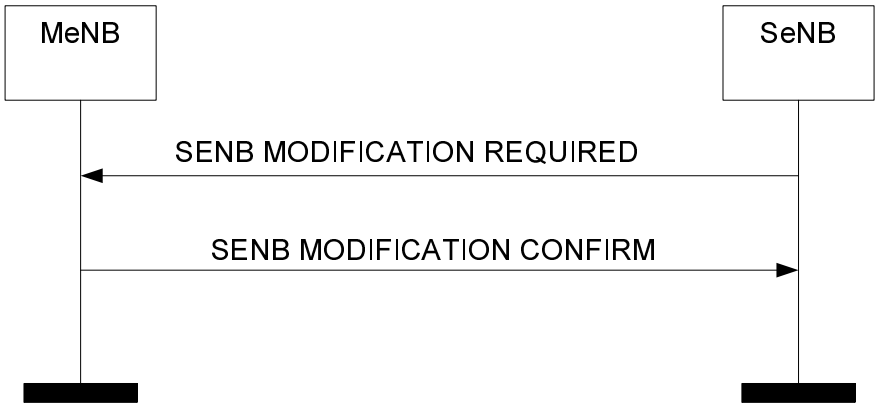



**Figure 8.6.4.2-1: SeNB initiated SeNB Modification, successful operation.**

The SeNB initiates the procedure by sending the SENB MODIFICATION REQUIRED message to the MeNB. When the SeNB sends the SENB MODIFICATION REQUIRED message, it shall start the timer $T_{DCoverall.}$

The SENB MODIFICATION REQUIRED message may contain

- the *SeNB to MeNB Container* IE.
- E-RABs to be released within the *E-RABs To Be Released Item* IE;
- the *SCG Change Indication* IE.

If the MeNB receives a SENB MODIFICATION REQUIRED message containing the *SCG Change Indication* IE, the MeNB shall act as specified in TS 36.300 [15].

If the MeNB is able to perform the modifications requested by the SeNB, the MeNB shall send the SENB MODIFICATION CONFIRM message to the SeNB. The SENB MODIFICATION CONFIRM message may contain the *MeNB to SeNB Container* IE.

Upon reception of the SENB MODIFICATION CONFIRM message the SeNB shall stop the timer $T_{DCoverall}$.

**Interaction with the MeNB initiated SeNB Modification Preparation procedure:**

If applicable, as specified in TS 36.300 [15], the SeNB may receive, after having initiated the SeNB initiated SeNB Modification procedure, the SENB MODIFICATION REQUEST message including the *DL Forwarding GTP Tunnel Endpoint* IE and the *UL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released List* IE.

If applicable, as specified in TS 36.300 [15], the SeNB may receive, after having initiated the SeNB initiated SeNB Modification procedure, the SENB MODIFICATION REQUEST message including the *SeNB Security Key* IE within the *UE Context Information* IE.

If the SeNB has initiated the SeNB initiated SeNB Modification procedure with the SENB MODIFICATION REQUIRED message including the *E-RABs To Be Released Item* IE, it may receive the SENB MODIFICATION REQUEST message including the *SCG Change Indication* IE, upon which the SeNB shall provide respective information in the *SeNB to MeNB Container* IE within the SENB MODIFICATION REQUEST ACKNOWLEDGMENT message, as specified in TS 36.300 [15].

### 8.6.4.3 Unsuccessful Operation

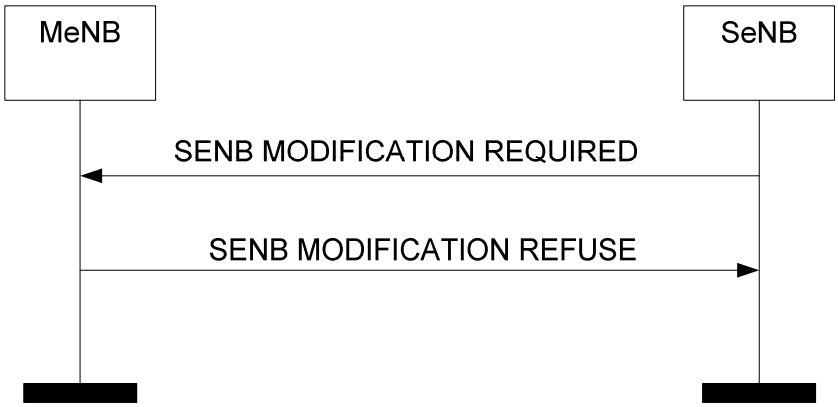



**Figure 8.6.4.3-1: SeNB initiated SeNB Modification, unsuccessful operation.**

In case the request modification cannot be performed successfully the MeNB shall respond with the SENB MODIFICATION REFUSE message to the SeNB with an appropriate cause value in the *Cause* IE.

The MeNB may also provide configuration information in the *MeNB to SeNB Container* IE.

### 8.6.4.4 Abnormal Conditions

If the timer $T_{DCoverall}$ expires before the SeNB has received the SENB MODIFICATION CONFIRM or the SENB MODIFICATION REFUSE message, the SeNB shall regard the requested modification as failed and may take further actions like triggering the SeNB initiated SeNB Release procedure to release all SeNB resources allocated for the UE.

If the MeNB is aware that the SeNB didn't receive the latest configuration information concerning the MCG, the MeNB may respond with the SENB MODIFICATION REFUSE message to the SeNB with an appropriate cause value in the *Cause* IE.

If the value received in the *E-RAB ID* IE of any of the *E-RABs To Be Released Items* IE is not known at the MeNB, the MeNB shall regard the procedure as failed and may take appropriate actions like triggering the MeNB initiated SeNB Release procedure.

**Interaction with the MeNB initiated SeNB Modification Preparation procedure:**

If the SeNB, after having initiated the SeNB initiated SeNB Modification procedure, receives the SENB MODIFICATION REQUEST message including other IEs than an applicable *SeNB Security Key* IE and/or applicable forwarding addresses and/or the *SCG Change Indication* IE the SeNB shall

- regard the SeNB initiated SeNB Modification Procedure as being failed,
- stop the $T_{DCoverall}$, which was started to supervise the SeNB initiated SeNB Modification procedure,
- be prepared to receive the SENB MODIFICATION REFUSE message from the MeNB and
- continue with the MeNB initiated SeNB Modification Preparation procedure as specified in section 8.6.3.

## 8.6.5 MeNB initiated SeNB Release

### 8.6.5.1 General

The MeNB initiated SeNB Release procedure is triggered by the MeNB to initiate the release of the resources for a specific UE.

The procedure uses UE-associated signalling.

### 8.6.5.2 Successful Operation




**Figure 8.6.5.2-1: MeNB initiated SeNB Release, successful operation**

The MeNB initiates the procedure by sending the SENB RELEASE REQUEST message. Upon reception of the SENB RELEASE REQUEST message the SeNB shall stop providing user data to the UE. The *SeNB UE X2AP ID* IE and, if available, the *SeNB UE X2AP ID Extension* IE shall be included if it has been obtained from the SeNB. The MeNB may provide appropriate information within the *Cause* IE.

If the bearer context in the SeNB was configured with the SCG bearer option, for each SCG bearer for which the MeNB requests forwarding of uplink/downlink data, the MeNB includes the *UL Forwarding GTP Tunnel Endpoint*/ *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE of the SENB RELEASE REQUEST message to indicate that the SeNB should perform data forwarding of uplink/downlink packets for that SCG bearer.

If the bearer context in the SeNB was configured with the split bearer option, for each Split bearer for which the MeNB requests forwarding of downlink data, the MeNB includes the *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE of the SENB RELEASE REQUEST message to indicate that the SeNB should perform data forwarding of downlink packets for that split bearer.

Upon reception of the SENB RELEASE REQUEST message containing *UE Context Kept Indicator* IE set to "True", the SeNB shall, if supported, only initiate the release of the resources related to the UE-associated signalling connection between the MeNB and the SeNB.

Upon reception of the SENB RELEASE REQUEST message containing *MakeBeforeBreak Indicator* IE set to "True", the SeNB shall, if supported, perform Make-Before-Break SeNB change as specified in TS 36.300 [15].

### 8.6.5.3 Unsuccessful Operation

Not applicable.

#### 8.6.5.4 Abnormal Conditions

Should the SENB RELEASE REQUEST message refer to a context that does not exist, the SeNB shall ignore the message.

When the MeNB has initiated the procedure and did not include the *SeNB UE X2AP ID* IE the MeNB shall regard the resources for the UE at the SeNB as being fully released.

### 8.6.6 SeNB initiated SeNB Release

#### 8.6.6.1 General

This procedure is triggered by the SeNB to initiate the release of the resources for a specific UE.

The procedure uses UE-associated signalling.

#### 8.6.6.2 Successful Operation

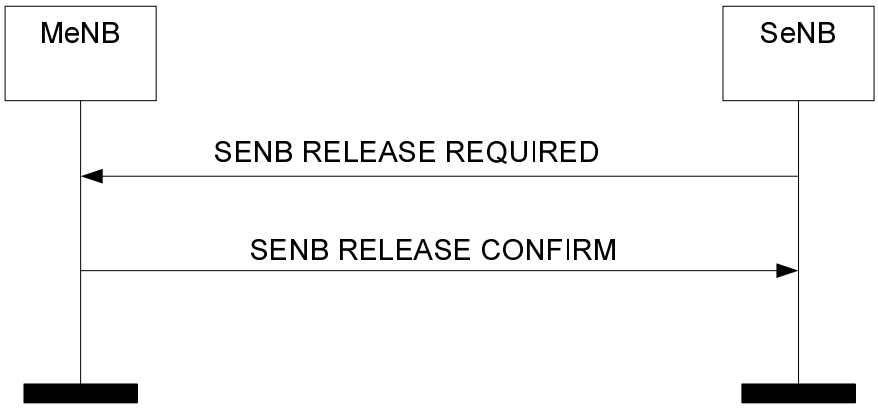



**Figure 8.6.6.2-1: SeNB initiated SeNB Release, successful operation.**

The SeNB initiates the procedure by sending the SENB RELEASE REQUIRED message to the MeNB.

Upon reception of the SENB RELEASE REQUIRED message, the MeNB replies with the SENB RELEASE CONFIRM message. For each E-RAB configured with the SCG bearer option, the MeNB may include the *DL Forwarding GTP Tunnel Endpoint* IE and the *UL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE to indicate that it requests data forwarding of uplink and downlink packets to be performed for that bearer. For each E-RAB configured with the split bearer option, the MeNB may include the *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE to indicate that it requests data forwarding of downlink packets to be performed for that bearer.

The SeNB may start data forwarding and stop providing user data to the UE upon reception of the SENB RELEASE CONFIRM message,

#### 8.6.6.3 Unsuccessful Operation

Not applicable.

#### 8.6.6.4 Abnormal Conditions

Void.

## 8.6.7 SeNB Counter Check

### 8.6.7.1 General

This procedure is initiated by the SeNB to request the MeNB to execute a counter check procedure to verify the value of the PDCP COUNTs associated with SCG bearers established in the SeNB.

The procedure uses UE-associated signalling.

### 8.6.7.2 Successful Operation




**Figure 8.6.7.2-1: SeNB Counter Check procedure, successful operation.**

The SeNB initiates the procedure by sending the SENB COUNTER CHECK REQUEST message to the MeNB.

Upon reception of the SENB COUNTER CHECK REQUEST message, the MeNB may perform the RRC counter check procedure as defined in TS 33.401 [18].

### 8.6.7.3 Unsuccessful Operation

Not applicable.

### 8.6.7.4 Abnormal Conditions

Not applicable.

# 8.7 Procedures for E-UTRAN-NR Dual Connectivity

## 8.7.1 EN-DC X2 Setup

### 8.7.1.1 General

The purpose of the EN-DC X2 Setup procedure is to exchange application level configuration data needed for eNB and en-gNB to interoperate correctly over the X2 interface. This procedure erases any existing application level configuration data in the two nodes and replaces it by the one received. This procedure also resets the X2 interface like a Reset procedure would do.

NOTE 1: If X2-C signalling transport is shared among multiple X2-C interface instances, one EN-DC X2 Setup procedure is issued per X2-C interface instance to be setup, i.e. several X2 Setup procedures may be issued via the same TNL association after that TNL association has become operational.

NOTE 2: Exchange of application level configuration data also applies between eNB and en-gNB in case the SN (i.e. the en-gNB) does not broadcast system information other than for radio frame timing and SFN, as specified in the TS 37.340 [32]. How to use this information when this option is used is not explicitly specified.

The procedure uses non UE-associated signalling.

### 8.7.1.2 Successful Operation

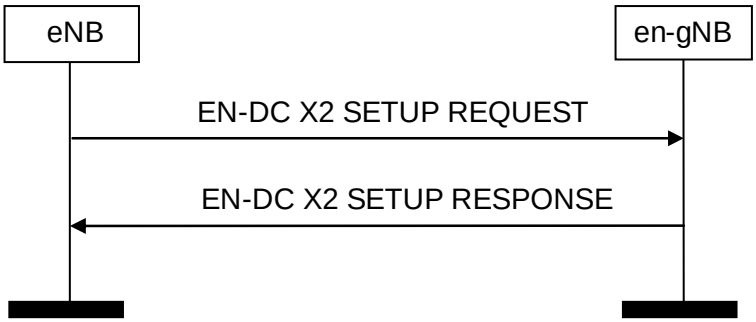



**Figure 8.7.1.2-1: eNB Initiated EN-DC X2 Setup, successful operation**

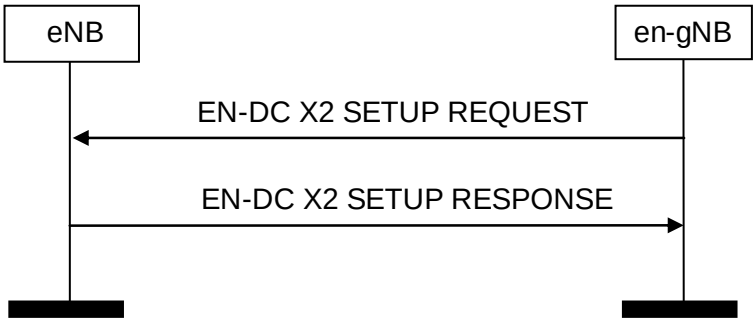



**Figure 8.7.1.2-2: en-gNB Initiated EN-DC X2 Setup, successful operation**

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC X2 SETUP REQUEST message and the EN-DC X2 SETUP RESPONSE message shall include the *Interface Instance Indication* IE to identify the corresponding interface instance. In the current version of this specification an eNB shall not include the *Interface Instance Indication* IE in the *Initiating NodeType* IE in the EN-DC X2 SETUP REQUEST message.

If the *SFN Offset* IE is included in the EN-DC X2 SETUP REQUEST or EN-DC X2 SETUP RESPONSE message, the receiving node shall, if supported, use this information to deduce the SFN0 time offset of the reported cell. The receiving node shall consider the received *SFN Offset* IE content valid until reception of an update of the IE for the same cell(s).

**eNB initiated EN-DC X2 Setup:**

An eNB initiates the procedure by sending the EN-DC X2 SETUP REQUEST message to a candidate en-gNB. The candidate en-gNB replies with the EN-DC X2 SETUP RESPONSE message. The initiating eNB shall transfer the complete list of its served cells to the candidate en-gNB. The candidate en-gNB shall reply with the complete list of its served cells or if supported, a partial list of its served cells together with the *Partial List Indicator* IE, according to the received information in *Cell and Capacity Assistance Information* IE in EN-DC X2 SETUP REQUEST message. If Supplementary Uplink is configured at the candidate en-gNB, the candidate en-gNB shall include in the EN-DC X2 SETUP RESPONSE message the *SUL Information* IE and the *Supported SUL band List* IE for each served cell where supplementary uplink is configured.

If the EN-DC X2 SETUP REQUEST message contains the *Protected E-UTRA Resource Indication* IE, the receiving en-gNB should take this into account for cell-level resource coordination with the eNB. The en-gNB shall consider the received *Protected E-UTRA Resource Indication* IE content valid until reception of a new update of the IE for the same eNB.

The protected resource pattern indicated in the *Protected E-UTRA Resource Indication* IE is not valid in subframes indicated by the *Reserved Subframes* IE, as well as in the non-control region of the MBSFN subframes i.e. it is valid only in the control region therein. The size of the control region of MBSFN subframes is indicated in the *Protected E-UTRA Resource Indication* IE.

If the *Partial List Indicator* IE is set to "partial" in the EN-DC X2 SETUP RESPONSE message from the en-gNB, the eNB shall, if supported, assume that the en-gNB has included in the *List of Served Cells NR* IE a partial list of cells.

If the EN-DC X2 SETUP REQUEST message contains the *TNL Transport Layer Address info* IE, the receiving en-gNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the EN-DC X2 SETUP RESPONSE message contains the *TNL Transport Layer Address info* IE, the receiving eNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the *NR Cell PRACH Configuration* IE is included in the *Served NR Cell Information* IE contained in the EN-DC X2 SETUP RESPONSE message, the eNB may store the information.

If the *CSI-RS Transmission Indication* IE is contained in the EN-DC X2 SETUP REQUEST message, the en-gNB may use this information for neighbour NR cell's CSI-RS measurement.

If the *Additional Measurement Timing Configuration List* IE is contained in the EN-DC X2 SETUP REQUEST message, the en-gNB may use this information for neighbour NR cell's CSI-RS measurement.

If the *Served Cell Specific Info Request* IE is included in the EN-DC X2 SETUP REQUEST message, the en-gNB shall, if supported, include the *Additional Measurement Timing Configuration List* IE for the requested NR cells in the EN-DC SETUP RESPONSE message.

If the *Intended TDD DL-UL Configuration NR* IE is contained in the *NR Neighbour Information* IE in the EN-DC X2 SETUP REQUEST message, en-gNB should take this information into account for cross-link interference management. The en-gNB shall consider the received *Intended TDD DL-UL Configuration NR* IE content valid until reception of an update of the IE for the same cell(s).

**Interaction with the eNB Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC X2 SETUP RESPONSE message to neighbouring eNBs by triggering the eNB Configuration Update procedure.

**Interaction with the EN-DC Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC X2 SETUP RESPONSE message to neighbouring en-gNBs by triggering the EN-DC Configuration Update procedure.

**en-gNB initiated EN-DC X2 Setup:**

An en-gNB initiates the procedure by sending the EN-DC X2 SETUP REQUEST message to a candidate eNB. The candidate eNB replies with the EN-DC X2 SETUP RESPONSE message. The initiating en-gNB shall transfer the complete list of its served cells or if supported, a partial list of its served cells together with the *Partial List Indicator* IE in the EN-DC X2 SETUP REQUEST message to the candidate eNB. The candidate eNB shall reply with the complete list of its served cells.

If Supplementary Uplink is configured at the en-gNB, the en-gNB shall include in the EN-DC X2 SETUP REQUEST message the *SUL Information* IE and the *Supported SUL band List* IE for each served cell where supplementary uplink is configured.

If the EN-DC X2 SETUP RESPONSE message contains the *Protected E-UTRA Resource Indication* IE, the receiving en-gNB should take this into account for cell-level resource coordination with the eNB. The en-gNB shall consider the received *Protected E-UTRA Resource Indication* IE content valid until reception of a new update of the IE for the same eNB.

If the *Partial List Indicator* IE is set to "partial" in the EN-DC X2 SETUP REQUEST message from the en-gNB, the eNB shall, if supported, assume that the en-gNB has included in the *List of Served Cells NR* IE a partial list of cells.

If the *Cell and Capacity Assistance Information* IE is present in the EN-DC X2 SETUP RESPONSE message from the eNB, the en-gNB shall, if supported, store the collected information to be used for future interface management.

If the EN-DC X2 SETUP REQUEST message contains the *TNL Transport Layer Address info* IE, the receiving eNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the EN-DC X2 SETUP RESPONSE message contains the *TNL Transport Layer Address info* IE, the receiving en-gNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the *NR Cell PRACH Configuration* IE is included in the *Served NR Cell Information* IE contained in the EN-DC X2 SETUP REQUEST message, the eNB may store the information.

If the *CSI-RS Transmission Indication* IE is contained in the EN-DC X2 SETUP REQUEST message, the eNB should take it into account when forwarding neighbour NR cell's CSI-RS configuration.

If the *Additional Measurement Timing Configuration List* IE is contained in the EN-DC X2 SETUP REQUEST message, the eNB should take it into account when forwarding neighbour NR cell's CSI-RS configuration.

If the *Served Cell Specific Info Request* IE is included in the EN-DC X2 SETUP REQUEST message, the eNB shall, if supported, include the *Additional Measurement Timing Configuration List* IE for the requested NR neighbour cells, capable of performing EN-DC with the served E-UTRA cell in the EN-DC X2 SETUP RESPONSE message.

If the *Intended TDD DL-UL Configuration NR* IE is contained in the *NR Neighbour Information* IE in the EN-DC X2 SETUP RESPONSE message, en-gNB should take this information into account for cross-link interference management. The en-gNB shall consider the received *Intended TDD DL-UL Configuration NR* IE content valid until reception of an update of the IE for the same cell(s).

**Interaction with the eNB Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC X2 SETUP REQUEST message to neighbouring eNBs by triggering the eNB Configuration Update procedure.

**Interaction with the EN-DC Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC X2 SETUP REQUEST message to neighbouring en-gNBs by triggering the EN-DC Configuration Update procedure.

### 8.7.1.3 Unsuccessful Operation

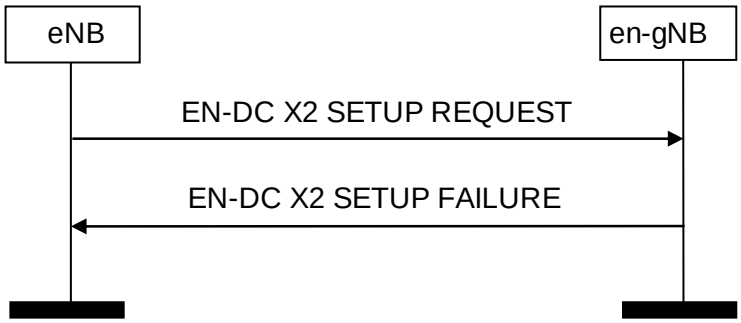



**Figure 8.7.1.3-1: eNB Initiated EN-DC X2 Setup, unsuccessful operation**

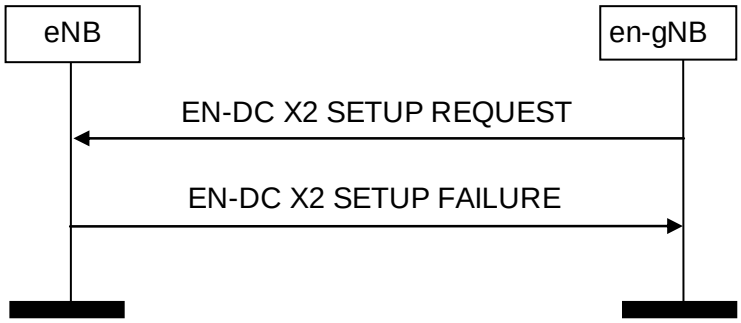



**Figure 8.7.1.3-2: en-gNB Initiated EN-DC X2 Setup, unsuccessful operation**

If the candidate receving node cannot accept the setup it shall respond with an EN-DC X2 SETUP FAILURE message with appropriate cause value.

If the *Message Oversize Notification* IE is included in the EN-DC X2 SETUP FAILURE, the initiating node shall, if supported, deduce that the failure is due to a too large EN-DC X2 SETUP REQUEST message and ensure that the total

number of served cells in following EN-DC X2 SETUP REQUEST message is equal to or lower than the value of the *Message Oversize Notification* IE.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC X2 SETUP REQUEST message and the EN-DC X2 SETUP FAILURE message shall include the *Interface Instance Indication* IE to identify the corresponding interface instance.

#### 8.7.1.4 Abnormal Conditions

If the first message received for a specific TNL association is not an EN-DC X2 SETUP REQUEST, EN-DC X2 SETUP RESPONSE, or EN-DC X2 SETUP FAILURE message then this shall be treated as a logical error.

If the initiating node does not receive either EN-DC X2 SETUP RESPONSE message or EN-DC X2 SETUP FAILURE message, the initiating node may reinitiate the EN-DC X2 Setup procedure towards the same candidate node, provided that the content of the EN-DC X2 SETUP REQUEST message is identical to the content of the previously unacknowledged EN-DC X2 SETUP REQUEST message.

If the EN-DC X2 SETUP FAILURE message includes the *Time To Wait* IE the initiating node shall wait at least for the indicated time before reinitiating the EN-DC X2 Setup procedure towards the same peer node.

If the initiating node receives an EN-DC X2 SETUP REQUEST message from the peer entity on the same X2 interface:

- In case the initiating node answers with an EN-DC X2 SETUP RESPONSE message and receives a subsequent EN-DC X2 SETUP FAILURE message, the initiating node shall consider the X2 interface as non operational and the procedure as unsuccessfully terminated according to sub clause 8.7.1.3.

- In case the initiating node answers with an EN-DC X2 SETUP FAILURE message and receives a subsequent EN-DC X2 SETUP RESPONSE message, the initiating node shall ignore the EN-DC X2 SETUP RESPONSE message and consider the X2 interface as non operational.

### 8.7.2 EN-DC Configuration Update

#### 8.7.2.1 General

The purpose of the EN-DC Configuration Update procedure is to update application level configuration data needed for eNB and en-gNB to interoperate correctly over the X2 interface.

NOTE: Update of application level configuration data also applies between eNB and en-gNB in case the SN (i.e. the en-gNB) does not broadcast system information other than for radio frame timing and SFN, as specified in the TS 37.340 [32]. How to use this information when this option is used is not explicitly specified.

The procedure uses non UE-associated signalling.

#### 8.7.2.2 Successful Operation

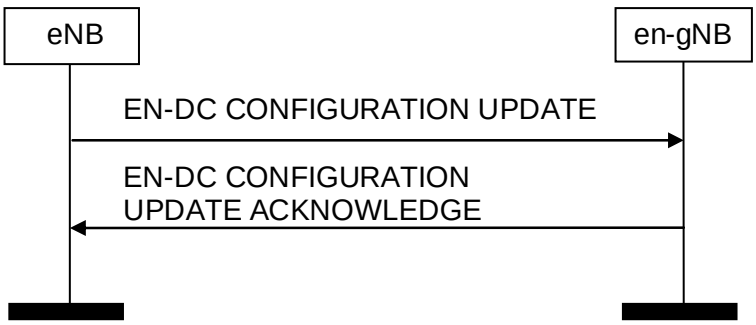



**Figure 8.7.2.2-1: eNB Initiated EN-DC Configuration Update, successful operation**

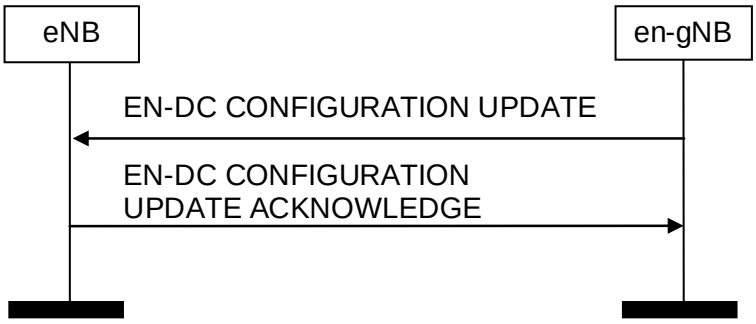



**Figure 8.7.2.2-2: en-gNB Initiated EN-DC Configuration Update, successful operation**

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC CONFIGURATION UPDATE message and the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message shall include the *Interface Instance Indication* IE to identify the corresponding interface instance.

If the *SFN Offset* IE is included in the EN-DC CONFIGURATION UPDATE or EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message, the receiving node shall, if supported, use this information to update the SFN0 time offset of the reported cell.

**eNB initiated EN-DC Configuration Update:**

An eNB initiates the procedure by sending an EN-DC CONFIGURATION UPDATE message to a peer en-gNB.

After successful update of requested information, en-gNB shall reply with the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message to inform the initiating eNB that the requested update of application data was performed successfully.

If the *Cell Assistance Information* IE is present, the en-gNB shall, if supported, use it to generate the *List of Served NR Cells* IE and include the list in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message.

If the EN-DC CONFIGURATION UPDATE REQUEST message contains the Protected E-UTRA Resource Indication IE, the receiving en-gNB should take this into account for cell-level resource coordination with the eNB. The en-gNB shall consider the received Protected E-UTRA Resource Indication IE content valid until reception of a new update of the IE for the same eNB. The protected resource pattern indicated in the Protected E-UTRA Resource Indication IE is not valid in subframes indicated by the Reserved Subframes IE, as well as in the non-control region of the MBSFN subframes i.e. it is valid only in the control region therein. The size of the control region of MBSFN subframes is indicated in the Protected E-UTRA Resource Indication IE.

The eNB may initiate a further EN-DC Configuration Update procedure only after a previous EN-DC Configuration Update procedure has been completed.

If Supplementary Uplink is configured at the en-gNB, the en-gNB shall include in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message the *SUL Information* IE and the *Supported SUL band List* IE for each cell added in the Served NR Cells To Add IE and in the Served NR Cells To Modify IE.

If the EN-DC CONFIGURATION UPDATE message contains the *TNL Transport Layer Address info* IE, the receiving en-gNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message contains the *TNL Transport Layer Address info* IE, the receiving eNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the *NR Cell PRACH Configuration* IE is included in the *Served NR Cell Information* IE contained in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message, the eNB may update the information.

If the C*SI-RS Transmission Indication* IE is contained in the EN-DC CONFIGURATION UPDATE message, the en-gNB may use this information for neighbour NR cell's CSI-RS measurement.

If the *Additional Measurement Timing Configuration List* IE is contained in the EN-DC CONFIGURATION UPDATE message, the en-gNB may use this information for neighbour NR cell's CSI-RS measurement.

If the *Served Cell Specific Info Request* IE is included in the EN-DC CONFIGURATION UPDATE message, the en-gNB shall, if supported, include the *Additional Measurement Timing Configuration List* IE for the requested NR cells in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message.

If the *Intended TDD DL-UL Configuration NR* IE is contained in the *NR Neighbour Information* IE in the EN-DC CONFIGURATION UPDATE message, en-gNB should take this information into account for cross-link interference management. The en-gNB shall consider the received *Intended TDD DL-UL Configuration NR* IE content valid until reception of an update of the IE for the same cell(s).

**Interaction with the eNB Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message to neighbouring eNBs by triggering the eNB Configuration Update procedure.

**Interaction with the EN-DC Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message to neighbouring en-gNBs by triggering the EN-DC Configuration Update procedure.

**en-gNB initiated EN-DC Configuration Update:**

An en-gNB initiates the procedure by sending an EN-DC CONFIGURATION UPDATE message to an eNB.

If Supplementary Uplink is configured at the en-gNB, the en-gNB shall include in the EN-DC CONFIGURATION UPDATE message the *SUL Information* IE and the *Supported SUL band List* IE for each served cell added in the Served NR Cells To Add IE and in the Served NR Cells To Modify IE.

If the Deactivation Indication IE is contained in the *Served NR Cells To Modify* IE, it indicates that the concerned NR cell was switched off to lower energy consumption, and is available for activation on request from the eNB, as described in TS 36.300 [15].

After successful update of requested information, eNB shall reply with the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message to inform the initiating en-gNB that the requested update of application data was performed successfully. In case the eNB receives an EN-DC CONFIGURATION UPDATE without any IE except for *Message Type* IE it shall reply with EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message without performing any updates to the existing configuration.

Upon reception of an EN-DC CONFIGURATION UPDATE message, eNB shall update the information for en-gNB as follows:

**Update of Served NR Cell Information:**

- If *Served NR Cells To Add* IE is contained in the EN-DC CONFIGURATION UPDATE message, eNB shall add cell information according to the information in the *Served NR Cell Information* IE.

- If *Served NR Cells To Modify* IE is contained in the EN-DC CONFIGURATION UPDATE message, eNB shall modify information of cell indicated by *Old NR-CGI* IE according to the information in the *Served NR Cell Information* IE.

- If *Served NR Cells To Delete* IE is contained in the EN-DC CONFIGURATION UPDATE message, eNB shall delete information of cell indicated by *Old NR-CGI* IE.

The en-gNB may initiate a further EN-DC Configuration Update procedure only after a previous EN-DC Configuration Update procedure has been completed.

If the EN-DC CONFIGURATION UPDATE message contains the *TNL Transport Layer Address info* IE, the receiving eNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message contains the *TNL Transport Layer Address info* IE, the receiving en-gNB shall, if supported, take this into account for IPSEC tunnel establishment.

If the *NR Cell PRACH Configuration* IE is included in the *Served NR Cell Information* IE contained in the EN-DC CONFIGURATION UPDATE message, the eNB may update the information.

If the *CSI-RS Transmission Indication* IE is contained in the EN-DC CONFIGURATION UPDATE message, the eNB should take it into account when forwarding neighbour NR cell's CSI-RS configuration.

If the *Additional Measurement Timing Configuration List* IE is contained in the EN-DC CONFIGURATION UPDATE message, the eNB should take it into account when forwarding neighbour NR cell's CSI-RS configuration.

**Update of SCTP associations:**

If the *TNL Association to Add List* IE is included in the EN-DC CONFIGURATION UPDATE message, the receiving eNB shall, if supported, use it to establish the TNL association(s) with the en-gNB. The eNB shall report to the en-gNB, in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message, the successful establishment of the TNL association(s) with the en-gNB as follows:

- A list of successfully established TNL associations shall be included in the *TNL Association Setup List* IE;
- A list of TNL associations that failed to be established shall be included in the *TNL Association Failed to Setup List* IE.

If the *TNL Association to Remove List* IE is included in the EN-DC CONFIGURATION UPDATE message, the receiving eNB shall, if supported, initiate removal of the TNL association(s) indicated by the received Transport Layer information towards the en-gNB.

If the *TNL Association to Update List* IE is included in the EN-DC CONFIGURATION UPDATE message the receiving eNB shall, if supported, update the TNL association(s) indicated by the received Transport Layer information towards the en-gNB.

If the *Intended TDD DL-UL Configuration NR* IE is contained in the *NR Neighbour Information* IE in the EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message, en-gNB should take this information into account for cross-link interference management. The en-gNB shall consider the received *Intended TDD DL-UL Configuration NR* IE content valid until reception of an update of the IE for the same cell(s).

**Interaction with the eNB Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC CONFIGURATION UPDATE message to neighbouring eNBs by triggering the eNB Configuration Update procedure.

**Interaction with the EN-DC Configuration Update procedure:**

The receiving eNB may forward the *Intended TDD DL-UL Configuration NR* IE received in the *Served NR Cell Information* IE in the EN-DC CONFIGURATION UPDATE message to neighbouring en-gNBs by triggering the EN-DC Configuration Update procedure.

### 8.7.2.3 Unsuccessful Operation

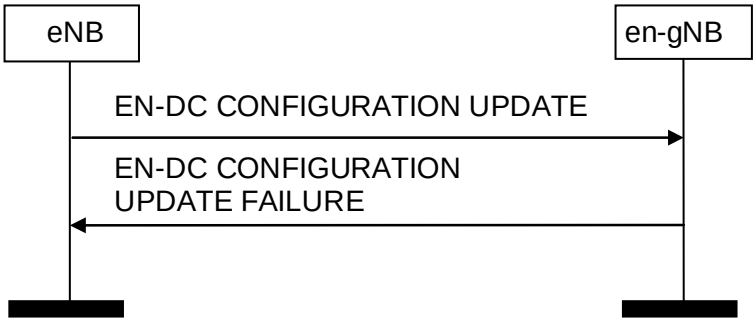



**Figure 8.7.2.3-1: eNB Initiated EN-DC Configuration Update, unsuccessful operation**

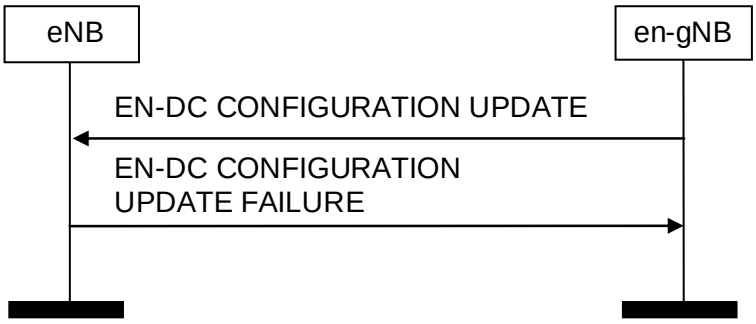



**Figure 8.7.2.3-2: en-gNB Initiated EN-DC Configuration Update, unsuccessful operation**

If the candidate receving node can not accept the update it shall respond with an EN-DC CONFIGURATION UPDATE FAILURE message and appropriate cause value.

If the EN-DC CONFIGURATION UPDATE FAILURE message includes the *Time To Wait* IE the initiating node shall wait at least for the indicated time before reinitiating the EN-DC Configuration Update procedure towards the same peer node. Both nodes shall continue to operate the X2 with their existing configuration data.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC CONFIGURATION UPDATE message and the EN-DC CONFIGURATION UPDATE FAILURE message shall include the *Interface Instance Indication* IE to identify the corresponding interface instance.

### 8.7.2.4 Abnormal Conditions

If the initiating node after initiating EN-DC Configuration Update procedure receives neither EN-DC CONFIGURATION UPDATE ACKNOWLEDGE message nor EN-DC CONFIGURATION UPDATE FAILURE message, the initiating node may reinitiate the EN-DC Configuration Update procedure towards the same candidate receving node, provided that the content of the EN-DC CONFIGURATION UPDATE message is identical to the content of the previously unacknowledged EN-DC CONFIGURATION UPDATE message.

## 8.7.3 EN-DC Cell Activation

### 8.7.3.1 General

The purpose of the EN-DC Cell Activation procedure is to enable an eNB to request a neighbouring en-gNB to switch on one or more cells, previously reported as inactive due to energy saving reasons.

The procedure uses non UE-associated signalling.

### 8.7.3.2 Successful Operation

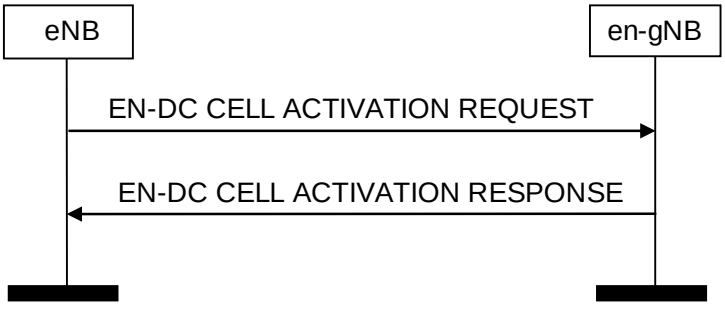



**Figure 8.7.3.2-1: EN-DC Cell Activation, successful operation**

An eNB initiates the procedure by sending a EN-DC CELL ACTIVATION REQUEST message to a peer en-gNB.

Upon receipt of this message, the en-gNB should activate the cell(s) indicated in the EN-DC CELL ACTIVATION REQUEST message and shall indicate in the EN-DC CELL ACTIVATION RESPONSE message for which cells the request was fulfilled.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC CELL ACTIVATION REQUEST message and the EN-DC CELL ACTIVATION RESPONSE message shall contain the *Interface Instance Indication* IE to identify the corresponding interface instance.

**Interactions with EN-DC Configuration Update procedure:**

The en-gNB shall not send an EN-DC CONFIGURATION UPDATE message to the eNB just for the reason of the cell(s) indicated in the EN-DC CELL ACTIVATION REQUEST message changing cell activation state, as the receipt of the EN-DC CELL ACTIVATION RESPONSE message by the eNB is used to update the information about the activation state of en-gNB cells in the eNB.

#### 8.7.3.3 Unsuccessful Operation

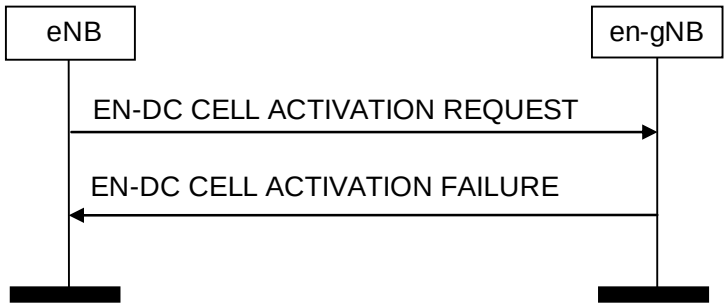



**Figure 8.7.3.3-1: EN-DC Cell Activation, unsuccessful operation**

If the en-gNB cannot activate any of the cells indicated in the EN-DC CELL ACTIVATION REQUEST message, it shall respond with a EN-DC CELL ACTIVATION FAILURE message with an appropriate cause value.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC CELL ACTIVATION REQUEST message and the EN-DC CELL ACTIVATION FAILURE message shall contain the *Interface Instance Indication* IE to identify the corresponding interface instance.

#### 8.7.3.4 Abnormal Conditions

Not applicable.

### 8.7.4 SgNB Addition Preparation

#### 8.7.4.1 General

The purpose of the SgNB Addition Preparation procedure is to request the en-gNB to allocate resources for EN-DC connectivity operation for a specific UE.

The procedure uses UE-associated signalling.

#### 8.7.4.2 Successful Operation

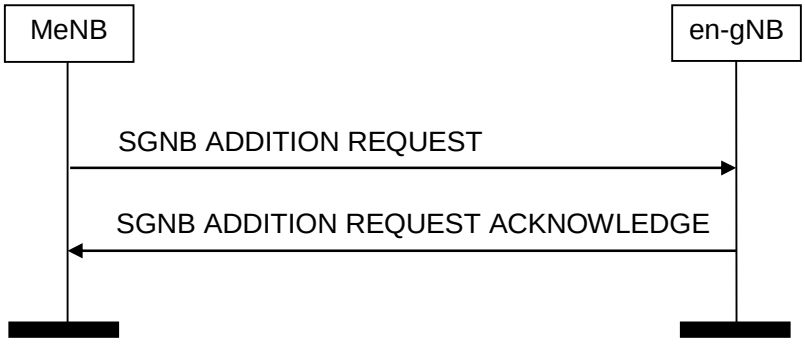



**Figure 8.7.4.2-1: SgNB Addition Preparation, successful operation**

The MeNB initiates the procedure by sending the SGNB ADDITION REQUEST message to the en-gNB. When the MeNB sends the SGNB ADDITION REQUEST message, it shall start the timer $T_{DCprep}$.

The allocation of resources according to the values of the *Allocation and Retention Priority* IE included in the *Full E-RAB Level QoS Parameters* IE or in the *Requested MCG E-RAB Level QoS Parameters IE* or in the *Requested SCG E-RAB Level QoS Parameters* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [4].

If the SGNB ADDITION REQUEST message contains the *Serving PLMN* IE, the en-gNB may use it for RRM purposes.

If the SGNB ADDITION REQUEST message contains the *Expected UE Behaviour* IE, the en-gNB shall, if supported, store this information and may use it to optimize resource allocation.

If the SGNB ADDITION REQUEST message contains the *Handover Restriction List* IE, the en-gNB node shall, if supported, store this information and use it to select an appropriate NR cell.

If the SGNB ADDITION REQUEST message contains the *MeNB Resource Coordination Information* IE, the en-gNB should forward it to lower layers and it may use it for the purpose of resource coordination with the MeNB, or to coordinate with sidelink resources used in the MeNB. The en-gNB shall consider the received *UL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. The en-gNB shall consider the received *DL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. If the *MeNB Coordination Assistance Information* IE is contained in the *MeNB Resource Coordination Information* IE, the en-gNB shall, if supported, use the information to determine further coordination of resource utilisation between the en-gNB and the MeNB.

The en-gNB shall choose the ciphering algorithm based on the information in the *NR UE Security Capabilities* IE and locally configured priority list of AS encryption algorithms and apply the key indicated in the *SgNB Security Key* IE as specified in the TS 33.401 [18].

If the SGNB ADDITION REQUEST message contains the *Subscriber Profile ID for RAT/Frequency Priority* IE, the en-gNB may use it for RRM purposes.

If the SGNB ADDITION REQUEST message contains the *Additional RRM Policy Index* IE, the en-gNB may use it for RRM purposes.

The en-gNB shall search for the target NR cell among the NR neighbour cells of the E-UTRAN cell indicated in *MeNB Cell ID* IE, as specified in the TS 37.340 [32].

If the *Masked IMEISV* IE is contained in the SGNB ADDITION REQUEST message the en-gNB shall, if supported, use it to determine the characteristics of the UE for subsequent handling.

The en-gNB shall report to the MeNB, in the SGNB ADDITION REQUEST ACKNOWLEDGE message, the result for all the requested E-RABs in the following way:

- a list of E-RABs which are successfully established shall be included in the *E-RABs Admitted To Be Added List* IE;
- a list of E-RABs which failed to be established shall be included in the *E-RABs Not Admitted List* IE.

NOTE: The MeNB may trigger the SgNB Addition Preparation procedure in the course of the Inter-MeNB handover without SgNB change procedure as described in TS 37.340 [32]. The deleted E-RABs are not included in the *E-RABs To Be Added List* IE in the SGNB ADDITION REQUEST message, from MeNB point of view. If the en-gNB reports a certain E-RAB to be successfully established, respective SCG resources, from an en-gNB point of view, may be actually successfully established or modified or kept; if a certain E-RAB is reported to be failed to be established, respective SCG resources, from an en-gNB point of view, may be actually failed to be established or modified or kept.

For each E-RAB successfully established in the en-gNB, the en-gNB shall report to the MeNB, in the SGNB ADDITION REQUEST ACKNOWLEDGE message, the same value in the *EN-DC Resource Configuration* IE as received in the SGNB ADDITION REQUEST message.

For each E-RAB for which allocation of the PDCP entity is requested at the en-gNB:

- the MeNB may propose to apply forwarding of downlink data by including the *DL Forwarding* IE within the *E-RABs To be Added Item* IE of the SGNB ADDITION REQUEST message. For each E-RAB that it has decided

to admit, the en-gNB may include the *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs Admitted To Be Added Item* IE of the SGNB ADDITION REQUEST ACKNOWLEDGE message to indicate that it accepts the proposed forwarding of downlink data for this bearer. This GTP tunnel endpoint may be different from the corresponding GTP tunnel endpoint, i.e the information contained in the *Transport Layer Address* IE and the *DL GTP TEID* IE in the *E-RAB To Be Modified List* IE of the E-RAB MODIFICATION INDICATION message (see TS 36.413 [4]) depending on implementation choice;

- the en-gNB may include for each bearer in the *E-RABs Admitted To Be Added List* IE the *UL Forwarding GTP Tunnel Endpoint* IE to indicate that it requests data forwarding of uplink packets to be performed for that bearer.

- the en-gNB shall use the *S1 UL GTP Tunnel Endpoint* IE of the SGNB ADDITION REQUEST message as the UL S1-U address.

- the MeNB shall use the *SgNB UL GTP Tunnel Endpoint at PDCP* IE of the SGNB ADDITION REQUEST ACKNOWLEDGE message as the UL X2-U address.

- if the SGNB ADDITION REQUEST message contains for an E-RAB to be added which is requested to be configured with MCG resources the *MeNB DL GTP Tunnel Endpoint at MCG* IE the en-gNB shall use it as the DL X2-U address for delivery of DL PDCP PDUs.

- the en-gNB shall include in the SGNB ADDITION REQUEST ACKNOWLEDGE message the *S1 DL GTP Tunnel Endpoint at the SgNB* IE.

- the en-gNB shall include in the SGNB ADDITION REQUEST ACKNOWLEDGE message the *RLC Mode* IE.

- the en-gNB may include for each bearer in the *E-RABs Admitted To Be Added List* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE the *PDCP SN Length* IE to indicate the PDCP SN length for that bearer.

- If the *RLC Mode* IE is included for an E-RAB within the *E-RABs To be Added List* IE in the SGNB ADDITION REQUEST message, it indicates the mode that the MeNB used for the E-RAB when it was hosted at the MeNB.

- If the *Bearer Type* IE for the concerned E-RAB is received by the en-gNB and is set to "non IP", the en-gNB shall, if supported, not perform IP header compression for the concerned E-RAB.

- If the *Ethernet Type* IE for the concerned E-RAB is received by the en-gNB and is set to "True", the en-gNB shall, if supported, take this into account to perform header compression appropriately for the concerned E-RAB.

Upon reception of the SGNB ADDITION REQUEST ACKNOWLEDGE message the MeNB shall stop the timer $T_{DCprep}$.

If the SGNB ADDITION ACKNOWLEDGE message contains the *SgNB Resource Coordination Information* IE, the MeNB may use it for the purpose of resource coordination with the en-gNB. The MeNB shall consider the received *UL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. The MeNB shall consider the received *DL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. If the *SgNB Coordination Assistance Information* IE is contained in the *SgNB Resource Coordination Information* IE, the MeNB shall, if supported, use the information to determine further coordination of resource utilisation between the en-gNB and the MeNB.

If the *SgNB UE X2AP ID* IE is contained in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, store this information and use it as defined in TS 37.340 [32].

If the SGNB ADDITION REQUEST message contains the *SGNB Addition Trigger Indication*, the en-gNB shall include the *RRC config indication* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message to inform the MeNB if the en-gNB applied full or delta configuration, as specified in TS 37.340 [32].

If the en-gNB receives for an E-RAB for which the PDCP entiy is allocated at the MeNB the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE and the *Duplication Activation* IE in the SGNB ADDITION REQUEST message, it may provide the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE and the *LCID* IE to the MeNB in the SGNB ADDITION REQUEST ACKNOWLEDGE message if PDCP duplication is configured at the en-gNB.

If the SGNB ADDITION REQUEST message contains the *UL PDCP SN Length* IE and the *DL PDCP SN Length* IE, the en-gNB shall, if supported, store this information and use it for lower layer configuration of the concerned MN terminated bearer.

The SgNB may include the *Location Information at SgNB* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message, if respective information is available at the SgNB.

If the *Location Information at SgNB Reporting* IE set to "pscell" is included in the SGNB ADDITION REQUEST, the SgNB shall start providing information about the current location of the UE. If the *Location Information at SgNB* IE is included in the SGNB ADDITION REQUEST ACKNOWLEDGE, the MeNB shall store the included information so that it may be transferred towards the MME.

If *Trace Activation* IE has previously been received for this UE, it shall be included in the SGNB ADDITION REQUEST message. If the *Trace Activation* IE is included in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, initiate the requested trace function as described in TS 32.422 [6]. If the *Trace Activation* IE includes the *MDT Configuration NR* IE, the en-gNB shall take it into account for MDT function as described in TS 37.320 [31].

If the *Management Based MDT Allowed* IE only or the *Management Based MDT Allowed* IE and the *Management Based MDT PLMN List* IE is contained in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, store the received information in the UE context, and use this information to allow subsequent selection of the UE for management based MDT defined in TS 32.422 [6].

The MeNB shall, if supported and available in the UE context, include the *Management Based MDT Allowed* IE and the *Management Based MDT PLMN List* IE in the SGNB ADDITION REQUEST message.

If the *UE Context Reference at Source NG-RAN* IE is contained in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, store this information and use it for UE context retrieval and allocate data forwarding resources as specified in TS 37.340 [32].

If the *Requested Fast MCG recovery via SRB3* IE set to "true" is included in the SGNB ADDITION REQUEST message and the en-gNB decides to configure fast MCG link recovery via SRB3 as specified in TS 37.340 [32], the en-gNB shall, if supported, include the *Available fast MCG recovery via SRB3* IE set to "true" in the SGNB ADDITION REQUEST ACKNOWLEDGE message.

If the *UE Radio Capability ID* IE is contained in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, store this information and use it as specified in TS 23.401 [12].

If the SGNB ADDITION REQUEST message contains the *IAB Node Indication* IE, the en-gNB shall, if supported, consider that the request is for an IAB node.

For each requested E-RAB configured as MN-terminated split bearer/SCG bearer, if the *QoS Mapping Information* IE is contained in the *GTP Tunnel Endpoint* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, use it to set DSCP and/or flow label fields for the downlink IP packets which are transmitted from MeNB to en-gNB through the GTP tunnels indicated by the *GTP Tunnel Endpoint* IE.

If the *Source NG-RAN Node ID* IE is included in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, use it to decide the direct data forwarding path availability with the indicated source NG-RAN node, and if the direct data forwarding path is available, include the *Direct Forwarding Path Availability* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message.

For each E-RAB for which the *Security Indication* IE is included in the *E-RABs To Be Added Item* IE of the SGNB ADDITION REQUEST message:

- if the *Integrity Protection Indication* IE is set to "required", the en-gNB shall, if supported, perform user plane integrity protection for the concerned E-RAB as specified in TS 33.401 [18], otherwise it shall reject the addition of the concerned E-RAB with an appropriate cause value.

- if the *Integrity Protection Indication* IE is set to "preferred", the en-gNB should perform user plane integrity protection for the concerned E-RAB as specified in TS 33.401 [18], and it shall notify the MeNB whether it performed the user plane integrity protection by including the *Integrity Protection result* IE in the *Security Result* IE of the SGNB ADDITION REQUEST ACKNOWLEDGE message.

- if the Integrity Protection Indication IE is set to "not needed", the en-gNB shall not perform user plane integrity protection for the concerned E-RAB.

If the *SCG Activation Request* IE is included in the SGNB ADDITION REQUEST message, the en-gNB may use it to configure SCG resources as specified in TS 37.340 [32], and shall, if supported, include the *SCG Activation Status* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message. If the *SCG Activation Request* IE in the SGNB ADDITION REQUEST message is set to "Activate SCG", the en-gNB shall, if supported, activate the SCG resources

and set the *SCG Activation Status* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message to "SCG activated".

If the *Source DL Forwarding IP Address* IE or both *Source DL Forwarding IP Address* IE and S*ource Node DL Forwarding IP Address* IE is included in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If the *Source DL Forwarding IP Address* IE is included in the SGNB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions to identify source TNL address for data forwarding in case of subsequent handover preparation, if such ACL functionality is deployed.

Upon reception of the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, start collecting the SCG information and continue for as long as the UE stays in one of its cells.

If the *UE History Information from the UE* IE is included in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, store this information.

If the *PScell Change History* IE set to “reporting full history” is included in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, signal the latest SCG UE History Information upon each PSCell change, to the MeNB, using the SgNB initiated SgNB Modification procedure.

If the *CHO Information SN Addition* IE is included in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, consider that the SgNB Addition Preparation procedure has been triggered as part of a conditional handover. It may use the *Source eNB ID* IE and the *Source eNB UE X2AP ID* IE to identify other active SgNB Addition Preparations related to this UE. If the *Estimated Arrival Probability* IE is contained in the *CHO Information SN Addition* IE included in the SGNB ADDITION REQUEST message, then the en-gNB may use the information to allocate necessary resources for the UE.

If the *Conditional PSCell Addition Information Request* IE is included in the SGNB ADDITION REQUEST message, the en-gNB shall, if supported, consider that the request concerns CPAC, as described in TS 37.340 [32]. Accordingly, the en-gNB shall, if supported, include the *Conditional PSCell Addition Information Acknowledge* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message.

If the *CG-CandidateList* is included in the *SgNB to MeNB Container* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, use it for the purpose of CPAC.

If the *Estimated Arrival Probability* IE is contained in the *Conditional PSCell Addition Information Request* IE included in the SGNB ADDITION REQUEST message, then the candidate target en-gNB may use the information to allocate necessary resources for the incoming CPAC procedure.

**Interactions with the MeNB initiated SgNB Modification procedure:**

If the en-gNB provides for an E-RAB for which the PDCP entiy is allocated at the MeNB the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE to the MeNB in the SGNB ADDITION REQUEST ACKNOWLEDGE message and the MeNB has not provided the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE in the SGNB ADDITION REQUEST message, the MeNB shall trigger the MeNB initiated SgNB Modification procedure to provide the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE to the SgNB.

**Interactions with the SgNB Reconfiguration Completion procedure:**

If the en-gNB admits at least one E-RAB, the en-gNB shall start the timer $T_{DCoverall}$ when sending the SGNB ADDITION REQUEST ACKNOWLEDGE message to the MeNB except for a request for conditional configuration. The reception of the SGNB RECONFIGURATION COMPLETE message shall stop the timer $T_{DCoverall}$ if $T_{DCoverall}$ is running.

**Interaction with the Activity Notification procedure**

Upon receiving an SGNB ADDITION REQUEST message containing the *Desired Activity Notification Level* IE, the en-gNB shall, if supported, use this information to decide whether to trigger subsequent SgNB Activitity Notification procedures according to the requested notification level.

### 8.7.4.3 Unsuccessful Operation

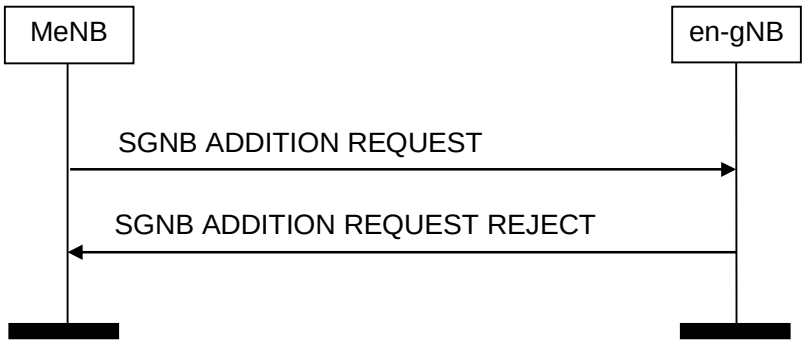



**Figure 8.7.4.3-1: SgNB Addition Preparation, unsuccessful operation**

If the en-gNB is not able to accept any of the bearers or a failure occurs during the SgNB Addition Preparation, the en-gNB sends the SGNB ADDITION REQUEST REJECT message with an appropriate cause value to the MeNB.

### 8.7.4.4 Abnormal Conditions

If the en-gNB receives a SGNB ADDITION REQUEST message containing multiple *E-RAB ID* IEs (in the *E-RABs To Be Added List* IE) set to the same value, the en-gNB shall consider the establishment of the corresponding E-RAB as failed.

If the en-gNB receives a SGNB ADDITION REQUEST message containing a *E-RAB Level QoS Parameters* IE which contains a *QCI* IE indicating a GBR bearer (as defined in TS 23.203 [13]), and which does not contain the *GBR QoS Information* IE, the en-gNB shall consider the establishment of the corresponding E-RAB as failed.

If the supported algorithms for encryption defined in the *NR Encryption Algorithms* IE in the *NR UE Security Capabilities* IE, plus the mandated support of NEA0 in all UEs (TS 33.401 [18]), do not match any algorithms defined in the configured list of allowed encryption algorithms in the en-gNB (TS 33.401 [18]), the en-gNB shall reject the procedure using the SGNB ADDITION REQUEST REJECT message.

If the supported algorithms for integrity defined in the *NR Integrity Protection Algorithms* IE in the *NR UE Security Capabilities* IE do not match any algorithms defined in the configured list of allowed integrity protection algorithms in the en-gNB (TS 33.401 [18]), the en-gNB shall reject the procedure using the SGNB ADDITION REQUEST REJECT message.

If the en-gNB receives a SGNB ADDITION REQUEST message containing a *SgNB UE X2AP ID* IE that does not match any existing UE Context that has such ID, the en-gNB shall reject the procedure using the SGNB ADDITION REQUEST REJECT message.

If the MeNB has provided the en-gNB for an E-RAB for which the PDCP entiy is allocated at the MeNB the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE and the *Duplication Activation* IE in the SGNB ADDITION REQUEST message, and the en-gNB does not provide the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE and the *LCID* IE to the MeNB in the SGNB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall assume that PDCP duplication was not configured at the en-gNB and releases duplication resources.

If the en-gNB provides for an E-RAB for which the PDCP entiy is allocated at the MeNB the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE and the *LCID* IE to the MeNB in the SGNB ADDITION REQUEST ACKNOWLEDGE message and the MeNB has not provided the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE and the *Duplication Activation* IE in the SGNB ADDITION REQUEST message, and the MeNB does not trigger the MeNB initiated SgNB Modification procedure to provide the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE and the *Duplication Activation* IE to the SgNB the en-gNB before the SgNB Reconfigurationi Completion procedure was triggered, the en-gNB shall trigger the release of the concerned E-RAB.

**Interactions with the SgNB Reconfiguration Completion and SgNB initiated SgNB Release procedure:**

If the timer $T_{DCoverall}$ expires before the en-gNB has received the SGNB RECONFIGURATION COMPLETE or the SGNB RELEASE REQUEST message, the en-gNB shall regard the requested RRC connection reconfiguration as being not applied by the UE and shall trigger the SgNB initiated SgNB Release procedure.

**Interactions with the MeNB initiated SgNB Release procedure:**

If the timer $T_{DCprep}$ expires before the MeNB has received the SGNB ADDITION REQUEST ACKNOWLEDGE message, the MeNB shall regard the SgNB Addition Preparation procedure as being failed and shall trigger the MeNB initiated SgNB Release procedure.

## 8.7.5 SgNB Reconfiguration Completion

### 8.7.5.1 General

The purpose of the SgNB Reconfiguration Completion procedure is to provide information to the en-gNB whether the requested configuration was successfully applied by the UE.

The procedure uses UE-associated signalling.

### 8.7.5.2 Successful Operation




**Figure 8.7.5.2-1: SgNB Reconfiguration Complete procedure, successful operation.**

The MeNB initiates the procedure by sending the SGNB RECONFIGURATION COMPLETE message to the en-gNB.

The SGNB RECONFIGURATION COMPLETE message may contain information that

- either the UE has successfully applied the configuration requested by the en-gNB. The MeNB may also provide NR *RRCReconfigurationComplete* message in the *MeNB to SgNB Container* IE.
- or the configuration requested by the en-gNB has been rejected. The MeNB shall provide information with sufficient precision in the included *Cause* IE to enable the en-gNB to know the reason for an unsuccessful reconfiguration.

Upon reception of the SGNB RECONFIGURATION COMPLETE message the en-gNB shall stop the timer $T_{DCoverall}$ if $T_{DCoverall}$ is running.

### 8.7.5.3 Abnormal Conditions

Void.

## 8.7.6 MeNB initiated SgNB Modification Preparation

### 8.7.6.1 General

This procedure is used to enable an MeNB to request an en-gNB to modify the UE context at the en-gNB, or to query the current SCG configuration for supporting delta signalling in MeNB initiated SgNB change, or to provide the S-RLF-related information to the en-gNB.

The procedure uses UE-associated signalling.

### 8.7.6.2 Successful Operation

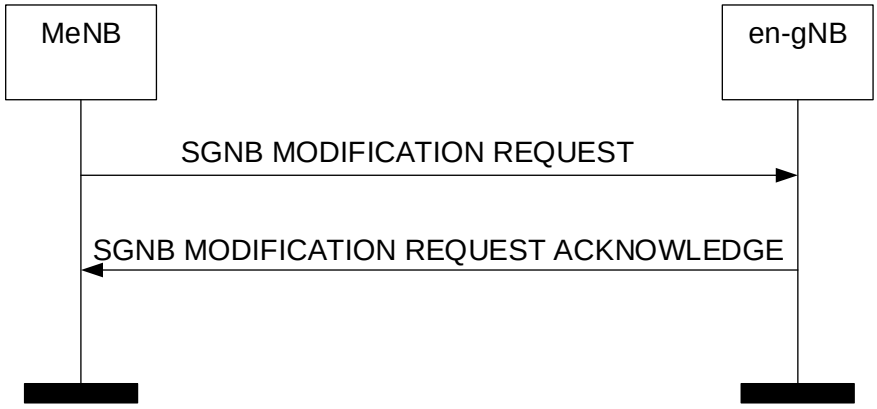



**Figure 8.7.6.2-1: MeNB initiated SgNB Modification Preparation, successful operation**

The MeNB initiates the procedure by sending the SGNB MODIFICATION REQUEST message to the en-gNB. When the MeNB sends the SGNB MODIFICATION REQUEST message, it shall start the timer $T_{DCprep}$.

The SGNB MODIFICATION REQUEST message may contain:

- within the *UE Context Information* IE (if the modification of the UE context at the en-gNB is requested);
  - E-RABs to be added within the *E-RABs To Be Added Item* IE;
  - E-RABs to be modified within the *E-RABs To Be Modified Item* IE;
  - E-RABs to be released within the *E-RABs To Be Released Item* IE;
  - the *SgNB UE Aggregate Maximum Bit Rate* IE;
- the *MeNB to SgNB Container* IE;
- the *SCG Configuration Query* IE;
- the *MeNB Resource Coordination Information* IE;
- the *Requested split SRBs IE*;
- the *Requested split SRBs release* IE;
- the *Requested fast MCG recovery via SRB3 IE*;
- the *Requested fast MCG recovery via SRB3 Release* IE.

If the SGNB MODIFICATION REQUEST message contains the *Serving PLMN* IE, the en-gNB may use it for RRM purposes.

If the SGNB MODIFICATION REQUEST message contains the *Handover Restriction List* IE, the en-gNB shall

- replace the previously provided Handover Restriction List by the received Handover Restriction List in the UE context;
- use this information to select an appropriate NR cell.

If the *SgNB UE Aggregate Maximum Bit Rate* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall:

- replace the previously provided SgNB UE Aggregate Maximum Bit Rate by the received SgNB UE Aggregate Maximum Bit Rate in the UE context;
- use the received SgNB UE Aggregate Maximum Bit Rate for non-GBR Bearers for the concerned UE as defined in TS 37.340 [32].

The allocation of resources according to the values of the *QCI* IE, *Allocation and Retention Priority* IE or *GBR QoS Information* IE included in the *Full E-RAB Level QoS Parameters* IE or in the *Requested SCG E-RAB Level QoS Parameters* IE shall follow the principles described for the E-RAB Setup procedure in TS 36.413 [4].

If the SGNB MODIFICATION REQUEST message contains the *MeNB Resource Coordination Information* IE, the en-gNB should forward it to lower layers and it may use it for the purpose of resource coordination with the MeNB, or to coordinate with sidelink resources used in the MeNB. The en-gNB shall consider the received *UL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. The en-gNB shall consider the received *DL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. If the *MeNB Coordination Assistance Information* IE is contained in the *MeNB Resource Coordination Information* IE, the en-gNB shall, if supported, use the information to determine further coordination of resource utilisation between the en-gNB and the MeNB.

If at least one of the requested modifications is admitted by the en-gNB, the en-gNB shall modify the related part of the UE context accordingly and send the SGNB MODIFICATION REQUEST ACKNOWLEDGE message back to the MeNB.

The en-gNB shall include the E-RABs for which resources have been either added or modified or released at the en-gNB either in the *E-RABs Admitted To Be Added List* IE or the *E-RABs Admitted To Be Modified List* IE or the *E-RABs Admitted To Be Released List* IE. The en-gNB shall include the E-RABs that have not been admitted in the *E-RABs Not Admitted List* IE with an appropriate cause value.

For each E-RAB successfully established or modified or released in the en-gNB, the en-gNB shall report to the MeNB, in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, the same value in the *EN-DC Resource Configuration* IE as received in the SGNB MODIFICATION REQUEST message.

The en-gNB shall, if included, choose the ciphering algorithm based on the information in the *NR UE Security Capabilities* IE and locally configured priority list of AS encryption algorithms and apply the key indicated in the *SgNB Security Key* IE as specified in the TS 33.401 [18].

For each E-RAB for which allocation of the PDCP entity is requested at the en-gNB:

- if applicable, the MeNB may propose to apply forwarding of downlink data by including the *DL Forwarding* IE within the *E-RABs To Be Added Item* IE of the SGNB MODIFICATION REQUEST message. For each E-RAB that it has decided to admit, the en-gNB may include the *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs Admitted To Be Added Item* IE of the SGNB MODIFICATION REQUEST ACKNOWLEDGE message to indicate that it accepts the proposed forwarding of downlink data for this bearer. The MeNB may also provide for an applicable E-RAB to be released the *DL Forwarding GTP Tunnel Endpoint* IE and the *UL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE of the SGNB MODIFICATION REQUEST message.

- if applicable, the en-gNB may include for each bearer in the *E-RABs Admitted To Be Added List* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message the *UL Forwarding GTP Tunnel Endpoint* IE to indicate that it requests data forwarding of uplink packets to be performed for that bearer.

- if applicable, the en-gNB may include for each bearer in the *E-RABs Admitted To Be Modified* List IE which is configured with the SN terminated split bearer option in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message the *UL Configuration* IE to indicate that the MCG UL configuration of the UE has changed.

- if applicable, the en-gNB may include for each bearer in the *E-RABs Admitted To Be Added List* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message the *UL PDCP SN Length* IE and the *DL PDCP SN Length* IE to indicate the PDCP SN length for that bearer.

- If the *Bearer Type* IE for the concerned E-RAB is received by the en-gNB and is set to"non IP", then the en-gNB shall, if supported, not perform IP header compression for the concerned E-RAB.

- If the *Ethernet Type* IE for the concerned E-RAB is received by the en-gNB and is set to "True", the en-gNB shall take this into account to perform header compression appropriately for the concerned E-RAB.

For each E-RAB configured with SCG resources and the PDCP entity is hosted by the MeNB and

- requested to be modified,

- if the SGNB MODIFICATION REQUEST message includes the *MeNB UL GTP Tunnel Endpoint at PDCP* IE in the *E-RABs To Be Modified Item* IE, the en-gNB shall act as specified in TS 37.340 [32].

- if the SGNB MODIFICATION REQUEST message contains the *MeNB UL GTP Tunnel Endpoint at PDCP* IE the en-gNB shall use it as the new UL X2-U address.

- the en-gNB may include in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message the *SgNB DL GTP Tunnel Endpoint at SCG* IE.

If, dependent on the configured bearer type, the *Full E-RAB Level QoS Parameters* IE or the *Maximum MCG admittable E-RAB Level QoS Parameters* IE or the *Requested SCG E-RAB level QoS Parameters* IE are included in the SGNB MODIFICATION REQUEST message for an E-RAB to be modified the en-gNB shall allocate respective resources and provide corresponding radio configuration information within the *SgNB to MeNB Container* IE as described in TS 37.340 [32].

If the SGNB MODIFICATION REQUEST message contains, for an E-RAB to be modified which is configured with the PDCP entity in the en-gNB, the *S1 UL GTP Tunnel Endpoint* IE, the en-gNB shall use it as the new UL S1-U address.

If the SGNB MODIFICATION REQUEST message contains an E-RAB to be modified which is configured with the MN terminated split bearer option, the MeNB may include the *UL Configuration* IE to indicate that the SCG UL configuration of the UE has changed.

If the SGNB MODIFICATION REQUEST message contains for an E-RAB to be modified which is configured with the PDCP enitiy in the en-gNB and MCG resources the *MeNB DL GTP Tunnel Endpoint at MCG* IE the en-gNB shall use it as the DL X2-U address.

If the SGNB MODIFICATION REQUEST message contains the *Subscriber Profile ID for RAT/Frequency Priority* IE, the en-gNB may use it for RRM purposes.

If the SGNB MODIFICATION REQUEST message contains the *Additional RRM Policy Index* IE, the en-gNB may use it for RRM purposes.

For an E-RAB to be modified which is configured with the PDCP entity in the en-gNB the en-gNB may include in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message the *S1 DL GTP Tunnel Endpoint at the SgNB* IE.

If the SGNB MODIFICATION REQUEST ACKNOWLEDGE message contains the *SgNB Resource Coordination Information* IE, the MeNB may use it for the purpose of resource coordination with the en-gNB. The MeNB shall consider the received *UL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. The MeNB shall consider the received *DL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. If the *SgNB Coordination Assistance Information* IE is contained in the *SgNB Resource Coordination Information* IE, the MeNB shall, if supported, use the information to determine further coordination of resource utilisation between the en-gNB and the MeNB.

Upon reception of the SGNB MODIFICATION REQUEST ACKNOWLEDGE message the MeNB shall stop the timer $T_{DCprep}$. If the SGNB MODIFICATION REQUEST ACKNOWLEDGE message has included the *SgNB to MeNB Container* IE the MeNB is then defined to have a Prepared SgNB Modification for that X2 UE-associated signalling.

If the *SCG Configuration Query* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall provide corresponding radio configuration information within the *SgNB to MeNB Container* IE as described in TS 37.340 [32].

If the SGNB MODIFICATION REQUEST message contains the *Requested split SRBs* IE, the en-gNB may use it to add split SRBs. If the SGNB MODIFICATION REQUEST message contains the *Requested split SRBs release* IE, the en-gNB may use it to release split SRBs.

If the *Requested Fast MCG recovery via SRB3* IE set to "true" is included in the SGNB MODIFICATION REQUEST message and the en-gNB decides to configure fast MCG link recovery via SRB3 as specified in TS 37.340 [32], the en-gNB shall, if supported, include the *Available fast MCG recovery via SRB3* IE set to "true" in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message. If the *Requested Fast MCG recovery via SRB3 Release* IE set to "true" is included in the SGNB MODIFICATION REQUEST message and the en-gNB decides to release fast MCG link recovery via SRB3, the en-gNB shall, if supported, include the *Release fast MCG recovery via SRB3* IE set to "true" in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message.

If the en-gNB receives for an E-RAB to be setup for which the PDCP entiy is allocated at the MeNB the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE and the *Duplication Activation* IE in the SGNB MODIFICATION REQUEST message, it may provide the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE and the *LCID* IE to the MeNB in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message if PDCP duplication is configured at the en-gNB.

If the SGNB MODIFICATION REQUEST message contains the *RLC Status* IE, the en-gNB shall assume that RLC has been reestablished at the MeNB and may trigger PDCP data recovery.

If the en-gNB applied a full configuration or delta configuration, e.g. as part of a mobility procedure involving a change of DU, the en-gNB shall inform the MeNB by including the *RRC config indication* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message.

If SGNB MODIFICATION REQUEST message contains the *UL PDCP SN Length* IE and the *DL PDCP SN Length* IE, the en-gNB shall, if supported, store this information and use it for lower layer configuration of the concerned MN terminated bearer.

If the *RLC Mode* IE is included for an E-RAB within the *E-RABs To be Added List* IE in the SGNB MODIFICATION REQUEST message, it indicates the mode that the MeNB used for the E-RAB when it was hosted at the MeNB.

If the SGNB MODIFICATION REQUEST message contains the *MeNB Cell ID* IE, the en-gNB may search for the target NR cell among the NR neighbour cells of the E-UTRAN cell indicated in *MeNB Cell ID* IE, as specified in the TS 37.340 [32].

If the SGNB MODIFICATION REQUEST ACKNOWLEDGE message contains the *RLC Status* IE, the MeNB shall assume that RLC has been reestablished at the en-gNB and may trigger PDCP data recovery.

The en-gNB may include the *Location Information at SgNB* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, if respective information is available at the en-gNB.

If the *Location Information at* en-gNB *Reporting* IE set to "pscell" is included in the SGNB MODIFICATION REQUEST, the SgNB shall start providing information about the current location of the UE. If the *Location Information at SgNB* IE is included in the SGNB MODIFICATION REQUEST ACKNOWLEDGE, the MeNB shall store the included information so that it may be transferred towards the MME.

If the *Lower Layer presence status change* IE set to "release lower layers" is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall act as specified in TS 37.340 [32].

If the *Lower Layer presence status change* IE set to "re-establish lower layers" is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall act as specified in TS 37.340 [32].

If the *Lower Layer presence status change* IE set to "suspend lower layers" is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall act as specified in TS 37.340 [32].

If the *Lower Layer presence status change* IE set to "resume lower layers" is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall act as specified in TS 37.340 [32].

If the SGNB MODIFICATION REQUEST message contains the *IAB Node Indication* IE, the en-gNB shall, if supported, consider that the request is for an IAB node.

For each E-RAB for which the *Security Indication* IE is included in the *E-RABs To Be Added Item* IE of the SGNB MODIFICATION REQUEST message:

- if the *Integrity Protection Indication* IE is set to "required", the en-gNB shall, if supported, perform user plane integrity protection for the concerned E-RAB as specified in TS 33.401 [18], and otherwise it shall reject the addition of the concerned E-RAB with an appropriate cause value.

- if the *Integrity Protection Indication* IE is set to "preferred", the en-gNB should perform user plane integrity protection for the concerned E-RAB as specified in TS 33.401 [18], and it shall notify the MeNB whether it performed the user plane integrity protection by including the *Integrity Protection result* IE in the *Security Result* IE of the SGNB MODIFICATION REQUEST ACKNOWLEDGE message.

- if the Integrity Protection Indication IE is set to "not needed", the en-gNB shall not perform user plane integrity protection for the concerned E-RAB.

For each requested E-RAB configured as MN-terminated split bearer/SCG bearer, if the *QoS Mapping Information* IE is contained in the *GTP Tunnel Endpoint* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, use it to set DSCP and/or flow label fields for the downlink IP packets which are transmitted from MeNB to SgNB through the GTP tunnels indicated by the *GTP Tunnel Endpoint* IE.

If the *PSCell History Information Retrieve* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall, if supported, use this information as specified in TS 37.340 [32].

If the *UE History Information from the UE* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall, if supported, store this information.

If the *CHO Information SN Modification* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall, if supported, consider that the MeNB initiated SgNB Modification Preparation procedure has been triggered as part of a conditional handover. If the *Estimated Arrival Probability* IE is contained in the *CHO Information SN Modification* IE included in the SGNB MODIFICATION REQUEST message, then the en-gNB may use the information to allocate necessary resources for the UE.

If the *SCG Activation Request* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB may use it to configure SCG resources as specified in TS 37.340 [32], and shall, if supported, include the *SCG Activation Status* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message.

If the *Conditional PSCell Change Information Update* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall, if supported, consider that the request provides the list of PSCells prepared at the target en-gNB, as described in TS 37.340 [32].

If the *Conditional PSCell Addition Information Modification Request* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall, if supported, consider that the request concerns an update of the previous CPAC preparation, as described in TS 37.340 [32]. Accordingly, the en-gNB shall, if supported, include the *Conditional PSCell Addition Information Modification Acknowledge* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message.

If the *CG-CandidateList* is included in the *SgNB to MeNB Container* IE in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, use it for the purpose of CPAC.

If the *Estimated Arrival Probability* IE is contained in the *Conditional PSCell Information Modification Request* IE included in the SGNB MODIFICATION REQUEST message, then the candidate target en-gNB node may use the information to allocate necessary resources for the incoming CPAC procedure.

If the *Source DL Forwarding IP Address* IE is included in the SGNB MODIFICATION REQUEST message, the en-gNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

If the *Source DL Forwarding IP Address* IE is included in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall, if supported, store this information and use it as part of its ACL functionality configuration actions to identify source TNL address for data forwarding in case of subsequent handover preparation, if such ACL functionality is deployed.

**Interactions with the MeNB initiated SgNB Modification procedure:**

If the en-gNB provides for an E-RAB to be setup for which the PDCP entiy is allocated at the MeNB the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE and the *LCID* IE to the MeNB in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message and the MeNB has not provided the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE and the *Duplication Activation* IE in the SGNB MODIFICATION REQUEST message, the MeNB shall trigger the MeNB initiated SgNB Modification procedure to provide the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE and the *Duplication Activation* IE to the SgNB.

**Interactions with the SgNB Reconfiguration Completion procedure:**

If the en-gNB admits a modification of the UE context requiring the MeNB to report about the success of the RRC connection reconfiguration procedure, the en-gNB shall start the timer $T_{DCoverall}$ when sending the SGNB MODIFICATION REQUEST ACKNOWLEDGE message to the MeNB except for a request for conditional configuration. The reception of the SGNB RECONFIGURATION COMPLETE message shall stop the timer $T_{DCoverall}$ if $T_{DCoverall}$ is running.

**Interaction with the Activity Notification procedure**

Upon receiving an SGNB MODIFICATION REQUEST message containing the *Desired Activity Notification Level* IE, the en-gNB shall, if supported, use this information to decide whether to trigger subsequent SgNB Activity Notification procedures, or stop or modify ongoing triggering of these procedures due to a previous request.

**Interaction with the SgNB initiated SgNB Modification Preparation procedure:**

If the MeNB receives the SGNB MODIFICATION REQUIRED message and the requested SN modification procedure needs further information from MeNB, the MeNB shall send SGNB MODIFICATION REQUEST message to en-gNB in response to a previously SgNB initiated SgNB Modification procedure.

### 8.7.6.3 Unsuccessful Operation

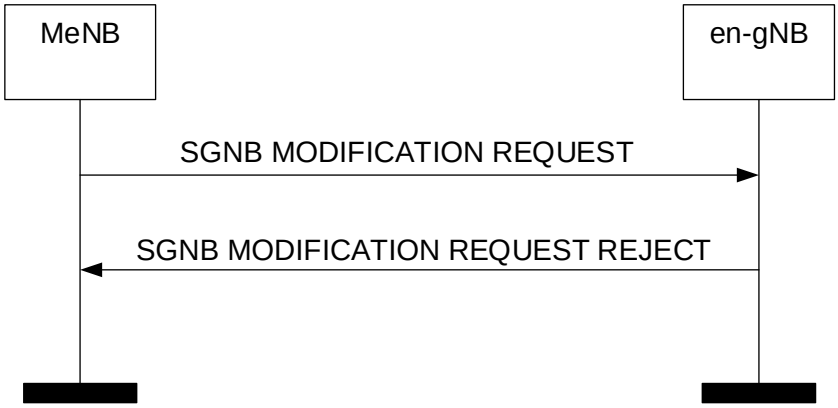



**Figure 8.7.6.3-1: MeNB initiated SgNB Modification Preparation, unsuccessful operation**

If the en-gNB does not admit any modification requested by the MeNB, or a failure occurs during the MeNB initiated SgNB Modfication Preparation, the en-gNB shall send the SGNB MODIFICATION REQUEST REJECT message to the MeNB. The message shall contain the *Cause* IE with an appropriate value.

If the en-gNB receives a SGNB MODIFICATION REQUEST message containing the *MeNB to SgNB Container* IE that does not include required information as specified in TS 38.331 [31], the en-gNB shall send the SGNB MODIFICATION REQUEST REJECT message to the MeNB.

### 8.7.6.4 Abnormal Conditions

If the en-gNB receives a SGNB MODIFICATION REQUEST message containing multiple *E-RAB ID* IEs (in the *E-RABs To Be Added List* IE and/or the *E-RABs To Be Modified List* IE) set to the same value, the en-gNB shall not admit the action requested for the corresponding E-RABs.

If the en-gNB receives an SGNB MODIFICATION REQUEST message containing multiple *E-RAB ID* IEs (in the *E-RAB To Be Released List* IE) set to the same value, the en-gNB shall initiate the release of one corresponding E-RAB and ignore the duplication of the instances of the selected corresponding E-RABs.

If the en-gNB receives a SGNB MODIFICATION REQUEST message containing, dependent on the configured bearer type, the *Full E-RAB Level QoS Parameters* IE or the *Requested SCG E-RAB Level QoS Parameters* IE which contains a *QCI* IE indicating a GBR bearer (as defined in TS 23.203 [13]), and which does not contain the *GBR QoS Information* IE, the en-gNB shall not admit the corresponding E-RAB.

If the supported algorithms for encryption defined in the *NR Encryption Algorithms* IE in the *NR UE Security Capabilities* IE in the *UE Context Information* IE, plus the mandated support of NEA0 in all UEs (TS 33.401 [18]), do not match any algorithms defined in the configured list of allowed encryption algorithms in the en-gNB (TS 33.401 [18]), the en-gNB shall reject the procedure using the SGNB MODIFICATION REQUEST REJECT message.

If the supported algorithms for integrity defined in the *NR Integrity Protection Algorithms* IE in the *NR UE Security Capabilities* IE in the *UE Context Information* IE do not match any algorithms defined in the configured list of allowed integrity protection algorithms in the en-gNB (TS 33.401 [18]), the en-gNB shall reject the procedure using the SGNB MODIFICATION REQUEST REJECT message.

If the timer $T_{DCprep}$ expires before the MeNB has received the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall regard the MeNB initiated SgNB Modification Preparation procedure as being failed and shall release the UE Context at the en-gNB.

If the MeNB has provided the en-gNB for an E-RAB to be setupr which the PDCP entiy is allocated at the MeNB the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE in the SGNB MODIFICATION REQUEST message, and the en-gNB does not provide the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE to the MeNB in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall assume that PDCP duplication was not configured at the en-gNB and releases duplication resources.

If the en-gNB provides for an E-RAB to be setup for which the PDCP entiy is allocated at the MeNB the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE to the MeNB in the SGNB MODIFICATION REQUEST ACKNOWLEDGE message and the MeNB has not provided the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE in the SGNB MODIFICATION REQUEST message, and the MeNB does not trigger the MeNB initiated SgNB Modification procedure to provide the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE to the SgNB the en-gNB before the SgNB Reconfigurationi Completion procedure was triggered, the en-gNB shall trigger the release of the concerned E-RAB.

**Interactions with the SgNB Reconfiguration Completion and SgNB initiated SgNB Release procedure:**

If the timer $T_{DCoverall}$ expires before the en-gNB has received the SGNB RECONFIGURATION COMPLETE or the SGNB RELEASE REQUEST message, the en-gNB shall regard the requested modification RRC connection reconfiguration as being not applied by the UE and shall trigger the SgNB initiated SgNB Release procedure.

**Interaction with the SgNB initiated SgNB Modification Preparation procedure:**

If the MeNB, after having initiated the MeNB initiated SgNB Modification procedure, receives the SGNB MODIFICATION REQUIRED message, the MeNB shall refuse the SgNB initiated SgNB Modification procedure with an appropriate cause value in the *Cause* IE.

If the MeNB has a Prepared SgNB Modification and receives the SGNB MODIFICATION REQUIRED message, the MeNB shall respond with the SGNB MODIFICATION REFUSE message to the en-gNB with an appropriate cause value in the *Cause* IE.

**Interactions with the MeNB initiated SgNB Release procedure:**

If the timer $T_{DCprep}$ expires before the MeNB has received the SGNB MODIFICATION REQUEST ACKNOWLEDGE message, the MeNB shall regard the SgNB Modification Preparation procedure as being failed and may trigger the MeNB initiated SgNB Release procedure.

## 8.7.7 SgNB initiated SgNB Modification

### 8.7.7.1 General

This procedure is used by the en-gNB to modify the UE context in the en-gNB.

The procedure uses UE-associated signalling.

### 8.7.7.2 Successful Operation

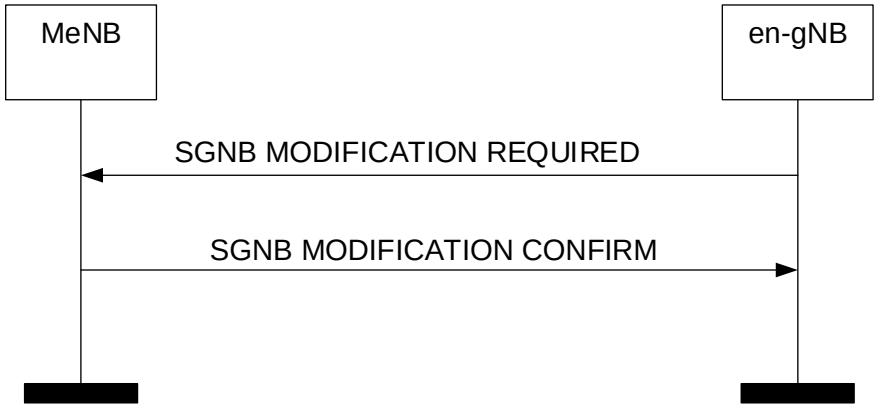



**Figure 8.7.7.2-1: SgNB initiated SgNB Modification, successful operation.**

The en-gNB initiates the procedure by sending the SGNB MODIFICATION REQUIRED message to the MeNB. When the en-gNB sends the SGNB MODIFICATION REQUIRED message, it shall start the timer $T_{DCoverall}$.

The SGNB MODIFICATION REQUIRED message may contain

- the *PDCP Change Indication* IE;
- the *SgNB to MeNB Container* IE.
- E-RABs to be modified within the *E-RABs To Be Modified Item* IE;
- E-RABs to be released within the *E-RABs To Be Released Item* IE;
- the *SgNB Resource Coordination Information* IE.

For the SN terminated split bearers, the en-gNB may include in the SGNB MODIFICATION REQUIRED message the *UL Configuration* IE to indicate that the MCG UL configuration of the UE has changed.

The en-gNB may include for each bearer in the *E-RABs to Be Modified List* IE in the SGNB MODIFICATION REQUIRED message the *New DRB ID Request* IE to request the MeNB to assign a new DRB ID for that bearer.

If the MeNB is able to perform the change requested by the en-gNB, the MeNB shall send the SGNB MODIFICATION CONFIRM message to the en-gNB. The SGNB MODIFICATION CONFIRM message may contain the *MeNB to SgNB Container* IE.

If the SGNB MODIFICATION REQUIRED message contains the *SgNB Resource Coordination Information* IE, the MeNB may use it for the purpose of resource coordination with the en-gNB. The MeNB shall consider the received *UL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. The MeNB shall consider the received *DL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. If the *SgNB Coordination Assistance Information* IE is contained in the *SgNB Resource Coordination Information* IE, the MeNB shall, if supported, use the information to determine further coordination of resource utilisation between the en-gNB and the MeNB.

If the en-gNB applied a full configuration or delta configuration, e.g. as part of a mobility procedure involving a change of DU, the en-gNB shall inform the MeNB by including the *RRC config indication* IE in the SGNB MODIFICATION REQUIRED message.

For each E-RAB successfully modified as requested by the en-gNB, the MeNB shall inform the en-gNB, in the SGNB MODIFICATION CONFIRM message, the same value in the *EN-DC Resource Configuration* IE as received in the SGNB MODIFICATION REQUIRED message.

If the *SCG resources* IE in the *EN-DC Resource Configuration* IE in the SGNB MODIFICATION REQUIRED message for all the E-RABs of the UE are set to “not present”, the MeNB shall, if supported, deduce that the SCG resources are removed.

Upon reception of the SGNB MODIFICATION CONFIRM message the en-gNB shall stop the timer $T_{DCoverall}$.

If the SGNB MODIFICATION CONFIRM message contains the *MeNB Resource Coordination Information* IE, the en-gNB should forward it to lower layers and it may use it for the purpose of resource coordination with the MeNB, or to coordinate with sidelink resources used in the MeNB. The en-gNB shall consider the received *UL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. The en-gNB shall consider the received *DL Coordination Information* IE value valid until reception of a new update of the IE for the same UE. If the *MeNB Coordination Assistance Information* IE is contained in the *MeNB Resource Coordination Information* IE, the en-gNB shall, if supported, use the information to determine further coordination of resource utilisation between the en-gNB and the MeNB.

If the MeNB receives for an E-RAB for which the PDCP entiy is allocated at the MeNB the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE in the SGNB MODIFICATION REQUIRED message, it shall provide the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE to the en-gNB in the SGNB MODIFICATION CONFIRM message. If the *LCID* IE is included in the SGNB MODIFICATION REQUIRED message, the MeNB should take it into account.

If the SGNB MODIFICATION REQUIRED message contains the *RLC Status* IE, the MeNB shall assume that RLC has been reestablished at the en-gNB and may trigger PDCP data recovery.

If the *RLC Mode* IE is included for an E-RAB within the *E-RABs To Be Released List* IE (for E-RABs hosted at the en-gNB) in the SGNB MODIFICATION REQUIRED message, it indicates the mode that the en-gNB used for the E-RAB when it was hosted at the en-gNB.

The MeNB shall include only E-RABs with the following IE in *E-RABs Admitted To Be Modified List* IE:

- the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE.

If the *Location Information at SgNB* IE is included in the SGNB MODIFICATION REQUIRED, the MeNB shall store the included information so that it may be transferred towards the MME.

For each requested E-RAB configured as MN-terminated split bearer/SCG bearer, if the *QoS Mapping Information* IE is included in the *GTP Tunnel Endpoint* IE in the SGNB MODIFICATION REQUIRED message, the MeNB shall, if supported, use it to set DSCP and/or flow label fields for the downlink IP packets which are transmitted from MeNB to SgNB through the GTP tunnels indicated by the *GTP Tunnel Endpoint* IE.

If the SGNB MODIFICATION REQUIRED message includes the *SCG UE History Information* IE, the MeNB shall, if supported, use the information to update the UE History Information with PSCell history.

If the *SCG Activation Request* IE is included in the SGNB MODIFICATION REQUIRED message, the MeNB shall, if supported, consider that the en-gNB node is about to reconfigure the SCG resources as specified in TS 37.340 [32].

If the *CPAC Information Required* IE is included in the SGNB MODIFICATION REQUIRED message, the MeNB shall, if supported, consider that the request provides the configuration update for the list of PSCells prepared at the target en-gNB, as described in TS 37.340 [32].

If the *CG-CandidateList* is included in the *SgNB to MeNB Container* IE in the SGNB MODIFICATION REQUIRED message, the MeNB shall, if supported, use it for the purpose of CPAC.

If the *SCG Reconfiguration Notification* IE is included in the SGNB MODIFICATION REQUIRED message, the MeNB shall, if supported, consider the request is sent to coordinate CHO or MN-initiated CPC with SCG reconfigurations:

- If the *SCG Reconfiguration Notification* IE is set to "executed", the MeNB shall, if supported, consider that a reconfiguration of the SCG resources using SRB3 has been executed. If the *SgNB to MeNB Container* IE is also included in the SGNB MODIFICATION REQUIRED message, the MeNB shall, if supported, consider that the received SCG configuration has already been applied in the UE and should not be forwarded to the UE.

- If the *SCG Reconfiguration Notification* IE is set to "executed-deleted", the MeNB shall, if supported, consider that a reconfiguration with sync of the SCG resources has been executed and earlier CHO or MN-initiated CPC configuration has been deleted in the UE. If the *SgNB to MeNB Container* IE is also included in the SGNB MODIFICATION REQUIRED message, the MeNB shall, if supported, consider that the received SCG configuration has already been applied in the UE and should not be forwarded to the UE.

- If the *SCG Reconfiguration Notification* IE is set to "deleted", the MeNB shall, if supported, consider that an earlier CHO or MN-initiated CPC configuration will be deleted in the UE when the SCG configuration provided in the *SgNB to MeNB Container* IE is delivered to the UE and executed.

**Interaction with the MeNB initiated SgNB Modification Preparation procedure:**

If applicable, as specified in TS 37.340 [32], the en-gNB may receive, after having initiated the SgNB initiated SgNB Modification procedure, the SGNB MODIFICATION REQUEST message including the *DL Forwarding GTP Tunnel Endpoint* IE and the *UL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released List* IE.

If applicable, as specified in TS 37.340 [32], the en-gNB may receive, after having initiated the SgNB initiated SgNB Modification procedure, the SGNB MODIFICATION REQUEST message including the *SgNB Security Key* IE within the *UE Context Information* IE.

If applicable, as specified in TS 37.340 [32], the en-gNB may receive, after having initiated the SgNB initiated SgNB Modification procedure, the SGNB MODIFICATION REQUEST message including the *measGapConfig* contained in the *CG-ConfigInfo* message as defined in TS 38.331 [31] within the *MeNB to SgNB Container* IE.

The en-gNB may receive, after having initiated the SgNB initiated SgNB modification procedure including the *New DRB ID Request* IE for an SN terminated bearer within the *E-RABs To Be Modified List* IE, the SGNB MODIFICATION REQUEST message to release and add the same bearer with a new DRB ID or with the same DRB ID but together with the *SgNB Security Key* IE within the *UE Context Information* IE.

The en-gNB may receive, after having initiated the SgNB initiated SgNB modification procedure, the SGNB MODIFICATION REQUEST message including the *SN triggered* IE.

### 8.7.7.3 Unsuccessful Operation

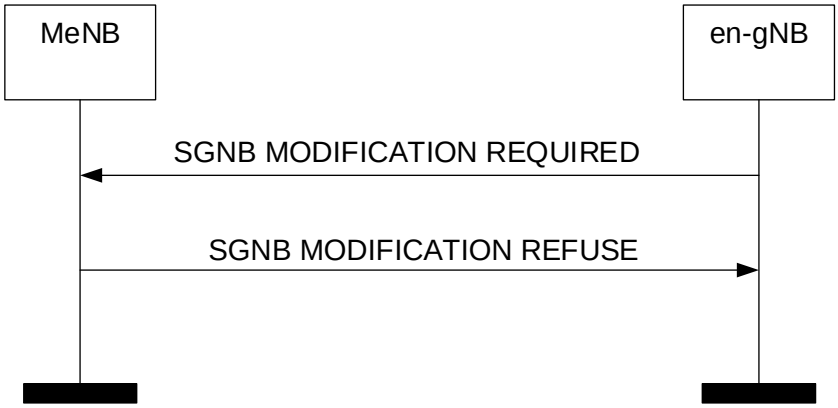



**Figure 8.7.7.3-1: SgNB initiated SgNB Modification, unsuccessful operation.**

In case the requested modification cannot be performed successfully the MeNB shall respond with the SGNB MODIFICATION REFUSE message to the en-gNB with an appropriate cause value in the *Cause* IE.

The MeNB may also provide configuration information in the *MeNB to SgNB Container* IE.

### 8.7.7.4 Abnormal Conditions

If the timer $T_{DCoverall}$ expires before the en-gNB has received the SGNB MODIFICATION CONFIRM or the SGNB MODIFICATION REFUSE message, the en-gNB shall regard the requested modification as failed and may take further actions like triggering the SgNB initiated SgNB Release procedure to release all en-gNB resources allocated for the UE.

If the value received in the *E-RAB ID* IE of any of the *E-RABs To Be Released Items* IE is not known at the MeNB, the MeNB shall regard the procedure as failed and may take appropriate actions like triggering the MeNB initiated SgNB Release procedure.

If the en-gNB does not receives for an E-RAB for which the PDCP entiy is allocated at the MeNB the *Secondary MeNB UL GTP Tunnel Endpoint at PDCP* IE to the en-gNB in the SGNB MODIFICATION CONFIRM message although the *Secondary SgNB DL GTP Tunnel Endpoint at SCG* IE was provided to the MeNB in the SGNB MODIFICATION REQUIRED message, it shall assume the setup of the secondary X2-U bearer as being failed.

**Interaction with the MeNB initiated SgNB Modification Preparation procedure:**

If the en-gNB, after having initiated the SgNB initiated SgNB Modification procedure, receives the SGNB MODIFICATION REQUEST message including other IEs than an applicable *SgNB Security Key* IE and/or applicable forwarding addresses or applicable measurement gap pattern or information applicable to release and add the same bearer with different DRB ID and/or the *SN triggered* IE set to "True", the en-gNB shall

- regard the SgNB initiated SgNB Modification Procedure as being failed;
- stop the $T_{DCoverall}$, which was started to supervise the SgNB initiated SgNB Modification procedure;
- be prepared to receive the SGNB MODIFICATION REFUSE message from the MeNB and;
- continue with the MeNB initiated SgNB Modification Preparation procedure as specified in section 8.7.6.

**Interaction with the MeNB initiated handover procedure:**

If the MeNB, after having initiated the handover procedure, receives the SGNB MODIFICATION REQUIRED message, the MeNB shall refuse the SgNB modification procedure with an appropriate cause value in the *Cause* IE.

## 8.7.8 SgNB Change

### 8.7.8.1 General

This procedure is used by the en-gNB to change to another en-gNB.

The procedure uses UE-associated signalling.

### 8.7.8.2 Successful Operation

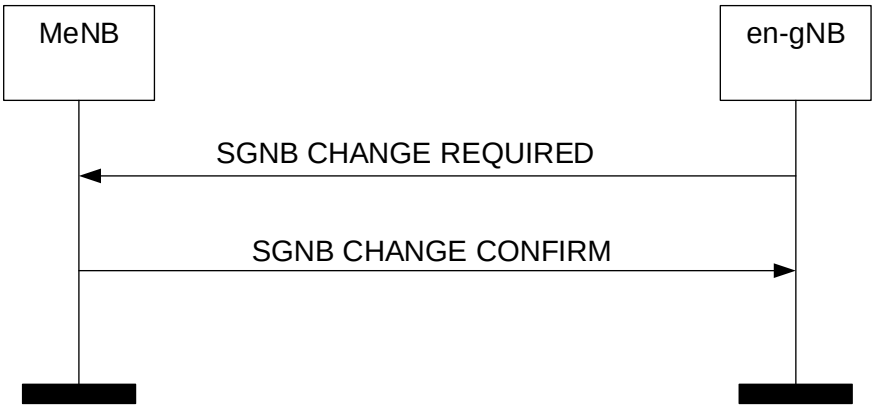



**Figure 8.7.8.2-1: SgNB Change, successful operation.**

The en-gNB initiates the procedure by sending the SGNB CHANGE REQUIRED message to the MeNB including the *Target SgNB ID Information IE*. When the en-gNB sends the SGNB CHANGE REQUIRED message, it shall start the timer $T_{DCoverall}$.

The SGNB CHANGE REQUIRED message may contain

- the *SgNB to MeNB Container* IE.

If the MeNB is able to perform the change requested by the en-gNB, the MeNB shall send the SGNB CHANGE CONFIRM message to the en-gNB. For each E-RAB configured with the PDCP entity in the en-gNB, the MeNB may include the *DL Forwarding GTP Tunnel Endpoint* IE and the *UL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE to indicate that it requests data forwarding of uplink and downlink packets to be performed for that bearer.

The en-gNB may start data forwarding and stop providing user data to the UE and shall stop the timer $T_{DCoverall}$ upon reception of the SGNB CHANGE CONFIRM message.

If the SGNB CHANGE REQUIRED message includes the *SCG UE History Information* IE, the MeNB shall, if supported, use the information to update UE History Information with PSCell history.

If the *Conditional PSCell Change Information Required* IE is included in the SGNB CHANGE REQUIRED message, the MeNB shall, if supported, consider that the requirement concerns CPAC, as described in TS 37.340 [32]. The *SgNB to MeNB Container* IE within the *Conditional PSCell Change Information Required* IE contains at least the suggested PSCell list for each candidate target SgNB. Accordingly, the MeNB may include the *Conditional PSCell Change Information Confirm* IE in the SGNB CHANGE CONFIRM message.

If the *Estimated Arrival Probability* IE is contained in the *Conditional PSCell Change Information Required* IE included in the SGNB CHANGE REQUIRED message, the MeNB shall, if supported, forward this information to the candidate target en-gNB, then the candidate target en-gNB may use the information to allocate necessary resources for the incoming CPAC procedure.

If the *Multiple Target SgNB Node List* IE is included in the SGNB CHANGE REQUIRED message, if multiple target en-gNBs are prepared, the MeNB may include the *Additional List of Forwarding GTP Tunnel Endpoint* IE in the SGNB CHANGE CONFIRM message to provide different data forwarding addresses for different target en-gNBs.

**Interaction with MeNB initiated SgNB Release:**

If the MeNB receives the SGNB CHANGE REQUIRED message releasing target en-gNB and cancelling all the prepared PSCells in the target en-gNB(s), the MeNB shall, if supported, trigger the MeNB initiated en-gNB release procedure to the target en-gNB(s) and cancel all the prepared PSCells at the target en-gNB(s).

#### 8.7.8.3 Unsuccessful Operation




**Figure 8.7.8.3-1: SgNB Change, unsuccessful operation.**

In case the request change cannot be performed successfully the MeNB shall respond with the SGNB CHANGE REFUSE message to the en-gNB with an appropriate cause value in the *Cause* IE.

#### 8.7.8.4 Abnormal Conditions

If the timer $T_{DCoverall}$ expires before the en-gNB has received the SGNB CHANGE CONFIRM or the SGNB CHANGE REFUSE message, the en-gNB shall regard the requested change as failed and may take further actions like triggering the SgNB initiated SgNB Release procedure to release all en-gNB resources allocated for the UE.

**Interaction with the MeNB initiated handover procedure:**

If the MeNB, after having initiated the handover procedure, receives the SGNB CHANGE REQUIRED message, the MeNB shall refuse the SgNB change procedure with an appropriate cause value in the Cause IE.

### 8.7.9 MeNB initiated SgNB Release

#### 8.7.9.1 General

The MeNB initiated SgNB Release procedure is triggered by the MeNB to initiate the release of the resources for a specific UE.

The procedure uses UE-associated signalling.

### 8.7.9.2 Successful Operation

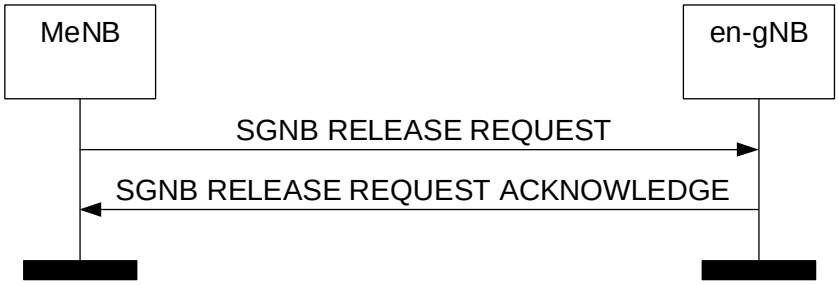



**Figure 8.7.9.2-1: MeNB initiated SgNB Release, successful operation**

The MeNB initiates the procedure by sending the SGNB RELEASE REQUEST message. Upon reception of the SGNB RELEASE REQUEST message the en-gNB shall stop providing user data to the UE. The *SgNB UE X2AP ID* IE shall be included if it has been obtained from the en-gNB.

If the bearer context in the en-gNB was configured with the PDCP entity in the en-gNB, for E-RAB for which the MeNB requests forwarding of uplink/downlink data, the MeNB includes the *UL Forwarding GTP Tunnel Endpoint*/ *DL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE of the SGNB RELEASE REQUEST message to indicate that the en-gNB should perform data forwarding of uplink/downlink packets for that E-RAB.

Upon reception of the SGNB RELEASE REQUEST message containing *UE Context Kept Indicator* IE set to "True", the en-gNB shall, if supported, only initiate the release of the resources related to the UE-associated signalling connection between the MeNB and the en-gNB.

If the en-gNB confirms the request to release en-gNB resources it shall send the SGNB RELEASE REQUEST ACKNOWLEDGE message to the MeNB.

If the *RLC Mode* IE is included for an E-RAB within the *E-RABs Admitted To Be Released List* IE (for E-RABs hosted at the en-gNB) in the SGNB RELEASE REQUEST ACKNOWLEDGE message, it indicates the mode that the en-gNB used for the E-RAB when it was hosted at the en-gNB.

If the MeNB did not include the *SgNB UE X2AP ID* IE in the SGNB RELEASE REQUEST message, the MeNB shall ignore the *SgNB UE X2AP ID* IE in the SGNB RELEASE REQUEST ACKNOWLEDGE message.

Upon successful completion of the procedure, the MeNB shall start counting time, so that information regarding time since Secondary Node Release may be transferred towards the MME as specified in TS 36.413 [4].

If the SGNB RELEASE REQUEST ACKNOWLEDGE message includes the *SCG UE History Information* IE, the MeNB shall, if supported, use the information to update UE History Information with PSCell history.

**Interaction with SN Status Transfer procedure:**

If the *UE Context Kept Indicator* IE set to "True" and the *E-RABs transferred to MeNB* IE are included in the SGNB RELEASE REQUEST message, then the en-gNB shall, if supported, include the uplink/downlink PDCP SN and HFN status for the listed E-RABs, as specified in TS 37.340 [32].

**Interaction with Data Forwarding Address Indication procedure:**

If the SGNB RELEASE REQUEST message concerns UE for which a CPC has been triggered, the en-gNB shall, if supported, consider that the triggered Conditional PSCell Change has been executed, and act as specified in TS 37.340 [32].

### 8.7.9.3 Unsuccessful Operation

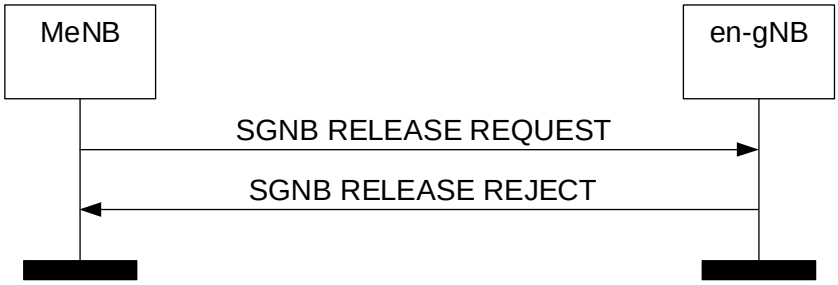



**Figure 8.7.9.3-1: MeNB initiated SgNB Release, unsuccessful operation**

If the en-gNB cannot confirm the request to release en-gNB resources it shall send the SGNB RELEASE REQUEST REJECT message to the MeNB with an appropriate cause indicated in the *Cause* IE.

If the MeNB did not include the *SgNB UE X2AP ID* IE in the SGNB RELEASE REQUEST message, the MeNB shall ignore the *SgNB UE X2AP ID* IE in the SGNB RELEASE REQUEST REJECT message.

### 8.7.9.4 Abnormal Conditions

If the SGNB RELEASE REQUEST message refer to a context that does not exist, the en-gNB shall ignore the message.

When the MeNB has initiated the procedure and did not include the *SgNB UE X2AP ID* IE the MeNB shall regard the resources for the UE at the en-gNB as being fully released.

**Interactions with the UE Context Release procedure:**

If the MeNB does not receive the reply from the en-gNB before it has to relase the EN-DC connection, or it receives SGNB RELEASE REQUEST REJECT, it may trigger the UE Context Release procedure. If the en-gNB received the UE CONTEXT RELEASE right after receiving the SGNB RELEASE REQUEST (and before or after responding to it), the en-gNB shall consider the related MeNB initiated SgNB Release procedure as being the resolution of abnormal conditions and release the related UE context immediately.

## 8.7.10 SgNB initiated SgNB Release

### 8.7.10.1 General

This procedure is triggered by the en-gNB to initiate the release of the resources for a specific UE.

The procedure uses UE-associated signalling.

### 8.7.10.2 Successful Operation

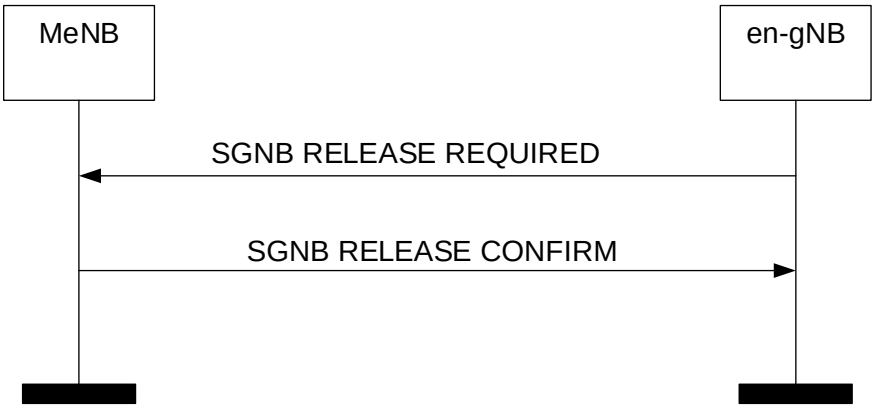



**Figure 8.7.10.2-1: SgNB initiated SgNB Release, successful operation.**

The en-gNB initiates the procedure by sending the SGNB RELEASE REQUIRED message to the MeNB.

Upon reception of the SGNB RELEASE REQUIRED message, the MeNB replies with the SGNB RELEASE CONFIRM message. For each E-RAB configured with the PDCP entity in the en-gNB, the MeNB may include the *DL Forwarding GTP Tunnel Endpoint* IE and the *UL Forwarding GTP Tunnel Endpoint* IE within the *E-RABs To Be Released Item* IE to indicate that it requests data forwarding of uplink and downlink packets to be performed for that bearer.

If the *RLC Mode* IE is included for an E-RAB within the *E-RABs To Be Released List* IE (for E-RABs hosted at the en-gNB) in the SGNB RELEASE REQUIRED message, it indicates the mode that the en-gNB used for the E-RAB when it was hosted at the en-gNB.

If the *SgNB to MeNB Container* IE is included in the SGNB RELEASE REQUIRED message, the MeNB may use the contained information to apply delta configuration.

The en-gNB may start data forwarding and stop providing user data to the UE upon reception of the SGNB RELEASE CONFIRM message.

Upon successful completion of the procedure, the MeNB shall start counting time, so that information regarding time since Secondary Node Release may be transferred towards the MME as specified in TS 36.413 [4].

If the SGNB RELEASE REQUIRED message includes the *SCG UE History Information* IE, the MeNB shall, if supported, use the information to update UE History Information with PSCell history.

#### 8.7.10.3 Unsuccessful Operation

Not applicable.

#### 8.7.10.4 Abnormal Conditions

Void.

### 8.7.11 SgNB Counter Check

#### 8.7.11.1 General

This procedure is initiated by the en-gNB to request the MeNB to execute a counter check procedure to verify the value of the PDCP COUNTs associated with SN terminated bearers.

The procedure uses UE-associated signalling.

#### 8.7.11.2 Successful Operation




**Figure 8.7.11.2-1: SgNB Counter Check procedure, successful operation.**

The en-gNB initiates the procedure by sending the SGNB COUNTER CHECK REQUEST message to the MeNB.

Upon reception of the SGNB COUNTER CHECK REQUEST message, the MeNB may perform the RRC counter check procedure as defined in TS 33.401 [18].

### 8.7.11.3 Unsuccessful Operation

Not applicable.

### 8.7.11.4 Abnormal Conditions

Not applicable.

## 8.7.12 RRC Transfer

### 8.7.12.1 General

The purpose of the RRC Transfer procedure is to deliver a PDCP-C PDU encapsulating an LTE RRC message to the en-gNB so that it may then be forwarded to the UE, or from the en-gNB, if it was received from the UE. Delivery status may also be provided from the en-gNB to the MeNB using the RRC Transfer.

The procedure is also to enable transfer of the NR RRC message container with the NR measurements from the MeNB to the en-gNB, when received from the UE.

The procedure is also to enable transfer of the NR RRC message container with the NR failure information from the MeNB to the en-gNB, when received from the UE.

The procedure is also used to enable transfer of the NR RRC message container with an IAB information from the MeNB to the en-gNB, when received from the IAB-MT.

The procedure is also to enable transfer of the NR RRC message container with the NR *RRCReconfigurationComplete* message from the MeNB to the en-gNB, when received from the UE.

The procedure is also to enable transfer of the NR RRC message container with the UE Assistance information from the MeNB to the en-gNB, when received from the UE.

The procedure uses UE-associated signalling.

### 8.7.12.2 Successful Operation

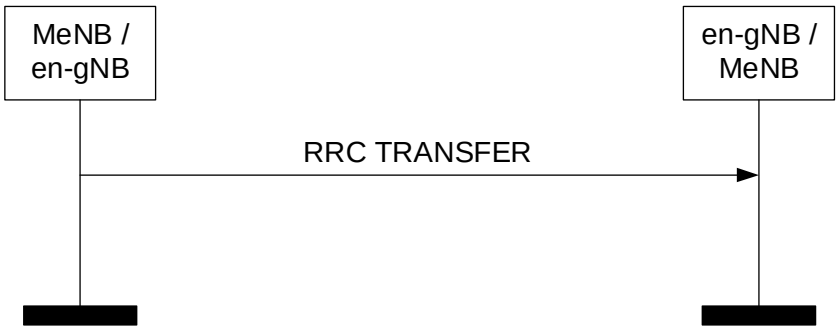



**Figure 8.7.12.2-1: RRC Transfer procedure, successful operation.**

Either the MeNB initiates the procedure by sending the RRC TRANSFER message to the en-gNB or the en-gNB initiates the procedure by sending the RRC TRANSFER message to the MeNB.

If the en-gNB receives an RRC TRANSFER message which does not include the *RRC Container* IE in the *Split SRB* IE, or the *RRC container* IE in *NR UE Report* IE, or the *RRC Container* IE in the *Fast MCG Recovery via SRB3 from MN to SN* IE, or the *RRC Container* IE in the *Fast MCG Recovery via SRB3 from SN to MN* IE, it shall ignore the message. If the en-gNB receives an RRC TRANSFER message with the Delivery Status IE, it shall ignore the message. If the en-gNB receives the *RRC Container* IE in the *Split SRB* IE, it shall deliver the contained PDCP-C PDU encapsulating an RRC message to the UE. If the en-gNB receives the *RRC Container* IE in the *Fast MCG Recovery from MeNB to SgNB* IE, the en-gNB shall, if supported, deliver the contained RRC Container encapsulating an RRC message to the UE.

If the MeNB receives the *Delivery Status* IE in the *split SRB* IE the MeNB shall consider RRC messages up to the indicated NR PDCP SN as having been successfully delivered (as defined in TS 36.322 [40]) to the UE by the en-gNB.

If the MeNB receives the *RRC Container* IE in the *Fast MCG Recovery from SgNB to MeNB* IE, the MeNB shall, if supported, consider MCG link failure detected at the UE as specified in TS 37.340 [32].

#### 8.7.12.3 Abnormal Conditions

In case of the split SRBs, the receiving node may ignore the message, if the MeNB has not indicated possibility of RRC transfer at the bearer setup.

### 8.7.13 Secondary RAT Data Usage Report

#### 8.7.13.1 General

This procedure is initiated by the en-gNB to report secondary RAT data volume.

The procedure uses UE-associated signalling.

#### 8.7.13.2 Successful Operation




**Figure 8.7.13.2-1: Secondary RAT Data Usage Report procedure, successful operation.**

The en-gNB initiates the procedure by sending the SECONDARY RAT DATA USAGE REPORT message to the MeNB.

#### 8.7.13.3 Unsuccessful Operation

Not applicable.

#### 8.7.13.4 Abnormal Conditions

Not applicable.

### 8.7.14 Partial reset of EN-DC

#### 8.7.14.1 General

This procedure is triggered by the en-gNB or the MeNB to initiate the reset of the resources for selected UEs.

The procedure uses non UE-associated signalling.

### 8.7.14.2 Successful Operation

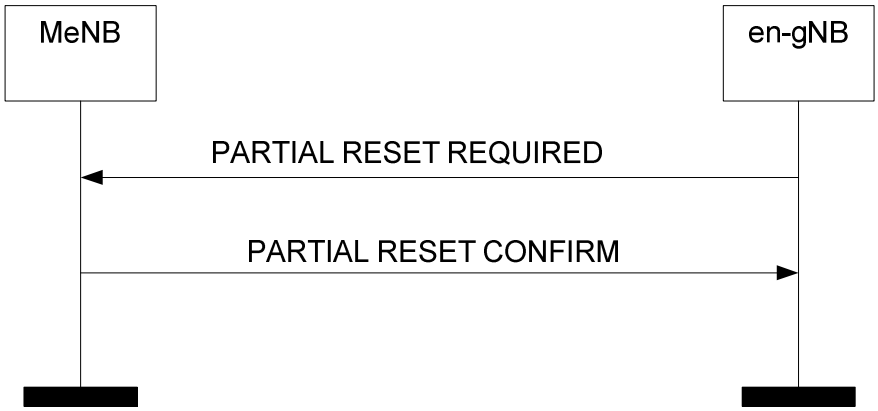



**Figure 8.7.14.2-1: en-gNB initiated Partial Reset of EN-DC, successful operation.**

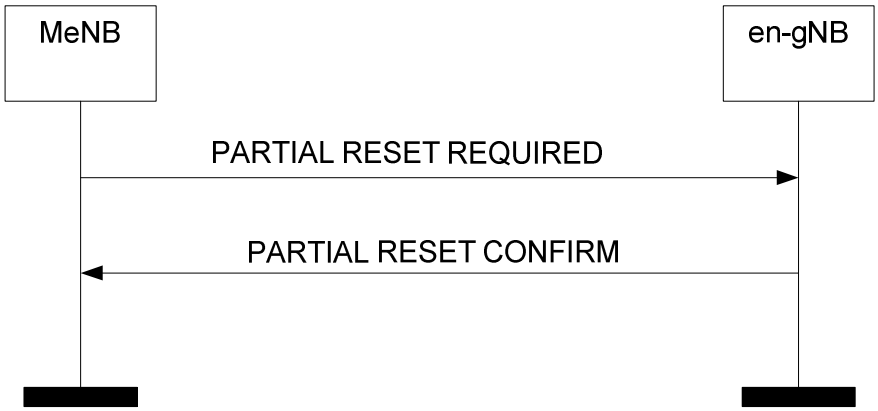



**Figure 8.7.14.2-2: eNB initiated Partial Reset of EN-DC, successful operation.**

The en-gNB or MeNB initiates the procedure by sending the PARTIAL RESET REQUIRED message to the MeNB or the en-gNB.

In case of the eNB-initiated Partial Reset, at reception of the PARTIAL RESET REQUIRED message, the en-gNB node shall release all allocated resources on X2 and Uu related to the UE association(s) indicated in the PARTIAL RESET REQUIRED message and remove the indicated UE contexts including X2AP ID.

In case of the en-gNB-initiated Partial Reset, at reception of the PARTIAL RESET REQUIRED message, the MeNB may decide to release all allocated resources on X2 and Uu related to the UE association(s) indicated in the PARTIAL RESET REQUIRED message and remove the indicated UE contexts including X2AP ID, or to reconfigure the UEs for MN-terminated MCG bearers.

After the receiving node has released or reconfigured all assigned X2 resources and the UE X2AP IDs for all indicated UE associations which can be used for new UE-associated logical X2-connections over the X2 interface, the receiving node shall respond with the PARTIAL RESET CONFIRM message. The node receiving the request does not need to wait for the release or reconfiguration of radio resources to be completed before returning the PARTIAL RESET CONFIRM message.

The node initiating the procedure shall include the *SgNB UE X2AP ID* IE in the PARTIAL RESET REQUIRED message if it has already been allocated for the UE. The node receiving the request shall use the *SgNB UE X2AP ID* IE (if included) and/or the *MeNB UE S1AP ID* IE (and the *MeNB UE S1AP ID Extension* IE, if included) to identify the UE association(s) to be released. If the *SgNB UE X2AP ID* IE was included in the PARTIAL RESET REQUIRED message, the receiving node shall include it also in the PARTIAL RESET CONFIRM message.

The node receiving the request shall include in the PARTIAL RESET CONFIRM message, for each UE association to be released, the same list of UE-associated logical X2-connections over X2. The list shall be in the same order as received in the PARTIAL RESET REQUIRED message and shall include also unknown UE-associated logical X2-connections.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the PARTIAL RESET REQUIRED message and the PARTIAL RESET CONFIRM message shall contain the *Interface Instance Indication* IE to identify the corresponding interface instance.

**Interactions with other procedures:**

If the PARTIAL RESET REQUIRED message is received, any other ongoing procedure (except for a Reset or another Partial Reset of EN-DC procedures) on the same X2 interface related to a UE association, indicated in the PARTIAL RESET REQUIRED message, shall be aborted.

### 8.7.14.3 Unsuccessful Operation

Not applicable.

### 8.7.14.4 Abnormal Conditions

Void.

## 8.7.15 E-UTRA – NR Cell Resource Coordination

### 8.7.15.1 General

The purpose of the E-UTRA – NR Cell Resource Coordination procedure is to enable coordination of radio resource allocation between an eNB and an en-gNB that are sharing spectrum and whose coverage areas are fully or partially overlapping. During the procedure, the eNB and en-gNB shall exchange their intended resource allocations for data traffic, and, if possible, converge to a shared resource. The procedure is only to be used for the purpose of E-UTRA – NR spectrum sharing.

The procedure uses non-UE-associated signalling.

### 8.7.15.2 Successful Operation

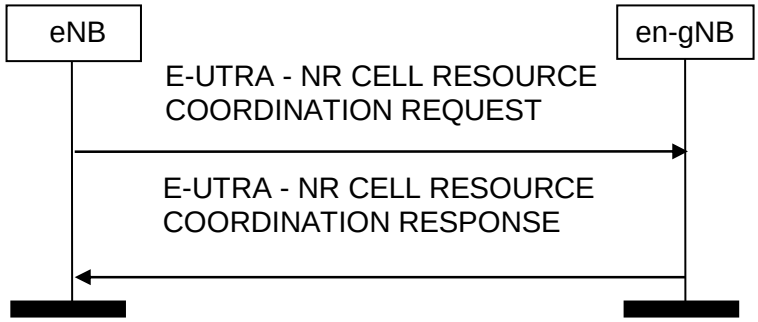



**Figure 8.7.15.2-1: eNB-initiated E-UTRA – NR Cell Resource Coordination request, successful operation**

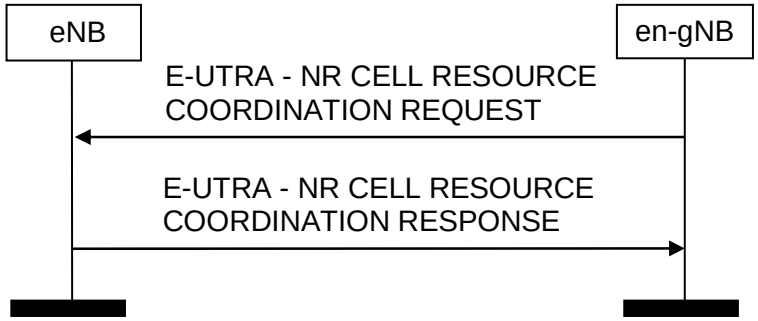



**Figure 8.7.15.2-2: en-gNB-initiated E-UTRA – NR Cell Resource Coordination request, successful operation**

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the E-UTRA – NR CELL RESOURCES COORDINATION REQUEST message and the E-UTRA – NR CELL RESOURCES COORDINATION RESPONSE message shall contain the *Interface Instance Indication* IE to identify the corresponding interface instance.

**eNB initiated E-UTRA – NR Cell Resource Coordination:**

An eNB initiates the procedure by sending the E-UTRA – NR CELL RESOURCE COORDINATION REQUEST message to an en-gNB over the X2 interface. The en-gNB extracts the *Data Traffic Resource Indication* IE and it replies by sending the E-UTRA – NR CELL RESOURCE COORDINATION RESPONSE message. The en-gNB shall calculate the full eNB resource allocation by combining the *Data Traffic Resource Indication* IE and the *Protected E-UTRA Resource Indication* IE that were most recently received from the eNB.

In case of conflict between the most recently received *Data Traffic Resource Indication* IE and the most recently received *Protected E-UTRA Resource Indication* IE, the en-gNB shall give priority to the *Protected E-UTRA Resource Indication* IE.

If the *Initiating Node Type* is eNB, then the E-UTRA – NR CELL RESOURCE COORDINATION REQUEST message shall contain at least one *EUTRA Cell ID* in the List of E-UTRA Cells in NR Coordination Request. If the *Initiating Node Type* is en-gNB, then the E-UTRA – NR CELL RESOURCE COORDINATION REQUEST message shall contain at least one NR-Cell ID in the List of NR Cells in NR Coordination Request.

**en-gNB initiated E-UTRA – NR Cell Resource Coordination:**

An en-gNB initiates the procedure by sending the E-UTRA – NR CELL RESOURCE COORDINATION REQUEST message to an eNB. The eNB replies with the E-UTRA – NR CELL RESOURCE COORDINATION RESPONSE message. The en-gNB shall calculate the full eNB resource allocation by combining the *Data Traffic Resource Indication* IE and the *Protected E-UTRA Resource Indication* IE that were most recently received from the eNB.

In case of conflict between the most recently received *Data Traffic Resource Indication* IE and the most recently received *Protected E-UTRA Resource Indication* IE, the en-gNB shall give priority to the *Protected E-UTRA Resource Indication* IE.

## 8.7.16 SgNB Activity Notification

### 8.7.16.1 General

The purpose of the SgNB Activity Notification procedure is to allow an en-gNB to send a notification to an eNB concerning user data traffic activity of already established E-RABs. The procedure uses UE-associated signalling.

### 8.7.16.2 Successful Operation

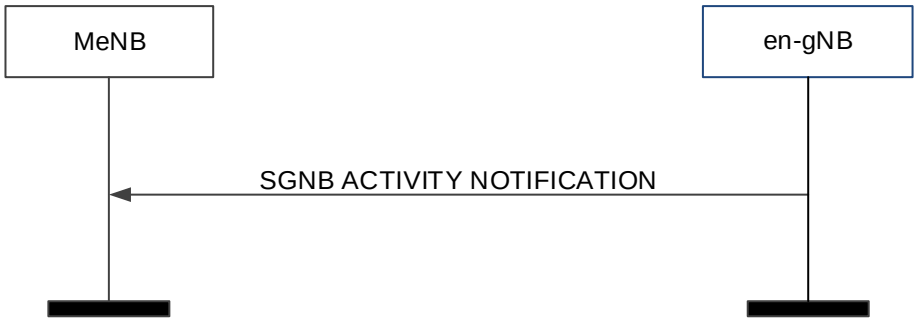



**Figure 8.7.16.2-1: Activity Notification procedure, successful operation**

The en-gNB initiates the procedure by sending an SGNB ACTIVITY NOTIFICATION message to the MeNB.

The SGNB ACTIVITY NOTIFICATION message may contain notification for UE context level user plane activity in the *UE Context level user plane activity report* IE.

The SGNB ACTIVITY NOTIFICATION message may contain notification for activity of E-RABs.

### 8.7.16.3 Abnormal Conditions

Void.

## 8.7.17 gNB Status Indication

### 8.7.17.1 General

The purpose of the gNB Status Indication procedure is to inform the eNB that the en-gNB is overloaded so that overload reduction actions can be applied. The procedure uses non-UE associated signalling.

### 8.7.17.2 Successful Operation

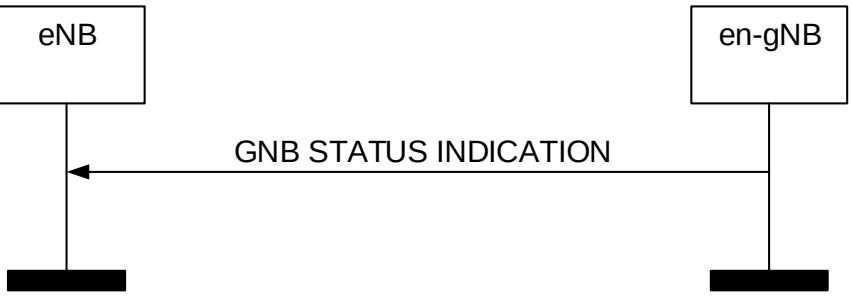



**Figure 8.7.17.2-1: gNB Status Indiciation procedure, successful operation**

If the *gNB Overload Information* IE in the GNB STATUS INDICATION message is set to "overloaded", the eNB shall apply overload reduction actions until it receives a subsequent GNB STATUS INDICATION message with *gNB Overload Information* IE set to "not-overloaded".

The detailed overload reduction policy is up to eNB implementation.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the GNB STATUS INDICATION message shall contain the *Interface Instance Indication* IE to identify the corresponding interface instance.

### 8.7.17.3 Abnormal Conditions

Void.

## 8.7.18 EN-DC Configuration Transfer

### 8.7.18.1 General

The purpose of the EN-DC Configuration Transfer procedure is to transfer the EN-DC SON Configuration container, either from the eNB to the en-gNB or from the en-gNB to the eNB, in the context of en-gNB X2 TNL address discovery as described in TS 36.300 [15].

The procedure uses non UE-associated signalling.

### 8.7.18.2 Successful Operation

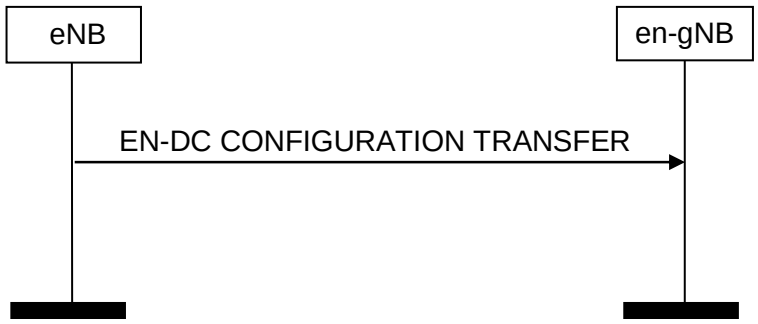



**Figure 8.7.18.2-1: eNB initiated EN-DC Configuration Transfer, successful operation**

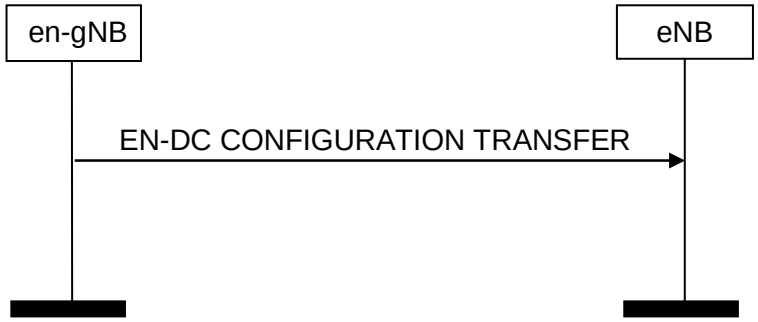



**Figure 8.7.18.2-2: en-gNB initiated EN-DC Configuration Transfer, successful operation**

The eNB initiates the procedure by sending the EN-DC CONFIGURATION TRANSFER message to an en-gNB.

If the en-gNB receives, in the *EN-DC SON Configuration Transfer* IE, the *SON Information* IE containing the *SON Information Request* IE, it may transfer back the requested information towards the eNB indicated in the *Source eNB-ID* IE of the *EN-DC SON Configuration Transfer* IE by initiating the EN-DC Configuration Transfer procedure.

If the en-gNB receives, in the *EN-DC SON Configuration Transfer* IE, the *X2 TNL Configuration Info* IE containing the *eNB X2 Extended Transport Layer Addresses* IE, it may use it as part of its ACL functionality configuration actions, if such ACL functionality is deployed.

In case the *IP-Sec Transport Layer Address* IE is present and the *GTP Transport Layer Addresses* IE within the *eNB X2 Extended Transport Layer Addresses* IE is not empty, GTP traffic is conveyed within an IP-Sec tunnel terminated at the IP-Sec tunnel end point given in by the *IP-Sec Transport Layer Address* IE.

In case the *IP-Sec Transport Layer Address* IE is not present, GTP traffic is terminated at the end points given by the list of addresses in *eNB GTP Transport Layer Addresses* IE within the *eNB X2 Extended Transport Layer Addresses* IE.

In case the *eNB GTP Transport Layer Addresses* IE is empty and the *IP-Sec Transport Layer Address* IE is present, SCTP traffic is conveyed within an IP-Sec tunnel terminated at the IP-Sec tunnel end point given in the *IP-Sec Transport Layer Address* IE, within the *eNB X2 Extended Transport Layer Addresses* IE.

If the en-gNB is configured to use one IPsec tunnel for EN-DC X2 traffic (IPsec star topology) then the traffic to the peer eNB shall be routed through this IPsec tunnel and the *IP-Sec Transport Layer Address* IE shall be ignored.

The en-gNB initiates the procedure by sending the EN-DC CONFIGURATION TRANSFER message to an eNB.

If case of network sharing with multiple cell ID broadcast with shared X2-C signalling transport, as specified in TS 36.300 [15], the EN-DC CONFIGURATION TRANSFER message shall contain the *Interface Instance Indication* IE to identify the corresponding interface instance.

### 8.7.18.3 Abnormal Conditions

Void.

## 8.7.19 Trace Start

### 8.7.19.1 General

The purpose of the Trace Start procedure is to allow the MeNB to request the en-gNB to initiate a trace session for a UE. The procedure uses UE-associated signalling.

### 8.7.19.2 Successful Operation




**Figure 8.7.19.2-1: Trace Start, successful operation**

The Trace Start procedure is initiated by the MeNB sending the TRACE START message to the en-gNB for that specific UE. Upon reception of the TRACE START message, the en-gNB shall initiate the requested trace session as described in TS 32.422 [6]. If the *Trace Activation* IE includes the *MDT Configuration NR* IE, the en-gNB shall take it into account for MDT function as described in TS 37.320 [31].

### 8.7.19.3 Abnormal Conditions

Void.

## 8.7.20 Deactivate Trace

### 8.7.20.1 General

The purpose of the Deactivate Trace procedure is to allow the MeNB to request the en-gNB to stop the trace session for the indicated trace reference. The procedure uses UE-associated signalling.

### 8.7.20.2 Successful Operation




**Figure 8.7.20.2-1: Deactivate Trace, successful opration**

The Deactivate Trace procedure is initiated by the MeNB by sending the DEACTIVATE TRACE to the en-gNB for that specific UE. Upon reception of the DEACTIVATE TRACE message, the en-gNB shall stop the trace session for the indicated trace reference in the *E-UTRAN Trace ID* IE.

### 8.7.20.3 Abnormal Conditions

Void.

## 8.7.21 EN-DC Resource Status Reporting Initiation

### 8.7.21.1 General

This procedure is used by the eNB to request the reporting of load measurements to the en-gNB .

The procedure uses non UE-associated signalling.

### 8.7.21.2 Successful Operation

#### 8.7.21.2.1 Successful Operation - eNB-initiated

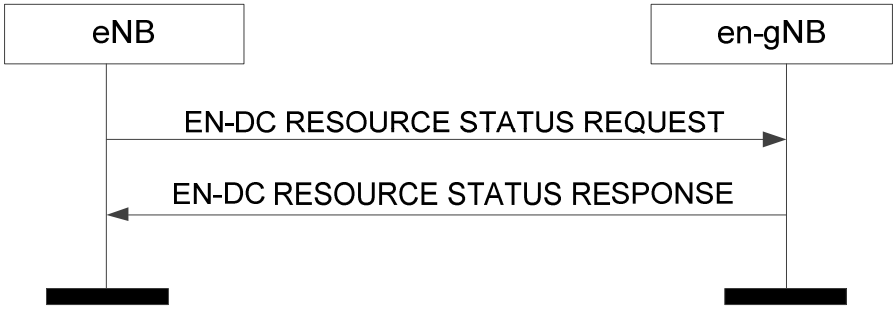



**Figure 8.7.21.2-1: EN-DC Resource Status Reporting Initiation, successful operation - eNB-initiated**

The procedure is initiated with an EN-DC RESOURCE STATUS REQUEST message sent from the eNB to the en-gNB to start a measurement, stop a measurement, add cells to report for a measurement.

If the *Report Characteristics EN-DC* IE is included in the EN-DC RESOURCE STATUS REQUEST message and indicates cell specific measurements, the *NR Cell To Report EN-DC List* IE shall be included.

Upon receipt of the EN-DC RESOURCE STATUS REQUEST message, the en-gNB:

- shall initiate the requested measurement according to the parameters given in the request in case the *Registration Request EN-DC* IE set to "start"; or
- shall stop all cells measurements and terminate the reporting in case the *Registration Request EN-DC* IE is set to "stop"; or
- shall add cells indicated in the *NR Cell To Report EN-DC List* IE list to the measurements initiated before for the given measurement IDs, in case the *Registration Request EN-DC* IE is set to "add". If measurements are already initiated for a cell indicated in the *NR Cell To Report EN-DC List* IE, this information shall be ignored.

The en-gNB shall send an EN-DC RESOURCE STATUS RESPONSE message to the eNB to indicate that all of the requested measurement objects the measurement can be initiated.

**Interaction with other procedures**

When starting a measurement, the *Report Characteristics EN-DC* IE in the EN-DC RESOURCE STATUS REQUEST indicates the type of objects en-gNB shall perform measurements on. For each cell, the en-gNB shall include in the EN-DC RESOURCE STATUS UPDATE message:

- the *NR Radio Resource Status* IE, if the first bit, "PRB Periodic" of the *Report Characteristics EN-DC* IE included in the EN-DC RESOURCE STATUS REQUEST message is set to "1". If the cell for which *NR Radio Resource Status* IE is requested to be reported supports more than one SSB, the *NR Radio Resource Status* IE for such cell shall include the *SSB Area Radio Resource Status Item* IE for all SSB areas supported by the cell. If the *SSB To Report List* IE is included for a cell, the *NR Radio Resource Status* IE for such cell shall include the requested *SSB Area Radio Resource Status List* IE.

- the *TNL Capacity Indicator* IE, if the second bit, "TNL Capacity Ind Periodic" of the *Report Characteristics EN-DC* IE included in the EN-DC RESOURCE STATUS REQUEST message is set to "1". The received *TNL Capacity Indicator* IE represents the lowest TNL capacity available for the cell, only taking into account interfaces providing user plane transport.

- the *NR Composite Available Capacity Group* IE, if the third bit, "Composite Available Capacity Periodic" of the *Report Characteristics EN-DC* IE included in the EN-DC RESOURCE STATUS REQUEST message is set to "1". If *Cell Capacity Class Value* IE is included within the *NR Composite Available Capacity Group* IE, this IE is used to assign weights to the available capacity indicated in the *Capacity Value* IE. If the cell for which *NR Composite Available Capacity Group* IE is requested to be reported supports more than one SSB, and if the *SSB To Report List* IE is included for a cell, the *NR Composite Available Capacity Group* IE for such cell shall include the requested *SSB Area Capacity Value List* IE, providing the SSB area capacity with respect to the Cell Capacity Class Value.

If the *Reporting Periodicity EN-DC* IE in the EN-DC RESOURCE STATUS REQUEST is present, this indicates the periodicity for the reporting of periodic measurements. The en-gNB shall only report more than once if the *Reporting Periodicity EN-DC* IE is included.

#### 8.7.21.2.2 Successful Operation - en-gNB-initiated

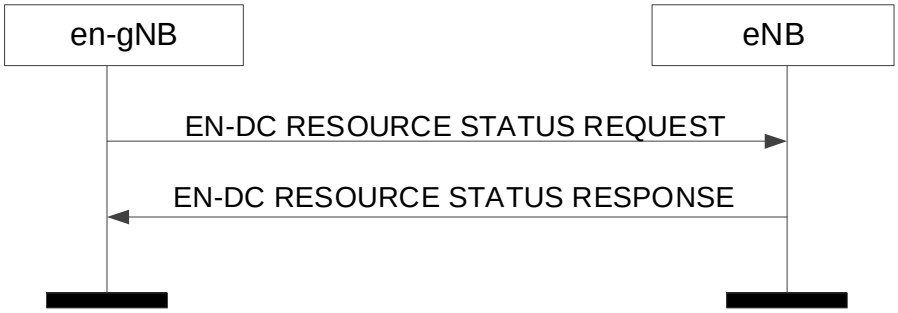



**Figure 8.7.21.2-2: EN-DC Resource Status Reporting Initiation, successful operation - en-gNB-initiated**

The procedure is initiated with an EN-DC RESOURCE STATUS REQUEST message sent from the en-gNB to the eNB to start a measurement, stop a measurement, add cells to report for a measurement.

If the *Report Characteristics EN-DC* IE is included in the EN-DC RESOURCE STATUS REQUEST message and indicates cell specific measurements, the *E-UTRA Cell To Report EN-DC List* IE shall be included.

Upon receipt of the EN-DC RESOURCE STATUS REQUEST message, the eNB:

- shall initiate the requested measurement according to the parameters given in the request in case the *Registration Request EN-DC* IE set to "start"; or

- shall stop all cells measurements and terminate the reporting in case the *Registration Request EN-DC* IE is set to "stop"; or

- shall add cells indicated in the *E-UTRA Cell To Report EN-DC List* IE list to the measurements initiated before for the given measurement IDs, in case the *Registration Request EN-DC* IE is set to "add". If measurements are already initiated for a cell indicated in the *E-UTRA Cell To Report EN-DC List* IE, this information shall be ignored. The eNB shall ignore the *E-UTRA Cell To Report EN-DC List* IE, if included.

The eNB shall send an EN-DC RESOURCE STATUS RESPONSE message to the en-gNB to indicate that all of the requested measurement objects the measurement can be initiated.

**Interaction with other procedures, en-gNB-initiated operation**

When starting a measurement, the *Report Characteristics EN-DC* IE in the EN-DC RESOURCE STATUS REQUEST indicates the type of objects eNB shall perform measurements on. For each cell, the eNB shall include in the EN-DC RESOURCE STATUS UPDATE message:

- the *Radio Resource Status* IE, if the first bit, "PRB Periodic" of the *Report Characteristics EN-DC* IE included in the RESOURCE STATUS REQUEST message is set to "1";

- the *S1 TNL Load Indicator* IE, if the second bit, "TNL Load Ind Periodic" of the *Report Characteristics EN-DC* IE included in the RESOURCE STATUS REQUEST message is set to "1";
- the *Hardware Load Indicator* IE, if the third bit, "HW Load Ind Periodic" of the *Report Characteristics EN-DC* IE included in the RESOURCE STATUS REQUEST message is set to "1";
- the *Composite Available Capacity Group* IE, if the fourth bit, "Composite Available Capacity Periodic" of the *Report Characteristics EN-DC* IE included in the RESOURCE STATUS REQUEST message is set to "1". If the *Cell Capacity Class Value* IE is included within the *Composite Available Capacity Group* IE, this IE is used to assign weights to the available capacity indicated in the *Capacity Value* IE;

If the *Reporting Periodicity EN-DC* IE in the EN-DC RESOURCE STATUS REQUEST is present, this indicates the periodicity for the reporting of periodic measurements. The eNB shall only report more than once if the *Reporting Periodicity EN-DC* IE is included.

### 8.7.21.3 Unsuccessful Operation

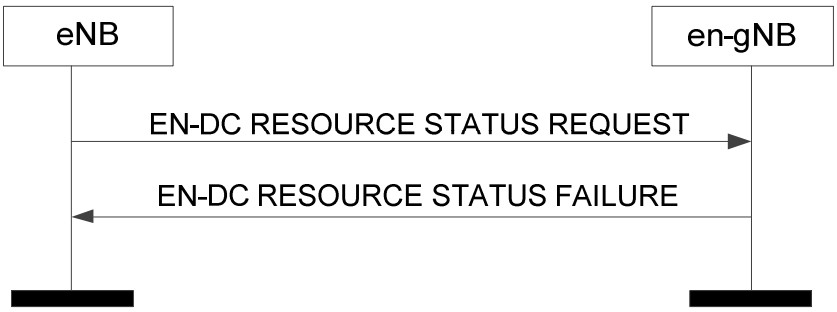



**Figure 8.7.21.3-1: EN-DC Resource Status Reporting Initiation, unsuccessful operation - eNB-initiated**

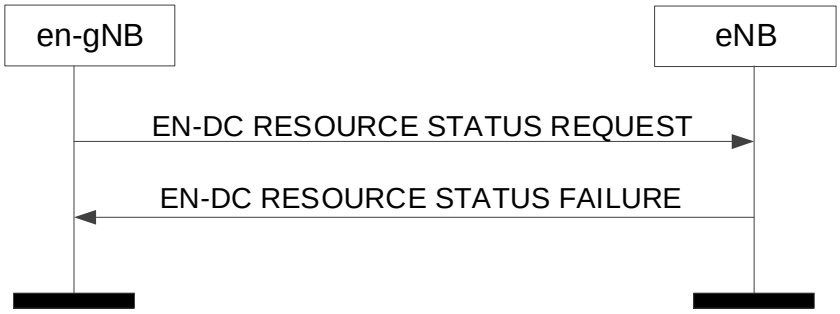



**Figure 8.7.21.3-2: EN-DC Resource Status Reporting Initiation, unsuccessful operation - en-gNB-initiated**

If any of the requested measurements cannot be initiated, the en-gNB or the eNB shall send an EN-DC RESOURCE STATUS FAILURE message.

### 8.7.21.4 Abnormal Conditions

Void.

## 8.7.22 EN-DC Resource Status Reporting

### 8.7.22.1 General

This procedure is initiated by the en-gNB or by the eNB to report the result of measurements admitted by the en-gNB or by the eNB following a successful EN-DC Resource Status Reporting Initiation procedure.

The procedure uses non UE-associated signalling.

### 8.7.22.2 Successful Operation

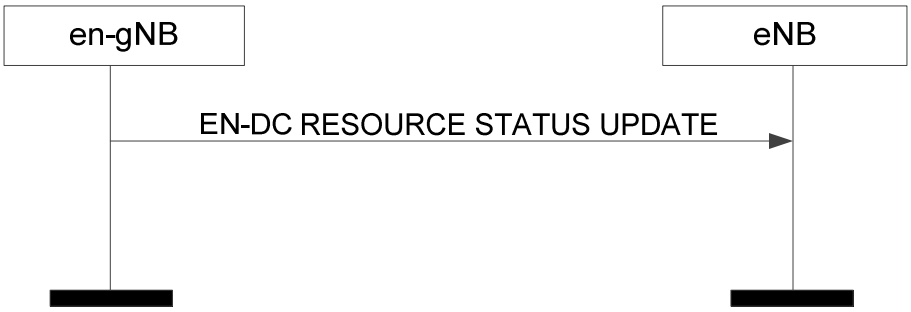



**Figure 8.7.22.2-1: EN-DC Resource Status Reporting, successful operation - en-gNB-initiated**

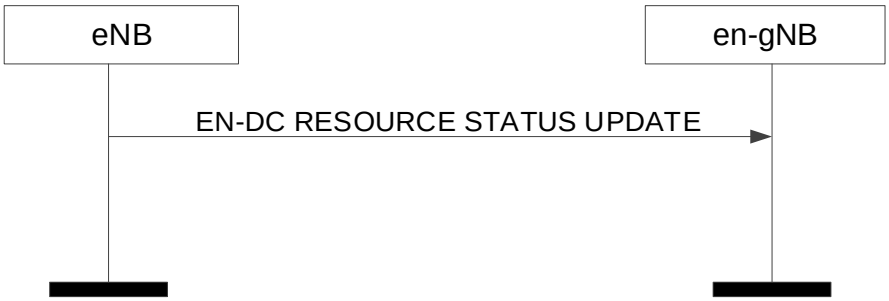



**Figure 8.7.22.2-2: EN-DC Resource Status Reporting, successful operation - eNB-initiated**

The en-gNB or the eNB shall report the results of the admitted measurements in the EN-DC RESOURCE STATUS UPDATE message. The admitted measurements are the measurements that were successfully initiated during the preceding EN-DC Resource Status Reporting Initiation procedure.

### 8.7.22.3 Unsuccessful Operation

Not applicable.

### 8.7.22.4 Abnormal Conditions

Void.

## 8.7.23 Cell Traffic Trace

### 8.7.23.1 General

The purpose of the Cell Traffic Trace procedure is to send the allocated Trace Recording Session Reference and the Trace Reference to the MeNB. The procedure uses UE-associated signalling.

### 8.7.23.2 Successful Operation




**Figure 8.7.23.2-1: Cell Traffic Trace procedure. Successful operation.**

The procedure is initiated with a CELL TRAFFIC TRACE message sent from the en-gNB to the MeNB.

If the *Privacy Indicator* IE is included in the message, the MeNB shall take the information into account for anonymisation of MDT data as specified in TS 32.422 [6].

## 8.7.24 UE Radio Capability ID Mapping

### 8.7.24.1 General

The purpose of the UE Radio Capability ID Mapping procedure is to enable an en-gNB to request a connected eNB to provide the UE Radio Capability information that maps to a specific UE Radio Capability ID.

The procedure uses non-UE-associated signalling.

### 8.7.24.2 Successful Operation

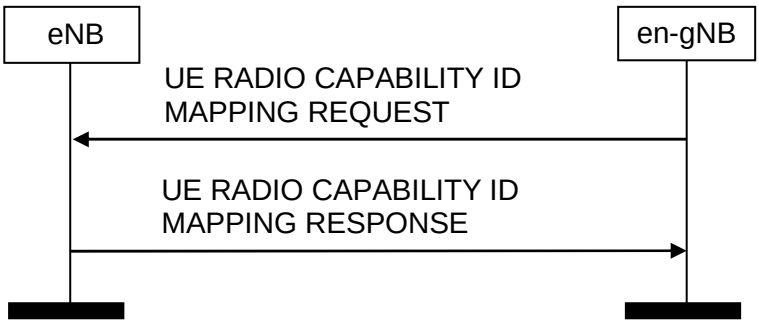



**Figure 8.7.24.2-1: UE Radio Capability ID Mapping procedure. Successful operation**

The en-gNB initiates the procedure by sending a UE RADIO CAPABILITY ID MAPPING REQUEST message to the eNB.

Upon receipt of the UE RADIO CAPABILITY ID MAPPING REQUEST message, the eNB shall include the UE Radio Capability information that maps to the UE Radio Capability ID indicated in the UE RADIO CAPABILITY ID MAPPING REQUEST message in the UE RADIO CAPABILITY ID MAPPING RESPONSE message.

### 8.7.24.3 Unsuccessful Operation

Not applicable.

## 8.7.25 Conditional PSCell Change Cancel

### 8.7.25.1 General

This procedure is used by the MeNB to inform the source en-gNB that all of the prepared PSCells are cancelled in the target en-gNB during a Conditional PSCell Change.

The procedure uses UE-associated signalling.

### 8.7.25.2 Successful Operation




**Figure 8.7.25.2-1: Conditional PSCell Change Cancel**

The MeNB initiates the procedure by sending the CONDITIONAL PSCELL CHANGE CANCEL message to the en-gNB including the *Target en-gNB ID* IE.

### 8.7.25.3 Unsuccessful Operation

Not applicable.

### 8.7.25.4 Abnormal Conditions

Void.

## 8.8 IAB Procedures

## 8.8.1 F1-C Traffic Transfer

### 8.8.1.1 General

The purpose of the F1-C Traffic Transfer procedure is to deliver F1-C traffic to the MeNB so that it is then forwarded to the IAB-node, or deliver F1-C traffic to the en-gNB, if it was received from the IAB-node.

The procedure uses UE-associated signalling.

### 8.8.1.2 Successful Operation




**Figure 8.8.1.2-1: F1-C Traffic Transfer procedure, successful operation.**

Either the MeNB initiates the procedure by sending the F1-C TRAFFIC TRANSFER message including the received F1-C traffic to the en-gNB, or the en-gNB initiates the procedure by sending the F1-C TRAFFIC TRANSFER message to the MeNB.

Upon reception of the F1-C TRAFFIC TRANSFER message, the MeNB shall deliver the contained F1-C traffic to the IAB-node as specified in TS 36.331 [9].

Upon reception of the F1-C TRAFFIC TRANSFER message, the en-gNB shall handle the received F1-C traffic as specified in TS 38.473 [44] and TS 38.472 [48].

#### 8.8.1.3 Unsuccessful Operation

Not applicable.

#### 8.8.1.4 Abnormal Conditions

Not Applicable.

# 9 Elements for X2AP Communication

## 9.0 General

Sub clauses 9.1 and 9.2 describe the structure of the messages and information elements required for the X2AP protocol in tabular format. Sub clause 9.3 provides the corresponding ASN.1 definition.

The following attributes are used for the tabular description of the messages and information elements: Presence, Range Criticality and Assigned Criticality. Their definition and use can be found in TS 36.413 [4].

NOTE: The messages have been defined in accordance to the guidelines specified in TR 25.921 [30].

## 9.1 Message Functional Definition and Content

### 9.1.1 Messages for Basic Mobility Procedures

#### 9.1.1.1 HANDOVER REQUEST

This message is sent by the source eNB to the target eNB to request the preparation of resources for a handover.

Direction: source eNB → target eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the source eNB | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Target Cell ID | M | | ECGI 9.2.14 | | YES | reject |
| GUMMEI | M | | 9.2.16 | | YES | reject |
| **UE Context Information** | | *1* | | | YES | reject |
| >MME UE S1AP ID | M | | INTEGER (0..$2^{32}$ -1) | MME UE S1AP ID allocated at the MME | – | |
| >UE Security Capabilities | M | | 9.2.29 | | – | |
| >AS Security Information | M | | 9.2.30 | | – | |
| >UE Aggregate Maximum Bit Rate | M | | 9.2.12 | | – | |
| >Subscriber Profile ID for RAT/Frequency priority | O | | 9.2.25 | | – | |
| **>E-RABs To Be Setup List** | | *1* | | | – | |
| **>>E-RABs To Be Setup Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>E-RAB Level QoS Parameters | M | | 9.2.9 | Includes necessary QoS parameters | – | |
| >>>DL Forwarding | O | | 9.2.5 | | – | |
| >>>UL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1 transport bearer. For delivery of UL PDUs. | – | |
| >>>Bearer Type | O | | 9.2.92 | | YES | reject |
| >>>DAPS Request Information | O | | 9.2.154 | | YES | ignore |
| >>>Ethernet Type | O | | 9.2.157 | | YES | ignore |
| >>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| >>>Security Indication | O | | 9.2.181 | | YES | reject |
| >RRC Context | M | | OCTET STRING | Includes the *HandoverPreparationInformation* message as defined in subclause 10.2.2 of TS 36.331 [9], or the *HandoverPreparationInformation-NB* message as defined in 10.6.2 of TS 36.331 [9]. | – | |
| >Handover Restriction List | O | | 9.2.3 | | – | |
| >Location Reporting Information | O | | 9.2.21 | Includes the necessary parameters for location reporting | – | |
| >Management Based MDT Allowed | O | | 9.2.59 | | YES | ignore |
| >Management Based MDT PLMN List | O | | MDT PLMN List 9.2.64 | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >UE Sidelink Aggregate Maximum Bit Rate | O | | 9.2.97 | This IE applies only if the UE is authorized for V2X services. | YES | ignore |
| >EPC Handover Restriction List Container | O | | 9.2.153 | | YES | ignore |
| >Additional RRM Policy Index | O | | 9.2.25a | | YES | ignore |
| >NR UE Sidelink Aggregate Maximum Bit Rate | O | | 9.2.159 | This IE applies only if the UE is authorized for NR V2X services. | YES | ignore |
| >UE Radio Capability ID | O | | 9.2.171 | | YES | reject |
| >IMS voice EPS fallback from 5G | O | | ENUMERATED (true, ...) | | YES | ignore |
| UE History Information | M | | 9.2.38 | Same definition as in TS 36.413 [4] | YES | ignore |
| Trace Activation | O | | 9.2.2 | | YES | ignore |
| SRVCC Operation Possible | O | | 9.2.33 | | YES | ignore |
| CSG Membership Status | O | | 9.2.52 | | YES | reject |
| Mobility Information | O | | BIT STRING (SIZE (32)) | Information related to the handover; the source eNB provides it in order to enable later analysis of the conditions that led to a wrong HO. | YES | ignore |
| Masked IMEISV | O | | 9.2.69 | | YES | ignore |
| UE History Information from the UE | O | | OCTET STRING | Includes the *VisitedCellInfoList* IE contained in the *UEInformationResponse* message (TS 36.331 [9]) | YES | ignore |
| Expected UE Behaviour | O | | 9.2.70 | | YES | ignore |
| ProSe Authorized | O | | 9.2.78 | | YES | ignore |
| UE Context Reference at the SeNB | O | | | | YES | ignore |
| >Global SeNB ID | M | | Global eNB ID 9.2.22 | | – | |
| >SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | – | |
| >SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | – | |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the source eNB | YES | reject |
| V2X Services Authorized | O | | 9.2.93 | | YES | ignore |
| UE Context Reference at the WT | O | | | | YES | ignore |
| >WT ID | M | | 9.2.95 | | – | |
| >WT UE XwAP ID | M | | 9.2.96 | | – | |
| NR UE Security Capabilities | O | | 9.2.107 | | YES | ignore |
| UE Context Reference at the SgNB | O | | | | YES | ignore |
| >Global en-gNB ID | M | | 9.2.112 | | – | |
| >SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the SgNB. | – | |
| Aerial UE subscription information | O | | 9.2.129 | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| Subscription Based UE Differentiation Information | O | | 9.2.136 | | YES | ignore |
| **Conditional Handover Information Request** | O | | | | YES | reject |
| >CHO Trigger | M | | ENUMERATED (CHO-initiation, CHO-replace, …) | | – | |
| >New eNB UE X2AP ID | C-ifCHOmod | | eNB UE X2AP ID<br><br>9.2.24 | Allocated at the target eNB | – | |
| >New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID<br><br>9.2.86 | Allocated at the target eNB | – | |
| >Estimated Arrival Probability | O | | INTEGER (1..100) | | – | |
| NR V2X Services Authorized | O | | 9.2.158 | | YES | ignore |
| PC5 QoS Parameters | O | | 9.2.160 | This IE applies only if the UE is authorized for NR V2X services. | YES | ignore |
| IAB Node Indication | O | | ENUMERATED (true, ...) | | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |
| maxnoofMDTPLMNs | PLMNs in the Management Based MDT PLMN list. Value is 16. |

| Condition | Explanation |
|---|---|
| ifCHOmod | This IE shall be present if the *CHO Trigger* IE is present and set to "CHO-replace". |

### 9.1.1.2 HANDOVER REQUEST ACKNOWLEDGE

This message is sent by the target eNB to inform the source eNB about the prepared resources at the target.

Direction: target eNB → source eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the source eNB | YES | ignore |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the target eNB | YES | ignore |
| **E-RABs Admitted List** | | *1* | | | YES | ignore |
| **>E-RABs Admitted Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>UL GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>DL GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer. used for forwarding of DL PDUs | – | |
| >>DAPS Response Information | O | | 9.2.155 | | YES | reject |
| E-RABs Not Admitted List | O | | E-RAB List 9.2.28 | A value for *E-RAB ID* shall only be present once in *E-RABs Admitted List* IE and in *E-RABs Not Admitted List* IE. | YES | ignore |
| Target eNB To Source eNB Transparent Container | M | | OCTET STRING | Includes the *HandoverCommand* message as defined in subclause 10.2.2 in TS 36.331 [9] | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| UE Context Kept Indicator | O | | 9.2.85 | | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the source eNB | YES | ignore |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the target eNB | YES | reject |
| WT UE Context Kept Indicator | O | | UE Context Kept Indicator 9.2.85 | Indicates that the WT has acknowledged to keep the UE context | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| E-RABs transferred to MeNB | O | | E-RAB List 9.2.28 | In case of EN-DC, indicates that SN Status is needed for the listed E-RABs from the SgNB.. | YES | ignore |
| **Conditional Handover Information Acknowledge** | O | | | | YES | reject |
| >Requested Target Cell ID | M | | ECGI 9.2.14 | Target cell indicated in the corresponding HANDOVER REQUEST message | – | |
| >Maximum Number of CHO Preparations | O | | 9.2.156 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.1.3 HANDOVER PREPARATION FAILURE

This message is sent by the target eNB to inform the source eNB that the Handover Preparation has failed.

Direction: target eNB → source eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the source eNB | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the source eNB | YES | ignore |
| Requested Target Cell ID | O | | ECGI 9.2.14 | Target cell indicated in the corresponding HANDOVER REQUEST message | YES | reject |

### 9.1.1.4 SN STATUS TRANSFER

This message is sent by the source eNB to the target eNB to transfer the uplink/downlink PDCP SN and HFN status during a handover or for EN-DC.

Direction: source eNB → target eNB (handover), eNB from which the E-RAB context is transferred → eNB to which the E-RAB context is transferred (RRC connection re-establishment or dual connectivity), MeNB/en-gNB from which the E-RAB context is transferred → en-gNB/MeNB to which the E-RAB context is transferred (EN-DC).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated for handover at the source eNB and for dual connectivity/EN-DC at the eNB from which the E-RAB context is transferred | YES | reject |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated for handover at the target eNB and for dual connectivity/EN-DC at the eNB to which the E-RAB context is transferred | YES | reject |
| **E-RABs Subject To Status Transfer List** | | *1* | | | YES | ignore |
| **>E-RABs Subject To Status Transfer Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>Receive Status Of UL PDCP SDUs | O | | BIT STRING (4096) | PDCP Sequence Number = (First Missing SDU Number + bit position) modulo 4096<br><br>0: PDCP SDU has not been received.<br>1: PDCP SDU has been received correctly. | – | |
| >>UL COUNT Value | M | | COUNT Value 9.2.15 | PDCP-SN and Hyper Frame Number of the first missing UL SDU in case of 12 bit long PDCP-SN | – | |
| >>DL COUNT Value | M | | COUNT Value 9.2.15 | PDCP-SN and Hyper frame number that the target eNB/en-gNB should assign for the next DL SDU not having an SN yet in case of 12 bit long PDCP-SN | – | |
| >>Receive Status Of UL PDCP SDUs Extended | O | | BIT STRING (1..16384) | The IE is used in case of 15 bit long PDCP-SN in this release.<br>The first bit indicates the status of the SDU after the First Missing UL PDCP SDU.<br>The $N^{th}$ bit indicates the status of the UL PDCP SDU in position (*N* + First Missing SDU Number) modulo (1 + the maximum value of the PDCP-SN).<br><br>0: PDCP SDU has not been received.<br>1: PDCP SDU has been received correctly. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>UL COUNT Value Extended | O | | COUNT Value Extended 9.2.66 | PDCP-SN and Hyper Frame Number of the first missing UL SDU in case of 15 bit long PDCP-SN | YES | ignore |
| >>DL COUNT Value Extended | O | | COUNT Value Extended 9.2.66 | PDCP-SN and Hyper Frame Number that the target eNB/en-gNB should assign for the next DL SDU not having an SN yet in case of 15 bit long PDCP-SN | YES | ignore |
| >>Receive Status Of UL PDCP SDUs for PDCP SN Length 18 | O | | BIT STRING (1..131072) | The IE is used in case of 18 bit long PDCP-SN.<br>The first bit indicates the status of the SDU after the First Missing UL PDCP SDU.<br>The Nth bit indicates the status of the UL PDCP SDU in position (N + First Missing SDU Number) modulo (1 + the maximum value of the PDCP-SN).<br><br>0: PDCP SDU has not been received.<br>1: PDCP SDU has been received correctly. | YES | ignore |
| >>UL COUNT Value for PDCP SN Length 18 | O | | COUNT Value for PDCP SN Length 18 9.2.82 | PDCP-SN and Hyper Frame Number of the first missing UL SDU in case of 18 bit long PDCP-SN | YES | ignore |
| >>DL COUNT Value for PDCP SN Length 18 | O | | COUNT Value for PDCP SN Length 18 9.2.82 | PDCP-SN and Hyper Frame Number that the target eNB/en-gNB should assign for the next DL SDU not having an SN yet in case of 18 bit long PDCP-SN | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for handover at the source eNB and for dual connectivity/EN-DC at the eNB from which the E-RAB context is transferred. | YES | reject |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for handover at the target eNB and for dual connectivity/EN-DC at the eNB to which the E-RAB context is transferred. | YES | reject |
| SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated for EN-DC at the SgNB. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256. |

### 9.1.1.5 UE CONTEXT RELEASE

This message is sent by the target eNB to the source eNB to indicate that resources can be released.

Direction: target eNB → source eNB (handover), MeNB → SeNB (dual connectivity), MeNB → en-gNB (EN-DC).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated for handover at the source eNB and for dual connectivity at the SeNB. | YES | reject |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated for handover at the target eNB and for dual connectivity/EN-DC at the MeNB. | YES | reject |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for handover at the source eNB and for dual connectivity at the SeNB. | YES | reject |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for handover at the source eNB and for dual connectivity/EN-DC at the MeNB. | YES | reject |
| SIPTO Bearer Deactivation Indication | O | | ENUMERATED (True, …) | Indicates that SIPTO@LN PDN connection deactivation is needed. | YES | ignore |
| SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated for EN-DC at the SgNB. | YES | ignore |

### 9.1.1.6 HANDOVER CANCEL

This message is sent by the source eNB to the target eNB to cancel an ongoing handover.

Direction: source eNB → target eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the source eNB | YES | reject |
| New eNB UE X2AP ID | O | | eNB UE X2AP ID 9.2.24 | Allocated at the target eNB | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the source eNB | YES | reject |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the target eNB | YES | ignore |
| **Candidate Cells To Be Cancelled List** | | *0 .. <maxnoofCellsinCHO>* | | | YES | reject |
| >Target Cell ID | M | | ECGI 9.2.14 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofCellsinCHO | Maximum no. cells that can be prepared for a conditional handover. Value is 8. |

### 9.1.1.7 HANDOVER SUCCESS

This message is sent by the target eNB to the source eNB to indicate the successful access of the UE toward the target eNB.

Direction: target eNB → source eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the source eNB | YES | reject |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the target eNB | YES | reject |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the source eNB | YES | ignore |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the target eNB | YES | ignore |
| Target Cell ID | M | | ECGI 9.2.14 | Target cell indicated in the corresponding Handover Preparation procedure | YES | reject |

### 9.1.1.8 CONDITIONAL HANDOVER CANCEL

This message is sent by the target eNB to the source eNB to cancel an ongoing conditional handover.

Direction: target eNB → source eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the source eNB | YES | ignore |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the target eNB | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the source eNB | YES | ignore |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the target eNB | YES | reject |
| **Candidate Cells To Be Cancelled List** | | 0 .. <maxnoofCellsinCHO> | | | YES | reject |
| >Target Cell ID | M | | ECGI 9.2.14 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofCellsinCHO | Maximum no. cells that can be prepared for a conditional handover. Value is 8. |

### 9.1.1.9 EARLY STATUS TRANSFER

This message is sent by the source eNB to the target eNB to transfer the COUNT value related to the forwarded downlink SDUs during DAPS Handover or Conditional Handover.

During a Conditional Handover with EN-DC or Dual Connectivity, or CPAC in EN-DC, this message is also used to transfer the COUNT value related to the forwarded downlink SDUs. In case of EN-DC, the COUNT value is transferred from the en-gNB to the eNB, while in case of Dual Connectivity, the COUNT value is transferred from the SeNB to the MeNB. In case of Conditional PSCell Addition, the COUNT value is transferred from the eNB to the en-gNB. In case of Conditional PSCell Change, the COUNT value is transferred from the source en-gNB to the eNB, and from eNB to the target en-gNB.

Direction: source eNB → target eNB (DAPS Handover or Conditional Handover).

Direction: en-gNB → MeNB (Conditional Handover with EN-DC), SeNB → MeNB (Conditional Handover with Dual Connectivity)

Direction: eNB → en-gNB (Conditional PSCell Addition)

Direction: source en-gNB → eNB, eNB → target en-gNB (Conditional PSCell Change)

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated for DAPS handover or Conditional handover at the source eNB | YES | reject |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated for DAPS handover or Conditional handover at the target eNB | YES | reject |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for DAPS handover or Conditional handover at the source eNB | YES | reject |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for DAPS handover or Conditional handover at the target eNB | YES | reject |
| CHOICE Procedure Stage | M | | | | YES | reject |
| *>First DL COUNT* | | | | | | |
| >>**E-RABs Subject To Early Status Transfer List** | | *1 .. <maxnoofBearers>* | | | – | |
| >>>**E-RABs Subject To Early Status Transfer Item** | | | | | – | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>FIRST DL COUNT Value | M | | COUNT Value 9.2.15 | PDCP-SN and Hyper frame number of the first DL SDU that the source eNB/SeNB/en-gNB forwards to the target eNB/MeNB in case of 12 bit long PDCP-SN | – | |
| >>>>FIRST DL COUNT Value Extended | O | | COUNT Value Extended 9.2.66 | PDCP-SN and Hyper frame number of the first DL SDU that the source eNB/SeNB/en-gNB forwards to the target eNB/MeNB in case of 15 bit long PDCP-SN | – | |
| >>>>FIRST DL COUNT Value for PDCP SN Length 18 | O | | COUNT Value for PDCP SN Length 18 9.2.82 | PDCP-SN and Hyper frame number of the first DL SDU that the source eNB/SeNB/en-gNB forwards to the target eNB/MeNB in case of 18 bit long PDCP-SN | – | |
| *>DL Discarding* | | | | | | |
| >>**E-RABs Subject To DL Discarding List** | M | *1* | | | – | |
| >>>**E-RABs Subject To DL Discarding Item** | | *1 .. <maxnoofBearers>* | | | – | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>DISCARD DL COUNT Value | M | | COUNT Value 9.2.15 | PDCP-SN and Hyper frame number for which the target eNB/MeNB should discard forwarded DL SDUs associated with lower values in case of 12 bit long PDCP-SN | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>DISCARD DL COUNT Value Extended | O | | COUNT Value Extended 9.2.66 | PDCP-SN and Hyper frame number for which the target eNB/MeNB should discard forwarded DL SDUs associated with lower values in case of 15 bit long PDCP-SN | – | |
| >>>>DISCARD DL COUNT Value for PDCP SN Length 18 | O | | COUNT Value for PDCP SN Length 18 9.2.82 | PDCP-SN and Hyper frame number for which the target eNB/MeNB should discard forwarded DL SDUs associated with lower values in case of 18 bit long PDCP-SN | – | |
| SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated for EN-DC at the en-gNB. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256. |

## 9.1.2 Messages for global procedures

### 9.1.2.1 LOAD INFORMATION

This message is sent by an eNB to neighbouring eNBs to transfer load and interference co-ordination information.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| **Cell Information** | M | | | | YES | ignore |
| **>Cell Information Item** | | *1 .. <maxCellineNB>* | | | EACH | ignore |
| >>Cell ID | M | | ECGI 9.2.14 | Id of the source cell | – | |
| >>UL Interference Overload Indication | O | | 9.2.17 | | – | |
| **>>UL High Interference Information** | | *0 .. <maxCellineNB>* | | | – | |
| >>>Target Cell ID | M | | ECGI 9.2.14 | Id of the cell for which the HII is meant | – | |
| >>>UL High Interference Indication | M | | 9.2.18 | | – | |
| >>Relative Narrowband Tx Power (RNTP) | O | | 9.2.19 | | – | |
| >>ABS Information | O | | 9.2.54 | | YES | ignore |
| >>Invoke Indication | O | | 9.2.55 | | YES | ignore |
| >>Intended UL-DL Configuration | O | | ENUMERATED(sa0, sa1, sa2, sa3, sa4, sa5, sa6,…) | One of the UL-DL configurations defined in TS 36.211 [10]. The UL subframe(s) in the indicated configuration is subset of those in SIB1 UL-DL configuration. This IE applies to TDD only. | YES | ignore |
| >>Extended UL Interference Overload Info | O | | 9.2.67 | This IE applies to TDD only. | YES | ignore |
| >>CoMP Information | O | | 9.2.74 | | YES | ignore |
| >>Dynamic DL transmission information | O | | 9.2.77 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

### 9.1.2.2 ERROR INDICATION

This message is used to indicate that some error has been detected in the eNB/en-gNB.

Direction: $eNB_1 \rightarrow eNB_2$ or eNB → en-gNB or en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Old eNB UE X2AP ID | O | | eNB UE X2AP ID 9.2.24 | Allocated for handover at the source eNB and for dual connectivity at the SeNB or for a SN Status Transfer procedure at the eNB from which the E-RAB context is transferred. | YES | ignore |
| New eNB UE X2AP ID | O | | eNB UE X2AP ID 9.2.24 | Allocated for handover at the target eNB and for dual connectivity/EN-DC at the MeNB or for a SN Status Transfer procedure at the eNB to which the E-RAB context is transferred. | YES | ignore |
| Cause | O | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for handover at the source eNB and for dual connectivity at the SeNB or for a SN Status Transfer procedure at the eNB from which the E-RAB context is transferred. | YES | ignore |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated for handover at the target eNB and for dual connectivity/EN-DC at the MeNB or for a SN Status Transfer procedure at the eNB to which the E-RAB context is transferred. | YES | ignore |
| Old en-gNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated for EN-DC at the en-gNB. | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.3 X2 SETUP REQUEST

This message is sent by an eNB to a neighbouring eNB to transfer the initialization information for a TNL association.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Global eNB ID | M | | 9.2.22 | | YES | reject |
| **Served Cells** | | *1 .. <maxCellineNB>* | | Complete list of cells served by the eNB | YES | reject |
| **>**Served Cell Information | M | | 9.2.8 | | – | |
| **>Neighbour Information** | | *0 .. <maxnoofNeighbours>* | | | – | |
| >>ECGI | M | | ECGI 9.2.14 | E-UTRAN Cell Global Identifier of the neighbour cell | – | |
| >>PCI | M | | INTEGER (0..503, …) | Physical Cell Identifier of the neighbour cell | – | |
| >>EARFCN | M | | 9.2.26 | DL EARFCN for FDD or EARFCN for TDD | – | |
| >>TAC | O | | OCTET STRING (2) | Tracking Area Code | YES | ignore |
| >>EARFCN Extension | O | | 9.2.65 | DL EARFCN for FDD or EARFCN for TDD. If this IE is present, the value signalled in the *EARFCN* IE is ignored. | YES | reject |
| >NR Neighbour Information | O | | 9.2.98 | NR neighbour, capable of performing EN-DC with the served E-UTRA cell | YES | ignore |
| >Served Cell Specific Info Request | O | | 9.2.180 | | YES | ignore |
| **GU Group Id List** | | *0 .. <maxfPools>* | | List of all the pools to which the eNB belongs | GLOBAL | reject |
| >GU Group Id | M | | 9.2.20 | | - | |
| LHN ID | O | | 9.2.83 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |
| maxnoofNeighbours | Maximum no. of neighbour cells associated to a given served cell. Value is 512. |
| maxPools | Maximum no. of pools an eNB can belong to. Value is 16. |

### 9.1.2.4 X2 SETUP RESPONSE

This message is sent by an eNB to a neighbouring eNB to transfer the initialization information for a TNL association.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Global eNB ID | M | | 9.2.22 | | YES | reject |
| **Served Cells** | | *1 .. <maxCellineNB>* | | Complete list of cells served by the eNB | GLOBAL | reject |
| >Served Cell Information | M | | 9.2.8 | | – | |
| **>Neighbour Information** | | *0 .. <maxnoofNeighbours>* | | | – | |
| >>ECGI | M | | ECGI 9.2.14 | E-UTRAN Cell Global Identifier of the neighbour cell | – | |
| >>PCI | M | | INTEGER (0..503, …) | Physical Cell Identifier of the neighbour cell | – | |
| >>EARFCN | M | | 9.2.26 | DL EARFCN for FDD or EARFCN for TDD | – | |
| >>TAC | O | | OCTET STRING (2) | Tracking Area Code | YES | ignore |
| >>EARFCN Extension | O | | 9.2.65 | DL EARFCN for FDD or EARFCN for TDD. If this IE is present, the value signalled in the *EARFCN* IE is ignored. | YES | reject |
| >NR Neighbour Information | O | | 9.2.98 | NR neighbour, capable of performing EN-DC with the served E-UTRA cell | YES | ignore |
| **GU Group Id List** | | *0 .. <maxPools>* | | List of all the pools to which the eNB belongs | GLOBAL | reject |
| >GU Group Id | M | | 9.2.20 | | - | |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| LHN ID | O | | 9.2.83 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |
| maxnoofNeighbours | Maximum no. of neighbour cells associated to a given served cell. Value is 512. |
| maxPools | Maximum no. of pools an eNB can belong to. Value is 16. |

### 9.1.2.5 X2 SETUP FAILURE

This message is sent by the eNB to indicate X2 Setup failure.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Time To Wait | O | | 9.2.32 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.6 RESET REQUEST

This message is sent from one eNB to another eNB/en-gNB or from en-gNB to an eNB and is used to request the X2 interface between the two eNB or between an eNB and an en-gNB to be reset.

Direction: $eNB_1 \rightarrow eNB_2$, eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.7 RESET RESPONSE

This message is sent by a eNB/en-gNB as a response to a RESET REQUEST message.

Direction: $eNB_2 \rightarrow eNB_1$, eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.8 ENB CONFIGURATION UPDATE

This message is sent by an eNB to a peer eNB to transfer updated information for a TNL association.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| **Served Cells To Add** | | *0 .. <maxCellineNB>* | | Complete list of added cells served by the eNB | GLOBAL | reject |
| >Served Cell Information | M | | 9.2.8 | | – | |
| **>Neighbour Information** | | *0 .. <maxnoofNeighbours>* | | | – | |
| >>ECGI | M | | ECGI 9.2.14 | E-UTRAN Cell Global Identifier of the neighbour cell | – | |
| >>PCI | M | | INTEGER (0..503, …) | Physical Cell Identifier of the neighbour cell | – | |
| >>EARFCN | M | | 9.2.26 | DL EARFCN for FDD or EARFCN for TDD | – | |
| >>TAC | O | | OCTET STRING (2) | Tracking Area Code | YES | ignore |
| >>EARFCN Extension | O | | 9.2.65 | DL EARFCN for FDD or EARFCN for TDD. If this IE is present, the value signalled in the *EARFCN* IE is ignored. | YES | reject |
| >NR Neighbour Information | O | | 9.2.98 | NR neighbour, capable of performing EN-DC with the served E-UTRA cell | YES | ignore |
| **Served Cells To Modify** | | *0 .. <maxCellineNB>* | | Complete list of modified cells served by the eNB | GLOBAL | reject |
| >Old ECGI | M | | ECGI 9.2.14 | Old E-UTRAN Cell Global Identifier | - | |
| >Served Cell Information | M | | 9.2.8 | | – | |
| **>Neighbour Information** | | *0 .. <maxnoofNeighbours>* | | | – | |
| >>ECGI | M | | ECGI 9.2.14 | E-UTRAN Cell Global Identifier of the neighbour cell | – | |
| >>PCI | M | | INTEGER (0..503, …) | Physical Cell Identifier of the neighbour cell | – | |
| >>EARFCN | M | | 9.2.26 | DL EARFCN for FDD or EARFCN for TDD | – | |
| >>TAC | O | | OCTET STRING (2) | Tracking Area Code | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>EARFCN Extension | O | | 9.2.65 | DL EARFCN for FDD or EARFCN for TDD. If this IE is present, the value signalled in the *EARFCN* IE is ignored. | YES | reject |
| >NR Neighbour Information | O | | 9.2.98 | NR neighbour, capable of performing EN-DC with the served E-UTRA cell | YES | ignore |
| >Deactivation Indication | O | | ENUMERATED(deactivated, …) | Indicates that the concerned cell is switched off for energy saving reasons | YES | ignore |
| **Served Cells To Delete** | | *0 .. <maxCellineNB>* | | Complete list of deleted cells served by the eNB | GLOBAL | reject |
| >Old ECGI | M | | ECGI 9.2.14 | Old E-UTRAN Cell Global Identifier of the cell to be deleted | - | |
| **GU Group Id To Add List** | | *0 .. <maxPools>* | | | GLOBAL | reject |
| >GU Group Id | M | | 9.2.20 | | - | |
| **GU Group Id To Delete List** | | *0 .. <maxPools>* | | | GLOBAL | reject |
| >GU Group Id | M | | 9.2.20 | | - | |
| Coverage Modification List | | *0 .. <maxCellineNB>* | | List of cells with modified coverage | GLOBAL | reject |
| >ECGI | M | | ECGI 9.2.14 | E-UTRAN Cell Global Identifier of the cell to be modified | - | |
| >Cell Coverage State | M | | INTEGER (0..15, …) | Value '0' indicates that the cell is inactive. Other values Indicates that the cell is active and also indicates the coverage configuration of the concerned cell | - | |
| >Cell Deployment Status Indicator | O | | ENUMERATED(pre-change-notification, ...) | Indicates the Cell Coverage State is planned to be used at the next reconfiguration | | |
| >Cell Replacing Info | C-ifCellDeploymentStatusIndicatorPresent | | | | | |
| >>Replacing Cells | | *0 .. <maxCellineNB>* | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>ECGI | | | ECGI 9.2.14 | E-UTRAN Cell Global Identifier of a cell that may replace all or part of the coverage of the cell to be modified | | |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |
| maxnoofNeighbours | Maximum no. of neighbour cells associated to a given served cell. Value is 512. |
| maxPools | Maximum no. of pools an eNB can belong to. Value is 16. |

| Condition | Explanation |
|---|---|
| ifCellDeploymentStatusIndicatorPresent | This IE shall be present if the *Cell Deployment Status Indicator* IE is present. |

### 9.1.2.9 ENB CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by an eNB to a peer eNB to acknowledge update of information for a TNL association.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.10 ENB CONFIGURATION UPDATE FAILURE

This message is sent by an eNB to a peer eNB to indicate eNB Configuration Update Failure.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Time To Wait | O | | 9.2.32 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.11 RESOURCE STATUS REQUEST

This message is sent by an $eNB_1$ to neighbouring $eNB_2$ to initiate the requested measurement according to the parameters given in the message.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| eNB1 Measurement ID | M | | INTEGER (1..4095,...) | Allocated by $eNB_1$ | YES | reject |
| eNB2 Measurement ID | C-ifRegistrationRequestStoporPartialStoporAdd | | INTEGER (1..4095,...) | Allocated by $eNB_2$ | YES | ignore |
| Registration Request | M | | ENUMERATED(start, stop, …, partial stop, add) | Type of request for which the resource status is required. | YES | reject |
| Report Characteristics | O | | BITSTRING (SIZE(32)) | Each position in the bitmap indicates measurement object the $eNB_2$ is requested to report.<br>First Bit = PRB Periodic,<br>Second Bit = TNL load Ind Periodic,<br>Third Bit = HW Load Ind Periodic,<br>Fourth Bit = Composite Available Capacity Periodic, this bit should be set to 1 if at least one of the First, Second or Third bits is set to 1,<br>Fifth Bit = ABS Status Periodic,<br>Sixth Bit = RSRP Measurement Report Periodic,<br>Seventh Bit = CSI Report Periodic, Eighth Bit = CAC for Possibly Aggregated Cells Periodic.<br>Other bits shall be ignored by the $eNB_2$. | YES | reject |
| **Cell To Report** | | *1* | | Cell ID list to which the request applies. | YES | ignore |
| **>Cell To Report Item** | | *1 .. <maxCellineNB>* | | | EACH | ignore |
| >>Cell ID | M | | ECGI 9.2.14 | | – | |
| Reporting Periodicity | O | | ENUMERATED(1000ms, 2000ms, 5000ms,10000ms, …) | Periodicity that can be used for reporting of PRB Periodic, TNL Load Ind Periodic, HW Load Ind Periodic, Composite Available Capacity Periodic or ABS Status Periodic. | YES | ignore |
| Partial Success Indicator | O | | ENUMERATED(partial success allowed, ...) | Included if partial success is allowed | YES | ignore |
| Reporting Periodicity of RSRP Measurement Report | O | | ENUMERATED(120ms, 240ms, 480ms, 640ms, ...) | Periodicity that can be used for the reporting of RSRP Measurement Report Periodic. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| Reporting Periodicity of CSI Report | O | | ENUMERATED(5ms, 10ms, 20ms, 40ms, 80ms, ...) | Periodicity that can be used for the reporting of CSI Report Periodic. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

| Condition | Explanation |
|---|---|
| ifRegistrationRequestStoporPartialStoporAdd | This IE shall be present if the *Registration Request* IE is set to the value "stop", "partial stop" or "add". |

### 9.1.2.12 RESOURCE STATUS RESPONSE

This message is sent by the $eNB_2$ to indicate that the requested measurement, for all or for a subset of the measurement objects included in the measurement is successfully initiated.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| eNB1 Measurement ID | M | | INTEGER (1..4095,...) | Allocated by $eNB_1$ | YES | reject |
| eNB2 Measurement ID | M | | INTEGER (1..4095,...) | Allocated by $eNB_2$ | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| **Measurement Initiation Result** | | *0..1* | | List of all cells in which measurement objects were requested, included when indicating partial success | YES | ignore |
| >**Measurement Initiation Result Item** | | *1 .. <maxCellineNB>* | | | EACH | ignore |
| >>Cell ID | M | | ECGI 9.2.14 | | – | |
| >>**Measurement Failure Cause List** | | *0..1* | | Indicates that $eNB_2$ could not initiate the measurement for at least one of the requested measurement objects in the cell | – | |
| >>>**Measurement Failure Cause Item** | | *1 .. <maxFailedMeasObjects>* | | | EACH | ignore |
| >>>>Measurement Failed Report Characteristics | M | | BITSTRING (SIZE(32)) | Each position in the bitmap indicates measurement object that failed to be initiated in the $eNB_2$. First Bit = PRB Periodic, Second Bit = TNL load Ind Periodic, Third Bit = HW Load Ind Periodic, Fourth Bit = Composite Available Capacity Periodic, Fifth Bit = ABS Status Periodic, Sixth Bit = RSRP Measurement Report Periodic, Seventh Bit = CSI Report Periodic, Eighth Bit = CAC for Possibly Aggregated Cells Periodic. Other bits shall be ignored by the $eNB_1$. | – | |
| >>>>Cause | M | | 9.2.6 | Failure cause for measurement objects for which the measurement cannot be initiated | – | |

| Range bound | Explanation |
|---|---|
| maxFailedMeasObjects | Maximum number of measurement objects that can fail per measurement. Value is 32. |
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

### 9.1.2.13 RESOURCE STATUS FAILURE

This message is sent by the $eNB_2$ to indicate that for none of the requested measurement objects the measurement can be initiated.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| eNB1 Measurement ID | M | | INTEGER (1..4095,...) | Allocated by $eNB_1$ | YES | reject |
| eNB2 Measurement ID | M | | INTEGER (1..4095,...) | Allocated by $eNB_2$ | YES | reject |
| Cause | M | | 9.2.6 | Ignored by the receiver when the Complete Failure Cause Information IE is included | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| **Complete Failure Cause Information** | | *0..1* | | Complete list of failure causes for all requested cells | YES | ignore |
| >**Complete Failure Cause Information Item** | | *1 .. <maxCellineNB>* | | | EACH | ignore |
| >>Cell ID | M | | ECGI 9.2.14 | | – | |
| >>**Measurement Failure Cause List** | | *1* | | | – | |
| >>>**Measurement Failure Cause Item** | | *1 .. <maxFailedMeasObjects>* | | | EACH | ignore |
| >>>>Measurement Failed Report Characteristics | M | | BITSTRING (SIZE(32)) | Each position in the bitmap indicates measurement object that failed to be initiated in the $eNB_2$. First Bit = PRB Periodic, Second Bit = TNL load Ind Periodic, Third Bit = HW Load Ind Periodic, Fourth Bit = Composite Available Capacity Periodic, Fifth Bit = ABS Status Periodic, Sixth Bit = RSRP Measurement Report Periodic, Seventh Bit = CSI Report Periodic, Eighth Bit = CAC for Possibly Aggregated Cells Periodic. Other bits shall be ignored by the $eNB_1$. | – | |
| >>>>Cause | M | | 9.2.6 | Failure cause for measurements that cannot be initiated | – | |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

| | |
|---|---|
| maxFailedMeasObjects | Max number of measurement objects that can fail per measurement. Value is 32. |

## 9.1.2.14 RESOURCE STATUS UPDATE

This message is sent by $eNB_2$ to neighbouring $eNB_1$ to report the results of the requested measurements.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| eNB1 Measurement ID | M | | INTEGER (1..4095,...) | Allocated by $eNB_1$ | YES | reject |
| eNB2 Measurement ID | M | | INTEGER (1..4095,...) | Allocated by $eNB_2$ | YES | reject |
| **Cell Measurement Result** | | *1* | | | YES | ignore |
| **>Cell Measurement Result Item** | | *1 .. <maxCellineNB>* | | | EACH | ignore |
| >>Cell ID | M | | ECGI 9.2.14 | | – | |
| >>Hardware Load Indicator | O | | 9.2.34 | | – | |
| >>S1 TNL Load Indicator | O | | 9.2.35 | | – | |
| >>Radio Resource Status | O | | 9.2.37 | | – | |
| >>Composite Available Capacity Group | O | | 9.2.44 | | YES | ignore |
| >>ABS Status | O | | 9.2.58 | | YES | ignore |
| >>RSRP Measurement Report List | O | | 9.2.76 | | YES | ignore |
| >>CSI Report | O | | 9.2.79 | | YES | ignore |
| >>Cell Reporting Indicator | O | | ENUMERATED(stop request, ...) | | YES | ignore |
| >>**Measurement Result for NR Cells Possibly Aggregated** | | *0..1* | | | YES | ignore |
| >>>**Measurement Result for NR Cells Possibly Aggregated Item** | | *1 .. < maxnoofReportedNRCellsPossiblyAggregated >* | | | – | |
| >>>>Cell ID | M | | NR CGI 9.2.111 | | – | |
| >>>>NR Composite Available Capacity Group | O | | 9.2.163 | | – | |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |
| *maxnoofReportedNRCellsPossiblyAggregated* | Maximum no. of reported NR cells that may be aggregated with the serving cell. Value is 16. |

## 9.1.2.15 MOBILITY CHANGE REQUEST

This message is sent by an $eNB_1$ to neighbouring $eNB_2$ to initiate adaptation of mobility parameters.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| eNB1 Cell ID | M | | ECGI 9.2.14 | | YES | reject |
| eNB2 Cell ID | M | | ECGI 9.2.14 | | YES | reject |
| eNB1 Mobility Parameters | O | | Mobility Parameters Information 9.2.48 | Configuration change in $eNB_1$ cell | YES | ignore |
| eNB2 Proposed Mobility Parameters | M | | Mobility Parameters Information 9.2.48 | Proposed configuration change in $eNB_2$ cell | YES | reject |
| Cause | M | | 9.2.6 | | YES | reject |

### 9.1.2.16 MOBILITY CHANGE ACKNOWLEDGE

This message is sent by the $eNB_2$ to indicate that the $eNB_2$ Proposed Mobility Parameter proposed by $eNB_1$ was accepted.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| eNB1 Cell ID | M | | ECGI 9.2.14 | | YES | reject |
| eNB2 Cell ID | M | | ECGI 9.2.14 | | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.17 MOBILITY CHANGE FAILURE

This message is sent by the $eNB_2$ to indicate that the $eNB_2$ Proposed Mobility Parameter proposed by $eNB_1$ was refused.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| eNB1 Cell ID | M | | ECGI 9.2.14 | | YES | ignore |
| eNB2 Cell ID | M | | ECGI 9.2.14 | | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Mobility Parameters Modification Range | O | | 9.2.49 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.18 RLF INDICATION

This message is sent by the $eNB_2$ to indicate an RRC re-establishment attempt or a reception of an RLF Report from a UE that suffered a connection failure at $eNB_1$.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Failure cell PCI | M | | INTEGER (0..503, …) | Physical Cell Identifier | YES | ignore |
| Re-establishment cell ECGI | M | | ECGI 9.2.14 | | YES | ignore |
| C-RNTI | M | | BIT STRING (SIZE (16)) | Corresponds to information provided in the *c-RNTI* contained in the *RRCConnection Reestablishment Request* message (TS 36.331 [9]) | YES | ignore |
| ShortMAC-I | O | | BIT STRING (SIZE (16)) | Corresponds to information provided in the *shortMAC-I* contained in the *RRCConnection Reestablishment Request* message (TS 36.331 [9]) | YES | ignore |
| UE RLF Report Container | O | | OCTET STRING | Includes the *rLF-Report-r9* contained in the *UEInformationResponse* message (TS 36.331 [9]) | YES | ignore |
| RRC Conn Setup Indicator | O | | ENUMERATED(RRC Conn Setup, ...) | Included if the RLF Report within the *UE RLF Report Container* IE is retrieved after an RRC connection setup or an incoming successful handover | YES | reject |
| RRC Conn Reestab Indicator | O | | ENUMERATED(reconfigurationFailure, handoverFailure, otherFailure, ...) | Corresponds to information provided in the *reestablishmentCause* in the *RRCConnection Reestablishment Request* message (TS 36.331 [9]) | YES | ignore |
| UE RLF Report Container for extended bands | O | | OCTET STRING | Includes the *rlf-Report-v9e0* contained in the *UEInformationResponse* message (TS 36.331 [9]) | YES | ignore |
| NB-IoT RLF Report Container | O | | OCTET STRING | Includes the *rlf-Report-NB* IE contained in the *UEInformationResponse-NB* message (TS 36.331 [9]) | YES | ignore |

### 9.1.2.19 HANDOVER REPORT

This message is sent by the $eNB_1$ to report a handover failure event or other critical mobility problem.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| Handover Report Type | M | | ENUMERATED (HO too early, HO to wrong cell, …, InterRAT ping-pong, Inter System ping-pong) | | YES | ignore |
| Handover Cause | M | | Cause 9.2.6 | Indicates handover cause employed for handover from $eNB_2$ | YES | ignore |
| Source cell ECGI | M | | ECGI 9.2.14 | ECGI of source cell for handover procedure (in $eNB_2$) | YES | ignore |
| Failure cell ECGI | M | | ECGI 9.2.14 | ECGI of target cell for handover procedure (in $eNB_1$) | YES | ignore |
| Re-establishment cell ECGI | C-ifHandoverReportType HoToWrongCell | | ECGI 9.2.14 | ECGI of cell where UE attempted re-establishment | YES | ignore |
| Target cell in UTRAN | C-ifHandoverReportType InterRATpingpong | | OCTET STRING | Encoded according to *UTRAN Cell ID* in the *Last Visited UTRAN Cell Information* IE, as defined in in TS 25.413 [24] | YES | ignore |
| Source cell C-RNTI | O | | BIT STRING (SIZE (16)) | Corresponds to information provided in the *sourceUE-Identity* (a C-RNTI) allocated at the source eNB (in $eNB_2$) contained in the *AS-Config* IE as specified in TS 36.331 [9]. | YES | ignore |
| Mobility Information | O | | BIT STRING (SIZE (32)) | Information provided in the HANDOVER REQUEST message from $eNB_2$. | YES | ignore |
| UE RLF Report Container | O | | OCTET STRING | Includes the *UE RLF Report Container* IE received in the RLF INDICATION message. | YES | ignore |
| UE RLF Report Container for extended bands | O | | OCTET STRING | Includes the *UE RLF Report Container for extended bands* IE received in the RLF INDICATION message. | YES | ignore |
| Target cell in NG-RAN | C-ifHandoverReportType Inter-systempingpong | | OCTET STRING | Encoded according to *NG-RAN CGI* IE in TS 38.413 [39]. | YES | ignore |

| Condition | Explanation |
|---|---|
| ifHandoverReportType HoToWrongCell | This IE shall be present if the *Handover Report Type* IE is set to the value "HO to wrong cell" |
| ifHandoverReportType InterRATpingpong | This IE shall be present if the *Handover Report Type* IE is set to the value "InterRAT ping-pong" |
| ifHandoverReportType Inter-system pingpong | This IE shall be present if the Handover Report Type IE is set to the value "Inter-system ping-pong" |

### 9.1.2.20 CELL ACTIVATION REQUEST

This message is sent by an eNB to a peer eNB to request a previously switched-off cell(s) to be re-activated.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| **Served Cells To Activate** | | *1 .. <maxCellineNB>* | | | GLOBAL | reject |
| >ECGI | M | | 9.2.14 | | - | |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

### 9.1.2.21 CELL ACTIVATION RESPONSE

This message is sent by an eNB to a peer eNB to indicate that one or more cell(s) previously switched-off has(have) been activated.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| **Activated Cell List** | | *1 .. <maxCellineNB>* | | | GLOBAL | ignore |
| >ECGI | M | | 9.2.14 | | - | |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

### 9.1.2.22 CELL ACTIVATION FAILURE

This message is sent by an eNB to a peer eNB to indicate cell activation failure.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.23 X2 RELEASE

This message is used to indicate that the signalling connection to an eNB is unavailable.

Direction: $eNB_1$ → $eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Global eNB ID | M | | 9.2.22 | | YES | reject |

### 9.1.2.24 X2AP MESSAGE TRANSFER

This message is used for indirect transport of an X2AP message (except the X2AP MESSAGE TRANSFER message) between two eNBs, and to allow an eNB to perform registration.

Direction: $eNB_1 \rightarrow eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| RNL Header | M | | 9.2.68 | | YES | reject |
| X2AP Message | O | | OCTET STRING | Includes any X2AP message except the X2AP MESSAGE TRANSFER message | YES | reject |

### 9.1.2.25 X2 REMOVAL REQUEST

This message is sent by an eNB to a neighbouring eNB to initiate the removal of the signaling connection.

Direction: $eNB_1$ → $eNB_2$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Global eNB ID | M | | 9.2.22 | | YES | reject |
| X2 Removal Threshold | O | | X2 Benefit Value 9.2.90 | | YES | reject |

### 9.1.2.26 X2 REMOVAL RESPONSE

This message is sent by an eNB to a neighbouring eNB to acknowledge the initiation of removal of the signaling connection.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Global eNB ID | M | | 9.2.22 | | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.27 X2 REMOVAL FAILURE

This message is sent by the eNB to indicate that removing the signaling connection cannot be accepted.

Direction: $eNB_2 \rightarrow eNB_1$.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.28 RETRIEVE UE CONTEXT REQUEST

This message is sent by the new eNB to request the old eNB to transfer the UE Context to the new eNB.

Direction: new eNB → old eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the new eNB | YES | reject |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the new eNB<br>See Note 1) | YES | reject |
| Resume ID | M | | 9.2.91 | | YES | reject |
| ShortMAC-I | M | | BIT STRING (SIZE (16)) | **RRC Resume:** Corresponds to information provided in the *shortResumeMAC-I* in the *RRCConnection ResumeRequest* message as defined in TS 36.331 [9]<br><br>**RRC Reestablishment:** Corresponds to information provided in the *shortMAC-I* in the *RRCConnectionReestablishmentRequest*message as defined in TS 36.331 [9]. | YES | reject |
| New E-UTRAN Cell Identifier | M | | BIT STRING (SIZE (28)) | **RRC Resume:** Corresponds to information provided in the *cellIdentity* within the *VarShortResumeMAC-Input* as specified in TS 36.331 [9].<br><br>**RRC Reestablishment:** Corresponds to information provided in the *cellIdentity* within the *VarShortMAC-Input* as specified in TS 36.331 [9]. | YES | reject |
| C-RNTI | O | | BIT STRING (SIZE (16)) | Corresponds to information provided in the c-RNTI contained in the *RRCReestablishmentRequest* message as specified in TS 36.331 [9]. If this IE is present, the *Resume ID* IE is ignored | YES | reject |
| Failure cell PCI | O | | INTEGER (0..503, …) | Physical Cell Identifier | YES | reject |

NOTE 1: The ASN.1 definition of the RETRIEVE UE CONTEXT REQUEST message contains the a wrong IE-Id, which references the *SeNB UE X2AP ID Extension* IE instead of the *New eNB UE X2AP ID Extension* IE. The old eNB interprets the content of this IE as the Extended eNB UE X2AP ID, which, together with the *New eNB UE X2AP ID* IE represents the eNB UE X2AP ID allocated at the new eNB.

### 9.1.2.29 RETRIEVE UE CONTEXT RESPONSE

This message is sent by the old eNB to transfer the UE context to the new eNB.

Direction: old eNB → new eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the new eNB | YES | ignore |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the new eNB | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the old eNB | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the old eNB | YES | ignore |
| GUMMEI | M | | 9.2.16 | | YES | reject |
| **UE Context Information** | | 1 | | | YES | reject |
| >MME UE S1AP ID | M | | INTEGER (0..$2^{32}$ - 1) | MME UE S1AP ID allocated at the MME | – | |
| >UE Security Capabilities | M | | 9.2.29 | | – | |
| >AS Security Information | M | | 9.2.30 | | – | |
| >UE Aggregate Maximum Bit Rate | M | | 9.2.12 | | – | |
| >Subscriber Profile ID for RAT/Frequency priority | O | | 9.2.25 | | – | |
| **>E-RABs To Be Setup List** | | 1 | | | – | |
| **>>E-RABs To Be Setup Item** | | 1 .. <maxnoofBearers> | | | EACH | ignore |
| >>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>E-RAB Level QoS Parameters | M | | 9.2.9 | Includes necessary QoS parameters | – | |
| >>>Bearer Type | O | | 9.2.92 | | – | |
| >>>UL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1 transport bearer. For delivery of UL PDUs. | YES | reject |
| >>>DL Forwarding | O | | 9.2.5 | | YES | ignore |
| >>>Ethernet Type | O | | 9.2.157 | | YES | ignore |
| >>>Security Indication | O | | 9.2.181 | | YES | reject |
| >RRC Context | M | | OCTET STRING | Includes either the *HandoverPreparation Information* message as defined in subclause 10.2.2 of TS 36.331 [9], or the *HandoverPreparationInformation-NB* message as defined in subclause 10.6.2 of TS 36.331 [9]. | – | |
| >Handover Restriction List | O | | 9.2.3 | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >Location Reporting Information | O | | 9.2.21 | Includes the necessary parameters for location reporting | – | |
| >Management Based MDT Allowed | O | | 9.2.59 | | – | |
| >Management Based MDT PLMN List | O | | MDT PLMN List 9.2.64 | | – | |
| >UE Sidelink Aggregate Maximum Bit Rate | O | | 9.2.97 | This IE applies only if the UE is authorized for V2X services. | YES | ignore |
| >Additional RRM Policy Index | O | | 9.2.25a | | YES | ignore |
| >EPC Handover Restriction List Container | O | | 9.2.153 | | YES | ignore |
| >NR UE Sidelink Aggregate Maximum Bit Rate | O | | 9.2.159 | This IE applies only if the UE is authorized for NR V2X services. | YES | ignore |
| >UE Radio Capability ID | O | | 9.2.171 | | YES | reject |
| >IMS voice EPS fallback from 5G | O | | ENUMERATED (true, …) | | YES | ignore |
| Trace Activation | O | | 9.2.2 | | YES | ignore |
| SRVCC Operation Possible | O | | 9.2.33 | | YES | ignore |
| Masked IMEISV | O | | 9.2.69 | | YES | ignore |
| Expected UE Behaviour | O | | 9.2.70 | | YES | ignore |
| ProSe Authorized | O | | 9.2.78 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| V2X Services Authorized | O | | 9.2.93 | | YES | ignore |
| Aerial UE subscription information | O | | 9.2.129 | | YES | ignore |
| Subscription Based UE Differentiation Information | O | | 9.2.136 | | YES | ignore |
| NR V2X Services Authorized | O | | 9.2.158 | | YES | ignore |
| PC5 QoS Parameters | O | | 9.2.160 | This IE applies only if the UE is authorized for NR V2X services. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.2.30 RETRIEVE UE CONTEXT FAILURE

This message is sent by the old eNB to inform the new eNB that the Retrieve UE Context procedure has failed.

Direction: old eNB → new eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the new eNB | YES | ignore |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the new eNB | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.2.31 EN-DC X2 SETUP REQUEST

This message is sent by an initiating node to a neighbouring node, both nodes able to interact for EN-DC, to transfer the initialization information for a TNL association.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE *Initiating NodeType* | M | | | | YES | reject |
| *>eNB* | | | | | | |
| >>Global eNB ID | M | | 9.2.22 | | YES | reject |
| **>>List of Served E-UTRA Cells** | | *1 .. <maxCellineNB>* | | Complete list of cells served by the eNB | YES | reject |
| >>>Served E-UTRA Cell Information | M | | Served Cell Information 9.2.8 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours | – | |
| >>Interface Instance Indication | O | | 9.2.143 | NOTE: In the current version of this specification this IE is not included in the *Initiating Node Type* IE. | YES | reject |
| >>Cell and Capacity Assistance Information | O | | 9.2.146 | | YES | ignore |
| *>en-gNB* | | | | | | |
| >>Global en-gNB ID | M | | 9.2.112 | | YES | reject |
| **>>List of Served NR Cells** | | *1 .. <maxCellinengNB>* | | List of cells served by the en-gNB. If a partial list of cells is signalled, it contains at least one cell per carrier configured at the gNB. | YES | reject |
| >>>Served NR Cell Information | M | | 9.2.110 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours. | – | |
| >>>Served Cell Specific Info Request | O | | 9.2.180 | | YES | ignore |
| >>Partial List Indicator | O | | ENUMERATED (partial, ...) | Value "partial" indicates that a partial list of cells is included in the *List of Served NR Cells* IE | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |
| TNL Transport Layer Address info | O | | 9.2.149 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |

### 9.1.2.32 EN-DC X2 SETUP RESPONSE

This message is sent by a neighbouring node to an initiating node, both nodes able to interact for EN-DC, to transfer the initialization information for a TNL association.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE *Responding NodeType* | M | | | | YES | reject |
| *>eNB* | | | | | | |
| >>Global eNB ID | M | | 9.2.22 | | YES | reject |
| **>>List of Served E-UTRA Cells** | | *1 .. <maxCellineNB>* | | Complete list of cells served by the eNB | YES | reject |
| >>>Served E-UTRA Cell Information | M | | Served Cell Information 9.2.8 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours | – | |
| >>Cell and Capacity Assistance Information | O | | 9.2.146 | | YES | ignore |
| *>en-gNB* | | | | | | |
| >>Global en-gNB ID | M | | 9.2.112 | | YES | reject |
| **>>List of Served NR Cells** | | *1 .. <maxCellinengNB>* | | List of cells served by the en-gNB. If a partial list of cells is signalled, it contains at least one cell per carrier configured at the gNB. | YES | reject |
| >>>Served NR Cell Information | M | | 9.2.110 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours | – | |
| >>Partial List Indicator | O | | ENUMERATED (partial, ...) | Value "partial" indicates that a partial list of cells is included in the *List of Served NR Cells* IE | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |
| TNL Transport Layer Address info | O | | 9.2.149 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |

### 9.1.2.33 EN-DC X2 SETUP FAILURE

This message is sent by the neighbouring node to indicate EN-DC X2 Setup failure.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Time To Wait | O | | 9.2.32 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |
| Message Oversize Notification | O | | 9.2.148 | | YES | ignore |

### 9.1.2.34 EN-DC CONFIGURATION UPDATE

This message is sent by an initiating node to a peer neighbouring node, both nodes able to interact for EN-DC, to transfer updated information for a TNL association.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE Initiating NodeType | M | | | | YES | reject |
| *>eNB* | | | | | | |
| >>Cell Assistance Information | O | | 9.2.115 | | YES | reject |
| **>>Served E-UTRA Cells To Add** | | *0 .. <maxCellineNB>* | | | GLOBAL | reject |
| >>>Served E-UTRA Cell Information | M | | Served Cell Information 9.2.8 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours | – | |
| **>>Served E-UTRA Cells To Modify** | | *0 .. <maxCellineNB>* | | Complete list of modified cells served by the eNB | GLOBAL | reject |
| >>>Old ECGI | M | | ECGI 9.2.14 | Old E-UTRAN Cell Global Identifier | – | |
| >>>Served E-UTRA Cell Information | M | | Served Cell Information 9.2.8 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours | – | |
| **>>Served E-UTRA Cells To Delete** | | *0 .. <maxCellineNB>* | | Complete list of deleted cells served by the eNB | GLOBAL | reject |
| >>>Old ECGI | M | | ECGI 9.2.14 | Old E-UTRAN Cell Global Identifier of the cell to be deleted | - | |
| *>en-gNB* | | | | | | |
| **>>Served NR Cells To Add** | | *0 .. <maxCellinengNB>* | | | GLOBAL | reject |
| >>>Served NR Cell Information | M | | 9.2.110 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours | – | |
| >>>Served Cell Specific Info Request | O | | 9.2.180 | | YES | ignore |
| **>>Served NR Cells To Modify** | | *0 .. <maxCellinengNB>* | | | GLOBAL | reject |
| >>>Old NR-CGI | M | | NR CGI 9.2.111 | | - | |
| >>>Served NR Cell Information | M | | 9.2.110 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours | – | |
| >>>NR Deactivation Indication | O | | ENUMERATED(deactivated, …) | Indicates that the concerned NR cell is switched off for energy saving reasons. If this IE is not included, indicates that the concerned cell is activated. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| **>>Served NR Cells To Delete** | | *0 .. <maxCellinengNB>* | | | GLOBAL | reject |
| >>>Old NR-CGI | M | | NR CGI 9.2.111 | | - | |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |
| TNL Transport Layer Address info | O | | 9.2.149 | | YES | ignore |
| **TNLA To Add List** | | *0..1* | | | YES | ignore |
| >**TNLA To Add Item IEs** | | *1..<maxnoofTNLAssociations>* | | | – | |
| >>TNLA Transport Layer Information | M | | 9.2.150 | CP Transport Layer Information of the en-gNB | - | - |
| >>TNLA Usage | M | | 9.2.151 | | - | - |
| **TNLA To Update List** | | *0..1* | | | YES | ignore |
| >**TNLA To Update Item IEs** | | *1..<maxnoofTNLAssociations>* | | | – | |
| >>TNLA Transport Layer Information | M | | 9.2.150 | CP Transport Layer Information of the en-gNB | - | - |
| >>TNLA Usage | O | | 9.2.151 | | - | - |
| **TNLA To Remove List** | | *0..1* | | | YES | ignore |
| >**TNLA To Remove Item IEs** | | *1..<maxnoofTNLAssociations>* | | | – | |
| >>TNLA Transport Layer Information | M | | 9.2.150 | CP Transport Layer Information of the en-gNB | - | - |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |
| maxnoofTNLAssociations | Maximum numbers of TNL Associations between the eNB and the en-gNB. Value is 32. |

### 9.1.2.35 EN-DC CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by a neighbouring node to a peer node, both nodes able to interact for EN-DC, to acknowledge update of information for a TNL association.

Direction: en-gNB → eNB, eNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE Responding NodeType | M | | | | YES | reject |
| >*eNB* | | | | | | |
| >*en-gNB* | | | | | | |
| **>>List of Served NR Cells** | | 0 .. <maxCellinengNB> | | Complete or limited list of cells served by the en-gNB, if requested by the eNB. | – | |
| >>>Served NR Cell Information | M | | 9.2.110 | | – | |
| >>>NR Neighbour Information | O | | 9.2.98 | NR neighbours. | – | |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| TNL Transport Layer Address info | O | | 9.2.149 | | YES | ignore |
| **TNLA Setup List** | | 0..1 | | | YES | ignore |
| >**TNLA Setup Item** | | 1..<maxnoofTNL Associations> | | | – | |
| >>TNLA Transport Layer Address | M | | 9.2.150 | CP Transport Layer Information of the en-gNB | – | |
| **TNLA Failed to Setup List** | | 0..1 | | | YES | ignore |
| >**TNLA Failed To Setup Item** | | 1..<maxnoofTNL Associations> | | | – | |
| >>TNLA Transport Layer Address | M | | 9.2.150 | CP Transport Layer Information of the en-gNB | – | |
| >>Cause | M | | 9.2.6 | | – | |

| Range bound | Explanation |
|---|---|
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |
| maxnoofTNLAssociations | Maximum numbers of TNL Associations between the eNB and the en-gNB. Value is 32. |

### 9.1.2.36 EN-DC CONFIGURATION UPDATE FAILURE

This message is sent by a neighbouring node to a peer node to indicate EN-DC eNB Configuration Update Failure.

Direction: en-gNB → eNB, eNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Time To Wait | O | | 9.2.32 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.37 EN-DC CELL ACTIVATION REQUEST

This message is sent by an eNB to a peer en-gNB to request a previously switched-off cell(s) to be re-activated.

Direction: eNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| **Served NR Cells To Activate** | | *1* | | | GLOBAL | reject |
| **>Served NR Cells To Activate Item** | | *1 .. < maxCellinengNB >* | | | | |
| >>NR CGI | M | | 9.2.111 | | - | |
| Activation ID | M | | INTEGER (0..255) | Allocated by the eNB | YES | reject |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

| Range bound | Explanation |
|---|---|
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |

### 9.1.2.38 EN-DC CELL ACTIVATION RESPONSE

This message is sent by an en-gNB to a peer eNB to indicate that one or more cell(s) previously switched-off has (have) been activated.

Direction: en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| **Activated NR Cell List** | | *1* | | | GLOBAL | ignore |
| **>Activated NR Cell Item** | | *1 .. < maxCellinengNB >* | | | | |
| >>NR CGI | M | | 9.2.111 | | - | |
| Activation ID | M | | INTEGER (0..255) | Allocated by the eNB | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

| Range bound | Explanation |
|---|---|
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |

### 9.1.2.39 EN-DC CELL ACTIVATION FAILURE

This message is sent by an en-gNB to a peer eNB to indicate cell activation failure.

Direction: en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Activation ID | M | | INTEGER (0..255) | Allocated by the eNB | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.40 EN-DC X2 REMOVAL REQUEST

This message is sent by an initiating node to a neighbouring node to initiate the removal of the interface instance.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE *Initiating Node Type* | M | | | | | |
| *>eNB* | | | | | | |
| >>Global eNB ID | M | | 9.2.22 | | YES | reject |
| *>en-gNB* | | | | | | |
| >>Global en-gNB ID | M | | 9.2.112 | | | |
| X2 Removal Threshold | O | | X2 Benefit Value 9.2.90 | | YES | reject |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.41 EN-DC X2 REMOVAL RESPONSE

This message is sent by an initiating node to a neighbouring node to acknowledge the initiation of removal of the interface instance.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE *Initiating Node Type* | M | | | | | |
| *>eNB* | | | | | | |
| >>Global eNB ID | M | | 9.2.22 | | YES | reject |
| *>en-gNB* | | | | | | |
| >>Global en-gNB ID | M | | 9.2.112 | | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.42 EN-DC X2 REMOVAL FAILURE

This message is sent by the initiating node to indicate that removing the interface instance cannot be accepted.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.43 DATA FORWARDING ADDRESS INDICATION

This message is sent by the new eNB to indicate to the old eNB forwarding addresses for each E-RAB for which it admits data forwarding.

During a Conditional Handover with EN-DC or Dual Connectivity or Conditional PSCell Change, this message is also used to provide data forwarding related information. In case of EN-DC, the data forwarding related information is transferred from the eNB to the en-gNB, while in case of Dual Connectivity, the data forwarding related information is transferred from the MeNB to the SeNB.

Direction: new eNB → old eNB.

Direction: MeNB → en-gNB (Conditional Handover with EN-DC, Conditional PSCell Change), MeNB → SeNB (Conditional Handover with Dual Connectivity)

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| New eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the new eNB | YES | ignore |
| New eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the new eNB | YES | ignore |
| Old eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the old eNB | YES | ignore |
| Old eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the old eNB | YES | ignore |
| **E-RABs Data Forwarding Address List** | | *1* | | This IE is ignored if the *CHO DC Indicator* IE is included and set to "coordination-only" or if the *CPC Data Forwarding Indicator* IE is included and set to "coordination-only". | YES | ignore |
| **> E-RABs Data Forwarding Address Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>DL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| CHO DC Indicator | O | | ENUMERATED (true, ..., coordination-only) | Indicating that the DATA FORWARDING ADDRESS INDICATION message is for a Conditional Handover. | YES | reject |
| CHO DC Early Data Forwarding Indicator | O | | ENUMERATED (stop, ...) | | YES | ignore |
| SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated for EN-DC at the en-gNB. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| CPC Data Forwarding Indicator | O | | ENUMERATED (triggered, early data transmission stop, ..., coordination-only) | Indicates that the DATA FORWARDING ADDRESS INDICATION message is for a Conditional PSCell Change. | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.2.44 EN-DC CONFIGURATION TRANSFER

This message is sent by an eNB in order to transfer the EN-DC SON Configuration container to an en-gNB, or it is sent by an en-gNB in order to transfer the EN-DC SON Configuration container to an eNB.

Direction: eNB → en-gNB or en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| EN-DC SON Configuration Transfer | O | | OCTET STRING | Contains the *EN-DC SON Configuration Transfer* IE as defined in TS 36.413 [4]. | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.45 EN-DC RESOURCE STATUS REQUEST

This message is sent by the eNB to the en-gNB or by the en-gNB to the eNB to initiate the requested measurement according to the parameters given in the message.

Direction: E-UTRAN node$_1$ → E-UTRAN node$_2$ (eNB → en-gNB, en-gNB → eNB).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.3.1 | | YES | reject |
| E-UTRAN node1 Measurement ID | M | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node1. | YES | reject |
| E-UTRAN node2 Measurement ID | C-ifRegistrationRequestStoporAdd | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node2. | YES | ignore |
| Registration Request EN-DC | M | | ENUMERATED (start, stop, add, …) | Type of request for which the resource status is required. | YES | reject |
| Reporting Periodicity EN-DC | O | | ENUMERATED (500ms, 1000ms, 2000ms, 5000ms,10000ms, …) | Periodicity that can be used for reporting of PRB Periodic, TNL Capacity Ind Periodic, Composite Available Capacity Periodic. Also used as the averaging window length for all measurement object if supported. | YES | ignore |
| Report Characteristics EN-DC | C-ifRegistrationRequestStart | | BITSTRING (SIZE(32)) | When sent by the eNB, each position in the bitmap indicates measurement object the en-gNB is requested to report. First Bit = PRB Periodic, Second Bit = TNL Capacity Ind Periodic, Third Bit = Composite Available Capacity Periodic, Fourth Bit = Number of Active UEs. Other bits shall be ignored by the en-gNB. When sent by the en-gNB, each position in the bitmap indicates measurement object the eNB is requested to report. First Bit = PRB Periodic, Second Bit = TNL load Ind Periodic, Third Bit = HW Load Ind Periodic, Fourth Bit = Composite Available Capacity Periodic, this bit should be set to 1 if at least one of the First, Second or Third bits is set to 1. Other bits shall be ignored by the eNB. | YES | ignore |
| **NR Cell To Report EN-DC List** | | *0..1* | | NR cell ID list to which the request applies. | YES | ignore |
| **>NR Cell To Report EN-DC Item** | | *1 .. <maxCellinengNB>* | | | EACH | ignore |
| >>NR Cell ID | M | | NR CGI 9.2.111 | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>**SSB To Report List** | | *0..1* | | SSB list to which the request applies. | YES | ignore |
| >>>**SSB To Report Item** | | *1 .. <maxnoofSSBAreas>* | | | EACH | ignore |
| >>>>SSB Index | M | | 9.2.167 | | – | |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |
| **E-UTRA Cell To Report EN-DC List** | | *0..1* | | E-UTRA cell ID list to which the request applies. | YES | ignore |
| **>E-UTRA Cell To Report EN-DC Item** | | *1 .. <maxCellineNB>* | | | EACH | ignore |
| >>E-UTRA Cell ID | M | | ECGI 9.2.14 | | – | |

| Condition | Explanation |
|---|---|
| ifRegistrationRequestStoporAdd | This IE shall be present if the *Registration Request EN-DC* IE is set to the value "stop", or "add". |
| ifRegistrationRequestStart | This IE shall be present if the Registration Request *EN-DC* IE is set to the value "start". |

| Range bound | Explanation |
|---|---|
| *maxCellinengNB* | Maximum no. cells that can be served by an en-gNB. Value is 16384. |
| *maxnoofSSBAreas* | Maximum no. SSB Areas that can be served by a NG-RAN node cell. Value is 64. |
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

### 9.1.2.46 EN-DC RESOURCE STATUS RESPONSE

This message is sent by the en-gNB or by the eNB to indicate that the requested measurement, for all of the measurement objects included in the measurement is successfully initiated.

Direction: E-UTRAN node$_2$ → E-UTRAN node$_1$ (en-gNB → eNB, eNB → en-gNB).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.3.1 | | YES | reject |
| E-UTRAN node1 Measurement ID | M | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node$_1$. | YES | reject |
| E-UTRAN node2 Measurement ID | M | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node$_2$. | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.47 EN-DC RESOURCE STATUS FAILURE

This message is sent by the en-gNB or by the eNB to indicate that for any of the requested measurement objects the measurement cannot be initiated.

Direction: E-UTRAN node$_2$ → E-UTRAN node$_1$ (en-gNB → eNB, eNB → en-gNB).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.3.1 | | YES | reject |
| E-UTRAN node1 Measurement ID | M | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node$_1$. | YES | reject |
| E-UTRAN node2 Measurement ID | M | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node$_2$. | YES | reject |
| Cause | M | | 9.2.6 | Ignored by the receiver when the Complete Failure Cause Information IE is included | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.2.48 EN-DC RESOURCE STATUS UPDATE

This message is sent by the en-gNB or by the eNB to the en-gNB to the eNB to report the results of the requested measurements.

Direction: E-UTRAN node$_2$ → E-UTRAN node$_1$ (en-gNB → eNB, eNB → en-gNB).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.3.1 | | YES | ignore |
| E-UTRAN node1 Measurement ID | M | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node$_1$. | YES | reject |
| E-UTRAN node2 Measurement ID | M | | INTEGER (1..4095, ...) | Allocated by the E-UTRAN node$_2$. | YES | reject |
| **NR Cell Measurement Result** | | *0..1* | | Concerned NR cells in the en-gNB. | YES | ignore |
| **>NR Cell Measurement Result Item** | | *1 .. <maxCellinengNB>* | | | EACH | ignore |
| >>NR Cell ID | M | | NR CGI 9.2.111 | | – | |
| >>NR Radio Resource Status | O | | 9.2.162 | | – | |
| >>TNL Capacity Indicator | O | | 9.2.161 | | – | |
| >>NR Composite Available Capacity Group | O | | 9.2.163 | | – | |
| >>Number of Active UEs | O | | INTEGER (0..16777215, ...) | As defined in TS 38.314 [45]. Value "1" is equivalent to 0.1 Active UEs, value "2" is equivalent to 0.2 Active UEs, value n is equivalent to n/10 Active UEs. | – | |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |
| **E-UTRA Cell Measurement Result** | | *0..1* | | Concerned E-UTRA cells in the eNB. | YES | ignore |
| **>E-UTRA Cell Measurement Result Item** | | *1 .. <maxCellineNB>* | | | EACH | ignore |
| >>E-UTRA Cell ID | M | | ECGI 9.2.14 | | – | |
| >>Hardware Load Indicator | O | | 9.2.34 | | – | |
| >>S1 TNL Load Indicator | O | | 9.2.35 | | – | |
| >>Radio Resource Status | O | | 9.2.37 | | – | |
| >>Composite Available Capacity Group | O | | 9.2.44 | | – | |

| Range bound | Explanation |
|---|---|
| *maxCellinengNB* | Maximum no. cells that can be served by an en-gNB. Value is 16384. |
| maxCellineNB | Maximum no. cells that can be served by an eNB. Value is 256. |

### 9.1.2.49 CELL TRAFFIC TRACE

This message is sent by en-gNB to transfer the trace information to the MeNB.

Direction: en-gNB → MeNB

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.1.1 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| E-UTRAN Trace ID | M | | OCTET STRING (SIZE(8)) | As per E-UTRAN Trace ID in *Trace Activation* IE | YES | ignore |
| Trace Collection Entity IP Address | M | | BIT STRING (1..160,…) | Defined in TS 32.422 [6] | YES | ignore |
| Privacy Indicator | O | | ENUMERATED (Immediate MDT, ...) | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

### 9.1.2.50 ACCESS AND MOBILITY INDICATION

This message is sent by the eNB to the en-gNB to transfer access and mobility related information.

Direction: eNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.3.1 | | YES | reject |
| **NR RACH Report List** | | *0..1* | | | YES | ignore |
| **>NR RACH Report List Item** | | *1 .. <maxnoofRACHReports>* | | | EACH | ignore |
| >>NR RACH Report Container | M | | OCTET STRING | Includes the *ra-ReportList* contained in the *UEInformationResponse* messageas defined in subclause 6.2.2 in TS 38.331 [31]. | – | |
| >>UE Assistant Identifier | O | | en-gNB UE X2AP ID 9.2.100 | | – | |

| Range bound | Explanation |
|---|---|
| *maxnoofRACHReports* | Maximum no. of RACH Reports, the maximum value is 64. |

## 9.1.3 Messages for Dual Connectivity Procedures

### 9.1.3.1 SENB ADDITION REQUEST

This message is sent by the MeNB to the SeNB to request the preparation of resources for dual connectivity operation for a specific UE

Direction: MeNB → SeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| UE Security Capabilities | C-ifSCGBearerOption | | 9.2.29 | | YES | reject |
| SeNB Security Key | C-ifSCGBearerOption | | 9.2.72 | The S-KeNB which is provided by the MeNB, see TS 33.401 [18]. | YES | reject |
| SeNB UE Aggregate Maximum Bit Rate | M | | UE Aggregate Maximum Bit Rate 9.2.12 | The UE Aggregate Maximum Bit Rate is split into MeNB UE Aggregate Maximum Bit Rate and SeNB UE Aggregate Maximum Bit Rate which are enforced by MeNB and SeNB respectively. | YES | reject |
| Serving PLMN | O | | PLMN Identity 9.2.4 | The serving PLMN of the SCG in the SeNB. | YES | ignore |
| **E-RABs To Be Added List** | | *1* | | | YES | reject |
| **>E-RABs To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | reject |
| >>CHOICE Bearer Option | M | | | | | |
| *>>>SCG Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>E-RAB Level QoS Parameters | M | | 9.2.9 | Includes necessary QoS parameters | – | |
| >>>>DL Forwarding | O | | 9.2.5 | | – | |
| >>>>S1 UL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1 transport bearer. For delivery of UL PDUs. | – | |
| >>>>Correlation ID | O | | Correlation ID 9.2.84 | | – | |
| >>>>SIPTO Correlation ID | O | | Correlation ID 9.2.84 | | – | |
| >>>>Bearer Type | O | | 9.2.92 | | YES | ignore |
| >>>>Ethernet Type | O | | 9.2.157 | | YES | ignore |
| >>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| *>>>Split Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>E-RAB Level QoS Parameters | M | | 9.2.9 | Includes necessary QoS parameters | – | |
| >>>>MeNB GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2 transport bearer. For delivery of UL PDUs. | – | |
| >>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| MeNB to SeNB Container | M | | OCTET STRING | Includes the *SCG-ConfigInfo* message as defined in TS 36.331 [9] | YES | reject |
| CSG Membership Status | O | | 9.2.52 | | YES | reject |
| SeNB UE X2AP ID | O | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |
| Expected UE Behaviour | O | | 9.2.70 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

| Condition | Explanation |
|---|---|
| ifSCGBearerOption | This IE shall be present if the *Bearer Option* IE is set to the value "SCG bearer". |

### 9.1.3.2 SENB ADDITION REQUEST ACKNOWLEDGE

This message is sent by the SeNB to confirm the MeNB about the SeNB addition preparation.

Direction: SeNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| **E-RABs Admitted To Be Added List** | | *1* | | | YES | ignore |
| **>E-RABs Admitted To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>CHOICE Bearer Option | M | | | | | |
| *>>>SCG Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>S1 DL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SeNB endpoint of the S1 transport bearer. For delivery of DL PDUs. | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| *>>>Split Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>SeNB GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | Endpoint of the X2 transport bearer at the SeNB. | – | |
| >>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| E-RABs Not Admitted List | O | | E-RAB List 9.2.28 | A value for *E-RAB ID* shall only be present once in *E-RABs Admitted List* IE and in *E-RABs Not Admitted List* IE. | YES | ignore |
| SeNB to MeNB Container | M | | OCTET STRING | Includes the *SCG-Config* message as defined in TS 36.331 [9] | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| GW Transport Layer Address | O | | BIT STRING (1..160, ...) | Indicating GW Transport Layer Address. | YES | ignore |
| SIPTO L-GW Transport Layer Address | O | | BIT STRING (1..160, ...) | Indicating SIPTO L-GW Transport Layer Address. | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |

| | | | | | | |
|---|---|---|---|---|---|---|
| Tunnel Information for BBF | O | | Tunnel Information 9.2.89 | Indicating eNB's Local IP Address assigned by the broadband access provider, UDP port Number. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.3.3 SENB ADDITION REQUEST REJECT

This message is sent by the SeNB to inform the MeNB that the SeNB Addition Preparation has failed.

Direction: SeNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |

### 9.1.3.4 SENB RECONFIGURATION COMPLETE

This message is sent by the MeNB to the SeNB to indicate whether the configuration requested by the SeNB was applied by the UE.

Direction: MeNB → SeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| **Response Information** | M | | | | YES | ignore |
| >CHOICE *Response Type* | M | | | | | |
| *>>Configuration successfully applied* | | | | | | |
| >>>MeNB to SeNB Container | O | | OCTET STRING | Includes the *SCG-ConfigInfo* message as defined in TS 36.331 [9] | - | |
| *>>Configuration rejected by the MeNB* | | | | | | |
| >>>Cause | M | | 9.2.6 | | - | |
| >>>MeNB to SeNB Container | O | | OCTET STRING | Includes the *SCG-ConfigInfo* message as defined in TS 36.331 [9] | - | |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |

### 9.1.3.5 SENB MODIFICATION REQUEST

This message is sent by the MeNB to the SeNB to request the preparation to modify SeNB resources for a specific UE.

Direction: MeNB → SeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| SCG Change Indication | O | | 9.2.73 | | YES | ignore |
| Serving PLMN | O | | PLMN Identity 9.2.4 | The serving PLMN of the SCG in the SeNB. | YES | ignore |
| **UE Context Information** | | *0..1* | | | YES | reject |
| >UE Security Capabilities | O | | 9.2.29 | | – | |
| >SeNB Security Key | O | | 9.2.72 | | – | |
| >SeNB UE Aggregate Maximum Bit Rate | O | | UE Aggregate Maximum Bit Rate 9.2.12 | | – | |
| **>E-RABs To Be Added List** | | *0..1* | | | – | |
| **>>E-RABs To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>CHOICE *Bearer Option* | M | | | | | |
| >>>>*SCG Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>E-RAB Level QoS Parameters | M | | 9.2.9 | Includes necessary QoS parameters | – | |
| >>>>>DL Forwarding | O | | 9.2.5 | | – | |
| >>>>>S1 UL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1 transport bearer. For delivery of UL PDUs. | – | |
| >>>>>Correlation ID | O | | Correlation ID 9.2.84 | | – | |
| >>>>>SIPTO Correlation ID | O | | Correlation ID 9.2.84 | | – | |
| >>>>>Bearer Type | O | | 9.2.92 | | YES | ignore |
| >>>>>Ethernet Type | O | | 9.2.157 | | YES | ignore |
| >>>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| >>>>*Split Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>E-RAB Level QoS Parameters | M | | 9.2.9 | Includes necessary QoS parameters | – | |
| >>>>>MeNB GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2 transport bearer. For delivery of UL PDUs. | – | |
| >>>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| **>E-RABs To Be Modified List** | | *0..1* | | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **>>E-RABs To Be Modified Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>CHOICE *Bearer Option* | M | | | | | |
| >>>>*SCG Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>E-RAB Level QoS Parameters | O | | 9.2.9 | Includes QoS parameters to be modified | – | |
| >>>>>S1 UL GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1 transport bearer. For delivery of UL PDUs. | – | |
| >>>>*Split Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>E-RAB Level QoS Parameters | O | | 9.2.9 | Includes QoS parameters to be modified | – | |
| >>>>>MeNB GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2 transport bearer. For delivery of UL PDUs. | – | |
| **>E-RABs To Be Released List** | | *0..1* | | | – | |
| **>>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>CHOICE *Bearer Option* | M | | | | | |
| >>>>*SCG Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| >>>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer. used for forwarding of UL PDUs | – | |
| >>>>*Split Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| MeNB to SeNB Container | O | | OCTET STRING | Includes the *SCG-ConfigInfo* message as defined in TS 36.331 [9] | YES | ignore |
| CSG Membership Status | O | | 9.2.52 | | YES | reject |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.3.6 SENB MODIFICATION REQUEST ACKNOWLEDGE

This message is sent by the SeNB to confirm the MeNB's request to modify the SeNB resources for a specific UE.

Direction: SeNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | ignore |
| **E-RABs Admitted List** | | *0..1* | | | YES | ignore |
| **>E-RABs Admitted To Be Added List** | | *1* | | | – | |
| **>>E-RABs Admitted To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>CHOICE *Bearer Option* | M | | | | | |
| >>>>*SCG Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>S1 DL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SeNB endpoint of the S1 transport bearer. For delivery of DL PDUs. | – | |
| >>>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| >>>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| >>>>*Split Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>SeNB GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | Endpoint of the X2 transport bearer at the SeNB. | – | |
| >>>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| **>E-RABs Admitted To Be Modified List** | | *0..1* | | | – | |
| **>>E-RABs Admitted To Be Modified Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>CHOICE *Bearer Option* | M | | | | | |
| >>>>*SCG Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>>S1 DL GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | SeNB endpoint of the S1 transport bearer. For delivery of DL PDUs. | – | |
| >>>>*Split Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>>SeNB GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Endpoint of the X2 transport bearer at the SeNB. | – | |
| **>E-RABs Admitted To Be Released List** | | *0..1* | | | – | |
| **>>E-RABs Admitted To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>CHOICE *Bearer Option* | M | | | | | |
| >>>>*SCG Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>*Split Bearer* | | | | | | |
| >>>>>E-RAB ID | M | | 9.2.23 | | – | |
| E-RABs Not Admitted List | O | | E-RAB List 9.2.28 | A value for *E-RAB ID* shall only be present once in *E-RABs Admitted List* IE and in *E-RABs Not Admitted List* IE. | YES | ignore |
| SeNB to MeNB Container | O | | OCTET STRING | Includes the *SCG-Config* message as defined in TS 36.331 [9] | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | Ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

## 9.1.3.7 SENB MODIFICATION REQUEST REJECT

This message is sent by the SeNB to inform the MeNB that the MeNB initiated SeNB Modification Preparation has failed.

Direction: SeNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | ignore |

### 9.1.3.8 SENB MODIFICATION REQUIRED

This message is sent by the SeNB to the MeNB to request the modification of SeNB resources for a specific UE.

Direction: SeNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| SCG Change Indication | O | | 9.2.73 | | YES | ignore |
| **E-RABs To Be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>Cause | M | | 9.2.6 | | – | |
| SeNB to MeNB Container | O | | OCTET STRING | Includes the *SCG-Config* message as defined in TS 36.331 [9] | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.3.9 SENB MODIFICATION CONFIRM

This message is sent by the MeNB to inform the SeNB about the successful modification.

Direction: MeNB → SeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | ignore |
| MeNB to SeNB Container | O | | OCTET STRING | Includes the *SCG-ConfigInfo* message as defined in TS 36.331 [9] | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | ignore |

### 9.1.3.10 SENB MODIFICATION REFUSE

This message is sent by the MeNB to inform the SeNB that the SeNB initiated SeNB Modification has failed.

Direction: MeNB → SeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| MeNB to SeNB Container | O | | OCTET STRING | Includes the *SCG-ConfigInfo* message as defined in TS 36.331 [9] | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | ignore |

### 9.1.3.11 SENB RELEASE REQUEST

This message is sent by the MeNB to the SeNB to request the release of resources.

Direction: MeNB → SeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID | O | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| Cause | O | | 9.2.6 | | YES | ignore |
| **E-RABs To Be Released List** | | *0..1* | | | YES | ignore |
| **> E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>CHOICE *Bearer Option* | M | | | | | |
| *>>>SCG Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer. used for forwarding of DL PDUs | – | |
| *>>>Split Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer. used for forwarding of DL PDUs | – | |
| UE Context Kept Indicator | O | | 9.2.85 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |
| MakeBeforeBreak Indicator | O | | ENUMERATED (True, …) | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.3.12 SENB RELEASE REQUIRED

This message is sent by the SeNB to request the release of all resources for a specific UE at the SeNB.

Direction: SeNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | reject |

### 9.1.3.13 SENB RELEASE CONFIRM

This message is sent by the MeNB to confirm the release of all resources for a specific UE at the SeNB.

Direction: MeNB → SeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | ignore |
| **E-RABs to be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | – | |
| >>CHOICE *Bearer Option* | M | | | | | |
| *>>>SCG Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| *>>>Split Bearer* | | | | | | |
| >>>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.3.14 SENB COUNTER CHECK REQUEST

This message is sent by the SeNB to request the verification of the value of the PDCP COUNTs associated with SCG bearers established in the SeNB.

Direction: SeNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the SeNB | YES | ignore |
| **E-RABs Subject to Counter Check List** | | *1* | | | YES | ignore |
| **>E-RABs Subject to Counter Check Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | - | |
| >>UL COUNT | M | *INTEGER(0.. 4294967295)* | | Indicates the value of uplink COUNT associated to this E-RAB. | - | |
| >>DL COUNT | M | *INTEGER(0.. 4294967295)* | | Indicates the value of downlink COUNT associated to this E-RAB. | - | |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| SeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the SeNB | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

## 9.1.4 Messages for E-UTRAN-NR Dual Connectivity Procedures

### 9.1.4.1 SGNB ADDITION REQUEST

This message is sent by the MeNB to the en-gNB to request the preparation of resources for EN-DC operation for a specific UE

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| NR UE Security Capabilities | M | | 9.2.107 | | YES | reject |
| SgNB Security Key | M | | 9.2.101 | The S-KgNB which is provided by the MeNB, see TS 33.401 [18]. | YES | reject |
| SgNB UE Aggregate Maximum Bit Rate | M | | UE Aggregate Maximum Bit Rate 9.2.12 | The UE Aggregate Maximum Bit Rate is split into MeNB UE Aggregate Maximum Bit Rate and SgNB UE Aggregate Maximum Bit Rate which are enforced by MeNB and en-gNB respectively. | YES | reject |
| Selected PLMN | O | | PLMN Identity 9.2.4 | The selected PLMN of the SCG in the en-gNB. | YES | ignore |
| Handover Restriction List | O | | 9.2.3 | | YES | ignore |
| **E-RABs To Be Added List** | | *1* | | | YES | reject |
| **>E-RABs To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | reject |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>DRB ID | M | | 9.2.122 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| *>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>Full E-RAB Level QoS Parameters | M | | E-RAB Level QoS Parameters 9.2.9 | Includes the E-RAB level QoS parameters as received on S1-MME. | – | |
| >>>>Maximum MCG admittable E-RAB Level QoS Parameters | C-ifMCGandSCGpresent_GBR | | GBR QoS Information 9.2.10 | Includes the GBR QoS Information admittable by the MCG. | – | |
| >>>>DL Forwarding | O | | 9.2.5 | | – | |
| >>>>MeNB DL GTP Tunnel Endpoint at MCG | C-ifMCGpresent | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer at MCG. For delivery of DL PDCP PDUs. | – | |
| >>>>S1 UL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1-U transport bearer. For delivery of UL PDUs from the en-gNB. | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>RLC Mode | O | | RLC Mode 9.2.119 | Indicates the RLC mode at the MeNB for PDCP transfer to en-gNB. | YES | ignore |
| >>>>Bearer Type | O | | 9.2.92 | | YES | ignore |
| >>>>Ethernet Type | O | | 9.2.157 | | YES | ignore |
| >>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source MN for data forwarding. | YES | ignore |
| >>>>Source Node DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | This IE is present only for the case of SA to EN-DC handover and it is used to identify the TNL address allocated by the source NG-RAN node for data forwarding. | YES | ignore |
| >>>>Security Indication | O | | 9.2.181 | | YES | reject |
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >>>>Requested SCG E-RAB Level QoS Parameters | M | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters requested to be provided by the SCG. | – | |
| >>>>MeNB UL GTP Tunnel Endpoint at PDCP | M | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer. For delivery of UL PDCP PDUs. | – | |
| >>>>Secondary MeNB UL GTP Tunnel Endpoint at PDCP | O | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer. For delivery of UL PDCP PDUs in case of PDCP duplication. | – | |
| >>>>RLC Mode | M | | RLC Mode 9.2.119 | Indicates the RLC mode to be used in the assisting node. | – | |
| >>>>UL Configuration | C-ifMCGandSCGpresent | | 9.2.118 | Information about UL usage in the en-gNB. | – | |
| >>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the UL. | YES | ignore |
| >>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the DL. | YES | ignore |
| >>>>Duplication activation | O | | 9.2.137 | Indicated the initial staus of PDCP duplication. | YES | ignore |
| MeNB to SgNB Container | M | | OCTET STRING | Includes the *CG-ConfigInfo* message as defined in TS 38.331 [31]. | YES | reject |

| | | | | | | |
|---|---|---|---|---|---|---|
| SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Expected UE Behaviour | O | | 9.2.70 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |
| Requested split SRBs | O | | ENUMERATED (srb1, srb2, srb1&2, ...) | Indicates that resources for Split SRB are requested. | YES | reject |
| MeNB Resource Coordination Information | O | | 9.2.116 | Information used to coordinate resources utilisation between MeNB and en-gNB. | YES | ignore |
| SGNB Addition Trigger Indication | O | | ENUMERATED (SN change, inter-eNB HO, intra-eNB HO, ...) | This IE indicates the trigger for SGNB Addition procedure. | YES | reject |
| Subscriber Profile ID for RAT/Frequency priority | O | | 9.2.25 | | YES | ignore |
| MeNB Cell ID | M | | ECGI 9.2.14 | Indicates the cell ID for PCell in MeNB. | YES | reject |
| Desired Activity Notification Level | O | | 9.2.141 | | YES | ignore |
| Trace Activation | O | | 9.2.2 | | YES | ignore |
| Location Information at SgNB reporting | O | | ENUMERATED (pscell, ...) | Indicates that the user's location information is to be provided. | YES | ignore |
| Masked IMEISV | O | | 9.2.69 | | YES | ignore |
| Additional RRM Policy Index | O | | 9.2.25a | | YES | ignore |
| Requested Fast MCG recovery via SRB3 | O | | ENUMERATED (true, ...) | Indicates that the resources for fast MCG recovery via SRB3 are requested. | YES | ignore |
| UE Context Reference at Source NG-RAN | O | | RAN UE NGAP ID 9.2.152 | | YES | ignore |
| Management Based MDT Allowed | O | | 9.2.59 | | YES | ignore |
| Management Based MDT PLMN List | O | | MDT PLMN List 9.2.64 | | YES | ignore |
| UE Radio Capability ID | O | | 9.2.171 | | YES | reject |
| IAB Node Indication | O | | ENUMERATED (true, ...) | | YES | reject |
| Source NG-RAN Node ID | O | | Global RAN Node ID 9.2.176 | | YES | ignore |
| UE History Information | O | | 9.2.38 | | YES | ignore |
| UE History Information from the UE | O | | OCTET STRING | Includes the V*isitedCellInfoList* contained in the *UEInformationResponse* message as defined TS 36.331 [9]. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| PScell Change History | O | | ENUMERATED (reporting full history, ...) | | YES | ignore |
| **CHO Information SN Addition** | O | | | | YES | reject |
| >Source eNB ID | M | | Global eNB ID 9.2.22 | | | |
| >Source eNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the source MeNB | | |
| >Source eNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the source MeNB | | |
| >Estimated Arrival Probability | O | | INTEGER (1..100) | | | |
| SCG Activation Request | O | | 9.2.179 | | YES | ignore |
| **Conditional PSCell Addition Information Request** | O | | | | YES | reject |
| >Maximum Number of PSCells To Prepare | M | | INTEGER (1..8, …) | Indicates the maximum number of PSCells that the target may prepare. | – | |
| >Estimated Arrival Probability | O | | INTEGER (1..100) | Indicates the arrival probability for the UE towards the candidate target en-gNB. | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256. |

| Condition | Explanation |
|---|---|
| ifMCGandSCGpresent | This IE shall be present if, for the E-RAB requested to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present". |
| ifMCGpresent | This IE shall be present if, for the E-RAB requested to be added, the *MCG resources* IE in the *EN-DC Resource Configuration* IE is set to the value "present". |
| C-ifMCGandSCGpresent_GBR | This IE shall be present if, for the E-RAB requested to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present", and *GBR QoS Information* IE is present in *Full E-RAB Level QoS Parameters* IE. |

### 9.1.4.2 SGNB ADDITION REQUEST ACKNOWLEDGE

This message is sent by the en-gNB to confirm the MeNB about the SgNB addition preparation.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| **E-RABs Admitted To Be Added List** | | *1* | | | YES | ignore |
| **>E-RABs Admitted To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| *>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>S1 DL GTP Tunnel Endpoint at the SgNB | M | | GTP Tunnel Endpoint 9.2.1 | en-gNB endpoint of the S1 transport bearer. For delivery of DL PDUs. | – | |
| >>>>SgNB UL GTP Tunnel Endpoint at PDCP | C-ifMCGpresent | | GTP Tunnel Endpoint 9.2.1 | en-gNB endpoint of the X2-U transport bearer at PDCP. For delivery of UL PDCP PDUs. | – | |
| >>>>RLC Mode | C-ifMCGpresent | | RLC Mode 9.2.119 | Indicates the RLC mode. | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the transport bearer used for forwarding of DL PDUs | – | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the transport bearer used for forwarding of UL PDUs | – | |
| >>>>Requested MCG E-RAB Level QoS Parameters | C-ifMCGandSCGpresent_GBRpresent | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters requested to be provided by the MCG. | – | |
| >>>>UL Configuration | C-ifMCGandSCGpresent | | 9.2.118 | Information about UL usage in the MeNB. | – | |
| >>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the UL. | YES | ignore |
| >>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the DL. | YES | ignore |
| >>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>Security Result | O | | 9.2.182 | | YES | ignore |
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >>>>SgNB DL GTP Tunnel Endpoint at SCG | M | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs. | – | |
| >>>>Secondary SgNB DL GTP Tunnel Endpoint at SCG | O | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs in case of PDCP duplication | – | |
| >>>>LCID | O | | 9.2.138 | LCID for the primary path in case of PDCP duplication | YES | ignore |
| E-RABs Not Admitted List | O | | E-RAB List 9.2.28 | A value for *E-RAB ID* shall only be present once in *E-RABs Admitted List* IE and in *E-RABs Not Admitted List* IE. | YES | ignore |
| SgNB to MeNB Container | M | | OCTET STRING | Includes the *CG-Config* message or the *CG-CandidateList* message, as defined in TS 38.331[31]. | YES | reject |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| Admitted split SRBs | O | | ENUMERATED (srb1, srb2, srb1&2, ...) | Indicates admitted SRBs | YES | reject |
| SgNB Resource Coordination Information | O | | 9.2.117 | Information used to coordinate resources utilisation between en-gNB and MeNB. | YES | ignore |
| RRC config indication | O | | 9.2.132 | Indicates the type of RRC configuration used at the en-gNB. | YES | reject |
| Location Information at SgNB | O | | 9.2.142 | Contains information to support localisation of the UE | YES | ignore |
| Available fast MCG recovery via SRB3 | O | | ENUMERATED (true, ...) | Indicates the fast MCG recovery via SRB3 is enabled. | YES | ignore |
| Direct Forwarding Path Availability | O | | ENUMERATED (direct path available, …) | Indicates direct forwarding path is available between the target en-gNB and source NG-RAN node for SA to EN-DC handover. | YES | ignore |
| SCG Activation Status | O | | 9.2.178 | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Conditional PSCell Addition Information Acknowledge** | O | | | | YES | ignore |
| **>Candidate PSCell List** | | *1* | | | – | |
| >>Candidate PSCell Item | | *1 .. <maxnoofPSCellCandidate>* | | | – | |
| >>>PSCell ID | M | | NR CGI 9.2.111 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |
| maxnoofPSCellCandidate | Maximum no. of PSCells for CPAC. Value is 8. |

| Condition | Explanation |
|---|---|
| ifMCGpresent | This IE shall be present if, for the E-RAB admitted to be added, the *MCG resources* IE in the *EN-DC Resource Configuration* IE is set to the value "present". |
| ifMCGandSCGpresent | This IE shall be present if, for the E-RAB admitted to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present". |
| C-ifMCGandSCGpresent_GBRpresent | This IE shall be present if, for the E-RAB admitted to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present", and the *GBR QoS Information* IE is present in the *Requested MCG E-RAB Level QoS Parameters* IE. |

### 9.1.4.3 SGNB ADDITION REQUEST REJECT

This message is sent by the en-gNB to inform the MeNB that the SgNB Addition Preparation has failed.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

### 9.1.4.4 SGNB RECONFIGURATION COMPLETE

This message is sent by the MeNB to the en-gNB to indicate whether the configuration requested by the en-gNB was applied by the UE.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| **Response Information** | M | | | | YES | ignore |
| >CHOICE *Response Type* | M | | | | | |
| *>>Configuration successfully applied* | | | | | | |
| >>>MeNB to SgNB Container | O | | OCTET STRING | Includes the *RRCReconfiguration Complete* message as defined in TS 38.331 [31]. | - | |
| *>>Configuration rejected* | | | | | | |
| >>>Cause | M | | 9.2.6 | | - | |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |

### 9.1.4.5 SGNB MODIFICATION REQUEST

This message is sent by the MeNB to the en-gNB to request the preparation to modify en-gNB resources for a specific UE, to query for the current SCG configuration, or to provide the S-RLF-related information to the en-gNB.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| Selected PLMN | O | | PLMN Identity 9.2.4 | The selected PLMN of the SCG in the en-gNB. | YES | ignore |
| Handover Restriction List | O | | 9.2.3 | | YES | ignore |
| SCG Configuration Query | O | | 9.2.103 | | YES | ignore |
| **UE Context Information** | | *0..1* | | | YES | reject |
| >NR UE Security Capabilities | O | | 9.2.107 | | – | |
| >SgNB Security Key | O | | 9.2.101 | | – | |
| >SgNB UE Aggregate Maximum Bit Rate | O | | UE Aggregate Maximum Bit Rate 9.2.12 | | – | |
| >Lower Layer presence status change | O | | 9.2.145 | | – | |
| **>E-RABs To Be Added List** | | *0..1* | | | – | |
| **>>E-RABs To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>DRB ID | M | | 9.2.122 | | – | |
| >>>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>>CHOICE *Resource Configuration* | M | | | | | |
| >>>>*PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | – | |
| >>>>>Full E-RAB Level QoS Parameters | M | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters as received on S1-MME. | – | |
| >>>>>Maximum MCG admittable E-RAB Level QoS Parameters | C-ifMCGandSCGpresent_GBR | | GBR QoS Information 9.2.10 | Includes the GBR QoS Information admittable by the MCG. | – | |
| >>>>>DL Forwarding | O | | 9.2.5 | | – | |
| >>>>>MeNB DL GTP Tunnel Endpoint at MCG | C-ifMCGpresent | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer at MCG. For delivery of DL PDCP PDUs. | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>>S1 UL GTP Tunnel Endpoint | M | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1-U transport bearer. For delivery of UL PDUs from the en-gNB. | – | |
| >>>>>RLC Mode | O | | RLC Mode 9.2.119 | Indicates the RLC mode at the MeNB for PDCP transfer to en-gNB. | YES | ignore |
| >>>>>Bearer Type | O | | 9.2.92 | | YES | ignore |
| >>>>>Ethernet Type | O | | 9.2.157 | | YES | ignore |
| >>>>>Security Indication | O | | 9.2.181 | | YES | reject |
| >>>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |
| >>>>*PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >>>>>Requested SCG E-RAB Level QoS Parameters | M | | E-RAB Level QoS Parameters 9.2.9 | Includes necessary E-RAB level QoS parameters requested to be provided by the SCG. | – | |
| >>>>>MeNB UL GTP Tunnel Endpoint at PDCP | M | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer. For delivery of UL PDCP PDUs. | – | |
| >>>>>Secondary MeNB UL GTP Tunnel Endpoint at PDCP | O | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer. For delivery of UL PDCP PDUs in case of PDCP duplication. | – | |
| >>>>>RLC Mode | M | | RLC Mode 9.2.119 | Indicates the RLC mode to be used in the assisting node. | – | |
| >>>>>UL Configuration | C-ifMCGandSCGpresent | | 9.2.118 | Information about UL usage in the en-gNB. | – | |
| >>>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the UL. | YES | ignore |
| >>>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the DL. | YES | ignore |
| >>>>>Duplication activation | O | | 9.2.137 | Indicated the initial staus of PDCP duplication. | YES | ignore |
| **>E-RABs To Be Modified List** | | *0..1* | | | – | |
| **>>E-RABs To Be Modified Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>>E-RAB ID | M | | 9.2.23 | | – | |
| >>>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>CHOICE *Resource Configuration* | M | | | | | |
| *>>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>>Full E-RAB Level QoS Parameters | O | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters to be modified as received on S1-MME | – | |
| >>>>>Maximum MCG admittable E-RAB Level QoS Parameters | O | | GBR QoS Information 9.2.10 | Includes the GBR QoS information admittable by the MCG | – | |
| >>>>>MeNB GTP Tunnel Endpoint at MCG | O | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer at MCG. For delivery of DL PDCP PDUs. | – | |
| >>>>>S1 UL GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | SGW endpoint of the S1-U transport bearer. For delivery of UL PDUs from the en-gNB. | – | |
| >>>>>RLC Status | O | | 9.2.131 | Indicates the RLC has been re-established.. | | |
| *>>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >>>>>Requested SCG E-RAB Level QoS Parameters | O | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters requested to be provided by the SCG. | – | |
| >>>>>MeNB UL GTP Tunnel Endpoint at PDCP | O | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer. For delivery of UL PDCP PDUs. | – | |
| >>>>>UL Configuration | O | | 9.2.118 | Information about UL usage in the en-gNB. | – | |
| >>>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the en-gNB if received. | YES | ignore |
| >>>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the en-gNB if received. | YES | ignore |
| >>>>>Secondary MeNB UL GTP Tunnel Endpoint at PDCP | O | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer. For delivery of UL PDCP PDUs in case of PDCP duplication. | YES | ignore |
| **>E-RABs To Be Released List** | | *0..1* | | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **>>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>>CHOICE *Resource Configuration* | M | | | | | |
| *>>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| >>>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer. used for forwarding of UL PDUs | – | |
| *>>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >Subscriber Profile ID for RAT/Frequency priority | O | | 9.2.25 | | YES | ignore |
| >Additional RRM Policy Index | O | | 9.2.25a | | YES | ignore |
| MeNB to SgNB Container | O | | OCTET STRING | Includes the *CG-ConfigInfo* message as defined in TS 38.331 [31]. | YES | reject |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| MeNB Resource Coordination Information | O | | 9.2.116 | Information used to coordinate resources utilisation between MeNB and en-gNB. | YES | ignore |
| Requested split SRBs | O | | ENUMERATED (srb1, srb2, srb1&2, ...) | Indicates that resources for Split SRB are requested. | YES | ignore |
| Requested split SRBs release | O | | ENUMERATED (srb1, srb2, srb1&2, ...) | Indicates that resources for Split SRB are requested to be released. | YES | ignore |
| Desired Activity Notification Level | O | | 9.2.141 | | YES | ignore |
| Location Information at SgNB reporting | O | | ENUMERATED (pscell, ...) | Indicates that the user's location information is to be provided. | YES | ignore |
| MeNB Cell ID | O | | ECGI 9.2.14 | Indicates the cell ID for PCell in MeNB. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| Requested Fast MCG recovery via SRB3 | O | | ENUMERATED (true, ...) | Indicates that the resources for fast MCG recovery via SRB3 are requested. | YES | ignore |
| Requested Fast MCG recovery via SRB3 Release | O | | ENUMERATED (true, ...) | Indicates that the resources for fast MCG recovery via SRB3 are requested to be released. | YES | ignore |
| SN triggered | O | | ENUMERATED (True, ...) | | YES | ignore |
| IAB Node Indication | O | | ENUMERATED (true, ...) | | YES | reject |
| PSCell History Information Retrieve | O | | ENUMERATED (query, ...) | Indicates that the SN UE history information is requested. | YES | ignore |
| UE History Information from the UE | O | | OCTET STRING | Includes the V*isitedCellInfoList* contained in the *UEInformationResponse* message as defined in TS 36.331 [9]. | YES | ignore |
| **CHO Information SN Modification** | O | | | | YES | ignore |
| >Conditional Reconfiguration | M | | ENUMERATED (intra-MN-CHO, ...) | | - | - |
| >Estimated Arrival Probability | O | | INTEGER (1..100) | | - | - |
| SCG Activation Request | O | | 9.2.179 | | YES | ignore |
| **Conditional PSCell Addition Information Modification Request** | O | | | This IE may be sent to the target en-gNB. | YES | ignore |
| >Maximum Number of PSCells To Prepare | O | | INTEGER (1..8, ...) | Indicates the maximum number of PSCells that the target en-gNB may prepare. | – | |
| >Estimated Arrival Probability | O | | INTEGER (1..100) | Indicates the arrival probability for the UE towards the candidate target en-gNB. | – | |
| **Conditional PSCell Change Information Update** | O | | | This IE may be sent to the source en-gNB. | YES | ignore |
| **>Multiple Target SgNB Node List** | | *1* | | | – | |
| >>Multiple Target SgNB Node Item | | *1 .. <maxnoofTargetSgNBs>* | | | – | |
| >>>Target SgNB ID | M | | Global en-gNB ID 9.2.112 | | – | |
| >>>Candidate PSCell List | | *1* | | | – | |
| >>>>Candidate PSCell Item | | *1 .. <maxnoofPSCellCandidate>* | | | – | |
| >>>>>PSCell ID | M | | NR CGI 9.2.111 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |
| maxnoofPSCellCandidate | Maximum no. of PSCells for CPAC. Value is 8. |
| maxnoofTargetSgNBs | Maximum no. of the target SgNB nodes. Value is 8 |

| Condition | Explanation |
|---|---|
| ifMCGandSCGpresent | This IE shall be present if, for the E-RAB requested to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present". |
| ifMCGpresent | This IE shall be present if, for the E-RAB requested to be added, the *MCG resources* IE in the *EN-DC Resource Configuration* IE is set to the value "present". |
| C-ifMCGandSCGpresent_GBR | This IE shall be present if, for the E-RAB requested to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present", and *GBR QoS Information* IE is present in *Full E-RAB Level QoS Parameters* IE. |

### 9.1.4.6 SGNB MODIFICATION REQUEST ACKNOWLEDGE

This message is sent by the en-gNB to confirm the MeNB's request to modify the en-gNB resources for a specific UE.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| **E-RABs Admitted To Be Added List** | | *0..1* | | | YES | ignore |
| **>E-RABs Admitted To Be Added Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| *>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>S1 DL GTP Tunnel Endpoint at the SgNB | M | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the S1 transport bearer. For delivery of DL PDUs. | – | |
| >>>>SgNB UL GTP Tunnel Endpoint at PDCP | C-ifMCGpresent | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at PDCP. For delivery of UL PDCP PDUs. | – | |
| >>>>RLC Mode | C-ifMCGpresent | | RLC Mode 9.2.119 | Indicates the RLC mode to be used at the assisting node. | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>>>Requested MCG E-RAB Level QoS Parameters | C-ifMCGandSCGpresent_GBRpresent | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters requested to be provided by the MCG. | – | |
| >>>>UL Configuration | C-ifMCGandSCGpresent | | 9.2.118 | Information about UL usage in the MeNB. | – | |
| >>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the UL. | YES | ignore |
| >>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Indicates the PDCP SN length of the bearer for the DL. | YES | ignore |
| >>>>Security Result | O | | 9.2.182 | | YES | ignore |
| >>>>Source DL Forwarding IP Address | O | | BIT STRING (1..160, ...) | Identifies the TNL address used by the source node for data forwarding. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >>>>SgNB DL GTP Tunnel Endpoint at SCG | M | | GTP Tunnel Endpoint 9.2.1 | Endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs. | – | |
| >>>>Secondary SgNB DL GTP Tunnel Endpoint at SCG | O | | GTP Tunnel Endpoint 9.2.1 | Endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs in case of PDCP duplication. | – | |
| >>>>LCID | O | | 9.2.138 | LCID for the primary path in case of PDCP duplication configured. | YES | ignore |
| **E-RABs Admitted To Be Modified List** | | *0..1* | | | YES | ignore |
| **>E-RABs Admitted To Be Modified Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| *>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>S1 DL GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the S1 transport bearer. For delivery of DL PDUs. | – | |
| >>>>SgNB UL GTP Tunnel Endpoint at PDCP | O | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at PDCP. For delivery of UL PDCP PDUs. | – | |
| >>>>Requested MCG E-RAB Level QoS Parameters | O | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters requested to be provided by the MCG. | – | |
| >>>>UL Configuration | O | | 9.2.118 | Information about UL usage in the MeNB. | – | |
| >>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the MeNB if received. | YES | ignore |
| >>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the MeNB if received. | YES | ignore |
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>SgNB DL GTP Tunnel Endpoint at SCG | O | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs. | – | |
| >>>>Secondary SgNB DL GTP Tunnel Endpoint at SCG | O | | GTP Tunnel Endpoint 9.2.1 | Endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs in case of PDCP duplication. | YES | ignore |
| >>>>RLC Status | O | | 9.2.131 | Indicates the RLC has been re-established. | YES | ignore |
| **E-RABs Admitted To Be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs Admitted To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | Note: no further information contained in the IE container | | |
| E-RABs Not Admitted List | O | | E-RAB List 9.2.28 | A value for *E-RAB ID* shall only be present once in *E-RABs Admitted List* IE and in *E-RABs Not Admitted List* IE. | YES | ignore |
| SgNB to MeNB Container | O | | OCTET STRING | Includes the *CG-Config* message or the *CG-CandidateList* message, as defined in TS 38.331 [31]. | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| SgNB Resource Coordination Information | O | | 9.2.117 | Information used to coordinate resources utilisation between en-gNB and MeNB. | YES | ignore |
| Admitted split SRBs | O | | ENUMERATED (srb1, srb2, srb1&2, ...) | Indicates admitted SRBs | YES | ignore |
| Admitted split SRBs release | O | | ENUMERATED (srb1, srb2, srb1&2, ...) | Indicates admitted SRBs release | YES | ignore |
| RRC config indication | O | | 9.2.132 | Indicates the type of RRC configuration used at the en-gNB. | YES | reject |
| Location Information at SgNB | O | | 9.2.142 | Contains information to support localisation of the UE | YES | ignore |
| Available fast MCG recovery via SRB3 | O | | ENUMERATED (true, ...) | Indicates the fast MCG recovery via SRB3 isenabled. | YES | ignore |
| Release fast MCG recovery via SRB3 | O | | ENUMERATED (true, ...) | Indicates the fast MCG recovery via SRB3 is released. | YES | ignore |
| SCG UE History Information | O | | 9.2.177 | | YES | ignore |
| SCG Activation Status | O | | 9.2.178 | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Conditional PSCell Addition Information Modification Acknowledge** | O | | | This IE may be sent from the target en-gNB. | YES | reject |
| **>Candidate PSCell List** | | *1* | | | – | |
| >>Candidate PSCell Item | | *1 .. <maxnoofPSCellCandidate>* | | | – | |
| >>>PSCell ID | M | | NR CGI 9.2.111 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |
| maxnoofPSCellCandidate | Maximum no. of PSCells for CPAC. Value is 8. |

| Condition | Explanation |
|---|---|
| ifMCGandSCGpresent | This IE shall be present if, for the E-RAB admitted to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present". |
| ifMCGpresent | This IE shall be present if, for the E-RAB admitted to be added, the *MCG resources* IE in the *EN-DC Resource Configuration* IE is set to the value "present". |
| C-ifMCGandSCGpresent_GBRpresent | This IE shall be present if, for the E-RAB admitted to be added, the *MCG resources* and *SCG resources* IEs in the *EN-DC Resource Configuration* IE are set to the value "present", and the *GBR QoS Information* IE is present in the *Requested MCG E-RAB Level QoS Parameters* IE. |

### 9.1.4.7 SGNB MODIFICATION REQUEST REJECT

This message is sent by the en-gNB to inform the MeNB that the MeNB initiated SgNB Modification Preparation has failed.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | ignore |

### 9.1.4.8 SGNB MODIFICATION REQUIRED

This message is sent by the en-gNB to the MeNB to request the modification of en-gNB resources for a specific UE.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| PDCP Change Indication | O | | 9.2.109 | | YES | ignore |
| **E-RABs To Be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>Cause | M | | 9.2.6 | | – | |
| >>RLC Mode | O | | RLC Mode 9.2.119 | Indicates the RLC mode at the en-gNB for PDCP transfer to MeNB. | YES | ignore |
| SgNB to MeNB Container | O | | OCTET STRING | Includes the *CG-Config* message or the *CG-CandidateList* message, as defined in TS 38.331 [31]. | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |
| **E-RABs To Be Modified List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Modified Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| >>>*PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>Requested MCG E-RAB Level QoS Parameters | O | | E-RAB Level QoS Parameters 9.2.9 | Includes E-RAB level QoS parameters requested to be provided by the MCG. | – | |
| >>>>UL Configuration | O | | 9.2.118 | Information about UL usage in the MeNB. | – | |
| >>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the MeNB if received. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the MeNB if received. | YES | ignore |
| >>>>SgNB UL GTP Tunnel Endpoint at PDCP | O | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at PDCP. For delivery of UL PDCP PDUs. | – | |
| >>>>S1 DL GTP Tunnel Endpoint at the SgNB | O | | GTP Tunnel Endpoint 9.2.1 | en-gNB endpoint of the S1 transport bearer. For delivery of DL PDUs. | – | |
| >>>>New DRB ID Request | O | | ENUMERATED (True, …) | | YES | ignore |
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >>>>SgNB DL GTP Tunnel Endpoint at SCG | O | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs. | – | |
| >>>>Secondary SgNB DL GTP Tunnel Endpoint at SCG | O | | GTP Tunnel Endpoint 9.2.1 | SgNB endpoint of the X2-U transport bearer at the SCG. For delivery of DL PDCP PDUs for PDCP duplication. | – | |
| >>>>RLC Status | O | | 9.2.131 | Indicates the RLC has been re-established.. | | |
| >>>>LCID | O | | 9.2.138 | Indicate the LCID of the primary path in case of PDCP duplication | YES | ignore |
| SgNB Resource Coordination Information | O | | 9.2.117 | Information used to coordinate resources utilisation between the en-gNB and the MeNB. | YES | ignore |
| RRC config indication | O | | 9.2.132 | Indicates the type of RRC configuration used at the en-gNB. | YES | reject |
| Location Information at SgNB | O | | 9.2.142 | Contains information to support localisation of the UE | YES | ignore |
| SCG UE History Information | O | | 9.2.177 | | YES | ignore |
| SCG Activation Request | O | | 9.2.179 | | YES | ignore |
| **CPAC Information Required** | O | | | This IE may be sent from the target en-gNB. | YES | ignore |
| **>Candidate PSCell List** | | *1* | | Indicates the full list of candidate PSCells prepared at the target en-gNB. | – | |
| >>Candidate PSCell Item | | *1 .. <maxnoofPSCellCandidate>* | | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>PSCell ID | M | | NR CGI 9.2.111 | | – | |
| SCG Reconfiguration Notification | O | | ENUMERATED (executed, ..., executed-deleted, deleted) | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |
| maxnoofPSCellCandidate | Maximum no. of PSCells for CPAC. Value is 8. |

### 9.1.4.9 SGNB MODIFICATION CONFIRM

This message is sent by the MeNB to inform the en-gNB about the successful modification.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| **E-RABs Admitted To Be Modified List** | | *0..1* | | | YES | ignore |
| **>E-RABs Admitted To Be Modified Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| >>>>Secondary MeNB UL GTP Tunnel Endpoint at PDCP | O | | GTP Tunnel Endpoint 9.2.1 | MeNB endpoint of the X2-U transport bearer at the PDCP. For delivery of UL PDCP PDUs for PDCP duplication. | – | |
| >>>>UL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the en-gNB if received. | YES | ignore |
| >>>>DL PDCP SN Length | O | | PDCP SN Length 9.2.133 | Shall be ignored by the en-gNB if received. | YES | ignore |
| MeNB to SgNB Container | O | | OCTET STRING | Includes the *RRCReconfigurationComplete* message as defined in TS 38.331 [31]. | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | ignore |
| MeNB Resource Coordination Information | O | | 9.2.116 | Information used to coordinate resources utilisation between the MeNB and the en-gNB. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.4.10 SGNB MODIFICATION REFUSE

This message is sent by the MeNB to inform the en-gNB that the SgNB initiated SgNB Modification has failed.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| MeNB to SgNB Container | O | | OCTET STRING | Includes the *CG-ConfigInfo* message as defined in TS 38.331 [31]. | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | ignore |

### 9.1.4.11 SGNB RELEASE REQUEST

This message is sent by the MeNB to the en-gNB to request the release of resources.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| **E-RABs To Be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| >>>*PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the transport bearer used for forwarding of UL PDUs | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the transport bearer used for forwarding of DL PDUs | – | |
| >>>*PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| UE Context Kept Indicator | O | | 9.2.85 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | reject |

| | | | | | | |
|---|---|---|---|---|---|---|
| MeNB to SgNB Container | O | | OCTET STRING | Includes the *CG-ConfigInfo* message as defined in TS 38.331 [31]. | YES | reject |
| E-RABs transferred to MeNB | O | | E-RAB List 9.2.28 | Indicates the target MeNB reconfigured the listed E-RABs as MN-terminated bearers. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.4.12 SGNB RELEASE REQUEST ACKNOWLEDGE

This message is sent by the en-gNB to the MeNB to confirme the request to release en-gNB resources.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |
| **E-RABs Admitted To Be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs Admitted To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>RLC Mode | M | | RLC Mode 9.2.119 | Indicates the RLC mode at the en-gNB for PDCP transfer to MeNB. | – | |
| SCG UE History Information | O | | 9.2.177 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.4.13 SGNB RELEASE REQUEST REJECT

This message is sent by the en-gNB to the MeNB to reject the request to release en-gNB resources.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

## 9.1.4.14 SGNB RELEASE REQUIRED

This message is sent by the en-gNB to request the release of all resources for a specific UE at the en-gNB.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |
| **E-RABs To Be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>RLC Mode | M | | RLC Mode 9.2.119 | Indicates the RLC mode at the en-gNB for PDCP transfer to MeNB. | – | |
| SgNB to MeNB Container | O | | OCTET STRING | Includes the *CG-Config* message as defined in TS 38.331 [31]. | YES | ignore |
| SCG UE History Information | O | | 9.2.177 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

## 9.1.4.15 SGNB RELEASE CONFIRM

This message is sent by the MeNB to confirm the release of all resources for a specific UE at the en-gNB.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| **E-RABs to be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | – | |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| *>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.4.16 SGNB COUNTER CHECK REQUEST

This message is sent by the en-gNB to request the verification of the value of the PDCP COUNTs associated with the bearers established in the en-gNB.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| **E-RABs Subject to Counter Check List** | | *1* | | | YES | ignore |
| **>E-RABs Subject to Counter Check Item** | | *1 .. <maxnoofBearers>* | | | EACH | ignore |
| >>E-RAB ID | M | | 9.2.23 | | - | |
| >>UL COUNT | M | *INTEGER(0.. 4294967295)* | | Indicates the value of uplink COUNT associated to this E-RAB. | - | |
| >>DL COUNT | M | *INTEGER(0.. 4294967295)* | | Indicates the value of downlink COUNT associated to this E-RAB. | - | |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.4.17 SGNB CHANGE REQUIRED

This message is sent by the en-gNB to the MeNB to request the change of en-gNB for a specific UE.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Target SgNB ID Information | M | | 9.2.102 | This IE shall be ignored if the *Conditional PSCell Change Information Required* IE is present. | YES | reject |
| Cause | M | | 9.2.6 | | YES | ignore |
| SgNB to MeNB Container | O | | OCTET STRING | Includes the *CG-Config* message as defined in TS 38.331 [31]. This IE shall be ignored if the *Conditional PSCell Change Information Required* IE is present. | YES | reject |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |
| SCG UE History Information | O | | 9.2.177 | | YES | ignore |
| **Conditional PSCell Change Information Required** | O | | | | YES | reject |
| **>Multiple Target SgNB Node List** | | *1* | | | – | |
| >>Multiple Target SgNB Node Item | | *1 .. <maxnoofTargetSgNBs>* | | | – | |
| >>>Target SgNB ID Information | M | | 9.2.102 | | – | |
| >>>CPC Indicator | M | | ENUMERATED (CPC-initiation, CPC-modification, CPC-cancel,...) | | – | |
| >>>Maximum Number of PSCells To Prepare | M | | INTEGER (1..8, …) | Indicates the maximum number of PSCells that the target may prepare. | – | |
| >>>Estimated Arrival Probability | O | | INTEGER (1..100) | Indicates the arrival probability for the UE towards the candidate target en-gNB. | – | |
| >>>SgNB to MeNB Container | M | | OCTET STRING | Includes the *CG-Config* message as defined in TS 38.331 [31]. | – | |

| Range bound | Explanation |
|---|---|
| maxnoofTargetSgNBs | Maximum no. of the target SgNB nodes. Value is 8 |

#### 9.1.4.18 SGNB CHANGE CONFIRM

This message is sent by the MeNB to inform the en-gNB about the successful change.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| **E-RABs to be Released List** | | *0..1* | | | YES | ignore |
| **>E-RABs To Be Released Item** | | *1 .. <maxnoofBearers>* | | | – | |
| >>E-RAB ID | M | | 9.2.23 | | – | |
| >>EN-DC Resource Configuration | M | | EN-DC Resource Configuration 9.2.108 | Indicates the PDCP and Lower Layer MCG/SCG configuration. | – | |
| >>CHOICE *Resource Configuration* | M | | | | | |
| *>>>PDCP present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "present". | | |
| >>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs | – | |
| >>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs | – | |
| >>>>Additional List of Forwarding GTP Tunnel Endpoint | | *0..1* | | This IE would be present only if multiple candidate target en-gNBs are prepared in case of SN initiated inter-SN CPC. | YES | ignore |
| >>>>>Additional List of Forwarding GTP Tunnel Endpoint-Item | | *1 .. <maxnoofTargetSgNBsMinusOne>* | | | - | |
| >>>>>>UL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of UL PDUs. | - | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>>>DL Forwarding GTP Tunnel Endpoint | O | | GTP Tunnel Endpoint 9.2.1 | Identifies the X2 transport bearer used for forwarding of DL PDUs. | - | |
| *>>>PDCP not present in SN* | | | | This choice tag is used if the *PDCP at SgNB* IE in the *EN-DC Resource Configuration* IE is set to the value "not present". | | |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB | YES | ignore |
| **Conditional PSCell Change Information Confirm** | O | | | | YES | ignore |
| **>Multiple Target SgNB Node List** | | *1* | | | – | |
| >>Multiple Target SgNB Node Item | | *1 .. <maxnoofTargetSgNBs>* | | | – | |
| >>>Target SgNB ID | M | | Global en-gNB ID 9.2.112 | | – | |
| >>>Candidate PSCell List | | *1* | | | – | |
| >>>>Candidate PSCell Item | | *1 .. <maxnoofPSCellCandidate>* | | | – | |
| >>>>>PSCell ID | M | | NR CGI 9.2.111 | | – | |
| MeNB to SgNB Container | O | | OCTET STRING | Includes the *RRCReconfigurationComplete* message as defined in subclause 6.2.2 of TS 38.331 [31]. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |
| maxnoofTargetSgNBs | Maximum no. of the target SgNB nodes. Value is 8 |
| maxnoofPSCellCandidate | Maximum no. of PSCells for CPAC. Value is 8. |
| maxnoofTargetSgNBsMinusOne | Maximum no. of the target SgNB nodes -1. Value is 7 |

### 9.1.4.19 SGNB CHANGE REFUSE

This message is sent by the MeNB to inform the en-gNB that the SgNB initiated SgNB Change has failed.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | ignore |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | ignore |
| Cause | M | | 9.2.6 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

### 9.1.4.20 SECONDARY RAT DATA USAGE REPORT

This message is sent by the en-gNB to report data volumes for secondary RAT.

Direction: en-gNB → MeNB

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Secondary RAT Usage Report List | M | | 9.2.120 | | YES | reject |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

### 9.1.4.21 RRC TRANSFER

This message is sent by the MeNB to the en-gNB or by the en-gNB to the MeNB to transfer an RRC message.

Direction: MeNB → en-gNB or en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| **Split SRB** | | *0..1* | | | YES | reject |
| >RRC Container | O | | OCTET STRING | Contains a PDCP-C PDU encapsulating an RRC message as defined in subclause 6.2.1 of TS 36.331 [9] and ciphered with the key of the MeNB | – | |
| >SRB Type | M | | ENUMERATED (srb1, srb2, ...) | The SRB type | – | |
| >Delivery Status | O | | 9.2.104 | DL RRC delivery status of split SRB | – | |
| **NR UE Report** | | *0..1* | | | YES | reject |
| >RRC Container | M | | OCTET STRING | Includes the *UL-DCCH-Message* as defined in subclause 6.2.1 of TS 38.331 [31] containing the *MeasurementReport* message, or the *FailureInformation* message, or the *RRCReconfigurationComplete* message, or the *UEAssistanceInformation* message, or the *IABOtherInformation* message. | – | |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |
| **Fast MCG Recovery via SRB3 from SgNB to MeNB** | | *0..1* | | | YES | ignore |
| >RRC Container | O | | OCTET STRING | Includes the *UL-DCCH-Message* as defined in subclause 6.2.1 of TS 36.331 [9] containing *MCGFailureInformation* message. | – | |
| **Fast MCG Recovery via SRB3 from MeNB to SgNB** | | *0..1* | | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >RRC Container | O | | OCTET STRING | Includes the *DL-DCCH-Message* as defined in subclause 6.2.1 of TS 36.331 [9] containing the *RRCConnectionReconfiguration* message, or the *RRCConnectionRelease* message, or the *MobilityfromEUTRACommand* message. | – | |

## 9.1.4.22 PARTIAL RESET REQUIRED

This message is sent by an initiating node to a neighbouring node, both nodes able to interact for EN-DC, to release all the resources for selected UEs.

Direction: en-gNB → MeNB, MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| **UEs to be Reset List** | | *1* | | | YES | reject |
| **>UEs To Be Reset Item** | | *1 .. <maxnoofUEs>* | | | | |
| >>MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | | |
| >>SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | | |
| >>MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | | |
| Cause | M | | 9.2.6 | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofUEs | Maximum no. of UEs. Value is 8192. |

## 9.1.4.23 PARTIAL RESET CONFIRM

This message is sent by an initiating node to a neighbouring node, both nodes able to interact for EN-DC, to confirm the release all the resources for selected UEs.

Direction: en-gNB → MeNB, MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| **UEs Admitted to be Reset List** | | *1* | | | YES | reject |
| **>UEs Admitted To Be Reset Item** | | *1 .. <maxnoofUEsinengNBDU>* | | | | |
| >>MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | | |
| >>SgNB UE X2AP ID | O | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | | |
| >>MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | | |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofUEsinengNBDU | Maximum no. of UEs. Value is 8192. |

### 9.1.4.24 E-UTRA – NR CELL RESOURCE COORDINATION REQUEST

Direction: eNB → en-gNB, en-gNB → eNB.

This message is sent by a neighbouring eNB to a peer en-gNB or by a neighbouring en-gNB to a peer eNB, both nodes able to interact for EN-DC, to express the desired resource allocation for data traffic, for the sake of E-UTRA - NR Cell Resource Coordination.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE *Initiating Node Type* | M | | | | - | |
| **>***eNB* | | | | | | |
| >>Data Traffic Resource Indication | M | | 9.2.126 | Indicates resource allocations for data traffic. | YES | reject |
| >>Spectrum Sharing Group ID | M | | INTEGER (1.. maxCellineNB) | Indicates the E-UTRA cells involved in resource coordination with the NR cells affiliated with the same ***Spectrum Sharing Group ID***. | YES | reject |
| **>>List of E-UTRA Cells in E-UTRA Coordination Request** | | *0 .. <maxCellineNB>* | | List of applicable E-UTRA cells. | YES | reject |
| >>>EUTRA Cell ID | M | | ECGI 9.2.14 | | - | |
| **>***en-gNB* | | | | | | |
| >>Data Traffic Resource Indication | M | | 9.2.126 | Indicates resource allocations for data traffic. | YES | reject |
| **>>List of E-UTRA Cells in NR Coordination Request** | | *1 .. <maxCellineNB>* | | List of applicable E-UTRA cells | YES | reject |
| >>>EUTRA Cell ID | M | | ECGI 9.2.14 | | – | |
| >>Spectrum Sharing Group ID | M | | INTEGER (1.. maxCellineNB) | Indicates the NR cells involved in resource coordination with the E-UTRA cells affiliated with the same ***Spectrum Sharing Group ID***. | YES | reject |
| **>>List of NR Cells in NR Coordination Request** | | *0 .. <* maxnoNRcellsSpectrumSharingwithE-UTRA > | | List of applicable NR cells | YES | reject |
| >>>NR-Cell ID | M | | NR-CGI 9.2.111 | | - | |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. of E-UTRA cells in eNB. Value is 256. |
| maxnoNRcellsSpectrumSharingwithE-UTRA | Maximum no. of NR cells affiliated to a Spectrum Sharing Group ID involved in cell resource coordination with a number of E-UTRA cells affiliated with the same Spectrum Sharing Group ID. Value is 64. |

### 9.1.4.25 E-UTRA – NR CELL RESOURCE COORDINATION RESPONSE

This message is sent by a neighbouring eNB to a peer en-gNB or by a neighbouring en-gNB to a peer eNB, both nodes able to interact for EN-DC, as a response to the E-UTRA – NR CELL RESOURCE COORDINATION REQUEST.

Direction: eNB → en-gNB, en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| CHOICE *Responding NodeType* | M | | | | - | |
| ***>eNB*** | | | | | | |
| >>Data Traffic Resource Indication | M | | 9.2.126 | Indicates resource allocations for data traffic. | YES | reject |
| >>Spectrum Sharing Group ID | M | | INTEGER (1.. maxCellineNB) | Indicates the E-UTRA cells involved in resource coordination with the NR cells affiliated with the same *Spectrum Sharing Group ID.* | YES | reject |
| **>>List of E-UTRA Cells in E-UTRA Coordination Response** | | *0 .. <maxCellineNB >* | | List of applicable E-UTRA cells | YES | reject |
| >>>EUTRA Cell ID | M | | ECGI 9.2.14 | | - | |
| ***>en-gNB*** | | | | | | |
| >>Data Traffic Resource Indication | M | | 9.2.126 | Indicates resource allocations for data traffic. | YES | reject |
| >>Spectrum Sharing Group ID | M | | INTEGER (1.. maxCellineNB) | Indicates the NR cells involved in resource coordination with the E-UTRA cells affiliated with the same *Spectrum Sharing Group ID.* | YES | reject |
| **>>List of NR Cells in NR Coordination Response** | | *0 .. < maxnoNRcellsSpectrumSharingwithE-UTRA >* | | List of applicable NR cells | YES | reject |
| >>>NR Cell ID | M | | NR-CGI 9.2.111 | | - | |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

| Range bound | Explanation |
|---|---|
| maxCellineNB | Maximum no. of E-UTRA cells in eNB. Value is 256. |
| maxnoNRcellsSpectrumSharingwithE-UTRA | Maximum no. of NR cells affiliated to a Spectrum Sharing Group ID involved in cell resource coordination with a number of E-UTRA cells affiliated with the same Spectrum Sharing Group ID. Value is 64. |

### 9.1.4.26 SGNB ACTIVITY NOTIFICATION

This message is sent by the en-gNB to inform the MeNB that resources for E-RABs controlled by the en-gNB have not been used or are in use again.

Direction: en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| UE Context level user plane activity report | O | | User plane traffic activity report 9.2.130 | | YES | ignore |
| **E-RAB Activity Notify Item List** | | *0..<maxnoofBearers>* | | | EACH | ignore |
| >E-RAB ID | M | | 9.2.23 | | – | |
| >User plane traffic activity report | M | | 9.2.130 | | – | |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256 |

### 9.1.4.27 GNB STATUS INDICATION

This message is sent by the en-gNB to indicate to the eNB its status of overload.

Direction: en-gNB → eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| gNB Overload Information | M | | ENUMERATED (overloaded, not-overloaded, ...) | | YES | ignore |
| Interface Instance Indication | O | | 9.2.143 | | YES | reject |

### 9.1.4.28 TRACE START

This message is sent by the MeNB to initiate a trace session for a UE.

Direction: MeNB → en-gNB

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.3.1.1 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| Trace Activation | M | | 9.2.2 | | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

### 9.1.4.29 DEACTIVATE TRACE

This message is sent by the MeNB to deactivate a trace session.

Direction: MeNB → en-gNB

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.3.1.1 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| E-UTRAN Trace ID | M | | OCTET STRING (SIZE(8)) | As per E-UTRAN Trace ID in *Trace Activation* IE | YES | ignore |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

### 9.1.4.30 UE Radio Capability ID Mapping Request

This message is sent by the en-gNB and is used to request the UE Radio Capability information that maps to a specific UE Radio Capability ID.

Direction: en-gNB → eNB

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| UE Radio Capability ID | M | | 9.2.171 | | YES | reject |

### 9.1.4.31 UE Radio Capability ID Mapping Response

This message is sent by the eNB and is used to provide the UE Radio Capability information that maps to a specific UE Radio Capability ID indicated in the UE RADIO CAPABILITY ID MAPPING REQUEST message.

Direction: eNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | reject |
| UE Radio Capability ID | M | | 9.2.171 | | YES | reject |
| UE Radio Capability | M | | 9.2.173 | | YES | ignore |
| Criticality Diagnostics | O | | 9.2.7 | | YES | ignore |

### 9.1.4.32 CONDITIONAL PSCELL CHANGE CANCEL

This message is sent by the MeNB to the source en-gNB to inform the cancellation of all of the prepared PSCells in the target en-gNB during a Conditional PSCell Change.

Direction: MeNB → en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |
| Cause | O | | 9.2.6 | | YES | ignore |
| Target en-gNB ID Information | M | | 9.2.102 | | YES | reject |

## 9.1.5 Messages for IAB Procedures

### 9.1.5.1 F1-C TRAFFIC TRANSFER

This message is sent by the en-gNB to the MeNB or by the MeNB to the en-gNB to transfer the F1-C traffic to and from an IAB-node.

Direction: MeNB → en-gNB or en-gNB → MeNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Message Type | M | | 9.2.13 | | YES | ignore |
| MeNB UE X2AP ID | M | | eNB UE X2AP ID 9.2.24 | Allocated at the MeNB. | YES | reject |
| SgNB UE X2AP ID | M | | en-gNB UE X2AP ID 9.2.100 | Allocated at the en-gNB. | YES | reject |
| F1-C Traffic Container | M | | OCTET STRING | Contains an F1-C interface SCTP CHUNK and IP header, or an IP packet to protect the traffic on the F1-C interface as defined in TS 33.501 [50]. This IE corresponds to the *dedicatedInfoF1c* defined in TS 36.331 [9]. | YES | reject |
| MeNB UE X2AP ID Extension | O | | Extended eNB UE X2AP ID 9.2.86 | Allocated at the MeNB. | YES | reject |

# 9.2 Information Element definitions

## 9.2.0 General

When specifying information elements which are to be represented by bit strings, if not otherwise specifically stated in the semantics description of the concerned IE or elsewhere, the following principle applies with regards to the ordering of bits:

- The first bit (leftmost bit) contains the most significant bit (MSB);
- The last bit (rightmost bit) contains the least significant bit (LSB);

- When importing bit strings from other specifications, the first bit of the bit string contains the first bit of the concerned information.

### 9.2.1 GTP Tunnel Endpoint

The *GTP Tunnel Endpoint* IE identifies an X2 transport bearer or the S-GW endpoint of the S1 transport bearer associated to an E-RAB. It contains a Transport Layer Address and a GTP Tunnel Endpoint Identifier. The Transport Layer Address is an IP address to be used for the X2 user plane transport (see TS 36.424 [8]) or for the S1 user plane transport (see TS 36.414 [19]). The GTP Tunnel Endpoint Identifier is to be used for the user plane transport. The QoS Mapping Information is used to set the IP header of packets in case that the en-gNB serves the IAB, and the packets belonging to MN-terminated split bearer/SCG bearer are transmitted from MeNB to en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Transport Layer Address | M | | BIT STRING (1..160, ...) | For details on the Transport Layer Address, see TS 36.424 [8], TS 36.414 [19] | – | |
| GTP TEID | M | | OCTET STRING (4) | For details and range, see TS 29.281 [26] | – | |
| QoS Mapping Information | O | | 9.2.172 | | YES | reject |

### 9.2.2 Trace Activation

Defines parameters related to trace activation.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| E-UTRAN Trace ID | M | | OCTET STRING (8) | The E-UTRAN Trace ID IE is composed of the following: Trace Reference defined in TS 32.422 [6] (leftmost 6 octets, with PLMN information coded as in 9.2.4), and Trace Recording Session Reference defined in TS 32.422 [6] (last 2 octets) | – | |
| Interfaces To Trace | M | | BIT STRING (8) | Each position in the bitmap represents a eNB interface: first bit =S1-MME, second bit =X2, third bit =Uu, fourth bit = F1-C, fifth bit = E1. Other bits reserved for future use. Value '1' indicates 'should be traced'. Value '0' indicates 'should not be traced'. | – | |
| Trace Depth | M | | ENUMERATED( minimum, medium, maximum, MinimumWithoutVendorSpecificExtension, MediumWithoutVendorSpecificExtension, MaximumWithoutVendorSpecificExtension, ...) | Defined in TS 32.421 [7] | – | |
| Trace Collection Entity IP Address | M | | BIT STRING (1..160,…) | For File based Reporting. Defined in TS 32.422 [6]. For details on the Transport Layer Address, see TS 36.424 [8], TS 36.414 [19]. This IE is ignored if the *Trace Collection Entity URI* IE is present. | – | |
| MDT Configuration | O | | 9.2.56 | | YES | ignore |
| UE Application layer measurement configuration | O | | 9.2.121 | | YES | ignore |
| MDT Configuration NR | O | | OCTET STRING | Defined in TS 38.413 [39]. | YES | ignore |
| Trace Collection Entity URI | O | | URI 9.2.174 | For Streaming based Reporting. Defined in TS 32.422 [11]. Replaces Trace Collection Entity IP Address if present. | YES | ignore |

### 9.2.3 Handover Restriction List

This IE defines roaming or access restrictions for subsequent mobility action for which the eNB provides information about the target of the mobility action towards the UE, e.g., handover and CCO, or for SCG selection during dual connectivity operation.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Serving PLMN | M | | PLMN Identity 9.2.4 | | – | |
| **Equivalent PLMNs** | | *0..<maxnoofEPLMNs>* | | Allowed PLMNs in addition to Serving PLMN. This list corresponds to the list of "equivalent PLMNs list" as defined in TS 24.301 [14]. This list is part of the roaming restriction information. Roaming restrictions apply to PLMNs other than the serving PLMN and Equivalent PLMNs. | – | |
| >PLMN Identity | M | | 9.2.4 | | – | |
| **Forbidden TAs** | | *0..<maxnoofEPLMNsPlusOne>* | | intra E-UTRAN roaming restrictions | – | |
| >PLMN Identity | M | | 9.2.4 | The PLMN of forbidden TACs | – | |
| **>Forbidden TACs** | | *1..<maxnoofForbTACs>* | | | – | |
| >>TAC | M | | OCTET STRING(2) | The forbidden TAC | – | |
| **Forbidden LAs** | | *0..<maxnoofEPLMNsPlusOne>* | | inter-3GPP RAT roaming restrictions | – | |
| >PLMN Identity | M | | 9.2.4 | | – | |
| **>Forbidden LACs** | | *1..<maxnoofForbLACs>* | | | – | |
| >>LAC | M | | OCTET STRING(2) | | – | |
| Forbidden inter RATs | O | | ENUMERATED(ALL, GERAN, UTRAN, CDMA2000, …,GERAN and UTRAN, CDMA2000 and UTRAN) | inter-3GPP and 3GPP2 RAT access restrictions. "ALL" means that all RATs mentioned in the enumeration of this IE are restricted. | – | |
| NR restriction in EPS as secondary RAT | O | | ENUMERATED(NRrestrictedinEPSasSecondaryRAT, …) | Restriction to use NR when the NR is used as secondary RAT in EN-DC. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Core Network Type Restrictions** | | *0..<maxnoofEPLMNsPlusOne>* | | Includes any of the Serving PLMN or any PLMN of the Equivalent PLMNs listed in the *Mobility Restriction List* IE for which core network type restriction applies as specified in TS 23.501 [38]. | YES | ignore |
| >PLMN Identity | M | | 9.2.4 | | – | |
| >Core Network Type | M | | ENUMERATED (5GCForbidden, …, EPCForbidden) | The indication indicates whether the UE is restricted to connect to 5GC or to EPC for this PLMN. | – | |
| NR Restriction in 5GS | O | | ENUMERATED(NRrestrictedin5GS, …) | Restriction to use NR when the NR connects to 5GS. | YES | ignore |
| Last NG-RAN PLMN Identity | O | | 9.2.4 | Indicates the NG-RAN PLMN from where the UE formerly handed over to EPS and which is preferred in case of subsequent mobility to 5GS. | YES | ignore |
| Unlicensed Spectrum Restriction | O | | ENUMERATED(UnlicensedRestricted, …) | Restriction to use unlicensed spectrum in the form of LAA or LWA/LWIP or NR-U as described in TS 23.401 [11]. | YES | ignore |
| **RAT Restrictions** | | *0..<maxnoofEPLMNsPlusOne>* | | This IE contains RAT restriction related information as specified in TS 23.401 [11]. | – | |
| >PLMN Identity | M | | 9.2.4 | | – | |
| >RAT Restriction Information | M | | BIT STRING { LEO (0), MEO (1), GEO (2), OTHERSAT (3)} (SIZE(8, …)) | Each position in the bitmap represents a RAT. If a bit is set to "1", the respective RAT is restricted for the UE. If a bit is set to "0", the respective RAT is not restricted for the UE. Bits 4-7 reserved for future use. | – | |

| Range bound | Explanation |
|---|---|
| maxnoofEPLMNs | Maximum no. of equivalent PLMN Ids. Value is 15. |
| maxnoofEPLMNsPlusOne | Maximum no. of equivalent PLMN Ids plus one. Value is 16. |
| maxnoofForbTACs | Maximum no. of forbidden Tracking Area Codes. Value is 4096. |
| maxnoofForbLACs | Maximum no. of forbidden Location Area Codes. Value is 4096. |

## 9.2.4 PLMN Identity

This information element indicates the PLMN Identity.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| PLMN Identity | M | | OCTET STRING (3) | - digits 0 to 9, encoded 0000 to 1001,<br>- 1111 used as filler digit,<br>two digits per octet,<br>- bits 4 to 1 of octet n encoding digit 2n-1<br>- bits 8 to 5 of octet n encoding digit 2n<br><br>-The PLMN identity consists of 3 digits from MCC followed by either<br>-a filler digit plus 2 digits from MNC (in case of 2 digit MNC) or<br>-3 digits from MNC (in case of a 3 digit MNC). |

## 9.2.5 DL Forwarding

This element indicates that the E-RAB is proposed for forwarding of downlink packets.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DL Forwarding | M | | ENUMERATED (DL forwarding proposed, …) | |

## 9.2.6 Cause

The purpose of the cause information element is to indicate the reason for a particular event for the whole protocol.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| CHOICE Cause Group | M | | | |
| *>Radio Network Layer* | | | | |
| >>Radio Network Layer Cause | M | | ENUMERATED (Handover Desirable for Radio Reasons, Time Critical Handover, Resource Optimisation Handover, Reduce Load in Serving Cell, Partial Handover, Unknown New eNB UE X2AP ID, Unknown Old eNB UE X2AP ID, Unknown Pair of UE X2AP ID, HO Target not Allowed, $TX2_{RELOCoverall}$ Expiry, $T_{RELOCprep}$ Expiry, Cell not Available, No Radio Resources Available in Target Cell, Invalid MME Group ID, Unknown MME Code, Encryption And/Or Integrity Protection Algorithms Not Supported, ReportCharacteristicsEmpty, NoReportPeriodicity, ExistingMeasurementID, Unknown eNB Measurement ID, Measurement Temporarily not Available, Unspecified,...,Load Balancing, Handover Optimisation, Value out of allowed range, Multiple E-RAB ID instances, Switch Off Ongoing, Not supported QCI value, Measurement not supported for the object,$T_{DCoverall}$ Expiry, $T_{DCprep}$ Expiry, Action Desirable for Radio Reasons, Reduce Load, Resource Optimisation, Time Critical action, Target not Allowed, No Radio Resources Available, Invalid QoS combination, Encryption Algorithms Not Supported, Procedure cancelled, RRM purpose, Improve user bit rate, User Inactivity, Radio Connection With UE Lost, Failure in the Radio Interface Procedure, Bearer Option not Supported, MCG Mobility, SCG Mobility, Count reaches max value, Unknown Old en-gNB UE X2AP ID, PDCP Overload, CHO-CPC resources to be changed, UE Power Saving, Insufficient UE Capabilities, Normal Release, Unknown E-UTRAN node Measurement ID, SCG activation deactivation failure, SCG deactivation failure due to data transmission, UP integrity protection not possible) | |

| | | | | |
|---|---|---|---|---|
| *>Transport Layer* | | | | |
| >>Transport Layer Cause | M | | ENUMERATED (Transport Resource Unavailable, Unspecified,...) | |
| *>Protocol* | | | | |
| >>Protocol Cause | M | | ENUMERATED (Transfer Syntax Error,Abstract Syntax Error (Reject),Abstract Syntax Error (Ignore and Notify),Message not Compatible with Receiver State,Semantic Error,Unspecified,Abstract Syntax Error (Falsely Constructed Message),...) | |
| *>Misc* | | | | |
| >>Miscellaneous Cause | M | | ENUMERATED (Control Processing Overload, Hardware Failure,O&M Intervention,Not enough User Plane Processing Resources,Unspecified,...) | |

The meaning of the different cause values is described in the following table. In general, "not supported" cause values indicate that the concerned capability is missing. On the other hand, "not available" cause values indicate that the concerned capability is present, but insufficient resources were available to perform the requested action.

| Radio Network Layer cause | Meaning |
|---|---|
| Cell not Available | The concerned cell is not available. |
| Handover Desirable for Radio Reasons | The reason for requesting handover is radio related. |
| Handover Target not Allowed | Handover to the indicated target cell is not allowed for the UE in question |
| Invalid MME Group ID | The target eNB doesn't belong to the same pool area of the source eNB i.e. S1 handovers should be attempted instead. |
| No Radio Resources Available in Target Cell | The target cell doesn't have sufficient radio resources available. |
| Partial Handover | Provides a reason for the handover cancellation. The target eNB did not admit all E-RABs included in the HANDOVER REQUEST and the source eNB estimated service continuity for the UE would be better by not proceeding with handover towards this particular target eNB. |
| Reduce Load in Serving Cell | Load in serving cell needs to be reduced. When applied to handover preparation, it indicates the handover is triggered due to load balancing. |
| Resource Optimisation Handover | The reason for requesting handover is to improve the load distribution with the neighbour cells. |
| Time Critical Handover | Handover is requested for time critical reason i.e. this cause value is reserved to represent all critical cases where the connection is likely to be dropped if handover is not performed. |
| $TX2_{RELOCoverall}$ Expiry | The reason for the action is expiry of timer $TX2_{RELOCoverall}$. |
| $T_{RELOCprep}$ Expiry | Handover Preparation procedure is cancelled when timer $T_{RELOCprep}$ expires. |
| Unknown MME Code | The target eNB belongs to the same pool area of the source eNB and recognizes the MME Group ID. However, the MME Code is unknown to the target eNB. |
| Unknown New eNB UE X2AP ID | The action failed because the New eNB UE X2AP ID or the MeNB UE X2AP ID is unknown. |
| Unknown Old eNB UE X2AP ID | The action failed because the Old eNB UE X2AP ID or the SeNB UE X2AP ID is unknown. |
| Unknown Pair of UE X2AP ID | The action failed because the pair of UE X2 AP IDs is unknown. |
| Encryption And/Or Integrity Protection Algorithms Not Supported | The target eNB is unable to support any of the encryption and/or integrity protection algorithms supported by the UE, or the en-gNB is unable to support any of the NR encryption and/or integrity protection algorithms supported by the UE for EN-DC operation. |
| ReportCharacteristicsEmpty | The action failed because there is no characteristic reported. |
| NoReportPeriodicity | The action failed because the periodicity is not defined. |
| ExistingMeasurementID | The action failed because measurement-ID is already used. |
| Unknown eNB Measurement ID | The action failed because some eNB Measurement-ID is unknown. |
| Measurement Temporarily not Available | The eNB can temporarily not provide the requested measurement object. |
| Load Balancing | The reason for mobility settings change is load balancing. |
| Handover Optimisation | The reason for mobility settings change is handover optimisation. |
| Value out of allowed range | The action failed because the proposed Handover Trigger parameter change in the $eNB_2$ Proposed Mobility Parameters IE is too low or too high. |
| Multiple E-RAB ID Instances | The action failed because multiple instances of the same E-RAB had been provided to the eNB. |
| Switch Off Ongoing | The reason for the action is an ongoing switch off i.e. the concerned cell will be switched off after offloading and not be available. It aides the receiving eNB in taking subsequent actions, e.g. selecting the target cell for subsequent handovers. |
| Not supported QCI value | The action failed because the requested QCI is not supported. |
| Unspecified | Sent when none of the above cause values applies but still the cause is Radio Network Layer related. |
| Measurement not Supported For The Object | At least one of the concerned cell(s) does not support the requested measurement. |
| $T_{DCoverall}$ Expiry | The reason for the action is expiry of timer $T_{DCoverall}$. |
| $T_{DCprep}$ Expiry | The reason for the action is expiry of timer $T_{DCprep}$. |
| Action Desirable for Radio Reasons | The reason for requesting the action is radio related.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Reduce Load | Load in the cell(group) served by the requesting node needs to be reduced.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |

| | |
|---|---|
| Resource Optimisation | The reason for requesting this action is to improve the load distribution with the neighbour cells.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Time Critical action | The action is requested for time critical reason i.e. this cause value is reserved to represent all critical cases where radio resources are likely to be dropped if the requested action is not performed.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Target not Allowed | Requested action towards the indicated target cell is not allowed for the UE in question.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| No Radio Resources Available | The cell(s) in the requested node don't have sufficient radio resources available.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Invalid QoS combination | The action was failed because of invalid QoS combination.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Encryption Algorithms Not Supported | The requested eNB is unable to support any of the encryption algorithms supported by the UE.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Procedure cancelled | The sending node cancelled the procedure due to other urgent actions to be performed.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| RRM purpose | The procedure is initiated due to node internal RRM purposes.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Improve User Bit Rate | The reason for requesting this action is to improve the user bit rate.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| User Inactivity | The action is requested due to user inactivity on all E-RABs, e.g., S1 is requested to be released in order to optimise the radio resources; or SeNB/en-gNB didn't see activity on the DRB recently.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Radio Connection With UE Lost | The action is requested due to losing the radio connection to the UE.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Failure in the Radio Interface Procedure | Radio interface procedure has failed.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Bearer Option not Supported | The requested bearer option is not supported by the sending node.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| MCG Mobility | The procedure is initiated due to mobility related at MCG radio resource. |
| SCG Mobility | The procedure is initiated due to mobility related at SCG radio resource. |
| Count reaches max value | Indicates the PDCP COUNT for UL or DL reached the max value and the bearer may be released. |
| Unknown Old en-gNB UE X2AP ID | The action failed because the Old en-gNB UE X2AP ID or the SgNB UE X2AP ID is unknown. |
| PDCP Overload | The procedure is initiated due to PDCP resource limitation. |
| CHO-CPC resources to be changed | The prepared resources for CHO or CPC for a UE are to be changed. |
| UE Power Saving | The procedure is initiated to accommodate the preference indicated by UE to release the SCG for UE power saving purpose.<br>In the current version of this specification applicable for Dual Connectivity and EN-DC only. |
| Insufficient UE Capabilities | The procedure can't proceed due to insufficient UE capabilities. |
| Normal Release | The release is due to normal reasons. |
| Unknown E-UTRAN node Measurement ID | The action failed because some E-UTRAN node Measurement-ID is unknown. |
| SCG activation deactivation failure | The action failed due to rejection of the SCG activation deactivation request. |

| | |
|---|---|
| SCG deactivation failure due to data transmission | The SCG deactivation failure due to ongoing or arriving data transmission. |
| UP integrity protection not possible | The E-RAB can't be accepted according to the required user plane integrity protection policy. |

| Transport Network Layer cause | Meaning |
|---|---|
| Transport resource unavailable | The required transport resources are not available. |
| Unspecified | Sent when none of the above cause values applies but still the cause is Transport Network Layer related |

| Protocol cause | Meaning |
|---|---|
| Abstract Syntax Error (Reject) | The received message included an abstract syntax error and the concerned criticality indicated "reject" (see sub clause 10.3 of TS 36.413 [4]). |
| Abstract Syntax Error (Ignore and Notify) | The received message included an abstract syntax error and the concerned criticality indicated "ignore and notify" (see sub clause 10.3 of TS 36.413 [4]). |
| Abstract syntax error (falsely constructed message) | The received message contained IEs or IE groups in wrong order or with too many occurrences (see sub clause 10.3 of TS 36.413 [4]). |
| Message not Compatible with Receiver State | The received message was not compatible with the receiver state (see sub clause 10.4 of TS 36.413 [4]). |
| Semantic Error | The received message included a semantic error (see sub clause 10.4 of TS 36.413 [4]). |
| Transfer Syntax Error | The received message included a transfer syntax error (see sub clause 10.2 of TS 36.413 [4]). |
| Unspecified | Sent when none of the above cause values applies but still the cause is Protocol related |

| Miscellaneous cause | Meaning |
|---|---|
| Control Processing Overload | eNB control processing overload |
| Hardware Failure | eNB hardware failure |
| Not enough User Plane Processing Resources | eNB has insufficient user plane processing resources available. |
| O&M Intervention | Operation and Maintenance intervention related to eNB equipment |
| Unspecified | Sent when none of the above cause values applies and the cause is not related to any of the categories Radio Network Layer, Transport Network Layer or Protocol |

## 9.2.7 Criticality Diagnostics

The *Criticality Diagnostics* IE is sent by the eNB/en-gNB when parts of a received message have not been comprehended or were missing, or if the message contained logical errors. When applicable, it contains information about which IEs were not comprehended or were missing.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Procedure Code | O | | INTEGER (0..255) | Procedure Code is to be used if Criticality Diagnostics is part of Error Indication procedure, and not within the response message of the same procedure that caused the error. |
| Triggering Message | O | | ENUMERATED(initiating message, successful outcome, unsuccessful outcome) | The Triggering Message is used only if the Criticality Diagnostics is part of Error Indication procedure. |
| Procedure Criticality | O | | ENUMERATED(reject, ignore, notify) | This Procedure Criticality is used for reporting the Criticality of the Triggering message (Procedure). |
| **Information Element Criticality Diagnostics** | | *0..<maxNrOfErrors>* | | |
| >IE Criticality | M | | ENUMERATED(reject, ignore, notify) | The IE Criticality is used for reporting the criticality of the triggering IE. The value "ignore"' shall not be used. |
| >IE ID | M | | INTEGER (0..65535) | The IE ID of the not understood or missing IE |
| >Type Of Error | M | | ENUMERATED(not understood, missing, …) | |

| Range bound | Explanation |
|---|---|
| maxNrOfErrors | Maximum no. of IE errors allowed to be reported with a single message. The value for maxnooferrors is 256. |

## 9.2.8 Served Cell Information

This IE contains cell configuration information of a cell that a neighbour eNB may need for the X2 AP interface.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PCI | M | | INTEGER (0..503, …) | Physical Cell ID | – | |
| Cell ID | M | | ECGI 9.2.14 | | – | |
| TAC | M | | OCTET STRING(2) | Tracking Area Code | – | |
| **Broadcast PLMNs** | | *1..<maxnoof BPLMNs>* | | Broadcast PLMNs in SIB1 associated to the E-UTRA Cell Identity in the *Cell ID* IE. | – | |
| >PLMN Identity | M | | 9.2.4 | | – | |
| CHOICE *EUTRA-Mode-Info* | M | | | | – | |
| *>FDD* | | | | | | |
| **>>FDD Info** | | *1* | | | – | |
| >>>UL EARFCN | M | | EARFCN 9.2.26 | Corresponds to $N_{UL}$ in TS 36.104 [16] for E-UTRA operating bands for which it is defined; ignored for E-UTRA operating bands for which $N_{UL}$ is not defined | – | |
| >>>DL EARFCN | M | | EARFCN 9.2.26 | Corresponds to $N_{DL}$ in TS 36.104 [16] | – | |
| >>>UL Transmission Bandwidth | M | | Transmission Bandwidth 9.2.27 | Same as DL Transmission Bandwidth in this release; ignored in case UL EARFCN value is ignored | – | |
| >>>DL Transmission Bandwidth | M | | Transmission Bandwidth 9.2.27 | | – | |
| >>>UL EARFCN Extension | O | | EARFCN Extension 9.2.65 | If this IE is present, the value signalled in the *UL EARFCN* IE is ignored. | YES | reject |
| >>>DL EARFCN Extension | O | | EARFCN Extension 9.2.65 | If this IE is present, the value signalled in the *DL EARFCN* IE is ignored. | YES | reject |
| >>>Offset of NB-IoT Channel Number to DL EARFCN | O | | Offset of NB-IoT Channel Number to EARFCN 9.2.94 | Corresponds to $M_{DL}$ in TS 36.104 [16] | YES | reject |
| >>>Offset of NB-IoT Channel Number to UL EARFCN | O | | Offset of NB-IoT Channel Number to EARFCN 9.2.94 | Corresponds to $M_{UL}$ in TS 36.104 [16] | YES | reject |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>NRS-NSSS-PowerOffset | O | | ENUMERATED (-3, 0, 3, …) | NRS to NSSS power ratio, as defined in TS6.213 [11]. | YES | Ignore |
| >>>NSSS-NumOccasionDifferentPrecoder | O | | ENUMERATED (2, 4, 8, …) | The number of consecutive NSSS occasions that use different precoders for NSSS transmission, as defined in TS6.213 [11]. | YES | ignore |
| *>TDD* | | | | | – | |
| **>>TDD Info** | | *1* | | | – | |
| >>>EARFCN | M | | 9.2.26 | Corresponds to $N_{DL}/N_{UL}$ in TS 36.104 [16] | – | |
| >>>Transmission Bandwidth | M | | Transmission Bandwidth 9.2.27 | | – | |
| >>>Subframe Assignment | M | | ENUMERATED (sa0, sa1, sa2, sa3, sa4, sa5, sa6,…) | Uplink-downlink subframe configuration information defined in TS 36.211 [10]. In NB-IOT, sa0 and sa6 are not applicable. | – | |
| **>>>Special Subframe Info** | | *1* | | Special subframe configuration information defined in TS 36.211 [10] | – | |
| >>>>Special Subframe Patterns | M | | ENUMERATED (ssp0, ssp1, ssp2, ssp3, ssp4, ssp5, ssp6, ssp7, ssp8, …) | | – | |
| >>>>Cyclic Prefix DL | M | | ENUMERATED (Normal, Extended,…) | | – | |
| >>>>Cyclic Prefix UL | M | | ENUMERATED (Normal, Extended,…) | | – | |
| **>>>Additional Special Subframe Info** | O | | | Special subframe configuration information defined in TS 36.211 [10]. Only for newly defined configuration of special subframe from Release 11. | YES | ignore |
| >>>>Additional Special Subframe Patterns | M | | ENUMERATED (ssp0, ssp1, ssp2, ssp3, ssp4, ssp5, ssp6, ssp7, ssp8, ssp9, …) | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>Cyclic Prefix DL | M | | ENUMERATED (Normal, Extended,…) | | – | |
| >>>>Cyclic Prefix UL | M | | ENUMERATED (Normal, Extended,…) | | – | |
| >>>EARFCN Extension | O | | 9.2.65 | If this IE is present, the value signalled in the *EARFCN* IE is ignored. | YES | reject |
| **>>>Additional Special Subframe Extension Info** | O | | | Special subframe configuration information defined in TS 36.211 [10]. Only for newly defined configuration of special subframe from Release 14. | YES | ignore |
| >>>>Additional Special Subframe Patterns Extension | M | | ENUMERATED (ssp10, …) | | – | |
| >>>>Cyclic Prefix DL | M | | ENUMERATED (Normal, Extended,…) | | – | |
| >>>>Cyclic Prefix UL | M | | ENUMERATED (Normal, Extended,…) | | – | |
| >>>Offset of NB-IoT Channel Number to DL EARFCN | O | | Offset of NB-IoT Channel Number to EARFCN 9.2.94 | Corresponds to $M_{DL}$ in TS 36.104 [16] | YES | reject |
| >>>NB-IoT UL DL Alignment Offset | O | | NB-IoT UL DL Alignment Offset 9.2.144 | | YES | reject |
| Number of Antenna Ports | O | | 9.2.43 | | YES | ignore |
| PRACH Configuration | O | | PRACH Configuration 9.2.50 | | YES | ignore |
| **MBSFN Subframe Info** | | *0..<maxnoof MBSFN>* | | Corresponds to information provided in the *MBSFN-SubframeConfig* IE as defined in TS 36.331 [9] | GLOBAL | ignore |
| >Radioframe Allocation Period | M | | ENUMERATED (n1, n2, n4, n8, n16, n32, …) | | – | |
| >Radioframe Allocation Offset | M | | INTEGER (0..7, ...) | | – | |
| >Subframe Allocation | M | | 9.2.51 | | – | |
| CSG ID | O | | 9.2.53 | | YES | ignore |
| **MBMS Service Area Identity List** | | *0..<maxnoof MBMSServiceAreaIdentities >* | | Supported MBMS Service Area Identities in the cell | GLOBAL | ignore |
| >MBMS Service Area Identity | | | OCTET STRING(2) | MBMS Service Area Identities as defined in TS 23.003 [29] | | |
| MultibandInfoList | O | | 9.2.60 | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| FreqBandIndicatorPriority | O | | ENUMERATED (not-broadcasted, broadcasted, ...) | This IE indicates that the eNodeB supports *FreqBandIndicationPriority*, and whether *FreqBandIndicatorPriority* is broadcasted in SIB 1 (see TS 36.331 [9]) | YES | ignore |
| BandwidthReducedSI | O | | ENUMERATED (scheduled, ...) | This IE indicates that the SystemInformationBlockType1-BR is scheduled in the cell (see TS 36.331 [9]) | YES | ignore |
| Protected E-UTRA Resource Indication | O | | 9.2.125 | This IE indicates which E-UTRA control/reference signal resources are protected and are not subject to E-UTRA - NR Cell Resource Coordination. | YES | ignore |
| **Broadcast PLMN Identity Info List E-UTRA** | | *0..<maxnoofBPLMNs>* | | This IE corresponds to information provided in the *cellAccessRelatedInfo* in the *SystemInformationBlockType1* message as specified in TS 36.331 [9]. All PLMN Identities and associated information contained in the *cellAccessRelatedInfo* IE are included and provided in the same order as broadcast in the *SystemInformationBlockType1* message. | YES | ignore |
| **>Broadcast PLMNs** | | *1..<maxnoofBPLMNs>* | | Broadcast PLMN IDs in the *SystemInformationBlockType1* message (SIB1) associated to the *E-UTRA Cell Identity* IE. | – | |
| >>PLMN Identity | M | | 9.2.4 | | – | |
| >TAC | M | | OCTET STRING(2) | | – | |
| >E-UTRA Cell Identity | M | | BIT STRING (28) | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| NPRACH Configuration | O | | NPRACH Configuration 9.2.170 | | YES | ignore |
| SFN Offset | O | | 9.2.175 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofBPLMNs | Maximum no. of Broadcast PLMN Ids. Value is 6. |
| maxnoofMBSFN | Maximum no. of MBSFN frame allocation with different offset. Value is 8. |
| maxnoofMBMSServiceAreaIdentities | Maximum no. of MBMS Service Area Identities. Value is 256. |

## 9.2.9 E-RAB Level QoS Parameters

This IE defines the QoS to be applied to an E-RAB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| QCI | M | | INTEGER (0..255) | QoS Class Identifier defined in TS 23.401 [12]. Logical range and coding specified in TS 23.203 [13]. | – | |
| Allocation and Retention Priority | M | | 9.2.31 | | – | |
| GBR QoS Information | O | | 9.2.10 | This IE applies to GBR bearers only and shall be ignored otherwise. | – | |
| Downlink Maximum Packet Loss Rate | O | | Packet Loss Rate 9.2.124 | This IE applies only to bearers with specific QCI (see TS 23.401 [12]) and indicates the maximum allowed packet loss rate for downlink as specified in TS 23.401 [12]. | YES | ignore |
| Uplink Maximum Packet Loss Rate | O | | Packet Loss Rate 9.2.124 | This IE applies only to bearers with specific QCI (see TS 23.401 [12]) and indicates the maximum allowed packet loss rate for uplink as specified in TS 23.401 [12]. | YES | ignore |

## 9.2.10 GBR QoS Information

This IE indicates the maximum and guaranteed bit rates of a GBR E-RAB for downlink and uplink.

NOTE: For LTE DC, the SeNB regards the *GBR QoS Information* IE as an E-RAB level parameter also for E-RABs configured with the split bearer option, although for the split bearer option the bitrates signalled by the MeNB are typically not equal to the bitrates signalled by the MME for that E-RAB (see TS 36.300 [15]).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| E-RAB Maximum Bit Rate Downlink | M | | Bit Rate 9.2.11 | Maximum Bit Rate in DL (i.e. from EPC to E-UTRAN) for the bearer.<br>Details in TS 23.401 [12].<br>If the *Extended E-RAB Maximum Bit Rate Downlink* IE is included, the *E-RAB Maximum Bit Rate Downlink* IE shall be ignored. | – | |
| E-RAB Maximum Bit Rate Uplink | M | | Bit Rate 9.2.11 | Maximum Bit Rate in UL (i.e. from E-UTRAN to EPC) for the bearer.<br>Details in TS 23.401 [12].<br>If the *Extended E-RAB Maximum Bit Rate Uplink* IE is included, the *E-RAB Maximum Bit Rate Uplink* IE shall be ignored. | – | |
| E-RAB Guaranteed Bit Rate Downlink | M | | Bit Rate 9.2.11 | Guaranteed Bit Rate (provided that there is data to deliver) in DL (i.e. from EPC to E-UTRAN) for the bearer.<br>Details in TS 23.401 [12].<br>If the *Extended E-RAB Guaranteed Bit Rate Downlink* IE is included, the *E-RAB Guaranteed Bit Rate Downlink* IE shall be ignored. | – | |
| E-RAB Guaranteed Bit Rate Uplink | M | | Bit Rate 9.2.11 | Guaranteed Bit Rate (provided that there is data to deliver) in UL (i.e. from E-UTRAN to EPC) for the bearer.<br>Details in TS 23.401 [12].<br>If the *Extended E-RAB Guaranteed Bit Rate Uplink* IE is included, the *E-RAB Guaranteed Bit Rate Uplink* IE shall be ignored. | – | |
| Extended E-RAB Maximum Bit Rate Downlink | O | | Extended Bit Rate 9.2.99 | Maximum Bit Rate in DL (i.e. from EPC to E-UTRAN) for the bearer.<br>Details in TS 23.401 [12]. | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Extended E-RAB Maximum Bit Rate Uplink | O | | Extended Bit Rate 9.2.99 | Maximum Bit Rate in UL (i.e. from E-UTRAN to EPC) for the bearer. Details in TS 23.401 [12]. | – | |
| Extended E-RAB Guaranteed Bit Rate Downlink | O | | Extended Bit Rate 9.2.99 | Guaranteed Bit Rate (provided that there is data to deliver) in DL (i.e. from EPC to E-UTRAN) for the bearer. Details in TS 23.401 [12]. | – | |
| Extended E-RAB Guaranteed Bit Rate Uplink | O | | Extended Bit Rate 9.2.99 | Guaranteed Bit Rate (provided that there is data to deliver) in UL (i.e. from E-UTRAN to EPC) for the bearer. Details in TS 23.401 [12]. | – | |

## 9.2.11 Bit Rate

This IE indicates the number of bits delivered by E-UTRAN in UL or to E-UTRAN in DL or by UE in sidelink within a period of time, divided by the duration of the period. It is used, for example, to indicate the maximum or guaranteed bit rate for a GBR E-RAB, or an aggregated maximum bit rate.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Bit Rate | M | | INTEGER (0..10,000,000,000) | The unit is: bit/s |

## 9.2.12 UE Aggregate Maximum Bit Rate

On Handover Aggregate Maximum Bitrate is transferred to the target eNB. In Dual Connectivity, UE Aggregate Maximum Bit Rate is split into MeNB UE Aggregate Maximum Bit Rate and SeNB UE Aggregate Maximum Bit Rate which are enforced by MeNB and SeNB respectively as specified in TS 36.300 [15]. The UE Aggregate Maximum Bitrate is applicable for all Non-GBR bearers per UE which is defined for the Downlink and the Uplink direction and provided by the MME to the eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| UE Aggregate Maximum Bit Rate Downlink | M | | Bit Rate 9.2.11 | If the *Extended UE Aggregate Maximum Bit Rate Downlink* IE is included, the *UE Aggregate Maximum Bit Rate Downlink* IE shall be ignored. | – | |
| UE Aggregate Maximum Bit Rate Uplink | M | | Bit Rate 9.2.11 | If the *Extended UE Aggregate Maximum Bit Rate Uplink* IE is included, the *UE Aggregate Maximum Bit Rate Uplink* IE shall be ignored. | – | |
| Extended UE Aggregate Maximum Bit Rate Downlink | O | | Extended Bit Rate 9.2.99 | UE Aggregate Maximum Bit Rate in DL. Details in TS 23.401 [12]. | – | |
| Extended UE Aggregate Maximum Bit Rate Uplink | O | | Extended Bit Rate 9.2.99 | UE Aggregate Maximum Bit Rate in UL. Details in TS 23.401 [12]. | – | |

## 9.2.13 Message Type

The *Message Type* IE uniquely identifies the message being sent. It is mandatory for all messages.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Procedure Code | M | | INTEGER (0..255) | |
| Type of Message | M | | CHOICE (Initiating Message, Successful Outcome, Unsuccessful Outcome, …) | |

## 9.2.14 ECGI

The E-UTRAN Cell Global Identifier (ECGI) is used to globally identify a cell (see TS 36.401 [2]).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PLMN Identity | M | | 9.2.4 | | – | |
| E-UTRAN Cell Identifier | M | | BIT STRING (28) | The leftmost bits of the *E-UTRAN Cell Identifier* IE value correspond to the value of the *eNB ID* IE contained in the *Global eNB ID* IE (defined in section 9.2.22) identifying the eNB that controls the cell. | – | |

## 9.2.15 COUNT Value

This information element indicates the 12 bit PDCP sequence number and the corresponding 20 bit Hyper frame number.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PDCP-SN | M | | INTEGER (0..4095) | | – | |
| HFN | M | | INTEGER (0..1048575) | | – | |

## 9.2.16 GUMMEI

This information element indicates the globally unique MME identity.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| GU Group Id | M | | 9.2.20 | | – | |
| MME code | M | | OCTET STRING (1) | | – | |

## 9.2.17 UL Interference Overload Indication

This IE provides, per PRB, a report on interference overload. The interaction between the indication of UL Interference Overload and UL High Interference is implementation specific.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| **UL Interference Overload Indication List** | | *1..<maxnoofPRBs>* | | |
| >UL Interference Overload Indication | M | | ENUMERATED (high interference, medium interference, low interference, …) | Each PRB is identified by its position in the list: the first element in the list corresponds to PRB 0, the second to PRB 1, etc. |

| Range bound | Explanation |
|---|---|
| maxnoofPRBs | Maximum no. Physical Resource Blocks. Value is 110. |

## 9.2.18 UL High Interference Indication

This IE provides, per PRB, a 2 level report on interference sensitivity. The interaction between the indication of UL Overload and UL High Interference is implementation specific.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| HII | M | | BIT STRING (1..110, …) | Each position in the bitmap represents a PRB (first bit=PRB 0 and so on), for which value '"1" indicates 'high interference sensitivity' and value "0" indicates 'low interference sensitivity'.<br>The maximum number of Physical Resource Blocks is 110. |

### 9.2.19 Relative Narrowband Tx Power (RNTP)

This IE provides an indication on DL power restriction per PRB or per subframe per PRB (Enhanced RNTP) in a cell and other information needed by a neighbour eNB for interference aware scheduling.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| RNTP Per PRB | M | | BIT STRING (6..110, …) | Each position in the bitmap represents a $n_{PRB}$ value (i.e. first bit=PRB 0 and so on), for which the bit value represents *RNTP ($n_{PRB}$)*, defined in TS 36.213 [11]. Value 0 indicates "Tx not exceeding RNTP threshold". Value 1 indicates "no promise on the Tx power is given". The IE is ignored if the *Enhanced RNTP* IE is included. | – | |
| RNTP Threshold | M | | ENUMERATED (-∞, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, …) | $RNTP_{threshold}$ is defined in TS 36.213 [11]. | – | |
| Number Of Cell-specific Antenna Ports | M | | ENUMERATED (1, 2, 4, …) | *P* (number of antenna ports for cell-specific reference signals) defined in TS 36.211 [10] | – | |
| P_B | M | | INTEGER (0..3, …) | $P_B$ is defined in TS 36.213 [11]. | – | |
| PDCCH Interference Impact | M | | INTEGER (0..4, …) | Measured by Predicted Number Of Occupied PDCCH OFDM Symbols (see TS 36.211 [10]).<br><br>Value 0 means "no prediction is available". | – | |
| Enhanced RNTP | O | | | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| *>Enhanced RNTP Bitmap* | M | | BIT STRING (12..8800, …) | Each position in the bitmap represents a PRB in a subframe; value "00" indicates "Tx not exceeding RNTP Threshold", value "01" indicates "Tx not exceeding RNTP High Power Threshold", value "11" indicates that "no promise on the Tx power is given". Value "10" is ignored by the receiver". Each position is applicable only in positions corresponding to DL subframes. The first 2 bits correspond to PRB 0 of the first subframe for which the IE is valid, the following 2 bits correspond to PRB 1 of the first subframe for which the IE is valid, and so on. The bit string may span across multiple contiguous subframes (maximum 40). The length of the bit string is an integer multiple of $2\times N_{\mathrm{RB}}^{\mathrm{DL}}$. $N_{\mathrm{RB}}^{\mathrm{DL}}$ is defined in TS 36.211 [10]. The Enhanced RNTP pattern is continuously repeated. | | |
| *>RNTP High Power Threshold* | M | | ENUMERATED (-∞, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, …) | Defined as the $RNTP_{threshold}$ in TS 36.213 [11]. | | |
| >Enhanced RNTP Start Time | | 0..1 | | | | |
| >>Start SFN | M | | INTEGER (0..1023, …) | SFN of the radio frame containing the first subframe when the *Enhanced RNTP* IE is valid. | | |
| >>Start Subframe Number | M | | INTEGER (0..9, …) | Subframe number, within the radio frame indicated by the *Start SFN* IE, of the first subframe when the *Enhanced RNTP* IE is valid. | | |

## 9.2.20 GU Group Id

The *GU Group Id* IE is the globally unique group id corresponding to a pool area.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PLMN Id | M | | PLMN Identity 9.2.4 | | – | |
| MME Group Id | M | | OCTET STRING(2) | | – | |

## 9.2.21 Location Reporting Information

This information element indicates how the location information should be reported.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Event | M | | ENUMERATED (Change of serving cell, …) | | – | |
| Report Area | M | | ENUMERATED (ECGI, …) | | – | |
| Additional Location Information | O | | ENUMERATED (Include PSCell, ...) | | YES | ignore |

## 9.2.22 Global eNB ID

This IE is used to globally identify an eNB (see TS 36.401 [2]).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PLMN Identity | M | | 9.2.4 | | – | |
| CHOICE *eNB ID* | M | | | | – | |
| >*Macro eNB ID* | M | | BIT STRING (20) | Equal to the 20 leftmost bits of the value of the *E-UTRAN Cell Identifier* IE contained in the *ECGI* IE (see section 9.2.14) identifying each cell controlled by the eNB | – | |
| >*Home eNB ID* | M | | BIT STRING (28) | Equal to the value of the *E-UTRAN Cell Identifier* IE contained in the *ECGI* IE (see section 9.2.14) identifying the cell controlled by the eNB | – | |
| >*Short Macro eNB ID* | M | | BIT STRING (SIZE(18)) | Equal to the 18 leftmost bits of the value of the *E-UTRAN Cell Identifier* IE contained in the *ECGI* IE (see section 9.2.14) identifying each cell controlled by the eNB. | – | |
| >*Long Macro eNB ID* | M | | BIT STRING (SIZE(21)) | Equal to the 21 leftmost bits of the value of the *E-UTRAN Cell Identifier* IE contained in the *ECGI* IE (see section 9.2.14) identifying each cell controlled by the eNB. | – | |

## 9.2.23 E-RAB ID

This IE uniquely identifies an E-RAB for a UE.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| E-RAB ID | M | | INTEGER (0..15, ...) | |

## 9.2.24 eNB UE X2AP ID

This information element, combined with the eNB UE X2AP ID Extension when present regardless its value, uniquely identifies an UE over the X2 interface within an eNB.

The usage of this IE is defined in TS 36.401 [2].

NOTE: If X2-C signalling transport is shared among multiple interface instances, the value of the eNB UE X2AP ID, combined with the eNB UE X2AP ID Extension, if applicable, is allocated so that it can be associated with an X2-C interface instance.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| eNB UE X2AP ID | M | | INTEGER (0..4095) | |

## 9.2.25 Subscriber Profile ID for RAT/Frequency priority

The *Subscriber Profile ID* IE for RAT/Frequency Selection Priority is used to define camp priorities in Idle mode and to control inter-RAT/inter-frequency handover in Active mode (TS 36.300 [15]).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Subscriber Profile ID for RAT/Frequency Priority | M | | INTEGER (1..256) | |

## 9.2.25a Additional RRM Policy Index

The *Additional RRM Policy Index* IE is used to provide additional information independent from the Subscriber Profile ID for RAT/Frequency priority as specified in TS 36.300 [15].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Additional RRM Policy Index | M | | BIT STRING (32) | |

## 9.2.26 EARFCN

The E-UTRA Absolute Radio Frequency Channel Number defines the carrier frequency used in a cell for a given direction (UL or DL) in FDD or for both UL and DL directions in TDD.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| EARFCN | M | | INTEGER (0..maxEARFCN) | The relation between EARFCN and carrier frequency (in MHz) are defined in TS 36.104 [16]. |

| Range bound | Explanation |
|---|---|
| maxEARFCN | Maximum value of EARFCNs. Value is 65535. |

## 9.2.27 Transmission Bandwidth

The *Transmission Bandwidth* IE is used to indicate the UL or DL transmission bandwidth expressed in units of resource blocks " $N_{RB}$ " (TS 36.104 [16]). The values bw1, bw6, bw15, bw25, bw50, bw75, bw100 correspond to the number of resource blocks "$N_{RB}$" 6, 15, 25, 50, 75, 100.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Transmission Bandwidth | **M** | | ENUMERATED (bw6, bw15, bw25, bw50, bw75, bw100,... , bw1) | |

## 9.2.28 E-RAB List

The IE contains a list of E-RAB identities with a cause value. It is used for example to indicate not admitted bearers.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| **E-RAB List Item** | | *1..<maxnoofBearers>* | | | EACH | ignore |
| >E-RAB ID | M | | 9.2.23 | | – | |
| >Cause | M | | 9.2.6 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBearers | Maximum no. of E-RABs. Value is 256. |

## 9.2.29 UE Security Capabilities

The *UE Security Capabilities* IE defines the supported algorithms for encryption and integrity protection in the UE. The eNBs store and send the complete bitmaps without modification or truncation as specified in TS 36.300 [15].

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Encryption Algorithms | M | | BIT STRING (16, ...) | Each position in the bitmap represents an encryption algorithm: "all bits equal to 0" - UE supports no other algorithm than EEA0 "first bit" - 128-EEA1, "second bit" - 128-EEA2, "third bit" - 128-EEA3, “fourth to seventh bit” correspond to bit 4 to bit 1 of octet 3 in the *UE Security Capability* IE defined in TS 24.301 [14], other bits reserved for future use. Value ‘1’ indicates support and value "0" indicates no support of the algorithm. Algorithms are defined in TS 33.401 [18]. |
| Integrity Protection Algorithms | M | | BIT STRING (16, ...) | Each position in the bitmap represents an integrity protection algorithm: all bits equal to 0" - UE supports no other algorithm than EIA0 (TS 33.401 [18]) "first bit" - 128-EIA1, "second bit" - 128-EIA2, "third bit" - 128-EIA3, “fourth to seventh bit” correspond to bit 4 to bit 1 of octet 4 in the *UE Security Capability* IE defined in TS 24.301 [14], other bits reserved for future use. Value '1' indicates support and value "0" indicates no support of the algorithm. Algorithms are defined in TS 33.401 [18]. |

## 9.2.30 AS Security Information

The *AS Security Information* IE is used to generate the key material to be used for AS security with the UE.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Key eNodeB Star | M | | BIT STRING (256) | KeNB* defined in TS 33.401 [18]. If the target cell belongs to multiple frequency bands, the source eNB selects the DL-EARFCN for KeNB* calculation as specified in section 10.3 of TS 36.331 [9]. |
| Next Hop Chaining Count | M | | INTEGER (0..7) | Next Hop Chaining Count (NCC) defined in TS 33.401 [18] |

## 9.2.31 Allocation and Retention Priority

This IE specifies the relative importance compared to other E-RABs for allocation and retention of the E-UTRAN Radio Access Bearer.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Priority Level | M | | INTEGER (0..15) | **Desc.:** This IE should be understood as "priority of allocation and retention" (see TS 23.401 [12]).<br>**Usage:**<br>Value 15 means "no priority".<br>Values between 1 and 14 are ordered in decreasing order of priority, i.e. 1 is the highest and 14 the lowest.<br>Value 0 shall be treated as a logical error if received. |
| Pre-emption Capability | M | | ENUMERATED(shall not trigger pre-emption, may trigger pre-emption) | **Descr.:** This IE indicates the pre-emption capability of the request on other E-RABs<br>**Usage:**<br>The E-RAB shall not pre-empt other E-RABs or, the E-RAB may pre-empt other E-RABs<br>The Pre-emption Capability indicator applies to the allocation of resources for an E-RAB and as such it provides the trigger to the pre-emption procedures/processes of the eNB. |
| Pre-emption Vulnerability | M | | ENUMERATED(not pre-emptable, pre-emptable) | **Desc.:** This IE indicates the vulnerability of the E-RAB to pre-emption of other E-RABs.<br>**Usage**:<br>The E-RAB shall not be pre-empted by other E-RABs or the E-RAB may be pre-empted by other RABs.<br>Pre-emption Vulnerability indicator applies for the entire duration of the E-RAB, unless modified, and as such indicates whether the E-RAB is a target of the pre-emption procedures/processes of the eNB. |

## 9.2.32 Time To Wait

This IE defines the minimum allowed waiting times.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Time To Wait | M | | ENUMERATED(1s, 2s, 5s, 10s, 20s, 60s, …) | |

## 9.2.33 SRVCC Operation Possible

The IE indicates that both the UE and the MME are SRVCC-capable. E-UTRAN behaviour on reception of this is specified in TS 23.216 [20].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| SRVCC Operation Possible | M | | ENUMERATED(Possible, …) | |

## 9.2.34 Hardware Load Indicator

The *Hardware Load Indicator* IE indicates the status of the Hardware Load experienced by the cell.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DL Hardware Load Indicator | M | | Load Indicator 9.2.36 | |
| UL Hardware Load Indicator | M | | Load Indicator 9.2.36 | |

## 9.2.35 S1 TNL Load Indicator

The *S1 TNL Load Indicator* IE indicates the status of the S1 Transport Network Load experienced by the cell.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DL S1TNL Load Indicator | M | | Load Indicator 9.2.36 | |
| UL S1TNL Load Indicator | M | | Load Indicator 9.2.36 | |

## 9.2.36 Load Indicator

The *Load Indicator* IE indicates the status of Load.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Load Indicator | M | | ENUMERATED (LowLoad, MediumLoad, HighLoad, Overload, ...) | |

## 9.2.37 Radio Resource Status

The *Radio Resource Status* IE indicates the usage of the PRBs for all traffic in Downlink and Uplink (TS 36.314 [22], TS 23.203 [13]) and the usage of PDCCH CCEs for Downlink and Uplink scheduling.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DL GBR PRB usage | M | | INTEGER (0..100) | |
| UL GBR PRB usage | M | | INTEGER (0..100) | |
| DL non-GBR PRB usage | M | | INTEGER (0..100) | |
| UL non-GBR PRB usage | M | | INTEGER (0..100) | |
| DL Total PRB usage | M | | INTEGER (0..100) | |
| UL Total PRB usage | M | | INTEGER (0..100) | |
| DL scheduling PDCCH CCE usage | O | | INTEGER (0..100) | |
| UL scheduling PDCCH CCE usage | O | | INTEGER (0..100) | |

## 9.2.38 UE History Information

The *UE History Information* IE contains information about cells that a UE has been served by in active state prior to the target cell. The overall mechanism is described in TS 36.300 [15].

NOTE: The definition of this IE is aligned with the definition of the *UE History Information* IE in TS 36.413 [4].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| **Last Visited Cell List** | | *1..<maxnoofCells>* | | Most recent information is added to the top of this list | – | |
| >Last Visited Cell Information | M | | 9.2.39 | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofCells | Maximum number of last visited cell information records that can be reported in the IE. Value is 16. |

## 9.2.39 Last Visited Cell Information

The Last Visited Cell Information may contain cell specific information.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| CHOICE *Last Visited Cell Information* | M | | | | - | |
| *>E-UTRAN Cell* | | | | | - | |
| >>Last Visited E-UTRAN Cell Information | M | | 9.2.40 | | - | |
| *>UTRAN Cell* | | | | | - | |
| >>Last Visited UTRAN Cell Information | M | | OCTET STRING | Defined in TS 25.413 [24] | | |
| *>GERAN Cell* | | | | | - | |
| >>Last Visited GERAN Cell Information | M | | 9.2.41 | | - | |
| *>NG-RAN Cell* | | | | | - | |
| >>Last Visited NG-RAN Cell Information | M | | OCTET STRING | Defined in TS 38.413 [39]. (see subclause 9.3.1.97). | | |

## 9.2.40 Last Visited E-UTRAN Cell Information

The Last Visited E-UTRAN Cell Information contains information about a cell that is to be used for RRM purposes.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Global Cell ID | M | | ECGI 9.2.14 | | - | |
| Cell Type | M | | 9.2.42 | | - | |
| Time UE stayed in Cell | M | | INTEGER (0..4095) | The duration of the time the UE stayed in the cell in seconds. If the UE stays in a cell more than 4095s, this IE is set to 4095. | - | |
| Time UE stayed in Cell Enhanced Granularity | O | | INTEGER (0..40950) | The duration of the time the UE stayed in the cell in 1/10 seconds. If the UE stays in a cell more than 4095s, this IE is set to 40950. | YES | ignore |
| HO Cause Value | O | | Cause 9.2.6 | The cause for the handover from the E-UTRAN cell. | YES | ignore |
| **Last Visited PSCell List** | | *0..<maxnoofPSCellsPerPrimaryCellinUEHistoryInfo>* | | List of cells configured as PSCells. Most recent PSCell related information is added to the top of the list. | YES | ignore |
| >Last Visited PSCell Information | M | | OCTET STRING | The PSCell related information. Defined in TS 36.413 [4]. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofPSCellsPerPrimaryCellinUEHistoryInfo | Maximum number of last visited PScell information records that can be reported in the IE. Value is 8. |

## 9.2.41 Last Visited GERAN Cell Information

The Last Visited Cell Information for GERAN is currently undefined.

NOTE: If in later Releases this is defined, the choice type may be extended with the actual GERAN specific information.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| CHOICE *Last Visited GERAN Cell Information* | M | | | | - | |
| >*Undefined* | M | | NULL | | - | |

## 9.2.42 Cell Type

The cell type provides the cell coverage area.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Cell Size | M | | ENUMERATED (verysmall, small, medium, large, ...) | | - | |

## 9.2.43 Number of Antenna Ports

The *Number of Antenna Ports* IE is used to indicate the number of cell specific antenna ports.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Number of Antenna Ports | | | ENUMERATED (an1, an2, an4,...) | an1 = One antenna port<br>an2 = Two antenna ports<br>an4 = Four antenna ports |

## 9.2.44 Composite Available Capacity Group

The *Composite Available Capacity Group* IE indicates the overall available resource level in the cell in Downlink and Uplink.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Composite Available Capacity Downlink | M | | Composite Available Capacity 9.2.45 | For the Downlink | - | |
| Composite Available Capacity Uplink | M | | Composite Available Capacity 9.2.45 | For the Uplink | - | |

## 9.2.45 Composite Available Capacity

The *Composite Available Capacity* IE indicates the overall available resource level in the cell in either Downlink or Uplink.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Cell Capacity Class Value | O | | 9.2.46 | | - | |
| Capacity Value | M | | 9.2.47 | '0' indicates no resource is available, Measured on a linear scale. | - | |

## 9.2.46 Cell Capacity Class Value

The *Cell Capacity Class Value* IE indicates the value that classifies the cell capacity with regards to the other cells. The *Cell Capacity Class Value* IE only indicates resources that are configured for traffic purposes.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Cell Capacity Class Value | M | | INTEGER (1..100,...) | Value 1 shall indicate the minimum cell capacity, and 100 shall indicate the maximum cell capacity. There should be a linear relation between cell capacity and Cell Capacity Class Value. | - | |

## 9.2.47 Capacity Value

The *Capacity Value* IE indicates the amount of resources that are available relative to the total E-UTRAN resources. The capacity value should be measured and reported so that the minimum E-UTRAN resource usage of existing services is reserved according to implementation. The *Capacity Value* IE can be weighted according to the ratio of cell capacity class values, if available.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Capacity Value | M | | INTEGER (0..100) | Value 0 shall indicate no available capacity, and 100 shall indicate maximum available capacity . Capacity Value should be measured on a linear scale. | - | |

## 9.2.48 Mobility Parameters Information

The *Mobility Parameters Information* IE contains the change of the Handover Trigger as compared to its current value. The Handover Trigger corresponds to the threshold at which a cell initialises the handover preparation procedure towards a specific neighbour cell. Positive value of the change means the handover is proposed to take place later.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Handover Trigger Change | M | | INTEGER (-20..20) | The actual value is IE value * 0.5 dB. |

## 9.2.49 Mobility Parameters Modification Range

The *Mobility Parameters Modification Range* IE contains the range of *Handover Trigger Change* values permitted by the $eNB_2$ at the moment the MOBILITY CHANGE FAILURE message is sent.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Handover Trigger Change Lower Limit | M | | INTEGER (-20..20) | The actual value is IE value * 0.5 dB. |
| Handover Trigger Change Upper Limit | M | | INTEGER (-20..20) | The actual value is IE value * 0.5 dB. |

## 9.2.50 PRACH Configuration

This IE indicates the PRACH resources used in neighbor cell.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| RootSequenceIndex | M | | INTEGER (0..837) | See section 5.7.2. in TS 36.211 [10] | – | |
| ZeroCorrelationZoneConfiguration | M | | INTEGER (0..15) | See section 5.7.2. in TS 36.211 [10] | – | |
| HighSpeedFlag | M | | BOOLEAN | TRUE corresponds to Restricted set and FALSE to Unrestricted set. See section 5.7.2 in TS 36.211 [10] | – | |
| PRACH-FrequencyOffset | M | | INTEGER (0..94) | See section 5.7.1 of TS 36.211 [10] | – | |
| PRACH-ConfigurationIndex | O | | INTEGER (0..63) | Mandatory for TDD, shall not be present for FDD. See section 5.7.1. in TS 36.211 [10] | – | |

## 9.2.51 Subframe Allocation

The *Subframe Allocation* IE is used to indicate the subframes that are allocated for MBSFN within the radio frame allocation period as defined in TS 36.331 [9].

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| CHOICE *Subframe Allocation* | M | | | |
| *>Oneframe* | M | | BITSTRING (SIZE(6)) | |
| *>Fourframes* | M | | BITSTRING (SIZE(24)) | |

## 9.2.52 CSG Membership Status

This element indicates the membership status of the UE to a particular CSG.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| CSG Membership Status | M | | ENUMERATED (member, not-member) | | - | |

## 9.2.53 CSG ID

This element indicates the identifier of the Closed Subscriber Group.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| CSG ID | M | | BIT STRING (SIZE (27)) | | - | |

## 9.2.54 ABS Information

This IE provides information about which sub frames the sending eNB is configuring as almost blank subframes and which subset of almost blank subframes are recommended for configuring measurements towards the UE. Almost blank subframes are subframes with reduced power on some physical channels and/or reduced activity.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CHOICE ABS Information | M | | | |
| *>FDD* | | | | |
| >>ABS Pattern Info | M | | BIT STRING (SIZE(40)) | Each position in the bitmap represents a DL subframe, for which value "1" indicates 'ABS' and value "0" indicates 'non ABS'. The first position of the ABS pattern corresponds to subframe 0 in a radio frame where *SFN* = 0. The ABS pattern is continuously repeated in all radio frames. The maximum number of subframes is 40. |
| >>Number Of Cell-specific Antenna Ports | M | | ENUMERATED (1, 2, 4, …) | *P* (number of antenna ports for cell-specific reference signals) defined in TS 36.211 [10] |
| >>Measurement Subset | M | | BIT STRING (SIZE(40)) | Indicates a subset of the ABS Pattern Info above, and is used to configure specific measurements towards the UE. |
| *>TDD* | | | | |
| >>ABS Pattern Info | M | | BIT STRING (1..70, ...) | Each position in the bitmap represents a subframe. Value "1" indicates 'ABS' and value "0" indicates 'non ABS' which is applicable only in positions corresponding to the DL direction. The maximum number of subframes depends on UL/DL subframe configuration. The maximum number of subframes is 20 for UL/DL subframe configuration 1~5; 60 for UL/DL subframe configuration 6; 70 for UL/DL subframe configuration 0. UL/DL subframe configuration defined in TS 36.211 [10]. The first position of the ABS pattern corresponds to subframe 0 in a radio frame where *SFN* = 0. The ABS pattern is continuously repeated in all radio frames, and restarted each time *SFN* = 0. |
| >>Number Of Cell-specific Antenna Ports | M | | ENUMERATED (1, 2, 4, …) | *P* (number of antenna ports for cell-specific reference signals) defined in TS 36.211 [10] |
| >>Measurement Subset | M | | BIT STRING (1..70, ...) | Indicates a subset of the ABS Pattern Info above, and is used to configure specific measurements towards the UE |

| | | | | |
|---|---|---|---|---|
| *>ABS Inactive* | M | | NULL | Indicates that interference coordination by means of almost blank sub frames is not active |

## 9.2.55 Invoke Indication

This IE provides an indication about which type of information the sending eNB would like the receiving eNB to send back.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Invoke Indication | M | | ENUMERATED (ABS Information, …, Start NAICS Information, Stop NAICS Information) | |

## 9.2.56 MDT Configuration

The IE defines the MDT configuration parameters.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| MDT Activation | M | | ENUMERATED(Immediate MDT only, Immediate MDT and Trace, …) | | – | |
| CHOICE *Area Scope of MDT* | M | | | | – | |
| >*Cell Based* | | | | | – | |
| >>**Cell ID List for MDT** | | *1..<maxnoofCellIDforMDT>* | | | – | |
| >>>ECGI | M | | 9.2.14 | | – | |
| >*TA Based* | | | | | – | |
| >>**TA List for MDT** | | *1..<maxnoofTAforMDT>* | | | – | |
| >>>TAC | M | | OCTET STRING (2) | Tracking Area Code. The TAI is derived using the current serving PLMN. | – | |
| >*PLMN Wide* | | | NULL | | – | |
| >*TAI based* | | | | | | |
| >>**TAI List for MDT** | | *1..<maxnoofTAforMDT>* | | | | |
| >>>TAC | M | | OCTET STRING (2) | Tracking Area Code | | |
| >>>PLMN Identity | M | | 9.2.4 | | | |
| Measurements to Activate | M | | BITSTRING (SIZE(8)) | Each position in the bitmap indicates a MDT measurement, as defined in TS 37.320 [25]. First Bit = M1, Second Bit = M2, Third Bit = M3, Fourth Bit = M4, Fifth Bit = M5, Sixth Bit = logging of M1 from event triggered measurement reports according to existing RRM configuration. Seventh Bit = M6, Eighth Bit = M7.<br><br>Value "1" indicates "activate" and value "0" indicates "do not activate". | – | |
| M1 Reporting Trigger | M | | ENUMERATED (periodic, A2event-triggered, …, A2event-triggered periodic) | This IE shall be ignored if the *Measurements to Activate* IE has the first bit set to "0". | – | |
| M1 Threshold Event A2 | C-ifM1A2trigger | | | Included in case of event-triggered or event-triggered periodic reporting for measurement M1 | – | |
| >CHOICE *Threshold* | M | | | | – | |
| >>*RSRP* | | | | | – | |
| >>>Threshold RSRP | M | | INTEGER (0..97) | Corresponds to information provided in the *RSRP-Range* IE as defined in TS 36.331 [9]. | – | |
| >>*RSRQ* | | | | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>Threshold RSRQ | M | | INTEGER (0..34) | Corresponds to information provided in the *RSRQ-Range* IE as defined in TS 36.331 [9]. | – | |
| M1 Periodic reporting | C-ifperiodic MDT | | | Included in case of periodic or event-triggered periodic reporting for measurement M1 | – | |
| >Report interval | M | | ENUMERATED ( ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, min60) | Corresponds to information provided in the *ReportInterval* IE as defined in TS 36.331 [9]. | – | |
| >Report amount | M | | ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity) | Number of reports | – | |
| M3 Configuration | C-ifM3 | | 9.2.61 | | YES | ignore |
| M4 Configuration | C-ifM4 | | 9.2.62 | | YES | ignore |
| M5 Configuration | C-ifM5 | | 9.2.63 | | YES | ignore |
| MDT Location Information | O | | BITSTRING(SIZE(8)) | Each position in the bitmap represents requested location information as defined in TS 37.320 [31].<br>First Bit = GNSS<br>Second Bit = E-CID information.<br>Other bits are reserved for future use and are ignored if received.<br>Value "1" indicates "activate" and value "0" indicates "do not activate".<br><br>The eNB shall ignore the first bit unless the *Measurements to Activate* IE has the first bit or the sixth bit set to "1". | YES | ignore |
| Signalling based MDT PLMN List | O | | MDT PLMN List 9.2.64 | | YES | ignore |
| M6 Configuration | C-ifM6 | | 9.2.87 | | YES | ignore |
| M7 Configuration | C-ifM7 | | 9.2.88 | | YES | ignore |
| Bluetooth Measurement Configuration | O | | 9.2.134 | | YES | ignore |
| WLAN Measurement Configuration | O | | 9.2.135 | | YES | ignore |
| Sensor Measurement Configuration | O | | 9.2.183 | | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxnoofCellIDforMDT | Maximum no. of Cell ID subject for MDT scope. Value is 32. |
| maxnoofTAforMDT | Maximum no. of TA subject for MDT scope. Value is 8. |

| Condition | Explanation |
|---|---|
| ifM1A2trigger | This IE shall be present if the *Measurements to Activate* IE has the first bit set to "1" and the *M1 Reporting Trigger* IE is set to "A2event-triggered" or to "A2event-triggered periodic". |
| ifperiodicMDT | This IE shall be present if the *M1 Reporting Trigger* IE is set to "periodic" or to "A2event-triggered periodic". |
| ifM3 | This IE shall be present if the *Measurements to Activate* IE has the third bit set to "1". |
| ifM4 | This IE shall be present if the *Measurements to Activate* IE has the fourth bit set to "1". |
| ifM5 | This IE shall be present if the *Measurements to Activate* IE has the fifth bit set to "1". |
| ifM6 | This IE shall be present if the *Measurements to Activate* IE has the seventh bit set to "1". |
| ifM7 | This IE shall be present if the *Measurements to Activate* IE has the eighth bit set to "1". |

## 9.2.57 Void

## 9.2.58 ABS Status

The *ABS Status* IE is used to aid the eNB designating ABS to evaluate the need for modification of the ABS pattern.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DL ABS status | M | | INTEGER (0..100) | Percentage of used ABS resources. The numerator of the percentage calculation consists of resource blocks within the ABS indicated in the *Usable ABS Pattern Info* IE allocated by the $eNB_2$ for DL traffic needing protection by ABS from inter-cell interference for DL scheduling, or allocated by the $eNB_2$ for other reasons (e.g. some control channels). The denominator of the percentage calculation is the total quantity of resource blocks within the ABS indicated in the *Usable ABS Pattern Info* IE. |
| CHOICE *Usable ABS Information* | M | | – | – |
| >*FDD* | | | – | – |
| >>Usable ABS Pattern Info | M | | BIT STRING (SIZE(40)) | Each position in the bitmap represents a subframe, for which value "1" indicates 'ABS that has been designated as protected from inter-cell interference by the $eNB_1$, and available to serve this purpose for DL scheduling in the $eNB_2$' and value "0" is used for all other subframes.<br>The pattern represented by the bitmap is a subset of, or the same as, the corresponding *ABS Pattern Info* IE conveyed in the LOAD INFORMATION message from the $eNB_1$. |
| >*TDD* | | | – | – |
| >>Usable ABS Pattern Info | M | | BIT STRING (1..70) | Each position in the bitmap represents a subframe, for which value "1" indicates 'ABS that has been designated as protected from inter-cell interference by the $eNB_1$, and available to serve this purpose for DL scheduling in the $eNB_2$' and value "0" is used for all other subframes.<br>The pattern represented by the bitmap is a subset of, or the same as, the corresponding *ABS Pattern Info* IE conveyed in the LOAD INFORMATION message from the $eNB_1$. |

## 9.2.59 Management Based MDT Allowed

This information element is used by the eNB to allow selection of the UE for management based MDT as described in TS 32.422 [6].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Management Based MDT Allowed | M | | ENUMERATED (Allowed, ...) | |

## 9.2.60 MultibandInfoList

The *MultibandInfoList* IE contains the additional frequency band indicators that a cell belongs to listed in decreasing order of preference, see TS 36.331 [9].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| **BandInfo** | | *1..<maxnoofBands>* | | | – | |
| >FrequencyBandIndicator | M | | INTEGER (1.. 256, ...) | E-UTRA operating band as defined in TS 36.101 [42, table 5.5-1] | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBands | Maximum number of frequency bands that a cell belongs to. The value is 16. |

## 9.2.61 M3 Configuration

This IE defines the parameters for M3 measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| M3 Collection Period | M | | ENUMERATED (ms100, ms1000, ms10000, …) | |

## 9.2.62 M4 Configuration

This IE defines the parameters for M4 measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| M4 Collection Period | M | | ENUMERATED (ms1024, ms2048, ms5120, ms10240, min1, …) | |
| M4 Links to log | M | | ENUMERATED(uplink, downlink, both-uplink-and-downlink, …) | |

## 9.2.63 M5 Configuration

This IE defines the parameters for M5 measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| M5 Collection Period | M | | ENUMERATED (ms1024, ms2048, ms5120, ms10240, min1, …) | |
| M5 Links to log | M | | ENUMERATED(uplink, downlink, both-uplink-and-downlink, …) | |

## 9.2.64 MDT PLMN List

The purpose of the *MDT PLMN List* IE is to provide the list of PLMNs allowed for MDT.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| **MDT PLMN List** | | *1..<maxnoof MDTPLMNs >* | | |
| >PLMN Identity | M | | 9.2.4 | |

| Range bound | Explanation |
|---|---|
| maxnoofMDTPLMNs | Maximum no. of PLMNs in the MDT PLMN list. Value is 16. |

## 9.2.65 EARFCN Extension

The E-UTRA Absolute Radio Frequency Channel Number Extension defines the carrier frequency used in a cell for a given direction (UL or DL) in FDD or for both UL and DL directions in TDD.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| EARFCN Extension | M | | INTEGER (maxEARFCN+1 .. newmaxEARFCN, ...) | The relation between EARFCN and carrier frequency (in MHz) are defined in TS 36.104 [16]. |

| Range bound | Explanation |
|---|---|
| maxEARFCN | Maximum value of EARFCNs. Value is 65535. |
| newmaxEARFCN | New maximum value of EARFCNs. Value is 262143. |

## 9.2.66 COUNT Value Extended

This information element indicates the 15 bit long PDCP SN and the corresponding 17 bit long Hyper Frame Number.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PDCP-SN Extended | M | | INTEGER (0..32767) | | - | |
| HFN Modified | M | | INTEGER (0..131071) | | - | |

## 9.2.67 Extended UL Interference Overload Info

This IE provides report on interference overload for the set of subframes that are subject to UL-DL subframe reconfiguration. This IE applies to TDD only.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Associated Subframes | M | | BITSTRING (SIZE(5)) | The set of subframe(s) to which the Extended UL interference overload indication is applicable. The bitmap from the least significant bit position to the most significant bit position represents subframes #{3, 4, 7, 8, 9} in a radio frame. Value "1" in a bit position indicates that the Extended UL interference overload indication is applicable to the corresponding subframe; and value "0" indicates otherwise. |
| Extended UL Interference Overload Indication | M | | UL Interference Overload Indication 9.2.17 | |

## 9.2.68 RNL Header

The *RNL Header* IE indicates the target eNB ID and source eNB ID.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Source eNB ID | M | | Global eNB ID 9.2.22 | | - | |
| Target eNB ID | O | | Global eNB ID 9.2.22 | | - | |

## 9.2.69 Masked IMEISV

This information element contains the IMEISV value with a mask, to identify a terminal model without identifying an individual Mobile Equipment.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Masked IMEISV | M | | BIT STRING (SIZE(64)) | Coded as the International Mobile station Equipment Identity and Software Version Number (IMEISV) defined in TS 23.003 [29] with the last 4 digits of the SNR masked by setting the corresponding bits to 1. |

## 9.2.70 Expected UE Behaviour

This IE defines the behaviour of a UE with predictable activity and/or mobility behaviour, to assist the eNB/en-gNB in determining the optimum RRC connection time.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Expected UE Activity Behaviour | M | | 9.2.71 | |
| Expected HO Interval | O | | ENUMERATED (sec15, sec30, sec60, sec90, sec120, sec180, long-time, …) | Indicates the expected time interval between inter-eNB handovers.<br>If "long-time" is included, the interval between inter-eNB handovers is expected to be longer than 180 seconds. |

## 9.2.71 Expected UE Activity Behaviour

Indicates information about the expected "UE activity behaviour" as defined in TS 23.401 [12].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Expected Activity Period | O | | INTEGER (1..30\|40\|50\|60\|80\|100\|120\|150\|180\|181, ...) | If this IE is set to "181" the expected activity time is longer than 180 seconds.<br>The remaining values indicate the expected activity time in [seconds]. |
| Expected Idle Period | O | | INTEGER (1..30\|40\|50\|60\|80\|100\|120\|150\|180\|181, ...) | If this IE is set to "181" the expected idle time is longer than 180 seconds.<br>The remaining values indicate the expected idle time in [seconds]. |
| Source of UE Activity Behaviour Information | O | | ENUMERATED (subscription information, statistics, ...) | If "subscription information" is indicated, the information contained in the *Expected Activity Period* IE and the *Expected Idle Period* IE, if present, is derived from subscription information.<br>If "statistics" is indicated, the information contained in the *Expected Activity Period* IE and the *Expected Idle Period* IE, if present, is derived from statistical information. |

## 9.2.72 SeNB Security Key

The *SeNB Security Key* IE is used to apply security in the SeNB as defined in TS 33.401 [18].

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| SeNB Security Key | M | | BIT STRING (SIZE(256)) | The S-KeNB which is provided by the MeNB, see TS 33.401 [18]. |

## 9.2.73 SCG Change Indication

The *SCG Change Indication* IE is either used to request the SeNB to prepare the SCG Change in the SeNB or to request the MeNB to initiate the SCG Change towards the UE (see TS 36.300 [15]).

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| SCG Change Indication | M | | ENUMERATED (PDCPCountWrapAround, PSCellChange, other, …) | |

## 9.2.74 CoMP Information

This IE provides the list of CoMP hypothesis sets, where each CoMP hypothesis set is the collection of CoMP hypothesis(es) of one or multiple cells and each CoMP hypothesis set is associated with a benefit metric.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CoMP Information Item | | *1 .. <maxnoofCoMPHypothesisSet>* | | |
| >CoMP Hypothesis Set | M | | 9.2.75 | |
| >Benefit Metric | M | | INTEGER (-101..100, …) | Value -100 indicates the maximum cost, and 100 indicates the maximum benefit.<br>Value -101 indicates unknown benefit.<br>Values from -100 to 100 should be calculated on a linear scale. |
| CoMP Information Start Time | | *0..1* | | |
| >Start SFN | M | | INTEGER (0..1023, …) | SFN of the radio frame containing the first subframe when the *CoMP Information* IE is valid. |
| >Start Subframe Number | M | | INTEGER (0..9, …) | Subframe number, within the radio frame indicated by the *Start SFN* IE, of the first subframe when the *CoMP Information* IE is valid. |

| Range bound | Explanation |
|---|---|
| maxnoofCoMPHypothesisSet | Maximum number of CoMP Hypothesis sets. The value is 256. |

## 9.2.75 CoMP Hypothesis Set

This IE provides a set of CoMP hypotheses. A CoMP hypothesis is hypothetical PRB-specific resource allocation information for a cell.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CoMP Hypothesis Set Item | | *1..<maxnoofCoMPCells>* | | |
| >Cell ID | M | | ECGI 9.2.14 | ID of the cell for which the *CoMP Hypothesis* IE is applied. |
| >CoMP Hypothesis | M | | BIT STRING (6..4400, …) | Each position in the bitmap represents a PRB in a subframe, for which value "1" indicates 'interference protected resource' and value "0" indicates 'resource with no utilization constraints,' which is applicable only in positions corresponding to the DL direction.<br>The first bit corresponds to PRB 0 of the first subframe for which the IE is valid, the second bit corresponds to PRB 1 of the first subframe for which the IE is valid, and so on.<br>The bit string may span across multiple contiguous subframes.<br>The length of the bit string is an integer (maximum 40) multiple of $N_{\mathrm{RB}}^{\mathrm{DL}}$. $N_{\mathrm{RB}}^{\mathrm{DL}}$ is defined in TS 36.211 [10].<br>The CoMP hypothesis pattern is continuously repeated. |

| Range bound | Explanation |
|---|---|
| maxnoofCoMPCells | Maximum number of cells in a CoMP hypothesis set. Value is 32. |

## 9.2.76 RSRP Measurement Report List

This IE provides RSRP measurement reports of UEs served by the sending eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| RSRP Measurement Report Item | | *1 .. <maxUEReport>* | | |
| >RSRP Measurement Result | | *1 .. <maxCellReport>* | | |
| >>RSRP Cell ID | M | | ECGI 9.2.14 | ID of the cell on which the RSRP is measured. |
| >>RSRP Measured | M | | INTEGER (0..97, ...) | Corresponds to information provided in the *RSRP-Range* IE as defined in TS 36.331 [9]. |
| >UE ID | O | | BIT STRING (SIZE(16)) | ID assigned by $eNB_2$ for the UE. |

| Range bound | Explanation |
|---|---|
| maxUEReport | Maximum number of UE measurement reports. Value is 128. |
| maxCellReport | Maximum number of reported cells. The value is 9. |

## 9.2.77 Dynamic DL transmission information

This IE contains assistance information for DL interference mitigation.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CHOICE *NAICS Information* | M | | | |
| *>NAICS Active* | | | | |
| >>Transmission Modes | O | | BIT STRING (SIZE(8)) | The set bits indicate some or all transmission modes: 1, 2, 3, 4, 6, 8, 9, 10, as defined in TS 36.213 [23, 7.1]. The first/ leftmost bit is for transmission mode 1, the second bit is for transmission mode 2, and so on. |
| >>P_B | O | | INTEGER (0..3) | See TS 36.213 [23, Table 5.2-1] |
| >>P_A_list | | *0 .. <maxnoofPA>* | | |
| >>>P_A | M | | ENUMERATED (dB-6, dB-4dot77, dB-3, dB-1dot77, dB0, dB1, dB2, dB3,...) | See $P_A$ TS 36.213 [23, 5.2]. Value dB-6 corresponds to -6 dB, dB-4dot77 corresponds to -4.77 dB etc. |
| *>NAICS Inactive* | | | NULL | |

| Range bound | Explanation |
|---|---|
| maxnoofPA | Maximum no of $P_A$ values that can be configured. Value is 3. |

## 9.2.78 ProSe Authorized

This IE provides information on the authorization status of the UE for ProSe service(s).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| ProSe Direct Discovery | O | | ENUMERATED (authorized, not authorized, ...) | Indicates whether the UE is authorized for ProSe Direct Discovery | - | |
| ProSe Direct Communication | O | | ENUMERATED (authorized, not authorized, ...) | Indicates whether the UE is authorized for ProSe Direct Communication | - | |
| ProSe UE-to-Network Relaying | O | | ENUMERATED (authorized, not authorized, ...) | Indicates whether the UE is authorized to act as ProSe UE-to-Network Relay | YES | ignore |

## 9.2.79 CSI Report

This IE provides CSI reports of UEs served by the cell for which the information is provided.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CSI Report per Cell | | *1 .. <maxUEReport>* | | |
| >UE ID | M | | BIT STRING (SIZE(16)) | ID assigned by eNB2 for the UE. |
| >CSI Report per CSI Process | | 1 .. <maxCSIProcess> | | |
| >>CSI Process Configuration Index | M | | INTEGER (1..7, …) | Indicates one of the possible CSI Process configurations in the serving cell. |
| >>CSI Report per CSI Process Item | | 1.. <maxCSIReport> | | |
| >>>RI | M | | INTEGER (1..8, …) | The RI corresponding to the CQI being reported for this CSI process item. Value defined in TS 36.213 [11]. |
| >>>Wideband CQI | M | | 9.2.80 | |
| >>>Subband Size | M | | ENUMERATED (2, 3, 4, 6, 8, …) | Corresponds to a value of subband size *k* defined in TS 36.213 [11] for the system bandwidth $N_{\mathrm{RB}}^{\mathrm{DL}}$. |
| >>>Subband CQI List | | *0 .. <maxSubband>* | | |
| >>>>Subband CQI | M | | 9.2.81 | |
| >>>>Subband Index | M | | INTEGER (0..27, …) | |

| Range bound | Explanation |
|---|---|
| maxUEReport | Maximum number of UE. Value is 128. |
| maxCSIProcess | Maximum number of CSI processes per UE. The value is 4. |
| maxCSIReport | Maximum number of CSI Reports per CSI Process. The value is 2. |
| maxSubband | Maximum number of subbands. The value is 14. |

## 9.2.80 Wideband CQI

This IE indicates the Wideband CQI as defined in TS 36.213 [11].

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Wideband CQI Codeword 0 | M | | INTEGER (0..15, …) | Value defined in TS 36.213 [11]. |
| CHOICE *Wideband CQI Codeword 1* | O | | | |
| *>4-bit CQI* | M | | INTEGER (0..15, …) | Value defined in TS 36.213 [11]. |
| *>3-bit spatial differential CQI* | M | | INTEGER (0..7, …) | Value defined in TS 36.213 [11]. |

## 9.2.81 Subband CQI

This IE indicates the Subband CQI as defined in TS 36.213 [11].

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| CHOICE *Subband CQI Codeword 0* | M | | | |
| *>4-bit CQI* | M | | INTEGER (0..15, …) | Value defined in TS 36.213 [11]. |
| *>2-bit Subband differential CQI* | M | | INTEGER (0..3, …) | Value defined in TS 36.213 [11]. |
| *>2-bit differential CQI* | M | | INTEGER (0..3, …) | Value defined in TS 36.213 [11]. |
| CHOICE *Subband CQI Codeword 1* | O | | | |
| *>4-bit CQI* | M | | INTEGER (0..15, …) | Value defined in TS 36.213 [11]. |
| *>3-bit spatial differential CQI* | M | | INTEGER (0..7, …) | Value defined in TS 36.213 [11]. |
| *>2-bit Subband differential CQI* | M | | INTEGER (0..3, …) | Value defined in TS 36.213 [11]. |
| *>2-bit differential CQI* | M | | INTEGER (0..3, …) | Value defined in TS 36.213 [11]. |

## 9.2.82 COUNT Value for PDCP SN Length 18

This information element indicates the 18 bit long PDCP SN and the corresponding 14 bit long Hyper Frame Number.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PDCP-SN Length 18 | M | | INTEGER (0..262143) | | – | |
| HFN for PDCP-SN Length 18 | M | | INTEGER (0..16383) | | – | |

## 9.2.83 LHN ID

The *LHN ID* IE is used to indicate the LHN ID of the eNB, as defined in TS 23.003 [29].

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Local Home Network ID | M | | OCTET STRING (SIZE (32..256)) | Identifies the Local Home Network. |

## 9.2.84 Correlation ID

This information element is the GTP Tunnel Endpoint Identifier or GRE key to be used for the user plane transport between eNB and the L-GW described in TS 23.401 [12].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Correlation ID | M | | OCTET STRING (SIZE(4)) | |

## 9.2.85 UE Context Kept Indicator

This IE indicates that the UE Context at the SeNB is kept in case of inter-MeNB handover without SeNB/SgNB Change procedure, as specified in TS 36.300 [15] or TS37.340 [32].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| UE Context Kept Indicator | M | | ENUMERATED (True, …) | |

## 9.2.86 eNB UE X2AP ID Extension

This information element combined with the eNB UE X2AP ID uniquely identifies an UE over the X2 interface within an eNB. If the setup of an UE associcated signalling connection was initiated including the eNB UE X2AP ID Extension, the eNB UE X2AP ID Extension shall be used by both peers for the life-time of the respective UE-associated signalling connection.

The usage of this IE is defined in TS 36.401 [2].

NOTE: If X2-C signalling transport is shared among multiple interface instances, the value of the eNB UE X2AP ID, combined with the eNB UE X2AP ID Extension, is allocated so that it can be associated with an X2-C interface instance.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| eNB UE X2AP ID Extension | M | | INTEGER (0..4095,…) | |

## 9.2.87 M6 Configuration

This IE defines the parameters for M6 measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| M6 Report Interval | M | | ENUMERATED (ms1024, ms2048, ms5120, ms10240, …) | |
| M6 Delay Threshold | C-ifUL | | ENUMERATED (ms30, ms40, ms50, ms60, ms70, ms80, ms90, ms100, ms150, ms300, ms500, ms750, …) | |
| M6 Links to log | M | | ENUMERATED(uplink, downlink, both-uplink-and-downlink, …) | |

| Condition | Explanation |
|---|---|
| ifUL | This IE shall be present if the *M6 Links to log* IE is set to "uplink" or to "both-uplink-and-downlink". |

## 9.2.88 M7 Configuration

This IE defines the parameters for M7 measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| M7 Collection Period | M | | INTEGER (1..60, …) | Unit: minutes |
| M7 Links to log | M | | ENUMERATED(uplink, downlink, both-uplink-and-downlink, …) | |

## 9.2.89 Tunnel Information

The *Tunnel Information* IE indicates the transport layer address and UDP port number.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Transport Layer Address | M | | BIT STRING (1..160, ...) | eNB's Transport Layer Address. |
| UDP Port Numbers | O | | OCTET STRING (SIZE(2)) | UDP Port Numbers if NAT/NAPT is deployed in the BBF access network. |

## 9.2.90 X2 Benefit Value

The *X2 Benefit Value* IE indicates the quantified benefit of the signalling connection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| X2 Benefit Value | M | | INTEGER (1..8, …) | Value 1 indicates low benefit, and 8 indicates high benefit. |

## 9.2.91 Resume ID

The *Resume ID* IE is used to address a suspended UE Context within an eNB.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| CHOICE *Resume ID* | M | | | |
| *>Resume ID not truncated* | | | | |
| >>Resume ID not truncated | M | | BIT STRING (SIZE (40)) | 40 bit Resume Resume Identity contained in the *RRCConnectionResumeRequest* message (TS 36.331 [9]).<br>The 20 most significant bits refer to the eNB ID of the eNB that allocated the Resume ID, the 20 least significant bits identify the UE Context stored at the eNB that allocated the Resume ID. |
| *>Resume ID truncated* | | | | |
| >>Resume ID truncated | M | | BIT STRING (SIZE (24)) | 24 bit Resume Identity contained in the *RRCConnectionResumeRequest* message (TS 36.331 [9]).<br>The 12 most significant bits refer to the 12 least significant bits of the eNB ID of the eNB that allocated the Resume ID.<br>The 12 least significant bits refer to the 12 least significant bits that identify the UE Context stored at the eNB that allocated the Resume ID. |

## 9.2.92 Bearer Type

This IE is used to support Non-IP data as specified in TS 23.401 [11].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Bearer Type | M | | ENUMERATED (non IP, …) | |

## 9.2.93 V2X Services Authorized

This IE provides information on the authorization status of the UE to use the sidelink for V2X services.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Vehicle UE | O | | ENUMERATED (authorized, not authorized, ...) | Indicates whether the UE is authorized as Vehicle UE | - | |
| Pedestrian UE | O | | ENUMERATED (authorized, not authorized, ...) | Indicates whether the UE is authorized as Pedestrian UE | - | |

## 9.2.94 Offset of NB-IoT Channel Number to EARFCN

This IE is used to indicate the offset of the NB-IoT Channel Number to the EARFCN (TS 36.104 [16]).

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Offset of NB-IoT Channel Number to EARFCN | M | | ENUMERATED (-10,-9,-8,-7,-6,-5,-4,-3,-2,-1,-0.5,0,1,2,3,4,5,6,7,8,9,... , -8.5, -4.5, 3.5, 7.5) | |

## 9.2.95 WT ID

This IE is used to identify a WT.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CHOICE *WT ID Type* | M | | | |
| *>WT ID Type 1* | | | | |
| >>PLMN ID | M | | PLMN Identity 9.2.4 | |
| >>Short WT ID | M | | BIT STRING (24) | |
| *>WT ID Type 2* | | | | |
| >>Long WT ID | M | | BIT STRING (48) | |

## 9.2.96 WT UE XwAP ID

The WT UE XwAP ID is allocated by the WT and uniquely identifies a UE over the Xw interface.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| WT UE XwAP ID | M | | OCTET STRING (SIZE(3)) | |

## 9.2.97 UE Sidelink Aggregate Maximum Bit Rate

This IE indicates the aggregate maximum bit rate for all radio bearers per UE in the sidelink for V2X services.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| UE Sidelink Aggregate Maximum Bit Rate | M | | Bit Rate 9.2.11 | Value 0 shall be considered as a logical error by the receiving eNB. |

### 9.2.98 NR Neighbour Information

This IE contains cell configuration information of NR cells that a neighbour node may need for the X2 AP interface.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| **NR Neighbour Information** | | *1 .. <maxnoofNR Neighbours>* | | | – | |
| >**NR Neighbour Information Item** | | | | | – | |
| >>NRPCI | M | | INTEGER (0..1007) | NR Physical Cell ID | – | |
| >>NR CGI | M | | 9.2.111 | | – | |
| >>5GS-TAC | O | | OCTET STRING (3) | Broadcast 5GS Tracking Area Code | – | |
| >>Configured TAC | O | | OCTET STRING (2) | This is the TAC configured in the en-gNB, different from the 5GS TAC broadcast in the NR cell and enables application of Roaming and Access Restrictions for EN-DC as specified in TS 37.340 [32]. | – | |
| >>Measurement Timing Configuration | M | | OCTET STRING | Includes the *MeasurementTimingConfiguration* inter-node message for the neighbour cell, as defined in TS 38.331 [31]. | – | |
| >>CHOICE *NR-Neighbour-Mode-Info* | M | | | | – | |
| >>>*FDD* | | | | | | |
| >>>>**FDD Info** | | *1* | | | – | |
| >>>>>UL ARFCNFreqInfo | M | | NR ARFCN Frequency Info 9.2.106 | This IE is ignored for NR operating bands for which uplink range of $N_{REF}$ is not defined in section 5.4.2.3 of TS 38.104 [37]. | – | |
| >>>>>DL ARFCNFreqInfo | M | | NR ARFCN Frequency Info 9.2.106 | | – | |
| >>>>>UL Carrier List | O | | NR Carrier List 9.2.168 | | YES | ignore |
| >>>*TDD* | | | | | | |
| >>>>**TDD Info** | | *1* | | | – | |
| >>>>>ARFCNNRFreqInfo | M | | NR ARFCN Frequency Info 9.2.106 | | – | |
| >>>>>Intended TDD DL-UL Configuration NR | O | | OCTET STRING | Contains the *Intended TDD DL-UL Configuration NR* IE as defined in TS 38.423 [49]. | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>>>TDD UL-DL Configuration Common NR | O | | OCTET STRING | Includes the *TDD-UL-DL-ConfigCommon* IE as defined in TS 38.331 [31]. | YES | ignore |
| >>>>>Carrier List | O | | NR Carrier List 9.2.168 | | YES | ignore |
| >>CSI-RS Transmission Indication | O | | ENUMERATED {activated, deactivated, ...} | This IE indicates the CSI-RS transmission status of the given cell.<br>If the *Additional Measurement Timing Configuration List* IE is present in the *NR Neighbour Information* IE, this IE is ignored. | YES | ignore |
| >>SSB Positions In Burst | O | | 9.2.169 | | YES | ignore |
| >>NR Cell PRACH Configuration | O | | OCTET STRING | Contains the NR Cell PRACH Configuration IE as defined in TS 38.473 [44]. | YES | ignore |
| >>Additional Measurement Timing Configuration List | O | *1 .. <maxnoofMTCItems>* | | | YES | ignore |
| >>>Measurement Timing Configuration Index | M | | INTEGER (0..16) | "0" refers to the configuration contained in the Measurement Timing Configuration IE.<br>Any value between "1" and "16" refers to a configuration within the *Additional Measurement Timing Configuration List* IE. | – | |
| >>>CSI-RS MTC Configuration List | M | *1 .. <maxnoofCSIRSconfigurations>* | | This list explicitly expresses the CSI-RS configurations contained in the MTC. | – | |
| >>>>CSI-RS Index | M | | INTEGER (0..95) | Index of CSI-RS as in MTC | – | |
| >>>>CSI-RS Status | M | | ENUMERATED (activated, deactivated, …) | This IE indicates the CSI-RS transmission status of the configuration. | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **>>>>CSI-RS Neighbour List** | O | *1 .. <maxnoofCSIRSneighbourCells>* | | This list expresses the cells and CSI-RSs neighbouring the CSI-RS in the CSI-RS Index IE. | – | |
| >>>>>NR CGI | M | | 9.2.111 | | – | |
| **>>>>>CSI-RS MTC Neighbour List** | O | *1 .. <maxnoofCSIRSneighbourCellsInMTC>* | | This list expresses the CSI-RSs served by the CGI, which are neighbouring the CSI-RS of the neighbour cell. | – | |
| >>>>>>CSI-RS Index | M | | INTEGER (0..95) | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofNRNeighbours | Maximum no. of neighbour NR cells associated to a given served cell. Value is 1024. |
| maxnoofMTCItems | Maximum no. of measurement timing configurations associated with the neighbour cell. Value is 16. |
| maxnoofCSIRSconfigurations | Maximum number of CSI-RS configurations reported in the MTC. Value is 96 |
| maxnoofCSIRSneighbourCells | Maximum number of cells neighbouring a CSI-RS coverage area. Value is 16 |
| maxnoofCSIRSneighbourCellsInMTC | Maximum number of CSI-RS coverage areas neighbouring a specific CSI-RS coverage area. Value is 16 |

## 9.2.99 Extended Bit Rate

This IE indicates the number of bits delivered by E-UTRAN in UL or to E-UTRAN in DL within a period of time, divided by the duration of the period. It is used, for example, to indicate the maximum or guaranteed bit rate for a GBR bearer, or an aggregated maximum bit rate.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Extended Bit Rate | | | INTEGER (10,000,000,001..4,000,000,000,000,...) | The unit is: bit/s |

## 9.2.100 en-gNB UE X2AP ID

This information element uniquely identifies an UE over the X2 interface within an en-gNB.

The usage of this IE is defined in TS 36.401 [2].

NOTE: If X2-C signalling transport is shared among multiple interface instances, the value of the en-gNB UE X2AP ID is allocated so that it can be associated with an X2-C interface instance.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| en-gNB UE X2AP ID | M | | INTEGER (0.. $2^{32}$ -1) | |

## 9.2.101 SgNB Security Key

The *SgNB Security Key* IE is used to apply security in the en-gNB as defined in TS 33.401 [18].

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| SgNB Security Key | M | | BIT STRING (SIZE(256)) | The S-KgNB which is provided by the MeNB, see TS 33.401 [18]. |

## 9.2.102 Target SgNB ID Information

This IE contains the target SgNB ID used by MeNB to find the target en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Target SgNB ID | M | | Global en-gNB ID 9.2.112 | |

## 9.2.103 SCG Configuration Query

The *SCG Configuration Query* IE is used to request the en-gNB to provide current SCG configuration.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| SCG Configuration Query | M | | ENUMERATED (True, …) | |

## 9.2.104 Delivery Status

This IE defines the Delivery Status IE of RRC Transfer message.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Highest successfully delivered NR PDCP Sequence Number | M | $0..2^{12}-1$ | INTEGER ($0..2^{12}-1$) | Highest successfully delivered NR PDCP SN, as defined in 38.323 [33]. | – | |

## 9.2.105 Void

Void

## 9.2.106 NR Frequency Info

The NR Frequency Info defines the carrier frequency and bands used in a cell for a given direction (UL or DL) in FDD or for both UL and DL directions in TDD.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| NRARFCN | M | | INTEGER (0..maxNRARFCN) | RF Reference Frequency as defined in TS 38.104 [37] section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A. | – | |
| **Frequency Band List** | | *1* | | | – | |
| **>Frequency Band Item** | | *1..<maxnoofNrCellBands>* | | | | |
| >>NR Frequency Band | M | | INTEGER (1.. 1024, ...) | Primary NR Operating Band as defined in TS38.104 [37] section 5.4.2.3.<br>The value 1 corresponds e n1, value 2 corresponds to NR operating band n2, etc. | – | |
| **>>Supported SUL band List** | | *0..<maxnoofNrCellBands>* | | | – | |
| >>>Supported SUL band Item | M | | INTEGER (1.. 1024, ...) | Supplementary NR Operating Band as defined in TS 38.104 [37] section 5.4.2.3 that can be used for SUL duplex mode as per TS 38.101-1 table 5.2.-1.<br>The value 80 corresponds to NR operating band n80, value 81 corresponds to NR operating band n81, etc. | – | |
| SUL Information | O | | 9.2.123 | | – | |
| Frequency Shift 7p5khz | O | | ENUMERATED (false, true, ...) | Indicate whether the value of $\Delta_{shift}$ is 0kHz or 7.5kHz when calculating $F_{REF,shift}$ as defined in Section 5.4.2.1 of TS 38.104 [37]. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxNRARFCN | Maximum value of NRARFCNs. Value is 3279165. |
| maxnoofNrCellBands | Maximum no. of frequency bands supported for a NR cell. Value is 32. |

## 9.2.107 NR UE Security Capabilities

This IE defines the supported algorithms for encryption and integrity protection in NR as defined in TS 33.401 [18]. The eNBs store and send the complete bitmaps without modification or truncation as specified in TS 36.300 [15].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| NR Encryption Algorithms | M | | BIT STRING (SIZE(16, …)) | Each position in the bitmap represents an encryption algorithm:<br>"all bits equal to 0" – UE supports no other NR algorithm than NEA0,<br>"first bit" – 128-NEA1,<br>"second bit" – 128-NEA2,<br>"third bit" – 128-NEA3,<br>“fourth to seventh bit” correspond to bit 4 to bit 1 of octet 3 in the *UE Additional Security Capability* IE defined in TS 24.301 [14],<br>“eighth to fifteenth bit” correspond to bit 8 to bit 1 of octet 4 in the *UE Additional Security Capability* IE defined in TS 24.301 [14],<br>other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm.<br>Algorithms are defined in TS 33.401 [18]. |
| NR Integrity Protection Algorithms | M | | BIT STRING (SIZE(16, …)) | Each position in the bitmap represents an integrity protection algorithm:<br>"all bits equal to 0" – UE supports no other NR algorithm than NIA0,<br>"first bit" – 128-NIA1,<br>"second bit" – 128-NIA2,<br>"third bit" – 128-NIA3,<br>“fourth to seventh bit” correspond to bit 4 to bit 1 of octet 5 in the *UE Additional Security Capability* IE defined in TS 24.301 [14],<br>“eighth to fifteenth bit” correspond to bit 8 to bit 1 of octet 6 in the *UE Additional Security Capability* IE defined in TS 24.301 [14],<br>other bits reserved for future use.<br>Value '1' indicates support and value '0' indicates no support of the algorithm.<br>Algorithms are defined in TS 33.401 [18]. |

## 9.2.108 EN-DC Resource Configuration

This IE contains the EN-DC resource configuration for an E-RAB, indicating the presence of PDCP at the en-gNB and Lower Layers at MCG and SCG.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PDCP at SgNB | M | | ENUMERATED (present, not present) | | – | |
| MCG resources | M | | ENUMERATED (present, not present) | | – | |
| SCG resources | M | | ENUMERATED (present, not present) | | – | |

## 9.2.109 PDCP Change Indication

The *PDCP Change Indication* IE is used to require the MeNB to either initiate the security key update or to perform PDCP data recovery towards the UE (see TS 37.340 [15]).

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| PDCP Change Indication | M | | ENUMERATED (S-KgNB update required, PDCP data recovery required, …) | The value of S-KgNB update required indicates that the security key in en-gNB needs to be updated.<br>The value of PDCP data recovery required indicates that MeNB needs to perform PDCP data recovery. |

## 9.2.110 Served NR Cell Information

This IE contains cell configuration information of an NR cell that a neighbour eNB may need for the X2 AP interface.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| NR-PCI | M | | INTEGER (0..1007) | NR Physical Cell ID | – | |
| Cell ID | M | | NR CGI 9.2.111 | | – | |
| 5GS-TAC | O | | OCTET STRING (3) | Broadcast 5GS Tracking Area Code. If this IE is included, the receiving node may assume that the NR cell provides 5GS service and is eligible as inter-system HO target candidate. | – | |
| Configured TAC | O | | OCTET STRING (2) | This is the TAC configured in the en-gNB, different from the 5GS TAC broadcast in the NR cell and enables application of Roaming and Access Restrictions for EN-DC as specified in TS 37.340 [32]. | – | |
| **Served PLMNs** | | *1..<maxnoofBPLMNs>* | | Broadcast PLMNs in SIB1 associated to the NR Cell Identity in the *Cell ID* IE. If more than maxnoofBPLMNs are needed for NR, they are provided by the *Additional PLMNs* IE. | – | |
| >PLMN Identity | M | | 9.2.4 | | – | |
| CHOICE *NR-Mode-Info* | M | | | | – | |
| *>FDD* | | | | | | |
| **>>FDD Info** | | *1* | | | – | |
| >>>UL FreqInfo | M | | NR Frequency Info 9.2.106 | This IE is ignored for NR operating bands for which uplink range of $N_{REF}$ is not defined in section 5.4.2.3 of TS 38.104 [37]. | – | |
| >>>DL FreqInfo | M | | NR Frequency Info 9.2.106 | | – | |
| >>>UL Transmission Bandwidth | M | | NR Transmission Bandwidth 9.2.114 | This IE is ignored for NR operating bands for which uplink range of $N_{REF}$ is not defined in section 5.4.2.3 of TS 38.104 [37]. | – | |
| >>>DL Transmission Bandwidth | M | | NR Transmission Bandwidth 9.2.114 | | – | |
| >>>UL Carrier List | O | | NR Carrier List 9.2.168 | If included, the *UL Transmission Bandwidth* IE shall be ignored. | YES | ignore |
| >>>DL Carrier List | O | | NR Carrier List 9.2.168 | If included, the *DL Transmission Bandwidth* IE shall be ignored. | YES | ignore |
| *>TDD* | | | | | | |
| **>>TDD Info** | | *1* | | | – | |
| >>>NRFreqInfo | M | | NR Frequency Info 9.2.106 | | – | |
| >>>Transmission Bandwidth | M | | NR Transmission Bandwidth 9.2.114 | | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>TDD UL-DL Configuration Common NR | O | | OCTET STRING | Includes the *TDD-UL-DL-ConfigCommon* IE as defined in TS 38.331 [31] | YES | ignore |
| >>>Carrier List | O | | NR Carrier List 9.2.168 | If included, the *Transmission Bandwidth* IE shall be ignored. | YES | ignore |
| >>>Intended TDD DL-UL Configuration NR | O | | OCTET STRING | Contains the *Intended TDD DL-UL Configuration NR* IE as defined in TS 38.423 [49]. | YES | ignore |
| Measurement Timing Configuration | M | | OCTET STRING | Includes the *MeasurementTimingConfiguration* inter-node message for the served cell, as defined in TS 38.331 [31]. | – | |
| **Additional PLMNs** | | *0..<maxnoofAdditionalPLMNs>* | | Additional PLMNs in addition to the Served PLMNs | YES | reject |
| >PLMN Identity | M | | 9.2.4 | | – | |
| **Broadcast PLMN Identity Info List NR** | | *0..<maxnoofextBPLMNs>* | | Corresponds to information provided in the *PLMN-IdentityInfoList* IE in the *SIB1* message as specified in TS 38.331 [31]. All PLMN Identities and associated information contained in the *PLMN-IdentityInfoList* IE are included and provided in the same order as broadcast in the *SIB1* message. | YES | ignore |
| **>Broadcast PLMNs** | | *1..<maxnoofextBPLMNs>* | | Broadcast PLMN IDs in SIB1 associated to the *NR Cell Identity* IE | – | |
| >>PLMN Identity | M | | 9.2.4 | | – | |
| >5GS-TAC | O | | OCTET STRING (3) | | – | |
| >NR Cell Identity | M | | BIT STRING (SIZE(36)) | | – | |
| SSB Positions In Burst | O | | 9.2.169 | | YES | ignore |
| NR Cell PRACH Configuration | O | | OCTET STRING | Containing 9.3.1.139 NR Cell PRACH Configuration as of TS 38.473 [44]. | YES | ignore |
| CSI-RS Transmission Indication | O | | ENUMERATED {activated, deactivated, ...} | This IE indicates the CSI-RS transmission status of the given cell. If the *Additional Measurement Timing Configuration List* IE is present, this IE is ignored. | YES | ignore |
| SFN Offset | O | | 9.2.175 | | YES | ignore |
| **Additional Measurement Timing Configuration List** | O | *1 .. <maxnoofMTCItems>* | | | YES | ignore |

| | | | | | | |
|---|---|---|---|---|---|---|
| >Measurement Timing Configuration Index | M | | INTEGER (0..16) | "0" refers to the configuration contained in the Measurement Timing Configuration IE. Any value between "1" and "16" refers to a configuration within the *Additional Measurement Timing Configuration List* IE. | – | |
| >CSI-RS MTC Configuration List | M | *1 .. <maxnoofCSIRSconfigurations>* | | This list explicitly expresses the CSI-RS configurations contained in the MTC. | – | |
| >>CSI-RS Index | M | | INTEGER (0..95) | Index of CSI-RS as in MTC | – | |
| >>CSI-RS Status | M | | ENUMERATED (activated, deactivated, …) | This IE indicates the CSI-RS transmission status of the configuration. | – | |
| >>**CSI-RS Neighbour List** | O | *1 .. <maxnoofCSIRSneighbourCells>* | | This list expresses the cells and CSI-RSs neighbouring the CSI-RS in the CSI-RS Index IE. | – | |
| >>>NR CGI | M | | 9.2.111 | | – | |
| >>>**CSI-RS MTC Neighbour List** | O | *1 .. <maxnoofCSIRSneighbourCellsInMTC>* | | This list expresses the CSI-RSs served by the CGI, which are neighbouring the CSI-RS of the neighbour cell. | – | |
| >>>>CSI-RS Index | M | | INTEGER (0..95) | | – | |

| Range bound | Explanation |
|---|---|
| maxnoofBPLMNs | Maximum no. of broadcast PLMN Ids. Value is 6. |
| maxnoofAdditionalPLMNs | Maximum no. additional PLMN Ids. Value is 6. |
| maxnoofextBPLMNs | Maximum no. of extended broadcast PLMN Ids. Value is 12. |
| maxnoofMTCItems | Maximum no. of measurement timing configurations associated with the neighbour cell. Value is 16. |
| maxnoofCSIRSconfigurations | Maximum number of CSI-RS configurations reported in the MTC. Value is 96 |
| maxnoofCSIRSneighbourCells | Maximum number of cells neighbouring a CSI-RS coverage area. Value is 16 |
| maxnoofCSIRSneighbourCellsInMTC | Maximum number of CSI-RS coverage areas neighbouring a specific CSI-RS coverage area. Value is 16 |

## 9.2.111 NR CGI

The NR Cell Global Identifier (NR CGI) is used to globally identify an NR cell (see TS 38.401 [34]).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| PLMN Identity | M | | 9.2.4 | |
| NR Cell Identity | M | | BIT STRING (36) | The leftmost bits of the *NR Cell Identity* IE value correspond to the value of the *en-gNB ID* IE contained in the *Global en-gNB ID* IE (defined in section 9.2.112) identifying the en-gNB that controls the cell. |

## 9.2.112 Global en-gNB ID

This IE is used to globally identify an en-gNB (see TS 37.340 [32]).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| PLMN Identity | M | | 9.2.4 | |
| CHOICE *en-gNB ID* | M | | | |
| *>en-gNB ID* | | | | |
| >>en-gNB ID | M | | BIT STRING (SIZE(22..32)) | Equal to the leftmost bits of the *NR Cell Identity* IE contained in the *NR CGI* IE of each cell served by the en-gNB. |

## 9.2.113 Void

## 9.2.114 NR Transmission Bandwidth

The *NR Transmission Bandwidth* IE is used to indicate the UL or DL transmission bandwidth.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| NR SCS | M | | ENUMERATED (scs15, scs30, scs60, scs120, …) | The values scs15, scs30, scs60 and scs120 corresponds to the sub carrier spacing in TS 38.104 [37]. |
| NR NRB | M | | ENUMERATED (nrb11, nrb18, nrb24, nrb25, nrb31, nrb32, nrb38, nrb51, nrb52, nrb65, nrb66, nrb78, nrb79, nrb93, nrb106, nrb107, nrb121, nrb132, nrb133, nrb135, nrb160, nrb162, nrb189, nrb216, nrb217, nrb245, nrb264, nrb270, nrb273, ...) | This IE is used to indicate the UL or DL transmission bandwidth expressed in units of resource blocks "$N_{RB}$" (TS 38.104 [37]). The values nrb11, nrb18, etc. correspond to the number of resource blocks "$N_{RB}$" 11, 18, etc. |

## 9.2.115 Cell Assistance Information

The *Cell Assistance Information* IE is used by the eNB to request information about NR cells.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| CHOICE Cell Assistance Type | M | | | This IE may be refined. |
| *>Limited List* | | | | |
| **>>List of Requested NR Cells** | | *1 .. < maxCellinengNB >* | | Included when the eNB requests a limited list of served NR cells. |
| >>>NR CGI | M | | 9.2.111 | NR cell for which served NR cell information is requested. |
| *>Full List* | | | | |
| >>Complete Information Request Indicator | M | | ENUMERATED (allServedNRCells, …) | Included when the eNB requests the complete list of served NR cells. |

| Range bound | Explanation |
|---|---|
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |

## 9.2.116 MeNB Resource Coordination Information

The *MeNB Resource Coordination Information* IE is LTE resource allocation at MeNB and used at the en-gNB to coordinate resource or sidelink resource utilisation between the MeNB and the en-gNB.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| EUTRA Cell ID | M | | ECGI 9.2.14 | This IE indicates the PCell. | – | |
| UL Coordination Information | M | | BIT STRING (6..4400, …) | Each position in the bitmap represents a PRB pair in a subframe; value "0" indicates "PCell resource not intended to be used for transmission by the sending node", value "1" indicates "PCell resource intended to be used for transmission by the sending node". The bit string spans from the first PRB pair of the first represented subframe to the last PRB pair of the same subframe and then moves to the following PRBs in the following subframes in the same order. Each position is applicable only in positions corresponding to UL subframes or SL subframes for sidelink transmission.<br>The bit string may span across multiple contiguous subframes (maximum 40).<br>The first position of the *UL Coordination Information* corresponds to subframe 0 in a radio frame where *SFN* = 0.<br>The length of the bit string is an integer multiple of $N_{\mathrm{RB}}^{\mathrm{UL}}$.<br>$N_{\mathrm{RB}}^{\mathrm{UL}}$ is defined in TS 36.211 [10].<br>The UL Coordination Information is continuously repeated. | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| DL Coordination Information | O | | BIT STRING (6..4400, …) | Each position in the bitmap represents a PRB pair in a subframe; value "0" indicates "PCell resource not intended to be used for transmission by the sending node", value "1" indicates "PCell resource intended to be used for transmission by the sending node". The bit string spans from the first PRB pair of the first represented subframe to the last PRB pair of the same subframe and then moves to the following PRBs in the following subframes in the same order. Each position is applicable only in positions corresponding to DL subframes.<br>The bit string may span across multiple contiguous subframes (maximum 40).<br>The first position of the *DL Coordination Information* corresponds to the receiving node's subframe 0 in a receiving node's radio frame where *SFN* = 0.<br>The length of the bit string is an integer multiple of $N_{RB}^{DL}$.<br>$N_{RB}^{DL}$ is defined in TS 36.211 [10].<br>The DL Coordination Information is continuously repeated. | – | |
| NR CGI | O | | 9.2.111 | This IE indicates the assumed PSCell. | YES | ignore |
| MeNB Coordination Assistance Information | O | | 9.2.139 | | YES | reject |

## 9.2.117 SgNB Resource Coordination Information

The *SgNB Resource Coordination Information* IE indicates resources within the bandwidth of the PCell which are not available for use by the MeNB and is used at the MeNB to coordinate resource utilisation between the en-gNB and the MeNB.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| NR CGI | M | | 9.2.111 | This IE indicates the PSCell. | – | |
| UL Coordination Information | M | | BIT STRING (6..4400, …) | Each position in the bitmap represents a PRB pair in a subframe; value "0" indicates "PCell resource not intended to be used for transmission by the sending node", value "1" indicates "PCell resource intended to be used for transmission by the sending node". The bit string spans from the first PRB pair of the first represented subframe to the last PRB pair of the same subframe and then moves to the following PRBs in the following subframes in the same order. Each position is applicable only in positions corresponding to UL subframes.<br>The bit string may span across multiple contiguous subframes (maximum 40).<br>The first position of the *UL Coordination Information* corresponds to the receiving node's subframe 0 in a receiving node's radio frame where *SFN* = 0.<br><br>The length of the bit string is an integer multiple of $N_{\mathrm{RB}}^{\mathrm{UL}}$. $N_{\mathrm{RB}}^{\mathrm{UL}}$ is defined in TS 36.211 [10].<br>The UL Coordination Information is continuously repeated. | – | |

| | | | | | | |
|---|---|---|---|---|---|---|
| DL Coordination Information | O | | BIT STRING (6..4400, …) | Each position in the bitmap represents a PRB pair in a subframe; value "0" indicates "PCell resource not intended to be used for transmission by the sending node", value "1" indicates "PCell resource intended to be used for transmission by the sending node". The bit string spans from the first PRB pair of the first represented subframe to the last PRB pair of the same subframe and then moves to the following PRBs in the following subframes in the same order. Each position is applicable only in positions corresponding to DL subframes.<br>The bit string may span across multiple contiguous subframes (maximum 40).<br>The first position of the *DL Coordination Information* corresponds to the receiving node's subframe 0 in a receiving node's radio frame where *SFN* = 0.<br>The length of the bit string is an integer multiple of $N_{RB}^{DL}$.<br>$N_{RB}^{DL}$ is defined in TS 36.211 [10].<br>The DL Coordination Information is continuously repeated. | – | |
| EUTRA Cell ID | O | | ECGI 9.2.14 | Reference cell for *UL Coordination Information* IE and *DL Coordination Information* IE. | YES | ignore |
| SgNB Coordination Assistance Information | O | | 9.2.140 | | YES | reject |

## 9.2.118 UL Configuration

This IE indicates how the UL PDCP is configured for the assisting node.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| UL UE Configuration | M | | ENUMERATED (no-data, shared, only, ...) | Indicates how the UE uses the UL at the assisting node. |

## 9.2.119 RLC Mode

The *RLC Mode* IE indicates the RLC Mode used for an E-RAB.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| RLC Mode | M | | ENUMERATED ( RLC-AM, RLC-UM-Bidirectional, RLC-UM-Unidirectional-UL, RLC-UM-Unidirectionall-DL, …) | |

## 9.2.120 Secondary RAT Usage Report List

This IE provides information on the NR resources used with EN-DC as specified in TS 37.340 [32].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| **Secondary RAT usage report Item** | | 1 .. < maxnoofbearers > | | | EACH | reject |
| >E-RAB ID | M | | 9.2.23 | | - | |
| >Secondary RAT Type | M | | ENUMERATED (nR, …, nR-unlicensed) | | - | |
| >**E-RAB Usage Report List** | | 1 | | | - | |
| >>**E-RAB Usage Report Item** | | 1.. <maxnooftimeperiods> | | | EACH | ignore |
| >>>Start timestamp | M | | OCTET STRING (SIZE(4)) | UTC time encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [35]. It indicates the start time of the collecting period of the included *Usage Count UL* IE and *Usage Count DL* IE. | - | |
| >>>End timestamp | M | | OCTET STRING (SIZE(4)) | UTC time encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [35]. It indicates the end time of the collecting period of the included *Usage Count UL* IE and *Usage Count DL* IE. | - | |
| >>>Usage count UL | M | | INTEGER (0..$2^{64}$-1) | The unit is: octets. | - | |
| >>>Usage count DL | M | | INTEGER (0..$2^{64}$-1) | The unit is: octets. | - | |

| Range bound | Explanation |
|---|---|
| maxnoofbearers | Maximum no. of E-RABs. Value is 256. |
| maxnooftimeperiods | Maximum no. of time reporting periods. Value is 2. |

## 9.2.121 UE Application layer measurement configuration

The IE defines configuration information for the QoE Measurement Collection (QMC) function.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigne Criticali |
|---|---|---|---|---|---|---|
| Container for application layer measurement configuration | M | | OCTET STRING (1..1000) | Indicates application layer measurement configuration, see Annex L in [36]. | - | |
| CHOICE *Area Scope of QMC* | M | | | | - | |
| >*Cell based* | | | | | | |
| >>**Cell ID List for QMC** | | *1 .. <maxnoofCellIDforQMC>* | | | | |
| >>>E-CGI | M | | 9.2.1.38 | | - | |
| >*TA based* | | | | | | |
| >>**TA List for QMC** | | *1 .. <maxnoofTAforQMC>* | | | | |
| >>>TAC | M | | 9.2.3.7 | The TAI is derived using the current serving PLMN. | - | |
| >*TAI based* | | | | | - | |
| >>**TAI List for QMC** | | *1 .. <maxnoofTAforQMC>* | | | - | |
| >>>TAI | M | | 9.2.3.16 | | - | |
| >*PLMN area based* | | | | | | |
| >>**PLMN List for QMC** | | *1 .. <maxnoofPLMNforQMC>* | | | | |
| >>>PLMN Identity | M | | 9.2.3.8 | | - | |
| Service Type | M | | ENUMERATED (QMC for streaming service, QMC for MTSI service, ...) | This IE indicates the service type of UE application layer measurements. | - | |

| Range bound | Explanation |
|---|---|
| maxnoofCellIDforQMC | Maximum no. of Cell ID subject for QMC scope. Value is 32. |
| maxnoofTAforQMC | Maximum no. of TA subject for QMC scope. Value is 8. |
| maxnoofPLMNforQMC | Maximum no. of PLMNs in the PLMN list for QMC scope. Value is 16. |

## 9.2.122 DRB ID

This information element uniquely identifies a DRB over the X2 interface within an en-gNB.

The usage of this IE is defined in TS 36.331 [9].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DRB ID | M | | INTEGER (1.. 32) | |

## 9.2.123 SUL Information

This IE provides information about the SUL carrier.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| SUL ARFCN | M | | INTEGER (0..maxNRARFCN) | RF Reference Frequency as defined in TS 38.104 [37] section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the SUL carrier. Its lowest subcarrier is also known as Point A. | – | |
| SUL Transmission Bandwidth | M | | NR Transmission Bandwidth 9.2.114 | | – | |
| Carrier List | O | | NR Carrier List 9.2.168 | If included, the SUL Transmission Bandwidth IE shall be ignored. | YES | ignore |
| Frequency Shift 7p5khz | O | | ENUMERATED (false, true, ...) | Indicate whether the value of $\Delta_{shift}$ is 0kHz or 7.5kHz when calculating $F_{REF,shift}$ as defined in Section 5.4.2.1 of TS 38.104 [37]. | YES | ignore |

| Range bound | Explanation |
|---|---|
| maxNRARFCN | Maximum value of NRARFCNs. Value is 3279165. |

## 9.2.124 Packet Loss Rate

This IE indicates the packet loss rate.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Packet Loss Rate | M | | INTEGER(0..1000) | Ratio of lost packets per number of packets sent, expressed in tenth of percent. | - | - |

## 9.2.125 Protected E-UTRA Resource Indication

This IE indicates the resources allocated for E-UTRA DL and UL reference and control signals (hereby referred to as protected resources). This information is used in the process of E-UTRA – NR Cell Resource Coordination.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Activation SFN | M | | INTEGER (0..1023) | Indicates from which SFN of the receiving node the resource allocation is valid. | | |
| **Protected Resource List** | | *1* | | The protected resource pattern is continuously repeated, and it is valid until stated otherwise or until replaced by a new pattern. The pattern does not apply in reserved subframes. | YES | ignore |
| **>Protected Resource List Item** | | *1..<maxnoofProtectedResourcePatterns>* | | Each item describes one transmission pattern. A pattern may comprise several control signals. | - | |
| >>Resource Type | M | | ENUMERATED (downlinknonCRS,CRS, uplink…) | Indicates whether the protected resource is E-UTRA DL non-CRS, E-UTRA CRS or E-UTRA UL. | - | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>Intra-PRB Protected Resource Footprint | M | | BIT STRING (84, ...) | The bitmap of REs occupied by the protected signal within one PRB. Each position in the bitmap represents an RE in one PRB; value "0" indicates "resource not protected", value "1" indicates "resource protected". The first bit of the string corresponds to the RE with the smallest time and frequency index in the PRB, where the indexing first goes into the frequency domain. The length of the bit string equals the product of $N_{RB}^{SC}$ and the length of PRB in time dimension, measured in REs. $N_{RB}^{SC}$ is defined in TS 36.211 [10]. The intra-PRB pattern consisting of all "1"s is equivalent to PRB-level granularity. | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>Protected Footprint Frequency Pattern | M | | BIT STRING(6..110, ...) | The bit string indicates in which PRBs inside carrier bandwidth the Intra-PRB Protected Resource Footprint applies. How often in time dimension this frequency pattern applies, depends on time periodicity of Intra-PRB Protected Resource Footprint. The first bit of the bit string corresponds to the PRB occupying the lowest subcarrier frequencies of the carrier bandwidth, where the indexing first goes into the frequency domain. Each position in the string represents a PRB; value "0" indicates " Intra-PRB Protected Resource Footprint does not appear in PRB", value "1" indicates "Intra-PRB Protected Resource Footprint appears in PRB". The length of the bit string equals the number of PRBs in the carrier bandwidth. | - | |
| **>>Protected Footprint Time Pattern** | M | | | The description of time periodicity of the Intra-PRB Protected Resource Footprint. | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| >>>Protected Footprint Time-periodicity | M | | INTEGER(1. .320, ...) | Periodicity with which the periodic Intra-PRB Protected Resource Footprint repeats in time-dimension (1= every PRB (i.e. slot), 2=every other PRB (i.e. slot) etc. | - | |
| >>>Protected Footprint Start Time | M | | INTEGER(1. .20, ...) | The time-position of the PRB inside the frame in which the periodic Intra-PRB Protected Resource Footprint appears for the first time. The value "1" corresponds to the receiving node's slot 0 in subframe 0 in the receiving node's radio frame where SFN = Activation SFN. | - | |
| MBSFN Control Region Length | O | | INTEGER(0. .3) | Length of control region in MBSFN subframes. Expressed in REs, in the time dimension. | | |
| PDCCH Region Length | M | | INTEGER(1. .3) | Length of PDCCH region in regular subframes. Expressed in REs, in the time dimension. | | |

| Range bound | Explanation |
|---|---|
| maxnoofProtectedResourcePatterns | Maximum no. protected resource patterns. Value is 16. |

## 9.2.126 Data Traffic Resource Indication

This IE indicates the intended data traffic resource allocation for E-UTRA - NR Cell Resource Coordination.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Activation SFN | M | | INTEGER (0..1023) | Indicates from which SFN of the receiving node the agreement is valid. | | |
| CHOICE *Shared Resource Type* | M | | | | - | |
| *>UL Only Sharing* | | | | | | |
| >>UL Resource Bitmap | M | | Data Traffic Resources 9.2.127 | | - | |
| *>UL and DL Sharing* | | | | | | |
| >>CHOICE *UL Resources* | M | | | | | |
| *>>>Unchanged* | | | NULL | | | |
| *>>>Changed* | | | | | | |
| >>>>UL Resource Bitmap | M | | Data Traffic Resources 9.2.127 | | | |
| >>CHOICE *DL* Resources | M | | | | | |
| >>>Unchanged | | | NULL | | | |
| >>>Changed | | | | | | |
| >>>>DL Resource Bitmap | M | | Data Traffic Resources 9.2.127 | | | |
| Reserved Subframe Pattern | O | | 9.2.128 | Indicates subframes in which the resource allocation does not hold. | | |

## 9.2.127 Data Traffic Resources

The *Data Traffic Resources* IE indicates the intended data traffic resource allocation for E-UTRA - NR Cell Resource Coordination.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Data Traffic Resources | M | | BIT STRING (6..17600) | The indication of resources allocated to E-UTRA PDSCH/PUSCH. Each position in the bit string represents a PRB pair in a subframe; value "0" indicates "resource not intended to be used for transmission", value "1" indicates "resource intended to be used for transmission ". The first bit of the bit string corresponds to the PRB pair occupying the lowest subcarrier frequencies of the carrier, where the indexing first goes into the frequency domain. The bit string may span across multiple contiguous subframes. The first position of the Data Traffic Resources corresponds to the receiving node's subframe 0 in a receiving node's radio frame where SFN = Activation SFN. **The length of the bit string is an integer multiple of $N_{RB}^{DL}$ or $N_{RB}^{UL}$, defined in TS 36.211 [10].** |

## 9.2.128 Reserved Subframe Pattern

The *Reserved Subframe Pattern* IE indicates the pattern of subframes in which the *Protected E-UTRA Resource Indication* and *Data Traffic Resource Indication* do not hold.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Subframe Type | M | | ENUMERATED( MBSFN,non-MBSFN, …) | Indicates what type of non-regular subframes the *Reserved Subframe Pattern* refers to (e.g. MBSFN). |
| Reserved Subframe Pattern | M | | BIT STRING (10..160) | Each position in the bitmap represents a subframe. Value ‘0’ indicates "regular subframe". Value '1' indicates "reserved subframe". For MBSFN subframes, the exception refers only to the non-control region of the subframe. The bit string may span across multiple contiguous subframes. The first position of the Subframe Configuration IE corresponds to the receiving node's subframe 0 in a receiving node's radio frame where SFN = Activation SFN. The IE is ignored if received by the eNB. |
| MBSFN Control Region Length | O | | INTEGER(0..3) | Length of control region in MBSFN subframes. Expressed in REs, in the time dimension. |

## 9.2.129 Aerial UE subscription information

This information element is used by the eNB to know if the UE is allowed to use aerial UE function, refer to TS 23.401[12].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Aerial UE subscription information | M | | ENUMERATED (allowed, not allowed, ...) | |

## 9.2.130 User plane traffic activity report

This IE is used to indicate user plane traffic activity.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| User plane traffic activity report | M | | ENUMERATED (inactive, re-activated, …) | "re-activated" shall be only set after "inactive" has been reported for the concerned reporting object |

## 9.2.131 RLC Status

This IE indicates about the RLC configuration change included in the container towards the UE.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Reestablishment Indication | O | | ENUMERATED (reestablished, ...) | Indicates that following the change of the radio status, the RLC has been re-established. |

## 9.2.132 RRC config indication

This IE is used to indicate the type of RRC configuration used at the en-gNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| RRC config indication | M | | ENUMERATED (full config, delta config, ...) | |

## 9.2.133 PDCP SN Length

The *PDCP SN Length* IE is used to indicate the PDCP SN length configuration of the bearer.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| PDCP SN Length | M | | ENUMERATED (12bits, 18bits, …) | This IE indicates the PDCP sequence number size. |

## 9.2.134 Bluetooth Measurement Configuration

This IE defines the parameters for Bluetooth measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Bluetooth Measurement Configuration | M | | ENUMERATED (Setup, …) | |
| **Bluetooth Measurement Configuration Name List** | | *0..1* | | This IE is present if the *Bluetooth Measurement Configuration* IE is set to "Setup". |
| **>Bluetooth Measurement Configuration Name Item** | | *1 .. <maxnoofBluetoothName>* | | |
| >>Bluetooth Measurement Configuration Name | M | | OCTET STRING (SIZE (1..248)) | |
| BT RSSI | O | | ENUMERATED (True, …) | In case of Immediate MDT, it corresponds to M8 measurement as defined in 37.320 [31]. |

| Range bound | Explanation |
|---|---|
| maxnoofBluetoothname | Maximum no. of Bluetooth local name used for Bluetooth measurement collection, the maximum value is 4. |

## 9.2.135 WLAN Measurement Configuration

This IE defines the parameters for WLAN measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| WLAN Measurement Configuration | M | | ENUMERATED (Setup, …) | |
| **WLAN Measurement Configuration Name List** | | *0..1* | | This IE is present if the *WLAN Measurement Configuration* IE is set to "Setup". |
| **>WLAN Measurement Configuration Name Item** | | *1 .. <maxnoofWLANName>* | | |
| >>WLAN Measurement Configuration Name | M | | OCTET STRING (SIZE (1..32)) | |
| WLAN RSSI | O | | ENUMERATED (True, …) | In case of Immediate MDT, it corresponds to M8 as defined in 37.320 [31]. |
| WLAN RTT | O | | ENUMERATED (True, …) | For Immediate MDT, it corresponds to M9 as defined in 37.320 [31]. |

| Range bound | Explanation |
|---|---|
| maxnoofWLANname | Maximum no. of WLAN SSID used for WLAN measurement collection, the maximum value is 4. |

## 9.2.136 Subscription Based UE Differentiation Information

This IE is generated by the MME based on the UE subscription information, it provides the Subscription Based UE differentiation Information.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Periodic Communication Indicator | O | | ENUMERATED(periodically, on demand, …) | This IE indicates whether the UE communicates periodically or not, e.g. only on demand. |
| Periodic Time | O | | INTEGER (1..3600, …) | This IE indicates the interval time of periodic communication, the unit is: second |
| **Scheduled Communication Time** | | *0..1* | | This IE indicates the time zone and day of the week when the UE is available for communication. |
| >Day of Week | O | | BIT STRING (SIZE(7)) | If Day-Of-Week is not provided this shall be interpreted as every day of the week.<br>Each position in the bitmap represents a day of the week: first bit = Mon, second bit =Tue, third bit =Wed, and so on. Value '1' indicates 'scheduled. Value '0' indicates 'not scheduled'. |
| >Time of Day Start | O | | INTEGER (0..86399, …) | This IE indicates the time to start of the day, each value represent the corresponding second since 00:00 of the day.<br>If Time-Of-Day-Start is not provided, starting time shall be set to start of the day(s) indicated by Day-Of-Week. |
| >Time of Day End | O | | INTEGER (0..86399, …) | This IE indicates the time to start of the day, each value represent the corresponding second since 00:00 of the day. The value of this IE should be bigger than the value of *Time of Day Start* IE.<br><br>If Time-Of-Day-End is not provided, ending time is end of the day(s) indicated by Day-Of-Week. |
| Stationary Indication | O | | ENUMERATED(stationary, mobile, …) | |
| Traffic Profile | O | | ENUMERATED(single packet, dual packets, multiple packets, …) | "single packet" indicates single packet transmission (UL or DL), "dual packets" indicates dual packet transmission (UL with subsequent DL, or DL with subsequent UL),<br>"multiple packets" indicates multiple packets transmission. |
| Battery Indication | O | | ENUMERATED(battery powered, battery powered not rechargeable or replaceable, not battery powered, …) | "battery powered" indicates that the UE is battery powered and the battery is rechargeable/replaceable, "battery powered not rechargeable or replaceable" indicates that the UE is battery powered but the battery is not rechargeable/replaceable, "not battery powered" indicates that the UE is not battery powered. |

## 9.2.137 Duplication activation

The *Duplication Activation* IE indicates whether UL PDCP Duplication is activated or not.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Duplication Activation | M | | ENUMERATED ( Active, Inactive, …) | |

## 9.2.138 LCID

This IE uniquely identifies a LCID for the associated DRB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| LCID | M | | INTEGER (1..32, ...) | Corresponds to information provided in the *LogicalChannelIdentity* IE as defined in TS 38.331 [8]. |

## 9.2.139 MeNB Coordination Assistance Information

The *MeNB Coordination Assistance Information* IE is provided by the MeNB and used by the SgNB to determine further coordination of resource utilisation between the en-gNB and the MeNB.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| MeNB Coordination Assistance Information | M | | ENUMERATED(Coordination Not Required, …) | The absence of this IE indicates that the resource coordination is required. |

## 9.2.140 SgNB Coordination Assistance Information

The *SgNB Coordination Assistance Information* IE is provided by the SgNB and used by the MeNB to determine further coordination of resource utilisation between the en-gNB and the MeNB.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| SgNB Coordination Assistance Information | M | | ENUMERATED(Coordination Not Required, …) | The absence of this IE indicates that the resource coordination is required. |

## 9.2.141 Desired Activity Notification Level

This IE contains information on which level activity notification shall be performed.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Desired Activity Notification Level | O | | ENUMERATED (None, E-RAB, UE-level, …) | |

## 9.2.142 Location Information at SgNB

The *Location Information at SgNB* IE enables the SgNB to provide the MeNB with information that supports localisation of the UE.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| PSCell ID | M | | NR CGI 9.2.111 | PSCell of the UE | – | |

## 9.2.143 Interface Instance Indication

The Interface Instance Indication identifies the interface instance the X2AP message is destined for.

NOTE: The Interface Instance Indication is allocated so that it can be associated with an X2-C interface instance. The Interface Instance Indication may identify more than one interface instance.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Interface Instance Indication | M | | INTEGER (0..255, ...) | |

## 9.2.144 NB-IoT UL DL Alignment Offset

This IE is used to indicate the offset between the UL carrier frequency center with respect to DL carrier frequency center.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| NB-IoT UL DL Alignment Offset | M | | ENUMERATED (-7.5, 0, 7.5, …) | Unit: kHz<br>Corresponds to information provided in the *TDD-UL-DL-AlignmentOffset-NB* as specified in TS 36.331 [9]. |

## 9.2.145 Lower Layer presence status change

This IE is used to indicate that lower layer resources' presence status shall be changed. If the presence status is set to "release lower layers" or "suspend lower layers", PDCP entities, X2-U bearer resources, S1-U bearer resources and UE context information shall be kept.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Lower Layer presence status change | M | | ENUMERATED (release lower layers, re-establish lower layers, suspend lower layers, resume lower layers ...) | "re-establish lower layers" shall be only set after "release lower layers" has been indicated.<br>"resume lower layers" shall restore SCG.<br>"resume lower layers" shall be only set after "suspend lower layers" has been indicated. |

## 9.2.146 Cell and Capacity Assistance Information

The *Cell and Capacity Assistance Information* IE is used by the eNB to request information about NR or E-UTRA cells and it includes information about cell list size capacity.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Maximum Cell List Size | O | | 9.2.147 | |
| Cell Assistance Information | O | | 9.2.115 | |

## 9.2.147 Maximum Cell List Size

This IE indicates the maximum size the sending node can handle for a given list.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Maximum Cell List Size | M | | INTEGER (0..16384, …) | |

## 9.2.148 Message Oversize Notification

This IE indicates the maximum number of cells that can be received in the *List of Served Cells NR* IE.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| Maximum Cell List Size | M | | 9.2.147 | | | |

## 9.2.149 TNL Transport Layer Address Info

This IE is used for signalling IP addresses of IP-Sec endpoints used for establishment of IP-Sec tunnels.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| **Transport UP Layer Addresses Info to Add List** | | *0..1* | | |
| **>Transport UP Layer Addresses Info to Add Item** | | *1..<maxnoofTLAs>* | | |
| >>IP-Sec Transport Layer Address | M | | BIT STRING(1..160, ...) | Transport Layer Addresses for IP-Sec endpoint. |
| **>>GTP Transport Layer Addresses To Add List** | | *0..1* | | |
| **>>>GTP Transport Layer Addresses To Add Item** | | *1..<maxnoofGTPTLAs>* | | |
| >>>>GTP Transport Layer Address Info | M | | BIT STRING (1..160, ...) | GTP Transport Layer Addresses for GTP end-points. |
| **Transport UP Layer Addresses Info to Remove List** | | *0..1* | | |
| **>Transport UP Layer Addresses Info to Remove Item** | | *1..<maxnoofTLAs>* | | |
| >>IP-Sec Transport Layer Address | M | | BIT STRING (1..160, ...) | Transport Layer Addresses for IP-Sec endpoint. |
| **>>GTP Transport Layer Addresses To Remove List** | | *0..1* | | |
| **>>>GTP Transport Layer Addresses To Remove Item** | | *1..<maxnoofGTPTLAs>* | | |
| >>>>GTP Transport Layer Address Info | M | | BIT STRING (1..160, ...) | GTP Transport Layer Addresses for GTP end-points. |

| Range bound | Explanation |
|---|---|
| maxnoofTLAs | Maximum no. of Transport Layer Addresses in the message. Value is 16 |
| maxnoofGTPTLAs | Maximum no. of GTP Transport Layer Addresses for a GTP end-point in the message. Value is 16. |

## 9.2.150 CP Transport Layer Information

This element is used to provide the transport layer information associated with EN-DC X2 control plane transport.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CHOICE *CP Transport Layer Information* | | | | |
| *>Endpoint-IP-address* | | | | |
| >>Endpoint IP Address | M | | BIT STRING (1..160, ...) | |
| *>Endpoint-IP-address-and-port* | | | | |
| >>Endpoint IP Address | M | | BIT STRING (1..160, ...) | |
| >>Port Number | M | | OCTET STRING (SIZE(2)) | |

## 9.2.151 TNL Association Usage

This IE indicates the usage of the TNL association.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| TNL Association Usage | O | | ENUMERATED (ue, non-ue, both, …) | Indicates whether the TNL association is only used for UE-associated signalling, or non-UE-associated signalling, or both. |

## 9.2.152 RAN UE NGAP ID

This IE uniquely identifies the UE association over the NG interface within the NG-RAN node, as specified in TS 38.413 [39].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| RAN UE NGAP ID | M | | INTEGER $(0..2^{32}-1)$ | |

## 9.2.153 EPC Handover Restriction List Container

This IE contains the *Handover Restriction List* IE specified in TS 36.413 [4] as received by the E-UTRAN from the EPC.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| EPC Handover Restriction List Container | M | | OCTET STRING | The octets of the OCTET STRING are encoded according to the specifications of the Handover Restriction List IE specified in TS 36.413 [4]. |

## 9.2.154 DAPS Request Information

The *DAPS Indicator* IE indicates that the source eNB requests a DAPS HO for the concered E-RAB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| DAPS Indicator | M | | ENUMERATED (DAPS HO required, …) | Indicates that DAPS HO is requested |

## 9.2.155 DAPS Response Information

The *DAPS Response Indicator* IE indicates the response to a requested DAPS Handover for the concerned E-RAB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DAPS Response Indicator | M | | ENUMERATED (DAPS HO accepted, DAPS HO not accepted, …) | Indicates that DAPS Handover is accepted or not. |

## 9.2.156 Maximum Number of CHO Preparations

This IE indicates the maximum number of concurrently prepared CHO candidate cells for a UE at a candidate target eNB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Maximum Number of CHO Preparations | M | | INTEGER (1..8, ...) | |

## 9.2.157 Ethernet Type

This IE is used to indicate that Ethernet data is expected.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Ethernet Type | M | | ENUMERATED (True, …) | |

## 9.2.158 NR V2X Services Authorized

This IE provides information on the authorization status of the UE to use the NR sidelink for V2X services.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Vehicle UE | O | | ENUMERATED (authorized, not authorized, ...) | Indicates whether the UE is authorized as Vehicle UE. |
| Pedestrian UE | O | | ENUMERATED (authorized, not authorized, ...) | Indicates whether the UE is authorized as Pedestrian UE. |

## 9.2.159 NR UE Sidelink Aggregate Maximum Bit Rate

This IE provides information on the Aggregate Maximum Bitrate of the UE's sidelink communication for NR V2X services.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| NR UE Sidelink Aggregate Maximum Bit Rate | M | | Bit Rate 9.2.97 | Value 0 shall be considered as a logical error by the receiving eNB. |

## 9.2.160 PC5 QoS Parameters

This IE provides information on the PC5 QoS parameters of the UE's sidelink communication for NR PC5.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| **PC5 QoS Flow List** | | *1* | | |
| **>PC5 QoS Flow Item** | | *1..<maxnoofPC5QoSFlows>* | | |
| >>PQI | M | | INTEGER (0..255, …) | PQI is a special 5QI as specified in TS 23.501 [9]. |
| >>**PC5 Flow Bit Rates** | O | | | Only applies for GBR QoS Flows. |
| >>>Guaranteed Flow Bit Rate | M | | Bit Rate 9.2.11 | Guaranteed Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9]. |
| >>>Maximum Flow Bit Rate | M | | Bit Rate 9.2.11 | Maximum Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9]. |
| >>Range | O | | ENUMERATED (m50, m80, m180, m200, m350, m400, m500, m700, m1000, …) | Only applies for groupcast. |
| PC5 Link Aggregated Bit Rates | O | | Bit Rate 9.2.11 | Only applies for non-GBR QoS Flows. |

| Range bound | Explanation |
|---|---|
| *maxnoofPC5QoSFlows* | Maximum no. of PC5 QoS flows allowed towards one UE. Value is 2048. |

## 9.2.161 TNL Capacity Indicator

The *TNL Capacity Indicator* IE indicates the available capacity of the Transport Network experienced by the NR cell.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DL TNL Maximum Offered Capacity | M | | INTEGER (1.. 16777216,...) | Maximum capacity offered by the transport portion of the cell in kbps |
| DL TNL Available Capacity | M | | INTEGER (0.. 100,...) | Available capacity over the transport portion serving the cell in percentage relative to the *DL TNL Maximum Offered Capacity*. Value 100 corresponds to the Maximum offered capacity. |
| UL TNL Maximum Offered Capacity | M | | INTEGER (1.. 16777216…) | Maximum capacity offered by the transport portion of the cell in kbps |
| UL TNL Available Capacity | M | | INTEGER (0.. 100,...) | Available capacity over the transport portion serving the cell in percentage relative to the *DL TNL Maximum Offered Capacity*. Value 100 corresponds to the Maximum offered capacity. |

## 9.2.162 NR Radio Resource Status

The *NR Radio Resource Status* IE indicates the usage of the PRBs per cell for MIMO and per SSB area for all traffic in Downlink and Uplink and the usage of PDCCH CCEs for Downlink and Uplink scheduling.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| **SSB Area Radio Resource Status List** | | *1* | | |
| **>SSB Area Radio Resource Status Item** | | *1..<maxnoofSSBAreas>* | | |
| >>SSB Index | M | | 9.2.167 | |
| >>SSB Area DL GBR PRB usage | M | | INTEGER (0..100) | Per SSB area DL GBR PRB usage |
| >>SSB Area UL GBR PRB usage | M | | INTEGER (0..100) | Per SSB area UL GBR PRB usage |
| >>SSB Area DL non-GBR PRB usage | M | | INTEGER (0..100) | Per SSB area DL non-GBR PRB usage |
| >>SSB Area UL non-GBR PRB usage | M | | INTEGER (0..100) | Per SSB area UL non-GBR PRB usage |
| >>SSB Area DL Total PRB usage | M | | INTEGER (0..100) | Per SSB area DL Total PRB usage |
| >>SSB Area UL Total PRB usage | M | | INTEGER (0..100) | Per SSB area UL Total PRB usage |
| >>DL scheduling PDCCH CCE usage | O | | INTEGER (0..100) | |
| >>UL scheduling PDCCH CCE usage | O | | INTEGER (0..100) | |
| **MIMO PRB usage Information** | O | | | |
| **>DL GBR PRB usage for MIMO** | M | | INTEGER (0..100) | Per cell DL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [45]. |
| **>UL GBR PRB usage for MIMO** | M | | INTEGER (0..100) | Per cell UL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [45]. |
| **>DL non-GBR PRB usage for MIMO** | M | | INTEGER (0..100) | Per cell DL non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [45]. |
| **>UL non-GBR PRB usage for MIMO** | M | | INTEGER (0..100) | Per cell UL non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [45]. |
| **>DL Total PRB usage for MIMO** | M | | INTEGER (0..100) | Per cell DL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [45]. |
| **>UL Total PRB usage for MIMO** | M | | INTEGER (0..100) | Per cell UL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [45]. |

| Range bound | Explanation |
|---|---|
| *maxnoofSSBAreas* | Maximum no. SSB Areas that can be served by a NG-RAN node cell. Value is 64. |

## 9.2.163 NR Composite Available Capacity Group

The *NR Composite Available Capacity Group* IE indicates the overall available resource level per cell and per SSB area in the cell in Downlink and Uplink.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Composite Available Capacity Downlink | M | | NR Composite Available Capacity 9.2.164 | For the Downlink |
| Composite Available Capacity Uplink | M | | NR Composite Available Capacity 9.2.164 | For the Uplink |

## 9.2.164 NR Composite Available Capacity

The *NR Composite Available Capacity* IE indicates the overall available resource level in the cell in either Downlink or Uplink.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Cell Capacity Class Value | O | | NR Cell Capacity Class Value 9.2.165 | |
| Capacity Value | M | | NR Capacity Value 9.2.166 | |

## 9.2.165 NR Cell Capacity Class Value

The *NR Cell Capacity Class Value* IE indicates the value that classifies the cell capacity with regards to the other cells. The *NR Cell Capacity Class Value* IE only indicates resources that are configured for traffic purposes.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| NR Capacity Class Value | M | | INTEGER (1..100,...) | Value 1 shall indicate the minimum cell capacity, and 100 shall indicate the maximum cell capacity. There should be a linear relation between cell capacity and Cell Capacity Class Value. |

## 9.2.166 NR Capacity Value

The *NR Capacity Value* IE indicates the amount of resources per cell and per SSB area that are available relative to the total en-gNB resources. The capacity value should be measured and reported so that the minimum en-gNB resource

usage of existing services is reserved according to implementation. The *NR Capacity Value* IE can be weighted according to the ratio of cell capacity class values, if available.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Capacity Value | M | | INTEGER (0..100) | Value 0 shall indicate no available capacity, and 100 shall indicate maximum available capacity with respect to the whole cell. Capacity Value should be measured on a linear scale. |
| **SSB Area Capacity Value List** | | *0..1* | | |
| **>SSB Area Capacity Value Item** | | *1..<maxnoofSSBAreas>* | | |
| >>SSB Index | M | | 9.2.167 | |
| >>SSB Area Capacity Value | M | | INTEGER (0..100) | Value 0 shall indicate no available capacity, and 100 shall indicate maximum available capacity . SSB Area Capacity Value should be measured on a linear scale. |

| Range bound | Explanation |
|---|---|
| *maxnoofSSBAreas* | Maximum no. SSB Areas that can be served by a NG-RAN node cell. Value is 64. |

## 9.2.167 SSB Index

The *SSB Index* IE identify an SSB area of an NR cell.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| SSB Index | M | | INTEGER (0..63) | |

## 9.2.168 NR Carrier List

This IE indicates the SCS-specific carriers per TDD, per DL, per UL or per SUL of an NR cell.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| **NR Carrier Item** | | *1..<maxnoofNRSCSs>* | | |
| >NR SCS | M | | ENUMERATED (scs15, scs30, scs60, scs120, …) | SCS for the corresponding carrier. |
| >Offset to Carrier | M | | INTEGER (0.. 2199, ...) | Offset in frequency domain between Point A (lowest subcarrier of common RB 0) and the lowest usable subcarrier on this carrier in number of PRBs (using the *NR SCS* IE defined for this carrier). The maximum value corresponds to 275×8−1. See TS 38.211 [42], clause 4.4.2. |
| >Carrier Bandwidth | M | | INTEGER (1.. maxnoofNRPhysicalResourceBlocks, ...) | Width of this carrier in number of PRBs (using the *NR SCS* IE defined for this carrier). See TS 38.211 [42], clause 4.4.2. |

| Range bound | Explanation |
|---|---|
| maxnoofNRSCSs | Maximum no. of SCS-specific carriers per TDD, per DL, per UL or per SUL of an NR cell. Value is 5. |
| maxnoofNRPhysicalResourceBlocks | Maximum no. of Physical Resource Blocks of an NR Cell. Value is 275. |

## 9.2.169 SSB Positions In Burst

Indicates the time domain positions of the transmitted SS-blocks in a half frame with SS/PBCH blocks as defined in TS 38.213 [43], clause 4.1.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| CHOICE *ssb-PositionsInBurst* | M | | | The first/ leftmost bit corresponds to SS/PBCH block index 0, the second bit corresponds to SS/PBCH block index 1, and so on. Value 0 in the bitmap indicates that the corresponding SS/PBCH block is not transmitted while value 1 indicates that the corresponding SS/PBCH block is transmitted. |
| *>ShortBitmap* | | | | |
| >>ShortBitmap | M | | BIT STRING (SIZE(4)) | |
| *>MediumBitmap* | | | | |
| >>MediumBitmap | M | | BIT STRING (SIZE(8)) | |
| *>LongBitmap* | | | | |
| >>LongBitmap | M | | BIT STRING (SIZE(64)) | |

## 9.2.170 NPRACH Configuration

This IE indicates the NPRACH Configuration.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CHOICE *FDDorTDD* | M | | | |
| >*FDD* | | | | |
| >>NPRACH-CP-Length | M | ENUMERATED {us66dot7, us266dot7, …} | | |
| >>Anchor Carrier NPRACH Configuration | M | | OCTET STRING | Includes the *NPRACH-ParametersList-NB-r13* IE as defined in 6.7.3.2 of TS 36.331 [9]. |
| >>Anchor Carrier EDT NPRACH Configuration | O | | OCTET STRING | Includes the *NPRACH-ParametersList-NB-r14* IE as defined in 6.7.3.2 of TS 36.331 [9]. |
| >>Anchor Carrier Format 2 NPRACH Configuration | O | | OCTET STRING | Includes the *NPRACH-ParametersListFmt2-NB-r15* IE as defined in 6.7.3.2 of TS 36.331 [9] |
| >>Anchor Carrier Format 2 EDT NPRACH Configuration | O | | OCTET STRING | Includes the *NPRACH-ParametersListFmt2-NB-r15* IE as defined in 6.7.3.2 of TS 36.331 [9] |
| >>Non Anchor Carrier NPRACH Configuration | O | | OCTET STRING | Includes the *UL-ConfigCommonList-NB-r14* IE as defined in 6.7.3.1 of TS 36.331 [9]. |
| >>Non Anchor Carrier Format 2 NPRACH Configuration | O | | OCTET STRING | Includes the *UL-ConfigCommonList-NB-v1530* IE as defined in 6.7.3.1 of TS 36.331 [9]. |
| >*TDD* | | | | |
| >>nprach-PreambleFormat | M | ENUMERATED {fmt0, fmt1, fmt2, fmt0-a, fmt1-a, …} | | |
| >>Anchor Carrier NPRACH Configuration TDD | M | | OCTET STRING | Includes the *NPRACH-ParametersListTDD-NB-r15* IE as defined in 6.7.3.2 of TS 36.331 [9] |
| **>>Non Anchor Carrier Frequency Configuration list** | | *0..< maxnoofNonAnchorCarrierFreqConfig>* | | |
| >>>Non Anchor Carrier Frequency | M | | OCTET STRING | Includes the *DL-CarrierConfigCommon-NB-r14* IE as defined in 6.7.3.2 of TS 36.331 [9] |
| >>Non Anchor Carrier NPRACH Configuration TDD | O | | OCTET STRING | Includes the *UL-ConfigCommonListTDD-NB-r15* IE as defined in 6.7.3.1 of TS 36.331 [9]. |

| Range bound | Explanation |
|---|---|
| maxnoofNonAnchorCarrierFreqConfig | Maximum no. of non-Anchor Carrier Frequency Configurations. Value is 15. |

## 9.2.171 UE Radio Capability ID

This IE contains the UE Capability ID as defined in TS 23.003[29].

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| UE Radio Capability ID | M | | OCTET STRING | |

## 9.2.172 QoS Mapping Information

This IE indicates the DSCP and/or IPv6 Flow Label field(s) of IP packets sent in the corresponding GTP-U tunnel for IAB.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| DSCP | O | | BIT STRING (SIZE(6)) | |
| Flow label | O | | BIT STRING (SIZE(20)) | |

## 9.2.173 UE Radio Capability

This IE contains UE Radio Capability information.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| UE Radio Capability | M | | OCTET STRING | Includes the *UERadioAccessCapabilityInformation* message as defined in 10.2.2 of TS 36.331 [9]. |

## 9.2.174 URI

This IE is a URI.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| URI | M | | VisibleString | String representing URI (Uniform Resource Identifier) |

## 9.2.175 SFN Offset

This IE contains the time offset between an absolute time reference and the SFN0 start. The IE is calculated assuming that the SFN transmission started at the absolute time reference. The absolute time reference chosen is 1980-01-06 T00:00:19 International Atomic Time (TAI).

| IE/Group Name | Presence | Range | IE type and reference | Semantics description | Criticality | Assigned Criticality |
|---|---|---|---|---|---|---|
| SFN Time Offset | M | | BIT STRING (SIZE(24)) | Time offset in microseconds between the absolute time reference "1980-01-06 T00:00:19 International Atomic Time (TAI)" and the SFN0 start. The maximum usable value is (1024*10^4-1). Values higher than the maximum are discarded. | | |

## 9.2.176 Global RAN Node ID

This IE is used to globally identify an NG-RAN node.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| CHOICE *NG-RAN node* | M | | | |
| *>gNB* | | | | |
| >>Global gNB ID | M | | Global en-gNB ID 9.2.112 | |

## 9.2.177 SCG UE History Information

This IE contains information about the PSCells served by the secondary node in an active state.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| **Last Visited PSCell List** | | *0..<maxnoofPSCellsPerSN>* | | List of cells configured as PSCells. Most recent PSCell related information is added to the top of the list. |
| >Last Visited PSCell Information | M | | OCTET STRING | The PSCell related information. Defined in TS 36.413 [4]. |

| Range bound | Explanation |
|---|---|
| maxnoofPSCellsPerSN | Maximum number of last visited PSCell information records that can be reported in the IE. Value is 8 |

## 9.2.178 SCG Activation Status

The *SCG Activation Status* IE indicates the status of the SCG resources.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| SCG Activation Status | M | | ENUMERATED (SCG activated, SCG deactivated, ...) | |

## 9.2.179 SCG Activation Request

The *SCG Activation Request* IE indicates whether the SCG resources are required to be activated or deactivated.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| SCG Activation Request | M | | ENUMERATED (Activate SCG, Deactivate SCG, ...) | |

## 9.2.180 Served Cell Specific Info Request

The *Served Cell Specific Info Request* IE is used by the NG-RAN node to request specific information about NR cells.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| **List of Requested NR Cells** | | *1* | | List of NR cells. |
| **>List of Requested NR Cells item** | | *1 .. < maxCellinengNB >* | | |
| >>NR CGI | M | | 9.2.111 | NR cell for which specific served NR cell information is requested. |
| >>Additional Measurement Timing Configuration List Request Indicator | O | | ENUMERATED (AdditionalMTCList Requested, …) | Included when the NG-RAN node requests the *Additional Measurement Timing Configuration List* IE to be included in the *Served Cell Information NR* IE for the requested cells. |

| Range bound | Explanation |
|---|---|
| maxCellinengNB | Maximum no. cells that can be served by an en-gNB. Value is 16384. |

## 9.2.181 Security Indication

This IE contains the user plane integrity protection indication which indicates the requirements on UP integrity protection for the corresponding E-RAB.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Integrity Protection Indication | M | | ENUMERATED (required, preferred, not needed, …) | Indicates whether UP integrity protection shall apply, should apply, or shall not apply for the concerned E-RAB. |

## 9.2.182 Security Result

This IE indicates whether the security policy indicated as "preferred" in the *Integrity Protection Indication* IE is performed or not.

| IE/Group Name | Presence | Range | IE Type and Reference | Semantics Description |
|---|---|---|---|---|
| Integrity Protection Result | M | | ENUMERATED (performed, not performed, …) | Indicates whether UP integrity protection is performed or not for the concerned E-RAB. |

## 9.2.183 Sensor Measurement Configuration

This IE defines the parameters for Sensor measurement collection.

| IE/Group Name | Presence | Range | IE type and reference | Semantics description |
|---|---|---|---|---|
| Sensor Measurement Configuration | M | | ENUMERATED (Setup, …) | |
| **Sensor Measurement Configuration Name List** | | *0..1* | | |
| **>Sensor Measurement Configuration Name Item** | | *1..<maxnoofSensorName>* | | |
| >>CHOICE *Sensor Name* | M | | | |
| *>>>Uncompensated Barometric* | | | | |
| >>>>Uncompensated Barometric Configuration | M | | ENUMERATED (true, …) | |

| Range bound | Explanation |
|---|---|
| maxnoofSensorName | Maximum no. of Sensor local name used for Sensor measurement collection. Value is 3. |

# 9.3 Message and Information Element Abstract Syntax (with ASN.1)

## 9.3.1 General

X2AP ASN.1 definition conforms to ITU-T Rec. X.680 [27] and ITU-T Rec. X.681 [28].

Sub clause 9.3 presents the Abstract Syntax of the X2AP protocol with ASN.1. In case there is contradiction between the ASN.1 definition in this sub clause and the tabular format in sub clause 9.1 and 9.2, the ASN.1 shall take precedence, except for the definition of conditions for the presence of conditional elements, in which the tabular format shall take precedence.

The ASN.1 definition specifies the structure and content of X2AP messages. X2AP messages can contain any IEs specified in the object set definitions for that message without the order or number of occurrence being restricted by ASN.1. However, for this version of the standard, a sending entity shall construct an X2AP message according to the PDU definitions module and with the following additional rules:

- IEs shall be ordered (in an IE container) in the order they appear in object set definitions.

- Object set definitions specify how many times IEs may appear. An IE shall appear exactly once if the presence field in an object has value "mandatory". An IE may appear at most once if the presence field in an object has value "optional" or "conditional". If in a tabular format there is multiplicity specified for an IE (i.e. an IE list) then in the corresponding ASN.1 definition the list definition is separated into two parts. The first part defines an IE container list in which the list elements reside. The second part defines list elements. The IE container list appears as an IE of its own. For this version of the standard an IE container list may contain only one kind of list elements.

NOTE: In the above, "IE" means an IE in the object set with an explicit ID. If one IE needs to appear more than once in one object set, then the different occurrences have different IE IDs.

If an X2AP message that is not constructed as defined above is received, this shall be considered as Abstract Syntax Error, and the message shall be handled as defined for Abstract Syntax Error in clause 10.

## 9.3.2 Usage of Private Message Mechanism for Non-standard Use

The private message mechanism for non-standard use may be used:

- for special operator (and/or vendor) specific features considered not to be part of the basic functionality, i.e. the functionality required for a complete and high-quality specification in order to guarantee multivendor inter-operability.

- by vendors for research purposes, e.g. to implement and evaluate new algorithms/features before such features are proposed for standardisation.

The private message mechanism shall not be used for basic functionality. Such functionality shall be standardised.

## 9.3.3 Elementary Procedure Definitions

```
-- ASN1START
```

```
-- **************************************************************
--
-- Elementary Procedure definitions
--
-- **************************************************************

X2AP-PDU-Descriptions {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
eps-Access (21) modules (3) x2ap (2) version1 (1) x2ap-PDU-Descriptions (0) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- IE parameter types from other modules.
--
-- **************************************************************

IMPORTS
    Criticality,
    ProcedureCode

FROM X2AP-CommonDataTypes

    CellActivationRequest,
    CellActivationResponse,
    CellActivationFailure,
    ENBConfigurationUpdate,
    ENBConfigurationUpdateAcknowledge,
    ENBConfigurationUpdateFailure,
    ErrorIndication,
    HandoverCancel,
    HandoverReport,
    HandoverPreparationFailure,
    HandoverRequest,
    HandoverRequestAcknowledge,
    LoadInformation,
    PrivateMessage,
    ResetRequest,
    ResetResponse,
    ResourceStatusFailure,
    ResourceStatusRequest,
    ResourceStatusResponse,
    ResourceStatusUpdate,
    RLFIndication,
    SNStatusTransfer,
    UEContextRelease,
    X2SetupFailure,
    X2SetupRequest,
    X2SetupResponse,
    MobilityChangeRequest,
    MobilityChangeAcknowledge,
```

```
    MobilityChangeFailure,
    X2Release,
    X2APMessageTransfer,
    SeNBAdditionRequest,
    SeNBAdditionRequestAcknowledge,
    SeNBAdditionRequestReject,
    SeNBReconfigurationComplete,
    SeNBModificationRequest,
    SeNBModificationRequestAcknowledge,
    SeNBModificationRequestReject,
    SeNBModificationRequired,
    SeNBModificationConfirm,
    SeNBModificationRefuse,
    SeNBReleaseRequest,
    SeNBReleaseRequired,
    SeNBReleaseConfirm,
    SeNBCounterCheckRequest,
    X2RemovalFailure,
    X2RemovalRequest,
    X2RemovalResponse,
    RetrieveUEContextRequest,
    RetrieveUEContextResponse,
    RetrieveUEContextFailure,
    SgNBAdditionRequest,
    SgNBAdditionRequestAcknowledge,
    SgNBAdditionRequestReject,
    SgNBReconfigurationComplete,
    SgNBModificationRequest,
    SgNBModificationRequestAcknowledge,
    SgNBModificationRequestReject,
    SgNBModificationRequired,
    SgNBModificationConfirm,
    SgNBModificationRefuse,
    SgNBReleaseRequest,
    SgNBReleaseRequestAcknowledge,
    SgNBReleaseRequestReject,
    SgNBReleaseRequired,
    SgNBReleaseConfirm,
    SgNBCounterCheckRequest,
    SgNBChangeRequired,
    SgNBChangeConfirm,
    SgNBChangeRefuse,
    RRCTransfer,
    ENDCX2SetupRequest,
    ENDCX2SetupResponse,
    ENDCX2SetupFailure,
    ENDCConfigurationUpdate,
    ENDCConfigurationUpdateAcknowledge,
    ENDCConfigurationUpdateFailure,
    SecondaryRATDataUsageReport,
    ENDCCellActivationRequest,
    ENDCCellActivationResponse,
    ENDCCellActivationFailure,
    ENDCPartialResetRequired,
```

```
    ENDCPartialResetConfirm,
    EUTRANRCellResourceCoordinationRequest,
    EUTRANRCellResourceCoordinationResponse,
    SgNBActivityNotification,
    ENDCX2RemovalRequest,
    ENDCX2RemovalResponse,
    ENDCX2RemovalFailure,
    DataForwardingAddressIndication,
    GNBStatusIndication,
    ENDCConfigurationTransfer,
    DeactivateTrace,
    TraceStart,
    HandoverSuccess,
    EarlyStatusTransfer,
    ConditionalHandoverCancel,
    ENDCResourceStatusRequest,
    ENDCResourceStatusResponse,
    ENDCResourceStatusFailure,
    ENDCResourceStatusUpdate,
    CellTrafficTrace,
    F1CTrafficTransfer,
    UERadioCapabilityIDMappingRequest,
    UERadioCapabilityIDMappingResponse,
    AccessAndMobilityIndication,
    CPC-cancel



FROM X2AP-PDU-Contents

    id-cellActivation,
    id-eNBConfigurationUpdate,
    id-errorIndication,
    id-handoverCancel,
    id-handoverReport,
    id-handoverPreparation,

    id-loadIndication,
    id-privateMessage,
    id-reset,

    id-resourceStatusReporting,
    id-resourceStatusReportingInitiation,
    id-rLFIndication,
    id-snStatusTransfer,
    id-uEContextRelease,
    id-x2Setup,
    id-mobilitySettingsChange,
    id-x2Release,
    id-x2APMessageTransfer,
    id-seNBAdditionPreparation,
    id-seNBReconfigurationCompletion,
    id-meNBinitiatedSeNBModificationPreparation,
```

```
    id-seNBinitiatedSeNBModification,
    id-meNBinitiatedSeNBRelease,
    id-seNBinitiatedSeNBRelease,
    id-seNBCounterCheck,
    id-x2Removal,
    id-retrieveUEContext,
    id-sgNBAdditionPreparation,
    id-sgNBReconfigurationCompletion,
    id-meNBinitiatedSgNBModificationPreparation,
    id-sgNBinitiatedSgNBModification,
    id-meNBinitiatedSgNBRelease,
    id-sgNBinitiatedSgNBRelease,
    id-sgNBChange,
    id-sgNBCounterCheck,
    id-rRCTransfer,
    id-endcX2Setup,
    id-endcConfigurationUpdate,
    id-secondaryRATDataUsageReport,
    id-endcCellActivation,
    id-endcPartialReset,
    id-eUTRANRCellResourceCoordination,
    id-SgNBActivityNotification,
    id-endcX2Removal,
    id-dataForwardingAddressIndication,
    id-gNBStatusIndication,
    id-endcConfigurationTransfer,
    id-deactivateTrace,
    id-traceStart,
    id-handoverSuccess,
    id-earlyStatusTransfer,
    id-conditionalHandoverCancel,
    id-endcresourceStatusReporting,
    id-endcresourceStatusReportingInitiation,
    id-cellTrafficTrace,
    id-f1CTrafficTransfer,
    id-UERadioCapabilityIDMapping,
    id-accessAndMobilityIndication,
    id-CPC-cancel



FROM X2AP-Constants;

-- ************************************************************
--
-- Interface Elementary Procedure Class
--
-- ************************************************************

X2AP-ELEMENTARY-PROCEDURE ::= CLASS {
    &InitiatingMessage              ,
    &SuccessfulOutcome              OPTIONAL,
    &UnsuccessfulOutcome                OPTIONAL,
```

```
    &procedureCode          ProcedureCode   UNIQUE,
    &criticality            Criticality     DEFAULT ignore
}
WITH SYNTAX {
    INITIATING MESSAGE      &InitiatingMessage
    [SUCCESSFUL OUTCOME     &SuccessfulOutcome]
    [UNSUCCESSFUL OUTCOME       &UnsuccessfulOutcome]
    PROCEDURE CODE          &procedureCode
    [CRITICALITY            &criticality]
}

-- **************************************************************
--
-- Interface PDU Definition
--
-- **************************************************************

X2AP-PDU ::= CHOICE {
    initiatingMessage   InitiatingMessage,
    successfulOutcome   SuccessfulOutcome,
    unsuccessfulOutcome UnsuccessfulOutcome,
    ...
}

InitiatingMessage ::= SEQUENCE {
    procedureCode   X2AP-ELEMENTARY-PROCEDURE.&procedureCode        ({X2AP-ELEMENTARY-PROCEDURES}),
    criticality     X2AP-ELEMENTARY-PROCEDURE.&criticality          ({X2AP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value           X2AP-ELEMENTARY-PROCEDURE.&InitiatingMessage    ({X2AP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

SuccessfulOutcome ::= SEQUENCE {
    procedureCode   X2AP-ELEMENTARY-PROCEDURE.&procedureCode        ({X2AP-ELEMENTARY-PROCEDURES}),
    criticality     X2AP-ELEMENTARY-PROCEDURE.&criticality          ({X2AP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value           X2AP-ELEMENTARY-PROCEDURE.&SuccessfulOutcome    ({X2AP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

UnsuccessfulOutcome ::= SEQUENCE {
    procedureCode   X2AP-ELEMENTARY-PROCEDURE.&procedureCode        ({X2AP-ELEMENTARY-PROCEDURES}),
    criticality     X2AP-ELEMENTARY-PROCEDURE.&criticality          ({X2AP-ELEMENTARY-PROCEDURES}{@procedureCode}),
    value           X2AP-ELEMENTARY-PROCEDURE.&UnsuccessfulOutcome  ({X2AP-ELEMENTARY-PROCEDURES}{@procedureCode})
}

-- **************************************************************
--
-- Interface Elementary Procedure List
--
-- **************************************************************

X2AP-ELEMENTARY-PROCEDURES X2AP-ELEMENTARY-PROCEDURE ::= {
    X2AP-ELEMENTARY-PROCEDURES-CLASS-1          |
    X2AP-ELEMENTARY-PROCEDURES-CLASS-2          ,
    ...
}
```

```
X2AP-ELEMENTARY-PROCEDURES-CLASS-1 X2AP-ELEMENTARY-PROCEDURE ::= {
    handoverPreparation                                        |
    reset                                                      |
    x2Setup                                                    |
    resourceStatusReportingInitiation                          |
    eNBConfigurationUpdate                                     |
    mobilitySettingsChange                                     |
    cellActivation                                             |
    seNBAdditionPreparation                                    |
    meNBinitiatedSeNBModificationPreparation                   |
    seNBinitiatedSeNBModification                              |
    seNBinitiatedSeNBRelease                                   |
    x2Removal                                                  |
    retrieveUEContext                                          |
    sgNBAdditionPreparation                                    |
    meNBinitiatedSgNBModificationPreparation                       |
    sgNBinitiatedSgNBModification                              |
    meNBinitiatedSgNBRelease                                       |
    sgNBinitiatedSgNBRelease                                       |
    sgNBChange                                                 |
    endcX2Setup                                                |
    endcConfigurationUpdate                                    |
    endcCellActivation                                         |
    endcPartialReset                                               |
    eUTRANRCellResourceCoordination                            |
    endcX2Removal                                              |
    endcresourceStatusReportingInitiation                      |
    uERadioCapabilityIDMapping                                 ,
    ...
}

X2AP-ELEMENTARY-PROCEDURES-CLASS-2 X2AP-ELEMENTARY-PROCEDURE ::= {
    snStatusTransfer                        |
    uEContextRelease                        |
    handoverCancel                          |
    errorIndication                         |
    resourceStatusReporting                 |
    loadIndication                          |
    privateMessage                          |
    rLFIndication                           |
    handoverReport                          |
    x2Release                               |
    x2APMessageTransfer                     |
    seNBReconfigurationCompletion           |
    meNBinitiatedSeNBRelease                |
    seNBCounterCheck                        |
    sgNBReconfigurationCompletion           |
    sgNBCounterCheck                           |
    rRCTransfer                             |
    secondaryRATDataUsageReport             |
    sgNBActivityNotification                   |
    dataForwardingAddressIndication         |
    gNBStatusIndication                     |
    endcConfigurationTransfer               |
```

```
    deactivateTrace                         |
    traceStart                              |
    handoverSuccess                         |
    earlyStatusTransfer                     |
    conditionalHandoverCancel               |
    endcresourceStatusReporting             |
    cellTrafficTrace                        |
    f1CTrafficTransfer                      |
    accessAndMobilityIndication             |
    cPC-cancel                             ,
    ...

}

-- **************************************************************
--
-- Interface Elementary Procedures
--
-- **************************************************************

handoverPreparation X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      HandoverRequest
    SUCCESSFUL OUTCOME      HandoverRequestAcknowledge
    UNSUCCESSFUL OUTCOME    HandoverPreparationFailure
    PROCEDURE CODE          id-handoverPreparation
    CRITICALITY             reject
}

snStatusTransfer X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SNStatusTransfer
    PROCEDURE CODE          id-snStatusTransfer
    CRITICALITY             ignore
}

uEContextRelease X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UEContextRelease
    PROCEDURE CODE          id-uEContextRelease
    CRITICALITY             ignore
}


handoverCancel X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      HandoverCancel
    PROCEDURE CODE          id-handoverCancel
    CRITICALITY             ignore
}

handoverReport X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      HandoverReport
    PROCEDURE CODE          id-handoverReport
    CRITICALITY             ignore
}

errorIndication X2AP-ELEMENTARY-PROCEDURE ::= {
```

```
    INITIATING MESSAGE      ErrorIndication
    PROCEDURE CODE          id-errorIndication
    CRITICALITY             ignore
}

reset   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ResetRequest
    SUCCESSFUL OUTCOME      ResetResponse
    PROCEDURE CODE          id-reset
    CRITICALITY             reject
}

x2Setup X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      X2SetupRequest
    SUCCESSFUL OUTCOME      X2SetupResponse
    UNSUCCESSFUL OUTCOME    X2SetupFailure
    PROCEDURE CODE          id-x2Setup
    CRITICALITY             reject
}


loadIndication X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      LoadInformation
    PROCEDURE CODE          id-loadIndication
    CRITICALITY             ignore
}

eNBConfigurationUpdate      X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENBConfigurationUpdate
    SUCCESSFUL OUTCOME      ENBConfigurationUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    ENBConfigurationUpdateFailure
    PROCEDURE CODE          id-eNBConfigurationUpdate
    CRITICALITY             reject
}

resourceStatusReportingInitiation   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE              ResourceStatusRequest
    SUCCESSFUL OUTCOME              ResourceStatusResponse
    UNSUCCESSFUL OUTCOME            ResourceStatusFailure
    PROCEDURE CODE                  id-resourceStatusReportingInitiation
    CRITICALITY                     reject
}

resourceStatusReporting X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ResourceStatusUpdate
    PROCEDURE CODE          id-resourceStatusReporting
    CRITICALITY             ignore
}

rLFIndication X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RLFIndication
    PROCEDURE CODE          id-rLFIndication
    CRITICALITY             ignore
}
```

```
privateMessage          X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      PrivateMessage
    PROCEDURE CODE          id-privateMessage
    CRITICALITY             ignore
}

mobilitySettingsChange  X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      MobilityChangeRequest
    SUCCESSFUL OUTCOME      MobilityChangeAcknowledge
    UNSUCCESSFUL OUTCOME    MobilityChangeFailure
    PROCEDURE CODE          id-mobilitySettingsChange
    CRITICALITY             reject
}

cellActivation  X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CellActivationRequest
    SUCCESSFUL OUTCOME      CellActivationResponse
    UNSUCCESSFUL OUTCOME    CellActivationFailure
    PROCEDURE CODE          id-cellActivation
    CRITICALITY             reject
}

x2Release X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      X2Release
    PROCEDURE CODE          id-x2Release
    CRITICALITY             reject
}

x2APMessageTransfer X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      X2APMessageTransfer
    PROCEDURE CODE          id-x2APMessageTransfer
    CRITICALITY             reject
}

seNBAdditionPreparation X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SeNBAdditionRequest
    SUCCESSFUL OUTCOME      SeNBAdditionRequestAcknowledge
    UNSUCCESSFUL OUTCOME    SeNBAdditionRequestReject
    PROCEDURE CODE          id-seNBAdditionPreparation
    CRITICALITY             reject
}

seNBReconfigurationCompletion   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SeNBReconfigurationComplete
    PROCEDURE CODE          id-seNBReconfigurationCompletion
    CRITICALITY             ignore
}

meNBinitiatedSeNBModificationPreparation    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SeNBModificationRequest
    SUCCESSFUL OUTCOME      SeNBModificationRequestAcknowledge
    UNSUCCESSFUL OUTCOME    SeNBModificationRequestReject
    PROCEDURE CODE          id-meNBinitiatedSeNBModificationPreparation
```

```
    CRITICALITY             reject
}

seNBinitiatedSeNBModification   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SeNBModificationRequired
    SUCCESSFUL OUTCOME      SeNBModificationConfirm
    UNSUCCESSFUL OUTCOME    SeNBModificationRefuse
    PROCEDURE CODE          id-seNBinitiatedSeNBModification
    CRITICALITY             reject
}

meNBinitiatedSeNBRelease    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SeNBReleaseRequest
    PROCEDURE CODE          id-meNBinitiatedSeNBRelease
    CRITICALITY             ignore
}

seNBinitiatedSeNBRelease    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SeNBReleaseRequired
    SUCCESSFUL OUTCOME      SeNBReleaseConfirm
    PROCEDURE CODE          id-seNBinitiatedSeNBRelease
    CRITICALITY             reject
}

seNBCounterCheck    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SeNBCounterCheckRequest
    PROCEDURE CODE          id-seNBCounterCheck
    CRITICALITY             reject
}

x2Removal   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      X2RemovalRequest
    SUCCESSFUL OUTCOME      X2RemovalResponse
    UNSUCCESSFUL OUTCOME    X2RemovalFailure
    PROCEDURE CODE          id-x2Removal
    CRITICALITY             reject
}

retrieveUEContext   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RetrieveUEContextRequest
    SUCCESSFUL OUTCOME      RetrieveUEContextResponse
    UNSUCCESSFUL OUTCOME    RetrieveUEContextFailure
    PROCEDURE CODE          id-retrieveUEContext
    CRITICALITY             reject
}

sgNBAdditionPreparation     X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBAdditionRequest
    SUCCESSFUL OUTCOME      SgNBAdditionRequestAcknowledge
    UNSUCCESSFUL OUTCOME    SgNBAdditionRequestReject
    PROCEDURE CODE          id-sgNBAdditionPreparation
    CRITICALITY             reject
}
```

```
sgNBReconfigurationCompletion   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBReconfigurationComplete
    PROCEDURE CODE          id-sgNBReconfigurationCompletion
    CRITICALITY             ignore
}

meNBinitiatedSgNBModificationPreparation    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBModificationRequest
    SUCCESSFUL OUTCOME      SgNBModificationRequestAcknowledge
    UNSUCCESSFUL OUTCOME    SgNBModificationRequestReject
    PROCEDURE CODE          id-meNBinitiatedSgNBModificationPreparation
    CRITICALITY             reject
}

sgNBinitiatedSgNBModification       X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBModificationRequired
    SUCCESSFUL OUTCOME      SgNBModificationConfirm
    UNSUCCESSFUL OUTCOME    SgNBModificationRefuse
    PROCEDURE CODE          id-sgNBinitiatedSgNBModification
    CRITICALITY             reject
}

meNBinitiatedSgNBRelease    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBReleaseRequest
    SUCCESSFUL OUTCOME      SgNBReleaseRequestAcknowledge
    UNSUCCESSFUL OUTCOME    SgNBReleaseRequestReject
    PROCEDURE CODE          id-meNBinitiatedSgNBRelease
    CRITICALITY             ignore
}

sgNBinitiatedSgNBRelease    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBReleaseRequired
    SUCCESSFUL OUTCOME      SgNBReleaseConfirm
    PROCEDURE CODE          id-sgNBinitiatedSgNBRelease
    CRITICALITY             reject
}

sgNBCounterCheck    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBCounterCheckRequest
    PROCEDURE CODE          id-sgNBCounterCheck
    CRITICALITY             reject
}

sgNBChange  X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBChangeRequired
    SUCCESSFUL OUTCOME      SgNBChangeConfirm
    UNSUCCESSFUL OUTCOME    SgNBChangeRefuse
    PROCEDURE CODE          id-sgNBChange
    CRITICALITY             reject
}

rRCTransfer X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RRCTransfer
    PROCEDURE CODE          id-rRCTransfer
```

```
    CRITICALITY             reject
}

endcX2Setup X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCX2SetupRequest
    SUCCESSFUL OUTCOME      ENDCX2SetupResponse
    UNSUCCESSFUL OUTCOME    ENDCX2SetupFailure
    PROCEDURE CODE          id-endcX2Setup
    CRITICALITY             reject
}


endcConfigurationUpdate     X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCConfigurationUpdate
    SUCCESSFUL OUTCOME      ENDCConfigurationUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    ENDCConfigurationUpdateFailure
    PROCEDURE CODE          id-endcConfigurationUpdate
    CRITICALITY             reject
}

secondaryRATDataUsageReport X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SecondaryRATDataUsageReport
    PROCEDURE CODE          id-secondaryRATDataUsageReport
    CRITICALITY             reject
}

endcCellActivation  X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCCellActivationRequest
    SUCCESSFUL OUTCOME      ENDCCellActivationResponse
    UNSUCCESSFUL OUTCOME    ENDCCellActivationFailure
    PROCEDURE CODE          id-endcCellActivation
    CRITICALITY             reject
}

endcPartialReset    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCPartialResetRequired
    SUCCESSFUL OUTCOME      ENDCPartialResetConfirm
    PROCEDURE CODE          id-endcPartialReset
    CRITICALITY             reject
}

eUTRANRCellResourceCoordination X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      EUTRANRCellResourceCoordinationRequest
    SUCCESSFUL OUTCOME      EUTRANRCellResourceCoordinationResponse
    PROCEDURE CODE          id-eUTRANRCellResourceCoordination
    CRITICALITY             reject
}

sgNBActivityNotification    X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      SgNBActivityNotification
    PROCEDURE CODE          id-SgNBActivityNotification
    CRITICALITY             reject
}
```

```
endcX2Removal   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCX2RemovalRequest
    SUCCESSFUL OUTCOME      ENDCX2RemovalResponse
    UNSUCCESSFUL OUTCOME    ENDCX2RemovalFailure
    PROCEDURE CODE          id-endcX2Removal
    CRITICALITY             reject
}

dataForwardingAddressIndication X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DataForwardingAddressIndication
    PROCEDURE CODE          id-dataForwardingAddressIndication
    CRITICALITY             ignore
}

gNBStatusIndication     X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      GNBStatusIndication
    PROCEDURE CODE          id-gNBStatusIndication
    CRITICALITY             ignore
}

endcConfigurationTransfer   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCConfigurationTransfer
    PROCEDURE CODE          id-endcConfigurationTransfer
    CRITICALITY             ignore
}

deactivateTrace X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      DeactivateTrace
    PROCEDURE CODE          id-deactivateTrace
    CRITICALITY             ignore
}

traceStart X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      TraceStart
    PROCEDURE CODE          id-traceStart
    CRITICALITY             ignore
}

handoverSuccess X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      HandoverSuccess
    PROCEDURE CODE          id-handoverSuccess
    CRITICALITY             ignore
}

earlyStatusTransfer     X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      EarlyStatusTransfer
    PROCEDURE CODE          id-earlyStatusTransfer
    CRITICALITY             ignore
}

conditionalHandoverCancel X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ConditionalHandoverCancel
    PROCEDURE CODE          id-conditionalHandoverCancel
```

```
    CRITICALITY             ignore
}

endcresourceStatusReportingInitiation   X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCResourceStatusRequest
    SUCCESSFUL OUTCOME      ENDCResourceStatusResponse
    UNSUCCESSFUL OUTCOME    ENDCResourceStatusFailure
    PROCEDURE CODE          id-endcresourceStatusReportingInitiation
    CRITICALITY             reject
}

endcresourceStatusReporting     X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ENDCResourceStatusUpdate
    PROCEDURE CODE          id-endcresourceStatusReporting
    CRITICALITY             ignore
}

cellTrafficTrace X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CellTrafficTrace
    PROCEDURE CODE          id-cellTrafficTrace
    CRITICALITY             ignore
}

f1CTrafficTransfer          X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      F1CTrafficTransfer
    PROCEDURE CODE          id-f1CTrafficTransfer
    CRITICALITY             ignore
}

uERadioCapabilityIDMapping X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      UERadioCapabilityIDMappingRequest
    SUCCESSFUL OUTCOME      UERadioCapabilityIDMappingResponse
    PROCEDURE CODE          id-UERadioCapabilityIDMapping
    CRITICALITY             reject
}

accessAndMobilityIndication X2AP-ELEMENTARY-PROCEDURE ::={
    INITIATING MESSAGE      AccessAndMobilityIndication
    PROCEDURE CODE          id-accessAndMobilityIndication
    CRITICALITY                 ignore
}


cPC-cancel X2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      CPC-cancel
    PROCEDURE CODE          id-CPC-cancel
    CRITICALITY             ignore
}


END
-- ASN1STOP
```

## 9.3.4 PDU Definitions

```
-- ASN1START
-- **************************************************************
--
-- PDU definitions for X2AP.
--
-- **************************************************************

X2AP-PDU-Contents {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
eps-Access (21) modules (3) x2ap (2) version1 (1) x2ap-PDU-Contents (1) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- IE parameter types from other modules.
--
-- **************************************************************

IMPORTS
    ABSInformation,
    ABS-Status,
    AS-SecurityInformation,
    BearerType,
    Cause,
    CompositeAvailableCapacityGroup,
    Correlation-ID,
    COUNTvalue,
    CellReportingIndicator,
    AerialUEsubscriptionInformation,
    CriticalityDiagnostics,
    CRNTI,
    CSGMembershipStatus,
    CSG-Id,
    DeactivationIndication,
    DL-Forwarding,
    DynamicDLTransmissionInformation,
    E-RABsSubjectToDLDiscarding-List,
    E-RABsSubjectToEarlyStatusTransfer-List,
    ECGI,
    E-RAB-ID,
    E-RAB-Level-QoS-Parameters,
    E-RAB-List,
    EUTRANTraceID,
    GlobalENB-ID,
    GTPtunnelEndpoint,
    GUGroupIDList,
    GUMMEI,
    HandoverReportType,
```

```
    HandoverRestrictionList,
    Masked-IMEISV,
    InvokeIndication,
    LocationReportingInformation,
    LowerLayerPresenceStatusChange,
    MDT-Configuration,
    ManagementBasedMDTallowed,
    MDTPLMNList,
    Neighbour-Information,
    PCI,
    PDCP-SN,
    PLMN-Identity,
    ReceiveStatusofULPDCPSDUs,
    Registration-Request,
    RelativeNarrowbandTxPower,
    RadioResourceStatus,
    RLC-Status,
    RRCConnReestabIndicator,
    RRCConnSetupIndicator,
    UE-RLF-Report-Container,
    UEAppLayerMeasConfig,
    RRC-Context,
    ServedCell-Information,
    ServedCells,
    ShortMAC-I,
    SRVCCOperationPossible,
    SubscriberProfileIDforRFP,
    TargetCellInUTRAN,
    TargeteNBtoSource-eNBTransparentContainer,
    TimeToWait,
    TraceActivation,
    TraceDepth,
    TransportLayerAddress,
    UEAggregateMaximumBitRate,
    UE-HistoryInformation,
    UE-HistoryInformationFromTheUE,
    UE-S1AP-ID,
    UESecurityCapabilities,
    UEsToBeResetList,
    UE-X2AP-ID,
    UL-HighInterferenceIndicationInfo,
    UL-InterferenceOverloadIndication,
    HWLoadIndicator,
    S1TNLLoadIndicator,
    Measurement-ID,
    ReportCharacteristics,
    MobilityParametersInformation,
    MobilityParametersModificationRange,
    ReceiveStatusOfULPDCPSDUsExtended,
    COUNTValueExtended,
    SubframeAssignment,
    ExtendedULInterferenceOverloadInfo,
    ExpectedUEBehaviour,
    SeNBSecurityKey,
```

```
    MeNBtoSeNBContainer,
    SeNBtoMeNBContainer,
    SCGChangeIndication,
    CoMPInformation,
    ReportingPeriodicityRSRPMR,
    RSRPMRList,
    UE-RLF-Report-Container-for-extended-bands,
    ProSeAuthorized,
    CoverageModificationList,
    ReportingPeriodicityCSIR,
    CSIReportList,
    ReceiveStatusOfULPDCPSDUsPDCP-SNlength18,
    COUNTvaluePDCP-SNlength18,
    LHN-ID,
    UE-ContextKeptIndicator,
    UE-X2AP-ID-Extension,
    SIPTOBearerDeactivationIndication,
    TunnelInformation,
    V2XServicesAuthorized,
    X2BenefitValue,
    ResumeID,
    EUTRANCellIdentifier,
    MakeBeforeBreakIndicator,
    WTID,
    WT-UE-XwAP-ID,
    UESidelinkAggregateMaximumBitRate,
    SgNBSecurityKey,
    MeNBtoSgNBContainer,
    SgNBtoMeNBContainer,
    SplitSRBs,
    RRCContainer,
    SRBType,
    GlobalGNB-ID,
    GNB-ID,
    SCGConfigurationQuery,
    SplitSRB,
    NRUeReport,
    EN-DC-ResourceConfiguration,
    TAC,
    NRFreqInfo,
    NRCGI,
    NRPCI,
    NRUESecurityCapabilities,
    PDCPChangeIndication,
    ULConfiguration,
    SgNB-UE-X2AP-ID,
    SecondaryRATUsageReportList,
    ActivationID,
    MeNBResourceCoordinationInformation,
    SgNBResourceCoordinationInformation,
    NR-TxBW,
    BroadcastPLMNs-Item,
    AdditionalPLMNs-Item,
    RLCMode,
```

```
    GBR-QosInformation,
    DRB-ID,
    FiveGS-TAC,
    SULInformation,
    Packet-LossRate,
    ResourceType,
    DataTrafficResourceIndication,
    SpectrumSharingGroupID,
    RRC-Config-Ind,
    SGNB-Addition-Trigger-Ind,
    UserPlaneTrafficActivityReport,
    ERABActivityNotifyItemList,
    PDCPSnLength,
    Subscription-Based-UE-DifferentiationInfo,
    LCID,
    DuplicationActivation,
    GNBOverloadInformation,
    NewDRBIDrequest,
    DesiredActNotificationLevel,
    LocationInformationSgNB,
    LocationInformationSgNBReporting,
    EndcSONConfigurationTransfer,
    NRNeighbour-Information,
    InterfaceInstanceIndication,
    BPLMN-ID-Info-NR,
    SNtriggered,
    EPCHandoverRestrictionListContainer,
    AdditionalRRMPriorityIndex,
    RequestedFastMCGRecoveryViaSRB3,
    AvailableFastMCGRecoveryViaSRB3,
    RequestedFastMCGRecoveryViaSRB3Release,
    ReleaseFastMCGRecoveryViaSRB3,
    FastMCGRecovery,
    PartialListIndicator,
    MaximumCellListSize,
    MessageOversizeNotification,
    TNLConfigurationInfo,
    TNLA-To-Add-List,
    TNLA-To-Update-List,
    TNLA-To-Remove-List,
    TNLA-Setup-List,
    TNLA-Failed-To-Setup-List,
    RAN-UE-NGAP-ID,
    CHOinformation-REQ,
    CHOinformation-ACK,
    CHOinformation-AddReq,
    CHOinformation-ModReq,
    DAPSRequestInfo,
    DAPSResponseInfo,
    CandidateCellsToBeCancelledList,
    CHO-DC-EarlyDataForwarding,
    CHO-DC-Indicator,
    Ethernet-Type,
    NRV2XServicesAuthorized,
```

```
    NRUESidelinkAggregateMaximumBitRate,
    PC5QoSParameters,
    TargetCellInNGRAN,
    Measurement-ID-ENDC,
    Registration-Request-ENDC,
    ReportCharacteristics-ENDC,
    NRRadioResourceStatus,
    TNLCapacityIndicator,
    NRCompositeAvailableCapacityGroup,
    SSBIndex,
    TDDULDLConfigurationCommonNR,
    NRCarrierList,
    SSB-PositionsInBurst,
    NRCellPRACHConfig,
    NBIoT-RLF-Report-Container,
    PrivacyIndicator,
    UERadioCapabilityID,
    CSI-RSTransmissionIndication,
    IABNodeIndication,
    F1CTrafficContainer,
    IntendedTDD-DL-ULConfiguration-NR,
    UERadioCapability,
    SFN-Offset,
    IMSvoiceEPSfallbackfrom5G,
    Global-RAN-NODE-ID,
    DirectForwardingPathAvailability,
    NRRACHReportInformation,
    SCG-UE-HistoryInformation,
    PSCellHistoryInformationRetrieve,

    PSCell-UE-HistoryInformation,
    PSCellChangeHistory,
    MeasurementResultforNRCellsPossiblyAggregated,
    SCGActivationStatus,
    SCGActivationRequest,
    CPAinformation-REQ,
    CPAinformation-REQ-ACK,
    CPAinformation-MOD,
    CPAinformation-MOD-ACK,
    CPACinformation-REQD,
    CPCinformation-REQD,
    CPCinformation-CONF,
    CPCinformation-NOTIFY,
    CPCupdate-MOD,
    Additional-Measurement-Timing-Configuration-List,
    ServedCellSpecificInfoReq-NR,
    SecurityIndication,
    SecurityResult,
    TraceCollectionEntityIPAddress,
    SCGreconfigNotification,
    AdditionalListofForwardingGTPTunnelEndpoint
```

```
FROM X2AP-IEs

    PrivateIE-Container{},
    ProtocolExtensionContainer{},
    ProtocolIE-Container{},
    ProtocolIE-ContainerList{},
    ProtocolIE-ContainerPair{},
    ProtocolIE-ContainerPairList{},
    ProtocolIE-Single-Container{},
    X2AP-PRIVATE-IES,
    X2AP-PROTOCOL-EXTENSION,
    X2AP-PROTOCOL-IES,
    X2AP-PROTOCOL-IES-PAIR
FROM X2AP-Containers

    id-ABSInformation,
    id-ActivatedCellList,
    id-BearerType,
    id-Cause,
    id-CellInformation,
    id-CellInformation-Item,
    id-CellMeasurementResult,
    id-CellMeasurementResult-NR-ENDC,
    id-CellMeasurementResult-Item,
    id-CellMeasurementResult-NR-ENDC-Item,
    id-CellMeasurementResult-E-UTRA-ENDC,
    id-CellMeasurementResult-E-UTRA-ENDC-Item,
    id-CellToReport,
    id-CellToReport-E-UTRA-ENDC,
    id-CellToReport-NR-ENDC,
    id-CellToReport-Item,
    id-CellToReport-E-UTRA-ENDC-Item,
    id-CellToReport-NR-ENDC-Item,
    id-CompositeAvailableCapacityGroup,
    id-AerialUEsubscriptionInformation,
    id-CriticalityDiagnostics,
    id-DeactivationIndication,
    id-DynamicDLTransmissionInformation,
    id-E-RABs-Admitted-Item,
    id-E-RABs-Admitted-List,
    id-E-RABs-NotAdmitted-List,
    id-E-RABs-SubjectToStatusTransfer-List,
    id-E-RABs-SubjectToStatusTransfer-Item,
    id-E-RABs-ToBeSetup-Item,
    id-GlobalENB-ID,
    id-GUGroupIDList,
    id-GUGroupIDToAddList,
    id-GUGroupIDToDeleteList,
    id-GUMMEI-ID,
    id-Masked-IMEISV,
    id-IMSvoiceEPSfallbackfrom5G,
    id-InvokeIndication,
    id-New-eNB-UE-X2AP-ID,
```

```
    id-Old-eNB-UE-X2AP-ID,
    id-Registration-Request,
    id-ReportingPeriodicity,
    id-RLC-Status,
    id-ServedCells,
    id-ServedCellsToActivate,
    id-ServedCellsToAdd,
    id-ServedCellsToModify,
    id-ServedCellsToDelete,
    id-SRVCCOperationPossible,
    id-TargetCell-ID,
    id-TargeteNBtoSource-eNBTransparentContainer,
    id-TimeToWait,
    id-TraceActivation,
    id-UE-ContextInformation,
    id-UE-HistoryInformation,
    id-UE-X2AP-ID,
    id-Measurement-ID,
    id-ReportCharacteristics,
    id-ENB1-Measurement-ID,
    id-ENB2-Measurement-ID,
    id-ENB1-Cell-ID,
    id-ENB2-Cell-ID,
    id-ENB2-Proposed-Mobility-Parameters,
    id-ENB1-Mobility-Parameters,
    id-ENB2-Mobility-Parameters-Modification-Range,
    id-FailureCellPCI,
    id-Re-establishmentCellECGI,
    id-FailureCellCRNTI,
    id-ShortMAC-I,
    id-SourceCellECGI,
    id-FailureCellECGI,
    id-HandoverReportType,
    id-UE-RLF-Report-Container,
    id-PartialSuccessIndicator,
    id-MeasurementInitiationResult-List,
    id-MeasurementInitiationResult-Item,
    id-MeasurementFailureCause-Item,
    id-CompleteFailureCauseInformation-List,
    id-CompleteFailureCauseInformation-Item,
    id-CSGMembershipStatus,
    id-CSG-Id,
    id-MDTConfiguration,
    id-ManagementBasedMDTallowed,
    id-ABS-Status,
    id-RRCConnSetupIndicator,
    id-RRCConnReestabIndicator,
    id-TargetCellInUTRAN,
    id-MobilityInformation,
    id-SourceCellCRNTI,
    id-ManagementBasedMDTPLMNList,
    id-ReceiveStatusOfULPDCPSDUsExtended,
    id-ULCOUNTValueExtended,
    id-DLCOUNTValueExtended,
```

```
    id-IntendedULDLConfiguration,
    id-ExtendedULInterferenceOverloadInfo,
    id-RNL-Header,
    id-x2APMessage,
    id-UE-HistoryInformationFromTheUE,
    id-ExpectedUEBehaviour,
    id-MeNB-UE-X2AP-ID,
    id-SeNB-UE-X2AP-ID,
    id-UE-SecurityCapabilities,
    id-SeNBSecurityKey,
    id-SeNBUEAggregateMaximumBitRate,
    id-ServingPLMN,
    id-E-RABs-ToBeAdded-List,
    id-E-RABs-ToBeAdded-Item,
    id-MeNBtoSeNBContainer,
    id-E-RABs-Admitted-ToBeAdded-List,
    id-E-RABs-Admitted-ToBeAdded-Item,
    id-SeNBtoMeNBContainer,
    id-ResponseInformationSeNBReconfComp,
    id-UE-ContextInformationSeNBModReq,
    id-E-RABs-ToBeAdded-ModReqItem,
    id-E-RABs-ToBeModified-ModReqItem,
    id-E-RABs-ToBeReleased-ModReqItem,
    id-E-RABs-Admitted-ToBeAdded-ModAckList,
    id-E-RABs-Admitted-ToBeModified-ModAckList,
    id-E-RABs-Admitted-ToBeReleased-ModAckList,
    id-E-RABs-Admitted-ToBeAdded-ModAckItem,
    id-E-RABs-Admitted-ToBeModified-ModAckItem,
    id-E-RABs-Admitted-ToBeReleased-ModAckItem,
    id-SCGChangeIndication,
    id-E-RABs-ToBeReleased-ModReqd,
    id-E-RABs-ToBeReleased-ModReqdItem,
    id-E-RABs-ToBeReleased-List-RelReq,
    id-E-RABs-ToBeReleased-RelReqItem,
    id-E-RABs-ToBeReleased-List-RelConf,
    id-E-RABs-ToBeReleased-RelConfItem,
    id-E-RABs-SubjectToCounterCheck-List,
    id-E-RABs-SubjectToCounterCheckItem,
    id-CoMPInformation,
    id-ReportingPeriodicityRSRPMR,
    id-RSRPMRList,
    id-UE-RLF-Report-Container-for-extended-bands,
    id-ProSeAuthorized,
    id-CoverageModificationList,
    id-ReportingPeriodicityCSIR,
    id-CSIReportList,
    id-ReceiveStatusOfULPDCPSDUsPDCP-SNlength18,
    id-ULCOUNTValuePDCP-SNlength18,
    id-DLCOUNTValuePDCP-SNlength18,
    id-LHN-ID,
    id-Correlation-ID,
    id-SIPTO-Correlation-ID,
    id-UE-ContextReferenceAtSeNB,
    id-UE-ContextReferenceAtWT,
```

```
    id-UE-ContextKeptIndicator,
    id-UEs-ToBeReset,
    id-UEs-Admitted-ToBeReset,
    id-WT-UE-ContextKeptIndicator,
    id-New-eNB-UE-X2AP-ID-Extension,
    id-Old-eNB-UE-X2AP-ID-Extension,
    id-MeNB-UE-X2AP-ID-Extension,
    id-SeNB-UE-X2AP-ID-Extension,
    id-SIPTO-BearerDeactivationIndication,
    id-Tunnel-Information-for-BBF,
    id-SIPTO-L-GW-TransportLayerAddress,
    id-GW-TransportLayerAddress,
    id-X2RemovalThreshold,
    id-CellReportingIndicator,
    id-V2XServicesAuthorized,
    id-resumeID,
    id-UE-ContextInformationRetrieve,
    id-E-RABs-ToBeSetupRetrieve-Item,
    id-NewEUTRANCellIdentifier,
    id-MakeBeforeBreakIndicator,
    id-UESidelinkAggregateMaximumBitRate,
    id-uL-GTPtunnelEndpoint,
    id-SgNBSecurityKey,
    id-SgNBUEAggregateMaximumBitRate,
    id-E-RABs-ToBeAdded-SgNBAddReqList,
    id-MeNBtoSgNBContainer,
    id-SgNB-UE-X2AP-ID,
    id-RequestedSplitSRBs,
    id-E-RABs-ToBeAdded-SgNBAddReq-Item,
    id-E-RABs-Admitted-ToBeAdded-SgNBAddReqAckList,
    id-SgNBtoMeNBContainer,
    id-AdmittedSplitSRBs,
    id-E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item,
    id-ResponseInformationSgNBReconfComp,
    id-UE-ContextInformation-SgNBModReq,
    id-E-RABs-ToBeAdded-SgNBModReq-Item,
    id-E-RABs-ToBeModified-SgNBModReq-Item,
    id-E-RABs-ToBeReleased-SgNBModReq-Item,
    id-E-RABs-Admitted-ToBeAdded-SgNBModAckList,
    id-E-RABs-Admitted-ToBeModified-SgNBModAckList,
    id-E-RABs-Admitted-ToBeReleased-SgNBModAckList,
    id-E-RABs-Admitted-ToBeAdded-SgNBModAck-Item,
    id-E-RABs-Admitted-ToBeModified-SgNBModAck-Item,
    id-E-RABs-Admitted-ToBeReleased-SgNBModAck-Item,
    id-E-RABs-Admitted-ToBeReleased-SgNBRelReqAckList,
    id-E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-Item,
    id-E-RABs-ToBeReleased-SgNBModReqdList,
    id-E-RABs-ToBeModified-SgNBModReqdList,
    id-E-RABs-ToBeReleased-SgNBModReqd-Item,
    id-E-RABs-ToBeModified-SgNBModReqd-Item,
    id-E-RABs-ToBeReleased-SgNBChaConfList,
    id-E-RABs-ToBeReleased-SgNBChaConf-Item,
    id-E-RABs-ToBeReleased-SgNBRelReqList,
    id-E-RABs-ToBeReleased-SgNBRelReq-Item,
```

```
    id-E-RABs-ToBeReleased-SgNBRelConfList,
    id-E-RABs-ToBeReleased-SgNBRelConf-Item,
    id-E-RABs-ToBeReleased-SgNBRelReqdList,
    id-E-RABs-ToBeReleased-SgNBRelReqd-Item,
    id-E-RABs-SubjectToSgNBCounterCheck-List,
    id-E-RABs-SubjectToSgNBCounterCheck-Item,
    id-Target-SgNB-ID,
    id-RRCContainer,
    id-SRBType,
    id-HandoverRestrictionList,
    id-SCGConfigurationQuery,
    id-SplitSRB,
    id-NRUeReport,
    id-InitiatingNodeType-EndcX2Setup,
    id-InitiatingNodeType-EndcConfigUpdate,
    id-RespondingNodeType-EndcX2Setup,
    id-RespondingNodeType-EndcConfigUpdate,
    id-NRUESecurityCapabilities,
    id-PDCPChangeIndication,
    id-ServedEUTRAcellsENDCX2ManagementList,
    id-ServedEUTRAcellsToModifyListENDCConfUpd,
    id-ServedEUTRAcellsToDeleteListENDCConfUpd,
    id-ServedNRcellsToModifyListENDCConfUpd,
    id-ServedNRcellsToDeleteListENDCConfUpd,
    id-CellAssistanceInformation,
    id-Globalen-gNB-ID,
    id-ServedNRcellsENDCX2ManagementList,
    id-Old-SgNB-UE-X2AP-ID,
    id-UE-ContextReferenceAtSgNB,
    id-SecondaryRATUsageReportList,
    id-ActivationID,
    id-ServedNRCellsToActivate,
    id-ActivatedNRCellList,
    id-MeNBResourceCoordinationInformation,
    id-SgNBResourceCoordinationInformation,
    id-UEAppLayerMeasConfig,
    id-SelectedPLMN,
    id-SubscriberProfileIDforRFP,
    id-InitiatingNodeType-EutranrCellResourceCoordination,
    id-RespondingNodeType-EutranrCellResourceCoordination,
    id-DataTrafficResourceIndication,
    id-SpectrumSharingGroupID,
    id-ListofEUTRACellsinEUTRACoordinationReq,
    id-ListofEUTRACellsinEUTRACoordinationResp,
    id-ListofEUTRACellsinNRCoordinationReq,
    id-ListofNRCellsinNRCoordinationReq,
    id-ListofNRCellsinNRCoordinationResp,
    id-RRCConfigIndication,
    id-SGNB-Addition-Trigger-Ind,
    id-RequestedSplitSRBsrelease,
    id-AdmittedSplitSRBsrelease,
    id-E-RABs-AdmittedToBeModified-SgNBModConfList,
    id-E-RABs-AdmittedToBeModified-SgNBModConf-Item,
    id-UEContextLevelUserPlaneActivity,
```

```
    id-ERABActivityNotifyItemList,
    id-MeNBCell-ID,
    id-InitiatingNodeType-EndcX2Removal,
    id-RespondingNodeType-EndcX2Removal,
    id-uLpDCPSnLength,
    id-dL-Forwarding,
    id-E-RABs-DataForwardingAddress-List,
    id-E-RABs-DataForwardingAddress-Item,
    id-Subscription-Based-UE-DifferentiationInfo,
    id-RLCMode-transferred,
    id-dLPDCPSnLength,
    id-secondarysgNBDLGTPTEIDatPDCP,
    id-secondarymeNBULGTPTEIDatPDCP,
    id-lCID,
    id-duplicationActivation,
    id-GNBOverloadInformation,
    id-new-drb-ID-req,
    id-NRNeighbourInfoToModify,
    id-DesiredActNotificationLevel,
    id-LocationInformationSgNB,
    id-LocationInformationSgNBReporting,
    id-endcSONConfigurationTransfer,
    id-EUTRANTraceID,
    id-additionalPLMNs-Item,
    id-InterfaceInstanceIndication,
    id-BPLMN-ID-Info-NR,
    id-SNtriggered,
    id-EPCHandoverRestrictionListContainer,
    id-ERABs-transferred-to-MeNB,
    id-AdditionalRRMPriorityIndex,
    id-LowerLayerPresenceStatusChange,
    id-FastMCGRecovery-SN-to-MN,
    id-FastMCGRecovery-MN-to-SN,
    id-RequestedFastMCGRecoveryViaSRB3,
    id-AvailableFastMCGRecoveryViaSRB3,
    id-RequestedFastMCGRecoveryViaSRB3Release,
    id-ReleaseFastMCGRecoveryViaSRB3,
    id-PartialListIndicator,
    id-MaximumCellListSize,
    id-MessageOversizeNotification,
    id-CellandCapacityAssistInfo,
    id-TNLConfigurationInfo,
    id-TNLA-To-Add-List,
    id-TNLA-To-Update-List,
    id-TNLA-To-Remove-List,
    id-TNLA-Setup-List,
    id-TNLA-Failed-To-Setup-List,
    id-UEContextReferenceatSourceNGRAN,
    id-CHOinformation-REQ,
    id-CHOinformation-ACK,
    id-CHOinformation-AddReq,
    id-CHOinformation-ModReq,
    id-DAPSRequestInfo,
    id-RequestedTargetCellID,
```

```
    id-CandidateCellsToBeCancelledList,
    id-DAPSResponseInfo,
    id-ProcedureStage,
    id-CHO-DC-EarlyDataForwarding,
    id-CHO-DC-Indicator,
    id-Ethernet-Type,
    id-NRV2XServicesAuthorized,
    id-NRUESidelinkAggregateMaximumBitRate,
    id-PC5QoSParameters,
    id-TargetCellInNGRAN,
    id-E-UTRAN-Node1-Measurement-ID,
    id-E-UTRAN-Node2-Measurement-ID,
    id-TDDULDLConfigurationCommonNR,
    id-CarrierList,
    id-ULCarrierList,
    id-SSB-PositionsInBurst,
    id-NRCellPRACHConfig,
    id-NBIoT-RLF-Report-Container,
    id-MDTConfigurationNR,
    id-PrivacyIndicator,
    id-TraceCollectionEntityIPAddress,
    id-UERadioCapabilityID,
    id-CSI-RSTransmissionIndication,
    id-DLCarrierList,
    id-IABNodeIndication,
    id-F1CTrafficContainer,
    id-IntendedTDD-DL-ULConfiguration-NR,
    id-UERadioCapability,
    id-SFN-Offset,
    id-DirectForwardingPathAvailability,
    id-sourceNG-RAN-node-id,
    id-SourceDLForwardingIPAddress,
    id-SourceNodeDLForwardingIPAddress,
    id-NRRACHReportInformation,
    id-SCG-UE-HistoryInformation,
    id-PSCellHistoryInformationRetrieve,
    id-MeasurementResultforNRCellsPossiblyAggregated,
    id-PSCellChangeHistory,
    id-SCGActivationStatus,
    id-SCGActivationRequest,
    id-CPAinformation-REQ,
    id-CPAinformation-REQ-ACK,
    id-CPAinformation-MOD,
    id-CPAinformation-MOD-ACK,
    id-CPACinformation-REQD,
    id-CPCinformation-REQD,
    id-CPCinformation-CONF,
    id-CPCinformation-NOTIFY,
    id-CPCupdate-MOD,
    id-Additional-Measurement-Timing-Configuration-List,
    id-ServedCellSpecificInfoReq-NR,
    id-SecurityIndication,
    id-SecurityResult,
    id-SCGreconfigNotification,
```

```
    id-AdditionalListofForwardingGTPTunnelEndpoint,
    maxCellineNB,
    maxnoofBearers,
    maxnoofPDCP-SN,
    maxFailedMeasObjects,
    maxnoofCellIDforMDT,
    maxnoofTAforMDT,
    maxCellinengNB,
    maxnoofCellIDforQMC,
    maxnoofTAforQMC,
    maxnoofPLMNforQMC,
    maxnoofProtectedResourcePatterns,
    maxnoNRcellsSpectrumSharingWithE-UTRA,
    maxnoofNrCellBands,
    maxnoofSSBAreas

FROM X2AP-Constants;

-- ************************************************************
--
-- HANDOVER REQUEST
--
-- ************************************************************

HandoverRequest ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container    {{HandoverRequest-IEs}},
    ...
}

HandoverRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                          CRITICALITY reject  TYPE UE-X2AP-ID                                 PRESENCE mandatory}|
    { ID id-Cause                                       CRITICALITY ignore  TYPE Cause                                      PRESENCE mandatory}|
    { ID id-TargetCell-ID                               CRITICALITY reject  TYPE ECGI                                       PRESENCE mandatory}|
    { ID id-GUMMEI-ID                                   CRITICALITY reject  TYPE GUMMEI                                     PRESENCE mandatory}|
    { ID id-UE-ContextInformation                       CRITICALITY reject  TYPE UE-ContextInformation                      PRESENCE mandatory}|
    { ID id-UE-HistoryInformation                       CRITICALITY ignore  TYPE UE-HistoryInformation                      PRESENCE mandatory}|
    { ID id-TraceActivation                             CRITICALITY ignore  TYPE TraceActivation                            PRESENCE optional}|
    { ID id-SRVCCOperationPossible                      CRITICALITY ignore  TYPE SRVCCOperationPossible                     PRESENCE optional}|
    { ID id-CSGMembershipStatus                         CRITICALITY reject  TYPE CSGMembershipStatus                        PRESENCE optional}|
    { ID id-MobilityInformation                         CRITICALITY ignore  TYPE MobilityInformation                        PRESENCE optional}|
    { ID id-Masked-IMEISV                               CRITICALITY ignore  TYPE Masked-IMEISV                              PRESENCE optional}|
    { ID id-UE-HistoryInformationFromTheUE              CRITICALITY ignore  TYPE UE-HistoryInformationFromTheUE             PRESENCE optional}|
    { ID id-ExpectedUEBehaviour                         CRITICALITY ignore  TYPE ExpectedUEBehaviour                        PRESENCE optional}|
    { ID id-ProSeAuthorized                             CRITICALITY ignore  TYPE ProSeAuthorized                            PRESENCE optional}|
    { ID id-UE-ContextReferenceAtSeNB                   CRITICALITY ignore  TYPE UE-ContextReferenceAtSeNB                  PRESENCE optional}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension                CRITICALITY reject  TYPE UE-X2AP-ID-Extension                       PRESENCE optional}|
    { ID id-V2XServicesAuthorized                       CRITICALITY ignore  TYPE V2XServicesAuthorized                      PRESENCE optional}|
    { ID id-UE-ContextReferenceAtWT                     CRITICALITY ignore  TYPE UE-ContextReferenceAtWT                    PRESENCE optional}|
    { ID id-NRUESecurityCapabilities                        CRITICALITY ignore  TYPE NRUESecurityCapabilities               PRESENCE optional}|
    { ID id-UE-ContextReferenceAtSgNB                   CRITICALITY ignore  TYPE UE-ContextReferenceAtSgNB                  PRESENCE optional}|
    { ID id-AerialUEsubscriptionInformation             CRITICALITY ignore  TYPE AerialUEsubscriptionInformation            PRESENCE optional}|
    { ID id-Subscription-Based-UE-DifferentiationInfo   CRITICALITY ignore  TYPE Subscription-Based-UE-DifferentiationInfo  PRESENCE optional}|
    { ID id-CHOinformation-REQ                          CRITICALITY ignore  TYPE CHOinformation-REQ                         PRESENCE optional}|
    { ID id-NRV2XServicesAuthorized                     CRITICALITY ignore  TYPE NRV2XServicesAuthorized                    PRESENCE optional}|
```

```
    { ID id-PC5QoSParameters                        CRITICALITY ignore  TYPE PC5QoSParameters                        PRESENCE optional }|
    { ID id-IABNodeIndication                       CRITICALITY reject  TYPE IABNodeIndication                       PRESENCE optional},
    ...
}

UE-ContextInformation ::= SEQUENCE {
    mME-UE-S1AP-ID                  UE-S1AP-ID,
    uESecurityCapabilities          UESecurityCapabilities,
    aS-SecurityInformation          AS-SecurityInformation,
    uEaggregateMaximumBitRate       UEAggregateMaximumBitRate,
    subscriberProfileIDforRFP       SubscriberProfileIDforRFP       OPTIONAL,
    e-RABs-ToBeSetup-List           E-RABs-ToBeSetup-List,
    rRC-Context                     RRC-Context,
    handoverRestrictionList         HandoverRestrictionList     OPTIONAL,
    locationReportingInformation    LocationReportingInformation    OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {UE-ContextInformation-ExtIEs} } OPTIONAL,
    ...
}

UE-ContextInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
{ ID id-ManagementBasedMDTallowed           CRITICALITY ignore  EXTENSION ManagementBasedMDTallowed             PRESENCE optional }|
{ ID id-ManagementBasedMDTPLMNList          CRITICALITY ignore  EXTENSION MDTPLMNList                          PRESENCE optional }|
{ ID id-UESidelinkAggregateMaximumBitRate   CRITICALITY ignore  EXTENSION UESidelinkAggregateMaximumBitRate     PRESENCE optional }|
{ ID id-EPCHandoverRestrictionListContainer CRITICALITY ignore  EXTENSION EPCHandoverRestrictionListContainer   PRESENCE optional }|
{ ID id-AdditionalRRMPriorityIndex          CRITICALITY ignore  EXTENSION AdditionalRRMPriorityIndex            PRESENCE optional}|
{ ID id-NRUESidelinkAggregateMaximumBitRate CRITICALITY ignore  EXTENSION NRUESidelinkAggregateMaximumBitRate   PRESENCE optional}|
{ ID id-UERadioCapabilityID                 CRITICALITY reject  EXTENSION UERadioCapabilityID                   PRESENCE optional }|
{ ID id-IMSvoiceEPSfallbackfrom5G                   CRITICALITY ignore  EXTENSION IMSvoiceEPSfallbackfrom5G                  PRESENCE optional},
    ...
}

E-RABs-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeSetup-ItemIEs} }

E-RABs-ToBeSetup-ItemIEs    X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeSetup-Item    CRITICALITY ignore     TYPE E-RABs-ToBeSetup-Item  PRESENCE mandatory },
    ...
}

E-RABs-ToBeSetup-Item ::= SEQUENCE {
    e-RAB-ID                    E-RAB-ID,
    e-RAB-Level-QoS-Parameters      E-RAB-Level-QoS-Parameters,
    dL-Forwarding                   DL-Forwarding                                           OPTIONAL,
    uL-GTPtunnelEndpoint            GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeSetup-ItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeSetup-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-BearerType      CRITICALITY reject  EXTENSION BearerType        PRESENCE optional}|
    { ID id-DAPSRequestInfo     CRITICALITY ignore  EXTENSION DAPSRequestInfo           PRESENCE optional}|
    { ID id-Ethernet-Type   CRITICALITY ignore  EXTENSION Ethernet-Type     PRESENCE optional}|
    { ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress     PRESENCE optional}|
    { ID id-SecurityIndication  CRITICALITY reject  EXTENSION SecurityIndication                PRESENCE optional},
    ...
```

```
}

MobilityInformation ::= BIT STRING (SIZE(32))

SourceDLForwardingIPAddress ::= BIT STRING (SIZE(1..160, ...))


UE-ContextReferenceAtSeNB ::= SEQUENCE {
    source-GlobalSeNB-ID        GlobalENB-ID,
    seNB-UE-X2AP-ID             UE-X2AP-ID,
    seNB-UE-X2AP-ID-Extension   UE-X2AP-ID-Extension,
    iE-Extensions               ProtocolExtensionContainer { {UE-ContextReferenceAtSeNB-ItemExtIEs} } OPTIONAL,
    ...
}

UE-ContextReferenceAtSeNB-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-ContextReferenceAtWT ::= SEQUENCE {
    wTID                    WTID,
    wT-UE-XwAP-ID           WT-UE-XwAP-ID,
    iE-Extensions           ProtocolExtensionContainer { {UE-ContextReferenceAtWT-ItemExtIEs} } OPTIONAL,
    ...
}

UE-ContextReferenceAtWT-ItemExtIEs  X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

UE-ContextReferenceAtSgNB ::= SEQUENCE {
    source-GlobalSgNB-ID            GlobalGNB-ID,
    sgNB-UE-X2AP-ID             SgNB-UE-X2AP-ID,
    iE-Extensions               ProtocolExtensionContainer { {UE-ContextReferenceAtSgNB-ItemExtIEs} } OPTIONAL,
    ...
}

UE-ContextReferenceAtSgNB-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- HANDOVER REQUEST ACKNOWLEDGE
--
-- **************************************************************

HandoverRequestAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{HandoverRequestAcknowledge-IEs}},
    ...
}

HandoverRequestAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                          CRITICALITY ignore  TYPE UE-X2AP-ID                                 PRESENCE mandatory}|
```

```
    { ID id-New-eNB-UE-X2AP-ID                      CRITICALITY ignore  TYPE UE-X2AP-ID                                 PRESENCE mandatory}|
    { ID id-E-RABs-Admitted-List                    CRITICALITY ignore  TYPE E-RABs-Admitted-List                      PRESENCE mandatory}|
    { ID id-E-RABs-NotAdmitted-List                 CRITICALITY ignore  TYPE E-RAB-List                              PRESENCE optional}|
    { ID id-TargeteNBtoSource-eNBTransparentContainer   CRITICALITY ignore  TYPE TargeteNBtoSource-eNBTransparentContainer   PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics                  PRESENCE optional}|
    { ID id-UE-ContextKeptIndicator                 CRITICALITY ignore  TYPE UE-ContextKeptIndicator                 PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension               CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                    PRESENCE optional}--
The id-SeNB-UE-X2AP-ID-Extension shall not be sent and shall be ignored, if received.--|
    { ID id-Old-eNB-UE-X2AP-ID-Extension            CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                    PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension            CRITICALITY reject  TYPE UE-X2AP-ID-Extension                    PRESENCE optional}|
    { ID id-WT-UE-ContextKeptIndicator              CRITICALITY ignore  TYPE UE-ContextKeptIndicator                 PRESENCE optional}|
    { ID id-ERABs-transferred-to-MeNB               CRITICALITY ignore  TYPE E-RAB-List                              PRESENCE optional}|
    { ID id-CHOinformation-ACK                      CRITICALITY ignore  TYPE CHOinformation-ACK
    PRESENCE optional},
    ...
}

E-RABs-Admitted-List       ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ItemIEs} }

E-RABs-Admitted-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-Item    CRITICALITY ignore  TYPE E-RABs-Admitted-Item   PRESENCE mandatory  }
}

E-RABs-Admitted-Item ::= SEQUENCE {
    e-RAB-ID                    E-RAB-ID,
    uL-GTP-TunnelEndpoint           GTPtunnelEndpoint                                       OPTIONAL,
    dL-GTP-TunnelEndpoint           GTPtunnelEndpoint                                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-Item-ExtIEs} }   OPTIONAL,
    ...
}

E-RABs-Admitted-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-DAPSResponseInfo                CRITICALITY reject  EXTENSION DAPSResponseInfo                          PRESENCE optional},
    ...
}

-- **************************************************************
--
-- HANDOVER PREPARATION FAILURE
--
-- **************************************************************

HandoverPreparationFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{HandoverPreparationFailure-IEs}},
    ...
}

HandoverPreparationFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID              CRITICALITY ignore  TYPE UE-X2AP-ID                PRESENCE mandatory}|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension    CRITICALITY ignore  TYPE UE-X2AP-ID-Extension       PRESENCE optional}|
    { ID id-RequestedTargetCellID           CRITICALITY reject  TYPE ECGI                       PRESENCE optional},
    ...
```

```
}

-- **************************************************************
--
-- HANDOVER REPORT
--
-- **************************************************************

HandoverReport ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{HandoverReport-IEs}},
    ...
}

HandoverReport-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-HandoverReportType                          CRITICALITY ignore  TYPE HandoverReportType                      PRESENCE mandatory}|
    { ID id-Cause                                       CRITICALITY ignore  TYPE Cause                                PRESENCE mandatory}|
    { ID id-SourceCellECGI                              CRITICALITY ignore  TYPE ECGI                                 PRESENCE mandatory}|
    { ID id-FailureCellECGI                             CRITICALITY ignore  TYPE ECGI                                 PRESENCE mandatory}|
    { ID id-Re-establishmentCellECGI                    CRITICALITY ignore  TYPE ECGI                                 PRESENCE conditional} -
- The IE shall be present if the Handover Report Type IE is set to “HO to Wrong Cell” -- |
    { ID id-TargetCellInUTRAN                           CRITICALITY ignore  TYPE TargetCellInUTRAN                     PRESENCE conditional} -
- The IE shall be present if the Handover Report Type IE is set to "InterRAT ping-pong" --|
    { ID id-SourceCellCRNTI                             CRITICALITY ignore  TYPE CRNTI                                PRESENCE optional}|
    { ID id-MobilityInformation                         CRITICALITY ignore  TYPE MobilityInformation                  PRESENCE optional}|
    { ID id-UE-RLF-Report-Container                     CRITICALITY ignore  TYPE UE-RLF-Report-Container              PRESENCE optional}|
    { ID id-UE-RLF-Report-Container-for-extended-bands  CRITICALITY ignore  TYPE UE-RLF-Report-Container-for-extended-bands PRESENCE optional}|
    { ID id-TargetCellInNGRAN                           CRITICALITY ignore  TYPE TargetCellInNGRAN                     PRESENCE conditional} -
- The IE shall be present if the Handover Report Type IE is set to "interSystemPingpong" --,
    ...
}

-- **************************************************************
--
-- EARLY STATUS TRANSFER
--
-- **************************************************************

EarlyStatusTransfer ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container    {{ EarlyStatusTransfer-IEs}},
    ...
}

EarlyStatusTransfer-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                  CRITICALITY reject      TYPE UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY reject      TYPE UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension        CRITICALITY reject      TYPE UE-X2AP-ID-Extension           PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension        CRITICALITY reject      TYPE UE-X2AP-ID-Extension           PRESENCE optional}|
    { ID id-ProcedureStage                      CRITICALITY reject      TYPE ProcedureStageChoice           PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY ignore      TYPE SgNB-UE-X2AP-ID                PRESENCE optional},
    ...
}

ProcedureStageChoice ::= CHOICE {
    first-dl-count                      FirstDLCount,
```

```
    dl-discarding                       DLDiscarding,
    choice-extension                    ProtocolIE-Single-Container { {ProcedureStageChoice-ExtIEs} }
}

ProcedureStageChoice-ExtIEs X2AP-PROTOCOL-IES ::= {
    ...
}

FirstDLCount ::= SEQUENCE {
    e-RABsSubjectToEarlyStatusTransfer      E-RABsSubjectToEarlyStatusTransfer-List,
    iE-Extension                                ProtocolExtensionContainer { {FirstDLCount-ExtIEs} } OPTIONAL,
    ...
}

FirstDLCount-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

DLDiscarding ::= SEQUENCE {
    e-RABsSubjectToDLDiscarding-List            E-RABsSubjectToDLDiscarding-List,
    iE-Extension                                ProtocolExtensionContainer { {DLDiscarding-ExtIEs} } OPTIONAL,
    ...
}

DLDiscarding-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
-- **************************************************************
--
-- SN STATUS TRANSFER
--
-- **************************************************************

SNStatusTransfer ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SNStatusTransfer-IEs}},
    ...
}

SNStatusTransfer-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                      CRITICALITY reject  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID                      CRITICALITY reject  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-E-RABs-SubjectToStatusTransfer-List     CRITICALITY ignore  TYPE E-RABs-SubjectToStatusTransfer-List    PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension            CRITICALITY reject  TYPE UE-X2AP-ID-Extension                   PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension            CRITICALITY reject  TYPE UE-X2AP-ID-Extension                   PRESENCE optional}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                           PRESENCE optional},
    ...
}

E-RABs-SubjectToStatusTransfer-List ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-SubjectToStatusTransfer-
ItemIEs} }

E-RABs-SubjectToStatusTransfer-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-SubjectToStatusTransfer-Item CRITICALITY ignore  TYPE E-RABs-SubjectToStatusTransfer-Item  PRESENCE mandatory   }
}
```

```
E-RABs-SubjectToStatusTransfer-Item ::= SEQUENCE {
    e-RAB-ID                                E-RAB-ID,

    receiveStatusofULPDCPSDUs               ReceiveStatusofULPDCPSDUs           OPTIONAL,
    uL-COUNTvalue                   COUNTvalue,
    dL-COUNTvalue                   COUNTvalue,
    iE-Extensions                           ProtocolExtensionContainer { {E-RABs-SubjectToStatusTransfer-ItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-SubjectToStatusTransfer-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-ReceiveStatusOfULPDCPSDUsExtended           CRITICALITY ignore  EXTENSION ReceiveStatusOfULPDCPSDUsExtended        PRESENCE optional}|
    { ID id-ULCOUNTValueExtended                        CRITICALITY ignore  EXTENSION COUNTValueExtended                       PRESENCE optional}|
    { ID id-DLCOUNTValueExtended                        CRITICALITY ignore  EXTENSION COUNTValueExtended                       PRESENCE optional}|
    { ID id-ReceiveStatusOfULPDCPSDUsPDCP-SNlength18    CRITICALITY ignore  EXTENSION ReceiveStatusOfULPDCPSDUsPDCP-SNlength18 PRESENCE optional}|
    { ID id-ULCOUNTValuePDCP-SNlength18                 CRITICALITY ignore  EXTENSION COUNTvaluePDCP-SNlength18                PRESENCE optional}|
    { ID id-DLCOUNTValuePDCP-SNlength18                 CRITICALITY ignore  EXTENSION COUNTvaluePDCP-SNlength18                PRESENCE optional},
    ...
}

-- **************************************************************
--
-- UE CONTEXT RELEASE
--
-- **************************************************************

UEContextRelease ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{UEContextRelease-IEs}},
    ...
}

UEContextRelease-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                  CRITICALITY reject  TYPE UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY reject  TYPE UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension        CRITICALITY reject  TYPE UE-X2AP-ID-Extension                PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension        CRITICALITY reject  TYPE UE-X2AP-ID-Extension                PRESENCE optional}|
    { ID id-SIPTO-BearerDeactivationIndication  CRITICALITY ignore  TYPE SIPTOBearerDeactivationIndication   PRESENCE optional}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                     PRESENCE optional},
    ...
}


-- **************************************************************
--
-- HANDOVER CANCEL
--
-- **************************************************************

HandoverCancel ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{HandoverCancel-IEs}},
    ...
}
```

```
HandoverCancel-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                  CRITICALITY reject  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY ignore  TYPE UE-X2AP-ID                             PRESENCE optional}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                                  PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension            CRITICALITY reject  TYPE UE-X2AP-ID-Extension               PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension            CRITICALITY ignore  TYPE UE-X2AP-ID-Extension               PRESENCE optional}|
    { ID id-CandidateCellsToBeCancelledList     CRITICALITY reject  TYPE CandidateCellsToBeCancelledList        PRESENCE optional},
    ...
}


-- **************************************************************
--
-- HANDOVER SUCCESS
--
-- **************************************************************

HandoverSuccess ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{HandoverSuccess-IEs}},
    ...
}

HandoverSuccess-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                  CRITICALITY reject TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY reject TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension        CRITICALITY ignore  TYPE UE-X2AP-ID-Extension   PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension        CRITICALITY ignore  TYPE UE-X2AP-ID-Extension   PRESENCE optional}|
    { ID id-TargetCell-ID                       CRITICALITY reject  TYPE ECGI                   PRESENCE mandatory},
    ...
}

-- **************************************************************
--
-- CONDITIONAL HANDOVER CANCEL
--
-- **************************************************************

ConditionalHandoverCancel ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ConditionalHandoverCancel-IEs}},
    ...
}

ConditionalHandoverCancel-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                  CRITICALITY reject  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE optional}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                   PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension        CRITICALITY reject  TYPE UE-X2AP-ID-Extension    PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension        CRITICALITY ignore  TYPE UE-X2AP-ID-Extension    PRESENCE optional}|
    { ID id-CandidateCellsToBeCancelledList     CRITICALITY reject  TYPE CandidateCellsToBeCancelledList        PRESENCE optional},
    ...
}

-- **************************************************************
--
```

```
-- ERROR INDICATION
--
-- **************************************************************

ErrorIndication ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ErrorIndication-IEs}},
    ...
}

ErrorIndication-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Old-eNB-UE-X2AP-ID                  CRITICALITY ignore  TYPE UE-X2AP-ID                    PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY ignore  TYPE UE-X2AP-ID                    PRESENCE optional}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                         PRESENCE optional}|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics        PRESENCE optional}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension        CRITICALITY ignore  TYPE UE-X2AP-ID-Extension          PRESENCE optional}|
    { ID id-New-eNB-UE-X2AP-ID-Extension        CRITICALITY ignore  TYPE UE-X2AP-ID-Extension          PRESENCE optional}|
    { ID id-Old-SgNB-UE-X2AP-ID                 CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID               PRESENCE optional}|
    { ID id-InterfaceInstanceIndication         CRITICALITY reject  TYPE InterfaceInstanceIndication   PRESENCE optional},
    ...
}

-- **************************************************************
--
-- RESET REQUEST
--
-- **************************************************************

ResetRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ResetRequest-IEs}},
    ...
}

ResetRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                         PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication     CRITICALITY reject  TYPE InterfaceInstanceIndication   PRESENCE optional},
    ...
}

-- **************************************************************
--
-- RESET RESPONSE
--
-- **************************************************************

ResetResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ResetResponse-IEs}},
    ...
}

ResetResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics       PRESENCE optional }|
    { ID id-InterfaceInstanceIndication     CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional},
    ...
}
```

```
-- **************************************************************
--
-- X2 SETUP REQUEST
--
-- **************************************************************

X2SetupRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2SetupRequest-IEs}},
    ...
}

X2SetupRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID            CRITICALITY reject  TYPE GlobalENB-ID           PRESENCE mandatory}|
    { ID id-ServedCells             CRITICALITY reject  TYPE ServedCells            PRESENCE mandatory}|
    { ID id-GUGroupIDList           CRITICALITY reject  TYPE GUGroupIDList          PRESENCE optional}|
    { ID id-LHN-ID                  CRITICALITY ignore  TYPE LHN-ID                 PRESENCE optional},
...
}



-- **************************************************************
--
-- X2 SETUP RESPONSE
--
-- **************************************************************

X2SetupResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2SetupResponse-IEs}},
    ...
}

X2SetupResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID                CRITICALITY reject  TYPE GlobalENB-ID           PRESENCE mandatory}|
    { ID id-ServedCells                 CRITICALITY reject  TYPE ServedCells            PRESENCE mandatory}|
    { ID id-GUGroupIDList               CRITICALITY reject  TYPE GUGroupIDList          PRESENCE optional}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE optional}|
    { ID id-LHN-ID                      CRITICALITY ignore  TYPE LHN-ID                 PRESENCE optional},
    ...
}



-- **************************************************************
--
-- X2 SETUP FAILURE
--
-- **************************************************************

X2SetupFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2SetupFailure-IEs}},
    ...
}
```

```
X2SetupFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                   CRITICALITY ignore      TYPE Cause                          PRESENCE mandatory} |
    { ID id-TimeToWait              CRITICALITY ignore      TYPE TimeToWait                     PRESENCE optional} |
    { ID id-CriticalityDiagnostics  CRITICALITY ignore      TYPE CriticalityDiagnostics         PRESENCE optional },

    ...
}



-- **************************************************************
--
-- LOAD INFORMATION
--
-- **************************************************************

LoadInformation ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{LoadInformation-IEs}},
    ...
}

LoadInformation-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellInformation             CRITICALITY ignore  TYPE CellInformation-List       PRESENCE mandatory} ,
    ...
}

CellInformation-List ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ProtocolIE-Single-Container { {CellInformation-ItemIEs} }

CellInformation-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellInformation-Item    CRITICALITY ignore  TYPE CellInformation-Item   PRESENCE mandatory  }
}

CellInformation-Item ::= SEQUENCE {
    cell-ID                         ECGI,
    ul-InterferenceOverloadIndication       UL-InterferenceOverloadIndication                               OPTIONAL,
    ul-HighInterferenceIndicationInfo       UL-HighInterferenceIndicationInfo                               OPTIONAL,
    relativeNarrowbandTxPower               RelativeNarrowbandTxPower                                       OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { {CellInformation-Item-ExtIEs} }    OPTIONAL,
    ...
}

CellInformation-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
{ ID id-ABSInformation                      CRITICALITY ignore  EXTENSION ABSInformation                    PRESENCE optional }|
{ ID id-InvokeIndication                    CRITICALITY ignore  EXTENSION InvokeIndication                  PRESENCE optional }|
{ ID id-IntendedULDLConfiguration           CRITICALITY ignore  EXTENSION SubframeAssignment                PRESENCE optional }|
{ ID id-ExtendedULInterferenceOverloadInfo  CRITICALITY ignore  EXTENSION ExtendedULInterferenceOverloadInfo PRESENCE optional }|
{ ID id-CoMPInformation                     CRITICALITY ignore  EXTENSION CoMPInformation                   PRESENCE optional }|
{ ID id-DynamicDLTransmissionInformation    CRITICALITY ignore  EXTENSION DynamicDLTransmissionInformation  PRESENCE optional },
    ...
}

-- **************************************************************
--
```

```
-- ENB CONFIGURATION UPDATE
--
-- **************************************************************

ENBConfigurationUpdate ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENBConfigurationUpdate-IEs}},
    ...
}

ENBConfigurationUpdate-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ServedCellsToAdd            CRITICALITY reject  TYPE ServedCells                   PRESENCE optional}|
    { ID id-ServedCellsToModify         CRITICALITY reject  TYPE ServedCellsToModify            PRESENCE optional}|
    { ID id-ServedCellsToDelete         CRITICALITY reject  TYPE Old-ECGIs                      PRESENCE optional}|
    { ID id-GUGroupIDToAddList          CRITICALITY reject  TYPE GUGroupIDList                  PRESENCE optional}|
    { ID id-GUGroupIDToDeleteList       CRITICALITY reject  TYPE GUGroupIDList                  PRESENCE optional}|
    { ID id-CoverageModificationList    CRITICALITY reject  TYPE CoverageModificationList       PRESENCE optional},
    ...
}

ServedCellsToModify::= SEQUENCE (SIZE (1..maxCellineNB)) OF ServedCellsToModify-Item

ServedCellsToModify-Item::= SEQUENCE {
    old-ecgi                        ECGI,
    servedCellInfo                  ServedCell-Information,
    neighbour-Info                  Neighbour-Information           OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {ServedCellsToModify-Item-ExtIEs} } OPTIONAL,
    ...
}

ServedCellsToModify-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-DeactivationIndication          CRITICALITY ignore  EXTENSION DeactivationIndication           PRESENCE optional }|
    { ID id-NRNeighbourInfoToModify         CRITICALITY ignore  EXTENSION NRNeighbour-Information          PRESENCE optional },
    ...
}


Old-ECGIs::= SEQUENCE (SIZE (1..maxCellineNB)) OF ECGI

-- **************************************************************
--
-- ENB CONFIGURATION UPDATE ACKNOWLEDGE
--
-- **************************************************************

ENBConfigurationUpdateAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENBConfigurationUpdateAcknowledge-IEs}},
    ...
}

ENBConfigurationUpdateAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional},
...
}
```

```
-- **************************************************************
--
-- ENB CONFIGURATION UPDATE FAIURE
--
-- **************************************************************

ENBConfigurationUpdateFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENBConfigurationUpdateFailure-IEs}},
    ...
}

ENBConfigurationUpdateFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory}|
    { ID id-TimeToWait                  CRITICALITY ignore  TYPE TimeToWait                 PRESENCE optional}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional},
    ...
}



-- **************************************************************
--
-- RESOURCE STATUS REQUEST
--
-- **************************************************************

ResourceStatusRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ResourceStatusRequest-IEs}},
    ...
}

ResourceStatusRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ENB1-Measurement-ID         CRITICALITY reject  TYPE Measurement-ID               PRESENCE mandatory}|
    { ID id-ENB2-Measurement-ID         CRITICALITY ignore  TYPE Measurement-ID               PRESENCE conditional}|-- The IE shall be present if
the Registration Request IE is set to “Stop”, “Partial stop” or to “Add”--
    { ID id-Registration-Request        CRITICALITY reject  TYPE Registration-Request         PRESENCE mandatory}|
    { ID id-ReportCharacteristics       CRITICALITY reject  TYPE ReportCharacteristics        PRESENCE optional}|
    { ID id-CellToReport                CRITICALITY ignore  TYPE CellToReport-List            PRESENCE mandatory}|
    { ID id-ReportingPeriodicity        CRITICALITY ignore  TYPE ReportingPeriodicity         PRESENCE optional}|
    { ID id-PartialSuccessIndicator     CRITICALITY ignore  TYPE PartialSuccessIndicator      PRESENCE optional}|
    { ID id-ReportingPeriodicityRSRPMR  CRITICALITY ignore  TYPE ReportingPeriodicityRSRPMR   PRESENCE optional}|
    { ID id-ReportingPeriodicityCSIR    CRITICALITY ignore  TYPE ReportingPeriodicityCSIR     PRESENCE optional},
    ...
}


CellToReport-List       ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ProtocolIE-Single-Container { {CellToReport-ItemIEs} }

CellToReport-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellToReport-Item   CRITICALITY ignore  TYPE CellToReport-Item  PRESENCE mandatory}
}

CellToReport-Item ::= SEQUENCE {
    cell-ID                             ECGI,
```

```
    iE-Extensions                           ProtocolExtensionContainer { {CellToReport-Item-ExtIEs} } OPTIONAL,
    ...
}

CellToReport-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


ReportingPeriodicity ::= ENUMERATED {
    one-thousand-ms,
    two-thousand-ms,
    five-thousand-ms,
    ten-thousand-ms,
...
}

PartialSuccessIndicator ::= ENUMERATED {
    partial-success-allowed,
...
}

-- **************************************************************
--
-- RESOURCE STATUS RESPONSE
--
-- **************************************************************

ResourceStatusResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ResourceStatusResponse-IEs}},
    ...
}

ResourceStatusResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ENB1-Measurement-ID                 CRITICALITY reject  TYPE Measurement-ID                    PRESENCE mandatory}|
    { ID id-ENB2-Measurement-ID                 CRITICALITY reject  TYPE Measurement-ID                    PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics            PRESENCE optional}|
    { ID id-MeasurementInitiationResult-List    CRITICALITY ignore  TYPE MeasurementInitiationResult-List  PRESENCE optional},
    ...
}


MeasurementInitiationResult-List ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ProtocolIE-Single-Container { {MeasurementInitiationResult-ItemIEs} }

MeasurementInitiationResult-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeasurementInitiationResult-Item    CRITICALITY ignore  TYPE MeasurementInitiationResult-Item  PRESENCE mandatory}
}

MeasurementInitiationResult-Item ::= SEQUENCE {
    cell-ID                                     ECGI,
    measurementFailureCause-List                MeasurementFailureCause-List    OPTIONAL,
    iE-Extensions                               ProtocolExtensionContainer { {MeasurementInitiationResult-Item-ExtIEs} }OPTIONAL,
    ...
}
```

```
MeasurementInitiationResult-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

MeasurementFailureCause-List ::= SEQUENCE (SIZE (1..maxFailedMeasObjects)) OF ProtocolIE-Single-Container { {MeasurementFailureCause-ItemIEs} }

MeasurementFailureCause-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeasurementFailureCause-Item    CRITICALITY ignore  TYPE MeasurementFailureCause-Item    PRESENCE mandatory}
}

MeasurementFailureCause-Item ::= SEQUENCE {
    measurementFailedReportCharacteristics          ReportCharacteristics,
    cause                                           Cause,
    iE-Extensions                                   ProtocolExtensionContainer { {MeasurementFailureCause-Item-ExtIEs} } OPTIONAL,
    ...
}

MeasurementFailureCause-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- RESOURCE STATUS FAILURE
--
-- **************************************************************

ResourceStatusFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ResourceStatusFailure-IEs}},
    ...
}

ResourceStatusFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ENB1-Measurement-ID                     CRITICALITY reject  TYPE Measurement-ID                      PRESENCE mandatory}|
    { ID id-ENB2-Measurement-ID                     CRITICALITY reject  TYPE Measurement-ID                      PRESENCE mandatory}|
    { ID id-Cause                                   CRITICALITY ignore  TYPE Cause                               PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics              PRESENCE optional}|
    { ID id-CompleteFailureCauseInformation-List    CRITICALITY ignore  TYPE CompleteFailureCauseInformation-List  PRESENCE optional},
    ...
}

CompleteFailureCauseInformation-List ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ProtocolIE-Single-Container { {CompleteFailureCauseInformation-
ItemIEs} }

CompleteFailureCauseInformation-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CompleteFailureCauseInformation-Item    CRITICALITY ignore  TYPE CompleteFailureCauseInformation-Item  PRESENCE mandatory}
}

CompleteFailureCauseInformation-Item ::= SEQUENCE {
    cell-ID                                         ECGI,
    measurementFailureCause-List                    MeasurementFailureCause-List,
    iE-Extensions                                   ProtocolExtensionContainer { {CompleteFailureCauseInformation-Item-ExtIEs} }  OPTIONAL,
    ...
```

```
}

CompleteFailureCauseInformation-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- RESOURCE STATUS UPDATE
--
-- **************************************************************

ResourceStatusUpdate ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ResourceStatusUpdate-IEs}},
    ...
}

ResourceStatusUpdate-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ENB1-Measurement-ID     CRITICALITY reject  TYPE Measurement-ID                PRESENCE mandatory}|
    { ID id-ENB2-Measurement-ID     CRITICALITY reject  TYPE Measurement-ID                PRESENCE mandatory}|
    { ID id-CellMeasurementResult   CRITICALITY ignore  TYPE CellMeasurementResult-List    PRESENCE mandatory},
    ...
}

CellMeasurementResult-List ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ProtocolIE-Single-Container { {CellMeasurementResult-ItemIEs} }

CellMeasurementResult-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellMeasurementResult-Item  CRITICALITY ignore  TYPE CellMeasurementResult-Item PRESENCE mandatory}
}

CellMeasurementResult-Item ::= SEQUENCE {
    cell-ID                     ECGI,
    hWLoadIndicator             HWLoadIndicator     OPTIONAL,
    s1TNLLoadIndicator          S1TNLLoadIndicator  OPTIONAL,
    radioResourceStatus         RadioResourceStatus OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {CellMeasurementResult-Item-ExtIEs} }       OPTIONAL,
    ...
}

CellMeasurementResult-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-CompositeAvailableCapacityGroup CRITICALITY ignore  EXTENSION CompositeAvailableCapacityGroup    PRESENCE optional}|
    { ID id-ABS-Status                      CRITICALITY ignore  EXTENSION ABS-Status                        PRESENCE optional}|
    { ID id-RSRPMRList                      CRITICALITY ignore  EXTENSION RSRPMRList                        PRESENCE optional}|
    { ID id-CSIReportList                   CRITICALITY ignore  EXTENSION CSIReportList                     PRESENCE optional}|
    { ID id-CellReportingIndicator          CRITICALITY ignore  EXTENSION CellReportingIndicator            PRESENCE optional}|
    { ID id-MeasurementResultforNRCellsPossiblyAggregated       CRITICALITY ignore  EXTENSION MeasurementResultforNRCellsPossiblyAggregated
    PRESENCE optional},
    ...
}


-- **************************************************************
--
-- PRIVATE MESSAGE
```

```
--
-- **************************************************************

PrivateMessage ::= SEQUENCE {
    privateIEs      PrivateIE-Container {{PrivateMessage-IEs}},
    ...
}

PrivateMessage-IEs X2AP-PRIVATE-IES ::= {
    ...
}

-- **************************************************************
--
-- MOBILITY CHANGE REQUEST
--
-- **************************************************************

MobilityChangeRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{MobilityChangeRequest-IEs}},
    ...
}

MobilityChangeRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ENB1-Cell-ID                        CRITICALITY reject  TYPE ECGI                                   PRESENCE mandatory}|
    { ID id-ENB2-Cell-ID                        CRITICALITY reject  TYPE ECGI                                   PRESENCE mandatory}|
    { ID id-ENB1-Mobility-Parameters            CRITICALITY ignore  TYPE MobilityParametersInformation          PRESENCE optional}|
    { ID id-ENB2-Proposed-Mobility-Parameters   CRITICALITY reject  TYPE MobilityParametersInformation          PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY reject  TYPE Cause                                  PRESENCE mandatory},
    ...
}

-- **************************************************************
--
-- MOBILITY CHANGE ACKNOWLEDGE
--
-- **************************************************************

MobilityChangeAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{MobilityChangeAcknowledge-IEs}},
    ...
}

MobilityChangeAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ENB1-Cell-ID            CRITICALITY reject  TYPE ECGI                       PRESENCE mandatory}|
    { ID id-ENB2-Cell-ID            CRITICALITY reject  TYPE ECGI                       PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional},
    ...
}


-- **************************************************************
--
-- MOBILITY CHANGE FAILURE
```

```
--
-- **************************************************************

MobilityChangeFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{MobilityChangeFailure-IEs}},
    ...
}

MobilityChangeFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ENB1-Cell-ID                                CRITICALITY ignore  TYPE ECGI                                   PRESENCE mandatory}|
    { ID id-ENB2-Cell-ID                                CRITICALITY ignore  TYPE ECGI                                   PRESENCE mandatory}|
    { ID id-Cause                                       CRITICALITY ignore  TYPE Cause                                  PRESENCE mandatory}|
    { ID id-ENB2-Mobility-Parameters-Modification-Range  CRITICALITY ignore  TYPE MobilityParametersModificationRange   PRESENCE optional}|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                 PRESENCE optional},
    ...
}

-- **************************************************************
--
-- RADIO LINK FAILURE INDICATION
--
-- **************************************************************

RLFIndication ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{RLFIndication-IEs}},
    ...
}

RLFIndication-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-FailureCellPCI                              CRITICALITY ignore  TYPE PCI                                    PRESENCE mandatory}|
    { ID id-Re-establishmentCellECGI                    CRITICALITY ignore  TYPE ECGI                                   PRESENCE mandatory}|
    { ID id-FailureCellCRNTI                            CRITICALITY ignore  TYPE CRNTI                                  PRESENCE mandatory}|
    { ID id-ShortMAC-I                                  CRITICALITY ignore  TYPE ShortMAC-I                             PRESENCE optional}|
    { ID id-UE-RLF-Report-Container                     CRITICALITY ignore  TYPE UE-RLF-Report-Container                PRESENCE optional}|
    { ID id-RRCConnSetupIndicator                       CRITICALITY reject  TYPE RRCConnSetupIndicator                  PRESENCE optional}|
    { ID id-RRCConnReestabIndicator                     CRITICALITY ignore  TYPE RRCConnReestabIndicator                PRESENCE optional}|
    { ID id-UE-RLF-Report-Container-for-extended-bands  CRITICALITY ignore  TYPE UE-RLF-Report-Container-for-extended-bands  PRESENCE optional}|
    { ID id-NBIoT-RLF-Report-Container                  CRITICALITY ignore  TYPE NBIoT-RLF-Report-Container             PRESENCE optional},
    ...
}


-- **************************************************************
--
-- CELL ACTIVATION REQUEST
--
-- **************************************************************

CellActivationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{CellActivationRequest-IEs}},
    ...
}

CellActivationRequest-IEs X2AP-PROTOCOL-IES ::= {
```

```
    { ID id-ServedCellsToActivate   CRITICALITY reject  TYPE ServedCellsToActivate      PRESENCE mandatory},
    ...
}

ServedCellsToActivate::= SEQUENCE (SIZE (1..maxCellineNB)) OF ServedCellsToActivate-Item

ServedCellsToActivate-Item::= SEQUENCE {
    ecgi                        ECGI,
    iE-Extensions                   ProtocolExtensionContainer { {ServedCellsToActivate-Item-ExtIEs} } OPTIONAL,
    ...
}

ServedCellsToActivate-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- CELL ACTIVATION RESPONSE
--
-- **************************************************************

CellActivationResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{CellActivationResponse-IEs}},
    ...
}

CellActivationResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ActivatedCellList       CRITICALITY ignore  TYPE ActivatedCellList              PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional},
    ...
}

ActivatedCellList ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ActivatedCellList-Item

ActivatedCellList-Item::= SEQUENCE {
    ecgi                            ECGI,
    iE-Extensions                   ProtocolExtensionContainer { {ActivatedCellList-Item-ExtIEs} } OPTIONAL,
    ...
}

ActivatedCellList-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

--**************************************************************
--
-- CELL ACTIVATION FAILURE
--
-- **************************************************************

CellActivationFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{CellActivationFailure-IEs}},
    ...
```

```
}

CellActivationFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional },
    ...
}

-- **************************************************************
--
-- X2 RELEASE
--
-- **************************************************************

X2Release ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2Release-IEs}},
    ...
}

X2Release-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID            CRITICALITY reject  TYPE GlobalENB-ID           PRESENCE mandatory},
...
}

-- **************************************************************
--
-- X2AP MESSAGE TRANSFER
--
-- **************************************************************

X2APMessageTransfer ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2APMessageTransfer-IEs}},
    ...
}

X2APMessageTransfer-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-RNL-Header  CRITICALITY reject  TYPE RNL-Header             PRESENCE mandatory}|
    { ID id-x2APMessage CRITICALITY reject  TYPE X2AP-Message           PRESENCE optional},
    ...
}

RNL-Header ::= SEQUENCE {
    source-GlobalENB-ID GlobalENB-ID,
    target-GlobalENB-ID GlobalENB-ID    OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {RNL-Header-Item-ExtIEs} } OPTIONAL,
    ...
}

RNL-Header-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

X2AP-Message ::= OCTET STRING
```

```
-- **************************************************************
--
-- SENB ADDITION REQUEST
--
-- **************************************************************

SeNBAdditionRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBAdditionRequest-IEs}},
    ...
}

SeNBAdditionRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-UE-SecurityCapabilities         CRITICALITY reject  TYPE UESecurityCapabilities         PRESENCE conditional}|
    -- This IE shall be present if the Bearer Option IE is set to the value “SCG bearer” --
    { ID id-SeNBSecurityKey                 CRITICALITY reject  TYPE SeNBSecurityKey                PRESENCE conditional}|
    -- This IE shall be present if the Bearer Option IE is set to the value “SCG bearer” --
    { ID id-SeNBUEAggregateMaximumBitRate   CRITICALITY reject  TYPE UEAggregateMaximumBitRate      PRESENCE mandatory}|
    { ID id-ServingPLMN                     CRITICALITY ignore  TYPE PLMN-Identity                  PRESENCE optional}|
    { ID id-E-RABs-ToBeAdded-List           CRITICALITY reject  TYPE E-RABs-ToBeAdded-List          PRESENCE mandatory}|
    { ID id-MeNBtoSeNBContainer             CRITICALITY reject  TYPE MeNBtoSeNBContainer            PRESENCE mandatory}|
    { ID id-CSGMembershipStatus             CRITICALITY reject  TYPE CSGMembershipStatus            PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension           PRESENCE optional}|
    { ID id-ExpectedUEBehaviour             CRITICALITY ignore  TYPE ExpectedUEBehaviour            PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension           PRESENCE optional},
    ...
}

E-RABs-ToBeAdded-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeAdded-ItemIEs} }

E-RABs-ToBeAdded-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeAdded-Item   CRITICALITY reject  TYPE E-RABs-ToBeAdded-Item      PRESENCE mandatory},
    ...
}

E-RABs-ToBeAdded-Item ::= CHOICE {
    sCG-Bearer      E-RABs-ToBeAdded-Item-SCG-Bearer,
    split-Bearer    E-RABs-ToBeAdded-Item-Split-Bearer,
    ...
}

E-RABs-ToBeAdded-Item-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    e-RAB-Level-QoS-Parameters      E-RAB-Level-QoS-Parameters,
    dL-Forwarding                   DL-Forwarding                                                   OPTIONAL,
    s1-UL-GTPtunnelEndpoint         GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-Item-SCG-BearerExtIEs} }  OPTIONAL,
    ...
}

E-RABs-ToBeAdded-Item-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-Correlation-ID              CRITICALITY ignore  EXTENSION Correlation-ID        PRESENCE optional}|
    { ID id-SIPTO-Correlation-ID        CRITICALITY ignore  EXTENSION Correlation-ID        PRESENCE optional}|
```

```
    { ID id-BearerType                 CRITICALITY ignore  EXTENSION BearerType           PRESENCE optional}|
    { ID id-Ethernet-Type              CRITICALITY ignore  EXTENSION Ethernet-Type        PRESENCE optional}|
    { ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress     PRESENCE optional},
    ...
}

E-RABs-ToBeAdded-Item-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                       E-RAB-ID,
    e-RAB-Level-QoS-Parameters     E-RAB-Level-QoS-Parameters,
    meNB-GTPtunnelEndpoint         GTPtunnelEndpoint,
    iE-Extensions                  ProtocolExtensionContainer { {E-RABs-ToBeAdded-Item-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeAdded-Item-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress     PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SENB ADDITION REQUEST ACKNOWLEDGE
--
-- **************************************************************

SeNBAdditionRequestAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBAdditionRequestAcknowledge-IEs}},
    ...
}

SeNBAdditionRequestAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                    CRITICALITY reject  TYPE UE-X2AP-ID                    PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                    CRITICALITY reject  TYPE UE-X2AP-ID                    PRESENCE mandatory}|
    { ID id-E-RABs-Admitted-ToBeAdded-List     CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-List PRESENCE mandatory}|
    { ID id-E-RABs-NotAdmitted-List            CRITICALITY ignore  TYPE E-RAB-List                    PRESENCE optional}|
    { ID id-SeNBtoMeNBContainer                CRITICALITY reject  TYPE SeNBtoMeNBContainer           PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics             CRITICALITY ignore  TYPE CriticalityDiagnostics        PRESENCE optional}|
    { ID id-GW-TransportLayerAddress           CRITICALITY ignore  TYPE TransportLayerAddress         PRESENCE optional}|
    { ID id-SIPTO-L-GW-TransportLayerAddress   CRITICALITY ignore  TYPE TransportLayerAddress         PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension          CRITICALITY reject  TYPE UE-X2AP-ID-Extension          PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension          CRITICALITY reject  TYPE UE-X2AP-ID-Extension          PRESENCE optional}|
    { ID id-Tunnel-Information-for-BBF         CRITICALITY ignore  TYPE TunnelInformation             PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-List ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeAdded-ItemIEs} }

E-RABs-Admitted-ToBeAdded-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeAdded-Item  CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-Item    PRESENCE mandatory}
}

E-RABs-Admitted-ToBeAdded-Item ::= CHOICE {
    sCG-Bearer      E-RABs-Admitted-ToBeAdded-Item-SCG-Bearer,
    split-Bearer    E-RABs-Admitted-ToBeAdded-Item-Split-Bearer,
```

```
    ...
}

E-RABs-Admitted-ToBeAdded-Item-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                                E-RAB-ID,
    s1-DL-GTPtunnelEndpoint                 GTPtunnelEndpoint,
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                                   OPTIONAL,
    uL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-Item-SCG-BearerExtIEs} }  OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-Item-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-SourceDLForwardingIPAddress             CRITICALITY ignore  EXTENSION TransportLayerAddress    PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-Item-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    seNB-GTPtunnelEndpoint          GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-Item-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-Item-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress    PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SENB ADDITION REQUEST REJECT
--
-- **************************************************************

SeNBAdditionRequestReject ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBAdditionRequestReject-IEs}},
    ...
}

SeNBAdditionRequestReject-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SENB RECONFIGURATION COMPLETE
--
```

```
-- **************************************************************

SeNBReconfigurationComplete ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBReconfigurationComplete-IEs}},
    ...
}

SeNBReconfigurationComplete-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
    { ID id-ResponseInformationSeNBReconfComp       CRITICALITY ignore  TYPE ResponseInformationSeNBReconfComp  PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY reject  TYPE UE-X2AP-ID-Extension               PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension               CRITICALITY reject  TYPE UE-X2AP-ID-Extension               PRESENCE optional},
    ...
}

ResponseInformationSeNBReconfComp ::= CHOICE {
    success             ResponseInformationSeNBReconfComp-SuccessItem,
    reject-by-MeNB      ResponseInformationSeNBReconfComp-RejectByMeNBItem,
    ...
}

ResponseInformationSeNBReconfComp-SuccessItem ::= SEQUENCE {
    meNBtoSeNBContainer             MeNBtoSeNBContainer OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {ResponseInformationSeNBReconfComp-SuccessItemExtIEs} } OPTIONAL,
    ...
}

ResponseInformationSeNBReconfComp-SuccessItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResponseInformationSeNBReconfComp-RejectByMeNBItem ::= SEQUENCE {
    cause                           Cause,
    meNBtoSeNBContainer             MeNBtoSeNBContainer                                                                 OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {ResponseInformationSeNBReconfComp-RejectByMeNBItemExtIEs} }  OPTIONAL,
    ...
}

ResponseInformationSeNBReconfComp-RejectByMeNBItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SENB MODIFICATION REQUEST
--
-- **************************************************************

SeNBModificationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ SeNBModificationRequest-IEs}},
    ...
}
```

```
SeNBModificationRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory}|
    { ID id-SCGChangeIndication                 CRITICALITY ignore  TYPE SCGChangeIndication            PRESENCE optional}|
    { ID id-ServingPLMN                         CRITICALITY ignore  TYPE PLMN-Identity                  PRESENCE optional}|
    { ID id-UE-ContextInformationSeNBModReq     CRITICALITY reject  TYPE UE-ContextInformationSeNBModReq PRESENCE optional}|
    { ID id-MeNBtoSeNBContainer                 CRITICALITY ignore  TYPE MeNBtoSeNBContainer            PRESENCE optional}|
    { ID id-CSGMembershipStatus                 CRITICALITY reject  TYPE CSGMembershipStatus            PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension           PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension           PRESENCE optional},
    ...
}

UE-ContextInformationSeNBModReq ::= SEQUENCE {
    uE-SecurityCapabilities         UESecurityCapabilities                                          OPTIONAL,
    seNB-SecurityKey                SeNBSecurityKey                                                 OPTIONAL,
    seNBUEAggregateMaximumBitRate   UEAggregateMaximumBitRate                                       OPTIONAL,
    e-RABs-ToBeAdded                E-RABs-ToBeAdded-List-ModReq                                    OPTIONAL,
    e-RABs-ToBeModified             E-RABs-ToBeModified-List-ModReq                                 OPTIONAL,
    e-RABs-ToBeReleased             E-RABs-ToBeReleased-List-ModReq                                 OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {UE-ContextInformationSeNBModReqExtIEs} }   OPTIONAL,
    ...
}

UE-ContextInformationSeNBModReqExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeAdded-List-ModReq ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeAdded-ModReqItemIEs} }

E-RABs-ToBeAdded-ModReqItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeAdded-ModReqItem CRITICALITY ignore  TYPE E-RABs-ToBeAdded-ModReqItem    PRESENCE mandatory},
    ...
}

E-RABs-ToBeAdded-ModReqItem ::= CHOICE {
    sCG-Bearer      E-RABs-ToBeAdded-ModReqItem-SCG-Bearer,
    split-Bearer    E-RABs-ToBeAdded-ModReqItem-Split-Bearer,
    ...
}

E-RABs-ToBeAdded-ModReqItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    e-RAB-Level-QoS-Parameters      E-RAB-Level-QoS-Parameters,
    dL-Forwarding                   DL-Forwarding                                                   OPTIONAL,
    s1-UL-GTPtunnelEndpoint         GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-ModReqItem-SCG-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeAdded-ModReqItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-Correlation-ID              CRITICALITY ignore  EXTENSION Correlation-ID        PRESENCE optional}|
    { ID id-SIPTO-Correlation-ID        CRITICALITY ignore  EXTENSION Correlation-ID        PRESENCE optional}|
```

```
    { ID id-BearerType                 CRITICALITY ignore  EXTENSION BearerType           PRESENCE optional}|
    { ID id-Ethernet-Type              CRITICALITY ignore  EXTENSION Ethernet-Type        PRESENCE optional}|
    { ID id-SourceDLForwardingIPAddress CRITICALITY ignore  EXTENSION TransportLayerAddress PRESENCE optional},
    ...
}

E-RABs-ToBeAdded-ModReqItem-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    e-RAB-Level-QoS-Parameters      E-RAB-Level-QoS-Parameters,
    meNB-GTPtunnelEndpoint          GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-ModReqItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeAdded-ModReqItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
{ ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress   PRESENCE optional},
    ...
}

E-RABs-ToBeModified-List-ModReq ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeModified-ModReqItemIEs} }

E-RABs-ToBeModified-ModReqItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeModified-ModReqItem  CRITICALITY ignore  TYPE E-RABs-ToBeModified-ModReqItem     PRESENCE mandatory},
    ...
}


E-RABs-ToBeModified-ModReqItem ::= CHOICE {
    sCG-Bearer      E-RABs-ToBeModified-ModReqItem-SCG-Bearer,
    split-Bearer    E-RABs-ToBeModified-ModReqItem-Split-Bearer,
    ...
}

E-RABs-ToBeModified-ModReqItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    e-RAB-Level-QoS-Parameters      E-RAB-Level-QoS-Parameters                                      OPTIONAL,
    s1-UL-GTPtunnelEndpoint         GTPtunnelEndpoint                                               OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeModified-ModReqItem-SCG-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeModified-ModReqItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeModified-ModReqItem-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    e-RAB-Level-QoS-Parameters      E-RAB-Level-QoS-Parameters                                          OPTIONAL,
    meNB-GTPtunnelEndpoint          GTPtunnelEndpoint                                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeModified-ModReqItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeModified-ModReqItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

E-RABs-ToBeReleased-List-ModReq ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-ModReqItemIEs} }

E-RABs-ToBeReleased-ModReqItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-ModReqItem  CRITICALITY ignore  TYPE E-RABs-ToBeReleased-ModReqItem     PRESENCE mandatory},
    ...
}

E-RABs-ToBeReleased-ModReqItem ::= CHOICE {
    sCG-Bearer      E-RABs-ToBeReleased-ModReqItem-SCG-Bearer,
    split-Bearer    E-RABs-ToBeReleased-ModReqItem-Split-Bearer,
    ...
}

E-RABs-ToBeReleased-ModReqItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    uL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-ModReqItem-SCG-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-ModReqItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-ModReqItem-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-ModReqItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-ModReqItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SENB MODIFICATION REQUEST ACKNOWLEDGE
--
-- **************************************************************

SeNBModificationRequestAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBModificationRequestAcknowledge-IEs}},
    ...
}

SeNBModificationRequestAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                             CRITICALITY ignore  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                             CRITICALITY ignore  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-E-RABs-Admitted-ToBeAdded-ModAckList        CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-ModAckList   PRESENCE optional}|
```

```
    { ID id-E-RABs-Admitted-ToBeModified-ModAckList     CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeModified-ModAckList PRESENCE optional}|
    { ID id-E-RABs-Admitted-ToBeReleased-ModAckList     CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeReleased-ModAckList PRESENCE optional}|
    { ID id-E-RABs-NotAdmitted-List                     CRITICALITY ignore  TYPE E-RAB-List                              PRESENCE optional}|
    { ID id-SeNBtoMeNBContainer                         CRITICALITY ignore  TYPE SeNBtoMeNBContainer                    PRESENCE optional}|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                 PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension                   CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                   PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension                   CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                   PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-ModAckList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeAdded-
ModAckItemIEs} }

E-RABs-Admitted-ToBeAdded-ModAckItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeAdded-ModAckItem    CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-ModAckItem     PRESENCE mandatory}
}

E-RABs-Admitted-ToBeAdded-ModAckItem ::= CHOICE {
    sCG-Bearer      E-RABs-Admitted-ToBeAdded-ModAckItem-SCG-Bearer,
    split-Bearer    E-RABs-Admitted-ToBeAdded-ModAckItem-Split-Bearer,
    ...
}

E-RABs-Admitted-ToBeAdded-ModAckItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    s1-DL-GTPtunnelEndpoint             GTPtunnelEndpoint,
    dL-Forwarding-GTPtunnelEndpoint     GTPtunnelEndpoint                                                   OPTIONAL,
    uL-Forwarding-GTPtunnelEndpoint     GTPtunnelEndpoint                                                   OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-ModAckItem-SCG-BearerExtIEs} }OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-ModAckItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
{ ID id-SourceDLForwardingIPAddress     CRITICALITY ignore  EXTENSION TransportLayerAddress     PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-ModAckItem-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                    E-RAB-ID,
    seNB-GTPtunnelEndpoint      GTPtunnelEndpoint,
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-ModAckItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-ModAckItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
{ ID id-SourceDLForwardingIPAddress     CRITICALITY ignore  EXTENSION TransportLayerAddress     PRESENCE optional},
    ...
}
E-RABs-Admitted-ToBeModified-ModAckList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeModified-
ModAckItemIEs} }

E-RABs-Admitted-ToBeModified-ModAckItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeModified-ModAckItem     CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeModified-ModAckItem PRESENCE mandatory}
}
```

```
E-RABs-Admitted-ToBeModified-ModAckItem ::= CHOICE {
    sCG-Bearer      E-RABs-Admitted-ToBeModified-ModAckItem-SCG-Bearer,
    split-Bearer    E-RABs-Admitted-ToBeModified-ModAckItem-Split-Bearer,
    ...
}

E-RABs-Admitted-ToBeModified-ModAckItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    s1-DL-GTPtunnelEndpoint         GTPtunnelEndpoint                                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeModified-ModAckItem-SCG-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeModified-ModAckItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-Admitted-ToBeModified-ModAckItem-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    seNB-GTPtunnelEndpoint          GTPtunnelEndpoint                                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeModified-ModAckItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeModified-ModAckItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-Admitted-ToBeReleased-ModAckList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeReleased-
ModAckItemIEs} }

E-RABs-Admitted-ToBeReleased-ModAckItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeReleased-ModAckItem     CRITICALITY ignore  TYPE E-RABs-Admitted-ToReleased-ModAckItem   PRESENCE mandatory}
}

E-RABs-Admitted-ToReleased-ModAckItem ::= CHOICE {
    sCG-Bearer      E-RABs-Admitted-ToBeReleased-ModAckItem-SCG-Bearer,
    split-Bearer    E-RABs-Admitted-ToBeReleased-ModAckItem-Split-Bearer,
    ...
}

E-RABs-Admitted-ToBeReleased-ModAckItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                    E-RAB-ID,
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-Admitted-ToBeReleased-ModAckItem-SCG-BearerExtIEs} }OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeReleased-ModAckItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-Admitted-ToBeReleased-ModAckItem-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                    E-RAB-ID,
```

```
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-Admitted-ToBeReleased-ModAckItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeReleased-ModAckItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SENB MODIFICATION REQUEST REJECT
--
-- **************************************************************

SeNBModificationRequestReject ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBModificationRequestReject-IEs}},
    ...
}

SeNBModificationRequestReject-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                   PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics  PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY ignore  TYPE UE-X2AP-ID-Extension    PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension           CRITICALITY ignore  TYPE UE-X2AP-ID-Extension    PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SENB MODIFICATION REQUIRED
--
-- **************************************************************

SeNBModificationRequired ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SeNBModificationRequired-IEs}},
    ...
}

SeNBModificationRequired-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                  PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                  PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                       PRESENCE mandatory}|
    { ID id-SCGChangeIndication                 CRITICALITY ignore  TYPE SCGChangeIndication         PRESENCE optional}|
    { ID id-E-RABs-ToBeReleased-ModReqd         CRITICALITY ignore  TYPE E-RABs-ToBeReleased-ModReqd  PRESENCE optional}|
    { ID id-SeNBtoMeNBContainer                 CRITICALITY ignore  TYPE SeNBtoMeNBContainer         PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension        PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension        PRESENCE optional},
    ...
}


E-RABs-ToBeReleased-ModReqd ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-ModReqdItemIEs} }
```

```
E-RABs-ToBeReleased-ModReqdItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-ModReqdItem  CRITICALITY ignore     TYPE E-RABs-ToBeReleased-ModReqdItem   PRESENCE mandatory },
    ...
}

E-RABs-ToBeReleased-ModReqdItem ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    cause                           Cause,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-ModReqdItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-ModReqdItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SENB MODIFICATION CONFIRM
--
-- **************************************************************

SeNBModificationConfirm ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBModificationConfirm-IEs}},
    ...
}

SeNBModificationConfirm-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                   CRITICALITY ignore  TYPE UE-X2AP-ID                PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                   CRITICALITY ignore  TYPE UE-X2AP-ID                PRESENCE mandatory}|
    { ID id-MeNBtoSeNBContainer               CRITICALITY ignore  TYPE MeNBtoSeNBContainer       PRESENCE optional}|
    { ID id-CriticalityDiagnostics            CRITICALITY ignore  TYPE CriticalityDiagnostics    PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension         CRITICALITY ignore  TYPE UE-X2AP-ID-Extension      PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension         CRITICALITY ignore  TYPE UE-X2AP-ID-Extension      PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SENB MODIFICATION REFUSE
--
-- **************************************************************

SeNBModificationRefuse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SeNBModificationRefuse-IEs}},
    ...
}

SeNBModificationRefuse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                   CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                   CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-Cause                             CRITICALITY ignore  TYPE Cause                   PRESENCE mandatory}|
    { ID id-MeNBtoSeNBContainer               CRITICALITY ignore  TYPE MeNBtoSeNBContainer     PRESENCE optional}|
```

```
    { ID id-CriticalityDiagnostics             CRITICALITY ignore  TYPE CriticalityDiagnostics   PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension          CRITICALITY ignore  TYPE UE-X2AP-ID-Extension     PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension          CRITICALITY ignore  TYPE UE-X2AP-ID-Extension     PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SENB RELEASE REQUEST
--
-- **************************************************************

SeNBReleaseRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SeNBReleaseRequest-IEs}},
    ...
}

SeNBReleaseRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                    CRITICALITY reject  TYPE UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                    CRITICALITY reject  TYPE UE-X2AP-ID                          PRESENCE optional}|
    { ID id-Cause                              CRITICALITY ignore  TYPE Cause                               PRESENCE optional}|
    { ID id-E-RABs-ToBeReleased-List-RelReq    CRITICALITY ignore  TYPE E-RABs-ToBeReleased-List-RelReq      PRESENCE optional}|
    { ID id-UE-ContextKeptIndicator            CRITICALITY ignore  TYPE UE-ContextKeptIndicator              PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension          CRITICALITY reject  TYPE UE-X2AP-ID-Extension                PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension          CRITICALITY reject  TYPE UE-X2AP-ID-Extension                PRESENCE optional}|
    { ID id-MakeBeforeBreakIndicator           CRITICALITY ignore  TYPE MakeBeforeBreakIndicator             PRESENCE optional},
    ...
}
E-RABs-ToBeReleased-List-RelReq ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-RelReqItemIEs} }

E-RABs-ToBeReleased-RelReqItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-RelReqItem     CRITICALITY ignore  TYPE E-RABs-ToBeReleased-RelReqItem    PRESENCE mandatory},
    ...
}

E-RABs-ToBeReleased-RelReqItem ::= CHOICE {
    sCG-Bearer      E-RABs-ToBeReleased-RelReqItem-SCG-Bearer,
    split-Bearer    E-RABs-ToBeReleased-RelReqItem-Split-Bearer,
    ...
}

E-RABs-ToBeReleased-RelReqItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    uL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-RelReqItem-SCG-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-RelReqItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-RelReqItem-Split-Bearer ::= SEQUENCE {
```

```
    e-RAB-ID                        E-RAB-ID,
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-RelReqItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-RelReqItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SENB RELEASE REQUIRED
--
-- **************************************************************

SeNBReleaseRequired ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBReleaseRequired-IEs}},
    ...
}

SeNBReleaseRequired-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SENB RELEASE CONFIRM
--
-- **************************************************************

SeNBReleaseConfirm ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBReleaseConfirm-IEs}},
    ...
}

SeNBReleaseConfirm-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
    { ID id-E-RABs-ToBeReleased-List-RelConf        CRITICALITY ignore  TYPE E-RABs-ToBeReleased-List-RelConf    PRESENCE optional}|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics             PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY ignore  TYPE UE-X2AP-ID-Extension               PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension               CRITICALITY ignore  TYPE UE-X2AP-ID-Extension               PRESENCE optional},
    ...
}
E-RABs-ToBeReleased-List-RelConf ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-RelConfItemIEs} }

E-RABs-ToBeReleased-RelConfItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-RelConfItem     CRITICALITY ignore      TYPE E-RABs-ToBeReleased-RelConfItem    PRESENCE mandatory},
```

```
    ...
}

E-RABs-ToBeReleased-RelConfItem ::= CHOICE {
    sCG-Bearer      E-RABs-ToBeReleased-RelConfItem-SCG-Bearer,
    split-Bearer    E-RABs-ToBeReleased-RelConfItem-Split-Bearer,
    ...
}

E-RABs-ToBeReleased-RelConfItem-SCG-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    uL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-RelConfItem-SCG-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-RelConfItem-SCG-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-RelConfItem-Split-Bearer ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-RelConfItem-Split-BearerExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-RelConfItem-Split-BearerExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SENB COUNTER CHECK REQUEST
--
-- **************************************************************

SeNBCounterCheckRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SeNBCounterCheckRequest-IEs}},
    ...
}

SeNBCounterCheckRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
    { ID id-E-RABs-SubjectToCounterCheck-List   CRITICALITY ignore  TYPE E-RABs-SubjectToCounterCheck-List   PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY ignore  TYPE UE-X2AP-ID-Extension               PRESENCE optional}|
    { ID id-SeNB-UE-X2AP-ID-Extension           CRITICALITY ignore  TYPE UE-X2AP-ID-Extension               PRESENCE optional},
    ...
}
E-RABs-SubjectToCounterCheck-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-SubjectToCounterCheckItemIEs} }

E-RABs-SubjectToCounterCheckItemIEs X2AP-PROTOCOL-IES ::= {
```

```
    { ID id-E-RABs-SubjectToCounterCheckItem    CRITICALITY ignore  TYPE E-RABs-SubjectToCounterCheckItem    PRESENCE mandatory},
    ...
}

E-RABs-SubjectToCounterCheckItem ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    uL-Count                        INTEGER (0..4294967295),
    dL-Count                        INTEGER (0..4294967295),
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-SubjectToCounterCheckItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-SubjectToCounterCheckItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- X2 REMOVAL REQUEST
--
-- **************************************************************

X2RemovalRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2RemovalRequest-IEs}},
    ...
}

X2RemovalRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID                CRITICALITY reject  TYPE GlobalENB-ID           PRESENCE mandatory}|
    { ID id-X2RemovalThreshold          CRITICALITY reject  TYPE X2BenefitValue         PRESENCE optional},
...
}



-- **************************************************************
--
-- X2 REMOVAL RESPONSE
--
-- **************************************************************

X2RemovalResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2RemovalResponse-IEs}},
    ...
}

X2RemovalResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID                CRITICALITY reject  TYPE GlobalENB-ID               PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional},
    ...
}
```

```
-- **************************************************************
--
-- X2 REMOVAL FAILURE
--
-- **************************************************************

X2RemovalFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{X2RemovalFailure-IEs}},
    ...
}

X2RemovalFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional},
    ...
}


-- **************************************************************
--
-- RETRIEVE UE CONTEXT REQUEST
--
-- **************************************************************

RetrieveUEContextRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ RetrieveUEContextRequest-IEs}},
    ...
}

RetrieveUEContextRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY reject  TYPE UE-X2AP-ID             PRESENCE mandatory}|
    { ID id-SeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension   PRESENCE optional}|
-- Allocated at the new eNB.
-- This IE contains an Extended eNB UE X2AP ID, which, together with the New eNB UE X2AP ID IE
-- represents the eNB UE X2AP ID allocated at the new eNB.
    { ID id-resumeID                            CRITICALITY reject  TYPE ResumeID               PRESENCE mandatory}|
    { ID id-ShortMAC-I                          CRITICALITY reject  TYPE ShortMAC-I             PRESENCE mandatory}|
    { ID id-NewEUTRANCellIdentifier             CRITICALITY reject  TYPE EUTRANCellIdentifier   PRESENCE mandatory}|
    {ID id-FailureCellCRNTI                     CRITICALITY reject  TYPE CRNTI                  PRESENCE optional}|
    {ID id-FailureCellPCI                       CRITICALITY reject  TYPE PCI                    PRESENCE optional},
    ...
}

-- **************************************************************
--
-- RETRIEVE UE CONTEXT RESPONSE
--
-- **************************************************************

RetrieveUEContextResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container    {{ RetrieveUEContextResponse-IEs}},
    ...
}
```

```
RetrieveUEContextResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-New-eNB-UE-X2AP-ID                          CRITICALITY ignore  TYPE UE-X2AP-ID                                 PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID-Extension                CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                       PRESENCE optional}|
    { ID id-Old-eNB-UE-X2AP-ID                          CRITICALITY ignore  TYPE UE-X2AP-ID                                 PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension                CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                       PRESENCE optional}|
    { ID id-GUMMEI-ID                                   CRITICALITY reject  TYPE GUMMEI                                     PRESENCE mandatory}|
    { ID id-UE-ContextInformationRetrieve               CRITICALITY reject  TYPE UE-ContextInformationRetrieve              PRESENCE mandatory}|
    { ID id-TraceActivation                             CRITICALITY ignore  TYPE TraceActivation                            PRESENCE optional}|
    { ID id-SRVCCOperationPossible                      CRITICALITY ignore  TYPE SRVCCOperationPossible                     PRESENCE optional}|
    { ID id-Masked-IMEISV                               CRITICALITY ignore  TYPE Masked-IMEISV                              PRESENCE optional}|
    { ID id-ExpectedUEBehaviour                         CRITICALITY ignore  TYPE ExpectedUEBehaviour                        PRESENCE optional}|
    { ID id-ProSeAuthorized                             CRITICALITY ignore  TYPE ProSeAuthorized                            PRESENCE optional}|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                     PRESENCE optional}|
    { ID id-V2XServicesAuthorized                       CRITICALITY ignore  TYPE V2XServicesAuthorized                      PRESENCE optional}|
    { ID id-AerialUEsubscriptionInformation             CRITICALITY ignore  TYPE AerialUEsubscriptionInformation            PRESENCE optional}|
    { ID id-Subscription-Based-UE-DifferentiationInfo   CRITICALITY ignore  TYPE Subscription-Based-UE-DifferentiationInfo  PRESENCE optional}|
    { ID id-NRV2XServicesAuthorized                     CRITICALITY ignore  TYPE NRV2XServicesAuthorized                    PRESENCE optional}|
    { ID id-PC5QoSParameters                            CRITICALITY ignore  TYPE PC5QoSParameters                           PRESENCE optional },
    ...
}

UE-ContextInformationRetrieve ::= SEQUENCE {
    mME-UE-S1AP-ID                      UE-S1AP-ID,
    uESecurityCapabilities              UESecurityCapabilities,
    aS-SecurityInformation              AS-SecurityInformation,
    uEaggregateMaximumBitRate           UEAggregateMaximumBitRate,
    subscriberProfileIDforRFP           SubscriberProfileIDforRFP       OPTIONAL,
    e-RABs-ToBeSetup-ListRetrieve       E-RABs-ToBeSetup-ListRetrieve,
    rRC-Context                         RRC-Context,
    handoverRestrictionList             HandoverRestrictionList         OPTIONAL,
    locationReportingInformation        LocationReportingInformation    OPTIONAL,
    managBasedMDTallowed                ManagementBasedMDTallowed       OPTIONAL,
    managBasedMDTPLMNList               MDTPLMNList                     OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {UE-ContextInformationRetrieve-ExtIEs} } OPTIONAL,
    ...
}

UE-ContextInformationRetrieve-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
{ ID id-UESidelinkAggregateMaximumBitRate   CRITICALITY ignore  EXTENSION UESidelinkAggregateMaximumBitRate     PRESENCE optional }|
{ ID id-AdditionalRRMPriorityIndex          CRITICALITY ignore  EXTENSION AdditionalRRMPriorityIndex            PRESENCE optional }|
{ ID id-EPCHandoverRestrictionListContainer CRITICALITY ignore  EXTENSION EPCHandoverRestrictionListContainer   PRESENCE optional }|
{ ID id-NRUESidelinkAggregateMaximumBitRate CRITICALITY ignore  EXTENSION NRUESidelinkAggregateMaximumBitRate   PRESENCE optional}|
{ ID id-UERadioCapabilityID                 CRITICALITY reject  EXTENSION UERadioCapabilityID                   PRESENCE optional }|
{ ID id-IMSvoiceEPSfallbackfrom5G           CRITICALITY ignore  EXTENSION IMSvoiceEPSfallbackfrom5G             PRESENCE optional },
    ...
}

E-RABs-ToBeSetup-ListRetrieve ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeSetupRetrieve-ItemIEs} }

E-RABs-ToBeSetupRetrieve-ItemIEs    X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeSetupRetrieve-Item   CRITICALITY ignore  TYPE E-RABs-ToBeSetupRetrieve-Item   PRESENCE mandatory},
    ...
}
```

```
E-RABs-ToBeSetupRetrieve-Item ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    e-RAB-Level-QoS-Parameters      E-RAB-Level-QoS-Parameters,
    bearerType                      BearerType  OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeSetupRetrieve-ItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeSetupRetrieve-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uL-GTPtunnelEndpoint        CRITICALITY reject  EXTENSION GTPtunnelEndpoint     PRESENCE mandatory}|
    { ID id-dL-Forwarding               CRITICALITY ignore  EXTENSION DL-Forwarding         PRESENCE optional}|
    { ID id-Ethernet-Type               CRITICALITY ignore  EXTENSION Ethernet-Type         PRESENCE optional}|
    { ID id-SecurityIndication          CRITICALITY reject  EXTENSION SecurityIndication            PRESENCE optional},
    ...
}

-- **************************************************************
--
-- RETRIEVE UE CONTEXT FAILURE
--
-- **************************************************************

RetrieveUEContextFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ RetrieveUEContextFailure-IEs}},
    ...
}

RetrieveUEContextFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-New-eNB-UE-X2AP-ID              CRITICALITY ignore  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID-Extension    CRITICALITY ignore  TYPE UE-X2AP-ID-Extension       PRESENCE optional}|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SGNB ADDITION REQUEST
--
-- **************************************************************

SgNBAdditionRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container {{SgNBAdditionRequest-IEs}},
    ...
}

SgNBAdditionRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-NRUESecurityCapabilities                    CRITICALITY reject  TYPE NRUESecurityCapabilities            PRESENCE mandatory}|
    { ID id-SgNBSecurityKey                         CRITICALITY reject  TYPE SgNBSecurityKey                         PRESENCE mandatory}|
    { ID id-SgNBUEAggregateMaximumBitRate           CRITICALITY reject  TYPE UEAggregateMaximumBitRate               PRESENCE mandatory}|
    { ID id-SelectedPLMN                                CRITICALITY ignore  TYPE PLMN-Identity                       PRESENCE optional}|
    { ID id-HandoverRestrictionList                 CRITICALITY ignore  TYPE HandoverRestrictionList                 PRESENCE optional}|
    { ID id-E-RABs-ToBeAdded-SgNBAddReqList         CRITICALITY reject  TYPE E-RABs-ToBeAdded-SgNBAddReqList         PRESENCE mandatory}|
```

```
    { ID id-MeNBtoSgNBContainer                    CRITICALITY reject  TYPE MeNBtoSgNBContainer                      PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                        CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                          PRESENCE optional}|
    { ID id-ExpectedUEBehaviour                    CRITICALITY ignore  TYPE ExpectedUEBehaviour                      PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension              CRITICALITY reject  TYPE UE-X2AP-ID-Extension                  PRESENCE optional}|
    { ID id-RequestedSplitSRBs                     CRITICALITY reject  TYPE SplitSRBs                             PRESENCE optional}|
    { ID id-MeNBResourceCoordinationInformation    CRITICALITY ignore  TYPE MeNBResourceCoordinationInformation   PRESENCE optional}|
    { ID id-SGNB-Addition-Trigger-Ind              CRITICALITY reject  TYPE SGNB-Addition-Trigger-Ind              PRESENCE optional}|
    { ID id-SubscriberProfileIDforRFP              CRITICALITY ignore  TYPE SubscriberProfileIDforRFP             PRESENCE optional}|
    { ID id-MeNBCell-ID                            CRITICALITY reject  TYPE ECGI                                  PRESENCE mandatory}|
    { ID id-DesiredActNotificationLevel            CRITICALITY ignore  TYPE DesiredActNotificationLevel           PRESENCE optional}|
    { ID id-TraceActivation                        CRITICALITY ignore  TYPE TraceActivation                       PRESENCE optional}|
    { ID id-LocationInformationSgNBReporting           CRITICALITY ignore  TYPE LocationInformationSgNBReporting        PRESENCE optional}|
    { ID id-Masked-IMEISV                          CRITICALITY ignore  TYPE Masked-IMEISV                         PRESENCE optional}|
    { ID id-AdditionalRRMPriorityIndex             CRITICALITY ignore  TYPE AdditionalRRMPriorityIndex            PRESENCE optional}|
    { ID id-RequestedFastMCGRecoveryViaSRB3        CRITICALITY ignore  TYPE     RequestedFastMCGRecoveryViaSRB3      PRESENCE optional}|
    { ID id-UEContextReferenceatSourceNGRAN        CRITICALITY ignore  TYPE RAN-UE-NGAP-ID                        PRESENCE optional}|
    { ID id-ManagementBasedMDTallowed              CRITICALITY ignore  TYPE ManagementBasedMDTallowed             PRESENCE optional }|
    { ID id-ManagementBasedMDTPLMNList             CRITICALITY ignore  TYPE MDTPLMNList                           PRESENCE optional }|
    { ID id-UERadioCapabilityID                    CRITICALITY reject  TYPE UERadioCapabilityID                   PRESENCE optional}|
    { ID id-IABNodeIndication                      CRITICALITY reject  TYPE IABNodeIndication                     PRESENCE optional}|
    { ID id-sourceNG-RAN-node-id                   CRITICALITY ignore      TYPE Global-RAN-NODE-ID                PRESENCE optional}|
    { ID id-UE-HistoryInformation                  CRITICALITY ignore  TYPE UE-HistoryInformation                 PRESENCE optional}|
    { ID id-UE-HistoryInformationFromTheUE         CRITICALITY ignore  TYPE UE-HistoryInformationFromTheUE        PRESENCE optional}|
    { ID id-PSCellChangeHistory            CRITICALITY ignore  TYPE PSCellChangeHistory                        PRESENCE optional}|
    { ID id-CHOinformation-AddReq                  CRITICALITY reject  TYPE CHOinformation-AddReq                 PRESENCE optional}|
    { ID id-SCGActivationRequest                   CRITICALITY ignore  TYPE SCGActivationRequest                  PRESENCE optional}|
    { ID id-CPAinformation-REQ                     CRITICALITY reject  TYPE CPAinformation-REQ                    PRESENCE optional},
    ...
}

E-RABs-ToBeAdded-SgNBAddReqList ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeAdded-SgNBAddReq-ItemIEs} }

E-RABs-ToBeAdded-SgNBAddReq-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeAdded-SgNBAddReq-Item       CRITICALITY reject  TYPE E-RABs-ToBeAdded-SgNBAddReq-Item    PRESENCE mandatory},
    ...
}

E-RABs-ToBeAdded-SgNBAddReq-Item ::= SEQUENCE {
    e-RAB-ID                                E-RAB-ID,
    drb-ID                                  DRB-ID,
    en-DC-ResourceConfiguration             EN-DC-ResourceConfiguration,
    resource-configuration                  CHOICE {
        sgNBPDCPpresent                         E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                      E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBAddReq-ItemExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBAddReq-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPpresent ::= SEQUENCE {
    full-E-RAB-Level-QoS-Parameters             E-RAB-Level-QoS-Parameters,
    max-MCG-admit-E-RAB-Level-QoS-Parameters    GBR-QosInformation                                                  OPTIONAL,
-- This IE shall be present if MCG resource and SCG resources IEs in the EN-DC Resource Configuration IE are set to “present” and GBR QoS
Information IE is present in Full E-RAB Level QoS Parameters IE --
    dL-Forwarding                               DL-Forwarding                                                       OPTIONAL,
    meNB-DL-GTP-TEIDatMCG                       GTPtunnelEndpoint                                                   OPTIONAL,
-- This IE shall be present if MCG resource IE in the EN-DC Resource Configuration IE is set to “present” --
    s1-UL-GTPtunnelEndpoint                     GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-RLCMode-transferred                 CRITICALITY ignore  EXTENSION RLCMode                   PRESENCE optional}|
    { ID id-BearerType                          CRITICALITY ignore  EXTENSION BearerType                PRESENCE optional}|
    { ID id-Ethernet-Type                       CRITICALITY ignore  EXTENSION Ethernet-Type             PRESENCE optional}|
    { ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress     PRESENCE optional}|
    { ID id-SecurityIndication                  CRITICALITY reject  EXTENSION SecurityIndication        PRESENCE optional}|
    { ID id-SourceNodeDLForwardingIPAddress     CRITICALITY ignore  EXTENSION TransportLayerAddress     PRESENCE optional},
    ...
}

E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    requested-SCG-E-RAB-Level-QoS-Parameters        E-RAB-Level-QoS-Parameters,
    meNB-UL-GTP-TEIDatPDCP                          GTPtunnelEndpoint,
    secondary-meNB-UL-GTP-TEIDatPDCP                GTPtunnelEndpoint   OPTIONAL,
    rlc-Mode                                        RLCMode,
    uL-Configuration                                ULConfiguration OPTIONAL,
-- This IE shall be present if MCG resource and SCG resources IEs in the EN-DC Resource Configuration IE are set to “present” --
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPnotpresentExtIEs} }     OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBAddReq-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength              PRESENCE optional}|
    { ID id-dLPDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength              PRESENCE optional}|
    { ID id-duplicationActivation                   CRITICALITY ignore  EXTENSION DuplicationActivation     PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SGNB ADDITION REQUEST ACKNOWLEDGE
--
-- **************************************************************

SgNBAdditionRequestAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBAdditionRequestAcknowledge-IEs}},
    ...
}

SgNBAdditionRequestAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory}|
```

```
    { ID id-SgNB-UE-X2AP-ID                              CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                              PRESENCE mandatory}|
    { ID id-E-RABs-Admitted-ToBeAdded-SgNBAddReqAckList CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-SgNBAddReqAckList   PRESENCE mandatory}|
    { ID id-E-RABs-NotAdmitted-List                      CRITICALITY ignore  TYPE E-RAB-List                                 PRESENCE optional}|
    { ID id-SgNBtoMeNBContainer                          CRITICALITY reject  TYPE SgNBtoMeNBContainer                       PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics                       CRITICALITY ignore  TYPE CriticalityDiagnostics                    PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension                    CRITICALITY reject  TYPE UE-X2AP-ID-Extension                      PRESENCE optional}|
    { ID id-AdmittedSplitSRBs                            CRITICALITY reject  TYPE SplitSRBs                                 PRESENCE optional}|
    { ID id-SgNBResourceCoordinationInformation          CRITICALITY ignore  TYPE SgNBResourceCoordinationInformation       PRESENCE optional}|
    { ID id-RRCConfigIndication                          CRITICALITY reject  TYPE RRC-Config-Ind                            PRESENCE optional}|
    { ID id-LocationInformationSgNB                      CRITICALITY ignore  TYPE LocationInformationSgNB                   PRESENCE optional}|
    { ID id-AvailableFastMCGRecoveryViaSRB3              CRITICALITY ignore  TYPE AvailableFastMCGRecoveryViaSRB3           PRESENCE optional }|
    { ID id-DirectForwardingPathAvailability             CRITICALITY ignore  TYPE DirectForwardingPathAvailability          PRESENCE optional }|
    { ID id-SCGActivationStatus                          CRITICALITY ignore  TYPE SCGActivationStatus                       PRESENCE optional}|
    { ID id-CPAinformation-REQ-ACK                       CRITICALITY ignore  TYPE CPAinformation-REQ-ACK                    PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBAddReqAckList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeAdded-
SgNBAddReqAck-ItemIEs} }

E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item    CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item    PRESENCE
mandatory}
}

E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBAddReqAck-ItemExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBAddReqAck-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPpresent ::= SEQUENCE {
    s1-DL-GTPtunnelEndpoint                 GTPtunnelEndpoint,
    sgNB-UL-GTP-TEIDatPDCP                  GTPtunnelEndpoint                                                                   OPTIONAL,
-- This IE shall be present if MCG resource IE in the EN-DC Resource Configuration IE is set to “present” --
    rlc-Mode                                RLCMode                                                                             OPTIONAL,
-- This IE shall be present if MCG resource IE in the EN-DC Resource Configuration IE is set to “present” --
    dL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                                                   OPTIONAL,
    uL-Forwarding-GTPtunnelEndpoint         GTPtunnelEndpoint                                                                   OPTIONAL,
    mCG-E-RAB-Level-QoS-Parameters          E-RAB-Level-QoS-Parameters                                                          OPTIONAL,
-- This IE shall be present if MCG resource and SCG resource IEs in the EN-DC Resource Configuration IE are set to “present” and the GBR QoS
Information IE is present in the Requested MCG E-RAB Level QoS Parameters IE --
    uL-Configuration                        ULConfiguration                                                                     OPTIONAL,
-- This IE shall be present if MCG resource and SCG resources IEs in the EN-DC Resource Configuration IE are set to “present” --
```

```
    iE-Extensions                           ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPpresentExtIEs} }
        OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional}|
    { ID id-dLPDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional}|
    { ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress    PRESENCE optional}|
    { ID id-SecurityResult                          CRITICALITY ignore  EXTENSION SecurityResult             PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    sgNB-DL-GTP-TEIDatSCG               GTPtunnelEndpoint,
    secondary-sgNB-DL-GTP-TEIDatSCG     GTPtunnelEndpoint       OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPnotpresentExtIEs} }
    OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-lCID                CRITICALITY ignore  EXTENSION LCID          PRESENCE optional},
    ...
}



-- ************************************************************
--
-- SGNB ADDITION REQUEST REJECT
--
-- ************************************************************

SgNBAdditionRequestReject ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBAdditionRequestReject-IEs}},
    ...
}

SgNBAdditionRequestReject-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                 CRITICALITY reject  TYPE SgNB-UE-X2AP-ID            PRESENCE optional}|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics     PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional},
    ...
}


-- ************************************************************
--
-- SGNB RECONFIGURATION COMPLETE
--
-- ************************************************************
```

```
SgNBReconfigurationComplete ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBReconfigurationComplete-IEs}},
    ...
}

SgNBReconfigurationComplete-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                              PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                        PRESENCE mandatory}|
    { ID id-ResponseInformationSgNBReconfComp       CRITICALITY ignore  TYPE ResponseInformationSgNBReconfComp      PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY reject  TYPE UE-X2AP-ID-Extension                   PRESENCE optional},
    ...
}

ResponseInformationSgNBReconfComp ::= CHOICE {
    success-SgNBReconfComp              ResponseInformationSgNBReconfComp-SuccessItem,
    reject-by-MeNB-SgNBReconfComp       ResponseInformationSgNBReconfComp-RejectByMeNBItem,
    ...
}

ResponseInformationSgNBReconfComp-SuccessItem ::= SEQUENCE {
    meNBtoSgNBContainer             MeNBtoSgNBContainer                 OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {ResponseInformationSgNBReconfComp-SuccessItemExtIEs} } OPTIONAL,
    ...
}

ResponseInformationSgNBReconfComp-SuccessItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResponseInformationSgNBReconfComp-RejectByMeNBItem ::= SEQUENCE {
    cause                           Cause,
    iE-Extensions                   ProtocolExtensionContainer { {ResponseInformationSgNBReconfComp-RejectByMeNBItemExtIEs} }     OPTIONAL,
    ...
}

ResponseInformationSgNBReconfComp-RejectByMeNBItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- **************************************************************
--
-- SGNB MODIFICATION REQUEST
--
-- **************************************************************

SgNBModificationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ SgNBModificationRequest-IEs}},
    ...
}

SgNBModificationRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                              PRESENCE mandatory}|
```

```
    { ID id-SgNB-UE-X2AP-ID                          CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-Cause                                    CRITICALITY ignore  TYPE Cause                                  PRESENCE mandatory}|
    { ID id-SelectedPLMN                             CRITICALITY ignore  TYPE PLMN-Identity                         PRESENCE optional}|
    { ID id-HandoverRestrictionList                  CRITICALITY ignore  TYPE HandoverRestrictionList               PRESENCE optional}|
    { ID    id-SCGConfigurationQuery                 CRITICALITY ignore  TYPE SCGConfigurationQuery                 PRESENCE optional}|
    { ID id-UE-ContextInformation-SgNBModReq         CRITICALITY reject  TYPE UE-ContextInformation-SgNBModReq      PRESENCE optional}|
    { ID id-MeNBtoSgNBContainer                      CRITICALITY reject  TYPE MeNBtoSgNBContainer                   PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension                CRITICALITY reject  TYPE UE-X2AP-ID-Extension                  PRESENCE optional}|
    { ID id-MeNBResourceCoordinationInformation      CRITICALITY ignore  TYPE MeNBResourceCoordinationInformation   PRESENCE optional}|
    { ID id-RequestedSplitSRBs                       CRITICALITY ignore  TYPE SplitSRBs                             PRESENCE optional}|
    { ID id-RequestedSplitSRBsrelease                CRITICALITY ignore  TYPE SplitSRBs                             PRESENCE optional}|
    { ID id-DesiredActNotificationLevel              CRITICALITY ignore  TYPE DesiredActNotificationLevel           PRESENCE optional}|
    { ID id-LocationInformationSgNBReporting         CRITICALITY ignore  TYPE LocationInformationSgNBReporting      PRESENCE optional}|
    { ID id-MeNBCell-ID                              CRITICALITY ignore  TYPE ECGI                                  PRESENCE optional}|
    { ID id-RequestedFastMCGRecoveryViaSRB3          CRITICALITY ignore  TYPE RequestedFastMCGRecoveryViaSRB3        PRESENCE optional}|
    { ID id-RequestedFastMCGRecoveryViaSRB3Release   CRITICALITY ignore  TYPE RequestedFastMCGRecoveryViaSRB3Release PRESENCE optional}|
    { ID id-SNtriggered                              CRITICALITY ignore  TYPE SNtriggered                           PRESENCE optional}|
    { ID id-IABNodeIndication                        CRITICALITY reject  TYPE IABNodeIndication                     PRESENCE optional}|
    { ID id-PSCellHistoryInformationRetrieve         CRITICALITY ignore  TYPE PSCellHistoryInformationRetrieve      PRESENCE optional}|
    { ID id-UE-HistoryInformationFromTheUE           CRITICALITY ignore  TYPE UE-HistoryInformationFromTheUE        PRESENCE optional}|
    { ID id-CHOinformation-ModReq                    CRITICALITY ignore  TYPE CHOinformation-ModReq                 PRESENCE optional}|
    { ID id-SCGActivationRequest                     CRITICALITY ignore  TYPE SCGActivationRequest                  PRESENCE optional}|
    { ID id-CPAinformation-MOD                       CRITICALITY ignore  TYPE CPAinformation-MOD                    PRESENCE optional}|
    { ID id-CPCupdate-MOD                            CRITICALITY ignore  TYPE CPCupdate-MOD                         PRESENCE optional},
    ...
}

UE-ContextInformation-SgNBModReq ::= SEQUENCE {
    nRUE-SecurityCapabilities       NRUESecurityCapability                                          OPTIONAL,
    sgNB-SecurityKey                SgNBSecurityKey                                                 OPTIONAL,
    sgNBUEAggregateMaximumBitRate   UEAggregateMaximumBitRate                                       OPTIONAL,
    e-RABs-ToBeAdded                E-RABs-ToBeAdded-SgNBModReq-List                                OPTIONAL,
    e-RABs-ToBeModified             E-RABs-ToBeModified-SgNBModReq-List                             OPTIONAL,
    e-RABs-ToBeReleased             E-RABs-ToBeReleased-SgNBModReq-List                             OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {UE-ContextInformationSgNBModReqExtIEs} }      OPTIONAL,
    ...
}

UE-ContextInformationSgNBModReqExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {   ID id-SubscriberProfileIDforRFP                CRITICALITY ignore  EXTENSION SubscriberProfileIDforRFP                      PRESENCE
optional}|
    { ID id-AdditionalRRMPriorityIndex              CRITICALITY ignore  EXTENSION AdditionalRRMPriorityIndex             PRESENCE optional}|
    {ID id-LowerLayerPresenceStatusChange           CRITICALITY ignore  EXTENSION LowerLayerPresenceStatusChange         PRESENCE optional},
    ...
}

E-RABs-ToBeAdded-SgNBModReq-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeAdded-SgNBModReq-ItemIEs} }

E-RABs-ToBeAdded-SgNBModReq-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeAdded-SgNBModReq-Item        CRITICALITY ignore  TYPE E-RABs-ToBeAdded-SgNBModReq-Item    PRESENCE mandatory},
    ...
}

E-RABs-ToBeAdded-SgNBModReq-Item ::= SEQUENCE {
```

```
    e-RAB-ID                                E-RAB-ID,
    drb-ID                                  DRB-ID,
    en-DC-ResourceConfiguration             EN-DC-ResourceConfiguration,
    resource-configuration                  CHOICE {
        sgNBPDCPpresent                     E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBModReq-ItemExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBModReq-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPpresent ::= SEQUENCE {
    full-E-RAB-Level-QoS-Parameters         E-RAB-Level-QoS-Parameters,
    max-MN-admit-E-RAB-Level-QoS-Parameters GBR-QosInformation                                              OPTIONAL,
-- This IE shall be present if MCG resource and SCG resources IEs in the EN-DC Resource Configuration IE are set to “present” and GBR QoS
Information IE is present in Full E-RAB Level QoS Parameters IE --
    dL-Forwarding                           DL-Forwarding                                                   OPTIONAL,
    meNB-DL-GTP-TEIDatMCG                   GTPtunnelEndpoint                                               OPTIONAL,
-- This IE shall be present if MCG resource IE in the EN-DC Resource Configuration IE is set to “present” --
    s1-UL-GTPtunnelEndpoint                 GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPpresentExtIEs} }    OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-RLCMode-transferred                 CRITICALITY ignore  EXTENSION RLCMode               PRESENCE optional}|
    { ID id-BearerType                          CRITICALITY ignore  EXTENSION BearerType            PRESENCE optional}|
    { ID id-Ethernet-Type                       CRITICALITY ignore  EXTENSION Ethernet-Type         PRESENCE optional}|
    { ID id-SecurityIndication                  CRITICALITY reject  EXTENSION SecurityIndication    PRESENCE optional}|
    { ID id-SourceDLForwardingIPAddress         CRITICALITY ignore  EXTENSION TransportLayerAddress PRESENCE optional},
    ...
}

E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    requested-SCG-E-RAB-Level-QoS-Parameters        E-RAB-Level-QoS-Parameters,
    meNB-UL-GTP-TEIDatPDCP                          GTPtunnelEndpoint,
    secondary-meNB-UL-GTP-TEIDatPDCP                GTPtunnelEndpoint       OPTIONAL,
    rlc-Mode                                        RLCMode,
    uL-Configuration                                ULConfiguration                                         OPTIONAL,
-- This IE shall be present if MCG resource and SCG resources IEs in the EN-DC Resource Configuration IE are set to “present” --
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPnotpresentExtIEs} }    OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBModReq-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                      CRITICALITY ignore  EXTENSION PDCPSnLength          PRESENCE optional}|
    { ID id-dLPDCPSnLength                      CRITICALITY ignore  EXTENSION PDCPSnLength          PRESENCE optional}|
    { ID id-duplicationActivation               CRITICALITY ignore  EXTENSION DuplicationActivation PRESENCE optional},
    ...
```

```
}


E-RABs-ToBeModified-SgNBModReq-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeModified-SgNBModReq-
ItemIEs} }

E-RABs-ToBeModified-SgNBModReq-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeModified-SgNBModReq-Item CRITICALITY ignore  TYPE E-RABs-ToBeModified-SgNBModReq-Item     PRESENCE mandatory},
    ...
}


E-RABs-ToBeModified-SgNBModReq-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeModified-SgNBModReq-ItemExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeModified-SgNBModReq-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPpresent ::= SEQUENCE {
    full-E-RAB-Level-QoS-Parameters         E-RAB-Level-QoS-Parameters                                                          OPTIONAL,
    max-MN-admit-E-RAB-Level-QoS-Parameters GBR-QosInformation                                                                  OPTIONAL,
    meNB-DL-GTP-TEIDatMCG                   GTPtunnelEndpoint                                                                   OPTIONAL,
    s1-UL-GTPtunnelEndpoint                 GTPtunnelEndpoint                                                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-RLC-Status          CRITICALITY ignore          EXTENSION RLC-Status                PRESENCE optional },
    ...
}

E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    requested-SCG-E-RAB-Level-QoS-Parameters        E-RAB-Level-QoS-Parameters                                                      OPTIONAL,
    meNB-UL-GTP-TEIDatPDCP                      GTPtunnelEndpoint                                                                   OPTIONAL,
    uL-Configuration                            ULConfiguration                                                                     OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPnotpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeModified-SgNBModReq-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength          PRESENCE optional}|
    { ID id-dLPDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength          PRESENCE optional}|
    { ID id-secondarymeNBULGTPTEIDatPDCP            CRITICALITY ignore  EXTENSION GTPtunnelEndpoint     PRESENCE optional},
```

```
    ...
}

E-RABs-ToBeReleased-SgNBModReq-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-SgNBModReq-
ItemIEs} }

E-RABs-ToBeReleased-SgNBModReq-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-SgNBModReq-Item CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBModReq-Item     PRESENCE mandatory},
    ...
}

E-RABs-ToBeReleased-SgNBModReq-Item ::= SEQUENCE {
    e-RAB-ID                                E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBModReq-ItemExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBModReq-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPpresent ::= SEQUENCE {
    dL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                                   OPTIONAL,
    uL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                                   OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPpresentExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPnotpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBModReq-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}



-- **************************************************************
--
-- SGNB MODIFICATION REQUEST ACKNOWLEDGE
--
-- **************************************************************
```

```
SgNBModificationRequestAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBModificationRequestAcknowledge-IEs}},
    ...
}

SgNBModificationRequestAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                             CRITICALITY ignore  TYPE UE-X2AP-ID                                    PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                             CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                                 PRESENCE
mandatory}|
    { ID id-E-RABs-Admitted-ToBeAdded-SgNBModAckList        CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-SgNBModAckList        PRESENCE
optional}|
    { ID id-E-RABs-Admitted-ToBeModified-SgNBModAckList CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeModified-SgNBModAckList   PRESENCE optional}|
    { ID id-E-RABs-Admitted-ToBeReleased-SgNBModAckList CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeReleased-SgNBModAckList   PRESENCE optional}|
    { ID id-E-RABs-NotAdmitted-List                     CRITICALITY ignore  TYPE E-RAB-List                                   PRESENCE optional}|
    { ID id-SgNBtoMeNBContainer                         CRITICALITY ignore  TYPE SgNBtoMeNBContainer                          PRESENCE optional}|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                       PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension                   CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                         PRESENCE optional}|
    { ID id-SgNBResourceCoordinationInformation         CRITICALITY ignore  TYPE SgNBResourceCoordinationInformation          PRESENCE optional}|
    { ID id-AdmittedSplitSRBs                           CRITICALITY ignore  TYPE SplitSRBs                                    PRESENCE optional}|
    { ID id-AdmittedSplitSRBsrelease                        CRITICALITY ignore  TYPE SplitSRBs                                PRESENCE
optional}|
    { ID id-RRCConfigIndication                         CRITICALITY reject  TYPE RRC-Config-Ind                               PRESENCE optional}|
    { ID id-LocationInformationSgNB                     CRITICALITY ignore  TYPE LocationInformationSgNB                      PRESENCE optional}|
    { ID id-AvailableFastMCGRecoveryViaSRB3             CRITICALITY ignore  TYPE AvailableFastMCGRecoveryViaSRB3              PRESENCE optional}|
    { ID id-ReleaseFastMCGRecoveryViaSRB3       CRITICALITY ignore  TYPE ReleaseFastMCGRecoveryViaSRB3                        PRESENCE optional}|
    { ID id-SCG-UE-HistoryInformation                   CRITICALITY ignore  TYPE SCG-UE-HistoryInformation                  PRESENCE optional}|
    { ID id-SCGActivationStatus                         CRITICALITY ignore  TYPE SCGActivationStatus                          PRESENCE optional}|
    { ID id-CPAinformation-MOD-ACK                      CRITICALITY reject  TYPE CPAinformation-MOD-ACK                       PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBModAckList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeAdded-
SgNBModAck-ItemIEs} }

E-RABs-Admitted-ToBeAdded-SgNBModAck-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeAdded-SgNBModAck-Item   CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeAdded-SgNBModAck-Item     PRESENCE mandatory}
}

E-RABs-Admitted-ToBeAdded-SgNBModAck-Item ::= SEQUENCE {
    e-RAB-ID                                E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-SgNBModAck-ItemExtIEs} }  OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBModAck-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPpresent ::= SEQUENCE {
    s1-DL-GTPtunnelEndpoint                GTPtunnelEndpoint,
    sgNB-UL-GTP-TEIDatPDCP                 GTPtunnelEndpoint                                                    OPTIONAL,
-- This IE shall be present if MCG resource IE in the EN-DC Resource Configuration IE are set to “present” --
    rlc-Mode                                   RLCMode                                                          OPTIONAL,
-- This IE shall be present if MCG resource IE in the EN-DC Resource Configuration IE are set to “present” --
    dL-Forwarding-GTPtunnelEndpoint        GTPtunnelEndpoint                                                    OPTIONAL,
    uL-Forwarding-GTPtunnelEndpoint        GTPtunnelEndpoint                                                    OPTIONAL,
    mCG-E-RAB-Level-QoS-Parameters         E-RAB-Level-QoS-Parameters                                           OPTIONAL,
-- This IE shall be present if MCG resource and SCG resource IEs in the EN-DC Resource Configuration IE are set to “present” and the GBR QoS
Information IE is present in the Requested MCG E-RAB Level QoS Parameters IE --
    uL-Configuration                       ULConfiguration                                                      OPTIONAL,
-- This IE shall be present if MCG resource and SCG resources IEs in the EN-DC Resource Configuration IE are set to “present” --
    iE-Extensions                  ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPpresentExtIEs} }    OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                     CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional}|
    { ID id-dLPDCPSnLength                     CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional}|
    { ID id-SecurityResult                     CRITICALITY ignore  EXTENSION SecurityResult          PRESENCE optional}|
    { ID id-SourceDLForwardingIPAddress        CRITICALITY ignore  EXTENSION TransportLayerAddress   PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    sgNB-DL-GTP-TEIDatSCG                  GTPtunnelEndpoint,
    secondary-sgNB-DL-GTP-TEIDatSCG        GTPtunnelEndpoint           OPTIONAL,
    iE-Extensions                  ProtocolExtensionContainer { {E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPnotpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeAdded-SgNBModAck-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {ID id-lCID            CRITICALITY ignore  EXTENSION LCID      PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeModified-SgNBModAckList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeModified-
SgNBModAck-ItemIEs} }

E-RABs-Admitted-ToBeModified-SgNBModAck-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeModified-SgNBModAck-Item       CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeModified-SgNBModAck-Item   PRESENCE
mandatory}
}

E-RABs-Admitted-ToBeModified-SgNBModAck-Item ::= SEQUENCE {
    e-RAB-ID                               E-RAB-ID,
    en-DC-ResourceConfiguration        EN-DC-ResourceConfiguration,
    resource-configuration             CHOICE {
        sgNBPDCPpresent                    E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                 E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPnotpresent,
        ...
    },
```

```
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeAdded-SgNBModAck-ItemExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeAdded-SgNBModAck-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPpresent ::= SEQUENCE {
    s1-DL-GTPtunnelEndpoint                 GTPtunnelEndpoint                                                                   OPTIONAL,
    sgNB-UL-GTP-TEIDatPDCP                  GTPtunnelEndpoint                                                                   OPTIONAL,
    mCG-E-RAB-Level-QoS-Parameters          E-RAB-Level-QoS-Parameters                                                          OPTIONAL,
    uL-Configuration                            ULConfiguration                                                                    OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional}|
    { ID id-dLPDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    sgNB-DL-GTP-TEIDatSCG                   GTPtunnelEndpoint
    OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPnotpresentExtIEs} }
    OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeModified-SgNBModAck-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-secondarysgNBDLGTPTEIDatPDCP            CRITICALITY ignore  EXTENSION GTPtunnelEndpoint          PRESENCE optional}|
    { ID id-RLC-Status                              CRITICALITY ignore          EXTENSION RLC-Status         PRESENCE optional },
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBModAckList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeReleased-
SgNBModAck-ItemIEs} }

E-RABs-Admitted-ToBeReleased-SgNBModAck-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeReleased-SgNBModAck-Item        CRITICALITY ignore  TYPE E-RABs-Admitted-ToReleased-SgNBModAck-Item     PRESENCE
mandatory}
}

E-RABs-Admitted-ToReleased-SgNBModAck-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBModAck-ItemExtIEs} }   OPTIONAL,
```

```
    ...
}

E-RABs-ToBeReleased-SgNBModAck-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPpresent ::= SEQUENCE {
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPpresentExtIEs} }     OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPnotpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBModAck-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- **************************************************************
--
-- SGNB MODIFICATION REQUEST REJECT
--
-- **************************************************************

SgNBModificationRequestReject ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBModificationRequestReject-IEs}},
    ...
}

SgNBModificationRequestReject-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID            PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                   PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics  PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY ignore  TYPE UE-X2AP-ID-Extension    PRESENCE optional},
    ...
}


-- **************************************************************
--
-- SGNB MODIFICATION REQUIRED
--
-- **************************************************************

SgNBModificationRequired ::= SEQUENCE {
```

```
    protocolIEs     ProtocolIE-Container        {{SgNBModificationRequired-IEs}},
    ...
}

SgNBModificationRequired-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-Cause                                   CRITICALITY ignore  TYPE Cause                               PRESENCE mandatory}|
    { ID id-PDCPChangeIndication                    CRITICALITY ignore  TYPE PDCPChangeIndication                PRESENCE optional}|
    { ID id-E-RABs-ToBeReleased-SgNBModReqdList     CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBModReqdList  PRESENCE optional}|
    { ID id-SgNBtoMeNBContainer                     CRITICALITY ignore  TYPE SgNBtoMeNBContainer                 PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY reject  TYPE UE-X2AP-ID-Extension                PRESENCE optional}|
    { ID id-E-RABs-ToBeModified-SgNBModReqdList     CRITICALITY ignore  TYPE E-RABs-ToBeModified-SgNBModReqdList  PRESENCE optional}|
    { ID id-SgNBResourceCoordinationInformation     CRITICALITY ignore  TYPE SgNBResourceCoordinationInformation  PRESENCE optional}|
    { ID id-RRCConfigIndication                     CRITICALITY reject  TYPE RRC-Config-Ind                      PRESENCE optional}|
    { ID id-LocationInformationSgNB                 CRITICALITY ignore  TYPE LocationInformationSgNB             PRESENCE optional}|
    { ID id-SCG-UE-HistoryInformation               CRITICALITY ignore  TYPE SCG-UE-HistoryInformation           PRESENCE optional}|
    { ID id-SCGActivationRequest                    CRITICALITY ignore  TYPE SCGActivationRequest                PRESENCE optional}|
    { ID id-CPACinformation-REQD                    CRITICALITY ignore  TYPE CPACinformation-REQD                PRESENCE optional}|
    { ID id-SCGreconfigNotification                 CRITICALITY ignore  TYPE SCGreconfigNotification             PRESENCE optional},
    ...
}


E-RABs-ToBeReleased-SgNBModReqdList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-SgNBModReqd-
ItemIEs} }

E-RABs-ToBeReleased-SgNBModReqd-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-SgNBModReqd-Item     CRITICALITY ignore     TYPE E-RABs-ToBeReleased-SgNBModReqd-Item    PRESENCE mandatory },
    ...
}

E-RABs-ToBeReleased-SgNBModReqd-Item ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    cause                       Cause,
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBModReqd-ItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBModReqd-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-RLCMode-transferred             CRITICALITY ignore  EXTENSION RLCMode           PRESENCE optional},
    ...
}

E-RABs-ToBeModified-SgNBModReqdList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeModified-SgNBModReqd-
ItemIEs} }

E-RABs-ToBeModified-SgNBModReqd-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeModified-SgNBModReqd-Item     CRITICALITY ignore     TYPE E-RABs-ToBeModified-SgNBModReqd-Item    PRESENCE mandatory },
    ...
}

E-RABs-ToBeModified-SgNBModReqd-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
```

```
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeModified-SgNBModReqd-ItemExtIEs} }  OPTIONAL,
    ...
}

E-RABs-ToBeModified-SgNBModReqd-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPpresent ::= SEQUENCE {
    requested-MCG-E-RAB-Level-QoS-Parameters        E-RAB-Level-QoS-Parameters                                      OPTIONAL,
    uL-Configuration                                ULConfiguration                                                 OPTIONAL,
    sgNB-UL-GTP-TEIDatPDCP                      GTPtunnelEndpoint                                                 OPTIONAL,
    s1-DL-GTP-TEIDatSgNB                            GTPtunnelEndpoint                                               OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPpresentExtIEs} }    OPTIONAL,
    ...
}

E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength          PRESENCE optional}|
    { ID id-dLPDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength          PRESENCE optional}|
    { ID id-new-drb-ID-req                          CRITICALITY ignore  EXTENSION NewDRBIDrequest       PRESENCE optional},
    ...
}

E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    sgNB-DL-GTP-TEIDatSCG               GTPtunnelEndpoint               OPTIONAL,
    secondary-sgNB-DL-GTP-TEIDatSCG     GTPtunnelEndpoint               OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPnotpresentExtIEs} }    OPTIONAL,
    ...
}

E-RABs-ToBeModified-SgNBModReqd-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-RLC-Status          CRITICALITY ignore          EXTENSION RLC-Status                PRESENCE optional}|
    { ID id-lCID                CRITICALITY ignore          EXTENSION LCID                      PRESENCE optional},
    ...
}


-- **************************************************************
--
-- SGNB MODIFICATION CONFIRM
--
-- **************************************************************

SgNBModificationConfirm ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBModificationConfirm-IEs}},
    ...
}
```

```
SgNBModificationConfirm-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                             CRITICALITY ignore  TYPE UE-X2AP-ID                                    PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                             CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                                   PRESENCE
mandatory}|
    { ID id-E-RABs-AdmittedToBeModified-SgNBModConfList CRITICALITY ignore  TYPE E-RABs-AdmittedToBeModified-SgNBModConfList   PRESENCE optional}|
    { ID id-MeNBtoSgNBContainer                         CRITICALITY ignore  TYPE MeNBtoSgNBContainer                         PRESENCE optional}|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                      PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension                   CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                        PRESENCE optional}|
    { ID id-MeNBResourceCoordinationInformation         CRITICALITY ignore  TYPE MeNBResourceCoordinationInformation          PRESENCE optional},
    ...
}

E-RABs-AdmittedToBeModified-SgNBModConfList ::= SEQUENCE (SIZE (1..maxnoofBearers)) OF ProtocolIE-Single-Container
    { {E-RABs-AdmittedToBeModified-SgNBModConf-ItemIEs} }

E-RABs-AdmittedToBeModified-SgNBModConf-ItemIEs X2AP-PROTOCOL-IES ::= {
{ ID id-E-RABs-AdmittedToBeModified-SgNBModConf-Item     CRITICALITY ignore TYPE E-RABs-AdmittedToBeModified-SgNBModConf-Item   PRESENCE
mandatory },
    ...
}

E-RABs-AdmittedToBeModified-SgNBModConf-Item ::= SEQUENCE {
    e-RAB-ID                                E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-AdmittedToBeModified-SgNBModConf-ItemExtIEs} }  OPTIONAL,
    ...
}

E-RABs-AdmittedToBeModified-SgNBModConf-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPpresent ::= SEQUENCE {
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPpresentExtIEs} }     OPTIONAL,
    ...
}

E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    secondary-meNB-UL-GTP-TEIDatPDCP                GTPtunnelEndpoint               OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPnotpresentExtIEs} }
    OPTIONAL,
    ...
}
```

```
E-RABs-AdmittedToBeModified-SgNBModConf-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-uLpDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional}|
    { ID id-dLPDCPSnLength                          CRITICALITY ignore  EXTENSION PDCPSnLength            PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SGNB MODIFICATION REFUSE
--
-- **************************************************************

SgNBModificationRefuse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SgNBModificationRefuse-IEs}},
    ...
}

SgNBModificationRefuse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID             PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                    PRESENCE mandatory}|
    { ID id-MeNBtoSgNBContainer                 CRITICALITY ignore  TYPE MeNBtoSgNBContainer          PRESENCE optional}|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics   PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY ignore  TYPE UE-X2AP-ID-Extension     PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SGNB RELEASE REQUEST
--
-- **************************************************************

SgNBReleaseRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SgNBReleaseRequest-IEs}},
    ...
}

SgNBReleaseRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                         PRESENCE optional}|
    { ID id-Cause                                   CRITICALITY ignore  TYPE Cause                                PRESENCE mandatory}|
    { ID id-E-RABs-ToBeReleased-SgNBRelReqList      CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBRelReqList    PRESENCE optional}|
    { ID id-UE-ContextKeptIndicator                 CRITICALITY ignore  TYPE UE-ContextKeptIndicator               PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY reject  TYPE UE-X2AP-ID-Extension                 PRESENCE optional}|
    { ID id-MeNBtoSgNBContainer                     CRITICALITY reject  TYPE MeNBtoSgNBContainer                     PRESENCE optional}|
    { ID id-ERABs-transferred-to-MeNB               CRITICALITY ignore  TYPE E-RAB-List                           PRESENCE optional},
    ...
}

E-RABs-ToBeReleased-SgNBRelReqList ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-SgNBRelReq-
ItemIEs} }

E-RABs-ToBeReleased-SgNBRelReq-ItemIEs X2AP-PROTOCOL-IES ::= {
```

```
    { ID id-E-RABs-ToBeReleased-SgNBRelReq-Item     CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBRelReq-Item    PRESENCE mandatory},
    ...
}

E-RABs-ToBeReleased-SgNBRelReq-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBRelReq-ItemExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBRelReq-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPpresent ::= SEQUENCE {
    uL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                   OPTIONAL,
    dL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                       OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPnotpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBRelReq-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SGNB RELEASE REQUEST ACKNOWLEDGE
--
-- **************************************************************

SgNBReleaseRequestAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SgNBReleaseRequestAcknowledge-IEs}},
    ...
}

SgNBReleaseRequestAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                             CRITICALITY ignore  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
```

```
    { ID id-SgNB-UE-X2AP-ID                                 CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                        PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics                          CRITICALITY ignore  TYPE CriticalityDiagnostics                PRESENCE optional }|
    { ID id-MeNB-UE-X2AP-ID-Extension                       CRITICALITY reject  TYPE UE-X2AP-ID-Extension                   PRESENCE optional }|
    { ID id-E-RABs-Admitted-ToBeReleased-SgNBRelReqAckList  CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeReleased-SgNBRelReqAckList  PRESENCE
optional }|
    { ID id-SCG-UE-HistoryInformation                   CRITICALITY ignore  TYPE SCG-UE-HistoryInformation                PRESENCE optional},
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBRelReqAckList ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF
                                    ProtocolIE-Single-Container { {E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-ItemIEs} }

E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-Item CRITICALITY ignore  TYPE E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-Item    PRESENCE
mandatory},
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    rlc-Mode-transferred            RLCMode,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-ItemExtIEs} }  OPTIONAL,
    ...
}

E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SGNB RELEASE REQUEST REJECT
--
-- **************************************************************

SgNBReleaseRequestReject ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SgNBReleaseRequestReject-IEs}},
    ...
}

SgNBReleaseRequestReject-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE UE-X2AP-ID                         PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                       PRESENCE mandatory}|
    { ID id-Cause                                   CRITICALITY ignore  TYPE Cause                              PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics             PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY reject  TYPE UE-X2AP-ID-Extension               PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SGNB RELEASE REQUIRED
--
-- **************************************************************
```

```
SgNBReleaseRequired ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBReleaseRequired-IEs}},
    ...
}

SgNBReleaseRequired-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY reject  TYPE UE-X2AP-ID                              PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                            PRESENCE mandatory}|
    { ID id-Cause                                   CRITICALITY ignore  TYPE Cause                                  PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY reject  TYPE UE-X2AP-ID-Extension                   PRESENCE optional }|
    { ID id-E-RABs-ToBeReleased-SgNBRelReqdList     CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBRelReqdList    PRESENCE optional }|
    { ID id-SgNBtoMeNBContainer                     CRITICALITY ignore  TYPE SgNBtoMeNBContainer                    PRESENCE optional }|
    { ID id-SCG-UE-HistoryInformation                   CRITICALITY ignore  TYPE SCG-UE-HistoryInformation           PRESENCE optional},
    ...
}

E-RABs-ToBeReleased-SgNBRelReqdList ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-SgNBRelReqd-
ItemIEs} }

E-RABs-ToBeReleased-SgNBRelReqd-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-SgNBRelReqd-Item        CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBRelReqd-Item    PRESENCE mandatory},
    ...
}

E-RABs-ToBeReleased-SgNBRelReqd-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    rlc-Mode-transferred            RLCMode,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBRelReqd-ItemExtIEs} }  OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBRelReqd-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SGNB RELEASE CONFIRM
--
-- **************************************************************

SgNBReleaseConfirm ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBReleaseConfirm-IEs}},
    ...
}

SgNBReleaseConfirm-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                            PRESENCE mandatory}|
    { ID id-E-RABs-ToBeReleased-SgNBRelConfList     CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBRelConfList   PRESENCE optional}|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics                 PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                   PRESENCE optional},
    ...
```

```
}

E-RABs-ToBeReleased-SgNBRelConfList ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-SgNBRelConf-
ItemIEs} }

E-RABs-ToBeReleased-SgNBRelConf-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-SgNBRelConf-Item        CRITICALITY ignore      TYPE E-RABs-ToBeReleased-SgNBRelConf-Item    PRESENCE mandatory},
    ...
}

E-RABs-ToBeReleased-SgNBRelConf-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPnotpresent,
        ...
    },
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBRelConf-ItemExtIEs} }  OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBRelConf-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPpresent ::= SEQUENCE {
    uL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                                   OPTIONAL,
    dL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPnotpresentExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBRelConf-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- **************************************************************
--
-- SGNB COUNTER CHECK REQUEST
--
-- **************************************************************

SgNBCounterCheckRequest ::= SEQUENCE {
```

```
    protocolIEs     ProtocolIE-Container    {{SgNBCounterCheckRequest-IEs}},
    ...
}

SgNBCounterCheckRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                             CRITICALITY reject  TYPE UE-X2AP-ID                              PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                             CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                            PRESENCE mandatory}|
    { ID id-E-RABs-SubjectToSgNBCounterCheck-List       CRITICALITY ignore  TYPE E-RABs-SubjectToSgNBCounterCheck-List   PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension                   CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                    PRESENCE optional},
    ...
}
E-RABs-SubjectToSgNBCounterCheck-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-SubjectToSgNBCounterCheck-
ItemIEs} }

E-RABs-SubjectToSgNBCounterCheck-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-SubjectToSgNBCounterCheck-Item       CRITICALITY ignore  TYPE E-RABs-SubjectToSgNBCounterCheck-Item   PRESENCE mandatory},
    ...
}

E-RABs-SubjectToSgNBCounterCheck-Item ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    uL-Count                        INTEGER (0..4294967295),
    dL-Count                        INTEGER (0..4294967295),
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-SubjectToSgNBCounterCheck-ItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-SubjectToSgNBCounterCheck-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SGNB CHANGE REQUIRED
--
-- **************************************************************

SgNBChangeRequired ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SgNBChangeRequired-IEs}},
    ...
}

SgNBChangeRequired-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY reject  TYPE SgNB-UE-X2AP-ID               PRESENCE mandatory}|
    { ID id-Target-SgNB-ID                      CRITICALITY reject  TYPE GlobalGNB-ID               PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                      PRESENCE mandatory}|
    { ID id-SgNBtoMeNBContainer                 CRITICALITY reject  TYPE SgNBtoMeNBContainer           PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional}|
    { ID id-SCG-UE-HistoryInformation           CRITICALITY ignore  TYPE SCG-UE-HistoryInformation  PRESENCE optional}|
    { ID id-CPCinformation-REQD                 CRITICALITY reject  TYPE CPCinformation-REQD        PRESENCE optional},
    ...
}
```

```
-- **************************************************************
--
-- ACCESS AND MOBILITY INDICATION
--
-- **************************************************************

AccessAndMobilityIndication ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container    {{ AccessAndMobilityIndication-IEs}},
    ...
}
AccessAndMobilityIndication-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-NRRACHReportInformation         CRITICALITY ignore      TYPE NRRACHReportInformation       PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SGNB CHANGE CONFIRM
--
-- **************************************************************

SgNBChangeConfirm ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{SgNBChangeConfirm-IEs}},
    ...
}

SgNBChangeConfirm-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE UE-X2AP-ID                             PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                         CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                           PRESENCE mandatory}|
    { ID id-E-RABs-ToBeReleased-SgNBChaConfList     CRITICALITY ignore  TYPE E-RABs-ToBeReleased-SgNBChaConfList   PRESENCE optional}|
    { ID id-CriticalityDiagnostics                  CRITICALITY ignore  TYPE CriticalityDiagnostics                PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension               CRITICALITY ignore  TYPE UE-X2AP-ID-Extension               PRESENCE optional}|
    { ID id-CPCinformation-CONF                     CRITICALITY ignore  TYPE CPCinformation-CONF                   PRESENCE optional}|
    { ID id-MeNBtoSgNBContainer                     CRITICALITY ignore  TYPE MeNBtoSgNBContainer                   PRESENCE optional},
    ...
}

E-RABs-ToBeReleased-SgNBChaConfList ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-ToBeReleased-SgNBChaConf-
ItemIEs} }

E-RABs-ToBeReleased-SgNBChaConf-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-ToBeReleased-SgNBChaConf-Item        CRITICALITY ignore      TYPE E-RABs-ToBeReleased-SgNBChaConf-Item   PRESENCE mandatory},
    ...
}

E-RABs-ToBeReleased-SgNBChaConf-Item ::= SEQUENCE {
    e-RAB-ID                            E-RAB-ID,
    en-DC-ResourceConfiguration         EN-DC-ResourceConfiguration,
    resource-configuration              CHOICE {
        sgNBPDCPpresent                     E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPpresent,
        sgNBPDCPnotpresent                  E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPnotpresent,
        ...
    },
```

```
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBChaConf-ItemExtIEs} }  OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBChaConf-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPpresent ::= SEQUENCE {
    uL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                                   OPTIONAL,
    dL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                                   OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPpresentExtIEs} } OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalListofForwardingGTPTunnelEndpoint             CRITICALITY ignore  EXTENSION AdditionalListofForwardingGTPTunnelEndpoint
                PRESENCE optional },
    ...
}

E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPnotpresent ::= SEQUENCE {
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPnotpresentExtIEs} }   OPTIONAL,
    ...
}

E-RABs-ToBeReleased-SgNBChaConf-Item-SgNBPDCPnotpresentExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- RRC TRANSFER
--
-- **************************************************************

RRCTransfer ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{RRCTransfer-IEs}},
    ...
}

RRCTransfer-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID             PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                 CRITICALITY reject  TYPE SgNB-UE-X2AP-ID          PRESENCE mandatory}|
    { ID id-SplitSRB                            CRITICALITY reject  TYPE SplitSRB             PRESENCE optional}|
    { ID id-NRUeReport                      CRITICALITY reject  TYPE NRUeReport             PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension   PRESENCE optional}|
    { ID id-FastMCGRecovery-SN-to-MN            CRITICALITY ignore  TYPE FastMCGRecovery          PRESENCE optional}|
    { ID id-FastMCGRecovery-MN-to-SN            CRITICALITY ignore  TYPE FastMCGRecovery          PRESENCE optional},
    ...
}

-- **************************************************************
--
```

```
-- SGNB CHANGE REFUSE
--
-- **************************************************************

SgNBChangeRefuse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SgNBChangeRefuse-IEs}},
    ...
}

SgNBChangeRefuse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE UE-X2AP-ID              PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID           PRESENCE mandatory}|
    { ID id-Cause                               CRITICALITY ignore  TYPE Cause                     PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics              CRITICALITY ignore  TYPE CriticalityDiagnostics  PRESENCE optional}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension    PRESENCE optional},
    ...
}

-- **************************************************************
--
-- EN-DC X2 SETUP REQUEST
--
-- **************************************************************

ENDCX2SetupRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCX2SetupRequest-IEs}},
    ...
}

ENDCX2SetupRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-InitiatingNodeType-EndcX2Setup          CRITICALITY reject  TYPE InitiatingNodeType-EndcX2Setup     PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication             CRITICALITY reject  TYPE InterfaceInstanceIndication        PRESENCE optional}|
    { ID id-TNLConfigurationInfo                    CRITICALITY ignore  TYPE TNLConfigurationInfo               PRESENCE optional},
    ...
}

InitiatingNodeType-EndcX2Setup ::= CHOICE {
    init-eNB                ProtocolIE-Container    {{ENB-ENDCX2SetupReqIEs}},
    init-en-gNB         ProtocolIE-Container    {{En-gNB-ENDCX2SetupReqIEs}},
    ...
}

ENB-ENDCX2SetupReqIEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID                            CRITICALITY reject  TYPE GlobalENB-ID                          PRESENCE mandatory}|
    { ID id-ServedEUTRAcellsENDCX2ManagementList    CRITICALITY reject  TYPE ServedEUTRAcellsENDCX2ManagementList  PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication             CRITICALITY reject  TYPE InterfaceInstanceIndication          PRESENCE optional }|
-- NOTE: In the current version of this specification the Interface Instance Indication IE is not included in the Initiating NodeType IE --
    { ID id-CellandCapacityAssistInfo               CRITICALITY ignore  TYPE CellandCapacityAssistInfo            PRESENCE optional },
    ...
}

ServedEUTRAcellsENDCX2ManagementList ::= SEQUENCE (SIZE (1.. maxCellineNB)) OF SEQUENCE {
    servedEUTRACellInfo                 ServedCell-Information,
    nrNeighbourInfo                     NRNeighbour-Information     OPTIONAL,
```

```
    iE-Extensions                       ProtocolExtensionContainer { {ServedEUTRAcellsENDCX2Management-ExtIEs} } OPTIONAL,
    ...
}

ServedEUTRAcellsENDCX2Management-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

En-gNB-ENDCX2SetupReqIEs X2AP-PROTOCOL-IES ::= {
    { ID id-Globalen-gNB-ID                         CRITICALITY reject  TYPE GlobalGNB-ID                          PRESENCE mandatory}|
    { ID id-ServedNRcellsENDCX2ManagementList       CRITICALITY reject  TYPE ServedNRcellsENDCX2ManagementList     PRESENCE mandatory}|
    { ID id-PartialListIndicator                        CRITICALITY ignore  TYPE PartialListIndicator                   PRESENCE optional },
    ...
}

ServedNRcellsENDCX2ManagementList ::= SEQUENCE (SIZE (1.. maxCellinengNB)) OF SEQUENCE {
    servedNRCellInfo                        ServedNRCell-Information,
    nRNeighbourInfo                     NRNeighbour-Information OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {En-gNBServedCells-ExtIEs} } OPTIONAL,
    ...
}

En-gNBServedCells-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-ServedCellSpecificInfoReq-NR    CRITICALITY ignore EXTENSION ServedCellSpecificInfoReq-NR   PRESENCE optional },
    ...
}

ServedNRCell-Information ::= SEQUENCE {
    nrpCI               NRPCI,
    nrCellID            NRCGI,
    fiveGS-TAC          FiveGS-TAC  OPTIONAL,
    configured-TAC      TAC         OPTIONAL,
    broadcastPLMNs      BroadcastPLMNs-Item,
    nrModeInfo          CHOICE {
        fdd     FDD-InfoServedNRCell-Information,
        tdd     TDD-InfoServedNRCell-Information,
        ...
    },
    measurementTimingConfiguration  OCTET STRING,
    iE-Extensions                       ProtocolExtensionContainer { {ServedNRCell-Information-ExtIEs} } OPTIONAL,
    ...
}

ServedNRCell-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-additionalPLMNs-Item            CRITICALITY ignore  EXTENSION AdditionalPLMNs-Item            PRESENCE optional}|
    { ID id-BPLMN-ID-Info-NR                CRITICALITY ignore  EXTENSION BPLMN-ID-Info-NR                PRESENCE optional}|
    { ID id-SSB-PositionsInBurst            CRITICALITY ignore  EXTENSION SSB-PositionsInBurst            PRESENCE optional}|
    { ID id-NRCellPRACHConfig               CRITICALITY ignore  EXTENSION NRCellPRACHConfig               PRESENCE optional}|
    { ID id-CSI-RSTransmissionIndication    CRITICALITY ignore  EXTENSION CSI-RSTransmissionIndication    PRESENCE optional}|
    { ID id-SFN-Offset                      CRITICALITY ignore EXTENSION SFN-Offset                       PRESENCE optional}|
    { ID id-Additional-Measurement-Timing-Configuration-List        CRITICALITY ignore  EXTENSION Additional-Measurement-Timing-Configuration-List
        PRESENCE optional },
    ...
}
```

```
FDD-InfoServedNRCell-Information ::= SEQUENCE {
    ul-NRFreqInfo               NRFreqInfo,
    dl-NRFreqInfo               NRFreqInfo,
    ul-NR-TxBW              NR-TxBW,
    dl-NR-TxBW              NR-TxBW,
    iE-Extensions           ProtocolExtensionContainer { {FDD-InfoServedNRCell-Information-ExtIEs} }    OPTIONAL,
    ...
}

FDD-InfoServedNRCell-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-ULCarrierList                   CRITICALITY ignore  EXTENSION NRCarrierList         PRESENCE optional }|
    { ID id-DLCarrierList                   CRITICALITY ignore  EXTENSION NRCarrierList         PRESENCE optional },
    ...
}

TDD-InfoServedNRCell-Information ::= SEQUENCE {
    nRFreqInfo              NRFreqInfo,
    nR-TxBW                 NR-TxBW,
    iE-Extensions           ProtocolExtensionContainer { {TDD-InfoServedNRCell-Information-ExtIEs} }    OPTIONAL,
    ...
}

TDD-InfoServedNRCell-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-TDDULDLConfigurationCommonNR    CRITICALITY ignore  EXTENSION TDDULDLConfigurationCommonNR  PRESENCE optional}|
    { ID id-CarrierList                     CRITICALITY ignore  EXTENSION NRCarrierList         PRESENCE optional}|
    { ID id-IntendedTDD-DL-ULConfiguration-NR   CRITICALITY ignore  EXTENSION IntendedTDD-DL-ULConfiguration-NR    PRESENCE optional},
    ...
}

CellandCapacityAssistInfo::= SEQUENCE {
    maximumCellListSize                         MaximumCellListSize         OPTIONAL,
    cellAssistanceInformation                   CellAssistanceInformation   OPTIONAL,
    iE-Extensions                               ProtocolExtensionContainer { {CellandCapacityAssistInfo-ExtIEs} } OPTIONAL,
    ...
}

CellandCapacityAssistInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellAssistanceInformation ::= CHOICE {
    limited-list                Limited-list,
    full-list               ENUMERATED {allServedNRcells, ...},
    ...
}


Limited-list    ::= SEQUENCE (SIZE (1..maxCellinengNB)) OF SEQUENCE {
    nrCellID            NRCGI,
        iE-Extensions                       ProtocolExtensionContainer { {Limited-list-ExtIEs} } OPTIONAL,
    ...
}
```

```
Limited-list-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- EN-DC X2 SETUP RESPONSE
--
-- **************************************************************

ENDCX2SetupResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCX2SetupResponse-IEs}},
    ...
}

ENDCX2SetupResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-RespondingNodeType-EndcX2Setup      CRITICALITY reject  TYPE RespondingNodeType-EndcX2Setup    PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication         CRITICALITY reject  TYPE InterfaceInstanceIndication      PRESENCE optional }|
    { ID id-TNLConfigurationInfo                    CRITICALITY ignore  TYPE TNLConfigurationInfo           PRESENCE optional },
    ...
}

RespondingNodeType-EndcX2Setup ::= CHOICE {
    respond-eNB         ProtocolIE-Container    {{ENB-ENDCX2SetupReqAckIEs}},
    respond-en-gNB      ProtocolIE-Container    {{En-gNB-ENDCX2SetupReqAckIEs}},
    ...
}

ENB-ENDCX2SetupReqAckIEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID                            CRITICALITY reject  TYPE GlobalENB-ID                          PRESENCE mandatory}|
    { ID id-ServedEUTRAcellsENDCX2ManagementList        CRITICALITY reject  TYPE ServedEUTRAcellsENDCX2ManagementList    PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication             CRITICALITY reject  TYPE InterfaceInstanceIndication             PRESENCE optional }|
-- NOTE: In the current version of this specification the Interface Instance Indication IE is not included in the Responding NodeType IE --
    { ID id-CellandCapacityAssistInfo               CRITICALITY ignore  TYPE CellandCapacityAssistInfo               PRESENCE optional },
    ...
}


En-gNB-ENDCX2SetupReqAckIEs X2AP-PROTOCOL-IES ::= {
    { ID id-Globalen-gNB-ID                         CRITICALITY reject  TYPE GlobalGNB-ID                          PRESENCE mandatory}|
    { ID id-ServedNRcellsENDCX2ManagementList       CRITICALITY reject  TYPE ServedNRcellsENDCX2ManagementList       PRESENCE mandatory}|
    { ID id-PartialListIndicator                    CRITICALITY ignore  TYPE PartialListIndicator                        PRESENCE optional },
    ...
}

-- **************************************************************
--
-- EN-DC X2 SETUP FAILURE
--
-- **************************************************************

ENDCX2SetupFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCX2SetupFailure-IEs}},
    ...
```

```
}

ENDCX2SetupFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                           CRITICALITY ignore      TYPE Cause                          PRESENCE mandatory} |
    { ID id-CriticalityDiagnostics          CRITICALITY ignore      TYPE CriticalityDiagnostics         PRESENCE optional } |
    { ID id-TimeToWait                      CRITICALITY ignore      TYPE TimeToWait                     PRESENCE optional } |
    { ID id-InterfaceInstanceIndication     CRITICALITY reject      TYPE InterfaceInstanceIndication    PRESENCE optional }|
    { ID id-MessageOversizeNotification     CRITICALITY ignore      TYPE MessageOversizeNotification    PRESENCE optional },
    ...
}

-- **************************************************************
--
-- EN-DC CONFIGURATION UPDATE
--
-- **************************************************************

ENDCConfigurationUpdate ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCConfigurationUpdate-IEs}},
    ...
}

ENDCConfigurationUpdate-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-InitiatingNodeType-EndcConfigUpdate         CRITICALITY reject  TYPE InitiatingNodeType-EndcConfigUpdate        PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication                 CRITICALITY reject  TYPE InterfaceInstanceIndication                PRESENCE optional}|
    { ID id-TNLConfigurationInfo                        CRITICALITY ignore  TYPE TNLConfigurationInfo                       PRESENCE optional }|
    { ID id-TNLA-To-Add-List                            CRITICALITY ignore  TYPE TNLA-To-Add-List                           PRESENCE optional }|
    { ID id-TNLA-To-Update-List                         CRITICALITY ignore  TYPE TNLA-To-Update-List                        PRESENCE optional }|
    { ID id-TNLA-To-Remove-List                         CRITICALITY ignore  TYPE TNLA-To-Remove-List                        PRESENCE optional },
    ...
}

InitiatingNodeType-EndcConfigUpdate::= CHOICE {
    init-eNB            ProtocolIE-Container    {{ENB-ENDCConfigUpdateIEs}},
    init-en-gNB         ProtocolIE-Container    {{En-gNB-ENDCConfigUpdateIEs}},
    ...
}

ENB-ENDCConfigUpdateIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellAssistanceInformation               CRITICALITY reject  TYPE CellAssistanceInformation                  PRESENCE optional }|
    { ID id-ServedEUTRAcellsENDCX2ManagementList        CRITICALITY reject  TYPE ServedEUTRAcellsENDCX2ManagementList       PRESENCE optional }|
    { ID id-ServedEUTRAcellsToModifyListENDCConfUpd CRITICALITY reject  TYPE ServedEUTRAcellsToModifyListENDCConfUpd     PRESENCE optional }|
    { ID id-ServedEUTRAcellsToDeleteListENDCConfUpd CRITICALITY reject  TYPE ServedEUTRAcellsToDeleteListENDCConfUpd     PRESENCE optional },
    ...
}

ServedEUTRAcellsToModifyListENDCConfUpd ::= SEQUENCE (SIZE (1.. maxCellineNB)) OF SEQUENCE {
    old-ECGI                                ECGI,
    servedEUTRACellInfo                 ServedCell-Information,
    nrNeighbourInfo                     NRNeighbour-Information     OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {ServedEUTRAcellsToModifyListENDCConfUpd-ExtIEs} } OPTIONAL,
    ...
}
```

```
ServedEUTRAcellsToModifyListENDCConfUpd-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ServedEUTRAcellsToDeleteListENDCConfUpd ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ECGI


En-gNB-ENDCConfigUpdateIEs X2AP-PROTOCOL-IES ::= {
    { ID id-ServedNRcellsENDCX2ManagementList       CRITICALITY reject  TYPE ServedNRcellsENDCX2ManagementList        PRESENCE optional }|
    { ID id-ServedNRcellsToModifyListENDCConfUpd        CRITICALITY reject  TYPE ServedNRcellsToModifyENDCConfUpdList    PRESENCE optional }|
    { ID id-ServedNRcellsToDeleteListENDCConfUpd        CRITICALITY reject  TYPE ServedNRcellsToDeleteENDCConfUpdList    PRESENCE optional },
    ...
}

ServedNRcellsToModifyENDCConfUpdList ::= SEQUENCE (SIZE (1..maxCellinengNB)) OF ServedNRCellsToModify-Item

ServedNRCellsToModify-Item::= SEQUENCE {
    old-nrcgi                       NRCGI,
    servedNRCellInformation         ServedNRCell-Information,
    nrNeighbourInformation          NRNeighbour-Information         OPTIONAL,
    nrDeactivationIndication            DeactivationIndication          OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {ServedNRCellsToModify-Item-ExtIEs} } OPTIONAL,
    ...
}

ServedNRCellsToModify-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ServedNRcellsToDeleteENDCConfUpdList ::= SEQUENCE (SIZE (1..maxCellinengNB)) OF NRCGI

-- **************************************************************
--
-- EN-DC CONFIGURATION UPDATE ACKNOWLEDGE
--
-- **************************************************************

ENDCConfigurationUpdateAcknowledge ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCConfigurationUpdateAcknowledge-IEs}},
    ...
}

ENDCConfigurationUpdateAcknowledge-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-RespondingNodeType-EndcConfigUpdate         CRITICALITY reject  TYPE RespondingNodeType-EndcConfigUpdate        PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication                 CRITICALITY reject  TYPE InterfaceInstanceIndication                PRESENCE optional }|
    { ID id-CriticalityDiagnostics                      CRITICALITY ignore  TYPE CriticalityDiagnostics                     PRESENCE optional }|
    { ID id-TNLConfigurationInfo                        CRITICALITY ignore  TYPE TNLConfigurationInfo                       PRESENCE optional }|
    { ID id-TNLA-Setup-List                             CRITICALITY ignore  TYPE TNLA-Setup-List                               PRESENCE optional }|
    { ID id-TNLA-Failed-To-Setup-List                   CRITICALITY ignore  TYPE TNLA-Failed-To-Setup-List                  PRESENCE optional },
    ...
}


RespondingNodeType-EndcConfigUpdate::= CHOICE {
```

```
    respond-eNB         ProtocolIE-Container    {{ENB-ENDCConfigUpdateAckIEs}},
    respond-en-gNB      ProtocolIE-Container    {{En-gNB-ENDCConfigUpdateAckIEs}},
    ...
}


ENB-ENDCConfigUpdateAckIEs X2AP-PROTOCOL-IES ::= {
    ...
}

En-gNB-ENDCConfigUpdateAckIEs X2AP-PROTOCOL-IES ::= {
    { ID id-ServedNRcellsENDCX2ManagementList       CRITICALITY reject  TYPE ServedNRcellsENDCX2ManagementList     PRESENCE optional},
    ...
}


-- **************************************************************
--
-- EN-DC CONFIGURATION UPDATE FAILURE
--
-- **************************************************************

ENDCConfigurationUpdateFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCConfigurationUpdateFailure-IEs}},
    ...
}

ENDCConfigurationUpdateFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics      PRESENCE optional }|
    { ID id-TimeToWait                      CRITICALITY ignore  TYPE TimeToWait                  PRESENCE optional }|
    { ID id-InterfaceInstanceIndication     CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional },
    ...
}


-- **************************************************************
--
-- EN-DC CELL ACTIVATION REQUEST
--
-- **************************************************************

ENDCCellActivationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCCellActivationRequest-IEs}},
    ...
}

ENDCCellActivationRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ServedNRCellsToActivate         CRITICALITY reject  TYPE ServedNRCellsToActivate           PRESENCE mandatory}|
    { ID id-ActivationID                    CRITICALITY reject  TYPE ActivationID                   PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication CRITICALITY reject  TYPE InterfaceInstanceIndication        PRESENCE optional },
    ...
}
```

```
ServedNRCellsToActivate::= SEQUENCE (SIZE (1.. maxCellinengNB)) OF ServedNRCellsToActivate-Item

ServedNRCellsToActivate-Item::= SEQUENCE {
    nrCellID                        NRCGI,
    iE-Extensions                   ProtocolExtensionContainer { {ServedNRCellsToActivate-Item-ExtIEs} } OPTIONAL,
    ...
}

ServedNRCellsToActivate-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- EN-DC CELL ACTIVATION RESPONSE
--
-- **************************************************************

ENDCCellActivationResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCCellActivationResponse-IEs}},
    ...
}

ENDCCellActivationResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ActivatedNRCellList         CRITICALITY ignore  TYPE ActivatedNRCellList             PRESENCE mandatory}|
    { ID id-ActivationID                    CRITICALITY reject  TYPE ActivationID                   PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional }|
    { ID id-InterfaceInstanceIndication CRITICALITY reject  TYPE InterfaceInstanceIndication     PRESENCE optional },
    ...
}

ActivatedNRCellList ::= SEQUENCE (SIZE (1.. maxCellinengNB)) OF ActivatedNRCellList-Item

ActivatedNRCellList-Item::= SEQUENCE {
    nrCellID                        NRCGI,
    iE-Extensions                       ProtocolExtensionContainer { {ActivatedNRCellList-Item-ExtIEs} } OPTIONAL,
    ...
}

ActivatedNRCellList-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

--**************************************************************
--
-- EN-DC CELL ACTIVATION FAILURE
--
-- **************************************************************

ENDCCellActivationFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCCellActivationFailure-IEs}},
    ...
}
```

```
ENDCCellActivationFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-ActivationID                    CRITICALITY reject  TYPE ActivationID              PRESENCE mandatory }|
    { ID id-Cause                           CRITICALITY ignore  TYPE Cause                     PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics    PRESENCE optional }|
    { ID id-InterfaceInstanceIndication     CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional },
    ...
}

-- **************************************************************
--
-- EN-DC RESOURCE STATUS REQUEST
--
-- **************************************************************

ENDCResourceStatusRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCResourceStatusRequest-IEs}},
    ...
}

ENDCResourceStatusRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-UTRAN-Node1-Measurement-ID          CRITICALITY reject  TYPE Measurement-ID-ENDC       PRESENCE mandatory  }|
    { ID id-E-UTRAN-Node2-Measurement-ID        CRITICALITY ignore  TYPE Measurement-ID-ENDC                    PRESENCE conditional}| -- The IE
shall be present if the Registration Request EN-DC IE is set to “Stop” or to “Add”
    { ID id-Registration-Request            CRITICALITY reject  TYPE Registration-Request-ENDC    PRESENCE mandatory  }|
    { ID id-ReportingPeriodicity            CRITICALITY ignore  TYPE ReportingPeriodicity-ENDC    PRESENCE optional   }|
    { ID id-ReportCharacteristics           CRITICALITY ignore  TYPE ReportCharacteristics-ENDC   PRESENCE conditional}| -- The IE shall be present
if the Registration Request EN-DC IE is set to “Start”
    { ID id-CellToReport-NR-ENDC        CRITICALITY ignore  TYPE CellToReport-NR-ENDC-List       PRESENCE optional   } |
    { ID id-InterfaceInstanceIndication CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional   }|
    { ID id-CellToReport-E-UTRA-ENDC    CRITICALITY ignore  TYPE CellToReport-E-UTRA-ENDC-List    PRESENCE optional},
    ...
}

ReportingPeriodicity-ENDC ::= ENUMERATED {ms500, ms1000, ms2000, ms5000, ms10000, ...}

CellToReport-NR-ENDC-List   ::= SEQUENCE (SIZE (1..maxCellinengNB)) OF ProtocolIE-Single-Container { {CellToReport-NR-ENDC-ItemIEs} }

CellToReport-NR-ENDC-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellToReport-NR-ENDC-Item           CRITICALITY ignore  TYPE CellToReport-NR-ENDC-Item     PRESENCE mandatory}
}

CellToReport-NR-ENDC-Item               ::= SEQUENCE {
    nr-cell-ID                              NRCGI,
    ssbToReport-List                    SSBToReport-List                                    OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {CellToReport-NR-ENDC-Item-ExtIEs} } OPTIONAL,
    ...
}

CellToReport-NR-ENDC-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellToReport-E-UTRA-ENDC-List       ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ProtocolIE-Single-Container { {CellToReport-E-UTRA-ENDC-Item-IEs} }
```

```
CellToReport-E-UTRA-ENDC-Item-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellToReport-E-UTRA-ENDC-Item   CRITICALITY ignore  TYPE CellToReport-E-UTRA-ENDC-Item   PRESENCE mandatory}
}

CellToReport-E-UTRA-ENDC-Item ::= SEQUENCE {
    e-utra-cell-ID                          ECGI,
    iE-Extensions                           ProtocolExtensionContainer { {CellToReport-E-UTRA-ENDC-Item-ExtIEs} } OPTIONAL,
    ...
}

CellToReport-E-UTRA-ENDC-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSBToReport-List    ::= SEQUENCE (SIZE (1.. maxnoofSSBAreas)) OF SSBToReport-Item

SSBToReport-Item    ::= SEQUENCE {
    ssbIndex                            SSBIndex,
    iE-Extensions                       ProtocolExtensionContainer { {SSBToReport-Item-ExtIEs} } OPTIONAL,
    ...
}

SSBToReport-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- EN-DC RESOURCE STATUS RESPONSE
--
-- **************************************************************

ENDCResourceStatusResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCResourceStatusResponse-IEs}},
    ...
}

ENDCResourceStatusResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-UTRAN-Node1-Measurement-ID        CRITICALITY reject  TYPE Measurement-ID-ENDC        PRESENCE mandatory}|
    { ID id-E-UTRAN-Node2-Measurement-ID    CRITICALITY reject  TYPE Measurement-ID-ENDC         PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional } |
    { ID id-InterfaceInstanceIndication CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional },
    ...
}

-- **************************************************************
--
-- EN-DC RESOURCE STATUS FAILURE
--
-- **************************************************************

ENDCResourceStatusFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCResourceStatusFailure-IEs}},
    ...
```

```
}

ENDCResourceStatusFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-UTRAN-Node1-Measurement-ID        CRITICALITY reject  TYPE Measurement-ID-ENDC        PRESENCE mandatory}|
    { ID id-E-UTRAN-Node2-Measurement-ID    CRITICALITY reject  TYPE Measurement-ID-ENDC        PRESENCE mandatory}|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics         PRESENCE optional }|
    { ID id-InterfaceInstanceIndication CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional },
    ...
}

-- **************************************************************
--
-- EN-DC RESOURCE STATUS UPDATE
--
-- **************************************************************

ENDCResourceStatusUpdate ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCResourceStatusUpdate-IEs}},
    ...
}

ENDCResourceStatusUpdate-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-UTRAN-Node1-Measurement-ID               CRITICALITY reject  TYPE Measurement-ID-ENDC                PRESENCE mandatory}|
    { ID id-E-UTRAN-Node2-Measurement-ID           CRITICALITY reject  TYPE Measurement-ID-ENDC                PRESENCE mandatory}|
    { ID id-CellMeasurementResult-NR-ENDC       CRITICALITY ignore  TYPE CellMeasurementResult-NR-ENDC-List     PRESENCE optional }|
    { ID id-InterfaceInstanceIndication         CRITICALITY reject  TYPE InterfaceInstanceIndication            PRESENCE optional }|
    { ID id-CellMeasurementResult-E-UTRA-ENDC   CRITICALITY ignore  TYPE CellMeasurementResult-E-UTRA-ENDC-List  PRESENCE optional},
    ...
}

CellMeasurementResult-NR-ENDC-List ::= SEQUENCE (SIZE (1..maxCellinengNB)) OF ProtocolIE-Single-Container { {CellMeasurementResult-NR-ENDC-ItemIEs}
}

CellMeasurementResult-NR-ENDC-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellMeasurementResult-NR-ENDC-Item  CRITICALITY ignore  TYPE CellMeasurementResult-NR-ENDC-Item   PRESENCE mandatory}
}

CellMeasurementResult-NR-ENDC-Item ::= SEQUENCE {
    nr-cell-ID                          NRCGI,
    nr-radioResourceStatus              NRRadioResourceStatus                               OPTIONAL,
    tnlCapacityIndicator                TNLCapacityIndicator                                OPTIONAL,
    nr-compositeAvailableCapacityGroup  NRCompositeAvailableCapacityGroup                   OPTIONAL,
    numberofActiveUEs                   INTEGER (0..16777215, ...)                          OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {CellMeasurementResult-NR-ENDC-Item-ExtIEs} } OPTIONAL,
    ...
}

CellMeasurementResult-NR-ENDC-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellMeasurementResult-E-UTRA-ENDC-List ::= SEQUENCE (SIZE (1..maxCellineNB))
                                               OF ProtocolIE-Single-Container { {CellMeasurementResult-E-UTRA-ENDC-ItemIEs} }
```

```
CellMeasurementResult-E-UTRA-ENDC-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-CellMeasurementResult-E-UTRA-ENDC-Item  CRITICALITY ignore  TYPE CellMeasurementResult-E-UTRA-ENDC-Item PRESENCE mandatory}
}

CellMeasurementResult-E-UTRA-ENDC-Item ::= SEQUENCE {
    e-utra-cell-ID              ECGI,
    hWLoadIndicator             HWLoadIndicator     OPTIONAL,
    s1TNLLoadIndicator          S1TNLLoadIndicator  OPTIONAL,
    radioResourceStatus         RadioResourceStatus OPTIONAL,
    compositeAvailableCapacityGroup CompositeAvailableCapacityGroup                                    OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {CellMeasurementResult-E-UTRA-ENDC-Item-ExtIEs} }   OPTIONAL,
    ...
}

CellMeasurementResult-E-UTRA-ENDC-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- SECONDARY RAT DATA USAGE REPORT
--
-- **************************************************************

SecondaryRATDataUsageReport ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SecondaryRATDataUsageReport-IEs}},
    ...
}

SecondaryRATDataUsageReport-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                   PRESENCE mandatory}|
    { ID id-SecondaryRATUsageReportList         CRITICALITY reject  TYPE SecondaryRATUsageReportList      PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension            PRESENCE optional},
    ...
}

-- **************************************************************
--
-- SGNB ACTIVITY NOTIFICATION
--
-- **************************************************************

SgNBActivityNotification ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{SgNBActivityNotification-IEs}},
    ...
}

SgNBActivityNotification-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                   PRESENCE mandatory}|
    { ID id-UEContextLevelUserPlaneActivity     CRITICALITY ignore  TYPE UserPlaneTrafficActivityReport   PRESENCE optional}|
    { ID id-ERABActivityNotifyItemList          CRITICALITY ignore  TYPE ERABActivityNotifyItemList       PRESENCE optional}|
```

```
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension             PRESENCE optional},
    ...
}

-- **************************************************************
--
-- EN-DC PARTIAL RESET REQUIRED
--
-- **************************************************************

ENDCPartialResetRequired ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCPartialResetRequired-IEs}},
    ...
}

ENDCPartialResetRequired-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-UEs-ToBeReset               CRITICALITY reject  TYPE UEsToBeResetList               PRESENCE mandatory}|
    { ID id-Cause                       CRITICALITY ignore  TYPE Cause                          PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication CRITICALITY reject  TYPE InterfaceInstanceIndication        PRESENCE optional},
    ...
}

-- **************************************************************
--
-- EN-DC PARTIAL RESET CONFIRM
--
-- **************************************************************

ENDCPartialResetConfirm ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCPartialResetConfirm-IEs}},
    ...
}

ENDCPartialResetConfirm-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-UEs-Admitted-ToBeReset              CRITICALITY reject  TYPE UEsToBeResetList              PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication         CRITICALITY reject  TYPE InterfaceInstanceIndication   PRESENCE optional},
    ...
}

-- **************************************************************
--
-- E-UTRA - NR CELL RESOURCE COORDINATION REQUEST
--
-- **************************************************************

EUTRANRCellResourceCoordinationRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{EUTRANRCellResourceCoordinationRequest-IEs}},
    ...
}

EUTRANRCellResourceCoordinationRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-InitiatingNodeType-EutranrCellResourceCoordination  CRITICALITY reject  TYPE InitiatingNodeType-EutranrCellResourceCoordination
    PRESENCE mandatory}|
```

```
    { ID id-InterfaceInstanceIndication                         CRITICALITY reject  TYPE InterfaceInstanceIndication
    PRESENCE optional },
    ...
}

InitiatingNodeType-EutranrCellResourceCoordination ::= CHOICE {
    initiate-eNB            ProtocolIE-Container    {{ENB-EUTRA-NRCellResourceCoordinationReqIEs}},
    initiate-en-gNB         ProtocolIE-Container    {{En-gNB-EUTRA-NRCellResourceCoordinationReqIEs}},
    ...
}


ENB-EUTRA-NRCellResourceCoordinationReqIEs X2AP-PROTOCOL-IES ::= {
    { ID id-DataTrafficResourceIndication           CRITICALITY reject  TYPE DataTrafficResourceIndication            PRESENCE mandatory}|
    { ID id-SpectrumSharingGroupID                  CRITICALITY reject  TYPE SpectrumSharingGroupID                   PRESENCE mandatory}|
    { ID id-ListofEUTRACellsinEUTRACoordinationReq  CRITICALITY reject  TYPE ListofEUTRACellsinEUTRACoordinationReq   PRESENCE mandatory},
    ...
}


En-gNB-EUTRA-NRCellResourceCoordinationReqIEs X2AP-PROTOCOL-IES ::= {
    { ID id-DataTrafficResourceIndication           CRITICALITY reject  TYPE DataTrafficResourceIndication         PRESENCE mandatory}|
    { ID id-ListofEUTRACellsinNRCoordinationReq     CRITICALITY reject  TYPE ListofEUTRACellsinNRCoordinationReq   PRESENCE mandatory }|
    { ID id-SpectrumSharingGroupID                  CRITICALITY reject  TYPE SpectrumSharingGroupID                PRESENCE mandatory}|
    { ID id-ListofNRCellsinNRCoordinationReq        CRITICALITY reject  TYPE ListofNRCellsinNRCoordinationReq      PRESENCE mandatory},
    ...
}


ListofEUTRACellsinEUTRACoordinationReq ::= SEQUENCE (SIZE (0..maxCellineNB)) OF ECGI
ListofEUTRACellsinNRCoordinationReq ::= SEQUENCE (SIZE (1..maxCellineNB)) OF ECGI
ListofNRCellsinNRCoordinationReq ::= SEQUENCE (SIZE (0..maxnoNRcellsSpectrumSharingWithE-UTRA)) OF NRCGI

-- ************************************************************
--
-- E-UTRA - NR CELL RESOURCE COORDINATION RESPONSE
--
-- ************************************************************

EUTRANRCellResourceCoordinationResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{EUTRANRCellResourceCoordinationResponse-IEs}},
    ...
}

EUTRANRCellResourceCoordinationResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-RespondingNodeType-EutranrCellResourceCoordination      CRITICALITY reject  TYPE RespondingNodeType-EutranrCellResourceCoordination
        PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional },
    ...
}

RespondingNodeType-EutranrCellResourceCoordination ::= CHOICE {
    respond-eNB         ProtocolIE-Container    {{ENB-EUTRA-NRCellResourceCoordinationReqAckIEs}},
    respond-en-gNB      ProtocolIE-Container    {{En-gNB-EUTRA-NRCellResourceCoordinationReqAckIEs}},
```

```
    ...
}


ENB-EUTRA-NRCellResourceCoordinationReqAckIEs X2AP-PROTOCOL-IES ::= {
    { ID id-DataTrafficResourceIndication              CRITICALITY reject  TYPE DataTrafficResourceIndication           PRESENCE mandatory}|
    { ID id-SpectrumSharingGroupID                     CRITICALITY reject  TYPE SpectrumSharingGroupID                  PRESENCE mandatory}|
    { ID id-ListofEUTRACellsinEUTRACoordinationResp    CRITICALITY reject  TYPE ListofEUTRACellsinEUTRACoordinationResp PRESENCE mandatory},
    ...
}


En-gNB-EUTRA-NRCellResourceCoordinationReqAckIEs X2AP-PROTOCOL-IES ::= {
    { ID id-DataTrafficResourceIndication      CRITICALITY reject  TYPE DataTrafficResourceIndication       PRESENCE mandatory}|
    { ID id-SpectrumSharingGroupID             CRITICALITY reject  TYPE SpectrumSharingGroupID              PRESENCE mandatory}|
    { ID id-ListofNRCellsinNRCoordinationResp  CRITICALITY reject  TYPE ListofNRCellsinNRCoordinationResp   PRESENCE mandatory},


    ...
}

ListofEUTRACellsinEUTRACoordinationResp ::= SEQUENCE (SIZE (0..maxCellineNB)) OF ECGI
ListofNRCellsinNRCoordinationResp ::= SEQUENCE (SIZE (0..maxnoNRcellsSpectrumSharingWithE-UTRA)) OF NRCGI

-- **************************************************************
--
-- EN-DC X2 REMOVAL REQUEST
--
-- **************************************************************

ENDCX2RemovalRequest ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCX2RemovalRequest-IEs}},
    ...
}

ENDCX2RemovalRequest-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-InitiatingNodeType-EndcX2Removal           CRITICALITY reject  TYPE InitiatingNodeType-EndcX2Removal     PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication                CRITICALITY reject  TYPE InterfaceInstanceIndication          PRESENCE optional },
    ...
}

InitiatingNodeType-EndcX2Removal ::= CHOICE {
    init-eNB                ProtocolIE-Container    {{ENB-ENDCX2RemovalReqIEs}},
    init-en-gNB         ProtocolIE-Container    {{En-gNB-ENDCX2RemovalReqIEs}},
    ...
}

ENB-ENDCX2RemovalReqIEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID                               CRITICALITY reject  TYPE GlobalENB-ID                          PRESENCE mandatory},
    ...
}

En-gNB-ENDCX2RemovalReqIEs X2AP-PROTOCOL-IES ::= {
    { ID id-Globalen-gNB-ID                        CRITICALITY reject  TYPE GlobalGNB-ID                          PRESENCE mandatory},
```

```
    ...
}

-- **************************************************************
--
-- EN-DC X2 REMOVAL RESPONSE
--
-- **************************************************************

ENDCX2RemovalResponse ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCX2RemovalResponse-IEs}},
    ...
}

ENDCX2RemovalResponse-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-RespondingNodeType-EndcX2Removal        CRITICALITY reject  TYPE RespondingNodeType-EndcX2Removal   PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication             CRITICALITY reject  TYPE InterfaceInstanceIndication          PRESENCE optional },
    ...
}

RespondingNodeType-EndcX2Removal ::= CHOICE {
    respond-eNB         ProtocolIE-Container    {{ENB-ENDCX2RemovalReqAckIEs}},
    respond-en-gNB      ProtocolIE-Container    {{En-gNB-ENDCX2RemovalReqAckIEs}},
    ...
}

ENB-ENDCX2RemovalReqAckIEs X2AP-PROTOCOL-IES ::= {
    { ID id-GlobalENB-ID                            CRITICALITY reject  TYPE GlobalENB-ID                        PRESENCE mandatory},
    ...
}

En-gNB-ENDCX2RemovalReqAckIEs X2AP-PROTOCOL-IES ::= {
    { ID id-Globalen-gNB-ID                         CRITICALITY reject  TYPE GlobalGNB-ID                         PRESENCE mandatory},
    ...
}

-- **************************************************************
--
-- EN-DC X2 REMOVAL FAILURE
--
-- **************************************************************

ENDCX2RemovalFailure ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCX2RemovalFailure-IEs}},
    ...
}

ENDCX2RemovalFailure-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-Cause                           CRITICALITY ignore      TYPE Cause                          PRESENCE mandatory} |
    { ID id-CriticalityDiagnostics          CRITICALITY ignore      TYPE CriticalityDiagnostics         PRESENCE optional }|
    { ID id-InterfaceInstanceIndication     CRITICALITY reject      TYPE InterfaceInstanceIndication    PRESENCE optional },
    ...
}
```

```
-- **************************************************************
--
-- DATA FORWARDING ADDRESS INDICATION
--
-- **************************************************************

DataForwardingAddressIndication ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{DataForwardingAddressIndication-IEs}},
    ...
}

DataForwardingAddressIndication-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-New-eNB-UE-X2AP-ID                  CRITICALITY ignore  TYPE UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-New-eNB-UE-X2AP-ID-Extension        CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                PRESENCE optional }|
    { ID id-Old-eNB-UE-X2AP-ID                  CRITICALITY ignore  TYPE UE-X2AP-ID                          PRESENCE mandatory}|
    { ID id-Old-eNB-UE-X2AP-ID-Extension        CRITICALITY ignore  TYPE UE-X2AP-ID-Extension                PRESENCE optional }|
    { ID id-E-RABs-DataForwardingAddress-List   CRITICALITY ignore  TYPE E-RABs-DataForwardingAddress-List   PRESENCE mandatory}|
    { ID id-CHO-DC-Indicator                    CRITICALITY reject  TYPE CHO-DC-Indicator                    PRESENCE optional}|
    { ID id-CHO-DC-EarlyDataForwarding          CRITICALITY ignore  TYPE CHO-DC-EarlyDataForwarding          PRESENCE optional}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY ignore  TYPE SgNB-UE-X2AP-ID                     PRESENCE optional}|
    { ID id-CPCinformation-NOTIFY               CRITICALITY reject  TYPE CPCinformation-NOTIFY               PRESENCE optional},
    ...
}

E-RABs-DataForwardingAddress-List ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RABs-DataForwardingAddress-ItemIEs} }

E-RABs-DataForwardingAddress-ItemIEs    X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABs-DataForwardingAddress-Item   CRITICALITY ignore  TYPE E-RABs-DataForwardingAddress-Item PRESENCE mandatory},
    ...
}

E-RABs-DataForwardingAddress-Item ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    dl-GTPtunnelEndpoint            GTPtunnelEndpoint,
    iE-Extensions                   ProtocolExtensionContainer { {E-RABs-DataForwardingAddress-ItemExtIEs} } OPTIONAL,
    ...
}

E-RABs-DataForwardingAddress-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- **************************************************************
--
-- GNB STATUS INDICATION
--
-- **************************************************************


GNBStatusIndication ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container    { { GNBStatusIndicationIEs} },
    ...
}
```

```
GNBStatusIndicationIEs X2AP-PROTOCOL-IES ::= {
    { ID id-GNBOverloadInformation          CRITICALITY ignore  TYPE GNBOverloadInformation      PRESENCE mandatory}|
    { ID id-InterfaceInstanceIndication     CRITICALITY reject  TYPE InterfaceInstanceIndication    PRESENCE optional },
    ...
}

-- **************************************************************
--
-- EN-DC CONFIGURATION TRANSFER
--
-- **************************************************************

ENDCConfigurationTransfer ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container    {{ENDCConfigurationTransfer-IEs}},
    ...
}

ENDCConfigurationTransfer-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-endcSONConfigurationTransfer    CRITICALITY ignore  TYPE EndcSONConfigurationTransfer      PRESENCE optional}|
    { ID id-InterfaceInstanceIndication     CRITICALITY reject  TYPE InterfaceInstanceIndication         PRESENCE optional },
    ...
}

-- **************************************************************
--
-- TRACE START
--
-- **************************************************************

TraceStart ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { {TraceStartIEs} },
    ...
}

TraceStartIEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID             CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory }|
    { ID id-SgNB-UE-X2AP-ID             CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                  PRESENCE mandatory   }|
    { ID id-TraceActivation             CRITICALITY ignore  TYPE TraceActivation                  PRESENCE mandatory   }|
    { ID id-MeNB-UE-X2AP-ID-Extension   CRITICALITY reject      TYPE UE-X2AP-ID-Extension        PRESENCE optional},
    ...
}

-- **************************************************************
--
-- DEACTIVATE TRACE
--
-- **************************************************************

DeactivateTrace ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { {DeactivateTraceIEs} },
    ...
}

DeactivateTraceIEs X2AP-PROTOCOL-IES ::= {
```

```
    { ID id-MeNB-UE-X2AP-ID            CRITICALITY reject  TYPE UE-X2AP-ID                    PRESENCE mandatory }|
    { ID id-SgNB-UE-X2AP-ID            CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                PRESENCE mandatory  }|
    { ID id-EUTRANTraceID              CRITICALITY ignore  TYPE EUTRANTraceID                  PRESENCE mandatory }|
    { ID id-MeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional},
    ...
}

-- **************************************************************
--
-- CELL TRAFFIC TRACE
--
-- **************************************************************

CellTrafficTrace ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        { {CellTrafficTraceIEs} },
    ...
}

CellTrafficTraceIEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                 CRITICALITY reject  TYPE UE-X2AP-ID                     PRESENCE mandatory  }|
    { ID id-SgNB-UE-X2AP-ID                 CRITICALITY reject  TYPE SgNB-UE-X2AP-ID                PRESENCE mandatory  }|
    { ID id-EUTRANTraceID                   CRITICALITY ignore  TYPE EUTRANTraceID                  PRESENCE mandatory  }|
    { ID id-TraceCollectionEntityIPAddress  CRITICALITY ignore  TYPE TraceCollectionEntityIPAddress PRESENCE mandatory  }|
    { ID id-PrivacyIndicator                CRITICALITY ignore  TYPE PrivacyIndicator               PRESENCE optional }|
    { ID id-MeNB-UE-X2AP-ID-Extension       CRITICALITY reject  TYPE UE-X2AP-ID-Extension           PRESENCE optional},
    ...
}

-- **************************************************************
--
-- F1-C TRAFFIC TRANSFER
--
-- **************************************************************

F1CTrafficTransfer ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{ F1CTrafficTransfer-IEs}},
    ...
}

F1CTrafficTransfer-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                     CRITICALITY reject  TYPE UE-X2AP-ID                 PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                     CRITICALITY reject  TYPE SgNB-UE-X2AP-ID            PRESENCE mandatory}|
    { ID id-F1CTrafficContainer                 CRITICALITY reject  TYPE F1CTrafficContainer        PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension           CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional},
    ...
}

-- **************************************************************
--
-- UE RADIO CAPABILITY ID MAPPING REQUEST
--
-- **************************************************************

UERadioCapabilityIDMappingRequest::= SEQUENCE {
```

```
    protocolIEs         ProtocolIE-Container       { { UERadioCapabilityIDMappingRequestIEs} },
    ...
}

UERadioCapabilityIDMappingRequestIEs X2AP-PROTOCOL-IES ::= {
    { ID id-UERadioCapabilityID     CRITICALITY reject  TYPE UERadioCapabilityID        PRESENCE mandatory },
    ...
}

-- **************************************************************
--
-- UE RADIO CAPABILITY ID MAPPING RESPONSE
--
-- **************************************************************

UERadioCapabilityIDMappingResponse ::= SEQUENCE {
    protocolIEs         ProtocolIE-Container       { { UERadioCapabilityIDMappingResponseIEs} },
    ...
}

UERadioCapabilityIDMappingResponseIEs X2AP-PROTOCOL-IES ::= {
    { ID id-UERadioCapabilityID             CRITICALITY reject  TYPE UERadioCapabilityID         PRESENCE mandatory }|
    { ID id-UERadioCapability               CRITICALITY ignore  TYPE UERadioCapability           PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics      PRESENCE optional  },
    ...
}


-- **************************************************************
--
-- CONDITIONAL PSCELL CHANGE CANCEL
--
-- **************************************************************

CPC-cancel ::= SEQUENCE {
    protocolIEs     ProtocolIE-Container        {{CPC-cancel-IEs}},
    ...
}

CPC-cancel-IEs X2AP-PROTOCOL-IES ::= {
    { ID id-MeNB-UE-X2AP-ID                   CRITICALITY reject  TYPE UE-X2AP-ID                PRESENCE mandatory}|
    { ID id-SgNB-UE-X2AP-ID                   CRITICALITY reject  TYPE SgNB-UE-X2AP-ID            PRESENCE mandatory}|
    { ID id-MeNB-UE-X2AP-ID-Extension         CRITICALITY reject  TYPE UE-X2AP-ID-Extension       PRESENCE optional}|
    { ID id-Cause                             CRITICALITY ignore  TYPE Cause                      PRESENCE optional}|
    { ID id-Target-SgNB-ID                    CRITICALITY reject  TYPE GlobalGNB-ID               PRESENCE mandatory},
    ...
}


END
-- ASN1STOP
```

## 9.3.5 Information Element definitions

```
-- ASN1START
-- **************************************************************
--
-- Information Element Definitions
--
-- **************************************************************

X2AP-IEs {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
eps-Access (21) modules (3) x2ap (2) version1 (1) x2ap-IEs (2) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

IMPORTS

    id-E-RAB-Item,
    id-Number-of-Antennaports,
    id-MBSFN-Subframe-Info,
    id-PRACH-Configuration,
    id-CSG-Id,
    id-MDTConfiguration,
    id-SignallingBasedMDTPLMNList,
    id-MultibandInfoList,
    id-FreqBandIndicatorPriority,
    id-NeighbourTAC,
    id-Time-UE-StayedInCell-EnhancedGranularity,
    id-MBMS-Service-Area-List,
    id-HO-cause,
    id-eARFCNExtension,
    id-DL-EARFCNExtension,
    id-UL-EARFCNExtension,
    id-M3Configuration,
    id-M4Configuration,
    id-M5Configuration,
    id-MDT-Location-Info,
    id-NRrestrictioninEPSasSecondaryRAT,
    id-NRrestrictionin5GS,
    id-AdditionalSpecialSubframe-Info,
    id-UEID,
    id-enhancedRNTP,
    id-ProSeUEtoNetworkRelaying,
    id-M6Configuration,
    id-M7Configuration,
    id-OffsetOfNbiotChannelNumberToDL-EARFCN,
    id-OffsetOfNbiotChannelNumberToUL-EARFCN,
    id-AdditionalSpecialSubframeExtension-Info,
    id-BandwidthReducedSI,
    id-extended-e-RAB-MaximumBitrateDL,
    id-extended-e-RAB-MaximumBitrateUL,
```

```
    id-extended-e-RAB-GuaranteedBitrateDL,
    id-extended-e-RAB-GuaranteedBitrateUL,
    id-extended-uEaggregateMaximumBitRateDownlink,
    id-extended-uEaggregateMaximumBitRateUplink,
    id-E-RABUsageReport-Item,
    id-SecondaryRATUsageReport-Item,
    id-UEAppLayerMeasConfig,
    id-DL-scheduling-PDCCH-CCE-usage,
    id-UL-scheduling-PDCCH-CCE-usage,
    id-DownlinkPacketLossRate,
    id-UplinkPacketLossRate,
    id-serviceType,
    id-ProtectedEUTRAResourceIndication,
    id-NRS-NSSS-PowerOffset,
    id-NSSS-NumOccasionDifferentPrecoder,
    id-CNTypeRestrictions,
    id-BluetoothMeasurementConfiguration,
    id-WLANMeasurementConfiguration,
    id-ECGI,
    id-NRCGI,
    id-MeNBCoordinationAssistanceInformation,
    id-SgNBCoordinationAssistanceInformation,
    id-NRNeighbourInfoToAdd,
    id-LastNG-RANPLMNIdentity,
    id-BPLMN-ID-Info-EUTRA,
    id-NBIoT-UL-DL-AlignmentOffset,
    id-UnlicensedSpectrumRestriction,
    id-CarrierList,
    id-FrequencyShift7p5khz,
    id-NPRACHConfiguration,
    id-MDTConfigurationNR,
    id-CSI-RSTransmissionIndication,
    id-QoS-Mapping-Information,
    id-IntendedTDD-DL-ULConfiguration-NR,
    id-TraceCollectionEntityURI,
    id-SFN-Offset,
    id-AdditionLocationInformation,
    id-SSB-PositionsInBurst,
    id-NRCellPRACHConfig,
    id-ULCarrierList,
    id-TDDULDLConfigurationCommonNR,
    id-MIMOPRBusageInformation,
    id-Additional-Measurement-Timing-Configuration-List,
    id-ServedCellSpecificInfoReq-NR,
    id-RAT-Restrictions,
    id-PSCell-UE-HistoryInformation,
    id-SensorMeasurementConfiguration,

    maxnoofBearers,
    maxCellineNB,
    maxEARFCN,
    maxEARFCNPlusOne,
    newmaxEARFCN,
    maxInterfaces,
```

```
    maxnoofBands,
    maxnoofBPLMNs,
    maxnoofAdditionalPLMNs,
    maxnoofCells,
    maxnoofEPLMNs,
    maxnoofEPLMNsPlusOne,
    maxnoofForbLACs,
    maxnoofForbTACs,
    maxnoofNeighbours,
    maxnoofPRBs,
    maxNrOfErrors,
    maxPools,
    maxnoofMBSFN,
    maxnoofTAforMDT,
    maxnoofCellIDforMDT,
    maxnoofMBMSServiceAreaIdentities,
    maxnoofMDTPLMNs,
    maxnoofCoMPHypothesisSet,
    maxnoofCoMPCells,
    maxUEReport,
    maxCellReport,
    maxnoofPA,
    maxCSIProcess,
    maxCSIReport,
    maxSubband,
    maxnooftimeperiods,
    maxnoofCellIDforQMC,
    maxnoofTAforQMC,
    maxnoofPLMNforQMC,
    maxUEsinengNBDU,
    maxnoofProtectedResourcePatterns,
    maxnoNRcellsSpectrumSharingWithE-UTRA,
    maxnoofNrCellBands,
    maxnoofBluetoothName,
    maxnoofWLANName,
    maxofNRNeighbours,
    maxnoofextBPLMNs,
    maxnoofTLAs,
    maxnoofGTPTLAs,
    maxnoofTNLAssociations,
    maxnoofCellsinCHO,  maxnoofPC5QoSFlows,
    maxnoofSSBAreas,
    maxnoofNRSCSs,
    maxnoofNRPhysicalResourceBlocks,
    maxnoofNonAnchorCarrierFreqConfig,
    maxnoofRACHReports,
    maxnoofPSCellsPerSN,
    maxnoofPSCellsPerPrimaryCellinUEHistoryInfo,
    maxnoofReportedNRCellsPossiblyAggregated,
    maxnoofPSCellCandidates,
    maxnoofTargetSgNBs,
    maxnoofMTCItems,
    maxnoofCSIRSconfigurations,
```

```
    maxnoofCSIRSneighbourCells,
    maxnoofCSIRSneighbourCellsInMTC,
    maxCellinengNB,
    maxnoofSensorName,
    maxnoofTargetSgNBsMinusOne


FROM X2AP-Constants

    Criticality,
    ProcedureCode,
    ProtocolIE-ID,
    TriggeringMessage
FROM X2AP-CommonDataTypes

    ProtocolExtensionContainer{},
    ProtocolIE-Single-Container{},

    X2AP-PROTOCOL-EXTENSION,
    X2AP-PROTOCOL-IES
FROM X2AP-Containers;

-- A

ABSInformation ::= CHOICE {
    fdd                 ABSInformationFDD,
    tdd                 ABSInformationTDD,
    abs-inactive        NULL,
    ...
}

ABSInformationFDD ::= SEQUENCE {
    abs-pattern-info                    BIT STRING (SIZE(40)),
    numberOfCellSpecificAntennaPorts    ENUMERATED {one, two, four, ...},
    measurement-subset                  BIT STRING (SIZE(40)),
    iE-Extensions                       ProtocolExtensionContainer { {ABSInformationFDD-ExtIEs} } OPTIONAL,
    ...
}

ABSInformationFDD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ABSInformationTDD ::= SEQUENCE {
    abs-pattern-info                    BIT STRING (SIZE(1..70, ...)),
    numberOfCellSpecificAntennaPorts    ENUMERATED {one, two, four, ...},
    measurement-subset                  BIT STRING (SIZE(1..70, ...)),
    iE-Extensions                       ProtocolExtensionContainer { {ABSInformationTDD-ExtIEs} } OPTIONAL,
    ...
}

ABSInformationTDD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
ABS-Status ::= SEQUENCE {
    dL-ABS-status                           DL-ABS-status,
    usableABSInformation                    UsableABSInformation,
    iE-Extensions                           ProtocolExtensionContainer { {ABS-Status-ExtIEs} } OPTIONAL,
    ...
}

ABS-Status-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ActivationID ::= INTEGER (0..255)

Additional-Measurement-Timing-Configuration-List ::= SEQUENCE (SIZE(1.. maxnoofMTCItems)) OF Additional-Measurement-Timing-Configuration-Item

Additional-Measurement-Timing-Configuration-Item ::= SEQUENCE {
    additionalMeasurementTimingConfiguration        INTEGER(0..16),
    csi-RS-MTC-Configuration-List       CSI-RS-MTC-Configuration-List,
    iE-Extensions                       ProtocolExtensionContainer { { Additional-Measurement-Timing-Configuration-Item-ExtIEs} } OPTIONAL,
    ...
}

Additional-Measurement-Timing-Configuration-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSI-RS-MTC-Configuration-List ::= SEQUENCE (SIZE(1.. maxnoofCSIRSconfigurations)) OF CSI-RS-MTC-Configuration-Item

CSI-RS-MTC-Configuration-Item ::= SEQUENCE {
    csi-RS-Index            INTEGER(0..95),
    csi-RS-Status           ENUMERATED {activated,deactivated, ...},
    csi-RS-Neighbour-List   CSI-RS-Neighbour-List OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { CSI-RS-MTC-Configuration-Item-ExtIEs} }   OPTIONAL,
    ...
}

CSI-RS-MTC-Configuration-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSI-RS-Neighbour-List ::= SEQUENCE (SIZE(1.. maxnoofCSIRSneighbourCells)) OF CSI-RS-Neighbour-Item

CSI-RS-Neighbour-Item ::= SEQUENCE {
    nr-cgi                          NRCGI,
    csi-RS-MTC-Neighbour-List   CSI-RS-MTC-Neighbour-List OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { CSI-RS-Neighbour-Item-ExtIEs} }  OPTIONAL,
    ...
}

CSI-RS-Neighbour-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
CSI-RS-MTC-Neighbour-List ::= SEQUENCE (SIZE(1.. maxnoofCSIRSneighbourCellsInMTC)) OF CSI-RS-MTC-Neighbour-Item

CSI-RS-MTC-Neighbour-Item ::= SEQUENCE {
    csi-RS-Index        INTEGER(0..95),
    iE-Extensions       ProtocolExtensionContainer { { CSI-RS-MTC-Neighbour-Item-ExtIEs} }  OPTIONAL,
    ...
}


CSI-RS-MTC-Neighbour-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalListofForwardingGTPTunnelEndpoint ::= SEQUENCE (SIZE(1..maxnoofTargetSgNBsMinusOne)) OF AdditionalListofForwardingGTPTunnelEndpoint-Item

AdditionalListofForwardingGTPTunnelEndpoint-Item ::= SEQUENCE {
    uL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                   OPTIONAL,
    dL-GTPtunnelEndpoint            GTPtunnelEndpoint                                                   OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { AdditionalListofForwardingGTPTunnelEndpoint-Item-ExtIEs} } OPTIONAL,
    ...
}

AdditionalListofForwardingGTPTunnelEndpoint-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionLocationInformation ::= ENUMERATED {
    includePSCell,
    ...
}

AdditionalRRMPriorityIndex ::= BIT STRING (SIZE(32))

AdditionalSpecialSubframe-Info ::=      SEQUENCE {
    additionalspecialSubframePatterns       AdditionalSpecialSubframePatterns,
    cyclicPrefixDL                          CyclicPrefixDL,
    cyclicPrefixUL                          CyclicPrefixUL,
    iE-Extensions                           ProtocolExtensionContainer { {AdditionalSpecialSubframe-Info-ExtIEs} } OPTIONAL,
    ...
}

AdditionalSpecialSubframe-Info-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalSpecialSubframePatterns ::= ENUMERATED {
    ssp0,
    ssp1,
    ssp2,
    ssp3,
    ssp4,
    ssp5,
    ssp6,
```

```
    ssp7,
    ssp8,
    ssp9,
    ...
}

AdditionalSpecialSubframeExtension-Info ::=     SEQUENCE {
    additionalspecialSubframePatternsExtension  AdditionalSpecialSubframePatternsExtension,
    cyclicPrefixDL                              CyclicPrefixDL,
    cyclicPrefixUL                              CyclicPrefixUL,
    iE-Extensions                               ProtocolExtensionContainer { {AdditionalSpecialSubframeExtension-Info-ExtIEs} } OPTIONAL,
    ...
}

AdditionalSpecialSubframeExtension-Info-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalSpecialSubframePatternsExtension ::= ENUMERATED {
    ssp10,
    ...
}

AvailableFastMCGRecoveryViaSRB3 ::= ENUMERATED {true,...}

AerialUEsubscriptionInformation ::= ENUMERATED {
    allowed,
    not-allowed,
    ...
}

AllocationAndRetentionPriority ::= SEQUENCE {
    priorityLevel               PriorityLevel,
    pre-emptionCapability       Pre-emptionCapability,
    pre-emptionVulnerability    Pre-emptionVulnerability,
    iE-Extensions               ProtocolExtensionContainer { {AllocationAndRetentionPriority-ExtIEs} } OPTIONAL,
    ...
}

AllocationAndRetentionPriority-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

AreaScopeOfMDT ::= CHOICE {
    cellBased                   CellBasedMDT,
    tABased                     TABasedMDT,
    pLMNWide                    NULL,
    ...,
    tAIBased                    TAIBasedMDT
}

AreaScopeOfQMC ::= CHOICE {
    cellBased                   CellBasedQMC,
    tABased                     TABasedQMC,
```

```
    tAIBased                    TAIBasedQMC,
    pLMNAreaBased               PLMNAreaBasedQMC,
    ...
}

AS-SecurityInformation ::= SEQUENCE {
    key-eNodeB-star     Key-eNodeB-Star,
    nextHopChainingCount            NextHopChainingCount,
    iE-Extensions                       ProtocolExtensionContainer { {AS-SecurityInformation-ExtIEs} } OPTIONAL,
    ...
}

AS-SecurityInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

AdditionalPLMNs-Item ::= SEQUENCE (SIZE(1..maxnoofAdditionalPLMNs)) OF PLMN-Identity


-- B

BandwidthReducedSI::= ENUMERATED {
    scheduled,
    ...
}

BearerType ::= ENUMERATED {
    non-IP,
    ...
}

BenefitMetric ::= INTEGER (-101..100, ...)

BitRate ::= INTEGER (0..10000000000)
BroadcastPLMNs-Item ::= SEQUENCE (SIZE(1..maxnoofBPLMNs)) OF PLMN-Identity

BluetoothMeasurementConfiguration ::= SEQUENCE {
    bluetoothMeasConfig             BluetoothMeasConfig,
    bluetoothMeasConfigNameList     BluetoothMeasConfigNameList         OPTIONAL,
    bt-rssi                         ENUMERATED {true, ...}              OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {BluetoothMeasurementConfiguration-ExtIEs} } OPTIONAL,
    ...
}

BluetoothMeasurementConfiguration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

BluetoothMeasConfigNameList ::= SEQUENCE (SIZE(1..maxnoofBluetoothName)) OF BluetoothName

BluetoothMeasConfig::= ENUMERATED {setup,...}

BluetoothName ::= OCTET STRING (SIZE (1..248))
```

```
BPLMN-ID-Info-EUTRA ::= SEQUENCE (SIZE(1..maxnoofBPLMNs)) OF BPLMN-ID-Info-EUTRA-Item

BPLMN-ID-Info-EUTRA-Item ::= SEQUENCE {
    broadcastPLMNs                  BroadcastPLMNs-Item,
    tac                             TAC,
    e-utraCI                        EUTRANCellIdentifier,
    iE-Extension                    ProtocolExtensionContainer { {BPLMN-ID-Info-EUTRA-Item-ExtIEs} } OPTIONAL,
    ...
}

BPLMN-ID-Info-EUTRA-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

BPLMN-ID-Info-NR ::= SEQUENCE (SIZE(1..maxnoofextBPLMNs)) OF BPLMN-ID-Info-NR-Item

BPLMN-ID-Info-NR-Item ::= SEQUENCE {
    broadcastPLMNs                  BroadcastextPLMNs,
    fiveGS-TAC                      FiveGS-TAC      OPTIONAL,
    nr-CI                           NRCellIdentifier,
    iE-Extension                    ProtocolExtensionContainer { {BPLMN-ID-Info-NR-Item-ExtIEs} } OPTIONAL,
    ...
}

BPLMN-ID-Info-NR-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

BroadcastextPLMNs ::= SEQUENCE (SIZE(1..maxnoofextBPLMNs)) OF PLMN-Identity

-- C

CapacityValue ::= INTEGER (0..100)

Cause ::= CHOICE {
    radioNetwork        CauseRadioNetwork,
    transport           CauseTransport,
    protocol            CauseProtocol,
    misc                CauseMisc,
    ...
}

CauseMisc ::= ENUMERATED {
    control-processing-overload,
    hardware-failure,
    om-intervention,
    not-enough-user-plane-processing-resources,
    unspecified,
    ...
}

CauseProtocol ::= ENUMERATED {
    transfer-syntax-error,
    abstract-syntax-error-reject,
```

```
    abstract-syntax-error-ignore-and-notify,
    message-not-compatible-with-receiver-state,
    semantic-error,
    unspecified,
    abstract-syntax-error-falsely-constructed-message,
    ...
}

CauseRadioNetwork ::= ENUMERATED {
    handover-desirable-for-radio-reasons,
    time-critical-handover,
    resource-optimisation-handover,
    reduce-load-in-serving-cell,
    partial-handover,
    unknown-new-eNB-UE-X2AP-ID,
    unknown-old-eNB-UE-X2AP-ID,
    unknown-pair-of-UE-X2AP-ID,
    ho-target-not-allowed,
    tx2relocoverall-expiry,
    trelocprep-expiry,
    cell-not-available,
    no-radio-resources-available-in-target-cell,
    invalid-MME-GroupID,
    unknown-MME-Code,
    encryption-and-or-integrity-protection-algorithms-not-supported,
    reportCharacteristicsEmpty,
    noReportPeriodicity,
    existingMeasurementID,
    unknown-eNB-Measurement-ID,
    measurement-temporarily-not-available,
    unspecified,
    ...,
    load-balancing,
    handover-optimisation,
    value-out-of-allowed-range,
    multiple-E-RAB-ID-instances,
    switch-off-ongoing,
    not-supported-QCI-value,
    measurement-not-supported-for-the-object,
    tDCoverall-expiry,
    tDCprep-expiry,
    action-desirable-for-radio-reasons,
    reduce-load,
    resource-optimisation,
    time-critical-action,
    target-not-allowed,
    no-radio-resources-available,
    invalid-QoS-combination,
    encryption-algorithms-not-supported,
    procedure-cancelled,
    rRM-purpose,
    improve-user-bit-rate,
    user-inactivity,
    radio-connection-with-UE-lost,
```

```
    failure-in-the-radio-interface-procedure,
    bearer-option-not-supported,
    mCG-Mobility,
    sCG-Mobility,
    count-reaches-max-value,
    unknown-old-en-gNB-UE-X2AP-ID,
    pDCP-Overload,
    cho-cpc-resources-tobechanged,
    ue-power-saving,
    insufficient-ue-capabilities,
    normal-release,
    unknown-E-UTRAN-Node-Measurement-ID,
    sCG-activation-deactivation-failure,
    sCG-deactivation-failure-due-to-data-transmission,
    up-integrity-protection-not-possible


}

CauseTransport ::= ENUMERATED {
    transport-resource-unavailable,
    unspecified,
    ...
}

CellBasedMDT::= SEQUENCE {
    cellIdListforMDT    CellIdListforMDT,
    iE-Extensions       ProtocolExtensionContainer { {CellBasedMDT-ExtIEs} } OPTIONAL,
    ...
}

CellBasedMDT-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellBasedQMC::= SEQUENCE {
    cellIdListforQMC        CellIdListforQMC,
    iE-Extensions       ProtocolExtensionContainer { {CellBasedQMC-ExtIEs} } OPTIONAL,
    ...
}

CellBasedQMC-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellCapacityClassValue ::= INTEGER (1..100, ...)

CellDeploymentStatusIndicator ::= ENUMERATED {pre-change-notification, ...}

CellIdListforMDT ::= SEQUENCE (SIZE(1..maxnoofCellIDforMDT)) OF ECGI

CellIdListforQMC ::= SEQUENCE (SIZE(1..maxnoofCellIDforQMC)) OF ECGI

CellReplacingInfo ::= SEQUENCE {
```

```
    replacingCellsList              ReplacingCellsList,
    iE-Extensions                   ProtocolExtensionContainer { {CellReplacingInfo-ExtIEs}}    OPTIONAL,
    ...
}

CellReplacingInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CellReportingIndicator ::= ENUMERATED {stop-request, ... }

Cell-Size ::= ENUMERATED {verysmall, small, medium, large, ... }


CellType ::= SEQUENCE {
    cell-Size                       Cell-Size,
    iE-Extensions                   ProtocolExtensionContainer { {CellType-ExtIEs}} OPTIONAL,
    ...
}

CellType-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPACcandidatePSCells-list ::= SEQUENCE (SIZE(1..maxnoofPSCellCandidates)) OF CPACcandidatePSCells-item

CPACcandidatePSCells-item ::= SEQUENCE {
    pscell-id                       NRCGI,
    iE-Extensions                   ProtocolExtensionContainer { {CPACcandidatePSCells-item-ExtIEs}} OPTIONAL,
    ...
}

CPACcandidatePSCells-item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPCindicator ::= ENUMERATED {cpc-initiation, cpc-modification, cpc-cancel, ...}


CPCdataforwarding ::= ENUMERATED {cpc-triggered, early-data-transmission-stop, ..., coordination-only}


CPAinformation-REQ ::= SEQUENCE {
    max-no-of-pscells               INTEGER (1..maxnoofPSCellCandidates),
    estimatedArrivalProbability     CHO-Probability                         OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {CPAinformation-REQ-ExtIEs} } OPTIONAL,
    ...
}

CPAinformation-REQ-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
CPAinformation-REQ-ACK ::= SEQUENCE {
    candidate-pscells                   CPACcandidatePSCells-list,
    iE-Extensions       ProtocolExtensionContainer { {CPAinformation-REQ-ACK-ExtIEs} } OPTIONAL,
    ...
}

CPAinformation-REQ-ACK-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPCinformation-REQD ::= SEQUENCE {
    cpc-target-sgnb-list        CPC-target-SgNB-reqd-list,
    iE-Extensions           ProtocolExtensionContainer { {CPCinformation-REQD-ExtIEs} } OPTIONAL,
    ...
}


CPCinformation-REQD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPC-target-SgNB-reqd-list ::= SEQUENCE (SIZE(1..maxnoofTargetSgNBs)) OF CPC-target-SgNB-reqd-item


CPC-target-SgNB-reqd-item ::= SEQUENCE {
    target-SgNB-ID                  GlobalGNB-ID,
    cpc-indicator                   CPCindicator,
    max-no-of-pscells               INTEGER (1..maxnoofPSCellCandidates),
    estimatedArrivalProbability     CHO-Probability                         OPTIONAL,
    sgNBtoMeNBContainer             SgNBtoMeNBContainer,
    iE-Extensions       ProtocolExtensionContainer { {CPC-target-SgNB-reqd-item-ExtIEs} } OPTIONAL,
    ...
}

CPC-target-SgNB-reqd-item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPCinformation-CONF ::= SEQUENCE {
    cpc-target-sgnb-list        CPC-target-SgNB-conf-list,
    iE-Extensions           ProtocolExtensionContainer { {CPCinformation-CONF-ExtIEs} } OPTIONAL,
    ...
}


CPCinformation-CONF-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
CPC-target-SgNB-conf-list ::= SEQUENCE (SIZE(1..maxnoofTargetSgNBs)) OF CPC-target-SgNB-conf-item


CPC-target-SgNB-conf-item ::= SEQUENCE {
    target-SgNB-ID                      GlobalGNB-ID,
    candidate-pscells                   CPACcandidatePSCells-list,
    iE-Extensions       ProtocolExtensionContainer { {CPC-target-SgNB-conf-item-ExtIEs} } OPTIONAL,
    ...
}

CPC-target-SgNB-conf-item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPCinformation-NOTIFY ::= SEQUENCE {
    cpc-indicator                   CPCdataforwarding,
    iE-Extensions       ProtocolExtensionContainer { {CPCinformation-NOTIFY-ExtIEs} } OPTIONAL,
    ...
}

CPCinformation-NOTIFY-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPAinformation-MOD ::= SEQUENCE {
    max-no-of-pscells               INTEGER (1..maxnoofPSCellCandidates)    OPTIONAL,
    estimatedArrivalProbability     CHO-Probability                         OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {CPAinformation-MOD-ExtIEs} } OPTIONAL,
    ...
}

CPAinformation-MOD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPCupdate-MOD ::= SEQUENCE {

    cpc-target-sgnb-list                    CPC-target-SgNB-mod-list,
    iE-Extensions                       ProtocolExtensionContainer { {CPCupdate-MOD-ExtIEs} } OPTIONAL,
    ...
}

CPCupdate-MOD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CPC-target-SgNB-mod-list ::= SEQUENCE (SIZE(1..maxnoofTargetSgNBs)) OF CPC-target-SgNB-mod-item

CPC-target-SgNB-mod-item ::= SEQUENCE {
    target-SgNB-ID                      GlobalGNB-ID,
```

```
    candidate-pscells                   CPACcandidatePSCells-list,
    iE-Extensions       ProtocolExtensionContainer { {CPC-target-SgNB-mod-item-ExtIEs} } OPTIONAL,
    ...
}

CPC-target-SgNB-mod-item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPAinformation-MOD-ACK ::= SEQUENCE {
    candidate-pscells                   CPACcandidatePSCells-list,
    iE-Extensions                       ProtocolExtensionContainer { {CPAinformation-MOD-ACK-ExtIEs} } OPTIONAL,
    ...
}

CPAinformation-MOD-ACK-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CPACinformation-REQD ::= SEQUENCE {
    candidate-pscells                   CPACcandidatePSCells-list,
    iE-Extensions                       ProtocolExtensionContainer { { CPACinformation-REQD-ExtIEs} } OPTIONAL,
    ...
}

CPACinformation-REQD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


CHO-DC-EarlyDataForwarding ::= ENUMERATED {stop, ...}

CHO-DC-Indicator ::= ENUMERATED {true, ..., coordination-only}

CNTypeRestrictions ::= SEQUENCE (SIZE(1.. maxnoofEPLMNsPlusOne)) OF CNTypeRestrictionsItem

CNTypeRestrictionsItem ::= SEQUENCE {
    plmn-Id             PLMN-Identity,
    cn-type             ENUMERATED {fiveGC-forbidden, ... , epc-forbidden},
    iE-Extensions       ProtocolExtensionContainer { {CNTypeRestrictionsItem-ExtIEs} } OPTIONAL,
    ...
}

CNTypeRestrictionsItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CoMPHypothesisSet ::= SEQUENCE (SIZE(1..maxnoofCoMPCells)) OF CoMPHypothesisSetItem

CoMPHypothesisSetItem ::= SEQUENCE {
    coMPCellID                      ECGI,
    coMPHypothesis                  BIT STRING (SIZE(6..4400, ...)),
```

```
    iE-Extensions                   ProtocolExtensionContainer { {CoMPHypothesisSetItem-ExtIEs} } OPTIONAL,
    ...
}

CoMPHypothesisSetItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CoMPInformation ::= SEQUENCE {
    coMPInformationItem                     CoMPInformationItem,
    coMPInformationStartTime                CoMPInformationStartTime,
    iE-Extensions                           ProtocolExtensionContainer { {CoMPInformation-ExtIEs} } OPTIONAL,
    ...
}

CoMPInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CoMPInformationItem ::= SEQUENCE (SIZE(1..maxnoofCoMPHypothesisSet)) OF
    SEQUENCE {
        coMPHypothesisSet                   CoMPHypothesisSet,
        benefitMetric                       BenefitMetric,
        iE-Extensions                       ProtocolExtensionContainer { {CoMPInformationItem-ExtIEs} } OPTIONAL,
        ...
    }

CoMPInformationItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CoMPInformationStartTime ::= SEQUENCE (SIZE(0..1)) OF
    SEQUENCE {
        startSFN                            INTEGER (0..1023, ...),
        startSubframeNumber                 INTEGER (0..9, ...),
        iE-Extensions                       ProtocolExtensionContainer { {CoMPInformationStartTime-ExtIEs} } OPTIONAL,
        ...
    }

CoMPInformationStartTime-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CompositeAvailableCapacity ::= SEQUENCE {
    cellCapacityClassValue                      CellCapacityClassValue              OPTIONAL,
    capacityValue                               CapacityValue,
    iE-Extensions                               ProtocolExtensionContainer { {CompositeAvailableCapacity-ExtIEs} } OPTIONAL,
    ...
}

CompositeAvailableCapacity-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
```

```
CompositeAvailableCapacityGroup ::= SEQUENCE {
    dL-CompositeAvailableCapacity                   CompositeAvailableCapacity,
    uL-CompositeAvailableCapacity                   CompositeAvailableCapacity,
    iE-Extensions                                   ProtocolExtensionContainer { {CompositeAvailableCapacityGroup-ExtIEs} } OPTIONAL,
    ...
}

CompositeAvailableCapacityGroup-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

Correlation-ID ::= OCTET STRING (SIZE (4))

COUNTvalue ::= SEQUENCE {
    pDCP-SN                 PDCP-SN,
    hFN                     HFN,
    iE-Extensions           ProtocolExtensionContainer { {COUNTvalue-ExtIEs} } OPTIONAL,
    ...
}

COUNTvalue-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

COUNTValueExtended ::= SEQUENCE {
    pDCP-SNExtended         PDCP-SNExtended,
    hFNModified             HFNModified,
    iE-Extensions           ProtocolExtensionContainer { {COUNTValueExtended-ExtIEs} } OPTIONAL,
    ...
}

COUNTValueExtended-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

COUNTvaluePDCP-SNlength18 ::= SEQUENCE {
    pDCP-SNlength18         PDCP-SNlength18,
    hFNforPDCP-SNlength18   HFNforPDCP-SNlength18,
    iE-Extensions           ProtocolExtensionContainer { {COUNTvaluePDCP-SNlength18-ExtIEs} } OPTIONAL,
    ...
}

COUNTvaluePDCP-SNlength18-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CoverageModificationList ::= SEQUENCE (SIZE (1..maxCellineNB)) OF CoverageModification-Item

CoverageModification-Item ::= SEQUENCE {
    eCGI                            ECGI,
    coverageState                   INTEGER (0..15, ...),
    cellDeploymentStatusIndicator   CellDeploymentStatusIndicator          OPTIONAL,
    cellReplacingInfo               CellReplacingInfo                      OPTIONAL,
-- Included in case the Cell Deployment Status Indicator IE is present
```

```
    ...
}

CPTransportLayerInformation     ::= CHOICE {
    endpointIPAddress                       TransportLayerAddress,
    endpointIPAddressAndPort                    TransportLayerAddressAndPort,
    ...
}

CriticalityDiagnostics ::= SEQUENCE {
    procedureCode                   ProcedureCode                   OPTIONAL,
    triggeringMessage               TriggeringMessage               OPTIONAL,
    procedureCriticality            Criticality                     OPTIONAL,
    iEsCriticalityDiagnostics       CriticalityDiagnostics-IE-List  OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {CriticalityDiagnostics-ExtIEs} }   OPTIONAL,
    ...
}


CriticalityDiagnostics-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CriticalityDiagnostics-IE-List ::= SEQUENCE (SIZE (1..maxNrOfErrors)) OF
    SEQUENCE {
        iECriticality           Criticality,
        iE-ID                   ProtocolIE-ID,
        typeOfError             TypeOfError,
        iE-Extensions           ProtocolExtensionContainer { {CriticalityDiagnostics-IE-List-ExtIEs} } OPTIONAL,
        ...
}

CriticalityDiagnostics-IE-List-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CRNTI ::= BIT STRING (SIZE (16))

CSGMembershipStatus ::= ENUMERATED {
    member,
    not-member
}

CSG-Id ::= BIT STRING (SIZE (27))

CSIReportList ::= SEQUENCE (SIZE(1..maxUEReport)) OF
    SEQUENCE {
        uEID                            UEID,
        cSIReportPerCSIProcess          CSIReportPerCSIProcess,
        iE-Extensions                   ProtocolExtensionContainer { {CSIReportList-ExtIEs} } OPTIONAL,
        ...
    }

CSIReportList-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

CSIReportPerCSIProcess ::= SEQUENCE (SIZE(1.. maxCSIProcess)) OF
    SEQUENCE {
        cSIProcessConfigurationIndex    INTEGER (1..7, ...),
        cSIReportPerCSIProcessItem      CSIReportPerCSIProcessItem,
        iE-Extensions                   ProtocolExtensionContainer { {CSIReportPerCSIProcess-ExtIEs} } OPTIONAL,
        ...
    }

CSIReportPerCSIProcess-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CSIReportPerCSIProcessItem ::= SEQUENCE (SIZE(1.. maxCSIReport)) OF
    SEQUENCE {
        rI                              INTEGER (1..8, ...),
        widebandCQI                     WidebandCQI,
        subbandSize                     SubbandSize,
        subbandCQIList                  SubbandCQIList  OPTIONAL,
        iE-Extensions                   ProtocolExtensionContainer { {CSIReportPerCSIProcessItem-ExtIEs} } OPTIONAL,
        ...
    }

CSIReportPerCSIProcessItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

CyclicPrefixDL ::= ENUMERATED {
    normal,
    extended,
    ...
}

CyclicPrefixUL ::= ENUMERATED {
    normal,
    extended,
    ...
}

CHOtrigger ::= ENUMERATED {
    cho-initiation,
    cho-replace,
    ...
}

CHOinformation-REQ ::= SEQUENCE {
    cho-trigger                     CHOtrigger,
    new-eNB-UE-X2AP-ID              UE-X2AP-ID                                      OPTIONAL
        -- This IE shall be present if the cho-trigger IE is present and set to "CHO-replace" --,
    new-eNB-UE-X2AP-ID-Extension    UE-X2AP-ID-Extension                            OPTIONAL,
    cHO-EstimatedArrivalProbability CHO-Probability                                 OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { { CHOinformation-REQ-ExtIEs} } OPTIONAL,
```

```
    ...
}

CHOinformation-REQ-ExtIEs X2AP-PROTOCOL-EXTENSION ::={
    ...
}

CHOinformation-ACK ::= SEQUENCE {
    requestedTargetCellID           ECGI,
    maxCHOpreparations              MaxCHOpreparations                                      OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { CHOinformation-ACK-ExtIEs} } OPTIONAL,
    ...
}

CHOinformation-ACK-ExtIEs X2AP-PROTOCOL-EXTENSION ::={
    ...
}


CandidateCellsToBeCancelledList ::= SEQUENCE (SIZE (1..maxnoofCellsinCHO)) OF ECGI

CHO-Probability ::= INTEGER (1..100)

CHOinformation-AddReq ::= SEQUENCE {
    source-eNB-ID               GlobalENB-ID,
    source-eNB-UE-X2AP-ID       UE-X2AP-ID,
    source-eNB-UE-X2AP-ID-Ext   UE-X2AP-ID-Extension                                    OPTIONAL,
    cHO-EstimatedArrivalProbability     CHO-Probability                                     OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { CHOinformation-AddReq-ExtIEs} }   OPTIONAL,
    ...
}

CHOinformation-AddReq-ExtIEs X2AP-PROTOCOL-EXTENSION ::={
    ...
}


CHOinformation-ModReq ::= SEQUENCE {
    conditionalReconfig                 ENUMERATED {intra-mn-cho, ...},
    cHO-EstimatedArrivalProbability     CHO-Probability                                     OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { CHOinformation-ModReq-ExtIEs} }   OPTIONAL,
    ...
}

CHOinformation-ModReq-ExtIEs X2AP-PROTOCOL-EXTENSION ::={
    ...
}


CSI-RSTransmissionIndication ::= ENUMERATED {
    activated,
    deactivated,
    ...
}
```

```
-- D


DataTrafficResources ::= BIT STRING (SIZE(6..17600))

DataTrafficResourceIndication ::= SEQUENCE {
    activationSFN                   INTEGER (0..1023),
    sharedResourceType              SharedResourceType,
    reservedSubframePattern         ReservedSubframePattern OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {DataTrafficResourceIndication-ExtIEs} } OPTIONAL,
...
}

DataTrafficResourceIndication-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

DAPSRequestInfo ::= SEQUENCE {
    dAPSIndicator               ENUMERATED {daps-HO-required, ...},
    iE-Extensions               ProtocolExtensionContainer { {DAPSRequestInfo-ExtIEs} } OPTIONAL,
    ...
}

DAPSRequestInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}
DAPSResponseInfo ::= SEQUENCE {
    dAPSResponseIndicator                   ENUMERATED { daps-HO-accepted, daps-HO-not-accepted,...},
    iE-Extensions               ProtocolExtensionContainer { { DAPSResponseInfo-ExtIEs} } OPTIONAL,
    ...
}

DAPSResponseInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

DeactivationIndication:= ENUMERATED {
    deactivated,
    ...
}

DeliveryStatus ::= SEQUENCE {
    highestSuccessDeliveredPDCPSN       INTEGER (0..4095),
    iE-Extensions       ProtocolExtensionContainer { {DeliveryStatus-ExtIEs} } OPTIONAL,
    ...
}

DeliveryStatus-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

DesiredActNotificationLevel ::= ENUMERATED {none, e-rab, ue-level, ...}
```

```
DirectForwardingPathAvailability ::= ENUMERATED {direct-path-available, ...}

DL-ABS-status::= INTEGER (0..100)

DL-Forwarding ::= ENUMERATED {
    dL-forwardingProposed,
    ...
}

DL-GBR-PRB-usage::= INTEGER (0..100)

DL-GBR-PRB-usage-for-MIMO::= INTEGER (0..100)

DL-non-GBR-PRB-usage::= INTEGER (0..100)

DL-non-GBR-PRB-usage-for-MIMO::= INTEGER (0..100)

DLResourceBitmapULandDLSharing ::= DataTrafficResources

DLResourcesULandDLSharing ::= CHOICE {
unchanged           NULL,
    changed             DLResourceBitmapULandDLSharing,
    ...
}

DL-scheduling-PDCCH-CCE-usage::= INTEGER (0..100)

DL-Total-PRB-usage::= INTEGER (0..100)

DL-Total-PRB-usage-for-MIMO::= INTEGER (0..100)

DRB-ID ::= INTEGER (1..32)

DuplicationActivation::= ENUMERATED {active, inactive, ...}

DynamicDLTransmissionInformation ::= CHOICE {
    naics-active            DynamicNAICSInformation,
    naics-inactive          NULL,
    ...
}

DynamicNAICSInformation ::= SEQUENCE {
    transmissionModes                   BIT STRING (SIZE(8))                                                OPTIONAL,
    pB-information                      INTEGER(0..3)                                                       OPTIONAL,
    pA-list                             SEQUENCE (SIZE(0..maxnoofPA)) OF PA-Values,
    iE-Extensions                       ProtocolExtensionContainer { {DynamicNAICSInformation-ExtIEs} } OPTIONAL,
    ...
}

DynamicNAICSInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- E
```

```
EARFCN ::= INTEGER (0..maxEARFCN)

EARFCNExtension ::= INTEGER(maxEARFCNPlusOne..newmaxEARFCN, ...)

ECGI ::= SEQUENCE {
    pLMN-Identity               PLMN-Identity,
    eUTRANcellIdentifier        EUTRANCellIdentifier,
    iE-Extensions               ProtocolExtensionContainer { {ECGI-ExtIEs} } OPTIONAL,
    ...
}

ECGI-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

EndcSONConfigurationTransfer ::= OCTET STRING

EnhancedRNTP ::= SEQUENCE {
    enhancedRNTPBitmap          BIT STRING (SIZE(12..8800, ...)),
    rNTP-High-Power-Threshold   RNTP-Threshold,
    enhancedRNTPStartTime       EnhancedRNTPStartTime OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {EnhancedRNTP-ExtIEs} } OPTIONAL,
    ...
}

EnhancedRNTP-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

EnhancedRNTPStartTime ::= SEQUENCE {
        startSFN                INTEGER (0..1023, ...),
        startSubframeNumber     INTEGER (0..9, ...),
        iE-Extensions           ProtocolExtensionContainer { {EnhancedRNTPStartTime-ExtIEs} } OPTIONAL,
        ...
    }

EnhancedRNTPStartTime-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ENB-ID ::= CHOICE {
    macro-eNB-ID    BIT STRING (SIZE (20)),
    home-eNB-ID     BIT STRING (SIZE (28)),
    ... ,
    short-Macro-eNB-ID      BIT STRING (SIZE(18)),
    long-Macro-eNB-ID       BIT STRING (SIZE(21))
}

EncryptionAlgorithms ::= BIT STRING (SIZE (16, ...))

EN-DC-ResourceConfiguration ::= SEQUENCE {
    pDCPatSgNB      ENUMERATED {present, not-present, ...},
    mCGresources    ENUMERATED {present, not-present, ...},
```

```
    sCGresources    ENUMERATED {present, not-present, ...},
    iE-Extensions                   ProtocolExtensionContainer { {EN-DC-ResourceConfigurationExtIEs} }  OPTIONAL,
    ...
}

EN-DC-ResourceConfigurationExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

EPCHandoverRestrictionListContainer ::= OCTET STRING
-- This octets of the OCTET STRING contain the Handover Restriction List IE as specified in TS 36.413 [4]. --

EPLMNs ::= SEQUENCE (SIZE(1..maxnoofEPLMNs)) OF PLMN-Identity

ERABActivityNotifyItemList ::= SEQUENCE (SIZE (0..maxnoofBearers)) OF ERABActivityNotifyItem

ERABActivityNotifyItem ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    activityReport                  UserPlaneTrafficActivityReport,
    iE-Extensions                   ProtocolExtensionContainer { {ERABActivityNotifyItem-ExtIEs} }   OPTIONAL,
    ...
}

ERABActivityNotifyItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RAB-ID ::= INTEGER (0..15, ...)

E-RAB-Level-QoS-Parameters ::= SEQUENCE {
    qCI                             QCI,
    allocationAndRetentionPriority  AllocationAndRetentionPriority,
    gbrQosInformation               GBR-QosInformation                                              OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {E-RAB-Level-QoS-Parameters-ExtIEs} }  OPTIONAL,
    ...
}

E-RAB-Level-QoS-Parameters-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
-- Extended for introduction of downlink and uplink packet loss rate for enhanced Voice performance –
    { ID id-DownlinkPacketLossRate          CRITICALITY ignore  EXTENSION Packet-LossRate       PRESENCE optional}|
    { ID id-UplinkPacketLossRate            CRITICALITY ignore  EXTENSION Packet-LossRate       PRESENCE optional},
    ...
}

E-RAB-List ::= SEQUENCE (SIZE(1.. maxnoofBearers)) OF ProtocolIE-Single-Container { {E-RAB-ItemIEs} }

E-RAB-ItemIEs   X2AP-PROTOCOL-IES ::= {
    { ID id-E-RAB-Item   CRITICALITY ignore     TYPE E-RAB-Item     PRESENCE mandatory },
    ...
}

E-RAB-Item ::= SEQUENCE {
    e-RAB-ID                E-RAB-ID,
    cause                       Cause,
```

```
    iE-Extensions               ProtocolExtensionContainer { {E-RAB-Item-ExtIEs} } OPTIONAL,
    ...
}

E-RAB-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABsSubjectToEarlyStatusTransfer-List ::= SEQUENCE (SIZE (1.. maxnoofBearers)) OF E-RABsSubjectToEarlyStatusTransfer-Item

E-RABsSubjectToEarlyStatusTransfer-Item ::= SEQUENCE {
    e-RAB-ID                                    E-RAB-ID,
    fIRST-DL-COUNTValue                         COUNTvalue,
    fIRST-DL-COUNTValueExtended                 COUNTValueExtended                                  OPTIONAL,
    fIRST-DL-COUNTValueforPDCPSNLength18        COUNTvaluePDCP-SNlength18                           OPTIONAL,
    iE-Extension        ProtocolExtensionContainer { { E-RABsSubjectToEarlyStatusTransfer-Item-ExtIEs} }   OPTIONAL,
    ...
}

E-RABsSubjectToEarlyStatusTransfer-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

E-RABsSubjectToDLDiscarding-List ::= SEQUENCE (SIZE (1.. maxnoofBearers)) OF E-RABsSubjectToDLDiscarding-Item

E-RABsSubjectToDLDiscarding-Item ::= SEQUENCE {
    e-RAB-ID                                    E-RAB-ID,
    dISCARD-DL-COUNTValue                       COUNTvalue,
    dISCARD-DL-COUNTValueExtended               COUNTValueExtended                                  OPTIONAL,
    dISCARD-DL-COUNTValueforPDCPSNLength18      COUNTvaluePDCP-SNlength18                           OPTIONAL,
    iE-Extension        ProtocolExtensionContainer { { E-RABsSubjectToDLDiscarding-Item-ExtIEs} }       OPTIONAL,
    ...
}

E-RABsSubjectToDLDiscarding-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


E-RABUsageReportList ::= SEQUENCE (SIZE(1..maxnooftimeperiods)) OF ProtocolIE-Single-Container { {E-RABUsageReport-ItemIEs} }

E-RABUsageReport-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-E-RABUsageReport-Item    CRITICALITY ignore     TYPE E-RABUsageReport-Item  PRESENCE mandatory },
    ...
}

E-RABUsageReport-Item ::= SEQUENCE {
    startTimeStamp              OCTET STRING (SIZE(4)),
    endTimeStamp                    OCTET STRING (SIZE(4)),
    usageCountUL                    INTEGER (0..18446744073709551615),
    usageCountDL                    INTEGER (0..18446744073709551615),
    iE-Extensions               ProtocolExtensionContainer { {E-RABUsageReport-Item-ExtIEs} } OPTIONAL,
...
}
```

```
E-RABUsageReport-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

Ethernet-Type ::= ENUMERATED {
    true,
    ...
}

EUTRA-Mode-Info ::= CHOICE {
    fDD     FDD-Info,
    tDD     TDD-Info,
    ...
}

EUTRANCellIdentifier ::= BIT STRING (SIZE (28))

EUTRANTraceID       ::= OCTET STRING (SIZE (8))

EventType ::= ENUMERATED{
    change-of-serving-cell,
    ...
}

ExpectedUEBehaviour ::= SEQUENCE {
    expectedActivity        ExpectedUEActivityBehaviour OPTIONAL,
    expectedHOInterval      ExpectedHOInterval          OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { {ExpectedUEBehaviour-ExtIEs} } OPTIONAL,
    ...
}

ExpectedUEBehaviour-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ExpectedUEActivityBehaviour ::= SEQUENCE {
    expectedActivityPeriod                  ExpectedActivityPeriod                  OPTIONAL,
    expectedIdlePeriod                      ExpectedIdlePeriod                      OPTIONAL,
    sourceofUEActivityBehaviourInformation  SourceOfUEActivityBehaviourInformation  OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {ExpectedUEActivityBehaviour-ExtIEs} } OPTIONAL,
    ...
}

ExpectedUEActivityBehaviour-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ExpectedActivityPeriod ::= INTEGER (1..30|40|50|60|80|100|120|150|180|181,...)

ExpectedIdlePeriod ::= INTEGER (1..30|40|50|60|80|100|120|150|180|181,...)

ExpectedHOInterval ::= ENUMERATED {
    sec15, sec30, sec60, sec90, sec120, sec180, long-time,
```

```
    ...
}

ExtendedULInterferenceOverloadInfo ::= SEQUENCE {
    associatedSubframes                         BIT STRING (SIZE (5)),
    extended-ul-InterferenceOverloadIndication  UL-InterferenceOverloadIndication,
    iE-Extensions                               ProtocolExtensionContainer { {ExtendedULInterferenceOverloadInfo-ExtIEs} } OPTIONAL,
    ...
}

ExtendedULInterferenceOverloadInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ExtendedBitRate ::= INTEGER (10000000001..4000000000000,...)

-- F

F1CTrafficContainer ::= OCTET STRING

FastMCGRecovery ::= SEQUENCE {
    rrcContainer                RRCContainer            OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {FastMCGRecovery-ExtIEs} } OPTIONAL,
    ...
}

FastMCGRecovery-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

FDD-Info ::= SEQUENCE {
    uL-EARFCN                       EARFCN,
    dL-EARFCN                       EARFCN,
    uL-Transmission-Bandwidth       Transmission-Bandwidth,
    dL-Transmission-Bandwidth       Transmission-Bandwidth,
    iE-Extensions               ProtocolExtensionContainer { {FDD-Info-ExtIEs} } OPTIONAL,
    ...
}

FDD-Info-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-UL-EARFCNExtension                      CRITICALITY reject  EXTENSION EARFCNExtension                       PRESENCE optional}|
    { ID id-DL-EARFCNExtension                      CRITICALITY reject  EXTENSION EARFCNExtension                       PRESENCE optional}|
    { ID id-OffsetOfNbiotChannelNumberToDL-EARFCN   CRITICALITY reject  EXTENSION OffsetOfNbiotChannelNumberToEARFCN     PRESENCE optional}|
    { ID id-OffsetOfNbiotChannelNumberToUL-EARFCN   CRITICALITY reject  EXTENSION OffsetOfNbiotChannelNumberToEARFCN     PRESENCE optional}|
    { ID id-NRS-NSSS-PowerOffset                    CRITICALITY ignore  EXTENSION NRS-NSSS-PowerOffset                  PRESENCE optional}|
    { ID id-NSSS-NumOccasionDifferentPrecoder       CRITICALITY ignore  EXTENSION NSSS-NumOccasionDifferentPrecoder     PRESENCE optional},
    ...
}

FDD-InfoNeighbourServedNRCell-Information ::= SEQUENCE {
    ul-NRFreqInfo           NRFreqInfo,
    dl-NRFreqInfo           NRFreqInfo,
    iE-Extensions           ProtocolExtensionContainer { {FDD-InfoNeighbourServedNRCell-Information-ExtIEs} }    OPTIONAL,
    ...
```

```
}

FDD-InfoNeighbourServedNRCell-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-ULCarrierList   CRITICALITY ignore  EXTENSION NRCarrierList         PRESENCE optional },
    ...
}

FiveQI ::= INTEGER (0..255, ...)

ForbiddenInterRATs ::= ENUMERATED {
    all,
    geran,
    utran,
    cdma2000,
    ...,
    geranandutran,
    cdma2000andutran

}

ForbiddenTAs ::= SEQUENCE (SIZE(1.. maxnoofEPLMNsPlusOne)) OF ForbiddenTAs-Item

ForbiddenTAs-Item ::= SEQUENCE {
    pLMN-Identity       PLMN-Identity,
    forbiddenTACs       ForbiddenTACs,
    iE-Extensions       ProtocolExtensionContainer { {ForbiddenTAs-Item-ExtIEs} } OPTIONAL,
    ...
}

ForbiddenTAs-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ForbiddenTACs ::= SEQUENCE (SIZE(1..maxnoofForbTACs)) OF TAC

ForbiddenLAs ::= SEQUENCE (SIZE(1..maxnoofEPLMNsPlusOne)) OF ForbiddenLAs-Item

ForbiddenLAs-Item ::= SEQUENCE {
    pLMN-Identity       PLMN-Identity,
    forbiddenLACs       ForbiddenLACs,
    iE-Extensions       ProtocolExtensionContainer { {ForbiddenLAs-Item-ExtIEs} } OPTIONAL,
    ...
}

ForbiddenLAs-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ForbiddenLACs ::= SEQUENCE (SIZE(1..maxnoofForbLACs)) OF LAC

Fourframes ::= BIT STRING (SIZE (24))

FreqBandIndicator ::= INTEGER (1..256, ...)
```

```
FreqBandIndicatorPriority ::= ENUMERATED {
    not-broadcasted,
    broadcasted,
    ...
}


FreqBandNrItem ::= SEQUENCE {
    freqBandIndicatorNr             INTEGER (1..1024,...),
    supportedSULBandList    SEQUENCE (SIZE(0..maxnoofNrCellBands)) OF SupportedSULFreqBandItem,
    iE-Extensions               ProtocolExtensionContainer { {FreqBandNrItem-ExtIEs} } OPTIONAL,
    ...
}

FreqBandNrItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

FrequencyShift7p5khz ::= ENUMERATED {false, true, ...}


-- G

GBR-QosInformation ::= SEQUENCE {
    e-RAB-MaximumBitrateDL          BitRate,
    e-RAB-MaximumBitrateUL          BitRate,
    e-RAB-GuaranteedBitrateDL       BitRate,
    e-RAB-GuaranteedBitrateUL       BitRate,
    iE-Extensions                   ProtocolExtensionContainer { {GBR-QosInformation-ExtIEs} } OPTIONAL,
    ...
}

GBR-QosInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
-- Extension for maximum bitrate > 10Gbps --
    { ID id-extended-e-RAB-MaximumBitrateDL     CRITICALITY ignore  EXTENSION ExtendedBitRate   PRESENCE optional}|
    { ID id-extended-e-RAB-MaximumBitrateUL     CRITICALITY ignore  EXTENSION ExtendedBitRate   PRESENCE optional}|
    { ID id-extended-e-RAB-GuaranteedBitrateDL  CRITICALITY ignore  EXTENSION ExtendedBitRate   PRESENCE optional}|
    { ID id-extended-e-RAB-GuaranteedBitrateUL  CRITICALITY ignore  EXTENSION ExtendedBitRate   PRESENCE optional},
    ...
}

GlobalENB-ID ::= SEQUENCE {
    pLMN-Identity           PLMN-Identity,
    eNB-ID                  ENB-ID,
    iE-Extensions           ProtocolExtensionContainer { {GlobalENB-ID-ExtIEs} } OPTIONAL,
    ...
}

GlobalENB-ID-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

GlobalGNB-ID ::= SEQUENCE {
    pLMN-Identity           PLMN-Identity,
```

```
    gNB-ID                  GNB-ID,
    iE-Extensions           ProtocolExtensionContainer { {GlobalGNB-ID-ExtIEs} } OPTIONAL,
    ...
}

GlobalGNB-ID-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

Global-RAN-NODE-ID ::= CHOICE {
    gNB                     GlobalGNB-ID,
    choice-extension        ProtocolIE-Single-Container { { Global-RAN-NODE-ID-ExtIEs} }
}

Global-RAN-NODE-ID-ExtIEs X2AP-PROTOCOL-IES ::= {
    ...
}

GNBOverloadInformation ::= ENUMERATED {overloaded, not-overloaded, ...}

GTPTLAs ::= SEQUENCE (SIZE(1.. maxnoofGTPTLAs)) OF  GTPTLA-Item


GTPTLA-Item ::= SEQUENCE {
    gTPTransportLayerAddresses          TransportLayerAddress,
    iE-Extensions                       ProtocolExtensionContainer { { GTPTLA-Item-ExtIEs } } OPTIONAL,
    ...
}

GTPTLA-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

GTPtunnelEndpoint ::= SEQUENCE {
    transportLayerAddress           TransportLayerAddress,
    gTP-TEID                        GTP-TEI,
    iE-Extensions                   ProtocolExtensionContainer { {GTPtunnelEndpoint-ExtIEs} } OPTIONAL,
    ...
}

GTPtunnelEndpoint-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {ID id-QoS-Mapping-Information  CRITICALITY reject  EXTENSION QoS-Mapping-Information   PRESENCE optional},
    ...
}

GTP-TEI                 ::= OCTET STRING (SIZE (4))

GUGroupIDList       ::= SEQUENCE (SIZE (1..maxPools)) OF GU-Group-ID


GU-Group-ID         ::= SEQUENCE {
    pLMN-Identity       PLMN-Identity,
    mME-Group-ID        MME-Group-ID,
    iE-Extensions       ProtocolExtensionContainer { {GU-Group-ID-ExtIEs} } OPTIONAL,
```

```
    ...
}

GU-Group-ID-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


GUMMEI          ::= SEQUENCE {

    gU-Group-ID     GU-Group-ID,
    mME-Code            MME-Code,
    iE-Extensions                   ProtocolExtensionContainer { {GUMMEI-ExtIEs} } OPTIONAL,
    ...
}

GUMMEI-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

GNB-ID ::= CHOICE {
    gNB-ID  BIT STRING (SIZE (22..32)),
    ...
}


-- H

HandoverReportType ::= ENUMERATED {
    hoTooEarly,
    hoToWrongCell,
    ...,
    interRATpingpong,
    interSystemPingpong
}

HandoverRestrictionList ::= SEQUENCE {
    servingPLMN                 PLMN-Identity,
    equivalentPLMNs             EPLMNs                  OPTIONAL,
    forbiddenTAs                ForbiddenTAs            OPTIONAL,
    forbiddenLAs                ForbiddenLAs            OPTIONAL,
    forbiddenInterRATs          ForbiddenInterRATs      OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {HandoverRestrictionList-ExtIEs} } OPTIONAL,
    ...
}

HandoverRestrictionList-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-NRrestrictioninEPSasSecondaryRAT            CRITICALITY ignore  EXTENSION NRrestrictioninEPSasSecondaryRAT      PRESENCE optional}|
    { ID id-CNTypeRestrictions                      CRITICALITY ignore  EXTENSION CNTypeRestrictions                    PRESENCE optional}|
    { ID id-NRrestrictionin5GS                      CRITICALITY ignore  EXTENSION NRrestrictionin5GS                    PRESENCE optional}|
    { ID id-LastNG-RANPLMNIdentity                  CRITICALITY ignore  EXTENSION PLMN-Identity                         PRESENCE optional}|
    { ID id-UnlicensedSpectrumRestriction           CRITICALITY ignore  EXTENSION UnlicensedSpectrumRestriction         PRESENCE optional}|
    { ID id-RAT-Restrictions                        CRITICALITY ignore  EXTENSION RAT-Restrictions                      PRESENCE optional},
    ...
```

```
}

HFN ::= INTEGER (0..1048575)

HFNModified ::= INTEGER (0..131071)

HFNforPDCP-SNlength18 ::= INTEGER (0..16383)

HWLoadIndicator ::= SEQUENCE {
    dLHWLoadIndicator           LoadIndicator,
    uLHWLoadIndicator           LoadIndicator,
    iE-Extensions               ProtocolExtensionContainer { {HWLoadIndicator-ExtIEs} } OPTIONAL,
    ...
}

HWLoadIndicator-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


-- I


IABNodeIndication ::= ENUMERATED {true,...}

IMSvoiceEPSfallbackfrom5G ::= ENUMERATED {
    true,
    ...
}

IntegrityProtectionAlgorithms ::= BIT STRING (SIZE (16, ...))

IntegrityProtectionIndication ::= ENUMERATED {
    required,
    preferred,
    notneeded,
    ...
}

IntegrityProtectionResult ::= ENUMERATED {
    performed,
    notperformed,
    ...
}

IntendedTDD-DL-ULConfiguration-NR ::= OCTET STRING

InterfaceInstanceIndication ::= INTEGER (0..255, ...)

InterfacesToTrace ::= BIT STRING (SIZE (8))

InvokeIndication ::= ENUMERATED{
    abs-information,
    ...,
```

```
    naics-information-start,
    naics-information-stop
}

-- J
-- K

Key-eNodeB-Star ::= BIT STRING (SIZE(256))

-- L

LastVisitedPSCell-Item ::= OCTET STRING

LAC                ::= OCTET STRING (SIZE (2)) --(EXCEPT ('0000'H|'FFFE'H))

LastVisitedCell-Item ::= CHOICE {
    e-UTRAN-Cell                    LastVisitedEUTRANCellInformation,
    uTRAN-Cell                      LastVisitedUTRANCellInformation,
    gERAN-Cell                      LastVisitedGERANCellInformation,
    ...,
    nG-RAN-Cell                     LastVisitedNGRANCellInformation
}

LastVisitedEUTRANCellInformation ::= SEQUENCE {
    global-Cell-ID                  ECGI,
    cellType                        CellType,
    time-UE-StayedInCell            Time-UE-StayedInCell,
    iE-Extensions                   ProtocolExtensionContainer { {LastVisitedEUTRANCellInformation-ExtIEs} } OPTIONAL,
    ...
}

LastVisitedEUTRANCellInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
-- Extension for Rel-11 to support enhanced granularity for time UE stayed in cell --
    { ID id-Time-UE-StayedInCell-EnhancedGranularity    CRITICALITY ignore  EXTENSION Time-UE-StayedInCell-EnhancedGranularity PRESENCE optional}|
    { ID id-HO-cause                                    CRITICALITY ignore  EXTENSION Cause                                    PRESENCE optional}|
    { ID id-PSCell-UE-HistoryInformation            CRITICALITY ignore  EXTENSION PSCell-UE-HistoryInformation                  PRESENCE
optional},
    ...
}

LastVisitedGERANCellInformation ::= CHOICE {
    undefined                       NULL,
    ...
}

LastVisitedNGRANCellInformation ::= OCTET STRING

LastVisitedUTRANCellInformation ::= OCTET STRING

LCID ::= INTEGER(1..32, ...)

LHN-ID ::= OCTET STRING(SIZE (32..256))

Links-to-log ::= ENUMERATED {uplink, downlink, both-uplink-and-downlink, ...}
```

```
LoadIndicator ::= ENUMERATED {
    lowLoad,
    mediumLoad,
    highLoad,
    overLoad,
    ...
}

LocationInformationSgNB ::= SEQUENCE {
    pSCell-id           NRCGI,
    iE-Extensions       ProtocolExtensionContainer { {LocationInformationSgNB-ExtIEs} } OPTIONAL,
    ...
}

LocationInformationSgNB-ExtIEs X2AP-PROTOCOL-EXTENSION ::={
    ...
}

LocationInformationSgNBReporting ::= ENUMERATED {
    pSCell,
    ...
}

LocationReportingInformation ::= SEQUENCE {
    eventType       EventType,
    reportArea      ReportArea,
    iE-Extensions       ProtocolExtensionContainer { {LocationReportingInformation-ExtIEs} } OPTIONAL,
    ...
}

LocationReportingInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::={
    { ID id-AdditionLocationInformation CRITICALITY ignore  EXTENSION AdditionLocationInformation PRESENCE optional},
    ...
}

LowerLayerPresenceStatusChange ::= ENUMERATED {
    release-lower-layers,
    re-establish-lower-layers,
    suspend-lower-layers,
    resume-lower-layers,
    ...
}

-- M

M1PeriodicReporting ::= SEQUENCE {
    reportInterval              ReportIntervalMDT,
    reportAmount                ReportAmountMDT,
    iE-Extensions               ProtocolExtensionContainer { {M1PeriodicReporting-ExtIEs} } OPTIONAL,
    ...
}

M1PeriodicReporting-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

M1ReportingTrigger::= ENUMERATED{
    periodic,
    a2eventtriggered,
    ...,
    a2eventtriggered-periodic
}

M1ThresholdEventA2 ::= SEQUENCE {
    measurementThreshold        MeasurementThresholdA2,
    iE-Extensions               ProtocolExtensionContainer { {M1ThresholdEventA2-ExtIEs} } OPTIONAL,
    ...
}

M1ThresholdEventA2-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

M3Configuration ::= SEQUENCE {
    m3period            M3period,
    iE-Extensions       ProtocolExtensionContainer { {M3Configuration-ExtIEs} } OPTIONAL,
    ...
}

M3Configuration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

M3period ::= ENUMERATED {ms100, ms1000, ms10000, ... }

M4Configuration ::= SEQUENCE {
    m4period            M4period,
    m4-links-to-log     Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { {M4Configuration-ExtIEs} } OPTIONAL,
    ...
}

M4Configuration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

M4period ::= ENUMERATED {ms1024, ms2048, ms5120, ms10240, min1, ... }


M5Configuration ::= SEQUENCE {
    m5period            M5period,
    m5-links-to-log     Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { {M5Configuration-ExtIEs} } OPTIONAL,
    ...
}

M5Configuration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

M5period ::= ENUMERATED {ms1024, ms2048, ms5120, ms10240, min1, ... }

M6Configuration ::= SEQUENCE {
    m6report-interval   M6report-interval,
    m6delay-threshold   M6delay-threshold   OPTIONAL,
-- This IE shall be present if the M6 Links to log IE is set to “uplink” or to “both-uplink-and-downlink” --
    m6-links-to-log     Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { {M6Configuration-ExtIEs} } OPTIONAL,
    ...
}

M6Configuration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

M6report-interval ::= ENUMERATED { ms1024, ms2048, ms5120, ms10240, ... }

M6delay-threshold ::= ENUMERATED { ms30, ms40, ms50, ms60, ms70, ms80, ms90, ms100, ms150, ms300, ms500, ms750, ... }

M7Configuration ::= SEQUENCE {
    m7period            M7period,
    m7-links-to-log     Links-to-log,
    iE-Extensions       ProtocolExtensionContainer { {M7Configuration-ExtIEs} } OPTIONAL,
    ...
}

M7Configuration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

M7period ::= INTEGER(1..60, ...)

MakeBeforeBreakIndicator::= ENUMERATED {true, ...}

ManagementBasedMDTallowed ::= ENUMERATED {allowed, ...}

Masked-IMEISV ::= BIT STRING (SIZE (64))

MaxCHOpreparations ::= INTEGER(1..8, ...)

MDT-Activation      ::= ENUMERATED {
    immediate-MDT-only,
    immediate-MDT-and-Trace,
    ...
}

MDT-Configuration ::= SEQUENCE {
    mdt-Activation              MDT-Activation,
    areaScopeOfMDT              AreaScopeOfMDT,
    measurementsToActivate      MeasurementsToActivate,
    m1reportingTrigger          M1ReportingTrigger,
```

```
    m1thresholdeventA2          M1ThresholdEventA2          OPTIONAL,
-- Included in case of event-triggered, or event-triggered periodic reporting for measurement M1
    m1periodicReporting         M1PeriodicReporting         OPTIONAL,
-- Included in case of periodic, or event-triggered periodic reporting for measurement M1
    iE-Extensions               ProtocolExtensionContainer { {MDT-Configuration-ExtIEs} } OPTIONAL,
    ...
}

MDT-Configuration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {ID id-M3Configuration                          CRITICALITY ignore  EXTENSION M3Configuration                    PRESENCE conditional}|
    {ID id-M4Configuration                          CRITICALITY ignore  EXTENSION M4Configuration                    PRESENCE conditional}|
    {ID id-M5Configuration                          CRITICALITY ignore  EXTENSION M5Configuration                    PRESENCE conditional}|
    {ID id-MDT-Location-Info                        CRITICALITY ignore  EXTENSION MDT-Location-Info                  PRESENCE optional}|
    {ID id-SignallingBasedMDTPLMNList               CRITICALITY ignore  EXTENSION MDTPLMNList                        PRESENCE optional}|
    {ID id-M6Configuration                          CRITICALITY ignore  EXTENSION M6Configuration                    PRESENCE conditional}|
    {ID id-M7Configuration                          CRITICALITY ignore  EXTENSION M7Configuration                    PRESENCE conditional}|
    { ID id-BluetoothMeasurementConfiguration       CRITICALITY ignore  EXTENSION BluetoothMeasurementConfiguration  PRESENCE optional}|
    { ID id-WLANMeasurementConfiguration            CRITICALITY ignore  EXTENSION WLANMeasurementConfiguration       PRESENCE optional}|
    { ID id-SensorMeasurementConfiguration          CRITICALITY ignore  EXTENSION SensorMeasurementConfiguration     PRESENCE optional},
    ...
}

MDTPLMNList ::= SEQUENCE (SIZE(1..maxnoofMDTPLMNs)) OF PLMN-Identity

MDT-Location-Info ::= BIT STRING (SIZE (8))

Measurement-ID  ::= INTEGER (1..4095, ...)

Measurement-ID-ENDC ::= INTEGER (1..4095, ...)


MeasurementsToActivate::= BIT STRING (SIZE (8))

MeasurementThresholdA2 ::= CHOICE {
    threshold-RSRP              Threshold-RSRP,
    threshold-RSRQ              Threshold-RSRQ,
    ...
}

MeNBCoordinationAssistanceInformation ::= ENUMERATED{
    coordination-not-required,
    ...
}

MeNBResourceCoordinationInformation ::= SEQUENCE {
    eUTRA-Cell-ID                   ECGI,
    uLCoordinationInformation       BIT STRING (SIZE(6..4400, ...)),
    dLCoordinationInformation       BIT STRING (SIZE(6..4400, ...))     OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {MeNBResourceCoordinationInformationExtIEs} }      OPTIONAL,
    ...
}

MeNBResourceCoordinationInformationExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-NRCGI                                   CRITICALITY ignore  EXTENSION NRCGI                                 PRESENCE optional}|
```

```
    { ID id-MeNBCoordinationAssistanceInformation       CRITICALITY reject  EXTENSION MeNBCoordinationAssistanceInformation PRESENCE optional},
    ...
}

MeNBtoSeNBContainer ::= OCTET STRING

MME-Group-ID    ::= OCTET STRING (SIZE (2))

MME-Code        ::= OCTET STRING (SIZE (1))

MBMS-Service-Area-Identity-List ::= SEQUENCE (SIZE(1.. maxnoofMBMSServiceAreaIdentities)) OF MBMS-Service-Area-Identity

MBMS-Service-Area-Identity ::= OCTET STRING (SIZE (2))

MBSFN-Subframe-Infolist::= SEQUENCE (SIZE(1.. maxnoofMBSFN)) OF MBSFN-Subframe-Info

MBSFN-Subframe-Info ::= SEQUENCE {
    radioframeAllocationPeriod      RadioframeAllocationPeriod,
    radioframeAllocationOffset      RadioframeAllocationOffset,
    subframeAllocation              SubframeAllocation,
    iE-Extensions                   ProtocolExtensionContainer { {MBSFN-Subframe-Info-ExtIEs} }   OPTIONAL,
    ...
}

MBSFN-Subframe-Info-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

MDT-ConfigurationNR ::= OCTET STRING

MobilityParametersModificationRange ::= SEQUENCE {
    handoverTriggerChangeLowerLimit     INTEGER (-20..20),
    handoverTriggerChangeUpperLimit     INTEGER (-20..20),
    ...
}

MobilityParametersInformation ::= SEQUENCE {
    handoverTriggerChange           INTEGER (-20..20),
    ...
}

MultibandInfoList ::= SEQUENCE (SIZE(1..maxnoofBands)) OF BandInfo

MessageOversizeNotification ::= SEQUENCE {
    maximumCellListSize                             MaximumCellListSize,
    iE-Extensions                   ProtocolExtensionContainer { {MessageOversizeNotification-ExtIEs} }    OPTIONAL,
    ...
}

MessageOversizeNotification-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

MaximumCellListSize ::= INTEGER(1..16384, ...)
```

```
BandInfo    ::= SEQUENCE {
    freqBandIndicator       FreqBandIndicator,
    iE-Extensions           ProtocolExtensionContainer { {BandInfo-ExtIEs} }    OPTIONAL,
    ...
}

BandInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

MeNBtoSgNBContainer ::= OCTET STRING

SplitSRBs ::= ENUMERATED {srb1, srb2, srb1and2, ...}

SplitSRB ::= SEQUENCE {
    rrcContainer                RRCContainer            OPTIONAL,
    srbType                     SRBType,
    deliveryStatus              DeliveryStatus          OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {SplitSRB-ExtIEs} } OPTIONAL,
    ...
}

SplitSRB-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- N

NBIoT-UL-DL-AlignmentOffset ::= ENUMERATED {
    khz-7dot5,
    khz0,
    khz7dot5,
    ...
}

NBIoT-RLF-Report-Container ::= OCTET STRING

Neighbour-Information ::= SEQUENCE (SIZE (0..maxnoofNeighbours)) OF SEQUENCE {
    eCGI                        ECGI,
    pCI                         PCI,
    eARFCN                      EARFCN,
    iE-Extensions               ProtocolExtensionContainer { {Neighbour-Information-ExtIEs} } OPTIONAL,
    ...
}

Neighbour-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-NeighbourTAC        CRITICALITY ignore  EXTENSION TAC               PRESENCE optional}|
    { ID id-eARFCNExtension     CRITICALITY reject  EXTENSION EARFCNExtension   PRESENCE optional},
    ...
}

NextHopChainingCount ::= INTEGER (0..7)
```

```
NewDRBIDrequest::= ENUMERATED {true, ...}

Number-of-Antennaports ::= ENUMERATED {
        an1,
        an2,
        an4,
        ...
}

NRCapacityValue                 ::= SEQUENCE {
    capacityValue                       INTEGER (0..100),
    ssbAreaCapacityValue-List           SSBAreaCapacityValue-List                           OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { NRCapacityValue-ExtIEs} } OPTIONAL,
    ...
}

NRCapacityValue-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRCarrierList ::= SEQUENCE (SIZE(1..maxnoofNRSCSs)) OF NRCarrierItem

NRCarrierItem ::= SEQUENCE {
    carrierSCS                  NRSCS,
    offsetToCarrier             INTEGER (0..2199, ...),
    carrierBandwidth            INTEGER (0..maxnoofNRPhysicalResourceBlocks, ...),
    iE-Extension        ProtocolExtensionContainer { {NRCarrierItem-ExtIEs} }       OPTIONAL,
    ...
}

NRCarrierItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRCellCapacityClassValue ::= INTEGER (1..100, ...)

NRCellPRACHConfig ::= OCTET STRING

NRCompositeAvailableCapacityGroup   ::= SEQUENCE {
    compositeAvailableCapacityDL        NRCompositeAvailableCapacity,
    compositeAvailableCapacityUL        NRCompositeAvailableCapacity,
    iE-Extensions                       ProtocolExtensionContainer { {NRCompositeAvailableCapacityGroup-ExtIEs} } OPTIONAL,
    ...
}

NRCompositeAvailableCapacityGroup-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRCompositeAvailableCapacity ::= SEQUENCE {
    cellCapacityClassValue              NRCellCapacityClassValue                            OPTIONAL,
    capacityValue                       NRCapacityValue,
    iE-Extensions                       ProtocolExtensionContainer { {NRCompositeAvailableCapacity-ExtIEs} } OPTIONAL,
    ...
```

```
}

NRCompositeAvailableCapacity-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRFreqInfo ::= SEQUENCE{
    nRARFCN         INTEGER (0.. 3279165),
    freqBandListNr  SEQUENCE (SIZE(1..maxnoofNrCellBands)) OF FreqBandNrItem,
    sULInformation  SULInformation      OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { {NRFreqInfo-ExtIEs} } OPTIONAL,

    ...
}

NRFreqInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-FrequencyShift7p5khz            CRITICALITY ignore  EXTENSION FrequencyShift7p5khz  PRESENCE optional},
    ...
}

NRCellIdentifier ::= BIT STRING (SIZE (36))

NRCGI ::= SEQUENCE {
    pLMN-Identity               PLMN-Identity,
    nRcellIdentifier                NRCellIdentifier,
    iE-Extensions               ProtocolExtensionContainer { {NRCGI-ExtIEs} } OPTIONAL,
    ...
}

NRCGI-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRRACHReportContainer   ::= OCTET STRING

NRRACHReportInformation ::= SEQUENCE (SIZE(1.. maxnoofRACHReports)) OF NRRACHReportList-Item

NRRACHReportList-Item   ::= SEQUENCE {
    nRRACHReport                        NRRACHReportContainer,
    uEAssitantIdentifier                SgNB-UE-X2AP-ID                                             OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { NRRACHReportList-Item-ExtIEs} }  OPTIONAL,
    ...
}

NRRACHReportList-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRNeighbour-Information ::= SEQUENCE (SIZE (1.. maxofNRNeighbours))OF SEQUENCE {
    nrpCI                           NRPCI,
    nrCellID                        NRCGI,
    fiveGS-TAC                      FiveGS-TAC  OPTIONAL,
    configured-TAC                  TAC         OPTIONAL,
    measurementTimingConfiguration  OCTET STRING,
```

```
    nRNeighbourModeInfo             CHOICE {
        fdd     FDD-InfoNeighbourServedNRCell-Information,
        tdd     TDD-InfoNeighbourServedNRCell-Information,
        ...
    },
    iE-Extensions                       ProtocolExtensionContainer { {NRNeighbour-Information-ExtIEs} } OPTIONAL,
    ...
}

NRNeighbour-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {ID id-CSI-RSTransmissionIndication     CRITICALITY ignore  EXTENSION CSI-RSTransmissionIndication  PRESENCE optional}|
    {ID id-SSB-PositionsInBurst             CRITICALITY ignore  EXTENSION SSB-PositionsInBurst            PRESENCE optional}|
    {ID id-NRCellPRACHConfig                CRITICALITY ignore  EXTENSION NRCellPRACHConfig               PRESENCE optional}|
    {ID id-Additional-Measurement-Timing-Configuration-List     CRITICALITY ignore  EXTENSION Additional-Measurement-Timing-Configuration-List
    PRESENCE optional },
    ...
}

NPRACHConfiguration::= SEQUENCE {
    fdd-or-tdd              CHOICE {
        fdd         NPRACHConfiguration-FDD,
        tdd         NPRACHConfiguration-TDD,
        ...
    },  iE-Extensions                       ProtocolExtensionContainer { { NPRACHConfiguration-ExtIEs} }   OPTIONAL,
...
}

NPRACHConfiguration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NPRACHConfiguration-FDD::= SEQUENCE {
    nprach-CP-length                            NPRACH-CP-Length,
    anchorCarrier-NPRACHConfig                  OCTET STRING,
    anchorCarrier-EDT-NPRACHConfig              OCTET STRING                        OPTIONAL,
    anchorCarrier-Format2-NPRACHConfig          OCTET STRING                        OPTIONAL,
    anchorCarrier-Format2-EDT-NPRACHConfig      OCTET STRING                        OPTIONAL,
    non-anchorCarrier-NPRACHConfig              OCTET STRING                        OPTIONAL,
    non-anchorCarrier-Format2-NPRACHConfig      OCTET STRING                        OPTIONAL,
    iE-Extensions     ProtocolExtensionContainer { { NPRACHConfiguration-FDD-ExtIEs} }  OPTIONAL,

...
}

NPRACHConfiguration-FDD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NPRACHConfiguration-TDD::= SEQUENCE {
    nprach-preambleFormat                       NPRACH-preambleFormat,
    anchorCarrier-NPRACHConfigTDD               OCTET STRING,
    non-anchorCarrierFequencyConfiglist         Non-AnchorCarrierFrequencylist      OPTIONAL,
    non-anchorCarrier-NPRACHConfigTDD           OCTET STRING                        OPTIONAL,
    iE-Extensions    ProtocolExtensionContainer { { NPRACHConfiguration-TDD-ExtIEs} }   OPTIONAL,
```

```
...
}

NPRACHConfiguration-TDD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NPRACH-CP-Length::=         ENUMERATED {
    us66dot7,
    us266dot7,
    ...
}

NPRACH-preambleFormat::=    ENUMERATED {fmt0,fmt1,fmt2,fmt0a,fmt1a,...}

Non-AnchorCarrierFrequencylist ::= SEQUENCE (SIZE(1..maxnoofNonAnchorCarrierFreqConfig)) OF
    SEQUENCE {
        non-anchorCarrioerFrquency      OCTET STRING,
        iE-Extensions                   ProtocolExtensionContainer { { Non-AnchorCarrierFrequencylist-ExtIEs} } OPTIONAL,
        ...
    }

Non-AnchorCarrierFrequencylist-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


NRPCI ::= INTEGER (0..1007)

NRrestrictioninEPSasSecondaryRAT ::= ENUMERATED {
    nRrestrictedinEPSasSecondaryRAT,
    ...
}

MeasurementResultforNRCellsPossiblyAggregated ::= SEQUENCE (SIZE(1.. maxnoofReportedNRCellsPossiblyAggregated)) OF
MeasurementResultforNRCellsPossiblyAggregated-Item

MeasurementResultforNRCellsPossiblyAggregated-Item ::= SEQUENCE {
    cellID                                  NRCGI,
    nrCompositeAvailableCapacityGroup       NRCompositeAvailableCapacityGroup       OPTIONAL,
    iE-Extension        ProtocolExtensionContainer { { MeasurementResultforNRCellsPossiblyAggregated-Item-ExtIEs} } OPTIONAL,
    ...
}

MeasurementResultforNRCellsPossiblyAggregated-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRRadioResourceStatus   ::= SEQUENCE {
    ssbAreaRadioResourceStatus-List     SSBAreaRadioResourceStatus-List,
    iE-Extensions                       ProtocolExtensionContainer { {NRRadioResourceStatus-ExtIEs} } OPTIONAL,
    ...
}
```

```
NRRadioResourceStatus-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-MIMOPRBusageInformation         CRITICALITY ignore  EXTENSION MIMOPRBusageInformation      PRESENCE optional },
    ...
}

MIMOPRBusageInformation ::= SEQUENCE {
    dl-GBR-PRB-usage-for-MIMO                   DL-GBR-PRB-usage-for-MIMO,
    ul-GBR-PRB-usage-for-MIMO                   UL-GBR-PRB-usage-for-MIMO,
    dl-non-GBR-PRB-usage-for-MIMO               DL-non-GBR-PRB-usage-for-MIMO,
    ul-non-GBR-PRB-usage-for-MIMO               UL-non-GBR-PRB-usage-for-MIMO,
    dl-Total-PRB-usage-for-MIMO                 DL-Total-PRB-usage-for-MIMO,
    ul-Total-PRB-usage-for-MIMO                 UL-Total-PRB-usage-for-MIMO,
    iE-Extensions                           ProtocolExtensionContainer { { MIMOPRBusageInformation-ExtIEs} } OPTIONAL,
    ...
}

MIMOPRBusageInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRrestrictionin5GS ::= ENUMERATED {
    nRrestrictedin5GS,
    ...
}

NRencryptionAlgorithms ::= BIT STRING (SIZE (16,...))
NRintegrityProtectionAlgorithms ::= BIT STRING (SIZE (16,...))

NR-TxBW ::= SEQUENCE {
    nRSCS   NRSCS,
    nRNRB   NRNRB,
    iE-Extensions               ProtocolExtensionContainer { {NR-TxBW-ExtIEs} } OPTIONAL,
    ...
}

NR-TxBW-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRNRB ::= ENUMERATED { nrb11, nrb18, nrb24, nrb25, nrb31, nrb32, nrb38, nrb51, nrb52, nrb65, nrb66, nrb78, nrb79, nrb93, nrb106, nrb107, nrb121,
nrb132, nrb133, nrb135, nrb160, nrb162, nrb189, nrb216, nrb217, nrb245, nrb264, nrb270, nrb273, ...}

NRSCS ::= ENUMERATED { scs15, scs30, scs60, scs120, ...}

NRS-NSSS-PowerOffset ::= ENUMERATED { minusThree, zero, three, ...}

FiveGS-TAC ::= OCTET STRING (SIZE (3))

NRUeReport ::= SEQUENCE {
    uENRMeasurements            RRCContainer,
    iE-Extensions           ProtocolExtensionContainer { { NRUeReport-ExtIEs} } OPTIONAL,
    ...
}
```

```
NRUeReport-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRUESidelinkAggregateMaximumBitRate ::= SEQUENCE {
    uESidelinkAggregateMaximumBitRate       BitRate,
    iE-Extensions                   ProtocolExtensionContainer { { NRUESidelinkAggregateMaximumBitRate-ExtIEs} } OPTIONAL,
    ...
}

NRUESidelinkAggregateMaximumBitRate-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NRUESecurityCapabilities ::= SEQUENCE {
    nRencryptionAlgorithms              NRencryptionAlgorithms,
    nRintegrityProtectionAlgorithms     NRintegrityProtectionAlgorithms,
    iE-Extensions                       ProtocolExtensionContainer { {NRUESecurityCapabilities-ExtIEs} }   OPTIONAL,
...
}

NRUESecurityCapabilities-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

NSSS-NumOccasionDifferentPrecoder ::= ENUMERATED { two, four, eight, ...}

NRV2XServicesAuthorized ::= SEQUENCE {
    vehicleUE           VehicleUE                                                   OPTIONAL,
    pedestrianUE        PedestrianUE                                                OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {NRV2XServicesAuthorized-ExtIEs} } OPTIONAL,
    ...
}

NRV2XServicesAuthorized-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- O

OffsetOfNbiotChannelNumberToEARFCN ::= ENUMERATED {
        minusTen,
        minusNine,
        minusEight,
        minusSeven,
        minusSix,
        minusFive,
        minusFour,
        minusThree,
        minusTwo,
        minusOne,
        minusZeroDotFive,
        zero,
```

```
        one,
        two,
        three,
        four,
        five,
        six,
        seven,
        eight,
        nine,
        ...,
        minusEightDotFive,
        minusFourDotFive,
        threeDotFive,
        sevenDotFive
}

Oneframe ::= BIT STRING (SIZE (6))

-- P

Packet-LossRate ::= INTEGER(0..1000)

PA-Values ::= ENUMERATED {
    dB-6,
    dB-4dot77,
    dB-3,
    dB-1dot77,
    dB0,
    dB1,
    dB2,
    dB3,
    ...
}

PC5QoSParameters ::= SEQUENCE {
    pc5QoSFlowList              PC5QoSFlowList,
    pc5LinkAggregatedBitRates   BitRate             OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { PC5QoSParameters-ExtIEs} }   OPTIONAL,
    ...
}

PC5QoSParameters-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

PC5QoSFlowList ::= SEQUENCE (SIZE(1..maxnoofPC5QoSFlows)) OF PC5QoSFlowItem

PC5QoSFlowItem::= SEQUENCE {
    pQI                         FiveQI,
    pc5FlowBitRates             PC5FlowBitRates             OPTIONAL,
    range                       Range                       OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { { PC5QoSFlowItem-ExtIEs} } OPTIONAL,
    ...
}
```

```
PC5QoSFlowItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
            ...
}

PC5FlowBitRates ::= SEQUENCE {
    guaranteedFlowBitRate       BitRate,
    maximumFlowBitRate          BitRate,
    iE-Extensions       ProtocolExtensionContainer { { PC5FlowBitRates-ExtIEs} }    OPTIONAL,
    ...
}

PC5FlowBitRates-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
            ...
}

PDCPChangeIndication ::= ENUMERATED {s-KgNB-update-required, pDCP-data-recovery-required,...}

PDCP-SN ::= INTEGER (0..4095)

PDCP-SNExtended ::= INTEGER (0..32767)

PDCP-SNlength18 ::= INTEGER (0..262143)

PDCPSnLength    ::= ENUMERATED {twelve-bits,eighteen-bits,...}

PCI ::= INTEGER (0..503, ...)

PLMN-Identity ::= OCTET STRING (SIZE(3))

Port-Number ::= OCTET STRING (SIZE (2))

PRACH-Configuration ::= SEQUENCE {
    rootSequenceIndex                       INTEGER (0..837),
    zeroCorrelationIndex                    INTEGER (0..15),
    highSpeedFlag                           BOOLEAN,
    prach-FreqOffset                        INTEGER (0..94),
    prach-ConfigIndex                       INTEGER (0..63)     OPTIONAL, -- present for TDD --
    iE-Extensions                           ProtocolExtensionContainer { {PRACH-Configuration-ExtIEs} } OPTIONAL,
    ...
}

PLMNAreaBasedQMC ::= SEQUENCE {
    plmnListforQMC      PLMNListforQMC,
    iE-Extensions       ProtocolExtensionContainer { {PLMNAreaBasedQMC-ExtIEs} } OPTIONAL,
    ...
}

PLMNAreaBasedQMC-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

PLMNListforQMC ::= SEQUENCE (SIZE(1..maxnoofPLMNforQMC)) OF PLMN-Identity
```

```
PRACH-Configuration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

Pre-emptionCapability ::= ENUMERATED {
    shall-not-trigger-pre-emption,
    may-trigger-pre-emption
}

Pre-emptionVulnerability ::= ENUMERATED {
    not-pre-emptable,
    pre-emptable
}

PriorityLevel               ::= INTEGER { spare (0), highest (1), lowest (14), no-priority (15) } (0..15)

ProSeAuthorized ::= SEQUENCE {
    proSeDirectDiscovery        ProSeDirectDiscovery            OPTIONAL,
    proSeDirectCommunication    ProSeDirectCommunication        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {ProSeAuthorized-ExtIEs} } OPTIONAL,
    ...
}

ProSeAuthorized-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-ProSeUEtoNetworkRelaying    CRITICALITY ignore  EXTENSION ProSeUEtoNetworkRelaying    PRESENCE optional},
    ...
}

ProSeDirectDiscovery ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

ProSeDirectCommunication ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

ProSeUEtoNetworkRelaying ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

ProtectedEUTRAResourceIndication::= SEQUENCE {
    activationSFN                           INTEGER (0..1023),
    protectedResourceList                   ProtectedResourceList,
    mBSFNControlRegionLength                INTEGER (0..3) OPTIONAL,
    pDCCHRegionLength                       INTEGER (1..3) OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { {ProtectedEUTRAResourceIndication-ExtIEs} }  OPTIONAL,
    ...
```

```
}

ProtectedEUTRAResourceIndication-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
} -- Rapporteur: missing extension --


ProtectedFootprintTimePattern ::= SEQUENCE {
    protectedFootprintTimePeriodicity               INTEGER (1..320, ...),
    protectedFootprintStartTime                     INTEGER (1..20, ...),
    iE-Extensions                                   ProtocolExtensionContainer { {ProtectedFootprintTimePattern-ExtIEs} }OPTIONAL,
    ...
}

ProtectedFootprintTimePattern-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ProtectedResourceList ::= SEQUENCE (SIZE(1.. maxnoofProtectedResourcePatterns)) OF ProtectedResourceList-Item

ProtectedResourceList-Item ::= SEQUENCE {
    resourceType                    ResourceType,
    intraPRBProtectedResourceFootprint      BIT STRING (SIZE(84, ...)),
    protectedFootprintFrequencyPattern      BIT STRING (SIZE(6..110, ...)),
    protectedFootprintTimePattern           ProtectedFootprintTimePattern,
    iE-Extensions                           ProtocolExtensionContainer { {ProtectedResourceList-Item-ExtIEs} }   OPTIONAL,
    ...
}

ProtectedResourceList-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

PartialListIndicator ::= ENUMERATED {partial, ...}

PrivacyIndicator ::= ENUMERATED {
    immediate-MDT,
    logged-MDT,
    ...
}

PSCellHistoryInformationRetrieve ::= ENUMERATED {query, ...}

PSCell-UE-HistoryInformation ::= SEQUENCE (SIZE(1.. maxnoofPSCellsPerPrimaryCellinUEHistoryInfo)) OF LastVisitedPSCell-Item

PSCellChangeHistory ::= ENUMERATED {reportingFullHistory,...}


-- Q

QCI ::= INTEGER (0..255)

QoS-Mapping-Information ::= SEQUENCE {
dscp                            BIT STRING (SIZE(6))            OPTIONAL,
```

```
flow-label                      BIT STRING (SIZE(20))       OPTIONAL,
iE-Extensions                   ProtocolExtensionContainer { {QoS-Mapping-Information-ExtIEs} } OPTIONAL,
...
}

QoS-Mapping-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

-- R

RadioframeAllocationOffset ::= INTEGER (0..7, ...)

RadioframeAllocationPeriod ::= ENUMERATED{
    n1,
    n2,
    n4,
    n8,
    n16,
    n32,
    ...
}


RadioResourceStatus ::= SEQUENCE {
    dL-GBR-PRB-usage                            DL-GBR-PRB-usage,
    uL-GBR-PRB-usage                            UL-GBR-PRB-usage,
    dL-non-GBR-PRB-usage                        DL-non-GBR-PRB-usage,
    uL-non-GBR-PRB-usage                        UL-non-GBR-PRB-usage,
    dL-Total-PRB-usage                          DL-Total-PRB-usage,
    uL-Total-PRB-usage                          UL-Total-PRB-usage,
    iE-Extensions                               ProtocolExtensionContainer { {RadioResourceStatus-ExtIEs} } OPTIONAL,
    ...
}

RadioResourceStatus-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {ID id-DL-scheduling-PDCCH-CCE-usage        CRITICALITY ignore  EXTENSION DL-scheduling-PDCCH-CCE-usage     PRESENCE optional}|
    {ID id-UL-scheduling-PDCCH-CCE-usage        CRITICALITY ignore  EXTENSION UL-scheduling-PDCCH-CCE-usage     PRESENCE optional},
    ...
}

Range ::= ENUMERATED {m50, m80, m180, m200, m350, m400, m500, m700, m1000, ...}

RAN-UE-NGAP-ID ::= INTEGER (0..4294967295)

RAT-Restrictions ::= SEQUENCE (SIZE(1..maxnoofEPLMNsPlusOne)) OF RAT-RestrictionsItem

RAT-RestrictionsItem ::= SEQUENCE {
    pLMN-Identity                   PLMN-Identity,
    rAT-RestrictionInformation      BIT STRING (SIZE(8, ...)),
    iE-Extensions                   ProtocolExtensionContainer { { RAT-RestrictionsItem-ExtIEs} } OPTIONAL,
    ...
}
```

```
RAT-RestrictionsItem-ExtIEs     X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ReceiveStatusofULPDCPSDUs ::= BIT STRING (SIZE(4096))

ReceiveStatusOfULPDCPSDUsExtended ::= BIT STRING (SIZE(1..16384))

ReceiveStatusOfULPDCPSDUsPDCP-SNlength18 ::= BIT STRING (SIZE(1..131072))

ReleaseFastMCGRecoveryViaSRB3 ::= ENUMERATED {true,...}

Reestablishment-Indication ::= ENUMERATED {
    reestablished,
    ...
}

Registration-Request    ::= ENUMERATED {
    start,
    stop,
    ...,
    partial-stop,
    add
}

Registration-Request-ENDC   ::= ENUMERATED {
    start,
    stop,
    add,
    ...
}


RelativeNarrowbandTxPower ::= SEQUENCE {

    rNTP-PerPRB                         BIT STRING (SIZE(6..110, ...)),
    rNTP-Threshold                      RNTP-Threshold,
    numberOfCellSpecificAntennaPorts    ENUMERATED {one, two, four, ...},
    p-B                                 INTEGER (0..3,...),
    pDCCH-InterferenceImpact            INTEGER (0..4,...),
    iE-Extensions                       ProtocolExtensionContainer { {RelativeNarrowbandTxPower-ExtIEs} } OPTIONAL,
    ...
}

RelativeNarrowbandTxPower-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-enhancedRNTP    CRITICALITY ignore  EXTENSION EnhancedRNTP      PRESENCE optional },
    ...
}

ReplacingCellsList ::= SEQUENCE (SIZE(0.. maxCellineNB)) OF ReplacingCellsList-Item

ReplacingCellsList-Item ::= SEQUENCE {
    eCGI            ECGI,
    ...
```

```
}

ReportAmountMDT ::= ENUMERATED{r1, r2, r4, r8, r16, r32, r64, rinfinity}

ReportArea ::= ENUMERATED{
    ecgi,
    ...
}

ReportCharacteristics   ::= BIT STRING (SIZE (32))

ReportingPeriodicityCSIR ::= ENUMERATED {
    ms5,
    ms10,
    ms20,
    ms40,
    ms80,
...
}

ReportCharacteristics-ENDC  ::= BIT STRING (SIZE (32))

ReportingPeriodicityRSRPMR ::= ENUMERATED {
    one-hundred-20-ms,
    two-hundred-40-ms,
    four-hundred-80-ms,
    six-hundred-40-ms,
...
}

ReportIntervalMDT ::= ENUMERATED {ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, min60}

RequestedFastMCGRecoveryViaSRB3 ::= ENUMERATED {true,...}

RequestedFastMCGRecoveryViaSRB3Release ::= ENUMERATED {true,...}

ReservedSubframePattern ::= SEQUENCE{
    subframeType                    SubframeType,
    reservedSubframePattern         BIT STRING (SIZE(10..160)),
    mBSFNControlRegionLength        INTEGER (0..3),
    iE-Extensions                   ProtocolExtensionContainer { {ReservedSubframePattern-ExtIEs} }  OPTIONAL,
    ...
}

ReservedSubframePattern-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ResourceType ::= ENUMERATED {
    downlinknonCRS,
    cRS,
    uplink,
    ...
}
```

```
ResumeID    ::= CHOICE {
    non-truncated   BIT STRING(SIZE(40)),
    truncated       BIT STRING(SIZE(24)),
    ...
}

RLCMode ::= ENUMERATED {
    rlc-am,
    rlc-um-bidirectional,
    rlc-um-unidirectional-ul,
    rlc-um-unidirectional-dl,
    ...
}

RLC-Status ::= SEQUENCE {
    reestablishment-Indication  Reestablishment-Indication,
    iE-Extensions               ProtocolExtensionContainer { {RLC-Status-ExtIEs} } OPTIONAL,
    ...
}

RLC-Status-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

RNTP-Threshold ::= ENUMERATED {
    minusInfinity,
    minusEleven,
    minusTen,
    minusNine,
    minusEight,
    minusSeven,
    minusSix,
    minusFive,
    minusFour,
    minusThree,
    minusTwo,
    minusOne,
    zero,
    one,
    two,
    three,
    ...
}

RRC-Config-Ind ::= ENUMERATED {
    full-config,
    delta-config,
    ...
}

RRC-Context ::= OCTET STRING

RRCConnReestabIndicator ::= ENUMERATED {
```

```
    reconfigurationFailure, handoverFailure, otherFailure, ...
}
-- The values correspond to the values of ReestablishmentCause reported from the UE in the RRCConnectionReestablishmentRequest, as defined in TS
36.331 [9]

RRCConnSetupIndicator::= ENUMERATED {
    rrcConnSetup,
    ...
}

RSRPMeasurementResult ::= SEQUENCE (SIZE(1..maxCellReport)) OF
    SEQUENCE {
        rSRPCellID                      ECGI,
        rSRPMeasured                    INTEGER (0..97, ...),
        iE-Extensions                   ProtocolExtensionContainer { {RSRPMeasurementResult-ExtIEs} } OPTIONAL,
        ...
    }

RSRPMeasurementResult-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

RSRPMRList ::= SEQUENCE (SIZE(1..maxUEReport)) OF
    SEQUENCE {
        rSRPMeasurementResult           RSRPMeasurementResult,
        iE-Extensions                   ProtocolExtensionContainer { {RSRPMRList-ExtIEs} } OPTIONAL,
        ...
    }

RSRPMRList-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-UEID    CRITICALITY ignore  EXTENSION UEID      PRESENCE optional},
    ...
}

RRCContainer ::= OCTET STRING

-- S

S1TNLLoadIndicator ::= SEQUENCE {
    dLS1TNLLoadIndicator            LoadIndicator,
    uLS1TNLLoadIndicator            LoadIndicator,
    iE-Extensions                   ProtocolExtensionContainer { {S1TNLLoadIndicator-ExtIEs} } OPTIONAL,
    ...
}

S1TNLLoadIndicator-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


SCGActivationStatus ::= ENUMERATED {scg-activated, scg-deactivated, ...}


SCGActivationRequest ::= ENUMERATED {activate-scg, deactivate-scg, ...}
```

```
SCGChangeIndication ::= ENUMERATED {pDCPCountWrapAround, pSCellChange, other, ...}

SCGreconfigNotification ::= ENUMERATED {executed, ..., executed-deleted, deleted }


SCG-UE-HistoryInformation ::= SEQUENCE (SIZE(1.. maxnoofPSCellsPerSN)) OF LastVisitedPSCell-Item

SecondaryRATUsageReportList ::= SEQUENCE (SIZE(1..maxnoofBearers)) OF ProtocolIE-Single-Container {{SecondaryRATUsageReport-ItemIEs}}

SecondaryRATUsageReport-ItemIEs X2AP-PROTOCOL-IES ::= {
    { ID id-SecondaryRATUsageReport-Item        CRITICALITY reject  TYPE SecondaryRATUsageReport-Item     PRESENCE mandatory},
    ...
}

SecondaryRATUsageReport-Item ::= SEQUENCE {
    e-RAB-ID                        E-RAB-ID,
    secondaryRATType                ENUMERATED {nr, ..., nR-unlicensed },
    e-RABUsageReportList            E-RABUsageReportList,
    iE-Extensions               ProtocolExtensionContainer { {SecondaryRATUsageReport-Item-ExtIEs} } OPTIONAL,
...
}

SecondaryRATUsageReport-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SecurityIndication ::= SEQUENCE {
    integrityProtectionIndication           IntegrityProtectionIndication,
    iE-Extensions                           ProtocolExtensionContainer { { SecurityIndication-ExtIEs } } OPTIONAL,
    ...
}

SecurityIndication-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SecurityResult ::= SEQUENCE {
    integrityProtectionResult           IntegrityProtectionResult,
    iE-Extensions                       ProtocolExtensionContainer { {SecurityResult-ExtIEs} }  OPTIONAL,
    ...
}

SecurityResult-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SeNBSecurityKey ::= BIT STRING (SIZE(256))

SeNBtoMeNBContainer ::= OCTET STRING

SensorMeasurementConfiguration ::=  SEQUENCE {
    sensorMeasConfig            SensorMeasConfig,
```

```
    sensorMeasConfigNameList    SensorMeasConfigNameList                                        OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {SensorMeasurementConfiguration-ExtIEs} }  OPTIONAL,
    ...
}

SensorMeasurementConfiguration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SensorMeasConfigNameList ::= SEQUENCE (SIZE(1..maxnoofSensorName)) OF SensorMeasConfigNameItem

SensorMeasConfigNameItem ::= SEQUENCE {
    sensorNameConfig        SensorNameConfig,
    iE-Extensions       ProtocolExtensionContainer { { SensorMeasConfigNameItem-ExtIEs } }  OPTIONAL,
    ...
}

SensorMeasConfigNameItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SensorMeasConfig::= ENUMERATED {setup,...}

SensorNameConfig ::= CHOICE {
    uncompensatedBarometricConfig       ENUMERATED {true, ...},
    choice-extension        ProtocolIE-Single-Container { {SensorNameConfig-ExtIEs} }
}

SensorNameConfig-ExtIEs X2AP-PROTOCOL-IES ::= {
    ...
}


ServedCells ::= SEQUENCE (SIZE (1.. maxCellineNB)) OF SEQUENCE {
    servedCellInfo                  ServedCell-Information,
    neighbour-Info                  Neighbour-Information           OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {ServedCell-ExtIEs} } OPTIONAL,
    ...
}

ServedCell-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-NRNeighbourInfoToAdd            CRITICALITY ignore  EXTENSION NRNeighbour-Information         PRESENCE optional }|
    { ID id-ServedCellSpecificInfoReq-NR    CRITICALITY ignore EXTENSION    ServedCellSpecificInfoReq-NR  PRESENCE optional },
    ...
}

ServedCell-Information ::= SEQUENCE {
    pCI                 PCI,
    cellId              ECGI,
    tAC                 TAC,
    broadcastPLMNs      BroadcastPLMNs-Item,
    eUTRA-Mode-Info     EUTRA-Mode-Info,
    iE-Extensions       ProtocolExtensionContainer { {ServedCell-Information-ExtIEs} } OPTIONAL,
    ...
```

```
}

ServedCell-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-Number-of-Antennaports              CRITICALITY ignore  EXTENSION Number-of-Antennaports            PRESENCE optional}|
    { ID id-PRACH-Configuration                 CRITICALITY ignore  EXTENSION PRACH-Configuration               PRESENCE optional}|
    { ID id-MBSFN-Subframe-Info                 CRITICALITY ignore  EXTENSION MBSFN-Subframe-Infolist          PRESENCE optional}|
    { ID id-CSG-Id                              CRITICALITY ignore  EXTENSION CSG-Id                           PRESENCE optional}|
    { ID id-MBMS-Service-Area-List              CRITICALITY ignore  EXTENSION MBMS-Service-Area-Identity-List  PRESENCE optional}|
    { ID id-MultibandInfoList                   CRITICALITY ignore  EXTENSION MultibandInfoList                PRESENCE optional}|
    { ID id-FreqBandIndicatorPriority           CRITICALITY ignore  EXTENSION FreqBandIndicatorPriority        PRESENCE optional}|
    { ID id-BandwidthReducedSI                  CRITICALITY ignore  EXTENSION BandwidthReducedSI               PRESENCE optional}|
    { ID id-ProtectedEUTRAResourceIndication    CRITICALITY ignore  EXTENSION ProtectedEUTRAResourceIndication PRESENCE optional}|
    { ID id-BPLMN-ID-Info-EUTRA                 CRITICALITY ignore  EXTENSION BPLMN-ID-Info-EUTRA               PRESENCE optional}|
    { ID id-NPRACHConfiguration                 CRITICALITY ignore  EXTENSION   NPRACHConfiguration            PRESENCE optional}|
    { ID id-SFN-Offset                          CRITICALITY ignore  EXTENSION SFN-Offset                       PRESENCE optional},
    ...
}

ServedCellSpecificInfoReq-NR    ::= SEQUENCE (SIZE(1.. maxCellinengNB)) OF ServedCellSpecificInfoReq-NR-Item

ServedCellSpecificInfoReq-NR-Item   ::= SEQUENCE {
    nRCGI                               NRCGI,
    additionalMTCListRequestIndicator   ENUMERATED {additionalMTCListRequested, ...}         OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { { ServedCellSpecificInfoReq-NR-Item-ExtIEs} }OPTIONAL,
    ...
}

ServedCellSpecificInfoReq-NR-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


ServiceType ::= ENUMERATED{
    qMC-for-streaming-service,
    qMC-for-MTSI-service,
    ...
}

SgNBCoordinationAssistanceInformation ::= ENUMERATED{
    coordination-not-required,
    ...
}

SgNBResourceCoordinationInformation ::= SEQUENCE {
    nR-CGI                          NRCGI,
    uLCoordinationInformation       BIT STRING (SIZE(6..4400, ...)),
    dLCoordinationInformation       BIT STRING (SIZE(6..4400, ...)) OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { {SgNBResourceCoordinationInformationExtIEs} }      OPTIONAL,
    ...
}

SgNBResourceCoordinationInformationExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-ECGI                                    CRITICALITY ignore  EXTENSION ECGI                                  PRESENCE optional}|
    { ID id-SgNBCoordinationAssistanceInformation   CRITICALITY reject  EXTENSION SgNBCoordinationAssistanceInformation PRESENCE optional},
```

```
    ...
}

SgNB-UE-X2AP-ID ::= INTEGER (0..4294967295)

SIPTOBearerDeactivationIndication ::= ENUMERATED {
    true,
    ...
}

SharedResourceType ::= CHOICE{
    uLOnlySharing       ULOnlySharing,
    uLandDLSharing      ULandDLSharing,
    ...
}

ShortMAC-I ::= BIT STRING (SIZE(16))


SGNB-Addition-Trigger-Ind    ::= ENUMERATED {
    sn-change,
    inter-eNB-HO,
    intra-eNB-HO,
    ...
}

SNtriggered ::=ENUMERATED{
    true,
    ...
}

SourceOfUEActivityBehaviourInformation ::= ENUMERATED {
    subscription-information,
    statistics,
    ...
}

SpecialSubframe-Info ::= SEQUENCE {
    specialSubframePatterns     SpecialSubframePatterns,
    cyclicPrefixDL              CyclicPrefixDL,
    cyclicPrefixUL              CyclicPrefixUL,
    iE-Extensions               ProtocolExtensionContainer { {SpecialSubframe-Info-ExtIEs} } OPTIONAL,
    ...
}

SpecialSubframe-Info-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SpecialSubframePatterns ::= ENUMERATED {
    ssp0,
    ssp1,
    ssp2,
    ssp3,
```

```
    ssp4,
    ssp5,
    ssp6,
    ssp7,
    ssp8,
    ...
}

SpectrumSharingGroupID ::= INTEGER (1..maxCellineNB)

SubbandCQI ::= SEQUENCE {
    subbandCQICodeword0         SubbandCQICodeword0,
    subbandCQICodeword1         SubbandCQICodeword1     OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {SubbandCQI-ExtIEs} } OPTIONAL,
    ...
}

Subscription-Based-UE-DifferentiationInfo ::= SEQUENCE {
    periodicCommunicationIndicator  ENUMERATED {periodically, ondemand, ...}        OPTIONAL,
    periodicTime                    INTEGER (1..3600, ...)                          OPTIONAL,
    scheduledCommunicationTime      ScheduledCommunicationTime                      OPTIONAL,
    stationaryIndication            ENUMERATED {stationary, mobile, ...}            OPTIONAL,
    trafficProfile                  ENUMERATED {single-packet, dual-packets, multiple-packets, ...}       OPTIONAL,
    batteryIndication               ENUMERATED {battery-powered, battery-powered-not-rechargeable-or-replaceable, not-battery-powered, ...}
    OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { Subscription-Based-UE-DifferentiationInfo-ExtIEs} } OPTIONAL,
    ...
}

Subscription-Based-UE-DifferentiationInfo-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ScheduledCommunicationTime ::= SEQUENCE {
    dayofWeek               BIT STRING (SIZE(7))                                    OPTIONAL,
    timeofDayStart          INTEGER (0..86399, ...)                                 OPTIONAL,
    timeofDayEnd            INTEGER (0..86399, ...)                                 OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { { ScheduledCommunicationTime-ExtIEs}}  OPTIONAL,
    ...
}

ScheduledCommunicationTime-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SRVCCOperationPossible ::= ENUMERATED {
    possible,
    ...
}

SSBAreaCapacityValue-List   ::= SEQUENCE (SIZE (1.. maxnoofSSBAreas)) OF SSBAreaCapacityValue-Item

SSBAreaCapacityValue-Item   ::=     SEQUENCE {
    ssbIndex                            SSBIndex,
```

```
    ssbAreaCapacityValue                INTEGER (0..100),
    iE-Extensions                       ProtocolExtensionContainer { {SSBAreaCapacityValue-ExtIEs} } OPTIONAL,
    ...
}

SSBAreaCapacityValue-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSBAreaRadioResourceStatus-List ::= SEQUENCE (SIZE (1.. maxnoofSSBAreas)) OF SSBAreaRadioResourceStatus-Item

SSBAreaRadioResourceStatus-Item ::= SEQUENCE {
    ssbIndex                            SSBIndex,
    ssbAreaDLGBRPRBUsage                INTEGER (0..100),
    ssbAreaULGBRPRBUsage                INTEGER (0..100),
    ssbAreaDLNonGBRPRBUsage             INTEGER (0..100),
    ssbAreaULNonGBRPRBUsage             INTEGER (0..100),
    ssbAreaDLTotalPRBUsage              INTEGER (0..100),
    ssbAreaULTotalPRBUsage              INTEGER (0..100),
    ssbAreaDLSchedulingPDCCHCCEUsage    INTEGER (0..100)                                    OPTIONAL,
    ssbAreaULSchedulingPDCCHCCEUsage    INTEGER (0..100)                                    OPTIONAL,
    iE-Extensions                       ProtocolExtensionContainer { {SSBAreaRadioResourceStatus-ExtIEs} } OPTIONAL,
    ...
}

SSBAreaRadioResourceStatus-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SSBIndex ::= INTEGER (0..63)

SSB-PositionsInBurst ::= CHOICE {
    shortBitmap                     BIT STRING (SIZE (4)),
    mediumBitmap                    BIT STRING (SIZE (8)),
    longBitmap                      BIT STRING (SIZE (64)),
    choice-extension                ProtocolIE-Single-Container { {SSB-PositionsInBurst-ExtIEs} }
}

SSB-PositionsInBurst-ExtIEs X2AP-PROTOCOL-IES ::= {
    ...
}


SubbandCQI-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SubbandCQICodeword0 ::= CHOICE {
    four-bitCQI                     INTEGER (0..15, ...),
    two-bitSubbandDifferentialCQI   INTEGER (0..3, ...),
    two-bitDifferentialCQI          INTEGER (0..3, ...),
    ...
}
SubbandCQICodeword1 ::= CHOICE {
```

```
    four-bitCQI                         INTEGER (0..15, ...),
    three-bitSpatialDifferentialCQI     INTEGER (0..7, ...),
    two-bitSubbandDifferentialCQI       INTEGER (0..3, ...),
    two-bitDifferentialCQI              INTEGER (0..3, ...),
    ...
}

SubbandCQIList ::= SEQUENCE (SIZE(1.. maxSubband)) OF SubbandCQIItem

SubbandCQIItem ::= SEQUENCE {
    subbandCQI          SubbandCQI,
    subbandIndex        INTEGER (0..27,...),
    iE-Extensions       ProtocolExtensionContainer { {SubbandCQIItem-ExtIEs} } OPTIONAL,
    ...
}

SubbandCQIItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SubbandSize ::= ENUMERATED {
    size2,
    size3,
    size4,
    size6,
    size8,
    ...
}

SubscriberProfileIDforRFP ::= INTEGER (1..256)

SubframeAllocation ::= CHOICE {
    oneframe                        Oneframe,
    fourframes                      Fourframes,
    ...
}

SubframeAssignment ::= ENUMERATED {
    sa0,
    sa1,
    sa2,
    sa3,
    sa4,
    sa5,
    sa6,
    ...
}

SubframeType ::= ENUMERATED{mbsfn,nonmbsfn,...}

SgNBSecurityKey ::= BIT STRING (SIZE(256))

SgNBtoMeNBContainer ::= OCTET STRING
```

```
SRBType ::= ENUMERATED {srb1, srb2, ...}
SCGConfigurationQuery ::= ENUMERATED {true,...}

SULInformation ::= SEQUENCE {
    sUL-ARFCN               INTEGER (0.. 3279165),
    sUL-TxBW                NR-TxBW,
    iE-Extensions           ProtocolExtensionContainer { {SULInformation-ExtIEs} }      OPTIONAL,
    ...
}


SupportedSULFreqBandItem ::= SEQUENCE {
    freqBandIndicatorNr             INTEGER (1..1024,...),
    iE-Extensions           ProtocolExtensionContainer { {SupportedSULFreqBandItem-ExtIEs} }     OPTIONAL,
    ...
}

SupportedSULFreqBandItem-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

SULInformation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-CarrierList                     CRITICALITY ignore  EXTENSION NRCarrierList         PRESENCE optional }|
    { ID id-FrequencyShift7p5khz            CRITICALITY ignore  EXTENSION FrequencyShift7p5khz  PRESENCE optional },
    ...
}

SFN-Offset ::= SEQUENCE {
    sFN-Time-Offset                 BIT STRING (SIZE(24)),
    iE-Extensions       ProtocolExtensionContainer { {SFN-Offset-ExtIEs} } OPTIONAL,
    ...
}

SFN-Offset-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
        ...
}

-- T

TABasedMDT::= SEQUENCE {
    tAListforMDT        TAListforMDT,
    iE-Extensions       ProtocolExtensionContainer { {TABasedMDT-ExtIEs} } OPTIONAL,
    ...
}

TABasedMDT-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TAC ::= OCTET STRING (SIZE (2))

TAIBasedMDT ::= SEQUENCE {
    tAIListforMDT           TAIListforMDT,
    iE-Extensions           ProtocolExtensionContainer { {TAIBasedMDT-ExtIEs} } OPTIONAL,
```

```
    ...
}

TAIBasedMDT-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIListforMDT ::= SEQUENCE (SIZE(1..maxnoofTAforMDT)) OF TAI-Item

TAI-Item ::= SEQUENCE {
    tAC                 TAC,
    pLMN-Identity       PLMN-Identity,
    iE-Extensions       ProtocolExtensionContainer { {TAI-Item-ExtIEs} } OPTIONAL,
    ...
}

TAI-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TAListforMDT ::= SEQUENCE (SIZE(1..maxnoofTAforMDT)) OF TAC

TABasedQMC ::= SEQUENCE {
    tAListforQMC        TAListforQMC,
    iE-Extensions       ProtocolExtensionContainer { {TABasedQMC-ExtIEs} } OPTIONAL,
    ...
}

TABasedQMC-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TAListforQMC ::= SEQUENCE (SIZE(1..maxnoofTAforQMC)) OF TAC

TAIBasedQMC ::= SEQUENCE {
    tAIListforQMC       TAIListforQMC,
    iE-Extensions       ProtocolExtensionContainer { {TAIBasedQMC-ExtIEs} } OPTIONAL,
    ...
}

TAIBasedQMC-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TAIListforQMC ::= SEQUENCE (SIZE(1..maxnoofTAforQMC)) OF TAI-Item

TargetCellInNGRAN ::= OCTET STRING

TargetCellInUTRAN ::= OCTET STRING -- This IE is to be encoded according to the UTRAN Cell ID in the Last Visited UTRAN Cell Information IE in TS
25.413 [24]

TargeteNBtoSource-eNBTransparentContainer   ::= OCTET STRING

TDD-Info ::= SEQUENCE {
```

```
    eARFCN                          EARFCN,
    transmission-Bandwidth          Transmission-Bandwidth,
    subframeAssignment              SubframeAssignment,
    specialSubframe-Info            SpecialSubframe-Info,
    iE-Extensions                   ProtocolExtensionContainer { {TDD-Info-ExtIEs} } OPTIONAL,
    ...
}

TDD-Info-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-AdditionalSpecialSubframe-Info          CRITICALITY ignore  EXTENSION AdditionalSpecialSubframe-Info         PRESENCE optional}|
    { ID id-eARFCNExtension                         CRITICALITY reject  EXTENSION EARFCNExtension                        PRESENCE optional}|
    { ID id-AdditionalSpecialSubframeExtension-Info CRITICALITY ignore  EXTENSION AdditionalSpecialSubframeExtension-Info PRESENCE optional}|
    { ID id-OffsetOfNbiotChannelNumberToDL-EARFCN   CRITICALITY reject  EXTENSION OffsetOfNbiotChannelNumberToEARFCN     PRESENCE optional}|
    { ID id-NBIoT-UL-DL-AlignmentOffset             CRITICALITY reject  EXTENSION NBIoT-UL-DL-AlignmentOffset            PRESENCE optional},
    ...
}

TDD-InfoNeighbourServedNRCell-Information ::= SEQUENCE {
    nRFreqInfo              NRFreqInfo,
    iE-Extensions           ProtocolExtensionContainer { {TDD-InfoNeighbourServedNRCell-Information-ExtIEs} }    OPTIONAL,
    ...
}

TDD-InfoNeighbourServedNRCell-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {ID id-IntendedTDD-DL-ULConfiguration-NR    CRITICALITY ignore  EXTENSION IntendedTDD-DL-ULConfiguration-NR    PRESENCE optional}|
    {ID id-TDDULDLConfigurationCommonNR         CRITICALITY ignore  EXTENSION TDDULDLConfigurationCommonNR         PRESENCE optional}|
    {ID id-CarrierList                          CRITICALITY ignore  EXTENSION NRCarrierList                        PRESENCE optional},
    ...
}

TDDULDLConfigurationCommonNR ::= OCTET STRING


Threshold-RSRP ::= INTEGER(0..97)

Threshold-RSRQ ::= INTEGER(0..34)

TimeToWait ::= ENUMERATED {
    v1s,
    v2s,
    v5s,
    v10s,
    v20s,
    v60s,
    ...
}

Time-UE-StayedInCell ::= INTEGER (0..4095)

Time-UE-StayedInCell-EnhancedGranularity ::= INTEGER (0..40950)

TNLA-To-Add-List ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF TNLA-To-Add-Item

TNLA-To-Add-Item ::= SEQUENCE {
```

```
    tNLAssociationTransportLayerAddress     CPTransportLayerInformation,
    tNLAssociationUsage                     TNLAssociationUsage,
    iE-Extensions                           ProtocolExtensionContainer { { TNLA-To-Add-Item-ExtIEs} } OPTIONAL
}

TNLA-To-Add-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TNLA-To-Update-List ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF TNLA-To-Update-Item

TNLA-To-Update-Item::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CPTransportLayerInformation,
    tNLAssociationUsage                     TNLAssociationUsage         OPTIONAL,
    iE-Extensions                           ProtocolExtensionContainer { { TNLA-To-Update-Item-ExtIEs} } OPTIONAL
}

TNLA-To-Update-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TNLA-To-Remove-List ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF TNLA-To-Remove-Item

TNLA-To-Remove-Item::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CPTransportLayerInformation,
    iE-Extensions                           ProtocolExtensionContainer { { TNLA-To-Remove-Item-ExtIEs} } OPTIONAL
}

TNLA-To-Remove-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TNLA-Setup-List ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF TNLA-Setup-Item

TNLA-Setup-Item ::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CPTransportLayerInformation,
    iE-Extensions                           ProtocolExtensionContainer { { TNLA-Setup-Item-ExtIEs} } OPTIONAL,
    ...
}

TNLA-Setup-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TNLA-Failed-To-Setup-List ::= SEQUENCE (SIZE(1..maxnoofTNLAssociations)) OF TNLA-Failed-To-Setup-Item

TNLA-Failed-To-Setup-Item ::= SEQUENCE {
    tNLAssociationTransportLayerAddress     CPTransportLayerInformation,
    cause                                   Cause,
    iE-Extensions                           ProtocolExtensionContainer { { TNLA-Failed-To-Setup-Item-ExtIEs} } OPTIONAL
}

TNLA-Failed-To-Setup-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
```

```
}

TNLAssociationUsage ::= ENUMERATED {
    ue,
    non-ue,
    both,
    ...
}

TNLCapacityIndicator ::= SEQUENCE {
    dlTNLMaximumOfferedCapacity         INTEGER (1..16777216, ...),
    dlTNLAvailableCapacity              INTEGER (0..100, ...),
    ulTNLMaximumOfferedCapacity         INTEGER (1..16777216, ...),
    ulTNLAvailableCapacity              INTEGER (0..100, ...),
    iE-Extensions                       ProtocolExtensionContainer { {TNLCapacityIndicator-ExtIEs} } OPTIONAL,
    ...
}

TNLCapacityIndicator-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


Transport-UP-Layer-Addresses-Info-To-Add-List   ::= SEQUENCE (SIZE(1.. maxnoofTLAs)) OF Transport-UP-Layer-Addresses-Info-To-Add-Item

Transport-UP-Layer-Addresses-Info-To-Add-Item ::= SEQUENCE {
    iP-SecTransportLayerAddress     TransportLayerAddress,
    gTPTransportLayerAddressesToAdd         GTPTLAs                         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { Transport-UP-Layer-Addresses-Info-To-Add-ItemExtIEs } }    OPTIONAL,
    ...
}

Transport-UP-Layer-Addresses-Info-To-Add-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

Transport-UP-Layer-Addresses-Info-To-Remove-List    ::= SEQUENCE (SIZE(1.. maxnoofTLAs)) OF Transport-UP-Layer-Addresses-Info-To-Remove-Item

Transport-UP-Layer-Addresses-Info-To-Remove-Item ::= SEQUENCE {
    iP-SecTransportLayerAddress     TransportLayerAddress,
    gTPTransportLayerAddressesToRemove          GTPTLAs                         OPTIONAL,
    iE-Extensions                   ProtocolExtensionContainer { { Transport-UP-Layer-Addresses-Info-To-Remove-ItemExtIEs } }    OPTIONAL,
    ...
}

Transport-UP-Layer-Addresses-Info-To-Remove-ItemExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


TNLConfigurationInfo  ::= SEQUENCE {
    transport-UP-Layer-Addresses-Info-To-Add-List       Transport-UP-Layer-Addresses-Info-To-Add-List               OPTIONAL,
    transport-UP-Layer-Addresses-Info-To-Remove-List    Transport-UP-Layer-Addresses-Info-To-Remove-List            OPTIONAL,
    iE-Extensions   ProtocolExtensionContainer { { TNLConfigurationInfo-ExtIEs } }                                 OPTIONAL,
```

```
    ...
}

TNLConfigurationInfo-ExtIEs     X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TraceActivation ::= SEQUENCE {
    eUTRANTraceID                   EUTRANTraceID,
    interfacesToTrace               InterfacesToTrace,
traceDepth                          TraceDepth,
traceCollectionEntityIPAddress      TraceCollectionEntityIPAddress,
    iE-Extensions                   ProtocolExtensionContainer { {TraceActivation-ExtIEs} } OPTIONAL,
    ...
}

TraceActivation-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-MDTConfiguration            CRITICALITY ignore  EXTENSION MDT-Configuration         PRESENCE optional}|
    { ID id-UEAppLayerMeasConfig        CRITICALITY ignore  EXTENSION UEAppLayerMeasConfig      PRESENCE optional}|
    { ID id-MDTConfigurationNR          CRITICALITY ignore  EXTENSION MDT-ConfigurationNR       PRESENCE optional}|
    { ID id-TraceCollectionEntityURI    CRITICALITY ignore  EXTENSION URI-Address               PRESENCE optional},
    ...
}

TraceCollectionEntityIPAddress ::= BIT STRING (SIZE(1..160, ...))

TraceDepth      ::= ENUMERATED {
    minimum,
    medium,
    maximum,
    minimumWithoutVendorSpecificExtension,
    mediumWithoutVendorSpecificExtension,
    maximumWithoutVendorSpecificExtension,
    ...
}

Transmission-Bandwidth ::= ENUMERATED {
        bw6,
        bw15,
        bw25,
        bw50,
        bw75,
        bw100,
        ...,
        bw1
}

TransportLayerAddress           ::= BIT STRING (SIZE(1..160, ...))

TransportLayerAddressAndPort        ::= SEQUENCE {
    endpointIPAddress               TransportLayerAddress,
    portnumber                      Port-Number
}
```

```
TunnelInformation ::= SEQUENCE {
    transportLayerAddress   TransportLayerAddress,
    uDP-Port-Number         Port-Number         OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { {Tunnel-Information-ExtIEs} } OPTIONAL,
    ...
}

Tunnel-Information-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

TypeOfError ::= ENUMERATED {
    not-understood,
    missing,
    ...
}


-- U

UEAggregateMaximumBitRate ::= SEQUENCE {
    uEaggregateMaximumBitRateDownlink   BitRate,
    uEaggregateMaximumBitRateUplink     BitRate,
    iE-Extensions                       ProtocolExtensionContainer { {UEAggregate-MaximumBitrate-ExtIEs} } OPTIONAL,
    ...
}

UEAggregate-MaximumBitrate-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    { ID id-extended-uEaggregateMaximumBitRateDownlink  CRITICALITY ignore  EXTENSION ExtendedBitRate   PRESENCE optional}|
    { ID id-extended-uEaggregateMaximumBitRateUplink    CRITICALITY ignore  EXTENSION ExtendedBitRate   PRESENCE optional},
    ...
}

UEAppLayerMeasConfig ::= SEQUENCE {
    containerForAppLayerMeasConfig          OCTET STRING (SIZE(1..1000)),
    areaScopeOfQMC      AreaScopeOfQMC,
    iE-Extensions       ProtocolExtensionContainer { {UEAppLayerMeasConfig-ExtIEs} } OPTIONAL,
    ...
}

UEAppLayerMeasConfig-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    {ID id-serviceType  CRITICALITY ignore  EXTENSION ServiceType   PRESENCE optional},
    ...
}

UE-ContextKeptIndicator ::= ENUMERATED {
    true,
    ...
}

UEID ::= BIT STRING (SIZE (16))

UE-HistoryInformation ::= SEQUENCE (SIZE(1..maxnoofCells)) OF LastVisitedCell-Item
```

```
UE-HistoryInformationFromTheUE ::= OCTET STRING
-- This IE is a transparent container and shall be encoded as the VisitedCellInfoList field contained in the UEInformationResponse message as
defined in TS 36.331 [9]

UE-S1AP-ID ::= INTEGER (0.. 4294967295)

UE-X2AP-ID ::= INTEGER (0..4095)

UE-X2AP-ID-Extension ::= INTEGER (0..4095, ...)

UERadioCapability ::= OCTET STRING

UERadioCapabilityID ::= OCTET STRING

UE-RLF-Report-Container::= OCTET STRING
-- This IE is a transparent container and shall be encoded as the RLF-Report-r9 field contained in the UEInformationResponse message as defined in
TS 36.331 [9]

UE-RLF-Report-Container-for-extended-bands ::= OCTET STRING
-- This IE is a transparent container and shall be encoded as the RLF-Report-v9e0 field contained in the UEInformationResponse message as defined
in TS 36.331 [9]

UESecurityCapabilities ::= SEQUENCE {
    encryptionAlgorithms                EncryptionAlgorithms,
    integrityProtectionAlgorithms       IntegrityProtectionAlgorithms,
    iE-Extensions                       ProtocolExtensionContainer { {UESecurityCapabilities-ExtIEs} }    OPTIONAL,
...
}

UESecurityCapabilities-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

UESidelinkAggregateMaximumBitRate ::= SEQUENCE {
    uESidelinkAggregateMaximumBitRate       BitRate,
    iE-Extensions                   ProtocolExtensionContainer { {UE-Sidelink-Aggregate-MaximumBitRate-ExtIEs} } OPTIONAL,
    ...
}

UE-Sidelink-Aggregate-MaximumBitRate-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

UEsToBeResetList ::= SEQUENCE (SIZE (1.. maxUEsinengNBDU)) OF UEsToBeResetList-Item

UEsToBeResetList-Item::= SEQUENCE {
    meNB-ID                 UE-X2AP-ID,
    meNB-ID-ext             UE-X2AP-ID-Extension                                                OPTIONAL,
    sgNB-ID                 SgNB-UE-X2AP-ID                                                     OPTIONAL,
    iE-Extensions           ProtocolExtensionContainer { {UEsToBeResetList-Item-ExtIEs} }       OPTIONAL,
    ...
}

UEsToBeResetList-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
```

```
    ...
}

ULandDLSharing ::= SEQUENCE{
    uLResourcesULandDLSharing              ULResourcesULandDLSharing,
    dLResourcesULandDLSharing              DLResourcesULandDLSharing,
    iE-Extensions           ProtocolExtensionContainer { {ULandDLSharing-ExtIEs} }             OPTIONAL,
    ...
}

ULandDLSharing-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ULConfiguration::= SEQUENCE {
    uL-PDCP         UL-UE-Configuration,
    iE-Extensions                   ProtocolExtensionContainer { {ULConfiguration-ExtIEs} } OPTIONAL,
    ...
}

ULConfiguration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-UE-Configuration::= ENUMERATED { no-data, shared, only, ... }

UL-GBR-PRB-usage::= INTEGER (0..100)

UL-GBR-PRB-usage-for-MIMO::= INTEGER (0..100)

UL-HighInterferenceIndicationInfo ::= SEQUENCE (SIZE(1..maxCellineNB)) OF UL-HighInterferenceIndicationInfo-Item

UL-HighInterferenceIndicationInfo-Item ::= SEQUENCE {
    target-Cell-ID                  ECGI,
    ul-interferenceindication       UL-HighInterferenceIndication,
    iE-Extensions                   ProtocolExtensionContainer { {UL-HighInterferenceIndicationInfo-Item-ExtIEs} } OPTIONAL,
    ...
}

UL-HighInterferenceIndicationInfo-Item-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

UL-HighInterferenceIndication ::= BIT STRING (SIZE(1..110, ...))

UL-InterferenceOverloadIndication ::= SEQUENCE (SIZE(1..maxnoofPRBs)) OF UL-InterferenceOverloadIndication-Item

UL-InterferenceOverloadIndication-Item ::= ENUMERATED {
    high-interference,
    medium-interference,
    low-interference,
    ...
}
```

```
UL-non-GBR-PRB-usage::= INTEGER (0..100)

UL-non-GBR-PRB-usage-for-MIMO::= INTEGER (0..100)

ULOnlySharing ::= SEQUENCE{
    uLResourceBitmapULOnlySharing   DataTrafficResources,
    iE-Extensions                   ProtocolExtensionContainer { {ULOnlySharing-ExtIEs} }           OPTIONAL,
    ...
}

ULOnlySharing-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

ULResourceBitmapULandDLSharing ::= DataTrafficResources


ULResourcesULandDLSharing ::= CHOICE {
    unchanged           NULL,
    changed             ULResourceBitmapULandDLSharing,
    ...
}

UL-scheduling-PDCCH-CCE-usage::= INTEGER (0..100)

UL-Total-PRB-usage::= INTEGER (0..100)

UL-Total-PRB-usage-for-MIMO::= INTEGER (0..100)

UnlicensedSpectrumRestriction ::= ENUMERATED {
    unlicensed-restricted,
    ...
}

URI-Address ::= VisibleString

UsableABSInformation ::= CHOICE {
    fdd                 UsableABSInformationFDD,
    tdd                 UsableABSInformationTDD,
    ...
}


UsableABSInformationFDD ::= SEQUENCE {
    usable-abs-pattern-info             BIT STRING (SIZE(40)),
    iE-Extensions                       ProtocolExtensionContainer { {UsableABSInformationFDD-ExtIEs} } OPTIONAL,
    ...
}

UsableABSInformationFDD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

UsableABSInformationTDD ::= SEQUENCE {
```

```
    usaable-abs-pattern-info           BIT STRING (SIZE(1..70, ...)),
    iE-Extensions                      ProtocolExtensionContainer { {UsableABSInformationTDD-ExtIEs} } OPTIONAL,
    ...
}

UsableABSInformationTDD-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}


UserPlaneTrafficActivityReport ::= ENUMERATED {inactive, re-activated, ...}

-- V

V2XServicesAuthorized ::= SEQUENCE {
    vehicleUE           VehicleUE                                                   OPTIONAL,
pedestrianUE            PedestrianUE                                                OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {V2XServicesAuthorized-ExtIEs} }   OPTIONAL,
    ...
}

V2XServicesAuthorized-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

VehicleUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

PedestrianUE ::= ENUMERATED {
    authorized,
    not-authorized,
    ...
}

-- W

WidebandCQI ::= SEQUENCE {
    widebandCQICodeword0        INTEGER (0..15, ...),
    widebandCQICodeword1        WidebandCQICodeword1        OPTIONAL,
    iE-Extensions               ProtocolExtensionContainer { {WidebandCQI-ExtIEs} } OPTIONAL,
    ...
}

WidebandCQI-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

WidebandCQICodeword1::= CHOICE {
    four-bitCQI                         INTEGER (0..15, ...),
    three-bitSpatialDifferentialCQI     INTEGER (0..7, ...),
    ...
```

```
}

WLANMeasurementConfiguration ::= SEQUENCE {
    wlanMeasConfig              WLANMeasConfig,
    wlanMeasConfigNameList      WLANMeasConfigNameList              OPTIONAL,
    wlan-rssi                   ENUMERATED {true, ...}              OPTIONAL,
    wlan-rtt                    ENUMERATED {true, ...}              OPTIONAL,
    iE-Extensions       ProtocolExtensionContainer { {WLANMeasurementConfiguration-ExtIEs} } OPTIONAL,
    ...
}

WLANMeasurementConfiguration-ExtIEs X2AP-PROTOCOL-EXTENSION ::= {
    ...
}

WLANMeasConfigNameList ::= SEQUENCE (SIZE(1..maxnoofWLANName)) OF WLANName

WLANMeasConfig::= ENUMERATED {setup,...}

WLANName ::= OCTET STRING (SIZE (1..32))

WTID ::= CHOICE {
    wTID-Type1          WTID-Type1,
    wTID-Type2          WTID-Long-Type2,
    ...
}

WTID-Type1 ::= SEQUENCE {
    pLMN-Identity                   PLMN-Identity,
    shortWTID                       BIT STRING (SIZE(24)),
    ...
}

WTID-Long-Type2 ::= BIT STRING (SIZE(48))

WT-UE-XwAP-ID ::= OCTET STRING (SIZE (3))

-- X

X2BenefitValue ::= INTEGER (1..8, ...)

-- Y
-- Z

END
-- ASN1STOP
```

## 9.3.6 Common definitions

```
-- ASN1START
-- **************************************************************
--
-- Common definitions
```

```
--
-- **************************************************************

X2AP-CommonDataTypes {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
eps-Access (21) modules (3) x2ap (2) version1 (1) x2ap-CommonDataTypes (3) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- Extension constants
--
-- **************************************************************

maxPrivateIEs                                   INTEGER ::= 65535
maxProtocolExtensions                           INTEGER ::= 65535
maxProtocolIEs                                  INTEGER ::= 65535

-- **************************************************************
--
-- Common Data Types
--
-- **************************************************************

Criticality     ::= ENUMERATED { reject, ignore, notify }

Presence        ::= ENUMERATED { optional, conditional, mandatory }

PrivateIE-ID    ::= CHOICE {
    local               INTEGER (0.. maxPrivateIEs),
    global              OBJECT IDENTIFIER
}

ProcedureCode       ::= INTEGER (0..255)


ProtocolIE-ID       ::= INTEGER (0..maxProtocolIEs)


TriggeringMessage   ::= ENUMERATED { initiating-message, successful-outcome, unsuccessful-outcome}

END
-- ASN1STOP
```

## 9.3.7 Constant definitions

```
-- ASN1START
-- **************************************************************
--
-- Constant definitions
```

```
--
-- **************************************************************

X2AP-Constants {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
eps-Access (21) modules (3) x2ap (2) version1 (1) x2ap-Constants (4) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

IMPORTS
    ProcedureCode,
    ProtocolIE-ID
FROM X2AP-CommonDataTypes;

-- **************************************************************
--
-- Elementary Procedures
--
-- **************************************************************

id-handoverPreparation                                  ProcedureCode ::= 0
id-handoverCancel                                       ProcedureCode ::= 1
id-loadIndication                                       ProcedureCode ::= 2
id-errorIndication                                      ProcedureCode ::= 3
id-snStatusTransfer                                     ProcedureCode ::= 4
id-uEContextRelease                                     ProcedureCode ::= 5
id-x2Setup                                              ProcedureCode ::= 6
id-reset                                                ProcedureCode ::= 7
id-eNBConfigurationUpdate                               ProcedureCode ::= 8
id-resourceStatusReportingInitiation                    ProcedureCode ::= 9
id-resourceStatusReporting                              ProcedureCode ::= 10
id-privateMessage                                       ProcedureCode ::= 11
id-mobilitySettingsChange                               ProcedureCode ::= 12
id-rLFIndication                                        ProcedureCode ::= 13
id-handoverReport                                       ProcedureCode ::= 14
id-cellActivation                                       ProcedureCode ::= 15
id-x2Release                                            ProcedureCode ::= 16
id-x2APMessageTransfer                                  ProcedureCode ::= 17
id-x2Removal                                            ProcedureCode ::= 18
id-seNBAdditionPreparation                              ProcedureCode ::= 19
id-seNBReconfigurationCompletion                        ProcedureCode ::= 20
id-meNBinitiatedSeNBModificationPreparation             ProcedureCode ::= 21
id-seNBinitiatedSeNBModification                        ProcedureCode ::= 22
id-meNBinitiatedSeNBRelease                             ProcedureCode ::= 23
id-seNBinitiatedSeNBRelease                             ProcedureCode ::= 24
id-seNBCounterCheck                                     ProcedureCode ::= 25
id-retrieveUEContext                                    ProcedureCode ::= 26
id-sgNBAdditionPreparation                              ProcedureCode ::= 27
id-sgNBReconfigurationCompletion                        ProcedureCode ::= 28
id-meNBinitiatedSgNBModificationPreparation             ProcedureCode ::= 29
id-sgNBinitiatedSgNBModification                        ProcedureCode ::= 30
id-meNBinitiatedSgNBRelease                             ProcedureCode ::= 31
```

```
id-sgNBinitiatedSgNBRelease                                 ProcedureCode ::= 32
id-sgNBCounterCheck                                         ProcedureCode ::= 33
id-sgNBChange                                               ProcedureCode ::= 34
id-rRCTransfer                                              ProcedureCode ::= 35
id-endcX2Setup                                              ProcedureCode ::= 36
id-endcConfigurationUpdate                                  ProcedureCode ::= 37
id-secondaryRATDataUsageReport                              ProcedureCode ::= 38
id-endcCellActivation                                       ProcedureCode ::= 39
id-endcPartialReset                                         ProcedureCode ::= 40
id-eUTRANRCellResourceCoordination                          ProcedureCode ::= 41
id-SgNBActivityNotification                                 ProcedureCode ::= 42
id-endcX2Removal                                            ProcedureCode ::= 43
id-dataForwardingAddressIndication                          ProcedureCode ::= 44
id-gNBStatusIndication                                      ProcedureCode ::= 45
id-deactivateTrace                                          ProcedureCode ::= 46
id-traceStart                                               ProcedureCode ::= 47
id-endcConfigurationTransfer                                ProcedureCode ::= 48
id-handoverSuccess                                          ProcedureCode ::= 49
id-conditionalHandoverCancel                                ProcedureCode ::= 50
id-earlyStatusTransfer                                      ProcedureCode ::= 51
id-cellTrafficTrace                                         ProcedureCode ::= 52
id-endcresourceStatusReporting                              ProcedureCode ::= 53
id-endcresourceStatusReportingInitiation                    ProcedureCode ::= 54
id-f1CTrafficTransfer                                       ProcedureCode ::= 55
id-UERadioCapabilityIDMapping                               ProcedureCode ::= 56
id-accessAndMobilityIndication                              ProcedureCode ::= 57
id-procedure-code-58-not-to-be-used                         ProcedureCode ::= 58
id-CPC-cancel                                               ProcedureCode ::= 59

-- ************************************************************
--
-- Lists
--
-- ************************************************************

maxEARFCN                                   INTEGER ::= 65535
maxEARFCNPlusOne                            INTEGER ::= 65536
newmaxEARFCN                                INTEGER ::= 262143
maxInterfaces                               INTEGER ::= 16
maxCellineNB                                INTEGER ::= 256
maxnoofBands                                INTEGER ::= 16
maxnoofBearers                              INTEGER ::= 256
maxNrOfErrors                               INTEGER ::= 256
maxnoofPDCP-SN                              INTEGER ::= 16
maxnoofEPLMNs                               INTEGER ::= 15
maxnoofEPLMNsPlusOne                        INTEGER ::= 16
maxnoofForbLACs                             INTEGER ::= 4096
maxnoofForbTACs                             INTEGER ::= 4096
maxnoofBPLMNs                               INTEGER ::= 6
maxnoofAdditionalPLMNs                      INTEGER ::= 6
maxnoofNeighbours                           INTEGER ::= 512
maxnoofPRBs                                 INTEGER ::= 110
maxPools                                    INTEGER ::= 16
maxnoofCells                                INTEGER ::= 16
```

```
maxnoofMBSFN                                INTEGER ::= 8
maxFailedMeasObjects                        INTEGER ::= 32
maxnoofCellIDforMDT                         INTEGER ::= 32
maxnoofTAforMDT                             INTEGER ::= 8
maxnoofMBMSServiceAreaIdentities            INTEGER ::= 256
maxnoofMDTPLMNs                             INTEGER ::= 16
maxnoofCoMPHypothesisSet                    INTEGER ::= 256
maxnoofCoMPCells                            INTEGER ::= 32
maxUEReport                                 INTEGER ::= 128
maxCellReport                               INTEGER ::= 9
maxnoofPA                                   INTEGER ::= 3
maxCSIProcess                               INTEGER ::= 4
maxCSIReport                                INTEGER ::= 2
maxSubband                                  INTEGER ::= 14
maxofNRNeighbours                           INTEGER ::= 1024
maxCellinengNB                              INTEGER ::= 16384
--  maxnoofNRCarriers                       INTEGER ::= 32
maxnooftimeperiods                          INTEGER ::= 2
maxnoofCellIDforQMC                         INTEGER ::= 32
maxnoofTAforQMC                             INTEGER ::= 8
maxnoofPLMNforQMC                           INTEGER ::= 16
maxUEsinengNBDU                             INTEGER ::= 8192
maxnoofProtectedResourcePatterns            INTEGER ::= 16
maxnoNRcellsSpectrumSharingWithE-UTRA       INTEGER ::= 64
maxnoofNrCellBands                          INTEGER ::= 32
maxnoofBluetoothName                        INTEGER ::= 4
maxnoofWLANName                             INTEGER ::= 4
maxnoofextBPLMNs                            INTEGER ::= 12
maxnoofTLAs                                 INTEGER ::= 16
maxnoofGTPTLAs                              INTEGER ::= 16
maxnoofTNLAssociations                      INTEGER ::= 32
maxnoofCellsinCHO                           INTEGER ::= 8
maxnoofPC5QoSFlows                          INTEGER ::= 2048
maxnoofSSBAreas                             INTEGER ::= 64
maxnoofNRSCSs                               INTEGER ::= 5
maxnoofNRPhysicalResourceBlocks             INTEGER ::= 275
maxnoofNonAnchorCarrierFreqConfig           INTEGER ::= 15
maxnoofRACHReports                          INTEGER ::= 64
maxnoofPSCellsPerSN                         INTEGER ::= 8
maxnoofPSCellsPerPrimaryCellinUEHistoryInfo INTEGER ::= 8
maxnoofReportedNRCellsPossiblyAggregated    INTEGER ::= 16
maxnoofPSCellCandidates                     INTEGER ::= 8
maxnoofTargetSgNBs                          INTEGER ::= 8
maxnoofMTCItems                             INTEGER ::= 16
maxnoofCSIRSconfigurations                  INTEGER ::= 96
maxnoofCSIRSneighbourCells                  INTEGER ::= 16
maxnoofCSIRSneighbourCellsInMTC             INTEGER ::= 16
maxnoofSensorName                           INTEGER ::= 3
maxnoofTargetSgNBsMinusOne                  INTEGER ::= 7

-- ************************************************************
--
-- IEs
--
```

```
-- **************************************************************

id-E-RABs-Admitted-Item                                    ProtocolIE-ID ::= 0
id-E-RABs-Admitted-List                                    ProtocolIE-ID ::= 1
id-E-RAB-Item                                              ProtocolIE-ID ::= 2
id-E-RABs-NotAdmitted-List                                 ProtocolIE-ID ::= 3
id-E-RABs-ToBeSetup-Item                                   ProtocolIE-ID ::= 4
id-Cause                                                   ProtocolIE-ID ::= 5
id-CellInformation                                         ProtocolIE-ID ::= 6
id-CellInformation-Item                                    ProtocolIE-ID ::= 7
id-New-eNB-UE-X2AP-ID                                      ProtocolIE-ID ::= 9
id-Old-eNB-UE-X2AP-ID                                      ProtocolIE-ID ::= 10
id-TargetCell-ID                                           ProtocolIE-ID ::= 11
id-TargeteNBtoSource-eNBTransparentContainer               ProtocolIE-ID ::= 12
id-TraceActivation                                         ProtocolIE-ID ::= 13
id-UE-ContextInformation                                   ProtocolIE-ID ::= 14
id-UE-HistoryInformation                                   ProtocolIE-ID ::= 15
id-UE-X2AP-ID                                              ProtocolIE-ID ::= 16
id-CriticalityDiagnostics                                  ProtocolIE-ID ::= 17
id-E-RABs-SubjectToStatusTransfer-List                     ProtocolIE-ID ::= 18
id-E-RABs-SubjectToStatusTransfer-Item                     ProtocolIE-ID ::= 19
id-ServedCells                                             ProtocolIE-ID ::= 20
id-GlobalENB-ID                                            ProtocolIE-ID ::= 21
id-TimeToWait                                              ProtocolIE-ID ::= 22
id-GUMMEI-ID                                               ProtocolIE-ID ::= 23
id-GUGroupIDList                                           ProtocolIE-ID ::= 24
id-ServedCellsToAdd                                        ProtocolIE-ID ::= 25
id-ServedCellsToModify                                     ProtocolIE-ID ::= 26
id-ServedCellsToDelete                                     ProtocolIE-ID ::= 27
id-Registration-Request                                    ProtocolIE-ID ::= 28
id-CellToReport                                            ProtocolIE-ID ::= 29
id-ReportingPeriodicity                                    ProtocolIE-ID ::= 30
id-CellToReport-Item                                       ProtocolIE-ID ::= 31
id-CellMeasurementResult                                   ProtocolIE-ID ::= 32
id-CellMeasurementResult-Item                              ProtocolIE-ID ::= 33
id-GUGroupIDToAddList                                      ProtocolIE-ID ::= 34
id-GUGroupIDToDeleteList                                   ProtocolIE-ID ::= 35
id-SRVCCOperationPossible                                  ProtocolIE-ID ::= 36
id-Measurement-ID                                          ProtocolIE-ID ::= 37
id-ReportCharacteristics                                   ProtocolIE-ID ::= 38
id-ENB1-Measurement-ID                                     ProtocolIE-ID ::= 39
id-ENB2-Measurement-ID                                     ProtocolIE-ID ::= 40
id-Number-of-Antennaports                                  ProtocolIE-ID ::= 41
id-CompositeAvailableCapacityGroup                         ProtocolIE-ID ::= 42
id-ENB1-Cell-ID                                            ProtocolIE-ID ::= 43
id-ENB2-Cell-ID                                            ProtocolIE-ID ::= 44
id-ENB2-Proposed-Mobility-Parameters                       ProtocolIE-ID ::= 45
id-ENB1-Mobility-Parameters                                ProtocolIE-ID ::= 46
id-ENB2-Mobility-Parameters-Modification-Range             ProtocolIE-ID ::= 47
id-FailureCellPCI                                          ProtocolIE-ID ::= 48
id-Re-establishmentCellECGI                                ProtocolIE-ID ::= 49
id-FailureCellCRNTI                                        ProtocolIE-ID ::= 50
id-ShortMAC-I                                              ProtocolIE-ID ::= 51
id-SourceCellECGI                                          ProtocolIE-ID ::= 52
```

```
id-FailureCellECGI                                  ProtocolIE-ID ::= 53
id-HandoverReportType                               ProtocolIE-ID ::= 54
id-PRACH-Configuration                              ProtocolIE-ID ::= 55
id-MBSFN-Subframe-Info                              ProtocolIE-ID ::= 56
id-ServedCellsToActivate                            ProtocolIE-ID ::= 57
id-ActivatedCellList                                ProtocolIE-ID ::= 58
id-DeactivationIndication                           ProtocolIE-ID ::= 59
id-UE-RLF-Report-Container                          ProtocolIE-ID ::= 60
id-ABSInformation                                   ProtocolIE-ID ::= 61
id-InvokeIndication                                 ProtocolIE-ID ::= 62
id-ABS-Status                                       ProtocolIE-ID ::= 63
id-PartialSuccessIndicator                          ProtocolIE-ID ::= 64
id-MeasurementInitiationResult-List                 ProtocolIE-ID ::= 65
id-MeasurementInitiationResult-Item                 ProtocolIE-ID ::= 66
id-MeasurementFailureCause-Item                     ProtocolIE-ID ::= 67
id-CompleteFailureCauseInformation-List             ProtocolIE-ID ::= 68
id-CompleteFailureCauseInformation-Item             ProtocolIE-ID ::= 69
id-CSG-Id                                           ProtocolIE-ID ::= 70
id-CSGMembershipStatus                              ProtocolIE-ID ::= 71
id-MDTConfiguration                                 ProtocolIE-ID ::= 72
id-ManagementBasedMDTallowed                        ProtocolIE-ID ::= 74
id-RRCConnSetupIndicator                            ProtocolIE-ID ::= 75
id-NeighbourTAC                                     ProtocolIE-ID ::= 76
id-Time-UE-StayedInCell-EnhancedGranularity         ProtocolIE-ID ::= 77
id-RRCConnReestabIndicator                          ProtocolIE-ID ::= 78
id-MBMS-Service-Area-List                           ProtocolIE-ID ::= 79
id-HO-cause                                         ProtocolIE-ID ::= 80
id-TargetCellInUTRAN                                ProtocolIE-ID ::= 81
id-MobilityInformation                              ProtocolIE-ID ::= 82
id-SourceCellCRNTI                                  ProtocolIE-ID ::= 83
id-MultibandInfoList                                ProtocolIE-ID ::= 84
id-M3Configuration                                  ProtocolIE-ID ::= 85
id-M4Configuration                                  ProtocolIE-ID ::= 86
id-M5Configuration                                  ProtocolIE-ID ::= 87
id-MDT-Location-Info                                ProtocolIE-ID ::= 88
id-ManagementBasedMDTPLMNList                       ProtocolIE-ID ::= 89
id-SignallingBasedMDTPLMNList                       ProtocolIE-ID ::= 90
id-ReceiveStatusOfULPDCPSDUsExtended                ProtocolIE-ID ::= 91
id-ULCOUNTValueExtended                             ProtocolIE-ID ::= 92
id-DLCOUNTValueExtended                             ProtocolIE-ID ::= 93
id-eARFCNExtension                                  ProtocolIE-ID ::= 94
id-UL-EARFCNExtension                               ProtocolIE-ID ::= 95
id-DL-EARFCNExtension                               ProtocolIE-ID ::= 96
id-AdditionalSpecialSubframe-Info                   ProtocolIE-ID ::= 97
id-Masked-IMEISV                                    ProtocolIE-ID ::= 98
id-IntendedULDLConfiguration                        ProtocolIE-ID ::= 99
id-ExtendedULInterferenceOverloadInfo               ProtocolIE-ID ::= 100
id-RNL-Header                                       ProtocolIE-ID ::= 101
id-x2APMessage                                      ProtocolIE-ID ::= 102
id-ProSeAuthorized                                  ProtocolIE-ID ::= 103
id-ExpectedUEBehaviour                              ProtocolIE-ID ::= 104
id-UE-HistoryInformationFromTheUE                   ProtocolIE-ID ::= 105
id-DynamicDLTransmissionInformation                 ProtocolIE-ID ::= 106
id-UE-RLF-Report-Container-for-extended-bands       ProtocolIE-ID ::= 107
```

```
id-CoMPInformation                                      ProtocolIE-ID ::= 108
id-ReportingPeriodicityRSRPMR                           ProtocolIE-ID ::= 109
id-RSRPMRList                                           ProtocolIE-ID ::= 110
id-MeNB-UE-X2AP-ID                                      ProtocolIE-ID ::= 111
id-SeNB-UE-X2AP-ID                                      ProtocolIE-ID ::= 112
id-UE-SecurityCapabilities                              ProtocolIE-ID ::= 113
id-SeNBSecurityKey                                      ProtocolIE-ID ::= 114
id-SeNBUEAggregateMaximumBitRate                        ProtocolIE-ID ::= 115
id-ServingPLMN                                          ProtocolIE-ID ::= 116
id-E-RABs-ToBeAdded-List                                ProtocolIE-ID ::= 117
id-E-RABs-ToBeAdded-Item                                ProtocolIE-ID ::= 118
id-MeNBtoSeNBContainer                                  ProtocolIE-ID ::= 119
id-E-RABs-Admitted-ToBeAdded-List                       ProtocolIE-ID ::= 120
id-E-RABs-Admitted-ToBeAdded-Item                       ProtocolIE-ID ::= 121
id-SeNBtoMeNBContainer                                  ProtocolIE-ID ::= 122
id-ResponseInformationSeNBReconfComp                    ProtocolIE-ID ::= 123
id-UE-ContextInformationSeNBModReq                      ProtocolIE-ID ::= 124
id-E-RABs-ToBeAdded-ModReqItem                          ProtocolIE-ID ::= 125
id-E-RABs-ToBeModified-ModReqItem                       ProtocolIE-ID ::= 126
id-E-RABs-ToBeReleased-ModReqItem                       ProtocolIE-ID ::= 127
id-E-RABs-Admitted-ToBeAdded-ModAckList                 ProtocolIE-ID ::= 128
id-E-RABs-Admitted-ToBeModified-ModAckList              ProtocolIE-ID ::= 129
id-E-RABs-Admitted-ToBeReleased-ModAckList              ProtocolIE-ID ::= 130
id-E-RABs-Admitted-ToBeAdded-ModAckItem                 ProtocolIE-ID ::= 131
id-E-RABs-Admitted-ToBeModified-ModAckItem              ProtocolIE-ID ::= 132
id-E-RABs-Admitted-ToBeReleased-ModAckItem              ProtocolIE-ID ::= 133
id-E-RABs-ToBeReleased-ModReqd                          ProtocolIE-ID ::= 134
id-E-RABs-ToBeReleased-ModReqdItem                      ProtocolIE-ID ::= 135
id-SCGChangeIndication                                  ProtocolIE-ID ::= 136
id-E-RABs-ToBeReleased-List-RelReq                      ProtocolIE-ID ::= 137
id-E-RABs-ToBeReleased-RelReqItem                       ProtocolIE-ID ::= 138
id-E-RABs-ToBeReleased-List-RelConf                     ProtocolIE-ID ::= 139
id-E-RABs-ToBeReleased-RelConfItem                      ProtocolIE-ID ::= 140
id-E-RABs-SubjectToCounterCheck-List                    ProtocolIE-ID ::= 141
id-E-RABs-SubjectToCounterCheckItem                     ProtocolIE-ID ::= 142
id-CoverageModificationList                             ProtocolIE-ID ::= 143
id-ReportingPeriodicityCSIR                             ProtocolIE-ID ::= 145
id-CSIReportList                                        ProtocolIE-ID ::= 146
id-UEID                                                 ProtocolIE-ID ::= 147
id-enhancedRNTP                                         ProtocolIE-ID ::= 148
id-ProSeUEtoNetworkRelaying                             ProtocolIE-ID ::= 149
id-ReceiveStatusOfULPDCPSDUsPDCP-SNlength18             ProtocolIE-ID ::= 150
id-ULCOUNTValuePDCP-SNlength18                          ProtocolIE-ID ::= 151
id-DLCOUNTValuePDCP-SNlength18                          ProtocolIE-ID ::= 152
id-UE-ContextReferenceAtSeNB                            ProtocolIE-ID ::= 153
id-UE-ContextKeptIndicator                              ProtocolIE-ID ::= 154
id-New-eNB-UE-X2AP-ID-Extension                         ProtocolIE-ID ::= 155
id-Old-eNB-UE-X2AP-ID-Extension                         ProtocolIE-ID ::= 156
id-MeNB-UE-X2AP-ID-Extension                            ProtocolIE-ID ::= 157
id-SeNB-UE-X2AP-ID-Extension                            ProtocolIE-ID ::= 158
id-LHN-ID                                               ProtocolIE-ID ::= 159
id-FreqBandIndicatorPriority                            ProtocolIE-ID ::= 160
id-M6Configuration                                      ProtocolIE-ID ::= 161
id-M7Configuration                                      ProtocolIE-ID ::= 162
```

```
id-Tunnel-Information-for-BBF                                ProtocolIE-ID ::= 163
id-SIPTO-BearerDeactivationIndication                        ProtocolIE-ID ::= 164
id-GW-TransportLayerAddress                                  ProtocolIE-ID ::= 165
id-Correlation-ID                                            ProtocolIE-ID ::= 166
id-SIPTO-Correlation-ID                                      ProtocolIE-ID ::= 167
id-SIPTO-L-GW-TransportLayerAddress                          ProtocolIE-ID ::= 168
id-X2RemovalThreshold                                        ProtocolIE-ID ::= 169
id-CellReportingIndicator                                    ProtocolIE-ID ::= 170
id-BearerType                                                ProtocolIE-ID ::= 171
id-resumeID                                                  ProtocolIE-ID ::= 172
id-UE-ContextInformationRetrieve                             ProtocolIE-ID ::= 173
id-E-RABs-ToBeSetupRetrieve-Item                             ProtocolIE-ID ::= 174
id-NewEUTRANCellIdentifier                                   ProtocolIE-ID ::= 175
id-V2XServicesAuthorized                                     ProtocolIE-ID ::= 176
id-OffsetOfNbiotChannelNumberToDL-EARFCN                     ProtocolIE-ID ::= 177
id-OffsetOfNbiotChannelNumberToUL-EARFCN                     ProtocolIE-ID ::= 178
id-AdditionalSpecialSubframeExtension-Info                   ProtocolIE-ID ::= 179
id-BandwidthReducedSI                                        ProtocolIE-ID ::= 180
id-MakeBeforeBreakIndicator                                  ProtocolIE-ID ::= 181
id-UE-ContextReferenceAtWT                                   ProtocolIE-ID ::= 182
id-WT-UE-ContextKeptIndicator                                ProtocolIE-ID ::= 183
id-UESidelinkAggregateMaximumBitRate                         ProtocolIE-ID ::= 184
id-uL-GTPtunnelEndpoint                                      ProtocolIE-ID ::= 185
id-DL-scheduling-PDCCH-CCE-usage                             ProtocolIE-ID ::= 193
id-UL-scheduling-PDCCH-CCE-usage                             ProtocolIE-ID ::= 194
id-UEAppLayerMeasConfig                                      ProtocolIE-ID ::= 195
id-extended-e-RAB-MaximumBitrateDL                           ProtocolIE-ID ::= 196
id-extended-e-RAB-MaximumBitrateUL                           ProtocolIE-ID ::= 197
id-extended-e-RAB-GuaranteedBitrateDL                        ProtocolIE-ID ::= 198
id-extended-e-RAB-GuaranteedBitrateUL                        ProtocolIE-ID ::= 199
id-extended-uEaggregateMaximumBitRateDownlink                ProtocolIE-ID ::= 200
id-extended-uEaggregateMaximumBitRateUplink                  ProtocolIE-ID ::= 201
id-NRrestrictioninEPSasSecondaryRAT                          ProtocolIE-ID ::= 202
id-SgNBSecurityKey                                           ProtocolIE-ID ::= 203
id-SgNBUEAggregateMaximumBitRate                             ProtocolIE-ID ::= 204
id-E-RABs-ToBeAdded-SgNBAddReqList                           ProtocolIE-ID ::= 205
id-MeNBtoSgNBContainer                                       ProtocolIE-ID ::= 206
id-SgNB-UE-X2AP-ID                                           ProtocolIE-ID ::= 207
id-RequestedSplitSRBs                                        ProtocolIE-ID ::= 208
id-E-RABs-ToBeAdded-SgNBAddReq-Item                          ProtocolIE-ID ::= 209
id-E-RABs-Admitted-ToBeAdded-SgNBAddReqAckList               ProtocolIE-ID ::= 210
id-SgNBtoMeNBContainer                                       ProtocolIE-ID ::= 211
id-AdmittedSplitSRBs                                         ProtocolIE-ID ::= 212
id-E-RABs-Admitted-ToBeAdded-SgNBAddReqAck-Item              ProtocolIE-ID ::= 213
id-ResponseInformationSgNBReconfComp                         ProtocolIE-ID ::= 214
id-UE-ContextInformation-SgNBModReq                          ProtocolIE-ID ::= 215
id-E-RABs-ToBeAdded-SgNBModReq-Item                          ProtocolIE-ID ::= 216
id-E-RABs-ToBeModified-SgNBModReq-Item                       ProtocolIE-ID ::= 217
id-E-RABs-ToBeReleased-SgNBModReq-Item                       ProtocolIE-ID ::= 218
id-E-RABs-Admitted-ToBeAdded-SgNBModAckList                  ProtocolIE-ID ::= 219
id-E-RABs-Admitted-ToBeModified-SgNBModAckList               ProtocolIE-ID ::= 220
id-E-RABs-Admitted-ToBeReleased-SgNBModAckList               ProtocolIE-ID ::= 221
id-E-RABs-Admitted-ToBeAdded-SgNBModAck-Item                 ProtocolIE-ID ::= 222
id-E-RABs-Admitted-ToBeModified-SgNBModAck-Item              ProtocolIE-ID ::= 223
```

```
id-E-RABs-Admitted-ToBeReleased-SgNBModAck-Item                ProtocolIE-ID ::= 224
id-E-RABs-ToBeReleased-SgNBModReqdList                         ProtocolIE-ID ::= 225
id-E-RABs-ToBeModified-SgNBModReqdList                         ProtocolIE-ID ::= 226
id-E-RABs-ToBeReleased-SgNBModReqd-Item                        ProtocolIE-ID ::= 227
id-E-RABs-ToBeModified-SgNBModReqd-Item                        ProtocolIE-ID ::= 228
id-E-RABs-ToBeReleased-SgNBChaConfList                         ProtocolIE-ID ::= 229
id-E-RABs-ToBeReleased-SgNBChaConf-Item                        ProtocolIE-ID ::= 230
id-E-RABs-ToBeReleased-SgNBRelReqList                          ProtocolIE-ID ::= 231
id-E-RABs-ToBeReleased-SgNBRelReq-Item                         ProtocolIE-ID ::= 232
id-E-RABs-ToBeReleased-SgNBRelConfList                         ProtocolIE-ID ::= 233
id-E-RABs-ToBeReleased-SgNBRelConf-Item                        ProtocolIE-ID ::= 234
id-E-RABs-SubjectToSgNBCounterCheck-List                       ProtocolIE-ID ::= 235
id-E-RABs-SubjectToSgNBCounterCheck-Item                       ProtocolIE-ID ::= 236
id-RRCContainer                                                ProtocolIE-ID ::= 237
id-SRBType                                                     ProtocolIE-ID ::= 238
id-Target-SgNB-ID                                              ProtocolIE-ID ::= 239
id-HandoverRestrictionList                                     ProtocolIE-ID ::= 240
id-SCGConfigurationQuery                                       ProtocolIE-ID ::= 241
id-SplitSRB                                                    ProtocolIE-ID ::= 242
id-NRUeReport                                                  ProtocolIE-ID ::= 243
id-InitiatingNodeType-EndcX2Setup                              ProtocolIE-ID ::= 244
id-InitiatingNodeType-EndcConfigUpdate                         ProtocolIE-ID ::= 245
id-RespondingNodeType-EndcX2Setup                              ProtocolIE-ID ::= 246
id-RespondingNodeType-EndcConfigUpdate                         ProtocolIE-ID ::= 247
id-NRUESecurityCapabilities                                    ProtocolIE-ID ::= 248
id-PDCPChangeIndication                                        ProtocolIE-ID ::= 249
id-ServedEUTRAcellsENDCX2ManagementList                        ProtocolIE-ID ::= 250
id-CellAssistanceInformation                                   ProtocolIE-ID ::= 251
id-Globalen-gNB-ID                                             ProtocolIE-ID ::= 252
id-ServedNRcellsENDCX2ManagementList                           ProtocolIE-ID ::= 253
id-UE-ContextReferenceAtSgNB                                       ProtocolIE-ID ::= 254
id-SecondaryRATUsageReport                                     ProtocolIE-ID ::= 255
id-ActivationID                                                ProtocolIE-ID ::= 256
id-MeNBResourceCoordinationInformation                         ProtocolIE-ID ::= 257
id-SgNBResourceCoordinationInformation                         ProtocolIE-ID ::= 258
id-ServedEUTRAcellsToModifyListENDCConfUpd                     ProtocolIE-ID ::= 259
id-ServedEUTRAcellsToDeleteListENDCConfUpd                     ProtocolIE-ID ::= 260
id-ServedNRcellsToModifyListENDCConfUpd                        ProtocolIE-ID ::= 261
id-ServedNRcellsToDeleteListENDCConfUpd                        ProtocolIE-ID ::= 262
id-E-RABUsageReport-Item                                       ProtocolIE-ID ::= 263
id-Old-SgNB-UE-X2AP-ID                                         ProtocolIE-ID ::= 264
id-SecondaryRATUsageReportList                                 ProtocolIE-ID ::= 265
id-SecondaryRATUsageReport-Item                                ProtocolIE-ID ::= 266
id-ServedNRCellsToActivate                                     ProtocolIE-ID ::= 267
id-ActivatedNRCellList                                         ProtocolIE-ID ::= 268
id-SelectedPLMN                                                ProtocolIE-ID ::= 269
id-UEs-ToBeReset                                               ProtocolIE-ID ::= 270
id-UEs-Admitted-ToBeReset                                      ProtocolIE-ID ::= 271
id-RRCConfigIndication                                         ProtocolIE-ID ::= 272
id-DownlinkPacketLossRate                                      ProtocolIE-ID ::= 273
id-UplinkPacketLossRate                                        ProtocolIE-ID ::= 274
id-SubscriberProfileIDforRFP                                   ProtocolIE-ID ::= 275
id-serviceType                                                 ProtocolIE-ID ::= 276
id-AerialUEsubscriptionInformation                             ProtocolIE-ID ::= 277
```

```
id-SGNB-Addition-Trigger-Ind                                    ProtocolIE-ID ::= 278
id-MeNBCell-ID                                                  ProtocolIE-ID ::= 279
id-RequestedSplitSRBsrelease                                    ProtocolIE-ID ::= 280
id-AdmittedSplitSRBsrelease                                     ProtocolIE-ID ::= 281
id-NRS-NSSS-PowerOffset                                         ProtocolIE-ID ::= 282
id-NSSS-NumOccasionDifferentPrecoder                            ProtocolIE-ID ::= 283
id-ProtectedEUTRAResourceIndication                             ProtocolIE-ID ::= 284
id-InitiatingNodeType-EutranrCellResourceCoordination           ProtocolIE-ID ::= 285
id-RespondingNodeType-EutranrCellResourceCoordination           ProtocolIE-ID ::= 286
id-DataTrafficResourceIndication                                ProtocolIE-ID ::= 287
id-SpectrumSharingGroupID                                       ProtocolIE-ID ::= 288
id-ListofEUTRACellsinEUTRACoordinationReq                       ProtocolIE-ID ::= 289
id-ListofEUTRACellsinEUTRACoordinationResp                      ProtocolIE-ID ::= 290
id-ListofEUTRACellsinNRCoordinationReq                          ProtocolIE-ID ::= 291
id-ListofNRCellsinNRCoordinationReq                             ProtocolIE-ID ::= 292
id-ListofNRCellsinNRCoordinationResp                            ProtocolIE-ID ::= 293
id-E-RABs-AdmittedToBeModified-SgNBModConfList                  ProtocolIE-ID ::= 294
id-E-RABs-AdmittedToBeModified-SgNBModConf-Item                 ProtocolIE-ID ::= 295
id-UEContextLevelUserPlaneActivity                              ProtocolIE-ID ::= 296
id-ERABActivityNotifyItemList                                   ProtocolIE-ID ::= 297
id-InitiatingNodeType-EndcX2Removal                             ProtocolIE-ID ::= 298
id-RespondingNodeType-EndcX2Removal                             ProtocolIE-ID ::= 299
id-RLC-Status                                                   ProtocolIE-ID ::= 300
id-CNTypeRestrictions                                           ProtocolIE-ID ::= 301
id-uLpDCPSnLength                                               ProtocolIE-ID ::= 302
id-BluetoothMeasurementConfiguration                            ProtocolIE-ID ::= 303
id-WLANMeasurementConfiguration                                 ProtocolIE-ID ::= 304
id-NRrestrictionin5GS                                           ProtocolIE-ID ::= 305
id-dL-Forwarding                                                ProtocolIE-ID ::= 306
id-E-RABs-DataForwardingAddress-List                            ProtocolIE-ID ::= 307
id-E-RABs-DataForwardingAddress-Item                            ProtocolIE-ID ::= 308
id-Subscription-Based-UE-DifferentiationInfo                    ProtocolIE-ID ::= 309
id-GNBOverloadInformation                                       ProtocolIE-ID ::= 310
id-dLPDCPSnLength                                               ProtocolIE-ID ::= 311
id-secondarysgNBDLGTPTEIDatPDCP                                 ProtocolIE-ID ::= 312
id-secondarymeNBULGTPTEIDatPDCP                                 ProtocolIE-ID ::= 313
id-lCID                                                         ProtocolIE-ID ::= 314
id-duplicationActivation                                        ProtocolIE-ID ::= 315
id-ECGI                                                         ProtocolIE-ID ::= 316
id-RLCMode-transferred                                          ProtocolIE-ID ::= 317
id-E-RABs-Admitted-ToBeReleased-SgNBRelReqAckList               ProtocolIE-ID ::= 318
id-E-RABs-Admitted-ToBeReleased-SgNBRelReqAck-Item              ProtocolIE-ID ::= 319
id-E-RABs-ToBeReleased-SgNBRelReqdList                          ProtocolIE-ID ::= 320
id-E-RABs-ToBeReleased-SgNBRelReqd-Item                         ProtocolIE-ID ::= 321
id-NRCGI                                                        ProtocolIE-ID ::= 322
id-MeNBCoordinationAssistanceInformation                        ProtocolIE-ID ::= 323
id-SgNBCoordinationAssistanceInformation                        ProtocolIE-ID ::= 324
id-new-drb-ID-req                                               ProtocolIE-ID ::= 325
id-endcSONConfigurationTransfer                                 ProtocolIE-ID ::= 326
id-NRNeighbourInfoToAdd                                         ProtocolIE-ID ::= 327
id-NRNeighbourInfoToModify                                      ProtocolIE-ID ::= 328
id-DesiredActNotificationLevel                                  ProtocolIE-ID ::= 329
id-LocationInformationSgNBReporting                             ProtocolIE-ID ::= 330
id-LocationInformationSgNB                                      ProtocolIE-ID ::= 331
```

```
id-LastNG-RANPLMNIdentity                                ProtocolIE-ID ::= 332
id-EUTRANTraceID                                         ProtocolIE-ID ::= 333
id-additionalPLMNs-Item                                  ProtocolIE-ID ::= 334
id-InterfaceInstanceIndication                           ProtocolIE-ID ::= 335
id-BPLMN-ID-Info-EUTRA                                   ProtocolIE-ID ::= 336
id-BPLMN-ID-Info-NR                                      ProtocolIE-ID ::= 337
id-NBIoT-UL-DL-AlignmentOffset                           ProtocolIE-ID ::= 338
id-ERABs-transferred-to-MeNB                             ProtocolIE-ID ::= 339
id-AdditionalRRMPriorityIndex                            ProtocolIE-ID ::= 340
id-LowerLayerPresenceStatusChange                        ProtocolIE-ID ::= 341
id-FastMCGRecovery-SN-to-MN                              ProtocolIE-ID ::= 342
id-RequestedFastMCGRecoveryViaSRB3                       ProtocolIE-ID ::= 343
id-AvailableFastMCGRecoveryViaSRB3                       ProtocolIE-ID ::= 344
id-RequestedFastMCGRecoveryViaSRB3Release                ProtocolIE-ID ::= 345
id-ReleaseFastMCGRecoveryViaSRB3                         ProtocolIE-ID ::= 346
id-FastMCGRecovery-MN-to-SN                              ProtocolIE-ID ::= 347
id-PartialListIndicator                                  ProtocolIE-ID ::= 348
id-MaximumCellListSize                                   ProtocolIE-ID ::= 349
id-MessageOversizeNotification                           ProtocolIE-ID ::= 350
id-CellandCapacityAssistInfo                             ProtocolIE-ID ::= 351
id-TNLConfigurationInfo                                  ProtocolIE-ID ::= 352
id-TNLA-To-Add-List                                      ProtocolIE-ID ::= 353
id-TNLA-To-Update-List                                   ProtocolIE-ID ::= 354
id-TNLA-To-Remove-List                                   ProtocolIE-ID ::= 355
id-TNLA-Setup-List                                       ProtocolIE-ID ::= 356
id-TNLA-Failed-To-Setup-List                             ProtocolIE-ID ::= 357
id-UnlicensedSpectrumRestriction                             ProtocolIE-ID ::= 358
id-UEContextReferenceatSourceNGRAN                       ProtocolIE-ID ::= 359
id-EPCHandoverRestrictionListContainer                   ProtocolIE-ID ::= 360
id-CHOinformation-REQ                                    ProtocolIE-ID ::= 361
id-CHOinformation-ACK                                    ProtocolIE-ID ::= 362
id-DAPSRequestInfo                                       ProtocolIE-ID ::= 363
id-RequestedTargetCellID                                 ProtocolIE-ID ::= 364
id-CandidateCellsToBeCancelledList                       ProtocolIE-ID ::= 365
id-DAPSResponseInfo                                      ProtocolIE-ID ::= 366
id-ProcedureStage                                        ProtocolIE-ID ::= 367
id-CHO-DC-Indicator                                      ProtocolIE-ID ::= 368
id-Ethernet-Type                                         ProtocolIE-ID ::= 369
id-NRV2XServicesAuthorized                               ProtocolIE-ID ::= 370
id-NRUESidelinkAggregateMaximumBitRate                   ProtocolIE-ID ::= 371
id-PC5QoSParameters                                      ProtocolIE-ID ::= 372
id-NPRACHConfiguration                                   ProtocolIE-ID ::= 373
id-NBIoT-RLF-Report-Container                            ProtocolIE-ID ::= 374
id-MDTConfigurationNR                                    ProtocolIE-ID ::= 375
id-PrivacyIndicator                                      ProtocolIE-ID ::= 376
id-TraceCollectionEntityIPAddress                        ProtocolIE-ID ::= 377
id-UERadioCapabilityID                                   ProtocolIE-ID ::= 378
id-SNtriggered                                           ProtocolIE-ID ::= 379
id-CSI-RSTransmissionIndication                          ProtocolIE-ID ::= 380
id-DLCarrierList                                         ProtocolIE-ID ::= 381
id-TargetCellInNGRAN                                     ProtocolIE-ID ::= 382
id-E-UTRAN-Node1-Measurement-ID                          ProtocolIE-ID ::= 383
id-E-UTRAN-Node2-Measurement-ID                          ProtocolIE-ID ::= 384
id-TDDULDLConfigurationCommonNR                          ProtocolIE-ID ::= 385
```

```
id-CarrierList                                                  ProtocolIE-ID ::= 386
id-ULCarrierList                                                ProtocolIE-ID ::= 387
id-FrequencyShift7p5khz                                         ProtocolIE-ID ::= 388
id-SSB-PositionsInBurst                                         ProtocolIE-ID ::= 389
id-NRCellPRACHConfig                                            ProtocolIE-ID ::= 390
id-CellToReport-NR-ENDC                                         ProtocolIE-ID ::= 391
id-CellToReport-NR-ENDC-Item                                    ProtocolIE-ID ::= 392
id-CellMeasurementResult-NR-ENDC                                ProtocolIE-ID ::= 393
id-CellMeasurementResult-NR-ENDC-Item                           ProtocolIE-ID ::= 394
id-IABNodeIndication                                            ProtocolIE-ID ::= 395
id-QoS-Mapping-Information                                      ProtocolIE-ID ::= 396
id-F1CTrafficContainer                                          ProtocolIE-ID ::= 397
id-IntendedTDD-DL-ULConfiguration-NR                            ProtocolIE-ID ::= 399
id-UERadioCapability                                            ProtocolIE-ID ::= 400
id-CellMeasurementResult-E-UTRA-ENDC                            ProtocolIE-ID ::= 401
id-CellMeasurementResult-E-UTRA-ENDC-Item                       ProtocolIE-ID ::= 402
id-CellToReport-E-UTRA-ENDC                                     ProtocolIE-ID ::= 403
id-CellToReport-E-UTRA-ENDC-Item                                ProtocolIE-ID ::= 404
id-TraceCollectionEntityURI                                     ProtocolIE-ID ::= 405
id-SFN-Offset                                                   ProtocolIE-ID ::= 406
id-CHO-DC-EarlyDataForwarding                                   ProtocolIE-ID ::= 407
id-IMSvoiceEPSfallbackfrom5G                                    ProtocolIE-ID ::= 408
id-AdditionLocationInformation                                  ProtocolIE-ID ::= 409
id-DirectForwardingPathAvailability                             ProtocolIE-ID ::= 410
id-sourceNG-RAN-node-id                                         ProtocolIE-ID ::= 411
id-SourceDLForwardingIPAddress                                  ProtocolIE-ID ::= 412
id-SourceNodeDLForwardingIPAddress                              ProtocolIE-ID ::= 413
id-NRRACHReportInformation                                      ProtocolIE-ID ::= 414
id-SCG-UE-HistoryInformation                                    ProtocolIE-ID ::= 415
id-PSCellHistoryInformationRetrieve                             ProtocolIE-ID ::= 416
id-MeasurementResultforNRCellsPossiblyAggregated                ProtocolIE-ID ::= 417
id-PSCell-UE-HistoryInformation                                 ProtocolIE-ID ::= 418
id-PSCellChangeHistory                                          ProtocolIE-ID ::= 419
id-CHOinformation-AddReq                                        ProtocolIE-ID ::= 420
id-CHOinformation-ModReq                                        ProtocolIE-ID ::= 421
id-SCGActivationStatus                                          ProtocolIE-ID ::= 422
id-SCGActivationRequest                                         ProtocolIE-ID ::= 423
id-CPAinformation-REQ                                           ProtocolIE-ID ::= 424
id-CPAinformation-REQ-ACK                                       ProtocolIE-ID ::= 425
id-CPAinformation-MOD                                           ProtocolIE-ID ::= 426
id-CPAinformation-MOD-ACK                                       ProtocolIE-ID ::= 427
id-CPACinformation-REQD                                         ProtocolIE-ID ::= 428
id-CPCinformation-REQD                                          ProtocolIE-ID ::= 429
id-CPCinformation-CONF                                          ProtocolIE-ID ::= 430
id-CPCinformation-NOTIFY                                        ProtocolIE-ID ::= 431
id-CPCupdate-MOD                                                ProtocolIE-ID ::= 432
id-Additional-Measurement-Timing-Configuration-List             ProtocolIE-ID ::= 433
id-ServedCellSpecificInfoReq-NR                                 ProtocolIE-ID ::= 434
id-SecurityIndication                                           ProtocolIE-ID ::= 435
id-SecurityResult                                               ProtocolIE-ID ::= 436
id-RAT-Restrictions                                             ProtocolIE-ID ::= 437
id-SCGreconfigNotification                                      ProtocolIE-ID ::= 438
id-MIMOPRBusageInformation                                      ProtocolIE-ID ::= 439
id-SensorMeasurementConfiguration                               ProtocolIE-ID ::= 440
```

```
id-AdditionalListofForwardingGTPTunnelEndpoint                  ProtocolIE-ID ::= 441

END
-- ASN1STOP
```

## 9.3.8 Container definitions

```
-- ASN1START
-- **************************************************************
--
-- Container definitions
--
-- **************************************************************

X2AP-Containers {
itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
eps-Access (21) modules (3) x2ap (2) version1 (1) x2ap-Containers (5) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--
-- IE parameter types from other modules.
--
-- **************************************************************

IMPORTS
    maxPrivateIEs,
    maxProtocolExtensions,
    maxProtocolIEs,
    Criticality,
    Presence,
    PrivateIE-ID,
    ProtocolIE-ID
FROM X2AP-CommonDataTypes;

-- **************************************************************
--
-- Class Definition for Protocol IEs
--
-- **************************************************************

X2AP-PROTOCOL-IES ::= CLASS {
    &id             ProtocolIE-ID           UNIQUE,
    &criticality    Criticality,
    &Value,
    &presence       Presence
}
WITH SYNTAX {
    ID              &id
    CRITICALITY     &criticality
```

```
    TYPE            &Value
    PRESENCE        &presence
}

-- **************************************************************
--
-- Class Definition for Protocol IEs
--
-- **************************************************************

X2AP-PROTOCOL-IES-PAIR ::= CLASS {
    &id                     ProtocolIE-ID       UNIQUE,
    &firstCriticality       Criticality,
    &FirstValue,
    &secondCriticality      Criticality,
    &SecondValue,
    &presence               Presence
}
WITH SYNTAX {
    ID                      &id
    FIRST CRITICALITY       &firstCriticality
    FIRST TYPE              &FirstValue
    SECOND CRITICALITY      &secondCriticality
    SECOND TYPE             &SecondValue
    PRESENCE                &presence
}

-- **************************************************************
--
-- Class Definition for Protocol Extensions
--
-- **************************************************************

X2AP-PROTOCOL-EXTENSION ::= CLASS {
    &id                 ProtocolIE-ID       UNIQUE,
    &criticality        Criticality,
    &Extension,
    &presence           Presence
}
WITH SYNTAX {
    ID                  &id
    CRITICALITY         &criticality
    EXTENSION           &Extension
    PRESENCE            &presence
}

-- **************************************************************
--
-- Class Definition for Private IEs
--
-- **************************************************************

X2AP-PRIVATE-IES ::= CLASS {
    &id                 PrivateIE-ID,
```

```
    &criticality        Criticality,
    &Value,
    &presence           Presence
}
WITH SYNTAX {
    ID                  &id
    CRITICALITY         &criticality
    TYPE                &Value
    PRESENCE            &presence
}

-- ************************************************************
--
-- Container for Protocol IEs
--
-- ************************************************************

ProtocolIE-Container {X2AP-PROTOCOL-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
    ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-Single-Container {X2AP-PROTOCOL-IES : IEsSetParam} ::=
    ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-Field {X2AP-PROTOCOL-IES : IEsSetParam} ::= SEQUENCE {
    id              X2AP-PROTOCOL-IES.&id                   ({IEsSetParam}),
    criticality     X2AP-PROTOCOL-IES.&criticality          ({IEsSetParam}{@id}),
    value           X2AP-PROTOCOL-IES.&Value                ({IEsSetParam}{@id})
}

-- ************************************************************
--
-- Container for Protocol IE Pairs
--
-- ************************************************************

ProtocolIE-ContainerPair {X2AP-PROTOCOL-IES-PAIR : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
    ProtocolIE-FieldPair {{IEsSetParam}}

ProtocolIE-FieldPair {X2AP-PROTOCOL-IES-PAIR : IEsSetParam} ::= SEQUENCE {
    id                  X2AP-PROTOCOL-IES-PAIR.&id                  ({IEsSetParam}),
    firstCriticality    X2AP-PROTOCOL-IES-PAIR.&firstCriticality    ({IEsSetParam}{@id}),
    firstValue          X2AP-PROTOCOL-IES-PAIR.&FirstValue          ({IEsSetParam}{@id}),
    secondCriticality   X2AP-PROTOCOL-IES-PAIR.&secondCriticality   ({IEsSetParam}{@id}),
    secondValue         X2AP-PROTOCOL-IES-PAIR.&SecondValue         ({IEsSetParam}{@id})
}

-- ************************************************************
--
-- Container Lists for Protocol IE Containers
--
-- ************************************************************
```

```
ProtocolIE-ContainerList {INTEGER : lowerBound, INTEGER : upperBound, X2AP-PROTOCOL-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (lowerBound..upperBound)) OF
    ProtocolIE-Container {{IEsSetParam}}

ProtocolIE-ContainerPairList {INTEGER : lowerBound, INTEGER : upperBound, X2AP-PROTOCOL-IES-PAIR : IEsSetParam} ::=
    SEQUENCE (SIZE (lowerBound..upperBound)) OF
    ProtocolIE-ContainerPair {{IEsSetParam}}

-- **************************************************************
--
-- Container for Protocol Extensions
--
-- **************************************************************

ProtocolExtensionContainer {X2AP-PROTOCOL-EXTENSION : ExtensionSetParam} ::=
    SEQUENCE (SIZE (1..maxProtocolExtensions)) OF
    ProtocolExtensionField {{ExtensionSetParam}}

ProtocolExtensionField {X2AP-PROTOCOL-EXTENSION : ExtensionSetParam} ::= SEQUENCE {
    id                  X2AP-PROTOCOL-EXTENSION.&id             ({ExtensionSetParam}),
    criticality         X2AP-PROTOCOL-EXTENSION.&criticality    ({ExtensionSetParam}{@id}),
    extensionValue      X2AP-PROTOCOL-EXTENSION.&Extension      ({ExtensionSetParam}{@id})
}

-- **************************************************************
--
-- Container for Private IEs
--
-- **************************************************************

PrivateIE-Container {X2AP-PRIVATE-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (1..maxPrivateIEs)) OF
    PrivateIE-Field {{IEsSetParam}}

PrivateIE-Field {X2AP-PRIVATE-IES : IEsSetParam} ::= SEQUENCE {
    id              X2AP-PRIVATE-IES.&id            ({IEsSetParam}),
    criticality     X2AP-PRIVATE-IES.&criticality   ({IEsSetParam}{@id}),
    value           X2AP-PRIVATE-IES.&Value         ({IEsSetParam}{@id})
}

END
-- ASN1STOP
```

## 9.4 Message transfer syntax

X2AP shall use the ASN.1 Basic Packed Encoding Rules (BASIC-PER) Aligned Variant as transfer syntax, as specified in ITU-T Rec. X.691 [5].

## 9.5 Timers

$T_{RELOCprep}$

- Specifies the maximum time for the Handover Preparation procedure in the source eNB.

$TX2_{RELOCoverall}$

- Specifies the maximum time for the protection of the overall handover procedure in the source eNB.

$T_{DCprep}$

- Specifies the maximum time for the SeNB Addition Preparation, MeNB initiated SeNB Modification Preparation, SgNB Addition Preparation, or MeNB initiated SgNB Modification Preparation procedure in the MeNB.

$T_{DCoverall}$

- Specifies the maximum time in the SeNB for either the SeNB initiated SeNB Modification procedure or the protection of the E-UTRAN actions necessary to configure UE resources at SeNB Addition or MeNB initiated SeNB Modification. Or specifies the maximum time in the SgNB for either the SgNB initiated SgNB Modification procedure or the protection of the E-UTRAN actions necessary to configure UE resources at SgNB Addition or MeNB initiated SgNB Modification.

# 10 Handling of unknown, unforeseen and erroneous protocol data

Section 10 of TS 36.413 [4] is applicable for the purposes of the present document.

# Annex A (informative): Change history

| TSG # | TSG Doc. | CR | Rev | Subject/Comment | New |
|---|---|---|---|---|---|
| 09/2009 | | | | Rel-9 version is created based on v.8.7.0 | 9.0.0 |
| 45 | RP-090787 | 0296 | 1 | Handling of Emergency Calls in Limited Service Mode | 9.0.0 |
| 45 | RP-090787 | 0297 | 1 | Emergency Calls Mobility Handling | 9.0.0 |
| 46 | RP-091192 | 0307 | | Introduction of signalling support for Composite Available Capacity with relative units | 9.1.0 |
| 46 | RP-091192 | 0308 | 2 | Configuration adaptation for MLB on X2 | 9.1.0 |
| 46 | RP-091183 | 0310 | 1 | Clarification on operational use of updated configuration data | 9.1.0 |
| 46 | RP-091192 | 0317 | 2 | Automatic PRACH information exchange over X2 for SON | 9.1.0 |
| 46 | RP-091192 | 0333 | 1 | Introduction of Radio Link Failure Indication procedure | 9.1.0 |
| 46 | RP-091192 | 0334 | 1 | Introduction of Handover Report procedure | 9.1.0 |
| 46 | RP-091192 | 0335 | | Introduction of signalling support for Composite Available Capacity with relative units | 9.1.0 |
| 47 | RP-100213 | 0337 | | Correction to the Resource Status Reporting Initiation procedure | 9.2.0 |
| 47 | RP-100229 | 0341 | 2 | Addition of MBSFN information on X2 interface | 9.2.0 |
| 47 | RP-100228 | 0344 | 4 | Cell pair identification for Mobility Settings Change procedure | 9.2.0 |
| 47 | RP-100213 | 0352 | | Addition of cause value for not admitted E-RAB | 9.2.0 |
| 47 | RP-100229 | 0355 | 1 | Rapporteur's update of X2AP protocol | 9.2.0 |
| 47 | RP-100230 | 0356 | 3 | RNL-based energy saving solution | 9.2.0 |
| 47 | RP-100228 | 0358 | 1 | Inclusion of UE RLF Report in RLF INDICATION message | 9.2.0 |
| 48 | RP-100599 | 0363 | 1 | Correction of RLF INDICATION message | 9.3.0 |
| 48 | RP-100599 | 0364 | 1 | Missing error cause for Not supported QCI on Handover | 9.3.0 |
| 48 | RP-100599 | 0370 | 1 | Introduction of PLMN-related abnormal conditions during X2 handover in network sharing scenarios. | 9.3.0 |
| 48 | RP-100599 | 0372 | 1 | Outcome of RAN3#68 review of X2AP | 9.3.0 |
| 48 | RP-100599 | 0373 | 1 | Correction of forbidden inter-RAT | 9.3.0 |
| 49 | RP-100908 | 0376 | 1 | Explicit PLMN coding in Trace IEs | 9.4.0 |
| 49 | RP-100906 | 0380 | 2 | The corrections for Last Visited Cell Information | 9.4.0 |
| 49 | RP-100906 | 0383 | 1 | Handover Restriction List | 9.4.0 |
| 49 | RP-100908 | 0384 | 1 | Complete list of served cells to be provided in X2 SETUP and eNB Configuration Update messages | 9.4.0 |
| 50 | RP-101271 | 0385 | | Clarification on Handover Restriction List | 9.5.0 |
| 50 | RP-101270 | 0403 | 3 | Correction of semantics description | 9.5.0 |
| 12/2010 | | | | Rel-10 version created based in v. 9.5.0 | 10.0.0 |
| 50 | RP-101304 | 0393 | 2 | Introduction of partial failure in Resource Status Reporting Initiation procedure including detailed reporting of failure cause | 10.0.0 |
| 50 | RP-101279 | 0407 | 4 | X2 handover support | 10.0.0 |
| SP-49 | SP-100629 | | | Clarification on the use of References (TS 21.801 CR#0030) | 10.1.0 |
| 51 | RP-110231 | 0408 | | Conditions for Enhanced X2 mobility | 10.1.0 |
| 51 | RP-110237 | 0409 | | Introduction of X2 signalling support for eICIC | 10.1.0 |
| 51 | RP-110222 | 0411 | 1 | Correction of the usage of optional ShortMAC-I IE in RLF INDICATION message | 10.1.0 |
| 51 | RP-110230 | 0413 | 2 | Support for MDT | 10.1.0 |
| 51 | RP-110226 | 0419 | 2 | Clarification on TEID value range for X2AP | 10.1.0 |
| 51 | RP-110231 | 0420 | | Clarify X2 Handover Scenarios | 10.1.0 |
| 51 | RP-110237 | 0427 | 1 | Enabling reporting of ABS resource status for eICIC purposes | 10.1.0 |
| 52 | RP-110695 | 0435 | 1 | MDT correction for TAI | 10.2.0 |
| 52 | RP-110698 | 0436 | 1 | Clarification on Radio Resource Status | 10.2.0 |
| 52 | RP-110700 | 0443 | 4 | X2 support of RLF Report extension for SON MRO defined in R10 | 10.2.0 |
| 52 | RP-110695 | 0447 | 3 | Support for MDT user consent | 10.2.0 |
| 52 | RP-110686 | 0451 | 2 | Rapporteur's proposal following review of TS 36.423 | 10.2.0 |
| 52 | RP-110689 | 0452 | 1 | Correction of the partial success mechanism in Resource Status Reporting | 10.2.0 |
| 52 | RP-110695 | 0453 | 2 | MDT amendments | 10.2.0 |
| 52 | RP-110685 | 0454 | | Reference review outcome in TS 36.423 | 10.2.0 |
| 52 | RP-110695 | 0456 | | Correction of trace function and trace session | 10.2.0 |
| 53 | RP-111196 | 0464 | 2 | Clarification of procedures defined for MLB purposes | 10.3.0 |
| 53 | RP-111196 | 0469 | 1 | ASN.1 definition conforms to ITU-T Recommendations | 10.3.0 |
| 53 | RP-111194 | 0476 | 2 | Updates of reported quantities for eICIC | 10.3.0 |
| 53 | RP-111195 | 0478 | 1 | Definition of value of bit in Measurements to Activate | 10.3.0 |
| 53 | RP-111197 | 0479 | | Clarification on PLMN Identity | 10.3.0 |
| 54 | RP-111648 | 0480 | 2 | Correction on ABS Information | 10.4.0 |
| 55 | RP-120234 | 0491 | 1 | Correct of reset | 10.5.0 |
| 03/2012 | | | | Rel-11 version created based in v. 10.5.0 | 11.0.0 |

| | | | | | |
|---|---|---|---|---|---|
| 55 | RP-120236 | 0487 | 1 | Addition of TAC to the neighbour information of a served cell for X2 setup and eNB update procedures | 11.0.0 |
| 56 | RP-120751 | 0496 | - | Introduction of the Security Algorithm (ZUC) | 11.1.0 |
| 56 | RP-120751 | 0498 | 2 | Clarification on TAC in X2 Setup | 11.1.0 |
| 56 | RP-120751 | 0501 | 3 | Adding RRC re-establishment cause to RLF indication | 11.1.0 |
| 56 | RP-120752 | 0513 | 1 | Correction on Emergency ARP Value | 11.1.0 |
| 56 | RP-120752 | 0516 | 1 | Improved granularity for the time UE stayed in cell | 11.1.0 |
| 57 | RP-121137 | 0520 | 2 | Support of MBMS Service Continuity | 11.2.0 |
| 57 | RP-121140 | 0527 | 3 | Multiband support per cell | 11.2.0 |
| 57 | RP-121135 | 0540 | 1 | Enhancement of HO REPORT to enable inter-RAT ping-pong detection and addition of HO cause value to the UE history information | 11.2.0 |
| 57 | RP-121139 | 0546 | | Support for new special subframe configurations | 11.2.0 |
| 58 | RP-121731 | 0548 | | Addition of Mobility Information | 11.3.0 |
| 58 | RP-121730 | 0549 | 3 | Introduction of new MDT measurements | 11.3.0 |
| 58 | RP-121732 | 0550 | 1 | HeNB Mobility enhancement when target is hybrid HeNB | 11.3.0 |
| 58 | RP-121730 | 0552 | 2 | Multi-PLMN MDT | 11.3.0 |
| 58 | RP-121731 | 0564 | | Clarification on successful handover for HO report procedure | 11.3.0 |
| 58 | RP-121737 | 0569 | 2 | X2AP Rapporteur Update | 11.3.0 |
| 59 | RP-130208 | 0572 | 3 | Correction on the Special Subframe Pattern | 11.4.0 |
| 59 | RP-130208 | 0580 | 2 | Support for Downlink-Only Bands | 11.4.0 |
| 59 | RP-130207 | 0581 | | Correction on use of Mobility Information | 11.4.0 |
| 59 | RP-130207 | 0582 | 1 | Correction on MRO procedures | 11.4.0 |
| 59 | RP-130237 | 0583 | 2 | Extending maxEARFCN | 11.4.0 |
| 59 | RP-130237 | 0584 | 1 | Extending Maximum Frequency Band Index | 11.4.0 |
| 59 | RP-130211 | 0585 | 1 | Rapporteur correction of X2AP | 11.4.0 |
| 59 | RP-130207 | 0586 | | Clarification on Signalling Based MDT PLMN List | 11.4.0 |
| 59 | RP-130210 | 0587 | 1 | X2AP modification for PDCP SN extension | 11.4.0 |
| 60 | RP-130643 | 0588 | | Correction on the Definition of Direct Neighbours | 11.5.0 |
| 60 | RP-130641 | 0589 | 1 | Correction for the MDT Location Information IE | 11.5.0 |
| 60 | RP-130640 | 0590 | 5 | Correction on RLF INDICATION procedure | 11.5.0 |
| 60 | RP-130643 | 0592 | 1 | Security key generation in case of MFBI | 11.5.0 |
| 60 | RP-130643 | 0593 | 2 | Correction on the Multiple Frequency Band Indicators | 11.5.0 |
| 61 | RP-131181 | 0598 | 1 | Correction on Handover Report procedure | 11.6.0 |
| 61 | RP-131179 | 0602 | 2 | Correction on ABS Information | 11.6.0 |
| 61 | RP-131183 | 0606 | 1 | Correction of terminology concerning the mobility restriction function | 11.6.0 |
| 62 | RP-131902 | 0609 | 3 | Correction of Handover Restriction List | 11.7.0 |
| 62 | RP-131902 | 0611 | 1 | Correction for Load Balancing Related cause value CR for 36423 | 11.7.0 |
| 62 | RP-131902 | 0623 | 2 | Correction for Load Balancing Related IE | 12.0.0 |
| 62 | RP-131909 | 0607 | 3 | Handling SIPTO@LN during UE Context Release procedure | 12.0.0 |
| 63 | RP-140294 | 0634 | | Correction to tabular of Served Cell Information IE | 12.1.0 |
| 64 | RP-140901 | 0629 | 4 | TDD eIMTA support on X2AP | 12.2.0 |
| 64 | RP-140906 | 0630 | 4 | Provide IMEISV to eNB to identify UE characteristics | 12.2.0 |
| 64 | RP-140905 | 0661 | 1 | Correction of SN STATUS TRANSFER | 12.2.0 |
| 64 | RP-140905 | 0676 | | Clarification of DL ABS status | 12.2.0 |
| 64 | RP-140897 | 0641 | 4 | Introduce X2GW procedures in Stage-3 | 12.2.0 |
| 65 | RP-141520 | 0663 | 3 | Introduction of the UE history reported from the UE | 12.3.0 |
| 65 | RP-141518 | 0690 | 2 | Introduction of an indication of the expected UE behaviour | 12.3.0 |
| 66 | RP-142089 | 0691 | 8 | Introduction of Dual Connectivity | 12.4.0 |
| 66 | RP-142090 | 0692 | 10 | Introduction of inter-eNB CoMP signalling | 12.4.0 |
| 66 | RP-142092 | 0748 | 5 | X2 support for Network Assisted Interference Cancellation | 12,4.0 |
| 66 | RP-142094 | 0754 | 2 | X2AP Rapporteur Update | 12.4.0 |
| 66 | RP-142094 | 0759 | 2 | Correction on RLF Report Container | 12.4.0 |
| 66 | RP-142094 | 0776 | 2 | Setting of Re-establishment Cell ID in RLF Indication message | 12.4.0 |
| 66 | RP-142094 | 0777 | 3 | X2 Removal Signaling | 12.4.0 |
| 12/2014 | | | | History table corrected | 12.4.1 |
| 12/2014 | | | | ASN.1 correction to make it compilable | 12.4.2 |
| 67 | RP-150353 | 0693 | 5 | ProSe authorized indication | 12.5.0 |
| 67 | RP-150351 | 0782 | 1 | Corrections on the usage of SeNB UE AMBR in dual connectivity | 12.5.0 |
| 67 | RP-150351 | 0790 | 1 | Corrections of Dual Connectivity in general | 12.5.0 |
| 67 | RP-150356 | 0797 | 1 | Correction on DC stage3 | 12.5.0 |
| 67 | RP-150348 | 0801 | 1 | Correction of the Usage of the MultibandInfoList IE | 12.5.0 |
| 67 | RP-150351 | 0802 | 1 | Introduction of Cause values for Dual Connectivity | 12.5.0 |
| 67 | RP-150356 | 0803 | 1 | ASN.1 Corrections for X2AP | 12.5.0 |
| 67 | RP-150351 | 0804 | 2 | Corrections for Dual Connectivity | 12.5.0 |

| | | | | | |
|---|---|---|---|---|---|
| 67 | RP-150356 | 0805 | | Miscellaneous Editorials for X2AP | 12.5.0 |
| 67 | RP-150351 | 0806 | 1 | Correction on SeNB behaviour for distinguishing uplink PDCP PDUs | 12.5.0 |
| 68 | RP-150943 | 0807 | 1 | Correction on the definition of SeNB Reconfiguration Complete | 12.6.0 |
| 68 | RP-150943 | 0827 | 1 | Introduction of a new DC cause for not supported configurations | 12.6.0 |
| 68 | RP-150943 | 0831 | | Clarification on UE-AMBR for split bearer | 12.6.0 |
| 06/2015 | | | | Rel-13 version created based in v. 12.6.0 | 13.0.0 |
| 68 | RP-150945 | 0808 | 8 | Addition of Intra-LTE notifications of AAS-based reconfigurations | 13.0.0 |
| 69 | RP-151455 | 0788 | 11 | Introduction of enhanced inter-eNB CoMP signalling | 13.1.0 |
| 69 | RP-151451 | 0854 | 1 | Correction on GBR parameters for dual connectivity | 13.1.0 |
| 69 | RP-151450 | 0877 | 1 | Handling of Unknown or Erroneous AP IDs in Dual Connectivity | 13.1.0 |
| 70 | RP-152100 | 0850 | 5 | UE-to-Network Relay authorization | 13.2.0 |
| 70 | RP-152099 | 0892 | 2 | Extension of PDCP SN | 13.2.0 |
| 70 | RP-152102 | 0901 | 4 | Adding CSG support to DC | 13.2.0 |
| 70 | RP-152086 | 0907 | | Correction on inter eNB CoMP | 13.2.0 |
| 70 | RP-152102 | 0910 | 5 | Support of SIPTO stand-alone architecture in dual connectivity | 13.2.0 |
| 70 | RP-152102 | 0911 | 2 | Support of SIPTO and LIPA in dual connectivity | 13.2.0 |
| 70 | RP-152102 | 0912 | 6 | Support of handover without SeNB change | 13.2.0 |
| 70 | RP-152102 | 0916 | 2 | Handling of User Inactivity in the SeNB | 13.2.0 |
| 70 | RP-152086 | 0918 | | Correction of Subband Index | 13.2.0 |
| 70 | RP-152085 | 0924 | 4 | Correction of intra cell handovers in multiband deployments | 13.2.0 |
| 70 | RP-152102 | 0927 | 2 | Extension of UE X2AP ID | 13.2.0 |
| 70 | RP-152102 | 0929 | 2 | SIPTO@LN and LIPA bearer deactivation for DC | 13.2.0 |
| 70 | RP-152103 | 0932 | 3 | Introduction of feMDT | 13.2.0 |
| 70 | RP-152108 | 0936 | 2 | Addition of the Cell Deployment Status Indicator and replacing cell information | 13.2.0 |
| 70 | RP-152102 | 0939 | 1 | Tunnel Information of BBAI in Dual Connectivity | 13.2.0 |
| 71 | RP-160449 | 0937 | 3 | Addition of X2 Removal Threshold to the X2 Removal Request message | 13.3.0 |
| 71 | RP-160449 | 0949 | 2 | Modification of an ongoing resource reporting procedure | 13.3.0 |
| 71 | RP-160448 | 0950 | 1 | Correction on SeNB Addition Preparation concerning inter-MeNB handover without SeNB change | 13.3.0 |
| 71 | RP-160448 | 0953 | 1 | Correction on usage of extended eNB UE X2AP ID | 13.3.0 |
| 71 | RP-160448 | 0954 | | Correction for SeNB Addition behaviour Abnormal | 13.3.0 |
| 71 | RP-160451 | 0959 | | Clarification on the abnormal condition for DC SIPTO@LN | 13.3.0 |
| 71 | RP-160449 | 0962 | 1 | Rapporteur's Update | 13.3.0 |
| 71 | RP-160448 | 0963 | 3 | Correction on Old/New eNB UE X2AP ID | 13.3.0 |
| 72 | RP-161042 | 0965 | 7 | Introduction of the inter-eNB UE Context Resume function | 13.4.0 |
| 72 | RP-161043 | 0968 | 1 | Correction on the DC function description | 13.4.0 |
| 72 | RP-161043 | 0969 | 3 | Correction on eNB UE X2AP ID Extension | 13.4.0 |
| 72 | RP-161043 | 0972 | 2 | Indication of Bearer Type for cIOT | 13.4.0 |
| 72 | RP-161047 | 0978 | | Correction of RSRP Measurement Report List | 13.4.0 |
| 73 | RP-161551 | 0989 | 1 | Correction on NB-IoT inter node RRC container | 13.5.0 |
| 73 | RP-161550 | 0998 | 2 | Correction on Security Related Information in UE Context Retrieval Request | 13.5.0 |
| 09/2016 | | | | Rel-14 version created based in v. 13.5.0 | 14.0.0 |
| 73 | RP-161552 | 0975 | 5 | Vehicular Authorization Signaling over X2 | 14.0.0 |
| 74 | RP-162337 | 1007 | | Clarification on V2X Services Authorized IE | 14.1.0 |
| 74 | RP-162340 | 1008 | 3 | Target cell selection for low complexity UEs and UEs in enhanced coverage | 14.1.0 |
| 74 | RP-162340 | 1011 | - | Correction to Served Cell Information for NB-IoT | 14.1.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | Meeting | TDoc | CR | Rev | Cat | Subject/Comment | New version |
| 03/2017 | RP-75 | RP-170535 | 1023 | | B | X2AP Support for Inter-eNB Mobility without WT Change | 14.2.0 |
| 03/2017 | RP-75 | RP-170537 | 1005 | 3 | B | Introduction of a new special subframe configuration | 14.2.0 |
| 03/2017 | RP-75 | RP-170538 | 1025 | | B | Support of V2X over X2 | 14.2.0 |
| 03/2017 | RP-75 | RP-170542 | 1026 | | B | Introduction of New types of eNB ID | 14.2.0 |
| 03/2017 | RP-75 | RP-170536 | 1024 | 1 | B | Introduction of eMOB Stage3 | 14.2.0 |
| 06/2017 | RP-75 | RP-171329 | 1033 | 1 | A | Introduction of UL TNL address in CIoT UP Solution | 14.3.0 |
| 09/2017 | RP-77 | RP-171974 | 1035 | 1 | F | Correction on NB-IoT UP mobility | 14.4.0 |
| 09/2017 | RP-77 | RP-171974 | 1037 | 1 | F | Correction of SeNB Release Confirm | 14.4.0 |
| 12/2017 | RP-78 | RP-172673 | 1044 | 2 | F | Introduction of new IEs in X2 for high performing load balance | 14.5.0 |
| 12/2017 | RP-78 | RP-172715 | 1046 | - | F | Correction of mismatched tabular and ASN.1 | 14.5.0 |
| 12/2017 | RP-78 | RP-172672 | 1041 | 6 | B | Baseline CR to TS 36.423 covering agreements of RAN3 #98 | 15.0.0 |
| 12/2017 | RP-78 | RP-172674 | 1045 | 1 | B | Introduction of QoE Measurement Collection for LTE | 15.0.0 |
| 03/2018 | RP-79 | RP-180468 | 1050 | - | B | X2AP corrections for agreed EN-DC BL CR | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1051 | 1 | F | Essential corrections for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1052 | 1 | F | Clarification on HRL for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1053 | - | F | Correction of counter Check procedure for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1054 | - | B | Support for supplementary UL carrier | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1056 | - | F | Correction on SgNB initiated SgNB Modification procedure | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1061 | 1 | F | Correction of mandatory/optional/Conditional IEs in 36.423 | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1067 | 2 | | Support for S-RLF | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1071 | 2 | F | Update of EN-DC X2 Setup and EN-DC Configuration Update | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1073 | - | F | Removal of wrong abnormal behaviour that does not exist in EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1078 | 1 | | CR for addition of cause | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1079 | 2 | F | Clarification and correction on X2 for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1081 | 1 | F | Corrections for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1082 | 1 | F | Resolve the remaining issues over X2 for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1083 | 1 | F | Introduction of DRB ID for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180314 | 1087 | 1 | F | Removing data forwarding from the corresponding node for EN-DC | 15.1.0 |
| 03/2018 | RP-79 | RP-180472 | 1092 | 1 | F | Rapporteur correction of 36.423 before NSA ASN.1 freeze | 15.1.0 |
| 03/2018 | RP-79 | RP-180473 | 1093 | - | A | Correction on Offset of NB-IoT Channel Number to EARFCN | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1094 | - | F | Correction of TAC for NG-RAN cells before NSA ASN.1 freeze | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1095 | - | F | Remove PDCP change indication in SN modification request message | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1096 | - | F | Change the presence of container in SgNB reconfiguration complete procedure | 15.1.0 |
| 03/2018 | RP-79 | RP-180468 | 1097 | - | F | Addition of Measurement Timing Configuration information | 15.1.0 |
| 06/2018 | RP-80 | RP-181241 | 1047 | 6 | B | Support of Enhanced VoLTE Performance | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1065 | 4 | F | X2 partial reset for EN-DC | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1068 | 1 | F | Clarification of the interactions with the UE Context Release | 15.2.0 |
| 06/2018 | RP-80 | RP-181241 | 1086 | 3 | C | Introduction of QMC for MTSI in EUTRAN | 15.2.0 |
| 06/2018 | RP-80 | RP-181237 | 1090 | 9 | B | Baseline CR for E-UTRA - NR Cell Resource Coordination for TS 36.423 covering agreements of RAN3#100 | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1104 | - | F | Correction of UL link configuration in TS36.423 | 15.2.0 |
| 06/2018 | RP-80 | RP-181410 | 1107 | 4 | F | Addition of the full config indicator | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1116 | 1 | F | Correction of the SeNB Reconfiguration Completion procedure | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1117 | 2 | F | Correction of abnormal conditions for EN-DC security algorithm selection | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1121 | 1 | F | Correction of reference in RRC Container | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1122 | - | F | Correction of condition presence of E-RAB Level QoS Parameters related | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1123 | 1 | F | Support of TEID change at SN | 15.2.0 |
| 06/2018 | RP-80 | RP-181237 | 1125 | 4 | B | X2AP CR for support of NR Multiple frequency band in EN-DC | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1130 | - | F | Correction of max NR ARFCN value | 15.2.0 |
| 06/2018 | RP-80 | RP-181243 | 1132 | 3 | B | Baseline CR: Introduction of the Aerial Usage Indication | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1133 | 1 | F | Use of SPID for EN-DC | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1134 | 1 | F | Correction of references to RRC containers for EN-DC | 15.2.0 |
| 06/2018 | RP-80 | RP-181238 | 1135 | - | F | Corrections on Tabular indentation and ASN.1 criticality | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1138 | - | F | Adding missing relation for the TEID | 15.2.0 |
| 06/2018 | RP-80 | RP-181241 | 1142 | 3 | B | Retrieve UE Context at UE Re-establishment | 15.2.0 |
| 06/2018 | RP-80 | RP-181241 | 1143 | - | D | Rapporteur's corrections on the specification | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1145 | - | F | Correction on the same terminology of "Split SRB" in TS36.423 | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1146 | 2 | F | Correction of Split SRB configuration in TS36.423 | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1149 | 3 | F | CR for Clarification on resource coordination | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1152 | - | F | Correction for PDCP Duplication | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1153 | - | F | Coordination of Inactivity for EN-DC | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1155 | - | C | Introduction of CN type restriction | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1158 | - | F | User Inactivity handling over X2 EN-DC | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1160 | 1 | F | Addition of Cause Value | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1161 | 2 | F | Addition of MeNB cell ID to solve the PCI confusion | 15.2.0 |

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| 06/2018 | RP-80 | RP-181239 | 1164 | - | F | Corrections on misalignment between tabular and ASN.1 | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1165 | 1 | F | Introduction of EN-DC X2 removal procedure | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1167 | - | F | Support of DL TEID change over S1 at SN | 15.2.0 |
| 06/2018 | RP-80 | RP-181242 | 1174 | 2 | B | Support of NB-IoT measurement enhancement and TDD Config | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1175 | - | F | ASN.1 correction for EN-DC support in TS 36.423 | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1176 | 1 | F | Introduction of a Configured TAC into the NR Neighbour Information IE and the Served NR Cell Information IE | 15.2.0 |
| 06/2018 | RP-80 | RP-181239 | 1178 | - | F | Correction of the Limited List IE encoding to enable extensibility | 15.2.0 |
| 09/2018 | RP-81 | RP-181920 | 1115 | 4 | F | Indication of the RLC re-establishment at the assisting node | 15.3.0 |
| 09/2018 | RP-81 | RP-181920 | 1190 | - | F | Addition of RRC config indication to SGNB MODIFICATION REQUIRED and SGNB MODIFICATION REQUEST ACKNOWLEDGE | 15.3.0 |
| 09/2018 | RP-81 | RP-181920 | 1191 | 2 | F | Clarification on secondary RAT data volume reporting | 15.3.0 |
| 09/2018 | RP-81 | RP-181920 | 1192 | 1 | F | Essential corrections for EN-DC | 15.3.0 |
| 09/2018 | RP-81 | RP-181920 | 1193 | 1 | F | Corrections on EN-DC Resource Configuration | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1196 | 3 | F | Notification of PDCP SN length change | 15.3.0 |
| 09/2018 | RP-81 | RP-181920 | 1198 | 1 | F | Corrections on E-UTRA – NR Cell Resource Coordination | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1201 | 2 | F | RLC Mode Indication over X2 – for 36.423 | 15.3.0 |
| 09/2018 | RP-81 | RP-181922 | 1202 | 4 | F | Baseline CR for TS 36.423 covering agreements of RAN3#AH1807 and RAN3#101 | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1203 | 2 | F | Correction of "Maximum MCG admittable E-RAB Level QoS Parameters" | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1206 | - | F | X2 Corrections for EN-DC | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1211 | 1 | F | Access Restriction Data for NR in EPC | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1214 | 2 | C | Extension of Data Traffic Resources IE for E-UTRA-NR Cell Resource Coordination | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1217 | - | F | Correction of 5GS TAC | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1221 | 1 | F | CR on clarification of successfully delivered for NR-U | 15.3.0 |
| 09/2018 | RP-81 | RP-181923 | 1226 | 3 | F | Data forwarding for Retrieve UE Context in case of RRC connection re-establishment | 15.3.0 |
| 09/2018 | RP-81 | RP-181924 | 1231 | 1 | F | CR to X2AP to introduce Bluetooth and WLAN measurement in MDT | 15.3.0 |
| 09/2018 | RP-81 | RP-182127 | 1233 | 4 | B | Introduction of Subscription based UE differentiation | 15.3.0 |
| 09/2018 | RP-81 | RP-181921 | 1235 | - | F | Correction of SgNB Activity Notification Procedure | 15.3.0 |
| 12/2018 | RP-82 | RP-182447 | 1237 | 4 | F | Addition of the RLC Mode information for PDCP transfer | 15.4.0 |
| 12/2018 | RP-82 | RP-182446 | 1243 | 3 | F | Correction on PDCP SN length | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1244 | 2 | F | Support of CA based PDCP duplication on X2 | 15.4.0 |
| 12/2018 | RP-82 | RP-182446 | 1245 | 1 | F | CR on Introduction of overload indication over X2 | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1246 | 1 | F | CR on alingment of terminology for eNB or MeNB | 15.4.0 |
| 12/2018 | RP-82 | RP-182446 | 1247 | 1 | F | Correction of SgNB Initiated SN Modification procedure for Measurement Gap | 15.4.0 |
| 12/2018 | RP-82 | RP-182446 | 1248 | - | F | ASN.1 corrections on NRNeighbour-Information IE and NRFreqInfo IE | 15.4.0 |
| 12/2018 | RP-82 | RP-182446 | 1250 | - | F | Correction on E-UTRA - NR resource coordination | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1253 | 3 | F | Corrections of MeNB/SgNB resource coordination | 15.4.0 |
| 12/2018 | RP-82 | RP-182446 | 1256 | 1 | F | Correction on SGNB ACTIVITY NOTIFICATION IE's | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1259 | 1 | F | Correction of PDCP SN Length Indication | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1264 | 2 | F | RLC reestablishment indication for TS36.423 | 15.4.0 |
| 12/2018 | RP-82 | RP-182504 | 1267 | 1 | F | Update on Retrieve UE Context Request message for TS36.423 | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1272 | 1 | F | Handling of RLC failure | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1273 | - | F | Add missing description on non-operational X2 interface for EN-DC | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1275 | 2 | F | Further corrections of MeNB/SgNB resource coordination | 15.4.0 |
| 12/2018 | RP-82 | RP-182447 | 1279 | 1 | F | Criticality Correction for X2AP UE-ID | 15.4.0 |
| 12/2018 | RP-82 | RP-182437 | 1280 | 2 | F | Allowing SgNB to request new DRB ID from MeNB in EN-DC for an already established SN terminated bearer | 15.4.0 |
| 03/2019 | RP-83 | RP-190555 | 1282 | 3 | F | Correction to RRC transfer | 15.5.0 |
| 03/2019 | RP-83 | RP-190555 | 1283 | 2 | F | Transfer of the PSCell information for LI purposes | 15.5.0 |
| 03/2019 | RP-83 | RP-190555 | 1285 | - | F | Enabling using Dual Connectivity cause values in EN-DC | 15.5.0 |
| 03/2019 | RP-83 | RP-190555 | 1287 | 1 | F | Desired Activity Notification Level | 15.5.0 |
| 03/2019 | RP-83 | RP-190555 | 1291 | - | F | Introduction of IMEISV to Addition Request over X2 | 15.5.0 |
| 03/2019 | RP-83 | RP-190555 | 1292 | 1 | F | Clarification on the usage of coordination assistance information | 15.5.0 |
| 03/2019 | RP-83 | RP-190556 | 1297 | 1 | F | Introducing NR Neighbour information in X2 Setup | 15.5.0 |
| 03/2019 | RP-83 | RP-190555 | 1298 | - | F | Rapporteur updates on version 15.4.0 | 15.5.0 |
| 03/2019 | RP-83 | RP-190523 | 1300 | 3 | F | Adding Trace Messages in X2AP | 15.5.0 |
| 03/2019 | RP-83 | RP-190556 | 1301 | 1 | F | Correction of EPC interworking | 15.5.0 |
| 03/2019 | RP-83 | RP-190555 | 1302 | - | F | Straighten-up SgNB's request to release and add the same SN-terminated bearer with different DRB ID | 15.5.0 |
| 03/2019 | RP-83 | RP-190561 | 1304 | 1 | F | Introduction of TNL Address discovery for EN-DC (using new container) | 15.5.0 |
| 2019-07 | RP-84 | RP-191395 | 1299 | 2 | F | Correction of MaxnoofBPLMNs for NR | 15.6.0 |
| 2019-07 | RP-84 | RP-191395 | 1307 | 1 | F | RRC config indication behaviour | 15.6.0 |
| 2019-07 | RP-84 | RP-191395 | 1308 | - | F | Transferring of NR RRC message in MeNB | 15.6.0 |

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| 2019-07 | RP-84 | RP-191396 | 1313 | 1 | F | PDCP SN length related clean-up over To Be Mofidified structure in SN initiated SN Modifcation procedure | 15.6.0 |
| 2019-07 | RP-84 | RP-191429 | 1314 | 5 | F | RAN sharing with multiple Cell ID broadcast | 15.6.0 |
| 2019-07 | RP-84 | RP-191395 | 1315 | - | F | SN Status Transfer applicability for Re-establishment | 15.6.0 |
| 2019-07 | RP-84 | RP-191395 | 1316 | 2 | F | Rapporteur's corrections to version 15.5.0 | 15.6.0 |
| 2019-07 | RP-84 | RP-191397 | 1317 | 1 | F | Correction of Core Network Type Restriction | 15.6.0 |
| 2019-07 | RP-84 | RP-191395 | 1318 | - | F | CR36423 for Addition of MN (MeNB) cell ID to solve the PCI confusion in SN(SgNB) modification Request message | 15.6.0 |
| 2019-07 | RP-84 | RP-191395 | 1321 | 1 | F | Updates on TS 36.423 for EN-DC TNL Address discovery | 15.6.0 |
| 2019-07 | RP-84 | RP-191394 | 1330 | 1 | F | PDCP SN length related clean-up over To Be Modified structure in MN initiated SN Modification procedure | 15.6.0 |
| 2019-09 | RP-85 | RP-192166 | 1322 | 2 | F | Correction of handling of the Location Information at the MeNB | 15.7.0 |
| 2019-09 | RP-85 | RP-192169 | 1336 | - | F | Correction on Data Forwarding Address Indication | 15.7.0 |
| 2019-09 | RP-85 | RP-192170 | 1339 | 1 | A | Correction on Handover Request Acknowledge | 15.7.0 |
| 2019-09 | RP-85 | RP-192169 | 1341 | 1 | F | Correction of NB-IoT TDD Cell Frequency info | 15.7.0 |
| 2019-09 | RP-85 | RP-192169 | 1359 | 1 | F | Non IP bearer support for Dual Connectivity | 15.7.0 |
| 2019-12 | RP-86 | RP-192916 | 1346 | 3 | F | Critical correction to the presence of the SgNB UE X2AP ID in the SgNB Release Request Reject | 15.8.0 |
| 2019-12 | RP-86 | RP-192916 | 1364 | 3 | F | Correction to SN Status Transfer considering EN-DC operations | 15.8.0 |
| 2019-12 | RP-86 | RP-192916 | 1380 | 2 | F | SN Status Transfer for bearer reconfiguration during HO with EN-DC | 15.8.0 |
| 2019-12 | RP-86 | RP-192915 | 1398 | - | F | Correction on the DL forwarding for MeNB terminated bearer in SgNB initiated SgNB Release | 15.8.0 |
| 2019-12 | RP-86 | RP-192916 | 1410 | 1 | F | CR36.423 for correction on EN-DC X2 SETUP REQUEST message | 15.8.0 |
| 2019-12 | RP-86 | RP-192916 | 1415 | 2 | F | Support of delta configuration in EN-DC | 15.8.0 |
| 2019-12 | RP-86 | RP-192913 | 1311 | 5 | B | Introduction of Additional RRM Policy Index (ARPI) | 16.0.0 |
| 2019-12 | RP-86 | RP-192910 | 1391 | 4 | B | Resuming SCG in RRC Resume | 16.0.0 |
| 2019-12 | RP-86 | RP-192910 | 1416 | 2 | B | Fast MCG link recovery via SRB3 | 16.0.0 |
| 2019-12 | RP-86 | RP-192913 | 1418 | 1 | F | Introduction of message size control for EN-DC X2 Setup | 16.0.0 |
| 2019-12 | RP-86 | RP-192692 | 1421 | 2 | F | Support for setting up IPSec a priori in X2 | 16.0.0 |
| 2020-03 | RP-87-e | RP-200425 | 1390 | 3 | B | Support for Multiple SCTP | 16.1.0 |
| 2020-03 | RP-87-e | RP-200422 | 1408 | 2 | B | Introduction of NR-U | 16.1.0 |
| 2020-03 | RP-87-e | RP-200429 | 1426 | - | A | Correction on Assigned Criticality for Bearer Type | 16.1.0 |
| 2020-03 | RP-87-e | RP-200419 | 1438 | 1 | B | SA to ENDC handover with shared SgNB/gNB | 16.1.0 |
| 2020-03 | RP-87-e | RP-200427 | 1448 | 1 | F | Cleanup for Fast MCG link Recovery with SRB3 | 16.1.0 |
| 2020-03 | RP-87-e | RP-200428 | 1452 | - | A | Correction of CR1380r2 to explicate procedural interaction | 16.1.0 |
| 2020-03 | RP-87-e | RP-200429 | 1454 | 1 | A | Correction of tabular representation of the RRC TRANSFER message | 16.1.0 |
| 2020-03 | RP-87-e | RP-200428 | 1456 | 1 | A | Correction of CR1415r2 – procedure text | 16.1.0 |
| 2020-03 | RP-87-e | RP-200428 | 1460 | - | A | Propagation of Roaming and Access Restriction information in E-UTRAN in non-homogenous eNB deployments | 16.1.0 |
| 2020-03 | RP-87-e | RP-200425 | 1462 | - | F | Correction of CR1418 on X2 Setup Message Size Control | 16.1.0 |
| 2020-03 | RP-87-e | RP-200425 | 1463 | - | F | Rapporteur Corrections Rel-16 | 16.1.0 |
| 2020-03 | RP-87-e | RP-200423 | 1468 | - | B | X2AP support for Radio Capability Signaling Optimization (The CR is not implemented. The CR was marked agreed by mistake while the WI is not yet complete) | 16.1.0 |
| 2020-07 | RP-88-e | RP-201077 | 1303 | 16 | B | BL CR to 36.423: Support for IAB | 16.2.0 |
| 2020-07 | RP-88-e | RP-201089 | 1331 | 15 | B | Baseline CR for introducing Rel-16 LTE further mobility enhancements | 16.2.0 |
| 2020-07 | RP-88-e | RP-201079 | 1340 | 9 | B | Support of Ethernet Type Bearer | 16.2.0 |
| 2020-07 | RP-88-e | RP-201074 | 1369 | 12 | B | Support of NR V2X over X2 | 16.2.0 |
| 2020-07 | RP-88-e | RP-201082 | 1373 | 12 | B | Addition of SON feature | 16.2.0 |
| 2020-07 | RP-88-e | RP-201088 | 1374 | 4 | B | CR for 36.423 on NB-IoT PRACH configuration exchange over X2AP | 16.2.0 |
| 2020-07 | RP-88-e | RP-201088 | 1427 | 4 | B | Support of RLF in NB-IoT | 16.2.0 |
| 2020-07 | RP-88-e | RP-201082 | 1440 | 6 | B | MDT support for EN-DC | 16.2.0 |
| 2020-07 | RP-88-e | RP-201078 | 1468 | 4 | B | X2AP support for Radio Capability Signaling Optimization | 16.2.0 |
| 2020-07 | RP-88-e | RP-201085 | 1472 | - | F | Correction of the criticality of the TNLA to add/update/remove list | 16.2.0 |
| 2020-07 | RP-88-e | RP-201090 | 1475 | 2 | A | Encoding PLMNs in served cell information NR | 16.2.0 |
| 2020-07 | RP-88-e | RP-201085 | 1478 | 1 | F | Rapporteur's Correction to X2AP version 16.1.0 | 16.2.0 |
| 2020-07 | RP-88-e | RP-201085 | 1479 | - | F | Wrong ASN.1 IE-Id for the New eNB UE X2AP ID Extension IE | 16.2.0 |
| 2020-07 | RP-88-e | RP-201091 | 1485 | - | A | CR 36.423 Correction to E-UTRA-NR Cell-level Resource Coordination | 16.2.0 |
| 2020-07 | RP-88-e | RP-201091 | 1489 | | F | Correction on nested SN modification procedure for EN-DC | 16.2.0 |
| 2020-07 | RP-88-e | RP-201090 | 1491 | - | A | Encoding PLMNs in served cell information IEs - semantics corrections | 16.2.0 |
| 2020-07 | RP-88-e | RP-201085 | 1492 | 1 | F | CR to TS36.423 on Correction of R3-202726 Agreed for EN-DC CSI-RS Transfer | 16.2.0 |
| 2020-07 | RP-88-e | RP-201090 | 1494 | 3 | A | Clarification on MIB only scenario | 16.2.0 |
| 2020-07 | RP-88-e | RP-201092 | 1502 | - | A | Tabular and ASN.1 correction of messages of the EN-DC X2 Setup and EN-DC Configuration Update procedures | 16.2.0 |
| 2020-07 | RP-88-e | RP-201076 | 1503 | - | B | Inter-RAT HO support for fast MCG recovery | 16.2.0 |
| 2020-07 | RP-88-e | RP-201085 | 1505 | 2 | F | Correction on RF parameters in NR cell information | 16.2.0 |

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| 2020-09 | RP-89-e | RP-201946 | 1504 | 2 | F | Further correction on fast MCG recovery via SRB3 | 16.3.0 |
| 2020-09 | RP-89-e | RP-201948 | 1511 | | B | Introduction of NR SCG Release for Power Saving | 16.3.0 |
| 2020-09 | RP-89-e | RP-201948 | 1513 | 1 | F | Support for intended TDD configuration transfer for EN-DC | 16.3.0 |
| 2020-09 | RP-89-e | RP-201948 | 1518 | 2 | F | Clarification of the TNL Capacity Indicator | 16.3.0 |
| 2020-09 | RP-89-e | RP-201951 | 1525 | - | F | Essential correction for Rel-16 LTE_feMob-Core WI | 16.3.0 |
| 2020-09 | RP-89-e | RP-201951 | 1526 | 1 | F | Rapporteur's corrections to TS 36.423 v16.2.0 | 16.3.0 |
| 2020-09 | RP-89-e | RP-201947 | 1528 | 1 | F | Rapporteur Corrections for NR SON MDT WI and IAB WI | 16.3.0 |
| 2020-09 | RP-89-e | RP-201952 | 1529 | - | F | Missing ASN.1 of TNL Transport Layer Address Info IE in the EN-DC X2 SETUP RESPONSE message | 16.3.0 |
| 2020-09 | RP-89-e | RP-201954 | 1530 | - | A | Missing MeNB UE X2AP ID Extension IE in Trace messages | 16.3.0 |
| 2020-09 | RP-89-e | RP-201954 | 1531 | - | A | Correction for the Interface Instance Indication in the EN-DC X2 SETUP RESPONSE message | 16.3.0 |
| 2020-09 | RP-89-e | RP-201948 | 1532 | 1 | B | Introducing UE Radio Capability ID Mapping procedure | 16.3.0 |
| 2020-09 | RP-89-e | RP-201951 | 1537 | - | F | Correction on the Maximum Number of CHO Preparations in X2AP | 16.3.0 |
| 2020-09 | RP-89-e | RP-201954 | 1539 | 2 | A | Clarification on TAC presence in Serving Cell Info over X2 | 16.3.0 |
| 2020-09 | RP-89-e | RP-201948 | 1542 | - | F | Correction on protocol IE for MDTConfigurationNR | 16.3.0 |
| 2020-12 | RP-90-e | RP-202314 | 1545 | 1 | F | Correction on CPC Complete Transfer | 16.4.0 |
| 2020-12 | RP-90-e | RP-202310 | 1550 | - | F | Correction of IAB related RRC Container in RRC TRANSFER message | 16.4.0 |
| 2020-12 | RP-90-e | RP-202310 | 1552 | 1 | F | Clarification for F1-C Traffic Container IE | 16.4.0 |
| 2020-12 | RP-90-e | RP-202313 | 1562 | - | F | Removal of duplicated imports | 16.4.0 |
| 2020-12 | RP-90-e | RP-202312 | 1563 | 1 | F | Load information for SN-initiated DC release | 16.4.0 |
| 2020-12 | RP-90-e | RP-202312 | 1565 | | F | Enabling URI configuration within Trace Activation over X2 | 16.4.0 |
| 2021-03 | RP-91-e | RP-210799 | 1367 | 8 | B | Introduction of SFN Offset per cell over X2 | 16.5.0 |
| 2021-03 | RP-91-e | RP-210239 | 1561 | 4 | F | Cause value on X2 for insufficient UE capabilities CR 36.423 | 16.5.0 |
| 2021-03 | RP-91-e | RP-210239 | 1580 | 1 | F | Cause value on X2 for normal release CR 38.423 | 16.5.0 |
| 2021-06 | RP-92-e | RP-211331 | 1543 | 3 | F | Clarification of the use of the max no of CHO preparations | 16.6.0 |
| 2021-06 | RP-92-e | RP-211331 | 1588 | 2 | F | 36.423 correction for CHO early data forwarding in MN to eNB Change scenario | 16.6.0 |
| 2021-06 | RP-92-e | RP-211317 | 1593 | 1 | F | Correction of en-gNB initiated EN-DC Resource Status Reporting | 16.6.0 |
| 2021-06 | RP-92-e | RP-211332 | 1594 | 1 | F | Correction of IMS Voice EPS Fallback | 16.6.0 |
| 2021-06 | RP-92-e | RP-211333 | 1600 | 2 | F | Correction on SN initiated SN Modification procedure for EN-DC | 16.6.0 |
| 2021-06 | RP-92-e | RP-211328 | 1603 | 1 | F | Addition of MeNB to SN sidelink resource coordination | 16.6.0 |
| 2021-09 | RP-93-e | RP-211874 | 1612 | 1 | F | Correction on EN-DC case | 16.7.0 |
| 2021-12 | RP-94-e | RP-212863 | 1630 | 1 | A | SN triggerred SCG release | 16.8.0 |
| 2021-12 | RP-94-e | RP-212863 | 1637 | - | A | Transfer of PSCell Location Reporting control information at X2 mobility | 16.8.0 |
| 2021-12 | RP-94-e | RP-212860 | 1645 | 1 | F | Direct data forwarding for mobility between DC and SA | 16.8.0 |
| 2021-12 | RP-94-e | RP-212864 | 1648 | - | A | Correction on EN-DC X2 Removal for RAN Sharing in Rel-16 | 16.8.0 |
| 2022-03 | RP-95-e | RP-220279 | 1639 | 3 | F | Dynamic ACL over X2 CR 36.423 | 16.9.0 |
| 2022-03 | RP-95-e | RP-220276 | 1658 | 1 | F | Correction of Qos Mapping Information IE in X2AP message for IAB (CR to TS36.423, R16) | 16.9.0 |
| 2022-03 | RP-95-e | RP-220275 | 1662 | 2 | A | Correction on UE X2AP ID in the ERROR INDICATION message | 16.9.0 |
| 2022-03 | RP-95-e | RP-220278 | 1664 | 2 | F | Correction of indirect data forwarding for MR-DC | 16.9.0 |
| 2022-03 | RP-95-e | RP-220279 | 1670 | 1 | F | Correction of frequency information for DL only cell for ENDC | 16.9.0 |
| 2022-03 | RP-95-e | RP-220279 | 1672 | | F | Correction on the ASN.1 for neighbour cell information | 16.9.0 |
| 2022-03 | RP-95-e | RP-220284 | 1679 | 1 | A | Misalignment on M8 and M9 measurement configurations | 16.9.0 |
| 2022-03 | RP-95-e | RP-220221 | 1564 | 6 | B | BLCR to 36.423: Support of MDT enhancement | 17.0.0 |
| 2022-03 | RP-95-e | RP-220221 | 1589 | 8 | B | BLCR to 36.423_Addition of SON features enhancement | 17.0.0 |
| 2022-03 | RP-95-e | RP-220237 | 1590 | 3 | C | Enabling CHO with SCG configuration [CHOwithDCkept] | 17.0.0 |
| 2022-03 | RP-95-e | RP-220218 | 1609 | 9 | B | SCG BL CR to TS 36.423 | 17.0.0 |
| 2022-03 | RP-95-e | RP-220218 | 1610 | 8 | B | CPAC BL CR to TS 36.423 | 17.0.0 |
| 2022-03 | RP-95-e | RP-220237 | 1614 | 1 | B | Signalling of Neighbour cell CSI-RS configuration information over X2 [CSIRSX2] | 17.0.0 |
| 2022-03 | RP-95-e | RP-220237 | 1624 | 3 | C | Support for mapping complete security capabilities from NAS [UE_Sec_Caps] | 17.0.0 |
| 2022-03 | RP-95-e | RP-220237 | 1642 | 2 | B | CSI-RS configuration request Indicator [CSIRSX2] | 17.0.0 |
| 2022-03 | RP-95-e | RP-220241 | 1663 | 6 | B | Supporting EPS User Plane Integrity Protection | 17.0.0 |
| 2022-03 | RP-95-e | RP-220216 | 1665 | 3 | B | (BL CR for 36.423) IoT over NTN | 17.0.0 |
| 2022-03 | RP-95-e | RP-220237 | 1673 | - | D | X2AP Rapporteur Corrections | 17.0.0 |
| 2022-06 | RP-96 | RP-221135 | 1681 | - | F | Alignment of the criticality of CPC Cancel with XnAP ASN.1 | 17.1.0 |
| 2022-06 | RP-96 | RP-221147 | 1682 | - | F | Correction on CHO Information SN Modification [CHOwithDCkept] | 17.1.0 |
| 2022-06 | RP-96 | RP-221135 | 1684 | 1 | F | Correction on CPAC | 17.1.0 |
| 2022-06 | RP-96 | RP-221135 | 1685 | | F | CPAC ASN.1 corrections over X2 | 17.1.0 |
| 2022-06 | RP-96 | RP-221150 | 1687 | | A | X2AP rapporteur's corrections | 17.1.0 |
| 2022-06 | RP-96 | RP-221135 | 1692 | 1 | F | Correction on CPAC | 17.1.0 |
| 2022-06 | RP-96 | RP-221150 | 1694 | 1 | A | Dynamic ACL over X2 CR 36.423 | 17.1.0 |
| 2022-06 | RP-96 | RP-221150 | 1697 | 1 | F | Correction of data forwarding for SA to EN-DC handover | 17.1.0 |
| 2022-06 | RP-96 | RP-221144 | 1700 | 2 | F | Correction of EPS Integrity Protection | 17.1.0 |
| 2022-06 | RP-96 | RP-221135 | 1703 | | F | ASN.1 Correction for Rel-17 SCG on SCG Activation Status IE | 17.1.0 |
| 2022-06 | RP-96 | RP-221135 | 1708 | | F | Rel-17 Correction for X2AP on the interaction with SN-intiated SCG (de)activation and SN Addition procedure | 17.1.0 |

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| 2022-09 | RP-97-e | RP-222194 | 1710 | 2 | F | Coordination of CHO and intra-SN SCG reconfiguration | 17.2.0 |
| 2022-09 | RP-97-e | RP-222197 | 1711 | - | F | Timer handling for CHO with SCG configuration [CHOwithDCkept] | 17.2.0 |
| 2022-09 | RP-97-e | RP-222196 | 1712 | 1 | F | Correction of the wrong constant definition in ASN.1 | 17.2.0 |
| 2022-09 | RP-97-e | RP-222185 | 1713 | 2 | B | CR for TS36.423 on Extending NR Operation to 71GHz | 17.2.0 |
| 2022-09 | RP-97-e | RP-222197 | 1717 | 1 | F | Correction of LTE MDT on Sensor information[LTE-Height-MDT] | 17.2.0 |
| 2022-09 | RP-97-e | RP-222196 | 1718 | 1 | F | Correction for TS 36.423 on UHI in MR-DC | 17.2.0 |
| 2022-12 | RP-98-e | RP-222878 | 1722 | 2 | F | Additional indicator for CHO-CPC coordination | 17.3.0 |
| 2022-12 | RP-98-e | RP-222878 | 1723 | 1 | F | Direct early data forwarding in SN initiated inter-SN CPC | 17.3.0 |
| 2023-03 | RAN#99 | RP-230582 | 1731 | 1 | F | Completion of the work on CHO-CPC coordination | 17.4.0 |
| 2023-03 | RAN#99 | RP-230593 | 1735 | 2 | F | X2AP corrections of references to RRC | 17.4.0 |
| 2023-03 | RAN#99 | RP-230582 | 1736 | 1 | F | Correction on coordination of CHO and CPC over X2 | 17.4.0 |

# History

| Document history | | |
|---|---|---|
| V17.0.0 | May 2022 | Publication |
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